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Applied Stochastic Control of Jump Diffusions

Third Edition

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Bernt Øksendal · Agnès Sulem

Applied Stochastic Control of Jump Diffusions

Third Edition

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To my family
Eva, Elise, Anders, and Karina
Bernt Øksendal

A tous ceux qui m'accompagnent
Agnès Sulem

Preface to the Third Edition

In this third edition, we have expanded and updated the second edition and included more recent developments within stochastic control and its applications.

Specifically, we have replaced Section 1.5 on application to finance by a more comprehensive presentation of financial markets modeled by jump diffusions (the new Chap. 2). We have added a new chapter on backward stochastic differential equations, convex risk measures, and recursive utilities (Chap. 4). Moreover, we have expanded the optimal stopping chapter (was Chap. 2, now Chap. 3) and the stochastic control chapter (was Chap. 3, now Chap. 5) and added a new chapter on stochastic differential games (Chap. 6). In addition, we have corrected errors and updated and improved the presentation throughout the book.

We wish to thank Nacira Agram, Kristina Dahl, Olfa Draouil, Michele Giordano, Huigi Guan, Gambarra Matteo, Sandun Perara, Andriy Pylypenko, and Yngve Willassen for their helpful comments.

Last but not least, we are very grateful to Martine Verneuille for her valuable help with the typing through many years.

Oslo, Norway — Bernt Øksendal
Paris, France — Agnès Sulem
2018

Preface to the Second Edition

In this second edition, we have added a chapter on optimal control of random jump fields (solutions of stochastic *partial* differential equations) and partial information control (Chap. 13). We have also added a section on optimal stopping with delayed information (Sect. 3.3). It has always been our intention to give a contemporary presentation of applied stochastic control, and we hope that the addition of these recent developments will contribute in this direction.

We have also made a number of corrections and other improvements, many of them based on helpful comments from our readers. In particular, we would like to thank Andreas Kyprianou for his valuable communications. We are also grateful to (in alphabetical order) Knut Aase, Jean-Philippe Chancelier, Inga Eide, Emil Framnes, Arne-Christian Lund, Jose-Luis Menaldi, Tamás K. Papp, Atle Seierstad, and Jens Arne Sukkestad for pointing out errors and suggesting improvements. Our special thanks go to Martine Verneuille for her skillful typing.

Oslo and Paris — Bernt Øksendal
November 2006 — Agnès Sulem

Preface to the First Edition

Jump diffusions are solutions of stochastic differential equations driven by Lévy processes. Since a Lévy process $\eta(t)$ can be written as a linear combination of t, a Brownian motion $B(t)$ and a pure jump process, jump diffusions represent a natural and useful generalization of Itô diffusions. They have received a lot of attention in the last years because of their many applications, particularly in economics.

There exist today several excellent monographs on Lévy processes. However, very few of them—if any—discuss the optimal control, optimal stopping, and impulse control of the corresponding jump diffusions, which is the subject of this book. Moreover, our presentation differs from these books in that it emphasizes the applied aspect of the theory. Therefore, we focus mostly on useful verification theorems, and we illustrate the use of the theory by giving examples and exercises throughout the text. Detailed solutions of some of the exercises are given at the end of the book. The exercises to which a solution is provided are marked with an asterisk $*$. It is our hope that this book will fill a gap in the literature and that it will be a useful text for students, researchers, and practitioners in stochastic analysis and its many applications. Although most of our results are motivated by examples in economics and finance, the results are general and can be applied in a wide variety of situations. To emphasize this, we have also included examples in biology and physics/engineering.

This book is partially based on courses given at the Norwegian School of Economics and Business Administration (NHH) in Bergen, Norway, during the Spring semesters 2000 and 2002, at INSEA in Rabat, Morocco in September 2000, at Odense University in August 2001, and at ENSAE in Paris in February 2002.

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exercises. We are grateful to Dina Haraldsson and Martine Verneuille for proficient typing and Eivind Brodal for his kind assistance. We acknowledge with gratitude the support by the French–Norwegian cooperation project Stochastic Control and Applications, Aur 99-050.

Oslo and Paris
August 2004

Bernt Øksendal
Agnès Sulem

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Chapter 1
Stochastic Calculus with Lévy Processes



1.1 Basic Definitions and Results on Lévy Processes

In this chapter we present the basic concepts and results needed for the applied calculus of jump diffusions. Since there are several excellent books which give a detailed account of this basic theory, we will just briefly review it here and refer the reader to these books for more information.

Definition 1.1 Let $(\Omega, \mathcal{F}, \mathbb{F} = \{\mathcal{F}_t\}_{t\geq 0}, P)$ be a filtered probability space. An $\mathbb{F}_t$-adapted process $\{\eta(t)\}_{t\geq 0} = \{\eta_t\}_{t\geq 0} \subset \mathbb{R}$ with $\eta_0 = 0$ a.s. is called a *Lévy process* if η_t is continuous in probability and has *stationary and independent* increments.

Theorem 1.2 *Let $\{\eta_t\}$ be a Lévy process. Then η_t has a* càdlàg *version (right continuous with left limits) which is also a Lévy process.*

Proof See, e.g., [P, S]. □

In view of this result we will from now on assume that the Lévy processes we work with are càdlàg.

The *jump* of η_t at $t \geq 0$ is defined by

$$\Delta\eta_t = \eta_t - \eta_{t^-}. \tag{1.1.1}$$

Let $\mathbf{B}_0$ be the family of Borel sets $U \subset \mathbb{R}$ whose closure $\bar{U}$ does not contain 0. For $U \in \mathbf{B}_0$ we define

$$N(t, U) = N(t, U, \omega) = \sum_{s:0<s\leq t} \mathcal{X}_U(\Delta\eta_s). \tag{1.1.2}$$

In other words, $N(t, U)$ is the number of jumps of size $\Delta\eta_s \in U$ which occur before or at time t. $N(t, U)$ is called the *Poisson random measure (or jump measure)* of $\eta(\cdot)$.

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Remark 1.3 Note that $N(t, U)$ is *finite* for all $U \in \mathbf{B}_0$. To see this we proceed as follows: Define

$$T_1(\omega) = \inf\{t > 0; \eta_t \in U\}.$$

We claim that $T_1(\omega) > 0$ a.s. To prove this note that by right continuity of paths we have

$$\lim_{t \to 0^+} \eta(t) = \eta(0) = 0 \quad \text{a.s.}$$

Therefore, for all $\varepsilon > 0$ there exists $t(\varepsilon) > 0$ such that $|\eta(t)| < \varepsilon$ for all $t < t(\varepsilon)$. This implies that $\eta(t) \notin U$ for all $t < t(\varepsilon)$, if $\varepsilon < \text{dist}(0, U)$.

Next define inductively

$$T_{n+1}(\omega) = \inf\{t > T_n(\omega); \Delta\eta_t \in U\}.$$

Then by the above argument $T_{n+1} > T_n$ a.s. We claim that

$$T_n \to \infty \quad \text{as} \quad n \to \infty, \text{a.s.}$$

Assume not. Then $T_n \to T < \infty$. But then

$$\lim_{s \to T^-} \eta(s) \quad \text{cannot exist,}$$

contradicting the existence of left limits of the paths.

It is well known that Brownian motion $\{B(t)\}_{t \geq 0}$ has stationary and independent increments. Thus $B(t)$ is a Lévy process. Another important example is the following.

Example 1.4 (***The Poisson Process***) The Poisson process $\pi(t)$ of intensity $\lambda > 0$ is a Lévy process taking values in $\mathbb{N} \cup \{0\}$ and such that

$$P[\pi(t) = n] = \frac{(\lambda t)^n}{n!} e^{-\lambda t}; \quad n = 0, 1, 2, \ldots$$

Theorem 1.5 ([P, Theorem 1.35])

1. *The set function $U \to N(t, U, \omega)$ defines a σ-finite measure on $\mathbf{B}_0$ for each fixed t, ω. The differential form of this measure is written $N(t, \mathrm{d}z)$.*
2. *The set function $[a, b) \times U \to N(b, U, \omega) - N(a, U, \omega)$; $[a, b) \subset [0, \infty)$, $U \in \mathbf{B}_0$ defines a σ-finite measure for each fixed ω. The differential form of this measure is written $N(\mathrm{d}t, \mathrm{d}z)$.*
3. *The set function*

$$\nu(U) = E[N(1, U)], \tag{1.1.3}$$

where $E = E_P$ denotes expectation with respect to P, also defines a σ-finite measure on $\mathbf{B}_0$, called the Lévy measure *of $\{\eta_t\}$.*

4. *Fix $U \in \mathbf{B}_0$. Then the process*

$$\pi_U(t) := \pi_U(t, \omega) := N(t, U, \omega)$$

is a Poisson process *of* intensity $\lambda = \nu(U)$.

Example 1.6 (***The Compound Poisson Process***) Let $X(n), n \in \mathbb{N}$ be a sequence of i.i.d. random variables taking values in $\mathbb{R}$ with common distribution $\mu_{X(1)} = \mu_X$ and let $\pi(t)$ be a Poisson process of intensity λ, independent of all the $X(n)$s.

The *compound Poisson process* $Y(t)$ is defined by

$$Y(t) = X(1) + \cdots + X(\pi(t)), \quad t \geq 0. \tag{1.1.4}$$

An increment of this process is given by

$$Y(s) - Y(t) = \sum_{k=\pi(t)+1}^{\pi(s)} X(k), \quad s > t.$$

This is independent of $X(1), \ldots, X(\pi(t))$, and its distribution depends only on the difference $(s - t)$ and on the distribution of $X(1)$. Thus $Y(t)$ is a Lévy process.

To find the Lévy measure ν of $Y(t)$ note that if $U \in \mathbf{B}_0$ then

$$\begin{aligned}\nu(U) = E[N(1, U)] &= E\left[\sum_{s;0<s\leq 1} \mathcal{X}_U(\Delta Y(s))\right] \\ &= E[(\text{number of jumps}) \times \chi_U(\text{jump})] = E[\pi(1)\mathcal{X}_U(X)] = \lambda\mu_X(U),\end{aligned}$$

by independence. We conclude that

$$\nu = \lambda\mu_X. \tag{1.1.5}$$

This shows that a Lévy process can be represented by a compound Poisson process if and only if its Lévy measure is finite. Note, however, that there are many interesting Lévy processes η_t with infinite Lévy measure ν, in fact even with $\int_{|z|<1} |z|\nu(\mathrm{d}z) = \infty$. See, e.g., [S]. In general, one can prove that for any fixed $R > 0$ the processes

$$M_t^{(k)} := \int_{1/k\leq|z|\leq R} z(N(t, \mathrm{d}z) - t\nu(\mathrm{d}z)), \quad k = 1, 2, \ldots$$

are $L^2(P)$-martingales and they converge in $L^2(P)$ to a martingale M_t denoted by

$$M_t = \int_{|z|\leq R} z(N(t, \mathrm{d}z) - t\nu(\mathrm{d}z)).$$

The following result gives a general description of Lévy processes:

Theorem 1.7 (***Itô–Lévy Decomposition*** [JS]) *Let $\{\eta_t\}$ be a Lévy process. Then η_t has the decomposition*

$$\eta_t = \alpha t + \sigma B(t) + \int_{|z|<R} z\widetilde{N}(t, \mathrm{d}z) + \int_{|z|\geq R} zN(t, \mathrm{d}z), \tag{1.1.6}$$

for some constants $\alpha \in \mathbb{R}$, $\sigma \in \mathbb{R}$, $R \in [0, \infty]$. Here

$$\widetilde{N}(\mathrm{d}t, \mathrm{d}z) = N(\mathrm{d}t, \mathrm{d}z) - \nu(\mathrm{d}z)\mathrm{d}t \tag{1.1.7}$$

is the compensated Poisson random measure *of $\eta(\cdot)$, and $B(t)$ is a Brownian motion independent of $\widetilde{N}(\mathrm{d}t, \mathrm{d}z)$. For each $A \in \mathbf{B}_0$ the process*

$$M_t := \widetilde{N}(t, A) \quad \textit{is a martingale}. \tag{1.1.8}$$

If $\alpha = 0$ and $R = \infty$, we call η_t a Lévy martingale. *Note that we can always choose $R = 1$.*

Theorem 1.8 *If*

$$E|\eta_t| < \infty \quad \textit{for all } t \geq 0,$$

then

$$\int_{|z|\geq 1} |z|\nu(\mathrm{d}z) < \infty$$

and hence we may choose $R = \infty$ and write

$$\eta_t = \alpha_1 t + \sigma B(t) + \int_{\mathbb{R}} z\widetilde{N}(t, \mathrm{d}z).$$

where

$$\alpha_1 = \alpha + \int_{|z|\geq 1} z\nu(dz).$$

(See [S, Theorem 25.3].)

Theorem 1.9 ([P]) *A Lévy process is a strong Markov process.*

Theorem 1.10 (***The Lévy–Khintchine formula*** [P]) *Let $\{\eta_t\}$ be a Lévy process with Lévy measure ν. Then $\int_{\mathbb{R}} \min(1, z^2)\nu(\mathrm{d}z) < \infty$ and*

$$E[\mathrm{e}^{\mathrm{i}u\eta_t}] = \mathrm{e}^{t\psi(u)}, \quad u \in \mathbb{R}, \tag{1.1.9}$$

where

$$\psi(u) = -\frac{1}{2}\sigma^2 u^2 + \mathrm{i}\alpha u + \int_{|z|<R} \{\mathrm{e}^{\mathrm{i}uz} - 1 - \mathrm{i}uz\}\nu(\mathrm{d}z) + \int_{|z|\geq R} (\mathrm{e}^{\mathrm{i}uz} - 1)\nu(\mathrm{d}z). \quad (1.1.10)$$

Conversely, given constants α, σ^2, and a measure ν on $\mathbf{B}_0$ s.t.

$$\int_{\mathbb{R}} \min(1, z^2)\nu(\mathrm{d}z) < \infty,$$

there exists a Lévy process $\eta(t)$ (unique in law) such that (1.1.9)–(1.1.10) hold.

Note It is possible that $\int_{|z|\leq R} |z|\nu(\mathrm{d}z) = \infty$.

Theorem 1.11 ([P, Corollary p. 48]) *A Lévy process is a* semimartingale.

Definition 1.12 ([P]) Let $\mathbf{D}_{\mathrm{ucp}}$ denote the space of càdlàg adapted processes, equipped with the topology of uniform convergence on compacts in probability (ucp) : $H_n \to H$ ucp if for all $t > 0 \sup_{0\leq s\leq t} |H_n(s) - H(s)| \to 0$ in probability ($A_n \to A$ in probability if for all $\varepsilon > 0$ there exists $n_\varepsilon \in \mathbb{N}$ such that $n \geq n_\varepsilon \Rightarrow P(|A_n - A| > \varepsilon) < \varepsilon$).

Let $\mathbf{L}_{\mathrm{ucp}}$ denote the space of adapted *càglàd* processes (left continuous with right limits), equipped with the ucp topology. If $H(t)$ is a step function of the form

$$H(t) = H_0 \mathcal{X}_{\{0\}}(t) + \sum_i H_i \mathcal{X}_{(T_i, T_{i+1}]}(t),$$

where $H_i \in \mathcal{F}_{T_i}$ and $0 = T_0 \leq T_1 \leq \cdots \leq T_{n+1} < \infty$ are $\mathbb{F}$-stopping times and X is càdlàg, we define

$$J_X H(t) := \int_0^t H_s \mathrm{d}X_s := H_0 X_0 + \sum_i H_i (X_{T_{i+1}\wedge t} - X_{T_i \wedge t}), \quad t \geq 0.$$

Theorem 1.13 ([P]) *Let X be a semimartingale. Then the mapping J_X can be extended to a continuous linear map*

$$J_X : \mathbf{L}_{\mathrm{ucp}} \to \mathbf{D}_{\mathrm{ucp}}.$$

This construction allows us to define stochastic integrals of the form

$$\int_0^t H(s)\mathrm{d}\eta_s$$

for all $H \in \mathbf{L}_{\mathrm{ucp}}$. (See also Remark 1.18.) In view of the decomposition (1.1.6) this integral can be split into integrals with respect to $\mathrm{d}s$, $\mathrm{d}B(s)$, $\widetilde{N}(\mathrm{d}s, \mathrm{d}z)$, and

$N(\mathrm{d}s, \mathrm{d}z)$. This makes it natural to consider the more general stochastic integrals of the form

$$X(t) = X(0) + \int_0^t \alpha(s,\omega)\mathrm{d}s + \int_0^t \beta(s,\omega)\mathrm{d}B(s) + \int_0^t \int_{\mathbb{R}} \gamma(s,z,\omega)\bar{N}(\mathrm{d}s, \mathrm{d}z), \tag{1.1.11}$$

where the integrands are $\mathbb{F}$-predictable and satisfy the growth condition

$$\int_0^t \left\{ |\alpha(s)| + \beta^2(s) + \int_{\mathbb{R}} \gamma^2(s,z)\nu(dz) \right\} ds < \infty \text{ a.s. for all } t > 0.$$

Hence we for simplicity have put

$$\bar{N}(\mathrm{d}s, \mathrm{d}z) = \begin{cases} N(\mathrm{d}s, \mathrm{d}z) - \nu(\mathrm{d}z)\mathrm{d}s & \text{if } |z| < R \\ N(\mathrm{d}s, \mathrm{d}z) & \text{if } |z| \geq R, \end{cases}$$

with R as in Theorem 1.7. As is customary we will use the following shorthand differential notation for processes $X(t)$ satisfying (1.1.11):

$$\mathrm{d}X(t) = \alpha(t)\mathrm{d}t + \beta(t)\mathrm{d}B(t) + \int_{\mathbb{R}} \gamma(t,z)\bar{N}(\mathrm{d}t, \mathrm{d}z). \tag{1.1.12}$$

We call such processes *Itô–Lévy processes*.

Recall that a semimartingale $M(t)$ is called a *local* martingale up to time T (with respect to P) if there exists an increasing sequence of $\mathcal{F}_t$-stopping times τ_n such that $\lim_{n\to\infty} \tau_n = T$ a.s. and

$$M(t \wedge \tau_n) \quad \text{is a martingale with respect to } P \text{ for all } n.$$

Note that

1. If

$$E\left[\int_0^T \int_{\mathbb{R}} \gamma^2(t,z)\nu(\mathrm{d}z)\mathrm{d}t\right] < \infty, \tag{1.1.13}$$

then the process

$$M(t) := \int_0^t \int_{\mathbb{R}} \gamma(s,z)\tilde{N}(\mathrm{d}s, \mathrm{d}z), \quad 0 \leq t \leq T$$

is a *martingale*.

2. If

$$\int_0^T \int_{\mathbb{R}} \gamma^2(t,z)\nu(\mathrm{d}z)\mathrm{d}t < \infty \text{ a.s.}, \tag{1.1.14}$$

then $M(t)$ is a *local* martingale, $0 \leq t \leq T$.

1.2 The Itô Formula and Related Results

We now come to the important Itô formula for Itô–Lévy processes:

If $X(t)$ is given by (1.1.12) and $f : \mathbb{R}^2 \to \mathbb{R}$ is a C^2 function, is the process $Y(t) := f(t, X(t))$ again an Itô–Lévy process and if so, how do we represent it in the form (1.1.12)?

If we argue heuristically and use our knowledge of the classical Itô formula it is easy to guess what the answer is:

Let $X^{(c)}(t)$ be the continuous part of $X(t)$, i.e., $X^{(c)}(t)$ is obtained by removing the jumps from $X(t)$. Then an increment in $Y(t)$ stems from an increment in $X^{(c)}(t)$ plus the jumps (coming from $N(\cdot, \cdot)$). Hence in view of the classical Itô formula we would guess that

$$\begin{aligned} \mathrm{d}Y(t) = &\frac{\partial f}{\partial t}(t, X(t))\mathrm{d}t + \frac{\partial f}{\partial x}(t, X(t))\mathrm{d}X^{(c)}(t) + \frac{1}{2}\frac{\partial^2 f}{\partial x^2}(t, X(t)) \cdot \beta^2(t)\mathrm{d}t \\ &+ \int_{\mathbb{R}} \{f(t, X(t^-) + \gamma(t, z)) - f(t, X(t^-))\} N(\mathrm{d}t, \mathrm{d}z). \end{aligned}$$

It can be proved that our guess is correct. Since

$$\mathrm{d}X^{(c)}(t) = \left(\alpha(t) - \int_{|z|<R} \gamma(t, z)\nu(\mathrm{d}z)\right)\mathrm{d}t + \beta(t)\mathrm{d}B(t),$$

this gives the following result.

Theorem 1.14 (***The One-Dimensional Itô Formula*** [BL, A, P]) *Suppose $X(t) \in \mathbb{R}$ is an Itô–Lévy process of the form*

$$\mathrm{d}X(t) = \alpha(t, \omega)\mathrm{d}t + \sigma(t, \omega)\mathrm{d}B(t) + \int_{\mathbb{R}} \gamma(t, z, \omega)\tilde{N}(\mathrm{d}t, \mathrm{d}z), \tag{1.2.1}$$

where

$$\tilde{N}(\mathrm{d}t, \mathrm{d}z) = \begin{cases} N(\mathrm{d}t, \mathrm{d}z) - \nu(\mathrm{d}z)\mathrm{d}t & \text{if } |z| < R \\ N(\mathrm{d}t, \mathrm{d}z) & \text{if } |z| \geq R \end{cases} \tag{1.2.2}$$

for some $R \in [0, \infty]$.

Let $f \in C^2(\mathbb{R}^2)$ and define $Y(t) = f(t, X(t))$. Then $Y(t)$ is again an Itô–Lévy process and

$$\begin{aligned} \mathrm{d}Y(t) = &\frac{\partial f}{\partial t}(t, X(t))\mathrm{d}t + \frac{\partial f}{\partial x}(t, X(t))[\alpha(t, \omega)\mathrm{d}t + \sigma(t, \omega)\mathrm{d}B(t)] \\ &+ \frac{1}{2}\sigma^2(t, \omega)\frac{\partial^2 f}{\partial x^2}(t, X(t))\mathrm{d}t \\ &+ \int_{|z|<R} \Big\{ f(t, X(t^-) + \gamma(t, z, \omega)) - f(t, X(t^-)) \end{aligned}$$

$$- \frac{\partial f}{\partial x}(t, X(t^-))\gamma(t, z, \omega)\Big\} \nu(\mathrm{d}z)\mathrm{d}t$$
$$+ \int_{\mathbb{R}} \{f(t, X(t^-) + \gamma(t, z, \omega)) - f(t, X(t^-))\}\bar{N}(\mathrm{d}t, \mathrm{d}z). \tag{1.2.3}$$

Example 1.15 (***The Geometric Lévy Process***) Consider the stochastic differential equation (SDE)

$$\mathrm{d}X(t) = X(t^-)\left[a\,\mathrm{d}t + b\,\mathrm{d}B(t) + \int_{\mathbb{R}} c(t, z)\bar{N}(\mathrm{d}t, \mathrm{d}z)\right], \tag{1.2.4}$$

where a, b are constants and $c(t, z) \geq -1$. To find the solution $X(t)$ of this equation we rewrite it as follows:

$$\frac{\mathrm{d}X(t)}{X(t^-)} = a\,\mathrm{d}t + b\,\mathrm{d}B(t) + \int_{\mathbb{R}} c(t, z)\bar{N}(\mathrm{d}t, \mathrm{d}z).$$

Now define

$$Y(t) = \ln X(t).$$

Then by Itô's formula,

$$\begin{aligned}
\mathrm{d}Y(t) &= \frac{X(t)}{X(t)}[a\,\mathrm{d}t + b\,\mathrm{d}B(t)] - \frac{1}{2}b^2X^{-2}(t)X^2(t)\mathrm{d}t \\
&\quad + \int_{|z|<R} \{\ln(X(t^-) + c(t, z)X(t^-)) - \ln(X(t^-)) \\
&\quad - X^{-1}(t^-)c(t, z)X(t^-)\}\nu(\mathrm{d}z)\mathrm{d}t \\
&\quad + \int_{\mathbb{R}} \{\ln(X(t^-) + c(t, z)X(t^-)) - \ln(X(t^-))\}\bar{N}(\mathrm{d}t, \mathrm{d}z) \\
&= \left(a - \frac{1}{2}b^2\right)\mathrm{d}t + b\,\mathrm{d}B(t) + \int_{|z|<R} \{\ln(1+c(t, z)) - c(t, z)\}\nu(\mathrm{d}z)\mathrm{d}t \\
&\quad + \int_{\mathbb{R}} \ln(1+c(t, z))\bar{N}(\mathrm{d}t, \mathrm{d}z).
\end{aligned}$$

Hence

$$\begin{aligned}
Y(t) = Y(0) + \left(a - \frac{1}{2}b^2\right)t + bB(t) + \int_0^t \int_{|z|<R} \{\ln(1 + c(s, z)) \\
- c(s, z)\}\nu(\mathrm{d}z)\mathrm{d}s + \int_0^t \int_{\mathbb{R}} \ln(1+c(s, z))\bar{N}(\mathrm{d}s, \mathrm{d}z)
\end{aligned}$$

and this gives the solution

$$X(t) = X(0)\exp\left\{\left(a - \frac{1}{2}b^2\right)t + bB(t) + \int_0^t\int_{|z|<R}\{\ln(1+c(s,z)) - c(s,z)\}\nu(\mathrm{d}z)\mathrm{d}s + \int_0^t\int_{\mathbb{R}}\ln(1+c(s,z))\tilde{N}(\mathrm{d}s,\mathrm{d}z)\right\}. \tag{1.2.5}$$

In analogy with the diffusion case ($N = 0$) we call this process $X(t)$ a *geometric Lévy process*. It is often used as a model for stock prices. See, e.g., [Eb].

Next we formulate the corresponding multidimensional version of Theorem 1.14.

Theorem 1.16 (***The Multidimensional Itô Formula***) *Let $X(t) \in \mathbb{R}^n$ be an Itô–Lévy process of the form*

$$\mathrm{d}X(t) = \alpha(t,\omega)\mathrm{d}t + \sigma(t,\omega)\mathrm{d}B(t) + \int_{\mathbb{R}^\ell}\gamma(t,z,\omega)\tilde{N}(\mathrm{d}t,\mathrm{d}z), \tag{1.2.6}$$

where $\alpha : [0,T]\times\Omega \to \mathbb{R}^n$, $\sigma : [0,T]\times\Omega \to \mathbb{R}^{n\times m}$, and $\gamma : [0,T]\times\mathbb{R}^\ell\times\Omega \to \mathbb{R}^{n\times\ell}$ are predictable processes such that the integrals exist. Here $B(t)$ is an m-dimensional Brownian motion and

$$\begin{aligned}\tilde{N}(\mathrm{d}t,\mathrm{d}z)^T &= (\tilde{N}_1(\mathrm{d}t,\mathrm{d}z_1),\dots,\tilde{N}_\ell(\mathrm{d}t,\mathrm{d}z_\ell))\\ &= (N_1(\mathrm{d}t,\mathrm{d}z_1) - \mathcal{X}_{|z_1|<R_1}\nu_1(\mathrm{d}z_1)\mathrm{d}t,\dots,N_\ell(\mathrm{d}t,\mathrm{d}z_\ell)\\ &\quad - \mathcal{X}_{|z_\ell|<R_\ell}\nu_\ell(\mathrm{d}z_\ell)\mathrm{d}t), \quad z = (z_1,z_2,\dots,z_\ell),\end{aligned}$$

where $\{N_j\}$ are independent Poisson random measures with Lévy measures ν_j coming from ℓ independent (one-dimensional) Lévy processes $\eta_1,\dots,\eta_\ell$.

Thus the integral on the right of (1.2.6) is just a shorthand matrix notation. When written out in detail component number i of $X(t)$ in (1.2.6), $X_i(t)$ gets the form

$$\mathrm{d}X_i(t) = \alpha_i(t,\omega)\mathrm{d}t + \sum_{j=1}^m\sigma_{ij}(t,\omega)\mathrm{d}B_j(t) + \sum_{j=1}^\ell\int_{\mathbb{R}}\gamma_{ij}(t,z,\omega)\tilde{N}_j(\mathrm{d}t,\mathrm{d}z_j), \quad 1\le i\le n. \tag{1.2.7}$$

Let $f \in C^{1,2}([0,T])\times\mathbb{R}^n;\mathbb{R})$. Put $Y(t) = f(t,X(t))$. Then, suppressing the ω for simplicity,

$$
\begin{aligned}
\mathrm{d}Y(t) = \frac{\partial f}{\partial t}(t, X(t))\mathrm{d}t + \sum_{i=1}^{n} \frac{\partial f}{\partial x_i}(t, X(t))(\alpha_i(t)\mathrm{d}t + \sigma_i(t)\mathrm{d}B(t)) \\
+ \frac{1}{2}\sum_{i,j=1}^{n} (\sigma\sigma^T)_{ij}(t)\frac{\partial^2 f}{\partial x_i \partial x_j}(t, X(t))\mathrm{d}t \\
+ \sum_{k=1}^{\ell} \int_{|z|<R_k} \Big\{ f(t, X(t^-) + \gamma^{(k)}(t, z)) - f(t, X(t^-)) \\
- \sum_{i=1}^{n} \gamma_i^{(k)}(t, z)\frac{\partial f}{\partial x_i}(X(t^-)) \Big\} \nu_k(\mathrm{d}z_k)\mathrm{d}t \\
+ \sum_{k=1}^{\ell} \int_{\mathbb{R}} \{ f(t, X(t^-) + \gamma^{(k)}(t, z)) - f(t, X(t^-)) \} \tilde{N}_k(\mathrm{d}t, \mathrm{d}z_k), \qquad (1.2.8)
\end{aligned}
$$

where $\gamma^{(k)} \in \mathbb{R}^n$ is column number k of the $n \times \ell$ matrix $\gamma = [\gamma_{ik}]$ and $\gamma_i^{(k)} = \gamma_{ik}$ is the coordinate number i of $\gamma^{(k)}$ and σ_i is row number i of σ.

Theorem 1.17 (***The Itô–Lévy Isometry***) *Let $X(t) \in \mathbb{R}^n$ be as in (1.2.6) but with $X(0) = 0$ and $\alpha = 0$. Then*

$$
\begin{aligned}
E[X^2(T)] &= E\left[\int_0^T \left\{ \sum_{i=1}^{n}\sum_{j=1}^{m} \sigma_{ij}^2(t) + \sum_{i=1}^{n}\sum_{j=1}^{\ell} \int_{\mathbb{R}} \gamma_{ij}^2(t, z)\nu_j(\mathrm{d}z_j) \right\} \mathrm{d}t \right] \\
&= \sum_{i=1}^{n} E\left[\int_0^T \left\{ \sum_{j=1}^{m} \sigma_{ij}^2(t) + \sum_{j=1}^{\ell} \int_{\mathbb{R}} \gamma_{ij}^2(t, z)\nu_j(\mathrm{d}z_j) \right\} \mathrm{d}t \right] \qquad (1.2.9)
\end{aligned}
$$

provided that the right-hand side is finite.

Proof This follows from the Itô formula applied to $f(t, x) = x^2 = |x|^2$. We omit the details. □

Remark 1.18 As a special case of Theorem 1.17 assume that

$$\mathrm{d}X(t) = \mathrm{d}\eta(t) = \int_{\mathbb{R}} z\tilde{N}(\mathrm{d}t, \mathrm{d}z) \in \mathbb{R}$$

with $E[X^2(T)] = T\int_{\mathbb{R}} z^2\nu(\mathrm{d}z) < \infty$. Then we get the isometry

$$E\left[\left(\int_0^T H(t)\mathrm{d}\eta(t) \right)^2 \right] = E\left[\int_0^T H^2(t)\mathrm{d}t \right] \int_{\mathbb{R}} z^2\nu(\mathrm{d}z)$$

for all $H \in \mathbf{L}_{\mathrm{ucp}}$ (see Definition 1.12) such that $H \in L^2([0, T] \times \Omega)$, i.e., such that

$$\|H\|^2_{L^2([0,T]\times\Omega)} := E\left[\int_0^T H^2(t)\mathrm{d}t\right] < \infty.$$

Using this we can in the usual way extend the definition of the integral

$$\int_0^T Y(t)\mathrm{d}\eta(t) \in L^2(\Omega)$$

to all processes $Y(t)$ which are limits in $L^2([0,T]\times\Omega)$ of processes $H_n(t) \in \mathbf{L}_{\text{ucp}} \cap L^2([0,T]\times\Omega)$. We will call such processes $Y(t)$ *predictable processes*.

1.3 Lévy Stochastic Differential Equations

From now on (and throughout the rest of this book) we assume that

$$E[\eta^2(t)] < \infty \text{ for all } t \geq 0.$$

This implies that we can choose $R = \infty$ and hence replace $\bar{N}$ by $\widetilde{N}$ (see Theorem 1.8).

The geometric Lévy process is an example of a *Lévy diffusion*, i.e., the solution of an SDE driven by Lévy processes.

Theorem 1.19 (***Existence and Uniqueness of Solutions of Lévy SDEs***) *Consider the following Lévy SDE in* $\mathbb{R}^n$*:* $X(0) = x_0 \in \mathbb{R}^n$ *and*

$$\mathrm{d}X(t) = \alpha(t, X(t))\mathrm{d}t + \sigma(t, X(t))\mathrm{d}B(t) + \int_{\mathbb{R}^n} \gamma(t, X(t^-), z)\widetilde{N}(\mathrm{d}t, \mathrm{d}z), \quad (1.3.1)$$

where $\alpha : [0,T]\times\mathbb{R}^n \to \mathbb{R}^n$, $\sigma : [0,T]\times\mathbb{R}^n \to \mathbb{R}^{n\times m}$ *and* $\gamma : [0,T]\times\mathbb{R}^n\times\mathbb{R}^n \to \mathbb{R}^{n\times\ell}$ *satisfy the following conditions*

(At most linear growth) There exists a constant $C_1 < \infty$ *such that*

$$\|\sigma(t,x)\|^2 + |\alpha(t,x)|^2 + \int_{\mathbb{R}} \sum_{k=1}^{\ell} |\gamma_k(t,x,z)|^2 \nu_k(\mathrm{d}z_k) \leq C_1(1+|x|^2)$$

for all $x \in \mathbb{R}^n$

(Lipschitz continuity) There exists a constant $C_2 < \infty$ *such that*

$$\|\sigma(t,x) - \sigma(t,y)\|^2 + |\alpha(t,x) - \alpha(t,y)|^2$$
$$+ \sum_{k=1}^{\ell} \int_{\mathbb{R}} |\gamma^{(k)}(t,x,z_k) - \gamma^{(k)}(t,y,z_k)|^2 \nu_k(\mathrm{d}z_k) \leq C_2|x-y|^2,$$

for all $x, y \in \mathbb{R}^n$.

Then there exists a unique càdlàg adapted solution $X(t)$ such that

$$E[|X(t)|^2] < \infty \quad \textit{for all } t.$$

Solutions of Lévy SDEs in the *time homogeneous* case, i.e., when $\alpha(t,x) = \alpha(x)$, $\sigma(t,x) = \sigma(x)$, and $\gamma(t,x,z) = \gamma(x,z)$ are called *jump diffusions* (or *Lévy diffusions*).

Theorem 1.20 *A jump diffusion is a strong Markov process.*

Proof See [P, Theorem V.32]. □

Definition 1.21 Let $X(t) \in \mathbb{R}^n$ be a jump diffusion. Then the generator A of X is defined on functions $f : \mathbb{R}^n \to \mathbb{R}$ by

$$Af(x) = \lim_{t \to 0^+} \frac{1}{t} \left\{ E^x[f(X(t))] - f(x) \right\} \quad \text{(if the limit exists)},$$

where $E^x[f(X(t))] = E[f(X^{(x)}(t))]$, $X^{(x)}(0) = x$.

Theorem 1.22 *Suppose $f \in C_0^2(\mathbb{R}^n)$. Then $Af(x)$ exists and is given by*

$$\begin{aligned} Af(x) = &\sum_{i=1}^{n} \alpha_i(x) \frac{\partial f}{\partial x_i}(x) + \frac{1}{2} \sum_{i,j=1}^{n} (\sigma\sigma^T)_{ij}(x) \frac{\partial^2 f}{\partial x_i \partial x_j}(x) \\ &+ \int_{\mathbb{R}} \sum_{k=1}^{\ell} \{ f(x + \gamma^{(k)}(x,z)) - f(x) - \nabla f(x) \cdot \gamma^{(k)}(x,z) \} \nu_k(\mathrm{d}z_k). \end{aligned} \tag{1.3.2}$$

From now on we *define* $Af(x)$ by the expression (1.3.2) for all f such that the partial derivatives of f and the integrals in (1.3.2) exist at x.

Theorem 1.23 (***The Dynkin Formula I***) *Let $X(t) \in \mathbb{R}^n$ be a jump diffusion and let $f \in C_0^2(\mathbb{R}^n)$. Let τ be a stopping time such that*

$$E^x[\tau] < \infty.$$

Then

$$E^x[f(X(\tau))] = f(x) + E^x \left[\int_0^\tau Af(X(s)) \mathrm{d}s \right].$$

Proof This follows by combining the Itô formula (1.2.8) with the formula (1.3.2) for A and taking expectation. □

This version is usually strong enough for applications in the case when there are no jumps ($N = 0$). However, for jump diffusions we need the following stronger, localized version.

Theorem 1.24 (***The Dynkin Formula II***) *Let $X(t) \in \mathbb{R}^n$ be a jump diffusion, $G \subset \mathbb{R}^n$ be an open set and let $f \in C^2(\mathbb{R}^n)$. Let $\tau < \infty$ be a stopping time. Suppose that*

$$\tau \leq \tau_G := \inf\{t > 0;\, X(t) \notin G\} \tag{1.3.3}$$

$$E^x\left[|f(X(\tau))| + \int_0^\tau |Af(X(t))|\mathrm{d}t\right] < \infty. \tag{1.3.4}$$

Then we have

$$E^x[f(X(\tau))] = f(x) + E^x\left[\int_0^\tau (Af)(X(t))\mathrm{d}t\right].$$

Definition 1.25 In general, if $\{\psi_m\}_{m=1}^\infty$ and g are functions defined on a set $G \subset \mathbb{R}^n$, we say that $\psi_m \to g$ *pointwise dominatedly* in G if $\psi_m(x) \to g(x)$ for all $x \in G$ and there exists a constant $C < \infty$ such that

$$|\psi_m(x)| \leq C|g(x)| \quad \text{for all } x \in G,\ m = 1, 2, \ldots$$

Proof of Theorem 1.24 Choose a sequence $\{G_k\}_{k=1}^\infty$ of open sets such that $\bar{G}_k$ is compact, $\bar{G}_k \subset G_{k+1}$ and $G = \bigcup_{k=1}^\infty G_k$.

For fixed k choose $f_m = f_m^{(k)} \in C_0^2(\mathbb{R}^n)$ such that $f_m \to f$ pointwise dominatedly in $\mathbb{R}^n$ and $\frac{\partial f_m}{\partial x_i} \to \frac{\partial f}{\partial x_i}$, $\frac{\partial^2 f_m}{\partial x_i \partial x_j} \to \frac{\partial^2 f}{\partial x_i \partial x_j}$, and $Af_m \to Af$ pointwise dominatedly in G_k as $m \to \infty$. Define

$$\tau_{G_k} = \inf\{t > 0\ ;\ X(t) \notin G_k\}\ ;\ k = 1, 2, \ldots$$

Then by Theorem 1.23, for $x \in G_k$,

$$E^x[f_m^{(k)}(X(\tau \wedge k \wedge \tau_{G_k}))] = f_m^{(k)}(x) + E^x\left[\int_0^{\tau \wedge k \wedge \tau_{G_k}} Af_m^{(k)}(X(s))ds\right].$$

Letting $m \to \infty$ we get

$$E^x[f(X(\tau \wedge k \wedge \tau_{G_k}))] = f(x) + E^x\left[\int_0^{\tau \wedge k \wedge \tau_{G_k}} Af(X(s))ds\right]\ ;\ x \in G_k.$$

We now use that $X(\cdot)$ is *quasi-left continuous* (i.e. left continuous over stopping times) (see [JS, Propositions I.2.26 and I.3.27]) to see that

$$\tau \wedge k \wedge \tau_{G_k} \to \tau \wedge \tau_G = \tau \text{ a.s.}$$

and

$$X(\tau \wedge k \wedge \tau_{G_k}) \to X(\tau) \text{ a.s., as } k \to \infty.$$

Hence

$$\begin{aligned} E^x[f(X(\tau))] &= \lim_{k\to\infty} E^x[f(X(\tau \wedge k \wedge \tau_{G_k}))] \\ &= f(x) + \lim_{k\to\infty} E^x\left[\int_0^{\tau\wedge k\wedge\tau_{G_k}} Af(X(s))ds\right] \\ &= f(x) + E^x\left[\int_0^{\tau} Af(X(s))ds\right] \text{ for all } x \in \bigcup_{k=1}^{\infty} G_k = G. \end{aligned}$$

□

1.4 The Girsanov Theorem and Applications

The Girsanov theorem and the related concept of an equivalent local martingale measure (ELMM) are important in the applications of stochastic analysis to finance.

In this chapter we first give a general semimartingale discussion and then we apply it to Itô–Lévy processes. We refer to [Ka] for more details.

Let $(\Omega, \mathcal{F}, \{\mathcal{F}_t\}_{t\geq 0}, P)$ be a filtered probability space. Let Q be another probability measure on $\mathcal{F}_T$. We say that Q is *equivalent* to $P \mid \mathcal{F}_T$ if $P \mid \mathcal{F}_T \ll Q$ and $Q \ll P \mid \mathcal{F}_T$, or, equivalently, if P and Q have the same zero sets in $\mathcal{F}_T$. By the Radon–Nikodym theorem this is the case if and only if we have

$$\mathrm{d}Q(\omega) = Z(T)\mathrm{d}P(\omega) \quad \text{and} \quad \mathrm{d}P(\omega) = Z^{-1}(T)\mathrm{d}Q(\omega) \quad \text{on } \mathcal{F}_T$$

for some $\mathcal{F}_T$-measurable random variable $Z(T) > 0$ a.s. P. In that case we also write

$$\frac{\mathrm{d}Q}{\mathrm{d}P} = Z(T) \quad \text{and} \quad \frac{\mathrm{d}P}{\mathrm{d}Q} = Z^{-1}(T) \quad \text{on } \mathcal{F}_T. \tag{1.4.1}$$

We first make a simple, but useful observation.

Lemma 1.26 *Suppose $Q \ll P$ with $\frac{\mathrm{d}Q}{\mathrm{d}P} = Z(T)$ on $\mathcal{F}_T$. Then*

$$Q \mid \mathcal{F}_t \ll P \mid \mathcal{F}_t \quad \textit{for all } t \in [0, T] \textit{ and}$$

$$Z(t) := \frac{\mathrm{d}(Q \mid \mathcal{F}_t)}{\mathrm{d}(P \mid \mathcal{F}_t)} = E_P[Z(T) \mid \mathcal{F}_t], \quad 0 \leq t \leq T. \tag{1.4.2}$$

In particular, $Z(t)$ is a P-martingale.

Proof Since $P(G) = 0 \Rightarrow Q(G) = 0$ for all $G \in \mathcal{F}_T \supseteq \mathcal{F}_t$, it is clear that $Q \mid \mathcal{F}_t \ll P \mid \mathcal{F}_t$. Choose $F \in \mathcal{F}_t$. Then

$$\begin{aligned} E_P\left[F \cdot E[Z(T) \mid \mathcal{F}_t]\right] &= E_P\left[E_P[FZ(T) \mid \mathcal{F}_t]\right] \\ &= E_P[FZ(T)] = E_Q[F] = E_P[F \cdot Z(t)]. \end{aligned}$$

Since this holds for all $F \in \mathcal{F}_t$ we conclude that

$$E_P[Z(T) \mid \mathcal{F}_t] = Z(t), \quad \text{as claimed.}$$

□

Lemma 1.27 *Suppose $Q \ll P$ with $\frac{\mathrm{d}Q}{\mathrm{d}P} = Z(T) > 0$ on $\mathcal{F}_T$. Let $X(t)$ be an adapted process such that $Z(t)X(t)$ is a martingale with respect to P. Then $X(t)$ is a martingale with respect to Q. Similarly, if $Z(t)X(t)$ is a* local *martingale with respect to P, then $X(t)$ is a* local *martingale with respect to Q.*

Proof We prove the last statement. Let $\tau \geq t$ be a stopping time, $\tau \leq T$. Then

$$\begin{aligned} E_Q[X(\tau) \mid \mathcal{F}_{t\wedge\tau}] &= \frac{E[Z(T)X(\tau) \mid \mathcal{F}_{t\wedge\tau}]}{E[Z(T) \mid \mathcal{F}_{t\wedge\tau}]} = \frac{E[Z(\tau)X(\tau) \mid \mathcal{F}_{t\wedge\tau}]}{Z(t\wedge\tau)} = \frac{Z(t\wedge\tau)X(t\wedge\tau)}{Z(t\wedge\tau)} \\ &= X(t\wedge\tau) \quad \text{for all } t \in [0,T]. \end{aligned}$$

□

Definition 1.28 Let $X(t), Y(t) \in \mathbb{R}^n$ be two càdlàg semimartingales. The *quadratic covariation* of $X(\cdot)$ and $Y(\cdot)$, denoted by $[X,Y](\cdot)$, is the unique semimartingale such that

$$X(t) \cdot Y(t) = X(0) \cdot Y(0) + \int_0^t X(s^-) \cdot \mathrm{d}Y(s) + \int_0^t Y(s^-) \cdot \mathrm{d}X(s) + [X,Y](t).$$

Example 1.29 Let

$$\mathrm{d}X_i(t) = \alpha_i(t,\omega)\mathrm{d}t + \sigma_i(t,\omega)\mathrm{d}B(t) + \int_{\mathbb{R}} \gamma_i(t,z)\widetilde{N}(\mathrm{d}t,\mathrm{d}z), \quad i = 1,2$$

be two Itô–Lévy processes. Then by the Itô formula (Theorem 1.16) we have (see Exercise 1.7)

$$\begin{aligned} \mathrm{d}(X_1(t)X_2(t)) &= X_1(t^-)\mathrm{d}X_2(t) + X_2(t^-)\mathrm{d}X_1(t) + \sigma_1(t)\sigma_2(t)\mathrm{d}t \\ &\quad + \int_{\mathbb{R}} \gamma_1(t,z)\gamma_2(t,z)N(\mathrm{d}t,\mathrm{d}z). \end{aligned}$$

Hence in this case the quadratic covariation is

$$
\begin{aligned}
[X_1, X_2](t) &= \int_0^t \sigma_1(s)\sigma_2(s)\mathrm{d}s + \int_0^t \int_{\mathbb{R}} \gamma_1(s,z)\gamma_2(s,z)N(\mathrm{d}s, \mathrm{d}z) \\
&= \int_0^t \left[\sigma_1(s)\sigma_2(s) + \int_{\mathbb{R}} \gamma_1(s,z)\gamma_2(s,z)\nu(\mathrm{d}z)\right]\mathrm{d}s \\
&\quad + \int_0^t \int_{\mathbb{R}} \gamma_1(s,z)\gamma_2(s,z)\widetilde{N}(\mathrm{d}s, \mathrm{d}z).
\end{aligned}
$$

Theorem 1.30 (***Girsanov Theorem for Semimartingales***) *Let Q be a probability measure on $\mathcal{F}_T$ and assume that Q is equivalent to P on $\mathcal{F}_T$, with*

$$\mathrm{d}Q(\omega) = Z(t)\mathrm{d}P(\omega) \quad \text{on } \mathcal{F}_t,\ t \in [0, T].$$

(See Lemma 1.26.) Let $M(t)$ be a local P-martingale. Then the process $\widehat{M}(t)$ defined by

$$\widehat{M}(t) := M(t) - \int_0^t \frac{\mathrm{d}[M, Z](s)}{Z(s)}$$

is a local Q-martingale.

Sketch of Proof Since we will give a complete proof of the Itô–Lévy process version of this below (Theorem 1.31) we settle with a proof of Theorem 1.30 in the case when $Z(t)$ is continuous. By integration by parts we have

$$
\begin{aligned}
\widehat{M}(t)Z(t) - \widehat{M}(0)Z(0) &= \int_0^t \widehat{M}(s^-)\mathrm{d}Z(s) + \int_0^t Z(s)\mathrm{d}\widehat{M}(s) + [M, Z](t) \\
&= \int_0^t \widehat{M}(s^-)\mathrm{d}Z(s) + \int_0^t Z(s)\mathrm{d}M(s) \\
&\quad - \int_0^t Z(s)\frac{\mathrm{d}[M, Z](s)}{Z(s)} + [M, Z](t) \\
&= \int_0^t \widehat{M}(s^-)\mathrm{d}Z(s) + \int_0^t Z(s)\mathrm{d}M(s).
\end{aligned}
$$

Therefore, since $Z(t)$ is a local P-martingale (Lemma 1.26) we conclude that $\widehat{M}(t)Z(t)$ is a local P-martingale. But then $\widehat{M}(t)$ is a local Q-martingale, by Lemma 1.27. We refer to [P, Theorem 35, p. 132] for a proof in the general case. □

Theorem 1.31 (***Girsanov Theorem I for Itô–Lévy Processes***) *Let $X(t)$ be an n-dimensional Itô–Lévy process of the form*

$$\mathrm{d}X(t) = \alpha(t,\omega)\mathrm{d}t + \sigma(t,\omega)\mathrm{d}B(t) + \int_{\mathbb{R}} \gamma(t,z,\omega)\widetilde{N}(\mathrm{d}t, \mathrm{d}z), \quad 0 \le t \le T. \tag{1.4.3}$$

Assume there exist predictable processes $\theta_0(t) = \theta_0(t,\omega) \in \mathbb{R}^m$ and $\theta_1(t,z) = \theta_1(t,z,\omega) \in \mathbb{R}^\ell$ such that

$$\sigma(t)\theta_0(t) + \int_{\mathbb{R}^\ell} \gamma(t,z)\theta_1(t,z)\nu(\mathrm{d}z) = \alpha(t) \quad \textit{for a.a. } (t,\omega) \in [0,T] \times \Omega \tag{1.4.4}$$

and such that the process

$$\begin{aligned} Z(t) := \exp\Bigg[&- \int_0^t \theta_0(s)\mathrm{d}B(s) - \frac{1}{2}\int_0^t \theta_0^2(s)\mathrm{d}s \\ &+ \sum_{j=1}^{\ell} \int_0^t \int_{\mathbb{R}} \ln(1-\theta_{1,j}(s,z))\widetilde{N}_j(\mathrm{d}s,\mathrm{d}z) \\ &+ \sum_{j=1}^{\ell} \int_0^t \int_{\mathbb{R}} \{\ln(1-\theta_{1,j}(s,z)) + \theta_{1,j}(s,z)\}\nu_j(\mathrm{d}z)\mathrm{d}s\Bigg], \quad 0 \le t \le T \end{aligned} \tag{1.4.5}$$

is well-defined and satisfies

$$E[Z(T)] = 1. \tag{1.4.6}$$

Define the probability measure Q on $\mathcal{F}_T$ by $\mathrm{d}Q(\omega) = Z(T)\mathrm{d}P(\omega)$. Then $X(t)$ is a local martingale with respect to Q.

Proof By Lemma 1.27 it suffices to prove that $Z(t)X(t)$ is a local martingale with respect to P. By Exercises 1.7 and 1.9 (extended to include the Brownian motion case) we have

$$dZ(t) = Z(t^-)\left[-\theta_0(t)dB(t) - \int_{\mathbb{R}} \theta_1(t,z)\tilde{N}(dt,dz)\right] \tag{1.4.7}$$

and hence

$$\begin{aligned} \mathrm{d}(Z(t)X(t)) &= Z(t^-)\mathrm{d}X(t) + X(t^-)\mathrm{d}Z(t) + \sigma(t)(-\theta_0(t))Z(t)\mathrm{d}t \\ &\quad + \int_{\mathbb{R}} \gamma(t,z)(-Z(t^-)\theta_1(t,z))N(\mathrm{d}t,\mathrm{d}z) \\ &= Z(t^-)\left(\alpha(t)\mathrm{d}t + \sigma(t)\mathrm{d}B(t) + \int_{\mathbb{R}} \gamma(t,z)\widetilde{N}(\mathrm{d}t,\mathrm{d}z)\right) \\ &\quad - X(t^-)Z(t^-)\int_{\mathbb{R}} \theta_1(t,z)\widetilde{N}(\mathrm{d}t,\mathrm{d}z) - \sigma(t)\theta_0(t)Z(t)\mathrm{d}t \\ &\quad - Z(t^-)\int_{\mathbb{R}} \gamma(t,z)\theta_1(t,z)\widetilde{N}(\mathrm{d}t,\mathrm{d}z) - Z(t^-)\int_{\mathbb{R}} \gamma(t,z)\theta_1(t,z)\nu(\mathrm{d}z)\mathrm{d}t \\ &= Z(t^-)\left(\alpha(t) - \left(\sigma(t)\theta_0(t) + \int_{\mathbb{R}} \gamma(t,z)\theta_1(t,z)\nu(\mathrm{d}z)\right)\right)\mathrm{d}t \\ &\quad + \text{local martingale terms.} \end{aligned}$$

By (1.4.4) $Z(t)X(t)$ is a local martingale. □

Remark 1.32 Such a measure Q is called an *equivalent local martingale measure* (ELMM) for $X(t)$. If $X(t)$ is a martingale with respect to Q, then Q is called an *equivalent martingale measure* (EMM) for $X(t)$.

By the above we deduce the following version, which is sometimes useful. For simplicity we only present the one-dimensional case.

Theorem 1.33 (***Girsanov Theorem II for Itô–Lévy processes***) *Let $\theta_0(t)$ and $\theta_1(t,z) \leq 1$ be predictable processes such that the process*

$$\begin{aligned} Z(t) := \exp\Big(&- \int_0^t \theta_0(s)\mathrm{d}B(s) - \frac{1}{2}\int_0^t \theta_0^2(s)\mathrm{d}s \\ &+ \int_0^t \int_{\mathbb{R}} \ln(1-\theta_1(s,z))\tilde{N}(\mathrm{d}s,\mathrm{d}z) \\ &+ \int_0^t \int_{\mathbb{R}} \{\ln(1-\theta_1(s,z)) + \theta_1(s,z)\}\nu(\mathrm{d}z)\mathrm{d}s\Big) \end{aligned} \tag{1.4.8}$$

exists for $0 \leq t \leq T$. Suppose that

$$E[Z(T)] = 1. \tag{1.4.9}$$

Define the probability measure Q on $\mathcal{F}_T$ by

$$\mathrm{d}Q(\omega) = Z(T)\mathrm{d}P(\omega).$$

Define the process $B_Q(t)$ and the random measure $\tilde{N}_Q(\mathrm{d}t,\mathrm{d}z)$ by

$$\mathrm{d}B_Q(t) = \theta_0(t)\mathrm{d}t + dB(t), \tag{1.4.10}$$

and

$$\tilde{N}_Q(\mathrm{d}t,\mathrm{d}z) = \theta_1(t,z)\nu(\mathrm{d}z)\mathrm{d}t + \tilde{N}(\mathrm{d}t,\mathrm{d}z). \tag{1.4.11}$$

(a) *Then $B_Q(\cdot)$ is a Brownian motion with respect to $\mathbb{F}$ and Q and $\tilde{N}_Q(\cdot,\cdot)$ is the $(\mathbb{F}, Q)$-compensated Poisson random measure of $N(\cdot,\cdot)$, in the sense that the process*

$$M(t) := \int_0^t \int_{\mathbb{R}} \gamma(s,z)\tilde{N}_Q(\mathrm{d}s,\mathrm{d}z), \quad 0 \leq t \leq T \tag{1.4.12}$$

is a local $(\mathbb{F}, Q)$-martingale, for all predictable processes $\gamma(s,z)$ such that

$$\sum_{k=1}^{\ell} \int_0^T \int_{\mathbb{R}} (\gamma^{(k)}(s,z))^2(1-\theta_{1,k}(s,z))\nu_k(\mathrm{d}z)\mathrm{d}s < \infty \text{ a.s.} \tag{1.4.13}$$

(b) *Moreover, the process*

$$R(t) := M^2(t) - \sum_{k=1}^{\ell} \int_0^t \int_{\mathbb{R}} (\gamma^{(k)}(s,z))^2 (1 - \theta_{1,k}(s,z)) \nu_k(\mathrm{d}z)\mathrm{d}s$$

is a local $(\mathbb{F}, Q)$*-martingale. In particular,*

$$E_Q[M^2(t)] = E_Q\left[\sum_{k=1}^{\ell} \int_0^t \int_{\mathbb{R}} (\gamma^{(k)}(s,z))^2 (1 - \theta_{1,k}(s,z)) \nu_k(\mathrm{d}z)\mathrm{d}s\right].$$

Proof **(a)** With $M(t)$ as in (1.4.12) and $\epsilon \in [0,1]$ define

$$X_\epsilon(t) := \epsilon B_Q(t) + M(t).$$

We have

$$\begin{aligned}\mathrm{d}X_\epsilon(t) &= \epsilon\theta_0(t)\mathrm{d}t + \epsilon \mathrm{d}B(t) + \int_{\mathbb{R}} \gamma(t,z)\theta_1(t,z)\nu(\mathrm{d}z)\mathrm{d}t + \int_{\mathbb{R}} \gamma(t,z)\tilde{N}(\mathrm{d}t,\mathrm{d}z) \\ &= \alpha_\epsilon(t)\mathrm{d}t + \epsilon \mathrm{d}B(t) + \int_{\mathbb{R}} \gamma(t,z)\tilde{N}(\mathrm{d}t,\mathrm{d}z),\end{aligned}$$

where

$$\alpha_\epsilon(t) = \epsilon\theta_0(t) + \int_{\mathbb{R}} \gamma(t,z)\theta_1(t,z)\nu(\mathrm{d}z).$$

By Theorem 1.31 $X_\epsilon(t)$ is a local $(\mathbb{F}, Q)$-martingale, for all $\epsilon \in [0,T]$. In particular, $X_0(t) = M(t)$ is a local $(\mathbb{F}, Q)$-martingale, but then $B_Q(t) = X_1(t) - M(t)$ is a local $(\mathbb{F}, Q)$-martingale also. Since $[B_Q, B_Q](t) = t$ we conclude that $B_Q(t)$ is an $(\mathbb{F}, Q)$-Brownian motion.

(b) For simplicity we consider only the 1-dimensional case. Define

$$\eta(t) = \int_0^t \int_{\mathbb{R}} \gamma(s,z)\tilde{N}(ds,dz),\ h(t) = \int_0^t \int_{\mathbb{R}} \gamma(s,z)\theta_1(s,z)\nu(dz)ds.$$

Then

$$M(t) = \eta(t) + h(t).$$

Hence, by the Itô formula

$$\begin{aligned}dR(t) &= d((\eta(t) + h(t))^2 - \int_{\mathbb{R}} \gamma^2(t,z)(1 - \theta_1(t,z))\nu(dz)dt \\ &= 2((\eta(t) + h(t)) \int_{\mathbb{R}} \gamma(t,z)\theta_1(t,z)\nu(dz)dt \\ &\quad + \int_{\mathbb{R}} \{(\eta(t) + h(t) + \gamma(t,z))^2 - (\eta(t) + h(t))^2\}\tilde{N}(dt,dz)\end{aligned}$$

$$
\begin{aligned}
&+\int_{\mathbb{R}}\{(\eta(t)+h(t)+\gamma(t,z))^2-(\eta(t)+h(t))^2\\
&\quad-2(\eta(t)+h(t))\gamma(t,z)\}\nu(dz)dt\\
&-\int_{\mathbb{R}}\gamma^2(t,z)(1-\theta_1(t,z))\nu(dz)dt\\
&=\{2(\eta(t)+h(t))\int_{\mathbb{R}}\gamma(t,z)\theta_1(t,z)\nu(dz)+\int_{\mathbb{R}}\gamma^2(t,z)\theta_1(t,z)\nu(dz)\}dt\\
&\quad+\int_{\mathbb{R}}\{2(\eta(t)+h(t))\gamma(t,z)+\gamma^2(t,z)\}\tilde{N}(dt,dz).
\end{aligned}
$$

This gives (see Example 1.29)

$$
\begin{aligned}
d(Z(t)R(t)) &= Z(t^-)dR(t)+R(t^-)dZ(t)+d[Z,R](t)\\
&= Z(t^-)[2(\eta(t)+h(t))\int_{\mathbb{R}}\gamma(t,z)\theta_1(t,z)\nu(dz)\\
&\quad-\int_{\mathbb{R}}\gamma^2(t,z)\theta_1(t,z)\nu(dz)]dt\\
&\quad+R(t^-)Z(t^-)[-\theta_0(t)dB(t)-\int_{\mathbb{R}}\theta_1(t,z)\tilde{N}(dt,dz)]\\
&\quad+\int_{\mathbb{R}}Z(t^-)(-\theta_1(t,z))\{2(\eta(t)+h(t))\gamma(t,z)+\gamma^2(t,z)\}N(dt,dz).
\end{aligned}
$$

Since $N(dt,dz)=\tilde{N}(dt,dz)+\nu(dz)dt$, we get

$$
\begin{aligned}
d(Z(t)R(t)) &= -R(t)Z(t)\theta_0(t)dB(t)\\
&\quad-Z(t^-)\int_{\mathbb{R}}\{R(t^-)+2(\eta(t)+h(t))\gamma(t,z)+\gamma^2(t,z)\}\tilde{N}(dt,dz),
\end{aligned}
$$

which is a local $(\mathbb{F},P)$-martingale.

Therefore $R(t)$ is a local $(\mathbb{F},Q)$-martingale by Lemma 1.27. □

Example 1.34

1. Suppose $X(t)=\pi(t):=\int_{\mathbb{R}}z\widetilde{N}(t,\mathrm{d}z)\in\mathbb{R}$ is a Poisson process with intensity $\lambda>0$. Then
$$\nu(\mathrm{d}z)=\lambda\delta_1(\mathrm{d}z)$$
where $\delta_1(\mathrm{d}z)$ is the unit point mass at 1 and condition (1.4.4) gets the form
$$\lambda\gamma(t,1)\theta_1(t,1)=\alpha(t),\quad t\in[0,T]. \tag{1.4.14}$$
Note that there is *at most one* solution $\theta_1(t,1)$ of (1.4.14) (unless $\alpha(t)=\gamma(t,1)=0$).

2. Next, suppose ν is supported on n points $z_1, \ldots, z_n$. Then (1.4.4) gets the form

$$\sum_{j=1}^{n} \gamma(t, z_j)\theta_1(t, z_j)\nu(\{z_j\}) = \alpha(t), \quad t \in [0, T].$$

For each $t \in [0, T]$ this is one linear equation in the n unknowns $\theta_1(t, z_1), \ldots, \theta_1(t, z_n)$. So unless in degenerate cases this equation will have infinitely many solutions and – under some conditions on $\{\gamma(t, z_j)\}_{j=1}^{n}$ – also infinitely many solutions satisfying $\theta_1(t, z_j) \leq 1$. This corresponds to infinitely many ELMMs, which again is equivalent to incompleteness of the associated financial market, according to the Second Fundamental Theorem of Asset Pricing (see, e.g., [DS, LS]).

Finally, we combine Theorems 1.31 and 1.33 to obtain the following

Theorem 1.35 (***Girsanov Theorem III for Itô–Lévy processes***) *Let $X(t)$ be as in (1.4.3) in Theorem 1.31. Let $\theta_0(t)$, $\theta_1(t, z)$ be $\mathbb{F}$-predictable processes satisfying (1.4.4) and (1.4.6). Let Q, B_Q and $\tilde{N}_Q$ be as defined in Theorem 1.33. Then in terms of B_Q and $\tilde{N}_Q$ the process $X(t)$ can be represented by*

$$dX(t) = \sigma(t, \omega)dB_Q(t) + \int_{\mathbb{R}} \gamma(t, z, \omega)\tilde{N}_Q(dt, dz).$$

Proof Using (1.4.3) and the definitions of B_Q and $\tilde{N}_Q$ we get

$$\begin{aligned} dX(t) &= \alpha(t)dt + \sigma(t)[dB_Q(t) - \theta_0(t)dt] \\ &\quad + \int_{\mathbb{R}} \gamma(t, z)[\tilde{N}_Q(dt, dz) - \theta_1(t, z)\nu(dz)dt] \\ &= \left[\alpha(t) - \{\sigma(t)\theta_0(t) + \int_{\mathbb{R}} \gamma(t, z)\theta_1(t, z)\nu(dz)\}\right]dt \\ &\quad + \sigma(t)dB_Q(t) + \int_{\mathbb{R}} \gamma(t, z)\tilde{N}_Q(dt, dz). \end{aligned}$$

□

We finish this section by giving a sufficient condition that condition (1.4.9) holds.

Theorem 1.36 (***The Novikov condition for Lévy processes***) *Let $Z(t)$ be as in (1.4.8). Assume that either*

$$E\left[\exp\left(\frac{1}{2}\int_0^T \theta_0^2(s)ds + \int_0^T \int_{\mathbb{R}} \{(1 - \theta(s, z))\ln(1 - \theta_1(s, z)) + \theta_1(s, z)\}\nu(dz)ds\right)\right] < \infty \quad (1.4.15)$$

or

$$E\left[\exp\left(\frac{1}{2}\int_0^T \theta_0^2(s)ds + \int_0^T\int_{\mathbb{R}} \theta_1^2(s,z))N(ds,dz)\right)\right] < \infty. \tag{1.4.16}$$

Then (1.4.9) *holds, i.e.*

$$E[Z(T)] = 1$$

and hence $\{Z(t)\}_{t\in[0,T]}$ *is a martingale and the measure* Q *defined by*

$$dQ(\omega) = Z(T)dP(\omega) \text{ on } \mathcal{F}_T$$

is a probability measure.

Remark 1.37 The sufficient of (1.4.15) is a special case of Theorem 3.2 in [KaSh] (see, in particular, Remark 3 p. 408 of that paper). We do not give the proof here, but refer to [KaSh] for proofs of this and more general results about uniform integrability of exponential local martingales.

The sufficiency of condition (1.4.16) follows from the more general semimartingale result in [PS, Theorem 6]. To see the connection, we first note that by (1.4.7)

$$Z(t) = Z(t^-)dM(t)$$

where $M(t)$ is the local martingale defined by

$$dM(t) = -\theta_0(t)dB(t) - \int_{\mathbb{R}_0} \theta_1(t,z)\tilde{N}(dt,dz).$$

Hence $Z(t)$ is the *Doléans exponential* of M, denoted by $\mathcal{E}(M)_t$. Moreover, by Exercise 1.8

$$dM^c(t) = -\theta_0(t)dB(t) \text{ and } dM^d(t) = -\int_{\mathbb{R}} \theta_1(t,z)\tilde{N}(dt,dz)$$

are the *continuous* and *purely discontinuous martingale parts* of M, respectively. And by Exercise 1.7

$$[M^c,M^c]_T = \int_0^T \theta_0^2(t)dt,\ [M^d,M^d]_T = \int_0^T\int_{\mathbb{R}} \theta_1^2(t,z)N(dt,dz).$$

Thus condition (1.4.16) can be written

$$E\left[\exp\left(\frac{1}{2}[M^c,M^c]_t + [M^d,M^d]_T\right)\right] < \infty$$

which is the condition in [PS, Theorem 6].

1.5 Exercises

Exercise* 1.1 Suppose

$$\mathrm{d}X(t) = \alpha\,\mathrm{d}t + \sigma\,\mathrm{d}B(t) + \int_{\mathbb{R}} \gamma(z)\bar{N}(\mathrm{d}t, \mathrm{d}z), \quad X(0) = x \in \mathbb{R},$$

where α, σ are constants, $\gamma : \mathbb{R} \to \mathbb{R}$ is a given function.

1. Use Itô's formula to find $\mathrm{d}Y(t)$ when

$$Y(t) = \exp(X(t)).$$

2. How do we choose α, σ, and $\gamma(z)$ if we want $Y(t)$ to solve the SDE

$$\mathrm{d}Y(t) = Y(t^-)\left[\beta\,\mathrm{d}t + \theta\,\mathrm{d}B(t) + \lambda\int_{\mathbb{R}} z\bar{N}(\mathrm{d}t, \mathrm{d}z)\right],$$

for given constants, β, θ, and λ?

Exercise* 1.2 Solve the following Lévy SDEs:

1. $\mathrm{d}X(t) = (m - X(t))\mathrm{d}t + \sigma\,\mathrm{d}B(t) + \gamma\int_{\mathbb{R}} z\bar{N}(\mathrm{d}t, \mathrm{d}z), \quad X(0) = x \in \mathbb{R}$

(m, σ, γ constants) (the mean-reverting Lévy–Ornstein–Uhlenbeck process)

2. $\mathrm{d}X(t) = \alpha\,\mathrm{d}t + \gamma\,X(t^-)\int_{\mathbb{R}} z\bar{N}(\mathrm{d}t, \mathrm{d}z), \quad X(0) = x \in \mathbb{R}$

(α, γ constants, $\gamma z > -1$ a.s. ν).

[Hint: Try to multiply the equation by

$$F(t) := \exp\left\{-\int_0^t\int_{\mathbb{R}} \theta(z)\bar{N}(\mathrm{d}s, \mathrm{d}z) + \int_{|z|<R} (\mathrm{e}^{\theta(z)} - 1 - \theta(z))\nu(\mathrm{d}z)\cdot t\right\},$$

for suitable $\theta(z)$.]

Exercise 1.3 (***Geometric Lévy Martingales***) Let $h \in L^2(\mathbb{R})$ be deterministic and define

$$Y(t) = \exp\left\{\int_0^t\int_{\mathbb{R}} h(s)z\tilde{N}(\mathrm{d}s, \mathrm{d}z) - \int_0^t\int_{\mathbb{R}} \left(\mathrm{e}^{h(s)z} - 1 - h(s)z\right)\nu(\mathrm{d}z)\mathrm{d}s\right\}.$$

Show that

$$\mathrm{d}Y(t) = Y(t^-)\int_{\mathbb{R}} \left(\mathrm{e}^{h(t)z} - 1\right)\tilde{N}(\mathrm{d}t, \mathrm{d}z).$$

Exercise 1.4 Find the generator A of the following jump diffusions:

(a) (The geometric Lévy process)

$$\mathrm{d}X(t) = X(t^-)\left[\mu\,\mathrm{d}t + \sigma\,\mathrm{d}B(t) + \gamma\int_{\mathbb{R}} z\widetilde{N}(\mathrm{d}t, \mathrm{d}z)\right].$$

(b) (The mean-reverting Lévy–Ornstein–Uhlenbeck process)

$$\mathrm{d}X(t) = (m - X(t))\mathrm{d}t + \sigma\,\mathrm{d}B(t) + \gamma\int_{\mathbb{R}} z\tilde{N}(\mathrm{d}t, \mathrm{d}z).$$

(c) (The graph of the geometric Lévy process)

$$\mathrm{d}Y(t) = \begin{bmatrix}\mathrm{d}t\\ \mathrm{d}X(t)\end{bmatrix}, \text{ where } X(t) \text{ is as in (a).}$$

(d) (The n-dimensional geometric Lévy process)

$$X(t) = \begin{bmatrix}X(t)\\ \vdots\\ X_n(t)\end{bmatrix},$$

where

$$\mathrm{d}X_i(t) = X_i(t^-)\left[\mu_i\mathrm{d}t + \sum_{j=1}^{n}\sigma_{ij}\mathrm{d}B_j(t) + \sum_{j=1}^{n}\gamma_{ij}\int_{\mathbb{R}} z_j\widetilde{N}_j(\mathrm{d}t, \mathrm{d}z_j)\right],$$
$$1 \le i \le n.$$

Exercise 1.5 (***The First Exit Time from a Ball***) Let $K = \{x \in \mathbb{R}^n, |x| \le R\}$ be the open ball of radius R in $\mathbb{R}^n$ and let

$$\eta(t) = (\eta_1(t), \ldots, \eta_n(t)),$$

where

$$\eta_i(t) = a_i + \int_0^t\int_{\mathbb{R}} z_i\widetilde{N}_i(\mathrm{d}t, \mathrm{d}z_i), \quad 1 \le i \le n$$

are independent one-dimensional pure jump Lévy processes and $a = (a_1, \ldots, a_n) \in K$ is constant. We assume that $0 < E\left[\sum_{i=1}^{n}(\eta_i(t) - a_i)^2\right] < \infty$ for all $t > 0$.

(a) Find the generator A of $\eta(\cdot)$ and show that if $f(x) = |x|^2$ then

$$Af(x) = \sum_{i=1}^{n}\int_{\mathbb{R}} |\zeta|^2\nu_i(\mathrm{d}\zeta) := \rho(n) \in (0, \infty) \quad \text{(constant)}.$$

(b) Let

$$\tau = \inf\{t > 0, \eta(t) \notin K\} \leq \infty$$

and put

$$\tau_k = \tau \wedge k, \quad k = 1, 2, \ldots$$

Let $\{f_m\}_{m=1}^{\infty}$ be a sequence of functions in $C^2(\mathbb{R}^n)$ such that
$f_m(x) = |x|^2$ for $|x| \leq R$, $0 \leq f_m(x) \leq 2R^2$ for all $x \in \mathbb{R}$,
$\text{supp } f_m \subset \left\{x \in \mathbb{R}^n, |x| \leq R + \frac{1}{m}\right\}$ for all m and
$f_m(x) \to |x|^2 \cdot \chi_K(x)$ as $m \to \infty$, for all $x \in \mathbb{R}^n$.
Use the Dynkin formula I to show that

$$E^a\left[f_m\left(\eta(\tau_k)\right)\right] = |a|^2 + \rho(n) \cdot E^a[\tau_k] \quad \text{for all } m, k.$$

(c) Show that

$$E^a[\tau] = \frac{1}{\rho(n)}\left(R^2 P^a[\eta(\tau) \in K] - |a|^2\right) \leq \frac{1}{\rho(n)}\left(R^2 - |a|^2\right).$$

In particular, $\tau < \infty$, a.s.

Remark If we replace $\eta(\cdot)$ by an n-dimensional Brownian motion $B(\cdot)$, then the corresponding exit time $\tau^{(B)}$ satisfies

$$E^a\left[\tau^{(B)}\right] = \frac{1}{n}\left(R^2 - |a|^2\right)$$

(see, e.g., [Ø1, Example 7.4.2]).

Exercise* 1.6 Show that, under some conditions on $\gamma(s, z)$ (deterministic),

$$E\left[\exp\left(\int_0^t \int_{\mathbb{R}} \gamma(s, z)\widetilde{N}(\mathrm{d}s, \mathrm{d}z)\right)\right] = \exp\left(\int_0^t \int_{\mathbb{R}} \{\mathrm{e}^{\gamma(s,z)} - 1 - \gamma(s, z)\}\nu(\mathrm{d}z)\mathrm{d}s\right).$$

Exercise* 1.7 Let

$$\mathrm{d}X_i(t) = \int_{\mathbb{R}} \gamma_i(t, z)\widetilde{N}(\mathrm{d}t, \mathrm{d}z), \quad i = 1, 2$$

be two one-dimensional Itô–Lévy processes. Use the two-dimensional Itô formula (Theorem 1.16) to prove the following *integration by parts formula*:

$$X_1(t)X_2(t) = X_1(0)X_2(0) + \int_0^t X_1(s^-)\mathrm{d}X_2(s) + \int_0^t X_2(s^-)\mathrm{d}X_1(s) + \int_0^t \int_{\mathbb{R}} \gamma_1(s, z)\gamma_2(s, z)N(\mathrm{d}s, \mathrm{d}z). \tag{1.5.1}$$

Remark The process

$$[X_1, X_2](t) := \int_0^t \int_{\mathbb{R}} \gamma_1(s,z)\gamma_2(s,z)N(\mathrm{d}s, \mathrm{d}z)$$
$$= \int_0^t \int_{\mathbb{R}} \gamma_1(s,z)\gamma_2(s,z)\nu(\mathrm{d}z)\mathrm{d}s + \int_0^t \int_{\mathbb{R}} \gamma_1(s,z)\gamma_2(s,z)\widetilde{N}(\mathrm{d}s, \mathrm{d}z) \tag{1.5.2}$$

is called the *quadratic covariation* of X_1 and X_2. See Definition 1.28.

Exercise 1.8 Let $M(t)$ be a local martingale. We say that $M^c(t)$ and $M^d(t)$ are the *continuous* and *purely discontinuous local martingale parts* of M if

$$M(t) = M^c(t) + M^d(t),$$

$M^c(t)$ is a continuous local martingale and $[M^c, M^d]_t$ is also a local martingale.

Suppose $M(t)$ is given by

$$dM(t) = \beta(t)dB(t) + \int_{\mathbb{R}} \gamma(t,z)\tilde{N}(dt, dz).$$

Show that in this case

$$dM^c(t) = \beta(t)dB(t) \text{ and } dM^d(t) = \int_{\mathbb{R}} \gamma(t,z)\tilde{N}(dt, dz).$$

Exercise* 1.9 Define, with suitable conditions on $\theta_1(s,z)$,

$$Z(t) = \exp\left(\int_0^t \int_{\mathbb{R}} \ln(1-\theta_1(s,z))\widetilde{N}(\mathrm{d}s, \mathrm{d}z) \right.$$
$$\left. + \int_0^t \int_{\mathbb{R}} \{\ln(1-\theta_1(s,z)) + \theta_1(s,z)\}\nu(\mathrm{d}z)\mathrm{d}s\right).$$

Show that

$$\mathrm{d}Z(t) = -Z(t^-)\int_{\mathbb{R}} \theta_1(t,z)\widetilde{N}(\mathrm{d}t, \mathrm{d}z).$$

In particular, $Z(t)$ is a local martingale.

Chapter 2
Financial Markets Modeled by Jump Diffusions



It has been argued (see, e.g., [EK, B-N, Sc, Eb, CT]) that Lévy processes are relevant in mathematical finance, in particular in the modeling of stock prices.

In this chapter we give a brief introduction to financial markets where the asset prices are represented by Itô–Lévy processes. For more information we refer to [CT].

2.1 Market Definitions and Arbitrage

As in Chap. 1 we fix an m-dimensional Brownian motion $B(t) = (B_1(t), \ldots, B_m(t))^T$ and ℓ independent compensated jump measures $\tilde{N}(\cdot, \cdot) = (\tilde{N}_1(\cdot, \cdot), \ldots, \tilde{N}_\ell(\cdot, \cdot))^T$. Recall that we have assumed that $E[\eta^2(t)] < \infty$ for all $t \geq 0$. We let $\mathcal{F}_t^B$ be the σ-algebra generated by $B(s)$; $s \leq t$ and we let $\mathcal{F}_t^N$ be the σ-algebra generated by $N(\mathrm{d}s, \mathrm{d}z)$; $s \leq t$. And we put $\mathcal{F}_t = \mathcal{F}_t^B \vee \mathcal{F}_t^N$, the σ-algebra generated by $\mathcal{F}_t^B$ and $\mathcal{F}_t^N$, and $\mathbb{F} = \{\mathcal{F}_t\}_{t \geq 0}$.

Suppose we have a financial market with the following $n + 1$ investment possibilities:

(i) A *risk free asset*, where the unit price $S_0(t)$ at time t is given by

$$\begin{aligned} \mathrm{d}S_0(t) &= r(t)S_0(t)\mathrm{d}t; \qquad t \in [0, T] \\ S_0(0) &= 1. \end{aligned} \tag{2.1.1}$$

(ii) n risky assets, where the unit price $S_i(t)$ at time t is given by

$$\mathrm{d}S_i(t) = \mu_i(t)\mathrm{d}t + \sigma_i(t)\mathrm{d}B(t) + \int_{\mathbb{R}} \gamma_i(t, z)\tilde{N}(\mathrm{d}t, \mathrm{d}z); \qquad t \in [0, T] \tag{2.1.2}$$
$$S_i(0) \in \mathbb{R}; \qquad 1 \leq i \leq n.$$

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Here $T > 0$ is a given constant, $r(t), \mu_i(t), \sigma_i(t)$ and $\gamma_i(t,z)$ are given $\mathbb{F}$-predictable stochastic processes, $\sigma_i(t) = (\sigma_{i1}(t), \dots, \sigma_{im}(t))$ is row number i of the $n \times m$-matrix

$$\sigma(t) = \left[\sigma_{ij}(t)\right]_{\substack{1 \le i \le n \\ 1 \le j \le m}}$$

and $\gamma_i(t,z) = (\gamma_{i1}(t,z), \dots, \gamma_{i\ell}(t,z))$ is row number i of the $n \times \ell$-matrix

$$\gamma(t,z) = \left[\gamma_{ij}(t,z)\right]_{\substack{1 \le i \le n \\ 1 \le j \le \ell}}.$$

We put $\mu(t) = [\mu_1(t), \dots, \mu_n(t)]^T$, $S(t) = [S_0(t), \dots, S_n(t)]^T$. Thus $\tilde{N}(\mathrm{d}t, \mathrm{d}z) = [\tilde{N}_1(\mathrm{d}t, \mathrm{d}z), \dots, \tilde{N}_\ell(\mathrm{d}t, \mathrm{d}z)]^T$ is a column vector of ℓ independent Poisson random measures. (See Theorem 1.16.)

A *portfolio* in this market is an $(n+1)$-dimensional $\mathbb{F}$-predictable process

$$\varphi(t) = (\varphi_0(t), \varphi_1(t), \dots, \varphi_n(t))$$

representing *the number of units held* at time t of the assets number $0, 1, \dots, n$, respectively. The *wealth process* $X(t) = X^\varphi(t)$ associated to the portfolio φ is defined by

$$X^\varphi(t) = \varphi(t)S(t) = \sum_{i=0}^{n} \varphi_i(t)S_i(t). \tag{2.1.3}$$

The portfolio φ is called *self-financing* if $\int_0^t \varphi(s)\mathrm{d}S(s)$ exists for all $t > 0$ and

$$X^\varphi(t) = X^\varphi(0) + \int_0^t \varphi(s)\mathrm{d}S(s) \tag{2.1.4}$$

or, in differential notation,

$$\mathrm{d}X^\varphi(t) = \varphi(t)\mathrm{d}S(t). \tag{2.1.5}$$

If, in addition, there exists a constant $K = K_\varphi > 0$ such that

$$X^\varphi(t) \ge -K_\varphi \qquad \text{a.s. for all } t \in [0,T] \tag{2.1.6}$$

then φ is called *admissible*. The set of all admissible portfolios is denoted by $\mathcal{A}$.

Definition 2.1 A portfolio $\varphi \in \mathcal{A}$ is called an *arbitrage* if

$$X^\varphi(0) = 0, \quad X^\varphi(T) \ge 0 \ \text{ a.s. and } \ P[X^\varphi(T) > 0] > 0. \tag{2.1.7}$$

In other words, an arbitrage is an opportunity to make money without risk. It is a basic equilibrium requirement in mathematical finance that arbitrages cannot exist. Therefore it is important to be able to decide when a general market of the form (2.1.1)–(2.1.2) allows arbitrage.

The non-existence of arbitrage turns out to be closely related to the existence of an equivalent local martingale measure (ELMM), defined as follows:

Definition 2.2 A probability measure Q on $\mathcal{F}_T$ is called an *equivalent local martingale measure* (ELMM) if $Q \sim P$ (i.e. $Q \ll P$ and $P \ll Q$) and the *normalized (discounted) price process*, $\bar{S}(t)$, defined by

$$\bar{S}(t) := \exp\Big(-\int_0^t r(u)\mathrm{d}u\Big)S(t); \qquad t \in [0, T] \tag{2.1.8}$$

is a *local martingale* with respect to Q.

For simplicity of notation we will from now on put

$$\xi(t) = \frac{1}{S_0(t)} = \exp\Big(-\int_0^t r(u)\mathrm{d}u\Big). \tag{2.1.9}$$

Note that

$$\mathrm{d}\xi(t) = -r(t)\xi(t)\mathrm{d}t; \qquad \xi(0) = 1, \tag{2.1.10}$$

and

$$\mathrm{d}\bar{S}(t) = \xi(t)[\mathrm{d}S(t) - r(t)S(t)\mathrm{d}t]; \qquad \bar{S}(0) = S(0). \tag{2.1.11}$$

Moreover, if we define the normalized (discounted) wealth process $\bar{X}^{\varphi}(t)$ by

$$\bar{X}^{\varphi}(t) = \xi(t)X^{\varphi}(t) = \xi(t)\varphi(t)S(t) = \varphi(t)\bar{S}(t), \tag{2.1.12}$$

then

$$\begin{aligned}\mathrm{d}\bar{X}^{\varphi}(t) &= \xi(t)[\mathrm{d}X^{\varphi}(t) - r(t)X^{\varphi}(t)\mathrm{d}t] \\ &= \xi(t)[\varphi(t)\mathrm{d}S(t) - r(t)\varphi(t)S(t)\mathrm{d}t] = \varphi(t)\mathrm{d}\bar{S}(t).\end{aligned} \tag{2.1.13}$$

Theorem 2.3 *Suppose there exists an ELMM. Then the market has no arbitrage.*

Proof Let Q be an ELMM. Suppose there exists an arbitrage φ. Then

$$X^{\varphi}(0) = 0, \quad X^{\varphi}(T) \geq 0 \quad \text{a.s. } P \text{ and} \quad P[X^{\varphi}(T) > 0] > 0.$$

Since $Q \sim P$ we deduce that

$$\bar{X}^{\varphi}(T) \geq 0 \quad \text{a.s. } Q \text{ and} \quad Q[\bar{X}^{\varphi}(T) > 0] > 0. \tag{2.1.14}$$

Since $\bar{S}(t)$ is a local Q-martingale, we also have that

$$\bar{X}^{\varphi}(t) = X^{\varphi}(0) + \int_0^t \varphi(u)\mathrm{d}\bar{S}(u)$$

is a local Q-martingale. Moreover, by (2.1.6) $\bar{X}^{\varphi}(t)$ is lower bounded. Therefore $\bar{X}^{\varphi}(t)$ is a *Q-supermartingale*, which implies that

$$0 = \bar{X}^{\varphi}(0) \geq E_Q[\bar{X}^{\varphi}(T)]. \tag{2.1.15}$$

This contradicts (2.1.14). This contradiction shows that arbitrages cannot exist. □

Remark 2.4 This result is related to the First Fundamental Theorem of Asset Pricing, which states the equivalence between

(i) the existence of an ELMM and
(ii) No Free Lunch with Vanishing Risk (NFLVR), a property which is slightly stronger than no arbitrage. See e.g. [DS], [LS] and [DS2].

In the following we put

$$\hat{S}(t) = [S_1(t), \ldots, S_n(t)]^T, \tag{2.1.16}$$

i.e. $\hat{S}(t)$ is the price vector with $S_0(t)$ removed.

We now combine Theorem 2.3 with the Girsanov theorem to get more explicit non-arbitrage conditions:

Theorem 2.5
(a) *Suppose there exist $\mathbb{F}$-predictable processes $\theta_0(t) \in \mathbb{R}^m$, $\theta_1(t,z) \in \mathbb{R}^\ell$ such that $\theta_{1,k} \leq 1$; $1 \leq k \leq \ell$, and*

$$\sigma(t)\theta_0(t) + \int_{\mathbb{R}^\ell} \gamma(t,z)\theta_1(t,z)\nu(\mathrm{d}z) = \mu(t) - r(t)\hat{S}(t^-) \quad \text{for a.a. } (t,\omega) \tag{2.1.17}$$

and

$$E\Big[\exp\Big(\int_0^T \Big\{\tfrac{1}{2}\theta_0^2(s) + \sum_{k=1}^{\ell}\int_{\mathbb{R}}\{(1-\theta_{1,k}(s,z))\ln(1-\theta_{1,k}(s,z)) + \theta_{1,k}(s,z)\}\nu_k(\mathrm{d}z)\Big\}\mathrm{d}s\Big)\Big] < \infty \text{ (the Novikov condition).} \tag{2.1.18}$$

Then the market has no arbitrage.

(b) *Conversely, suppose the market has no arbitrage. Then there exist $\mathcal{F}_t$-predictable processes $\theta_0(t) \in \mathbb{R}^m$, $\theta_1(t,z) \in \mathbb{R}^\ell$ such that (2.1.17) holds.*

Proof **(a)** With $\theta_0(t), \theta_1(t,z)$ as in (2.1.17), (2.1.18) define, with $\theta = (\theta_0, \theta_1)$,

$$Z(t) = Z_\theta(t) = \exp\Big(- \int_0^t \theta_0(s)\mathrm{d}B(s) - \tfrac{1}{2}\int_0^t \theta_0^2(s)\mathrm{d}s + \sum_{k=1}^{\ell} \int_0^t \int_{\mathbb{R}} \ln(1-\theta_{1,k}(s,z))\tilde{N}_k(\mathrm{d}s,\mathrm{d}z) + \sum_{k=1}^{\ell} \int_0^t \int_{\mathbb{R}} \{\ln(1-\theta_{1,k}(s,z)) + \theta_{1,k}(s,z)\}\nu_k(\mathrm{d}z)\mathrm{d}s\Big); \quad 0 \le t \le T. \tag{2.1.19}$$

Then by Theorem 1.36 we have $E[Z(T)] = 1$ and therefore the measure Q defined by

$$\mathrm{d}Q(\omega) = Z(T)\mathrm{d}P(\omega) \quad \text{on } \mathcal{F}_T \tag{2.1.20}$$

is a probability measure. By (2.1.11) we have

$$\mathrm{d}\bar{S}(t) = \xi(t)\Big[(\mu(t) - r(t)\hat{S}(t))\mathrm{d}t + \sigma(t)\mathrm{d}B(t) + \int_{\mathbb{R}} \gamma(t,z)\tilde{N}(\mathrm{d}t,\mathrm{d}z)\Big], \tag{2.1.21}$$

where $\hat{S}(t) = [S_1(t), \ldots, S_n(t)]^T$.

Therefore $\bar{S}(t)$ is a local martingale with respect to Q, by Theorem 1.31. So the conclusion follows from Theorem 2.3.

(b) For given $t \in [0,T]$, $\omega \in \Omega$ define

$$L = L_{t,\omega} = \{x \in \mathbb{R}^n; \text{ there exists } \theta_0 \in \mathbb{R}^m \text{ and } \theta_1(z) \in \mathbb{R}^\ell \text{ such that } \textstyle\int_{\mathbb{R}^\ell} |\gamma(t,z)|\,|\theta_1(z)|\nu(\mathrm{d}z) < \infty \text{ and } \sigma(t)\theta_0 + \int_{\mathbb{R}^\ell} \gamma(t,z)\theta_1(z)\nu(\mathrm{d}z) = x\}.$$

We see that L coincides with the linear span of the columns of $\sigma(t)$ and the columns of $\gamma(t,z)$; $z \in \mathrm{supp}_0\nu$, where $\mathrm{supp}_0\nu = \mathrm{supp}\nu \setminus \{0\}$, with $\mathrm{supp}\nu$ being the (smallest) closed support of ν.

Fix $t \in [0,T]$ and define

$$\begin{aligned} G = G_t &= \{\omega; \text{ the Eq. (2.1.17) has no solution } (u(t), \theta(t,z))\} \\ &= \{\omega; \mu(t) - r(t)\hat{S}(t^-) \notin L_{t,\omega}\} \\ &= \{\omega; \text{ there exists a } v = v(t) \in \mathbb{R}^{1\times n} \text{ such that } v \text{ is orthogonal to all the columns of } \sigma(t) \text{ and all } \gamma(t,z);\ z \in \mathrm{supp}_0\nu, \text{ and such that} \\ &\qquad v(t)(\mu(t) - r(t)\hat{S}(t^-)) \neq 0\} \\ &= \{\omega; \text{ there exists a } v = v(t) \in \mathbb{R}^{1\times n} \text{ such that } v(t)\sigma(t) = v(t)\gamma(t,z) = 0 \text{ for all } z \in \mathrm{supp}_0\nu \text{ and} \\ &\qquad v(t)(\mu(t) - r(t)\hat{S}(t^-)) \neq 0\}. \end{aligned}$$

Define

$$\varphi_i(t) = \begin{cases} \text{sign}(v(t)(\mu(t) - r(t)\hat{S}(t^-))v_i(t); & \omega \in G_t \\ 0; & \omega \notin G_t \end{cases}$$

for $i = 1, 2, \ldots, n$ and choose $\varphi_0(t)$ such that $\varphi(t) = (\varphi_0(t), \varphi_1(t), \ldots, \varphi_n(t))$ is self-financing and $X^{\varphi}(0) = 0$. Since $\sigma(t), \gamma(t, z)$ and $\mu(t) - r(t)\hat{S}(t^-)$ is $\mathbb{F}$-predictable, we can choose $\varphi(t)$ to be $\mathbb{F}$-predictable also. Moreover, with $\hat{\varphi}(s) = (\varphi_1(s), \ldots, \varphi_n(s))$, we have

$$\begin{aligned} \bar{X}^{\varphi}(t) &= \int_0^t \varphi(s)\mathrm{d}\bar{S}(s) \\ &= \int_0^t \hat{\varphi}(s)\Big[(\mu(s) - r(s)\hat{S}(s))\mathrm{d}s + \sigma(s)\mathrm{d}B(s) + \int_{\mathbb{R}^\ell} \gamma(s, z)\tilde{N}(\mathrm{d}s, \mathrm{d}z)\Big] \\ &= \int_0^t \sum_{i=1}^n |\mu_i(s) - r(s)S_i(s)|\chi_{G_s}(\omega)\mathrm{d}s \geq 0 \quad \text{a.s.} \end{aligned}$$

Hence $\bar{X}^{\varphi}(t)$ is lower bounded, and therefore $\varphi \in \mathcal{A}$. Since we have assumed that the market has no arbitrage, we conclude that $\chi_{G_s}(\omega) = 0$ for a.a. (s, ω). This is equivalent to saying that (2.1.17) has a solution for a.a. (t, ω). □

Example 2.6 Consider the market $(S_0(t), S_1(t))$ where

$$\mathrm{d}S_0(t) = r(t)S_0(t)\mathrm{d}t; \quad S_0(0) = 1$$

and

$$\mathrm{d}S_1(t) = S_1(t)\Big[a(t)\mathrm{d}t + b(t)\mathrm{d}B(t) + \int_{\mathbb{R}} c(t, z)\tilde{N}(\mathrm{d}t, \mathrm{d}z)\Big] \quad S_1(0) > 0,$$

where $c(t, z) > -1 + \delta$ for some $\delta > 0$. Then the Eq. (2.1.17) gets the form

$$S_1(t)b(t)\theta_0(t) + \int_{\mathbb{R}} S_1(t)c(t, z)\theta_1(t, z)\nu(\mathrm{d}z) = S_1(t)a(t) - S_1(t)r(t).$$

Since $S_1(t) > 0$ we can simplify the equation to

$$b(t)\theta_0(t) + \int_{\mathbb{R}} c(t, z)\theta_1(t, z)\nu(\mathrm{d}z) = a(t) - r(t).$$

If $b(t) \neq 0$ we can simply choose $\theta_1 = 0$ and

$$\theta_0(t) = \frac{a(t) - r(t)}{b(t)}.$$

If $b(t) = 0$ we get the equation

$$\int_{\mathbb{R}} c(t,z)\theta_1(t,z)\nu(\mathrm{d}z) = a(t) - r(t)$$

in the unknown $\theta_1(t,z)$, for each t, ω. If ν has a point mass $\lambda \geq |a(t) - r(t)|$ at $z = z_0$ then we can choose

$$\theta_1(t,z) = \chi_{\{z_0\}}(z)\frac{1}{\lambda}(a(t) - r(t)).$$

More generally, if

$$\int_{\mathbb{R}} |c(t,z)|\nu(\mathrm{d}z) \geq |a(t) - r(t)| \tag{2.1.22}$$

then there exists an $A \subset \mathbb{R}_0$ such that

$$|a(t) - r(t)| \geq \int_A |c(t,z)|\nu(\mathrm{d}z) =: K < \infty\,,$$

and then we can choose

$$\theta_1(t,z) = \frac{a(t) - r(t)}{K}\chi_A(z)\mathrm{sign}c(t,z).$$

Thus we see that if either $b(t) \neq 0$ or (2.1.22) holds, then the market has no arbitrage.

2.2 Hedging and Completeness

To make sure that the market has no arbitrage, we assume from now on that there exist predictable processes $\theta_0(t) \in \mathbb{R}^m$ and $\theta_1(t,z) \in \mathbb{R}^\ell$ such that $\theta_{1,k} \leq 1$; $1 \leq k \leq \ell$ and (2.1.17) and (2.1.18) hold. We fix such a pair $\theta = (\theta_0, \theta_1)$ and define $Z(t)$; Q, $\mathrm{d}B_Q = \theta_0(t)\mathrm{d}t + \mathrm{d}B(t)$ and $\tilde{N}_Q(\mathrm{d}t, \mathrm{d}z) = \theta_1(\mathrm{d}z)\mathrm{d}t + \tilde{N}(\mathrm{d}t, \mathrm{d}z)$ as in Theorem 1.35.

We now introduce the following terminology:

Definition 2.7

(a) A *claim* (or a T-claim) is a lower bounded $\mathcal{F}_T$-measurable random variable.

(b) A claim F is called *replicable* (or *hedgable* or *attainable*) if there exists a portfolio $\varphi \in \mathcal{A}$ and a constant $x \in \mathbb{R}$ such that

$$F = X_x^\varphi(T) := x + \int_0^T \varphi(s)\mathrm{d}S(s) \quad a.s.$$

and such that the normalized wealth process

$$\bar{X}_x^\varphi(t) := x + \int_0^t \varphi(s)\mathrm{d}\bar{S}(s)$$

is a Q-martingale.
(c) The market $\{S(t)\}_{t\in[0,T]}$ is called $\mathcal{F}_T$-*complete* if every bounded claim is replicable. Otherwise the market is *incomplete*.

What claims are replicable? When is the market complete? To answer such questions we need the following result: (For simplicity we formulate only the 1-dimensional case)

Theorem 2.8 (***The Extended Itô Representation Theorem***)

Let Q be a probability measure on $\mathcal{F}_T$ defined by

$$\mathrm{d}Q(\omega) = Z_\theta(T)\mathrm{d}P(\omega)$$

where

$$\mathrm{d}Z_\theta(t) = -Z_\theta(t^-)\left[\theta_0(t)\mathrm{d}B(t) + \int_{\mathbb{R}} \theta_1(t,z)\tilde{N}(\mathrm{d}t,\mathrm{d}z)\right]\ ;\ Z_\theta(0) = 1$$

for some $\theta = (\theta_0, \theta_1)$ with $\theta_1 > -1$ and

$$E\left[\int_0\left\{\theta_0^2(s) + \int_{\mathbb{R}} \theta_1^2(s,z)\nu(\mathrm{d}z)\right\}(\mathrm{d}z)\right] < \infty.$$

Let G be $\mathcal{F}_T$-measurable and assume that $Z_\theta(T)G \in L^2(P)$. Then there exists a unique pair $(h(t), \zeta(t,z))$ of $\mathbb{F}$-predictable processes such that

$$E_Q\Big[\int_0^T \Big\{h^2(t) + \int_{\mathbb{R}} \zeta^2(t,z)(1-\theta(t,z))\nu(\mathrm{d}z)\Big\}\mathrm{d}t\Big] < \infty \tag{2.2.1}$$

and

$$G = E_Q[G] + \int_0^T h(t)\mathrm{d}B_Q(t) + \int_0^T\int_{\mathbb{R}} \zeta(t,z)\tilde{N}_Q(\mathrm{d}t,\mathrm{d}z). \tag{2.2.2}$$

Remark 2.9

(i) Note that this result is not covered by the classical representation theorem of Itô [I], since the filtration $\tilde{\mathcal{F}}_t$ generated by $B_Q(s)$ and $\tilde{N}_Q(\mathrm{d}s,\mathrm{d}z); 0 \le s \le t, z \in \mathbb{R}_0$ may be strictly smaller than the filtration $\mathcal{F}_t$ generated by $B(s)$ and $\tilde{N}(\mathrm{d}s,\mathrm{d}z)$; $0 \le s \le t$, $z \in \mathbb{R}_0$. The stochastic integrals in (2.2.2) still make sense as semimartingale integrals, since $B_Q(t)$ and $\tilde{N}_Q(\mathrm{d}t,\mathrm{d}z)$ are semimartingales with respect to $\mathcal{F}_t$.

(ii) The condition (2.2.1) guarantees uniqueness of h and ζ. This follows from (2.2.2) by the Itô isometry for B_Q and $\tilde{N}_Q(\cdot,\cdot)$. See Theorem 1.35.

(iii) Under some additional assumptions, the integrands $h(t)$ and $\zeta(t,z)$ can in fact be expressed in terms of the *Malliavin–Hida generalized derivatives* of F. See

[Hue] and [Ok] for details. See also Theorem 12.22 in [DØP]. For a proof of Theorem 2.8 based on this theorem, see Exercise 2.6.

We can now give a characterization of replicable bounded claims:

Theorem 2.10 *Let F be a bounded claim. Define*

$$G = \bar{F} = \xi(T)F.$$

Suppose G has the representation (2.2.2). Then F is replicable if and only if there exists an $\mathcal{F}_t$-predictable process $\varphi(t)$ such that

$$h(t) = \xi(t)\varphi(t)\sigma(t) \tag{2.2.3}$$

and

$$\zeta(t,z) = \xi(t)\varphi(t)\gamma(t,z) \quad \textit{for a.a. } t, z, \omega. \tag{2.2.4}$$

Proof
(i) Suppose there exists an $\mathcal{F}_t$-predictable process $\varphi(t)$ such that (2.2.3) and (2.2.4) hold. Then by (2.2.2) we have

$$\begin{aligned}\bar{F} &= E_Q[\bar{F}] + \int_0^T h(t)\mathrm{d}B_Q(t) + \int_0^T \int_{\mathbb{R}} \zeta(t,z)\tilde{N}_Q(\mathrm{d}t,\mathrm{d}z)\\ &= E_Q|\bar{F}] + \int_0^T \xi(t)\varphi(t)\sigma(t)\mathrm{d}B_Q(t)\\ &\quad + \int_0^T \int_{\mathbb{R}} \xi(t)\varphi(t)\gamma(t,z)\tilde{N}_Q(\mathrm{d}t,\mathrm{d}z).\end{aligned}$$

By (2.1.21) and Theorem 1.35 this can be written

$$\begin{aligned}\bar{F} &= E_Q[\bar{F}] + \int_0^T \varphi(t)\mathrm{d}\bar{S}(t), \quad \text{i.e., using (2.1.13)},\\ F &= E_Q[\bar{F}] + \int_0^T \varphi(t)\mathrm{d}S(t).\end{aligned}$$

Moreover, with $x = E_Q[\bar{F}]$,

$$\begin{aligned}\bar{X}_x^{\varphi}(t) &= x + \int_0^t \varphi(s)\mathrm{d}\bar{S}(s)\\ &= x + \int_0^t h(s)\mathrm{d}B_Q(s) + \int_0^t \int_{\mathbb{R}} \zeta(s,z)\tilde{N}_Q(\mathrm{d}s,\mathrm{d}z); \qquad 0 \le t \le T.\end{aligned}$$

By (2.2.1) this is a Q-martingale. Hence F is replicable with φ as the replicating portfolio. **(ii)** Conversely, suppose F is replicable with replicating portfolio $\varphi \in \mathcal{A}$.

Then

$$\begin{aligned}\bar{F} = \bar{X}_x^{\varphi}(T) &= x + \int_0^T \varphi(s)\mathrm{d}\bar{S}(s) \\ &= x + \int_0^T \xi(s)\varphi(s)\Big[\sigma(s)\mathrm{d}B_Q(s) + \int_{\mathbb{R}} \gamma(s,z)\tilde{N}_Q(\mathrm{d}s,\mathrm{d}z)\Big]. \end{aligned} \tag{2.2.5}$$

Comparing (2.2.5) and (2.2.2) we get (2.2.3) and (2.2.4). □

Example 2.11 Consider the market $(S_0(t), S(t))$, with

$$\begin{aligned} S_0(t) &= 1; \\ \mathrm{d}S(t) &= b\,\mathrm{d}B(t) + \lambda\int_{\mathbb{R}} z\tilde{N}(\mathrm{d}t,\mathrm{d}z); \qquad t \in [0,T] \end{aligned}$$

where b, λ are constants. Here we can choose $Q = P$, $B_Q = B$ and $\tilde{N}_Q = \tilde{N}$. By the Itô formula

$$\begin{aligned} \mathrm{d}(S^2(t)) &= 2S(t)b\,\mathrm{d}B(t) + b^2\mathrm{d}t \\ &\quad + \int_{\mathbb{R}} \{(S(t^-) + \lambda z)^2 - S^2(t^-)\}\tilde{N}(\mathrm{d}t,\mathrm{d}z) \\ &\quad + \int_{\mathbb{R}} \{(S(t^-) + \lambda z)^2 - S^2(t) - 2S(t)\lambda z\}\nu(\mathrm{d}z)\mathrm{d}t \\ &= \Big\{b^2 + \int_{\mathbb{R}} \lambda^2 z^2 \nu(\mathrm{d}z)\Big\}\mathrm{d}t + 2S(t)b\,\mathrm{d}B(t) \\ &\quad + \int_{\mathbb{R}} \{2S(t^-)\lambda z + \lambda^2 z^2\}\tilde{N}(\mathrm{d}t,\mathrm{d}z). \end{aligned}$$

We conclude that the (2.2.2) representation of $F := S_1^2(T)$ is

$$F = x + \int_0^T h(t)\mathrm{d}B(t) + \int_0^T \int_{\mathbb{R}} \zeta(t,z)\tilde{N}(\mathrm{d}t,\mathrm{d}z),$$

with

$$x = \Big(b^2 + \int_{\mathbb{R}} \lambda^2 z^2 \nu(\mathrm{d}z)\Big)T, \quad h(t) = 2bS(t^-)$$

and

$$\zeta(t,z) = 2\lambda z S(t^-) + \lambda^2 z^2.$$

Thus by Theorem 2.10 F is replicable in this market if and only if there exists an $\mathbb{F}$-predictable process $\varphi(t)$ such that

$$2bS(t^-) = \varphi(t)b \quad \text{and} \quad 2\lambda z S(t^-)\lambda^2 z^2 = \varphi(t)\lambda z. \tag{2.2.6}$$

If $\lambda = 0$ then $S^2(T)$ is replicable with $\varphi(t) = 2S(t^-)$. If $b = 0$, $\lambda \neq 0$ then $S^2(T)$ is replicable if and only if ν is supported on one single point $\{z_0\}$, which corresponds to $S(t)$ being (essentially) a Poisson process with jump size z_0 and intensity $\Lambda = \lambda\nu(\{z_0\})$. If $b \neq 0$, $\lambda \neq 0$, then $S_1^2(T)$ is not replicable for any choice of Lévy measure ν.

This example illustrates that markets driven by Lévy processes with jumps are typically *incomplete*. Indeed, we have

Theorem 2.12
(a) *The market* $\{S(t)\}_{t\in[0,T]}$ *given in* (2.1.1)–(2.1.2) *is* $\mathcal{F}_T = \mathcal{F}_T^B \vee \mathcal{F}_T^N$-complete *if and only if the following hold:*

$$\begin{aligned} &\textit{There exists an } \mathcal{F}_t\textit{-adapted process } \Lambda(t) \in \mathbb{R}^{m\times n} \textit{ such that} \\ &\quad \Lambda(t)\sigma(t) = I_m \quad \textit{(the } m \times m \textit{ identity matrix)} \qquad (2.2.7) \\ &\textit{(i.e. } \operatorname{rank}\sigma(t) = m\textit{) for a.a. } t, \omega \\ &\nu = 0 \quad \textit{(i.e. there are no jumps).} \qquad (2.2.8) \end{aligned}$$

(b) *If* $\gamma = 0$, *then the market* $\{S(t)\}_{t\in[0,T]}$ *is* $\mathcal{F}_T^B$*-complete if and only if* (2.2.7) *holds.*

(c) *If* $\sigma = 0$, *then the market* $\{S(t)\}_{t\in[0,T]}$ *is* $\mathcal{F}_T^N$*-complete if and only if* ν *is supported on one point* $z_0 \in \mathbb{R}_0$, *in which case the underlying Lévy process*

$$\eta(t) = \int_0^t \int_{\mathbb{R}} z\tilde{N}(\mathrm{d}s, \mathrm{d}z) = z_0\tilde{N}((0,t], \{z_0\})$$

is a Poisson process with jump size z_0 *and intensity* $\nu(\{z_0\})$.

Proof **(a)** By Theorem 2.10 the market is $\mathcal{F}_T$-complete if and only if the two equations

$$\begin{aligned} \varphi(t)\sigma(t) &= h(t) \\ \varphi(t)\gamma(t,z) &= \zeta(t,z) \qquad (2.2.9) \end{aligned}$$

can be solved for $\varphi(t)$, for all choices of $h(t)$ and $\zeta(t,z)$, for a.a. t, z, ω. To be able to solve (2.2.9) for all $h(t)$ it is necessary and sufficient that $\sigma(t)$ has a left inverse $\Lambda(t)$, as in (2.2.7). Choosing $h(t) = 0$ we therefore get $\varphi(t) = 0$, but then (2.2.10) is only possible for all choices of $\zeta(t,z)$, for a.a. z with respect to ν if $\nu = 0$.

Conversely, if (2.2.8) and (2.2.9) hold, then it is well-known that the market is complete. See e.g. [Ø1, Theorem 12.2.5].
(b) If $\gamma = 0$ and we only consider $\mathcal{F}_T^B$-measurable claims, then we have $\zeta = 0$ in (2.2.2) and hence (2.2.10) is trivially satisfied. So the result follows from (a).
(c) If $\sigma = 0$ and we only consider $\mathcal{F}_T^N$-measurable claims, then we have $h(t) = 0$ in (2.2.2) and hence we are left with the necessary and sufficient condition (2.2.4). For this equation to hold for all choices of $\zeta(t,z)$ and for almost all z with respect to ν, we see that ν must be supported on one point only, say $z_0 \in \mathbb{R}_0$. □

The next result gives another useful characterization of replicable claims.

Theorem 2.13 *Let F be a bounded claim. Then F is replicable if and only if*

$$E_{\hat{Q}}[\bar{F}] = E_Q[\bar{F}] \tag{2.2.10}$$

for all EMMs $\hat{Q}$.

Proof (i) If F is replicable, then $\bar{F}$ is replicable and there exists a unique $x = E_Q[\bar{F}]$ and $\mathbb{F}$-predictable portfolio $\varphi(t)$ such that

$$x + \int_0^T \varphi(t)\mathrm{d}\bar{S}(t) = \bar{F}.$$

Hence $E_{\hat{Q}}[\bar{F}] = x$ for all EMMs $\hat{Q}$.
(ii) Conversely, suppose F is not replicable. Then $\bar{F}$ is not replicable, so by Theorem 2.10 there does not exist a portfolio $\varphi(t)$ such that (2.2.3) and (2.2.4) hold. In other words, if we define

$$H_t = \{\omega\ ;\ h(t) \text{ is not in the linear span of the rows of } \sigma(t)\} \tag{2.2.11}$$

and

$$K_{t,z} = \{\omega\ ;\ \zeta(t,z) \text{ is not in the linear span of the rows of } \gamma(t,z)\} \tag{2.2.12}$$

then

$$\int_0^T \left\{ P(H_t) + \int_{\mathbb{R}} P(K_{t,z})\nu(\mathrm{d}z) \right\} \mathrm{d}t > 0. \tag{2.2.13}$$

For $\omega \in H_t$ we can choose $\theta_0(t) \in \mathbb{R}^{1\times n}$ such that $\theta_0(t)$ is orthogonal to all the rows of $\sigma(t)$ but not to $h(t)$, i.e.,

$$\theta_0(t)\sigma(t,\omega) = 0 \text{ and } h(t)\cdot\theta_0(t) < 0, \tag{2.2.14}$$

where $\cdot$ denotes the scalar product.

If $\omega \notin H_t$ we put $\theta_0(t) = 0$.

For $\omega \in K_{t,z}$ we can choose $\theta_1(t,z) \in \mathbb{R}^{1\times n}$ such that

$$\theta_1(t,z)\gamma(t,z) = 0 \text{ and } \zeta(t,z)\cdot\theta_1(t,z) < 0. \tag{2.2.15}$$

If $\omega \notin K_{t,z}$ we put $\theta(t,z) = 0$.

We may choose $\theta_0(t)$ and $\theta_1(t,z)$ to be bounded, predictable processes, $\theta_{1,k}(t,z) \leq 1-\varepsilon$ for some constant $\varepsilon > 0$. Define the measure $\hat{Q}$ on $\mathcal{F}_T$ by

$$\mathrm{d}\hat{Q}(\omega) = Z_\theta(T)\mathrm{d}P(\omega),$$

where $\theta = (\theta_0, \theta_1)$ and $Z_\theta(t)$ is given by the equation

$$\begin{aligned}\mathrm{d}Z_\theta(t) &= -Z_\theta(t^-)\left[\theta_0(t)\mathrm{d}B_Q(t) + \int_{\mathbb{R}} \theta_1(t,z)\tilde{N}_Q(\mathrm{d}t,\mathrm{d}z)\right];\ t \leq T\\ Z_\theta(0) &= 1.\end{aligned}$$

Then $E_Q[Z_\theta(T)] = 1$ so $\hat{Q}$ is a probability measure. We claim that $\hat{Q}$ is an EMM. To see this, let $0 \leq t < t+h \leq T$ and put

$$\Delta\bar{S}_n(t) = \bar{S}_n(t+h) - \bar{S}_n(t).$$

Then

$$\begin{aligned}E_{\hat{Q}}[\Delta\bar{S}_n(t) \mid \mathcal{F}_t] &= \frac{E_Q[Z_\theta(T)\Delta\bar{S}_n(t) \mid \mathcal{F}_t]}{E_Q[Z_\theta(T) \mid \mathcal{F}_t]}\\ &= Z_\theta^{-1}(t)\cdot E_Q\left[\left(1 - \int_0^T Z_\theta(s)\theta_0(s)\mathrm{d}B_Q(s)\right.\right.\\ &\qquad \left. - \int_0^T\int_{\mathbb{R}} Z_\theta(s^-)\theta_1(s,z)\tilde{N}_Q(\mathrm{d}s,\mathrm{d}z)\right)\\ &\quad \cdot \left.\left(\int_t^{t+h}\sigma_n(s)\mathrm{d}B_Q(s) + \int_t^{t+h}\gamma_n(s,z)\tilde{N}_Q(\mathrm{d}s,\mathrm{d}z)\right)\right| \mathcal{F}_t\Bigg]\\ &= Z_\theta^{-1}(t)E_Q\left[-\int_t^{t+h} Z_\theta(s)\theta_0(s)\sigma_n(s)\mathrm{d}s\right.\\ &\quad \left. - \int_t^{t+h}\int_{\mathbb{R}} Z_\theta(s)\theta_1(s,z)\gamma_n(s,z)\nu(\mathrm{d}z)\mathrm{d}s \mid \mathcal{F}_t\right]\\ &= 0, \text{ by (2.2.14) and (2.2.15).}\end{aligned}$$

Hence $\hat{Q}$ is an EMM, as claimed.

Finally, we show that

$$E_{\hat{Q}}[\bar{F}] \neq E_Q[\bar{F}].$$

To this end, consider

$$\begin{aligned}E_{\hat{Q}}[\bar{F}] &= E_Q[Z_\theta(T)\bar{F}]\\ &= E_Q\left[\left(1 - \int_0^T Z_\theta(s)u(s)\mathrm{d}B_Q(s) - \int_0^T\int_{\mathbb{R}} Z_\theta(s)\theta(s,z)\tilde{N}(\mathrm{d}s,\mathrm{d}z)\right)\right.\end{aligned}$$

$$\cdot \left(E_Q[\bar{F}] + \int_0^T h(s)\mathrm{d}B_Q(s) + \int_0^T \int_{\mathbb{R}} \zeta(s,z)\tilde{N}_Q(\mathrm{d}s,\mathrm{d}z) \right) \Big]$$

$$= E_Q[\bar{F}] + E_Q\Big[-\int_0^T Z_\theta(s)\theta_0(s)\cdot h(s)\mathrm{d}s$$

$$- \int_0^T \int_{\mathbb{R}} Z_\theta(s)\theta_1(s,z)\cdot\zeta(s,z)\nu(\mathrm{d}z)\mathrm{d}s \Big]$$

$$= E_Q[\bar{F}] + E_Q\Big[\int_0^T \chi_{_{H_s}}(\omega) Z_\theta(s)|\theta_0(s)\cdot h(s)|\mathrm{d}s$$

$$+ \int_0^T \int_{\mathbb{R}} \chi_{_{K_{s,z}}}(\omega) Z_\theta(s)|\theta_1(s,z)\cdot\zeta(s,z)|\nu(\mathrm{d}z)\mathrm{d}s \Big] > E_Q[\bar{F}],$$

by (2.2.14) and (2.2.15). □

2.3 Option Pricing

2.3.1 European Options

Let F be a given claim. We may consider F to be the payoff at the terminal time T of a contract. Such a contract is called a *European option*. For example, if

$$F = (S_i(T) - K)^+$$

for some $i \in \{1,\dots,n\}$ and some constant $K > 0$, then F is the payoff of a *European call option*.

We now ask: What is the "right" price to pay at time $t = 0$ for a contract with the payoff $F(\omega)$ at time $t = T$? In general the answer depends on whether we ask the *buyer* or the *seller* of the contract. If we assume that neither of them want to risk losing money, then the answer is the following:

> *Buyer's price*, $p_b(F)$, is the maximum initial payment that allows the buyer to end up with a nonnegative terminal wealth a.s. by a proper choice of portfolio.

In other words:

$$p_b(F) = \sup\{x \in \mathbb{R};\ \text{there exists a } \varphi \in \mathcal{A} \text{ such that}$$

$$-x + \int_0^T \varphi(s)\mathrm{d}S(s) + F \geq 0 \text{ a.s.}\} \tag{2.3.1}$$

This price is also called the *lower hedging price*. Similarly,

Seller's price, $p_s(F)$, is the minimal initial received amount that allows the seller to end up with enough money at the terminal time to pay out $F(\omega)$ a.s., by a proper choice of portfolio.

In other words:

$$p_s(F) = \inf\{y \in \mathbb{R};\ \text{there exists a } \psi \in \mathcal{A} \text{ such that } y + \int_0^T \psi(s)\mathrm{d}S(s) \geq F \text{ a.s.}\} \tag{2.3.2}$$

This price is also called the *upper hedging price*. The relation between $p_b(F)$ and $p_s(F)$ is given in the following result:

Theorem 2.14 (***Pricing of European Options***)
(a) *Let F be a claim and let Q be an ELMM. Then, with $\xi(t) = \exp(-\int_0^t r(s)\mathrm{d}s)$ as before,*

$$p_b(F) \leq E_Q[\xi(T)F] \leq p_s(F). \tag{2.3.3}$$

(b) *Suppose F is replicable. Then*

$$p_b(F) = E_Q[\xi(T)F] = p_s(F) \tag{2.3.4}$$

for all ELMMs Q.

Proof **(a)** Choose $x \in \mathbb{R}$ and $\varphi \in \mathcal{A}$ such that

$$-x + \int_0^T \varphi(s)\mathrm{d}S(s) \geq -F \quad \text{a.s.} \tag{2.3.5}$$

Then by (2.1.13)

$$-x + \int_0^T \varphi(s)\mathrm{d}\bar{S}(s) \geq -\bar{F} \quad \text{a.s.}$$

Since $\bar{X}^{\varphi}_{-x}(t) = -x + \int_0^t \varphi(s)\mathrm{d}\bar{S}(s)$ is a lower bounded local Q-martingale (because $\varphi \in \mathcal{A}$), it is a supermartingale and hence

$$-x \geq E_Q\left[-x + \int_0^T \varphi(s)\mathrm{d}\bar{S}(s)\right] \geq E_Q[-\bar{F}]$$

i.e.

$$x \leq E_Q[\bar{F}].$$

Since this holds for all $x \in \mathbb{R}$ and $\varphi \in \mathcal{A}$ satisfying (2.3.5) we conclude that

$$p_b(F) \leq E_Q[\bar{F}].$$

Next, choose $y \in \mathbb{R}$ and $\psi \in \mathcal{A}$ such that

$$y + \int_0^T \psi(s)\mathrm{d}S(s) \geq F \quad \text{a.s.}$$

Then

$$y + \int_0^T \psi(s)\mathrm{d}\bar{S}(s) \geq \bar{F} \quad \text{a.s.}$$

Since $\bar{X}_y^{\psi}(t) := y + \int_0^T \psi(s)\mathrm{d}\bar{S}(s)$ is a lower bounded local Q-martingale, it is a Q-supermartingale and hence

$$y \geq E_Q\Big[y + \int_0^T \psi(s)\mathrm{d}\bar{S}(s)\Big] \geq E_Q[\bar{F}].$$

Since this holds for all such y, ψ we conclude that

$$p_s(F) \geq E_Q[\bar{F}].$$

(b) Suppose F is replicable. Then there exists $x_0 \in \mathbb{R}$ and $\varphi_0 \in \mathcal{A}$ such that

$$-x_0 + \int_0^T \varphi_0(s)\mathrm{d}\bar{S}(s) = -\bar{F} \quad \text{a.s.}$$

and such that

$$\bar{X}_{-x_0}^{\varphi_0}(t) := -x_0 + \int_0^t \varphi_0(s)\mathrm{d}\bar{S}(s) \tag{2.3.6}$$

is a Q-martingale. Then, by taking Q-expectation of (2.3.6) for $t = T$ we get

$$-x_0 = E_Q[\bar{X}_{-x_0}^{\varphi_0}(T)] = E_Q[-\bar{F}]$$

or

$$x_0 = E_Q[\bar{F}].$$

Hence

$$E_Q[\bar{F}] = x_0 \leq p_b(F).$$

Combined with (a) this gives

$$p_b(F) = x_0 = E_Q[\bar{F}].$$

A similar argument gives that $p_s(F) = E_Q[\bar{F}]$. □

Remark 2.15 Let $\mathcal{M}$ denote the set of all ELMMs Q. Then we have, under certain conditions

$$p_b(F) = \inf_{Q \in \mathcal{M}} E_Q[\xi(T)F] \tag{2.3.7}$$

and

$$p_s(F) = \sup_{Q \in \mathcal{M}} E_Q[\xi(T)F]. \tag{2.3.8}$$

See [Kr] and [Kun] for details.

2.3.2 American Options

Let $F(t), t \in [0, T]$, be a given $\mathcal{F}_t$-predictable lower bounded process. An American option on $F(\cdot)$ is a contract which gives the owner the right at any time $\tau \in [0, T]$ to exercise the contract, thereby receiving the payoff $F(\tau)$.

For example, if

$$F(t) = (K - S_i(t))^+; \qquad t \in [0, T]$$

for some $i = 1, \ldots, n$ and some constant $K > 0$, then the contract is called an *American put option*.

The time τ may be random, but must be based on what has happened in the market up to that time (not on future events). The mathematical way of expressing this is to say that τ must be an $\mathcal{F}_t$-*stopping time*, in the sense that

$$\{\omega; \tau(\omega) \leq t\} \in \mathcal{F}_t \qquad \text{for all } t \geq 0.$$

We let $\mathcal{T}$ denote the set of all $\mathcal{F}_t$-stopping times $\tau \leq T$.

Again we may ask: What is the "right" price $p^A(F)$ to pay at time $t = 0$ for such a contract? From the buyer's point of view the answer is:

$$\begin{aligned} p_b^A(F) = \sup\{x \in \mathbb{R};\ &\text{there exists } \varphi \in \mathcal{A} \text{ and } \tau \in \mathcal{T} \text{ such that} \\ &-x + \int_0^\tau \varphi(s)\mathrm{d}S(s) + F(\tau) \geq 0 \quad \text{a.s.}\} \end{aligned} \tag{2.3.9}$$

The seller, on the other hand, must be prepared at any time t to pay out the guaranteed amount $F(t)$. To achieve this, the seller must at least have the following initial amount:

$$\begin{aligned} p_s^A(F) = \inf\{y \in \mathbb{R};\ &\text{there exists a } \psi \in \mathcal{A} \text{ such that} \\ &y + \int_0^t \psi(s)\mathrm{d}S(s) \geq F(t) \quad \text{a.s. for } \textit{all } t \in [0, T]\}. \end{aligned} \tag{2.3.10}$$

The following concept is useful:

Define the *Snell envelope* $Y(t)$ by

$$Y(t) = \sup_{\tau \in \mathcal{T}, \tau \geq t} E_Q[\xi(\tau)F(\tau)|\mathcal{F}_t]; \qquad 0 \leq t \leq T. \tag{2.3.11}$$

Then $Y(t)$ is a Q-*supermartingale*. Therefore, by the Doob–Meyer decomposition theorem we can write

$$Y(t) = M(t) - A(t), \tag{2.3.12}$$

where $M(t)$ is a Q-martingale and $A(t)$ is an increasing process, $A(0) = 0$.

Note that

$$M(t) = Y(t) + A(t) \geq Y(t) \geq \xi(t)F(t) \tag{2.3.13}$$

and

$$M(0) = Y(0) = y_0, \tag{2.3.14}$$

where

$$y_0 = \sup_{\tau \in \mathcal{T}} E_Q[\xi(\tau)F(\tau)]. \tag{2.3.15}$$

We can now formulate the following analogue of Theorem 2.13:

Theorem 2.16 (***Pricing of American Options***)
(a) *Let Q be an ELMM. Then*

$$p_b^A(F) \leq \sup_{\tau \in \mathcal{T}} E_Q[\xi(\tau)F(\tau)] \leq p_s^A(F). \tag{2.3.16}$$

(b) *Suppose $F(\tau)$ is replicable for all $\tau \in \mathcal{T}$. Then*

$$p_b^A(F) = \sup_{\tau \in \mathcal{T}} E_Q[\xi(\tau)F(\tau)] = p_s^A(F). \tag{2.3.17}$$

(c) *Suppose $S_0(T)M(T)$ is replicable. Then*

$$p_s^A(F) = \sup_{\tau \in \mathcal{T}} E_Q[\xi(\tau)F(\tau)]. \tag{2.3.18}$$

Proof **(a)** Suppose $x \in \mathbb{R}$ is such that there exist $\tau_0 \in \mathcal{T}$ and $\varphi \in \mathcal{A}$ such that

$$-x + \int_0^{\tau_0} \varphi(s)\mathrm{d}S(s) + F(\tau_0) \geq 0 \qquad \text{a.s.}$$

Then

$$-x + \int_0^{\tau_0} \varphi(s)\mathrm{d}\bar{S}(s) + \xi(\tau_0)F(\tau_0) \geq 0 \qquad \text{a.s.}$$

Since $t \to \int_0^t \varphi(s)\mathrm{d}\bar{S}(s)$ is a Q-supermartingale we get by taking expectation that

$$x \leq E_Q[\xi(\tau_0)F(\tau_0)].$$

Since this holds for all such x we conclude that

$$p_b^A(F) \leq E_Q[\xi(\tau_0)F(\tau_0)].$$

Therefore

$$p_b^A(F) \leq \sup_{\tau \in \mathcal{T}} E_Q[\xi(\tau)F(\tau)].$$

Similarly, suppose $y \in \mathbb{R}$ is such that there exists a $\psi \in \mathcal{A}$ such that

$$y + \int_0^t \psi(s)\mathrm{d}S(s) \geq F(t) \qquad \text{a.s. for all } t \in [0, T].$$

Then

$$y + \int_0^\tau \psi(s)\mathrm{d}\bar{S}(s) \geq \xi(\tau)F(\tau) \qquad \text{a.s. for all } \tau \in \mathcal{T}.$$

Taking Q-expectations we get

$$y \geq E_Q[\xi(\tau)F(\tau)].$$

Since this holds for all such y we conclude that

$$p_s^A(F) \geq \sup_{\tau} E_Q[\xi(\tau)F(\tau)].$$

This proves (a).

(b) Choose $\tau \in \mathcal{T}$. Suppose $F(\tau)$ is replicable. Then we can find $x \in \mathbb{R}$ and $\varphi \in \mathcal{A}$ such that

$$-x + \int_0^\tau \varphi(s)\mathrm{d}S(s) = -F(\tau)$$

i.e.

$$-x + \int_0^\tau \varphi(s)\mathrm{d}\bar{S}(s) + \xi(\tau)F(\tau) = 0.$$

Moreover, $\int_0^t \varphi(s)\mathrm{d}\bar{S}(s)$ is a Q-martingale. Therefore, taking Q-expectation we get

$$x = E_Q[\xi(\tau)F(\tau)].$$

Hence, any payment x of the form $x = E_Q[\xi(\tau)F(\tau)]$ is acceptable for the buyer. It follows that

$$p_b^A(F) \geq \sup_{\tau \in \mathcal{T}} E_Q[\xi(\tau)F(\tau)].$$

Combined with (a) this completes the proof of (b).

(c) Suppose $S_0(T)M(T)$ is replicable. Then we can find $y \in \mathbb{R}$ and $\psi \in \mathcal{A}$ such that

$$y + \int_0^T \psi(s)\mathrm{d}S(s) = -S_0(T)M(T)$$

i.e.

$$y + \int_0^T \psi(s)\mathrm{d}\bar{S}(s) = \xi(T)S_0(T)M(T) = M(T).$$

Moreover, $\int_0^t \psi(s)\mathrm{d}\bar{S}(s)$ is a martingale. Therefore

$$y + \int_0^t \psi(s)\mathrm{d}\bar{S}(s) = M(t) \geq \xi(t)F(t); \qquad t \in [0, T].$$

Taking Q-expectations we get

$$y = E[M(t)] = M(0) = y_0.$$

This proves that with $y = y_0$ and $\psi \in \mathcal{A}$ as above we do indeed have

$$y_0 + \int_0^t \psi(s)\mathrm{d}S(s) \geq F(t) \qquad \text{for all } t \in [0, T].$$

Hence, by definition of $p_s^A(F)$,

$$p_s^A(F) \leq y_0.$$

Combined with (a) this proves (c). □

The problem to find

$$y_0 = \sup_{\tau \in \mathcal{T}} E_Q[\xi(\tau)F(\tau)]$$

and, if possible, a $\tau^* \in \mathcal{T}$ for which the supremum is attained, is called an *optimal stopping problem*. In the next chapter we will study this problem in a general jump diffusion context.

2.4 Exercises

Exercise* 2.1 Decide whether or not the following markets have arbitrages. If the market has an arbitrage, find one.

(a) $\mathrm{d}S_0(t) = 0$; $S_0(0) = 1$
$\mathrm{d}S_1(t) = S_1(t^-)[\alpha\,\mathrm{d}t + \int_{\mathbb{R}} z\tilde{N}(\mathrm{d}t, \mathrm{d}z)]$; $S_1(0) > 0$,
where $z > -1$ a.s. ν, $\alpha \in \mathbb{R}$ is a constant and

$$\int_{\mathbb{R}} |z|\nu(\mathrm{d}z) > |\alpha|.$$

(b) $\mathrm{d}S_0(t) = 0$; $S_0(0) = 1$
$\mathrm{d}S_1(t) = S_1(t^-)[-\mathrm{d}t - 2\mathrm{d}B(t) + 3\int_{\mathbb{R}} z\tilde{N}(\mathrm{d}t, \mathrm{d}z)]$
$\mathrm{d}S_2(t) = S_2(t^-)[4\mathrm{d}t + 4\mathrm{d}B(t) - 6\int_{\mathbb{R}} z\tilde{N}(\mathrm{d}t, \mathrm{d}z)]$
$S_i(0) > 0, i = 1, 2.$

Exercise* 2.2 Consider the following Lévy version of the Black–Scholes market:

$$\begin{aligned} &\text{(Bondprice)} \quad \mathrm{d}S_0(t) = rS_0(t)\mathrm{d}t; && S_0(0) = 1 \\ &\text{(Stockprice)} \quad \mathrm{d}S_1(t) = S_1(t^-)[\mu\,\mathrm{d}t + \gamma\,\mathrm{d}\eta(t)]; && S_1(0) = x > 0, \end{aligned}$$

where r, μ, and $\gamma \neq 0$ are constants and

$$\eta(t) = \int_0^t \int_{\mathbb{R}} z\tilde{N}(\mathrm{d}t, \mathrm{d}z)$$

is a pure jump Lévy martingale. To ensure that $S_1(t) \geq 0$ for all $t \geq 0$ we assume that $\gamma z \geq -1$ a.s. ν. Assume in addition that

$$\int_{\mathbb{R}} |z|\nu(\mathrm{d}z) > \left|\frac{\mu - r}{\gamma}\right|, \tag{2.4.1}$$

i.e., that the total mass of the jump measure (Lévy measure) ν exceeds $|\frac{\mu-r}{\gamma}|$.

The *normalized* stock price $\bar{S}_1(t)$ is given by

$$\bar{S}_1(t) = \frac{1}{S_0(t)} S_1(t) = \mathrm{e}^{-rt} S_1(t).$$

Note that

$$\mathrm{d}\bar{S}_1(t) = \bar{S}_1(t^-)[(\mu - r)\mathrm{d}t + \gamma\,\mathrm{d}\eta(t)], \quad \bar{S}_1(0) = x.$$

(a) Find an ELMM Q of the process $\bar{S}_1(t)$.
(b) We now discuss the concept of *arbitrage* in this market (see Definition 2.1).

A *portfolio* in this market is a predictable process $\phi(t) = (\phi_0(t), \phi_1(t)) \in \mathbb{R}^2$ such that

$$\int_0^T \{\phi_0^2(s) + \phi_1^2(s)\}\mathrm{d}s < \infty \text{ a.s. } P. \tag{2.4.2}$$

The corresponding *wealth process* $V^\phi(t)$ is defined by

$$V^\phi(t) = \phi_0(t)S_0(t) + \phi_1(t)S_1(t), \quad 0 \leq t \leq T. \tag{2.4.3}$$

We say that (ϕ_0, ϕ_1) is *self-financing* if $V^\phi(t)$ is also given by

$$V^\phi(t) = V^\phi(0) + \int_0^t \phi_0(s)\mathrm{d}S_0(s) + \int_0^t \phi_1(s)\mathrm{d}S_1(s). \tag{2.4.4}$$

If, in addition,

$$\left\{V^\phi(t)\right\}_{t\in[0,T]} \quad \text{is lower bounded} \tag{2.4.5}$$

we say that ϕ is *admissible* and write $\phi \in \mathcal{A}_0$. A portfolio $\phi \in \mathcal{A}_0$ is called an *arbitrage* if

$$V^\phi(0) = 0, \quad V^\phi(T) \geq 0, \quad \text{and} \quad P[V^\phi(T) > 0] > 0. \tag{2.4.6}$$

Does this market have an arbitrage?

Exercise* 2.3 Consider the following market

$$\begin{array}{lll}
\text{(Bondprice)} & \mathrm{d}S_0(t) = 0; & S_0(0) = 1 \\
\text{(Stockprice 1)} & \mathrm{d}S_1(t) = S_1(t^-)[\mu_1\mathrm{d}t + \gamma_{11}\mathrm{d}\eta_1(t) + \gamma_{12}\mathrm{d}\eta_2(t)]; & S_1(0) = x_1 > 0 \\
\text{(Stockprice 2)} & \mathrm{d}S_2(t) = S_2(t^-)[\mu_2\mathrm{d}t + \gamma_{21}\mathrm{d}\eta_1(t) + \gamma_{22}\mathrm{d}\eta_2(t)]; & S_2(0) = x_2 > 0
\end{array}$$

where μ_i and γ_{ij} are constants and $\eta_1(t)$, $\eta_2(t)$ are independent Lévy martingales of the form

$$\mathrm{d}\eta_i(t) = \int_{\mathbb{R}} z\widetilde{N}_i(\mathrm{d}t, \mathrm{d}z), \quad i = 1, 2.$$

Assume that the matrix $\gamma := \left[\gamma_{ij}\right]_{1\leq i,j\leq 2} \in \mathbb{R}^2$ is invertible, with inverse

$$\gamma^{-1} = \lambda = \left[\lambda_{ij}\right]_{1\leq i,j\leq 2}$$

and assume that

$$\int_{\mathbb{R}} |z|\nu_i(\mathrm{d}z) > |\lambda_{i1}\mu_1 + \lambda_{i2}\mu_2| \quad \text{for} \quad i = 1, 2. \tag{2.4.7}$$

Find an ELMM Q for $(S_1(t), S_2(t))$ and use this to deduce that there is no arbitrage in this market.

Exercise 2.4 Consider the market in Example 2.11.

(a) Find all ELMMs Q_θ of the form

$$\mathrm{d}Q_\theta = Z_\theta(T)\mathrm{d}P \text{ on } \mathcal{F}_T,$$

where

$$\begin{aligned} \mathrm{d}Z_\theta(t) &= Z_\theta(t^-)\left[-\theta_0(t)B(t) - \int_{\mathbb{R}} \theta_1(t,z)\tilde{N}(\mathrm{d}t,\mathrm{d}z)\right]\ ;\ t \in [0,T] \\ Z_\theta(0) &= 1 \end{aligned}$$

and $\theta = (\theta_0, \theta_1)$. (See e.g. Theorem 2.5).

(b) Show that

$$\begin{aligned} E_{Q_\theta}[S^2(T)] &= E[Z_\theta(T)S^2(T)] \\ &= S^2(0) + E\left[\int_0^T Z_\theta(t^-)\mathrm{d}(S^2(t)) + \int_0^T S^2(t^-)\mathrm{d}Z_\theta(t)\right. \\ &\quad \left. - \int_0^T Z_\theta(t)\theta_0(t)b\mathrm{d}t - \int_0^T \int_{\mathbb{R}} Z_\theta(t^-)\theta_1(t,z)\lambda z\nu(\mathrm{d}z)\mathrm{d}t\right] \\ &= S^2(0) + \left(b^2 + \int_{\mathbb{R}} \lambda^2 z^2 \nu(\mathrm{d}z)\right) E\left[\int_0^T Z_\theta(t)\mathrm{d}t\right] \\ &\quad - E\left[\int_0^T Z_\theta(t)\left\{b\theta_0(t) + \int_{\mathbb{R}} \lambda z\theta_1(t,z)\nu(\mathrm{d}z)\right\}\mathrm{d}t\right] \\ &= S^2(0) + T\left(b^2 + \int_{\mathbb{R}} \lambda^2 z^2 \nu(\mathrm{d}z)\right) \\ &\quad - E\left[\int_0^T Z_\theta(t)\left\{b\theta_0(t) + \int_{\mathbb{R}} \lambda z\theta_1(t,z)\nu(\mathrm{d}z)\right\}\mathrm{d}t\right]. \end{aligned}$$

(c) Use (b) combined with Theorem 2.13 to prove that if $b \neq 0$, then $S^2(T)$ is not replicable. This gives an alternative proof of the statement at the end of Example 2.11.

Exercise 2.5 (***Using the Girsanov Theorem in Stochastic Control***)

The following exercise illustrates a useful way of applying the Girsanov theorem to solve some stochastic control problems. The idea of this method goes back to [Fl]. See also [PR]. A presentation of this method in the Brownian motion case is given in [Ø1, Exercise 8.18].

Consider the following financial market

(Risk free asset)

$$\mathrm{d}S_0(t) = \rho(t)S_0(t)\mathrm{d}t\ ;\ S_0(0) = 1$$

(Risky asset)

$$\mathrm{d}S_1(t) = S_1(t^-)[\mu(t)\mathrm{d}t + \sigma(t)\mathrm{d}B(t) + \int_{\mathbb{R}} \gamma(t,z)\tilde{N}(\mathrm{d}t,\mathrm{d}z)]\ ;\ S_1(0) > 0.$$

Here $\rho(t)$, $\mu(t)$, $\sigma(t)$ and $\gamma(t,z)$ are *deterministic* functions such that

$$\int_0^T \left\{ |\rho(s)| + |\mu(s)| + \sigma^2(s) + \int_{\mathbb{R}} \gamma^2(t,z)\nu(\mathrm{d}z) \right\} \mathrm{d}s < \infty,$$

where $T > 0$ is a given constant.

Let $\pi(t) = \pi(t,\omega)$ be an $\mathbb{F}$-predictable process, representing the fraction of the total wealth invested in the risky asset at time t.

(a) Explain that, if we assume that $\pi(t)$ is self-financing, then the corresponding wealth process $X_\pi(t)$ satisfies the equations

$$\begin{aligned} \mathrm{d}X_\pi(t) = X_\pi(t^-)\,[\{\rho(t) + (\mu(t) - \rho(t))\pi(t)\}\mathrm{d}t \\ + \pi(t)\beta(t)\mathrm{d}B(t) + \pi(t)\int_{\mathbb{R}} \gamma(t,z)\tilde{N}(\mathrm{d}t,\mathrm{d}z)\Big]\ ;\ t \in [0,T]. \end{aligned} \tag{2.4.8}$$

We put $X_\pi(0) = 1$.

We call $\pi(\cdot)$ *admissible* and write $\pi \in \mathcal{A}$ if π is $\mathcal{F}_t$-predictable and self-financing and

$$\int_0^T \{|\mu(s) - \rho(s)||\pi(s)| + \pi^2(s)\sigma^2(s) + \int_{\mathbb{R}} \pi^2(s)\gamma^2(s,z)\nu(\mathrm{d}z)\}\mathrm{d}s < \infty \text{ a.s.}$$

(b) Use the Itô formula to show that the solution $X_\pi(t)$ of the Eq. (2.4.8) is

$$\begin{aligned} X_\pi(t) = \exp\Bigg(\int_0^t \Bigg\{ \rho(s) + (\mu(s) - \rho(s))\pi(s) - \frac{1}{2}\pi^2(s)\beta^2(s) \\ + \int_{\mathbb{R}} (\ln(1 + \pi(s)\gamma(s,z)) - \pi(s)\gamma(s,z))\nu(\mathrm{d}z) \Bigg\} \mathrm{d}s \\ + \int_0^t \pi(s)\beta(s)\mathrm{d}B(s) + \int_0^t \ln(1 + \pi(s)\gamma(s,z))\tilde{N}(\mathrm{d}s,\mathrm{d}z) \Bigg). \end{aligned} \tag{2.4.9}$$

(c) Let $\lambda \in (-\infty, 1)\backslash\{0\}$ be a given constant. We want to maximize

$$\frac{1}{\lambda} E[X_\pi^\lambda(T)]$$

over all admissible portfolios $\pi \in \mathcal{A}$.

(d) Choose $\theta_0(s)$ and $\theta_1(s,z)$ such that $\theta_0(s) = \lambda\pi(s)\beta(s)$ and $\ln(1 - \theta_1(s,z)) = \lambda \ln(1 + \pi(s)\gamma(s,z))$. Show that $X_\pi^\lambda(T)$ can be written

$$\begin{aligned} X_\pi^\lambda(T) = \exp\Bigg(\int_0^T \lambda \Bigg\{ \rho(s) + (\mu(s) - \rho(s))\pi(s) - \frac{1}{2}\pi^2(s)\beta^2(s) \\ + \int_{\mathbb{R}} (\ln(1 + \pi(s)\gamma(s,z)) - \pi(s)\gamma(s,z))\nu(\mathrm{d}z) \Bigg\} \mathrm{d}s \end{aligned}$$

$$+\int_0^T \theta_0(s)\mathrm{d}B(s)+\int_0^T\int_{\mathbb{R}}\ln(1-\theta_1(s,z))\tilde{N}(\mathrm{d}s,\mathrm{d}z)\Big). \quad (2.4.10)$$

(e) Define

$$\begin{aligned}Z_\pi(T)=\exp\Big(&\int_0^T\theta_0(s)\mathrm{d}B(s)-\frac{1}{2}\int_0^T\theta_0^2(s)\mathrm{d}s\\&+\int_0^T\int_{\mathbb{R}_0}\ln(1-\theta_1(s,z))\tilde{N}(\mathrm{d}s,\mathrm{d}z)\\&+\int_0^T\int_{\mathbb{R}}(\ln(1-\theta_1(s,z))+\theta_1(s,z))\nu(\mathrm{d}z)\mathrm{d}s\Big).\end{aligned}$$

Assume that

$$E[Z_\pi(T)]=1$$

and define the probability measure Q_π on $\mathcal{F}_T$ by

$$\mathrm{d}Q_\pi(\omega)=Z_\pi(T)\mathrm{d}P(\omega).$$

Show that

$$E[X_\pi^\lambda(T)]=E_{Q_\pi}\left[\exp\left(\int_0^T h_\pi(t)\mathrm{d}t\right)\right], \quad (2.4.11)$$

where

$$\begin{aligned}h_\pi(t)=&\lambda\rho(t)+\lambda(\mu(t)-\rho(t))\pi(t)+\frac{1}{2}\lambda(\lambda-1)\pi^2(t)\beta^2(t)\\&+\int_{\mathbb{R}}((1+\pi(t)\gamma(t,z))^\lambda-1-\lambda\pi(t)\gamma(t,z))\nu(\mathrm{d}z). \quad (2.4.12)\end{aligned}$$

(f) If we minimize or maximize the function $\pi\to h_\pi(t)$ for fixed t,ω, show that we get the following first-order equation for the optimal $\pi(t)$:

$$\mu(t)-\rho(t)-(1-\lambda)\pi(t)\beta^2(t)+\int_{\mathbb{R}}\gamma(t,z)[(1+\pi(t)\gamma(t,z))^{\lambda-1}-1]\nu(\mathrm{d}z)=0. \quad (2.4.13)$$

(g) Since the solution $\pi^*(t)$ of Eq. (2.4.13) is deterministic, conclude that, if $\lambda\in(0,1)$,

$$\begin{aligned}E[X_\pi^\lambda(T)]&=E_{Q_\pi}\left[\exp\left(\int_0^T h_\pi(t)\mathrm{d}t\right)\right]\\&\le E_{Q_\pi}\left[\exp\left(\int_0^T h_{\pi^*}(t)\mathrm{d}t\right)\right]=\exp\left(\int_0^T h_{\pi^*}(t)\mathrm{d}t\right)\end{aligned}$$

for all $\pi \in \mathcal{A}$. A similar argument works for $\lambda < 0$. *This proves that the optimal portfolio $\pi(t)$ for the problem in (c) is given as the solution of the equation* (2.4.13). Note that we have assumed that all the coefficients $\rho(t)$, $\mu(t)$, $\sigma(t)$ and $\gamma(t,z)$ are deterministic. Hence (2.4.13) shows, in particular, that in this case the optimal portfolio is also deterministic.

Exercise 2.6 **(*The Extended Itô Representation Theorem*)**

This exercise requires knowledge of Malliavin calculus for Lévy processes. We refer to [DØP] for information about this calculus.

By a result of F. Huenhe [Hue] and Y.Y. Okur [Ok] the conclusion of Theorem 2.8 holds if the random variable G and the processes $\theta_0(t)$, $\theta_1(t,z)$ satisfy certain conditions involving Malliavin derivatives. In that case we have the representation

$$G = E_Q[G] + \int_0^T \varphi_\theta(t)\mathrm{d}B_Q(t) + \int_0^T \int_{\mathbb{R}} \psi_\theta(t,z)\tilde{N}_Q(\mathrm{d}t,\mathrm{d}z), \tag{2.4.14}$$

where

$$\varphi_\theta(t) = E_Q\left[D_tG - G\int_t^T D_t\theta_0(s)\mathrm{d}B_Q(s) \mid \mathcal{F}_t\right] \tag{2.4.15}$$

and

$$\psi_\theta(t,z) = E_Q[F(\tilde{H}_{t,z} - 1) + \tilde{H}_{t,z}D_{t,z}F \mid \mathcal{F}_t], \tag{2.4.16}$$

where

$$\tilde{H}_{t,z} = \exp\left[\int_t^T \int_{\mathbb{R}} \left\{D_{t,z}\theta_1(s,x) + \ln\left(1 - \frac{D_{t,z}\theta_1(s,x)}{1-\theta_1(s,x)}\right)(1-\theta_1(s,x))\right\} \nu(\mathrm{d}x)\mathrm{d}s + \int_t^T \int_{\mathbb{R}} \ln\left(1 - \frac{D_{t,z}\theta_1(s,x)}{1-\theta_1(s,x)}\right)\tilde{N}_Q(\mathrm{d}s,\mathrm{d}x)\right]. \tag{2.4.17}$$

See also Theorem 2.22 in [DØP].

We proceed from this to the general situation in Theorem 2.8 in several steps:

Step 1

First assume that G is bounded and Malliavin differentiable. Then if θ_0, θ_1 are Malliavin differentiable use the product rule for Malliavin derivatives and the Bayes rule for conditional expectation to prove that

$$\varphi_\theta(t) = \frac{E[D_t(Z_TG) \mid \mathcal{F}_t]}{Z_t} + \theta_0(t)E_Q[G \mid \mathcal{F}_t] \tag{2.4.18}$$

and

$$\psi_\theta(t,z) = \frac{E[D_{t,z}(Z_TG) \mid \mathcal{F}_t]}{Z_t} + \theta_1(t,z)E_Q[G \mid \mathcal{F}_t]. \tag{2.4.19}$$

Step 2

Given θ_0, θ_1 as in Theorem 2.8 we can find a sequence of $\theta_0^{(k)}, \theta_1^{(k)}$ satisfying the conditions of [Hue] and [Ok] and such that

$$E\left[\int_0^T \{(\theta_0(s) - \theta_0^{(k)}(s))^2 + \int_{\mathbb{R}} (\theta_1(s,z) - \theta_1^{(k)}(s,z))^2 \nu(\mathrm{d}z)\}\mathrm{d}s\right] \to 0$$

when $k \to \infty$, and such that (2.4.14)–(2.4.17) hold for each $(\theta_0^{(k)}, \theta_1^{(k)})$; $k = 1, 2, \ldots$. Use (2.4.18) and (2.4.19) to show that the sequences $\{\varphi_{\theta_0^{(k)}}(t)\}_{k=1}^{\infty}$ and $\{\psi_{\theta_1^{(k)}}(t,z)\}_{k=1}^{\infty}$ and Cauchy sequences in $L^2(\mathrm{d}\lambda \times \mathrm{d}P)$ and $L^2(\mathrm{d}\lambda \times \mathrm{d}\nu \times \mathrm{d}P)$, respectively, and denote the limits by $\varphi(t)$ and $\psi(t,z)$, respectively.

Deduce that (2.2.2) holds with $h(t) = \varphi(t)$ and $\zeta(t,z) = \psi(t,z)$ when G is bounded and Malliavin differentiable.

Step 3

Finally, prove the result in Theorem 2.8 for a general $\mathcal{F}_T$-measurable G with $Z_T G \in L^2(P)$ by approximating G by bounded Malliavin differentiable G_n such that

$$Z_T G_n \to Z_T G \text{ in } L^2(P),$$

and using that the representation (2.4.14)–(2.4.17) holds for each G_n. Here $D_t(Z_T G)$ and $D_{t,z}(Z_T G)$ are the generalized Hida–Malliavin derivatives in $\mathcal{G}^*$. See [AaØU] and also [DØP, Theorems 6.4.1 and 13.26].

Chapter 3
Optimal Stopping of Jump Diffusions



3.1 A General Formulation and a Verification Theorem

Fix an open set $\mathcal{S} \subset \mathbb{R}^k$ (the *solvency region*) and let $Y(t)$ be a jump diffusion in $\mathbb{R}^k$ given by

$$\mathrm{d}Y(t) = b(Y(t))\mathrm{d}t + \sigma(Y(t))\mathrm{d}B(t) + \int_{\mathbb{R}^\ell} \gamma(Y(t^-), z)\tilde{N}(\mathrm{d}t, \mathrm{d}z), \quad Y(0) = y \in \mathbb{R}^\ell,$$

where $b : \mathbb{R}^k \to \mathbb{R}^k$, $\sigma : \mathbb{R}^k \to \mathbb{R}^{k\times m}$, and $\gamma : \mathbb{R}^k \times \mathbb{R}^\ell \to \mathbb{R}^{k\times\ell}$ are given functions such that a unique solution $Y(t)$ exists (see Theorem 1.19). Let

$$\tau_{\mathcal{S}} = \tau_{\mathcal{S}}(y, \omega) = \inf\{t > 0; Y(t) \notin \mathcal{S}\} \tag{3.1.1}$$

be the *bankruptcy time* and let $\mathcal{T}$ denote the set of all stopping times $\tau \le \tau_{\mathcal{S}}$.

The results below remain valid, with natural modifications, if we allow $\mathcal{S}$ to be any Borel set such that $\mathcal{S} \subset \overline{\mathcal{S}^0}$, where $\mathcal{S}^0$ denotes the interior of $\mathcal{S}$ and $\overline{\mathcal{S}^0}$ its closure.

Let $f : \mathbb{R}^k \to \mathbb{R}$ and $g : \mathbb{R}^k \to \mathbb{R}$ be continuous functions satisfying the conditions

$$E^y\left[\int_0^{\tau_{\mathcal{S}}} f^-(Y(t))\mathrm{d}t\right] < \infty \quad \text{for all } y \in \mathbb{R}^k. \tag{3.1.2}$$

$$\text{The family } \{g^-(Y(\tau)) \cdot \mathcal{X}_{\{\tau<\infty\}}, \tau \in \mathcal{T}\} \text{ is uniformly integrable, for all } y \in \mathbb{R}^k. \tag{3.1.3}$$

(If x is a real number, then $x^- := \max(-x, 0)$ denotes the *negative part* of x.)

The general *optimal stopping problem* is the following:

Find $\Phi(y)$ and $\tau^* \in \mathcal{T}$ such that

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$$\Phi(y) = \sup_{\tau \in \mathcal{T}} J^{\tau}(y) = J^{\tau^*}(y), \quad y \in \mathbb{R}^k,$$

where

$$J^{\tau}(y) = E^y\left[\int_0^{\tau} f(Y(t))\mathrm{d}t + g(Y(\tau)) \cdot \mathcal{X}_{\{\tau<\infty\}}\right], \quad \tau \in \mathcal{T}$$

is the *performance criterion*.

The function Φ is called the *value function* and the stopping time τ^* (if it exists) is called an *optimal stopping time*.

In the following we let A be the integrodifferential operator which coincides with the generator of $Y(t)$ on $C_0^2(\mathbb{R}^k)$, i.e.,

$$\begin{aligned} A\phi(y) = &\sum_{i=1}^{k} b_i(y)\frac{\partial \phi}{\partial y_i}(y) + \frac{1}{2}\sum_{i,j=1}^{k}(\sigma\sigma^T)_{ij}(y)\frac{\partial^2\phi}{\partial y_i \partial y_j}(y) \\ &+ \sum_{j=1}^{\ell}\int_{\mathbb{R}}\{\phi(y+\gamma^{(j)}(y,z_j)) \\ &- \phi(y) - \nabla\phi(y)\cdot\gamma^{(j)}(y,z_j)\}\nu_j(\mathrm{d}z_j) \end{aligned} \tag{3.1.4}$$

for all $\phi : \mathbb{R}^k \to \mathbb{R}$ and $y \in \mathbb{R}^k$ such that (3.1.4) exists. (See Theorems 1.22 and 1.24.)

We will need the following result. A proof of a related result (in the no jump case) can be found in [Ø1].

Theorem 3.1 (***Approximation Theorem***)

Let D be an open set, $D \subset \mathcal{S}$. Assume that $Y(\tau_{\mathcal{S}}) \in \partial\mathcal{S}$ a.s. on $\{\tau_{\mathcal{S}} < \infty\}$ and

$$\partial D \text{ is a Lipschitz surface} \tag{3.1.5}$$

(i.e., ∂D is locally the graph of a Lipschitz continuous function) and let $\varphi : \bar{\mathcal{S}} \to \mathbb{R}$ be a function with the following properties:

$$\varphi \in C^1(\mathcal{S}) \cap C(\bar{\mathcal{S}}) \tag{3.1.6}$$

and

$$\varphi \in C^2(\mathcal{S}\backslash\partial D) \tag{3.1.7}$$

and the second-order derivatives of φ are locally bounded near ∂D.

Then there exists a sequence $\{\varphi_m\}_{m=1}^{\infty} \subset C^2(\mathcal{S}) \cap C(\bar{\mathcal{S}})$ such that, with A as in (3.1.4),

$$\varphi_m \to \varphi \text{ pointwise dominatedly in } \bar{\mathcal{S}} \text{ as } m \to \infty \tag{3.1.8}$$

$$\frac{\partial \varphi_m}{\partial x_i} \to \frac{\partial \varphi}{\partial x_i} \text{ pointwise dominatedly in } \mathcal{S} \text{ as } m \to \infty \tag{3.1.9}$$

$$\frac{\partial^2 \varphi_m}{\partial x_i \partial x_j} \to \frac{\partial^2 \varphi}{\partial x_i \partial x_j} \text{ and } A\varphi_m \to A\varphi$$
$$\text{pointwise dominatedly in } \mathcal{S} \backslash \partial D \text{ as } m \to \infty. \tag{3.1.10}$$

We now formulate a set of sufficient conditions for a given function ϕ to coincide with the value function Φ and the corresponding stopping time, τ_D, to be optimal. The result is analogous to the variational inequality verification theorem for optimal stopping of continuous diffusions. See, e.g., [Ø1, Theorem 10.4.1].

Theorem 3.2 (***Integrovariational Inequalities for Optimal Stopping***)

(a) *Suppose we can find a function* $\phi : \bar{\mathcal{S}} \to \mathbb{R}$ *such that*

(i) $\phi \in C^1(\mathcal{S}) \cap C(\bar{\mathcal{S}})$.
(ii) $\phi \geq g$ *on* $\mathcal{S}$.
Define

$$D = \{y \in \mathcal{S};\ \phi(y) > g(y)\} \quad \textit{(the continuation region)}.$$

Suppose
(iii) $E^y\left[\int_0^{\tau_{\mathcal{S}}} \mathcal{X}_{\partial D}(Y(t))\mathrm{d}t\right] = 0.$
(iv) ∂D *is a Lipschitz surface.*
(v) $\phi \in C^2(\mathcal{S} \setminus \partial D)$ *with locally bounded derivatives near* ∂D.
(vi) $A\phi + f \leq 0$ *on* $\mathcal{S} \setminus \partial D$.
(vii) $\phi(y) = g(y)$ *for all* $y \notin \mathcal{S}$.
(viii) $E^y\left[|\phi(Y(\tau))| + \int_0^{\tau_{\mathcal{S}}} |A\phi(Y(t))|\mathrm{d}t\right] < \infty$ *for all* $\tau \in \mathcal{T}$.
Then $\phi(y) \geq \Phi(y)$ *for all* $y \in \bar{\mathcal{S}}$.

(b) *Moreover, assume*

(ix) $A\phi + f = 0$ *on* D.
(x) $\tau_D := \inf\{t > 0;\ Y(t) \notin D\} < \infty$ *a.s. for all* y.
Then

$$\phi(y) = \Phi(y)$$

and

$$\tau^* = \tau_D \text{ is an optimal stopping time.}$$

Proof of Theorem 3.2 (***Sketch***) (a) Let $\tau \leq \tau_{\mathcal{S}}$ be a stopping time. By Theorem 3.1 we can assume that $\phi \in C^2(\mathcal{S})$. Then by (vii) and (viii) and the Dynkin formula (Theorem 1.24) applied to $\tau_m := \min(\tau, m)$, $m = 1, 2, \ldots$ we have, by (vi),

$$E^y[\phi(Y(\tau_m))] = \phi(y) + E^y\left[\int_0^{\tau_m} A\phi(Y(t))\mathrm{d}t\right] \leq \phi(y) - E^y\left[\int_0^{\tau_m} f(Y(t))\mathrm{d}t\right].$$

Hence by (ii) and the Fatou lemma

$$\begin{aligned}\phi(y) &\geq \liminf_{m\to\infty} E^y\left[\int_0^{\tau_m} f(Y(t))\mathrm{d}t + \phi(Y(\tau_m))\right] \\ &\geq E^y\left[\int_0^{\tau} f(Y(t))\mathrm{d}t + g(Y(\tau))\chi_{\{\tau<\infty\}}\right] = J^{\tau}(y).\end{aligned}$$

Here we have used that $Y(\cdot)$ is quasi-left continuous (i.e. left continuous at increasing stopping times; see [JS, Propositions I.2.26 and I.3.27]).

Hence

$$\phi(y) \geq \Phi(y). \tag{3.1.11}$$

(b) If we apply the above argument to $\tau = \tau_D$, then by (ix)–(x) and the Dynkin formula II (Theorem 1.24),

$$\begin{aligned}E^y[\phi(Y(\tau_D))] &= \phi(y) + E^y\left[\int_0^{\tau_D} A\phi(Y(t))dt\right] \\ &= \phi(y) - E^y\left[\int_0^{\tau_D} f(Y(t))dt\right].\end{aligned}$$

Hence

$$\phi(y) = E^y\left[\int_0^{\tau_D} f(Y(t))dt + \phi(Y(\tau_D))\right] = J^{\tau_D}(y).$$

Combined with (a) this gives

$$\Phi(y) \leq \phi(y) = J^{\tau_D}(y) \leq \Phi(y)$$

and the proof is complete. □

The following result is sometimes helpful.

Proposition 3.3 *Suppose the conditions of Theorem 3.2 hold. Suppose $g \in C^2(\mathbb{R}^k)$ and that $\phi = g$ satisfies* (viii). *Define*

$$U = \{y \in \mathcal{S}; Ag(y) + f(y) > 0\}.$$

Suppose that for all $y \in U$ there exists a neighborhood W of y such that $\tau_W := \inf\{t > 0; Y(t) \notin W\} < \infty$ a.s. Then

$$U \subset \{y \in \mathcal{S}; \Phi(y) > g(y)\} = D.$$

Hence it is never optimal to stop while $Y(t) \in U$.

Proof Choose $y \in U$ and let $W \subset U$ be a neighborhood of y with $\tau_W < \infty$ a.s. Then by the Dynkin formula (Theorem 1.24)

$$E^y[g(Y(\tau_W))] = g(y) + E^y\left[\int_0^{\tau_W} Ag(Y(t))\mathrm{d}t\right] > g(y) - E^y\left[\int_0^{\tau_W} f(Y(t))\mathrm{d}t\right].$$

Hence

$$g(y) < E^y\left[\int_0^{\tau_W} f(Y(t))\mathrm{d}t + g(Y_{\tau_W})\right] \leq \Phi(y),$$

as claimed. □

Another useful observation is:

Proposition 3.4 *Let U be as in Proposition 3.3. Suppose $U = \emptyset$. Then*

$$\Phi(y) = g(y) \quad \textit{and} \quad \tau^* = 0 \quad \textit{is optimal.}$$

Proof If $U = \emptyset$ then $Ag(y) + f(y) \leq 0$ for all $y \in \mathcal{S}$. Hence the function $\phi = g$ satisfies all the conditions of Theorem 3.2. Therefore $D = \emptyset$, $g(y) = \Phi(y)$, and $\tau^* = 0$ is optimal. □

3.2 Applications and Examples

Example 3.5 (***The Optimal Time to Sell***) Suppose the price $X(t)$ at time t of an asset (a property, a stock...) is a geometric Lévy process given by

$$\mathrm{d}X(t) = X(t^-)\Big[\alpha\,\mathrm{d}t + \beta\,\mathrm{d}B(t) + \gamma\int_{\mathbb{R}} z\widetilde{N}(\mathrm{d}t, \mathrm{d}z)\Big], \quad X(0) = x > 0,$$

where α, β, and γ are constants, and we assume that

$$-1 < \gamma z \leq 0 \quad \text{a.s. } \nu. \tag{3.2.1}$$

If we sell the asset at time $s + \tau$ we get the expected discounted net payoff

$$J^\tau(s, x) := E^{s,x}\big[\mathrm{e}^{-\rho(s+\tau)}(X(\tau) - a)\mathcal{X}_{\{\tau<\infty\}}\big],$$

where $\rho > 0$ (the discounting exponent) and $a > 0$ (the transaction cost) are constants.

We seek the value function $\Phi(s,x)$ and an optimal stopping time $\tau^* \in \mathcal{T}$ such that

$$\Phi(s,x) = \sup_{\tau \in \mathcal{T}} J^{\tau}(s,x) = J^{\tau^*}(s,x). \tag{3.2.2}$$

We apply Theorem 3.2 to solve this problem as follows: Put $\mathcal{S} = \mathbb{R} \times (0,\infty)$ and

$$Y(t) = \begin{bmatrix} s+t \\ X(t) \end{bmatrix}, \quad t \geq 0.$$

Then

$$\mathrm{d}Y(t) = \begin{bmatrix} 1 \\ \alpha X(t) \end{bmatrix} \mathrm{d}t + \begin{bmatrix} 0 \\ \beta X(t) \end{bmatrix} \mathrm{d}B(t) + \begin{bmatrix} 0 \\ \gamma X(t^-) \int_{\mathbb{R}} z \widetilde{N}(\mathrm{d}t, \mathrm{d}z) \end{bmatrix}, \quad Y(0) = \begin{bmatrix} s \\ x \end{bmatrix}$$

and the generator A of $Y(t)$ is

$$\begin{aligned} A\phi(s,x) = \frac{\partial \phi}{\partial s} + \alpha x \frac{\partial \phi}{\partial x} + \frac{1}{2}\beta^2 x^2 \frac{\partial^2 \phi}{\partial x^2} \\ + \int_{\mathbb{R}} \left\{ \phi(s, x+\gamma x z) - \phi(s,x) - \gamma x z \frac{\partial \phi}{\partial x} \right\} \nu(\mathrm{d}z). \end{aligned} \tag{3.2.3}$$

If we try a function ϕ of the form

$$\phi(s,x) = \mathrm{e}^{-\rho s} x^{\lambda} \quad \text{for some constant } \lambda \in \mathbb{R}$$

we get

$$\begin{aligned} A\phi(s,x) &= \mathrm{e}^{-\rho s} \Big[-\rho x^{\lambda} + \alpha x \lambda x^{\lambda-1} + \frac{1}{2}\beta^2 x^2 \lambda(\lambda-1) x^{\lambda-2} \\ &\qquad + \int_{\mathbb{R}} \{ (x+\gamma x z)^{\lambda} - x^{\lambda} - \gamma x z \lambda x^{\lambda-1} \} \nu(\mathrm{d}z) \Big] \\ &= \mathrm{e}^{-\rho s} x^{\lambda} h(\lambda), \end{aligned}$$

where

$$h(\lambda) = -\rho + \alpha\lambda + \frac{1}{2}\beta^2 \lambda(\lambda-1) + \int_{\mathbb{R}} \{ (1+\gamma z)^{\lambda} - 1 - \lambda \gamma z \} \nu(\mathrm{d}z).$$

Note that

$$h(1) = \alpha - \rho \quad \text{and} \quad \lim_{\lambda \to \infty} h(\lambda) = \infty.$$

Therefore, if we assume that

$$\alpha < \rho, \tag{3.2.4}$$

then we get that there exists a $\lambda_1 \in (1, \rho/\alpha)$ such that

$$h(\lambda_1) = 0. \tag{3.2.5}$$

With this value of λ_1 we put

$$\phi(s,x) = \begin{cases} e^{-\rho s} C x^{\lambda_1} & \text{for} \quad (s,x) \in D \\ e^{-\rho s}(x-a) & \text{for} \quad (s,x) \notin D \end{cases} \tag{3.2.6}$$

for some constant C, to be determined.

To find a reasonable guess for the continuation region D we use Proposition 3.3. In this case we have $f = 0$ and $g(s,x) = e^{-\rho s}(x-a)$ and hence by (3.2.3)

$$Ag + f = e^{-\rho s}(-\rho(x-a) + \alpha x) = e^{-\rho s}((\alpha - \rho)x + \rho a).$$

Therefore

$$U = \{(s,x);\, (\alpha - \rho)x + \rho a > 0\}.$$

Case 1: $\alpha \geq \rho$. In this case $U = \mathbb{R}^2$ and it is easily seen that $\Phi = \infty$. We can get as high an expected payoff as we wish by waiting long enough before stopping.
Case 2: $\alpha < \rho$. In this case

$$U = \left\{(s,x);\, x < \frac{\rho a}{\rho - \alpha}\right\}. \tag{3.2.7}$$

Therefore, in view of Proposition 3.3 we now guess that the continuation region D has the form

$$D = \{(s,x);\, 0 < x < x^*\} \tag{3.2.8}$$

for some x^* such that $U \subseteq D$, i.e.,

$$x^* \geq \frac{\rho a}{\rho - \alpha}. \tag{3.2.9}$$

Hence, by (3.2.6) we now put

$$\phi(s,x) = \begin{cases} e^{-\rho s} C x^{\lambda_1} & \text{for} \quad 0 < x < x^* \\ e^{-\rho s}(x-a) & \text{for} \quad x^* \leq x, \end{cases} \tag{3.2.10}$$

for some constant $C > 0$ (to be determined). We guess that the value function is C^1 at $x = x^*$ and this gives the following "high contact" conditions

$$C(x^*)^{\lambda_1} = x^* - a \quad (\text{continuity at } x = x^*)$$

and

$$C\lambda_1(x^*)^{\lambda_1 - 1} = 1 \quad (\text{differentiability at } x = x^*).$$

It is easy to see that the solution of these equations is

$$x^* = \frac{\lambda_1 a}{\lambda_1 - 1}, \quad C = \frac{1}{\lambda_1}(x^*)^{1-\lambda_1}. \tag{3.2.11}$$

It remains to verify that with these values of x^* and C the function ϕ given by (3.2.10) satisfies all the conditions (i)–(xi) of Theorem 6.10.

To this end, first note that (i) and (ix) hold by construction of ϕ and by (3.2.1). Moreover, $\phi = g$ outside D. Therefore, to verify (ii) we only need to prove that $\phi \geq g$ on D, i.e., that

$$Cx^{\lambda_1} \geq x - a \quad \text{for} \quad 0 < x < x^*. \tag{3.2.12}$$

Define $k(x) = Cx^{\lambda_1} - x + a$. By our chosen values of C and x^* we have $k(x^*) = k'(x^*) = 0$. Moreover, $k''(x) = C\lambda_1(\lambda_1 - 1)x^{\lambda_1 - 2} > 0$ for $x < x^*$. Therefore $k(x) > 0$ for $0 < x < x^*$ and (3.2.12) holds and hence (ii) is proved.
(iii): In this case $\partial D = \{(s, x); x = x^*\}$ and hence

$$E^y\left[\int_0^\infty \mathcal{X}_{\partial D}(Y(t))\mathrm{d}t\right] = \int_0^\infty P^x[X(t) = x^*]\mathrm{d}t = 0.$$

(iv) and (v) are trivial.
(vi): Outside D we have $\phi(s, x) = \mathrm{e}^{-\rho s}(x - a)$ and therefore

$$\begin{aligned} A\phi + f(s, x) &= \mathrm{e}^{-\rho s}(-\rho(x - a) + \alpha x) \\ &\quad + \int_{\mathbb{R}} \left\{\phi(s, x + \gamma xz) - \phi(s, x) - \frac{\partial \phi}{\partial x}(s, x)\gamma xz\right\} \nu(\mathrm{d}z) \\ &= \mathrm{e}^{-\rho s}\Big((\alpha - \rho)x + \rho a \\ &\quad + \int_{x + \gamma xz < x^*} \{C(x + \gamma xz)^{\lambda_1} - (x - a) - \gamma xz\}\nu(\mathrm{d}z)\Big) \\ &\leq \mathrm{e}^{-\rho s}\left((\alpha - \rho)x + (\rho + \|\nu\|)a\right), \end{aligned}$$

where $\|\nu\| = \nu(\mathbb{R})$. By (3.2.4) we get that

$$
\begin{aligned}
&(\alpha-\rho)x+(\rho+\|\nu\|)a\le 0 \qquad \text{for all} \quad x\ge x^*\\
&\qquad\Updownarrow\\
&(\alpha-\rho)x^*+(\rho+\|\nu\|)a\le 0\\
&\qquad\Updownarrow\\
&x^*\ge\frac{(\rho+\|\nu\|)a}{\rho-\alpha}\\
&\qquad\Updownarrow\\
&\lambda_1\le\frac{\rho+\|\nu\|}{\alpha+\|\nu\|}.
\end{aligned}
\tag{3.2.13}
$$

(vii) holds since we have assumed that $\gamma z > -1$ a.s. ν.
For (viii) to hold it suffices that

$$
E^x\left[\int_0^\infty \mathrm{e}^{-2\rho t}\left\{X^2(t)+\gamma^2\int_{\mathbb{R}} z^2\nu(\mathrm{d}z)t\right\}\mathrm{d}t\right]<\infty.
$$

By the above this holds if

$$
2\alpha-2\rho+\beta^2+\gamma^2\int_{\mathbb{R}} z^2\nu(\mathrm{d}z)<0. \tag{3.2.14}
$$

(x): To check if $\tau_D < \infty$ a.s. we consider the solution $X(t)$ of (3.2.1), which by (1.2.5) is given by

$$
\begin{aligned}
X(t)=x\exp\Bigg\{&\Big(\alpha-\frac{1}{2}\beta^2-\gamma\int_{\mathbb{R}} z\nu(\mathrm{d}z)\Big)t\\
&+\int_0^t\int_{\mathbb{R}}\ln(1+\gamma z)N(\mathrm{d}t,\mathrm{d}z)+\beta B(t)\Bigg\}.
\end{aligned}
$$

By the law of iterated logarithm for Brownian motion (see the argument in [Ø1, Chapter 5]) we see that if

$$
\alpha>\frac{1}{2}\beta^2+\gamma\int_{\mathbb{R}} z\nu(\mathrm{d}z) \tag{3.2.15}
$$

and

$$
z\ge 0 \quad \text{a.s. } \nu \tag{3.2.16}
$$

then

$$
\lim_{t\to\infty} X(t)=\infty \quad \text{a.s.}
$$

and in particular $\tau_D < \infty$ a.s.
(xi): Since ϕ is bounded on $[0, x^*]$ it suffices to check that

$$\left\{e^{-\rho\tau}X(\tau)\right\}_{\tau\in\mathcal{T}} \quad \text{is uniformly integrable.}$$

For this to hold it suffices that there exists a constant K such that

$$E[e^{-2\rho\tau}X^2(\tau)] \le K \quad \text{for all } \tau \in \mathcal{T}. \tag{3.2.17}$$

By (3.2.13) and Exercise 1.6 we have

$$\begin{aligned}
E[e^{-2\rho T}X^2(\tau)] &= x^2E\Big[\exp\Big\{\Big(2\alpha - 2\rho - \beta^2 - 2\gamma\int_{\mathbb{R}} z\nu(dz)\Big)\tau \\
&\qquad + 2\int_0^\tau\int_{\mathbb{R}} \ln(1+\gamma z)N(dt,dz) + 2\beta B(\tau)\Big\}\Big] \\
&= x^2E\Big[\exp\Big\{\Big(2\alpha - 2\rho + \beta^2 + 2\int_{\mathbb{R}} (\ln(1+\gamma z) - \gamma z)\nu(dz)\Big)\tau \\
&\qquad + 2\int_0^\tau\int_{\mathbb{R}} \ln(1+\gamma z)\widetilde{N}(dt,dz)\Big\}\Big] \\
&= x^2E\Big[\exp\Big\{\Big(2\alpha - 2\rho + \beta^2 + \int_{\mathbb{R}} [2\ln(1+\gamma z) - 2\gamma z + (1+\gamma z)^2 \\
&\qquad - 1 - 2\ln(1+\gamma z)]\nu(dz)\Big)\tau\Big\}\Big] \\
&= x^2E\Big[\exp\Big\{\Big(2\alpha - 2\rho + \beta^2 + \int_{\mathbb{R}} [(1+\gamma z)^2 \\
&\qquad - 1 - 2\gamma z]\nu(dz)\Big)\tau\Big\}\Big].
\end{aligned}$$

We conclude that if

$$2\alpha - 2\rho + \beta^2 + \gamma^2\int_{\mathbb{R}} z^2\nu(dz) \le 0$$

then (3.2.17) holds and hence (xi) holds also.

We summarize what we have proved:

Theorem 3.6 *Suppose that* (3.2.1), (3.2.4), (3.2.13), (3.2.15), *and* (3.2.14) *hold. Then, with* λ_1, C, *and* x^* *given by* (3.2.5) *and* (3.2.11), *the function* ϕ *given by* (3.2.10) *coincides with the value function* Φ *of problem* (3.2.2) *and* $\tau^* = \tau_D$ *is an optimal stopping time, where* D *is given by* (3.2.8).

Remark 3.7

(1) If condition (3.2.1) is relaxed to

$$-1 < \gamma z \quad \text{a.s. } \nu$$

then the situation becomes more complicated. See [Mo] for a solution of a related problem in this case. See also [Ky].

(2) The C^1-property of the value function assumed in Theorem 6.10 (often called the "high contact" or "smooth pasting" assumption) need not hold in general, although we found that it holds in Example 3.5 (under some conditions). See [AKy] for a discussion of this. See also Sect. 12.1.
(3) For other applications of optimal stopping to jump diffusions we refer to [Ma].

3.3 Optimal Stopping with Delayed Information

This presentation is based on [Ø4]. Let $Y(t)$ be a jump diffusion in $\mathbb{R}^k$. Let $\delta \geq 0$ be a fixed constant. In this section we consider optimal stopping problems of the form

$$\Phi_\delta(y) := \sup_{\alpha \in \mathcal{T}_\delta} E^y \left[\int_0^\alpha f(Y(t))\mathrm{d}t + g(Y(\alpha)) \right], \tag{3.3.1}$$

where we interpret $g(Y(\alpha))$ as 0 if $\alpha = \infty$. Here $\mathcal{T}_\delta$ is the set of *δ-delayed stopping times*, defined as follows.

Definition 3.8 A function $\alpha : \Omega \to [\delta, \infty]$ is called a *δ-delayed stopping time* if

$$\{\omega;\, \alpha(\omega) \leq t\} \in \mathcal{F}_{t-\delta} \quad \text{for all } t \geq \delta \tag{3.3.2}$$

or, equivalently,

$$\{\omega;\, \alpha(\omega) \leq s + \delta\} \in \mathcal{F}_s \quad \text{for all } s \geq 0. \tag{3.3.3}$$

The set of all δ-delayed stopping times is denoted by $\mathcal{T}_\delta$.

In other words, if we interpret $\alpha(\omega)$ as the time to stop, then $\alpha \in \mathcal{T}_\delta$ if the decision whether or not to stop at or before time t is based on the information represented by $\mathcal{F}_{t-\delta}$. In particular, if $\delta = 0$ then $\mathcal{T}_\delta = \mathcal{T}_0$ is the family of classical stopping times and (3.3.1) becomes the classical optimal stopping problem, discussed in previous sections.

In the delayed case problem (3.3.1) models the situation when there is a delay $\delta > 0$ in the flow of information available to the agent searching for the optimal time to stop. An alternative way of stating this is that there is a delay $\delta > 0$ from the decided stopping time $\tau \in \mathcal{T}_0$ (based on the complete current information available from the system) to the time $\alpha = \tau + \delta \in \mathcal{T}_\delta$ when the system actually stops. This new formulation is based on the following simple observation.

Lemma 3.9

(i) $\tau \in \mathcal{T}_0 \iff \alpha := \tau + \delta \in \mathcal{T}_\delta$,
(ii) $\alpha \in \mathcal{T}_\delta \iff \tau := \alpha - \delta \in \mathcal{T}_0$.

Proof It suffices to prove (i).

First, assume $\tau \in \mathcal{T}_0$. Then, for $t \geq \delta$,

$$\{\omega; \tau(\omega) + \delta \leq t\} = \{\omega; \tau(\omega) \leq t - \delta\} \in \mathcal{F}_{t-\delta},$$

and hence $\alpha := \tau + \delta \in \mathcal{T}_\delta$.

Conversely, if $\alpha := \tau + \delta \in \mathcal{T}_\delta$ then

$$\{\omega; \tau(\omega) \leq t\} = \{\omega; \tau(\omega) + \delta \leq t + \delta\} = \{\omega; \alpha(\omega) \leq t + \delta\} \in \mathcal{F}_{(t+\delta)-\delta} = \mathcal{F}_t,$$

and hence $\tau \in \mathcal{T}_0$. □

Remark 3.10 In view of this result we see that it is possible to give another interpretation of problem (3.3.1), namely

$$\Phi_\delta(y) = \sup_{\tau \in \mathcal{T}_0} E^y\left[\int_0^{\tau+\delta} f(Y(t))\mathrm{d}t + g(Y(\tau+\delta))\right]. \tag{3.3.4}$$

In this formulation the problem appears as an optimal stopping problem over classical stopping times $\tau \in \mathcal{T}_0$, but with *delayed effect* of the stopping. If the stopping time $\tau \in \mathcal{T}_0$ is chosen, then the system itself is stopped at time $\tau + \delta$, i.e., after a delay $\delta > 0$.

Note that $\mathcal{T}_\delta \subset \mathcal{T}_0$ for $\delta > 0$ and hence

$$\Phi_\delta(y) \leq \Phi_0(y)$$

and we can interpret $\Phi_0(y) - \Phi_\delta(y)$ as the loss of value due to the delay of information.

In this section we show that the delayed optimal stopping problem (3.3.1) can be reduced to a classical optimal stopping problem by a simple transformation (Theorem 3.11).

We call $\alpha^* \in \mathcal{T}_\delta$ an *optimal stopping time* for the problem (3.3.1) if

$$\Phi_\delta(y) = E^y\left[\int_0^{\alpha^*} f(Y(t))\mathrm{d}t + g(Y(\alpha^*))\right]. \tag{3.3.5}$$

The result of this section may be regarded as a partial extension of [AK], where the geometric Brownian motion case is studied and solved (see Exercise 3.12), with a more general (Markovian) delay $\delta(X) \leq 0$. For a related type of problem involving impulse control with delivery lags, see [BS].

We are now ready to state and prove the main result of this section.

Theorem 3.11

(a) *Consider the two optimal stopping problems:*

$$\Phi_\delta(y) := \sup_{\alpha \in \mathcal{T}_\delta} E^y\left[\int_0^\alpha f(Y(t))\mathrm{d}t + g(Y(\alpha))\right], \tag{3.3.6}$$

$$\tilde{\Phi}(y) := \sup_{\tau \in \mathcal{T}_0} E^y\left[\int_0^\tau f(Y(t))\mathrm{d}t + \tilde{g}_\delta(Y(\tau))\right], \tag{3.3.7}$$

where

$$\tilde{g}_\delta(y) = E^y\left[\int_0^\delta f(Y(t))\mathrm{d}t + g(Y(\delta))\right]. \tag{3.3.8}$$

Then we have

$$\Phi_\delta(y) = \tilde{\Phi}(y) \quad \textit{for all } y \in \mathbb{R}^k,\ \delta \geq 0.$$

(b) *Moreover, $\alpha^* \in \mathcal{T}_\delta$ is an optimal stopping time for the delayed problem (3.3.6) if and only if*

$$\alpha^* := \tau^* + \delta, \tag{3.3.9}$$

where $\tau^* \in \mathcal{T}_0$ is an optimal stopping time for the nondelayed problem (3.3.7).

Proof (a) Define

$$J^{(\alpha)}(y) = E^y\left[\int_0^\alpha f(Y(t))\mathrm{d}t + g(Y(\alpha))\right], \quad \alpha \in \mathcal{T}_\delta, \tag{3.3.10}$$

and

$$\tilde{J}^{(\tau)}(y) = E^y\left[\int_0^\tau f(Y(t))\mathrm{d}t + \tilde{g}_\delta(Y(\tau))\right], \quad \tau \in \mathcal{T}_0. \tag{3.3.11}$$

Choose $\alpha \in \mathcal{T}_\delta$ and put

$$\tau = \alpha - \delta \in \mathcal{T}_0.$$

Then $\alpha = \tau + \delta$ and hence

$$\begin{aligned} J^{(\alpha)}(y) &= E^y\left[\int_0^\alpha f(Y(t))\mathrm{d}t + g(Y(\alpha))\right] \\ &= E^y\left[\int_0^{\tau+\delta} f(Y(t))\mathrm{d}t + g(Y(\tau+\delta))\right] \end{aligned}$$

$$\begin{aligned} &= E^y\left[\int_0^\tau f(Y(t))\mathrm{d}t + \int_\tau^{\tau+\delta} f(Y(t))\mathrm{d}t + g(Y(\tau+\delta))\right] \\ &= E^y\left[\int_0^\tau f(Y(t))\mathrm{d}t + E^y\left[\theta_\tau\left\{\int_0^\delta f(Y(t))\mathrm{d}t + g(Y(\delta))\right\}\right]\right], \end{aligned} \tag{3.3.12}$$

where θ_τ is the *shift operator*, defined by

$$\theta_\tau\{h(Y(s))\} = h(Y(\tau+s)) \quad \text{for } s \geq 0, \text{ for all measurable } h : \mathbb{R}^k \longrightarrow \mathbb{R},$$

and we have used that

$$\theta_\tau\left(\int_0^\delta f(Y(t))\mathrm{d}t\right) = \int_\tau^{\tau+\delta} f(Y(t))\mathrm{d}t.$$

We refer to [BG] for more information about Markov processes. By the strong Markov property we now get from (3.3.12) that

$$\begin{aligned} J^{(\alpha)}(y) &= E^y\left[\int_0^\tau f(Y(t))\mathrm{d}t + E^y\left[\theta_\tau\left\{\int_0^\tau f(Y(t))\mathrm{d}t + g(Y(\delta))\right\}\Big|\mathcal{F}_\tau\right]\right] \\ &= E^y\left[\int_0^\tau f(Y(t))\mathrm{d}t + E^{Y(\tau)}\left[\int_0^\delta f(Y(t))\mathrm{d}t + g(Y(\delta))\right]\right] \\ &= E^y\left[\int_0^\tau f(Y(t))\mathrm{d}t + \tilde{g}_\delta(Y(\tau))\right] = \tilde{J}^{(\tau)}(y). \end{aligned} \tag{3.3.13}$$

Hence, by Lemma 3.9 (ii),

$$\begin{aligned} \Phi_\delta(y) &= \sup_{\alpha\in\mathcal{T}_\delta} J^{(\alpha)}(y) = \sup_{\alpha-\delta\in\mathcal{T}_0} J^{(\alpha)}(y) \\ &= \sup_{\alpha-\delta\in\mathcal{T}_0} \tilde{J}^{(\alpha-\delta)}(y) = \sup_{\tau\in\mathcal{T}_0} \tilde{J}^{(\tau)}(y) = \tilde{\Phi}(y), \end{aligned}$$

as claimed.
(b) Suppose $\tau^* \in \mathcal{T}_0$ is optimal for (3.3.7). Define

$$\alpha^* := \tau^* + \delta.$$

Then $\alpha^* \in \mathcal{T}_\delta$ by Lemma 3.9 and by (3.3.13) combined with (a) we have

$$J^{(\alpha^*)}(y) = \tilde{J}^{(\tau^*)}(y) = \tilde{\Phi}(y) = \Phi_\delta(y).$$

Hence α^* is optimal for (3.3.6).

Conversely, if $\alpha^* \in \mathcal{T}_\delta$ is optimal for (3.3.6) a similar argument gives that $\tau^* := \alpha^* - \delta$ is optimal for (3.3.7). □

We illustrate Theorem 3.11 by solving the following problem.

In the no delay case the following example is discussed in [Ø1] (continuous case) and Exercise 3.2 (jump diffusion case). Our example models the situation when there is a time lag $\delta > 0$ between the decided stopping time $\tau \in \mathcal{T}_0$ and the time $\alpha = \tau + \delta \in \mathcal{T}_\delta$ when the result of the stopping decision comes into effect.

Example 3.12 (***Optimal Time to Stop Resource Extraction***) Suppose the price $P(t)$ at time t per unit of a resource (oil, gas, …) is given by

$$\mathrm{d}P(t) = P(t^-)\left[\mu\,\mathrm{d}t + \sigma\,\mathrm{d}B(t) + \int_{\mathbb{R}} z\tilde{N}(\mathrm{d}t, \mathrm{d}z)\right], \quad P(0) = p > 0, \tag{3.3.14}$$

where μ and σ are given constants and we assume that $z \geq 0$ a.s. with respect to ν.

Let $Q(t)$ denote the amount of remaining resources at time t. As long as the extraction field is open, we assume that the extraction rate is proportional to the remaining amount, i.e.,

$$\mathrm{d}Q(t) = -\lambda Q(t)\mathrm{d}t, \quad Q(0) = q > 0, \tag{3.3.15}$$

where $\lambda > 0$ is a known constant.

If we decide to stop the extraction and close the field at a (delayed) stopping time $\alpha \in \mathcal{T}_\delta$, then the expected total discounted net profit $J^\alpha(s, p, q)$ is assumed to have the form

$$J^\alpha(s, p, q) = E^{(s,p,q)}\left[\int_0^\alpha \mathrm{e}^{-\rho(s+t)}(\lambda P(t)Q(t) - K)\mathrm{d}t + \theta\mathrm{e}^{-\rho(s+\alpha)}P(\alpha)Q(\alpha)\right], \tag{3.3.16}$$

where $K > 0$ is the (constant) running cost rate and $\rho > 0$, $\theta > 0$ are other constants. The expectation $E^{(s,p,q)}$ is taken with respect to the probability law $P^{(s,p,q)}$ of the strong Markov process

$$Y(t) := \begin{bmatrix} s+t \\ P(t) \\ Q(t) \end{bmatrix}, \text{ which starts at } y = \begin{bmatrix} s \\ p \\ q \end{bmatrix} \text{ at time } t = 0. \tag{3.3.17}$$

The explanation of the quantity $J^\alpha(s, p, q)$ in (3.3.16) is the following.

As long as the field is open (i.e., as long as $t < \alpha$) the gross income rate from the production is price times production rate, i.e., $P(t)\lambda Q(t)$. Subtracting the running cost rate K we get the net profit rate

$$\lambda P(t)Q(t) - K \quad \text{for } 0 \leq t < \alpha.$$

If the field is closed at time α the net value of the remaining resources is estimated to be $\theta P(\alpha)Q(\alpha)$. Discounting and integrating/adding these quantities and taking expectation we get (3.3.16).

We want to find the value function $\Phi_\delta(s, p, q)$ and the corresponding optimal delayed stopping time $\alpha^* \in \mathcal{T}_\delta$ such that

$$\Phi_\delta(y) = \Phi_\delta(s, p, q) = \sup_{\alpha \in \mathcal{T}_\delta} J^\alpha(s, p, q) = J^{\alpha^*}(s, p, q). \tag{3.3.18}$$

In the case of *no delay* ($\delta = 0$) it is shown in Exercise 3.2 that if the following relations between the parameters hold

$$0 < \theta(\lambda + \rho - \mu) < \lambda \tag{3.3.19}$$

then the optimal stopping time $\tau_0^* \in \mathcal{T}_0$ is

$$\tau_0^* = \inf\{t > 0; P(t)Q(t) \leq w_0^*\}, \tag{3.3.20}$$

where

$$w_0^* = \frac{(-r_2)K(\lambda + \rho - \mu)}{(1 - r_2)\rho(\lambda - \theta(\lambda + \rho - \mu))}, \tag{3.3.21}$$

$r_2 < 0$ being the negative solution of the equation

$$h(r) := -\rho + (\mu - \lambda)r + \frac{1}{2}\sigma^2 r(r-1) + \int_{\mathbb{R}} \{(1+z)^r - 1 - rz\}\nu(\mathrm{d}z) = 0. \tag{3.3.22}$$

In this case we have

$$f(y) = f(s,p,q) = \mathrm{e}^{-\rho s}(\lambda pq - K)$$

and

$$g(y) = g(s,p,q) = \theta \mathrm{e}^{-\rho s} pq.$$

Thus

$$\begin{aligned}
\tilde{g}_\delta(y) &= E^y\left[\int_0^\delta \mathrm{e}^{-\rho(s+t)}(\lambda P(t)Q(t) - K)\mathrm{d}t\right] + E^y[\theta \mathrm{e}^{-\rho(s+\delta)}P(\delta)Q(\delta)] \\
&= \int_0^\delta \mathrm{e}^{-\rho(s+t)}(\lambda E[P(t)Q(t)] - K)\mathrm{d}t + \theta \mathrm{e}^{-\rho(s+\delta)}E^y[P(\delta)Q(\delta)] \\
&= \int_0^\delta \mathrm{e}^{-\rho(s+t)}(\lambda pq \mathrm{e}^{(\mu-\lambda)t} - K)\mathrm{d}t + \theta \mathrm{e}^{-\rho(s+\delta)} pq \mathrm{e}^{(\mu-\lambda)\delta} \\
&= \mathrm{e}^{-\rho s}\Big[\{(\lambda + \rho - \mu)^{-1}\lambda(1 - \mathrm{e}^{-(\lambda+\rho-\mu)\delta}) + \theta \mathrm{e}^{-(\lambda+\rho-\mu)\delta}\}pq \\
&\qquad - \frac{K}{\rho}(1 - \mathrm{e}^{-\rho\delta})\Big] \\
&= \mathrm{e}^{-\rho s}[F_1 pq - F_2],
\end{aligned} \tag{3.3.23}$$

where

$$F_1 = (\lambda + \rho - \mu)^{-1}\lambda(1 - \mathrm{e}^{-(\lambda+\rho-\mu)\delta}) + \theta \mathrm{e}^{-(\lambda+\rho-\mu)\delta} \tag{3.3.24}$$

and

$$F_2 = \frac{K}{\rho}(1 - \mathrm{e}^{-\rho\delta}). \tag{3.3.25}$$

Therefore, according to Theorem 3.11 we have

$$\Phi_\delta(y) = \sup_{\tau\in\mathcal{T}_0} E^y\left[\int_0^\tau \mathrm{e}^{-\rho(s+t)}(\lambda P(t)Q(t) - K)\mathrm{d}t\right] + E^y[\mathrm{e}^{-\rho(s+\tau)}(F_1P(\tau)Q(\tau) + F_2)]. \tag{3.3.26}$$

The method used in Exercise 3.2 to provide the solutions (3.3.20)–(3.3.22) in the no delay case can easily be modified to find the optimal stopping time τ^* for the problem (3.3.26). The result is

$$w_\delta^* = \frac{(-r_2)K(\lambda + \rho - \mu)\mathrm{e}^{(\lambda-\mu)\delta}}{(1 - r_2)\rho[\lambda - \theta(\lambda + \rho - \mu)]} = w_0^*\mathrm{e}^{(\lambda-\mu)\delta}. \tag{3.3.27}$$

We have proved.

Theorem 3.13 *The optimal stopping time $\alpha^* \in \mathcal{T}_\delta$ for the delayed optimal stopping problem (3.3.18) is*

$$\alpha^* = \tau_\delta^* + \delta, \tag{3.3.28}$$

where

$$\tau_\delta^* = \inf\{t > 0; P(t)Q(t) \le w_\delta^*\}, \tag{3.3.29}$$

with w_δ^* given by (3.3.27).

Remark 3.14 Note that the threshold w_δ^* for the decision to close down in the case of a time lag in the action only differs from the corresponding threshold w_0^* in the no delay case by the factor $\mathrm{e}^{(\lambda-\mu)\delta}$.

Assume, for example, that $\lambda > \mu$. Then we should decide to stop *sooner* in the delay case than in the no delay case, because of the anticipation that $P(t)Q(t)$ will probably decrease during the extra time δ it takes before the closing down actually takes place.

3.4 Exercises

Exercise* 3.1 Solve the optimal stopping problem

$$\Phi(s, x) = \sup_{\tau\in\mathcal{T}} E^{(s,x)}\left[\mathrm{e}^{-\rho(s+\tau)}(X(\tau) - a)\right],$$

where

$$\mathrm{d}X(t) = \mathrm{d}B(t) + \gamma \int_{\mathbb{R}} z\bar{N}(\mathrm{d}t, \mathrm{d}z), \quad X(0) = x \in \mathbb{R}$$

and $\rho > 0$, $a > 0$, and γ are constants, $\gamma z \leq 0$ a.s. ν.

Exercise* 3.2 (***An Optimal Resource Extraction Stopping Problem***) Suppose the price $P(t)$ per unit of a resource (oil, gas...) at time t is given by

(1) $\mathrm{d}P(t) = \alpha P(t)\mathrm{d}t + \beta P(t)\mathrm{d}B(t) + \gamma P(t^-) \int_{\mathbb{R}} z\widetilde{N}(\mathrm{d}t, \mathrm{d}z), \quad P(0) = p > 0$

and the remaining amount of resources $Q(t)$ at time t is

(2) $\mathrm{d}Q(t) = -\lambda Q(t)\mathrm{d}t, \quad Q(0) = q > 0,$

where $\lambda > 0$ is the (constant) relative extraction rate and α, β, γ are constants. We assume that $\gamma z \geq 0$ a.s. ν.

If we decide to stop extraction and close the field at a stopping time $\tau \geq 0$, the expected discounted total net profit $J^\tau(s, p, q)$ is given by

$$J^\tau(s, p, q) = E^{(s,p,q)}\left[\int_0^\tau \mathrm{e}^{-\rho(s+t)}(\lambda P(t)Q(t) - K)\mathrm{d}t + \theta \mathrm{e}^{-\rho(s+\tau)}P(\tau)Q(\tau)\right],$$

where $K > 0$ is the (constant) running cost rate, $\rho > 0$ is the (constant) discounting exponent, and $\theta > 0$ another constant. Find Φ and τ^* such that

$$\Phi(s, p, q) = \sup_{\tau \in \mathcal{T}} J^\tau(s, p, q) = J^{\tau^*}(s, p, q).$$

[Hint: Try $\phi(s, p, q) = \mathrm{e}^{-\rho s}\psi(p \cdot q)$ for some function $\psi : \mathbb{R} \to \mathbb{R}$.]

Exercise* 3.3 Solve the optimal stopping problem

$$\Phi(s, x) = \sup_{\tau \in \mathcal{T}} E^x[\mathrm{e}^{-\rho(s+\tau)}|X(\tau)|]$$

where

$$\mathrm{d}X(t) = \mathrm{d}B(t) + \int_{\mathbb{R}} \theta(X(t), z)\tilde{N}(\mathrm{d}t, \mathrm{d}z)$$

and $\rho > 0$ is a constant. Assume that there exists an $\xi > 0$ such that

$$\theta(x, z) = 0 \text{ for a.a. } z \text{ if } |x| < \xi \tag{3.4.1}$$

$$\theta(x, z)(x - \xi) \geq 0 \text{ for a.a. } z \text{ if } |x| \geq \xi \tag{3.4.2}$$

and

$$\operatorname{tgh}(\sqrt{2\rho}\,\xi) \geq \frac{1}{\sqrt{2\rho}\,\xi}. \tag{3.4.3}$$

Exercise* 3.4 **(*The Optimal Time with Delay to Sell an Asset*)**

This case (without the jump part) was first solved by [AK], with a more general (Markovian) delay $\delta(X) \geq 0$.

Suppose the value $X(t)$ of an asset at time t is modeled by a geometric Lévy process of the form

$$\mathrm{d}X(t) = X(t^-)\left[\mu\,\mathrm{d}t + \sigma\,\mathrm{d}B(t) + \int_{\mathbb{R}} z\tilde{N}(\mathrm{d}t, \mathrm{d}z)\right], \quad X(0) = x > 0, \tag{3.4.4}$$

where μ, σ, and x are constants.

We assume that

$$-1 < z < 0 \quad \text{a.s. } \nu. \tag{3.4.5}$$

This guarantees that $X(t)$ never jumps down to a negative value. For convenience, we also assume that

$$E[\eta^2(t)] < \infty \quad \text{for all } t \geq 0. \tag{3.4.6}$$

Then by the Itô formula for Lévy processes the solution of (3.4.4) is

$$\begin{aligned} X(t) = x\exp\Bigg[&\left(\mu - \frac{1}{2}\sigma^2\right)t + \sigma B(t) \\ &+ \int_0^t \int_{\mathbb{R}} \{\ln(1+z) - z\}\nu(\mathrm{d}z)\mathrm{d}s + \int_0^t \int_{\mathbb{R}} \ln(1+z)\tilde{N}(\mathrm{d}s, \mathrm{d}z)\Bigg], \\ &t \geq 0. \end{aligned} \tag{3.4.7}$$

Solve the following delayed optimal stopping problem

$$\Phi_\delta(s,x) = \sup_{\alpha \in \mathcal{T}_\delta} E^{s,x}[\mathrm{e}^{-\rho(s+\alpha)}(X(\alpha) - q)], \tag{3.4.8}$$

where $E^{s,x}$ denotes expectation with respect to the probability law $P^{s,x}$ of the time-space process

$$\mathrm{d}Y(t) = \begin{bmatrix} \mathrm{d}t \\ \mathrm{d}X(t) \end{bmatrix}, \quad Y(0) = \begin{bmatrix} s \\ x \end{bmatrix}$$

and $\rho > 0$, $q > 0$ are constants. We assume that

$$\rho > \mu. \tag{3.4.9}$$

One possible interpretation of this problem is that $\Phi_\delta(s,x)$ represents the maximal expected discounted net payment obtained by selling the asset at a δ-delayed stopping time (ρ is the discounting exponent and q is the transaction cost).

Chapter 4
Backward Stochastic Differential Equations and Risk Measures



In this chapter we give an introduction to backward stochastic differential equations (BSDEs) with jumps, and we relate them to the concepts of *recursive utilities* and *convex risk measures*. This section is based on the papers [ØS2] and [QS]. For a similar introduction in the Brownian motion case, we refer the reader to the survey paper on BSDEs by M.C. Quenez [Q].

Let $B(t)$ and $\tilde{N}(\mathrm{d}t, \mathrm{d}\zeta) := N(\mathrm{d}t, \mathrm{d}\zeta) - \nu(\mathrm{d}\zeta)\mathrm{d}t$ denote a Brownian motion and an independent compensated Poisson random measure, respectively, on a filtered probability space $(\Omega, \mathcal{F}, \mathbb{F} := \{\mathcal{F}_t\}_{0 \le t \le T}, P)$ satisfying the usual conditions, P is a reference probability measure and ν is the Lévy measure of N.

4.1 Examples

We first give some examples where BSDEs appear. For simplicity we do not include jumps in these examples. The more general versions with jumps will be discussed in the subsequent sections.

Example 4.1 (***Replicating Portfolio***) Consider a financial market with one risk free and one risky investment possibility, with prices $S_0(t)$, $S_1(t)$ per unit given by, respectively

$$\begin{cases} \mathrm{d}S_0(t) = S_0(t)r(t)\mathrm{d}t \ ; \ S_0(0) = 1 \\ \mathrm{d}S_1(t) = S_1(t)[\mu(t)\mathrm{d}t + \sigma(t)\mathrm{d}B(t)] \ ; \ S_1(0) > 0. \end{cases} \tag{4.1.1}$$

Here $r(t)$, $\mu(t)$ and $\sigma(t)$ are adapted and bounded processes and $\sigma(t)$ is different from 0.

Let $\pi(t)$ be a self-financing portfolio, representing the fraction of the total wealth $Y(t) = Y_\pi(t)$ invested in the risky asset at time t. The corresponding wealth process $Y(t)$ is given by

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B. Øksendal and A. Sulem, *Applied Stochastic Control of Jump Diffusions*,
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$$\begin{aligned} dY(t) &= \frac{(1-\pi(t))Y(t)}{S_0(t)}dS_0(t) + \frac{\pi(t)Y(t)}{S_1(t)}dS_1(t) \\ &= Y(t)[\{(1-\pi(t))r(t) + \pi(t)\mu(t)\}dt + \pi(t)\sigma(t)dB(t)]. \end{aligned} \tag{4.1.2}$$

Let $F \in L^2(\mathcal{F}_T, P)$ be a given T-claim. We want to find $Y(0) = y > 0$ and $\pi(t)$ such that

$$Y(T) = F \text{ a.s.} \tag{4.1.3}$$

Put

$$Z(t) = Y(t)\pi(t)\sigma(t). \tag{4.1.4}$$

Then

$$\pi(t) = \frac{Z(t)}{Y(t)\sigma(t)} \tag{4.1.5}$$

and (4.1.2) becomes

$$dY(t) = \left\{ r(t)Y(t) + \frac{Z(t)}{\sigma(t)}(\mu(t) - r(t)) \right\} dt + Z(t)dB(t) \; ; \; 0 \le t \le T. \tag{4.1.6}$$

The pair (4.1.6), (4.1.3) of equations is an example of a (linear) BSDE in the pair $(Y(t), Z(t))$ of unknown processes. If we can solve this equation for $Y(t)$, $Z(t)$, then the replicating portfolio $\pi(t)$ is given by (4.1.5).

Note that, in contrast to ordinary SDEs in one unknown process, this equation has *two* unknown processes and the *terminal* value $Y(T)$ of Y is given, not the initial value.

More generally, let

$$g(t, y, z, \omega) : [0, T] \times \mathbb{R} \times \mathbb{R} \times \Omega \to \mathbb{R}$$

be an $\mathbb{F}$-adapted stochastic process in (t, ω) for each y, z. Then the equation

$$\begin{cases} dY(t) = -g(t, Y(t), Z(t), \omega)dt + Z(t)dB(t) \; ; \; 0 \le t \le T \\ Y(T) = F \text{ a.s.} \end{cases} \tag{4.1.7}$$

is a BSDE in the unknown $\mathcal{F}_t$-adapted processes $(Y(t), Z(t))$ (driven by Brownian motion). (See Sect. 4.2 for a more comprehensive presentation.)

For simplicity of notation we suppress ω in the following.

Example 4.2 (***Recursive Utility***) ([DE, EZ, KP]) Let $g(t, y, c)$ be an $\mathcal{F}_t$-adapted process. Assume that $c \to g(t, y, c)$ is concave for all t, y, and let F be a claim in $L^2(\mathcal{F}_T)$. Then the *recursive utility* process of a given consumption process $c(\cdot) \ge 0$ is defined as the solution $Y(t) = Y_g(t)$ of the equation

$$Y(t) = E\left[\int_t^T g(s, Y(s), c(s))\mathrm{d}s \mid \mathcal{F}_t\right] \; ; \; 0 \le t \le T. \tag{4.1.8}$$

In Sect. 4.2 we shall see that (4.1.8) is equivalent to the following BSDE in (Y_g, Z_g):

$$\begin{cases} \mathrm{d}Y_g(t) = -g(t, Y(t), c(t))\mathrm{d}t + Z_g(t)\mathrm{d}B(t) \\ Y_g(T) = 0. \end{cases} \tag{4.1.9}$$

In particular, the (total) *recursive utility* $U(c)$ of a given consumption process $c(\cdot)$ is defined as

$$U(c) := Y_g(0). \tag{4.1.10}$$

Example 4.3 (***Convex Risk Measures***) ([FS1, FR])

Definition 4.4 Let $p \in [2, \infty]$. A map

$$\rho : \mathbb{F} := L^p(\mathcal{F}_T) \to \mathbb{R}$$

is called a *convex risk measure* if the following hold:

(i) (Convexity) $\rho(\lambda F_1 + (1-\lambda)F_2) \le \lambda\rho(F_1) + (1-\lambda)\rho(F_2)$ for all $\lambda \in [0, 1]$ and all $F_1, F_2 \in \mathbb{F}$.
(ii) (Monotonicity) If $F_1 \le F_2$ then $\rho(F_1) \ge \rho(F_2)$.
(iii) (Translation invariance) $\rho(F + a) = \rho(F) - a$ for all $F \in \mathbb{F}$ and all constants a.
(iv) (For convenience) $\rho(0) = 0$.

Interpretation:
$\rho(F)$ is the amount that has to be added to the financial standing F to make it "acceptable". Note that, by (iii), $\rho(F + \rho(F)) = 0$.

We shall see that convex risk measures are related to BSDEs as follows: Let $g(t, z)$ be a concave function of z. For given $F \in \mathbb{F}$ let $(Y_g^{(F)}(t), Z_g^{(F)}(t))$ be the solution of the BSDE

$$\begin{cases} \mathrm{d}Y_g^{(F)}(t) = -g(t, Z_g^{(F)}(t))dt + Z_g^{(F)}(t)dB(t) \; ; \; 0 \le t \le T \\ Y_g^{(F)}(T) = F. \end{cases} \tag{4.1.11}$$

Then

$$\rho(F) := -Y_g^{(F)}(0) \tag{4.1.12}$$

defines a convex risk measure. The proof will be given in the next section, after we have studied BSDEs in a more general context with jumps.

4.2 General BSDEs with Jumps

Let $g(t, y, z, k, \omega) : [0, T] \times \mathbb{R} \times \mathbb{R} \times \mathcal{R} \times \Omega \to \mathbb{R}$ be a given function such that $(t, \omega) \to g(t, y, z, k, \omega)$ is $\mathcal{F}_t$-predictable for each y, z, k. Here $\mathcal{R}$ is the set of functions $k \in L^2(\mathbb{R}_0, \nu)$, where $\mathbb{R}_0 := \mathbb{R}\backslash\{0\}$. Let $F \in L^2(\mathcal{F}_T)$. We seek a triple $(Y, Z, K) = (Y_g^{(F)}(t), Z_g^{(F)}(t), K_g^{(F)}(t))$ of stochastic processes such that Y is a càdlàg adapted process, Z and K are predictable and

$$\begin{cases} \mathrm{d}Y(t) &= -g(t, Y(t), Z(t), K(t, \cdot), \omega)\mathrm{d}t + Z(t)\mathrm{d}B(t) \\ &\quad + \int_{\mathbb{R}} K(t, \zeta)\tilde{N}(\mathrm{d}t, \mathrm{d}\zeta)\ ;\ 0 \le t \le T \\ Y(T) &= F. \end{cases} \tag{4.2.1}$$

The process $g(t, y, z, k) = g(t, y, z, k, \omega)$ is called the *driver* of the BSDE (4.2.1). We state the next result without proof. We refer to [TL], [QS, Theorem 2.3], for details.

Theorem 4.5 (***Existence and Uniqueness of Solutions of BSDEs***) *Suppose the following hold:*

(i) $E\left[\int_0^T g^2(t, 0, 0, 0)\mathrm{d}t\right] < \infty$.

(ii) *g is Lipschitz in y, z, k a.s., i.e. there exists a constant $C > 0$ such that for all y_i, z_i, k_i $|g(t, y_1, z_1, k_1) - g(t, y_2, z_2, k_2)| \le C(|y_1 - y_2|) + |z_1 - z_2| + \|k_1 - k_2\|)$ for a.e. t, ω, where $\|k\|^2 = \int_{\mathbb{R}} k^2(\zeta)\nu(\mathrm{d}\zeta)$.*

Then there exists a unique triple (Y, Z, K) solution of (4.2.1) such that Y is a càdlàg adapted process with $E(\sup_{0\le t\le T} |Y(t)|^2) < \infty$ and (Z, K) are predictable processes with

$$E\left[\int_0^T \left\{Z^2(t) + \int_{\mathbb{R}} K^2(t, \zeta)\nu(\mathrm{d}\zeta)\right\}\mathrm{d}t\right] < \infty. \tag{4.2.2}$$

This result can be extended to the case when the terminal time T is a stopping time τ with values in $[0, T]$ and the terminal condition is a random variable ξ in $L^2(\mathcal{F}_\tau)$. In this case, $(Y^{(\xi,\tau)}, Z^{(\xi,\tau)}, K^{(\xi,\tau)})$ is defined as the unique solution of the BSDE with driver $g(t, y, z, k)\mathbf{1}_{\{t\le\tau\}}$ and terminal conditions (T, ξ). Note that $Y^{(\xi,\tau)}(t) = \xi$, $Z^{(\xi,\tau)}(t) = K^{(\xi,\tau)}(t) = 0$ for $t \ge \tau$.

Lemma 4.6

(i) *Suppose (Y, Z, K) satisfies the BSDE (4.2.1). Then*

$$Y(t) = E\left[\left(\int_t^T g(s, Y(s), Z(s), K(s, \cdot))\mathrm{d}s + F\right) \mid \mathcal{F}_t\right]\ ;\ 0 \le t \le T. \tag{4.2.3}$$

(ii) Conversely, suppose the driver g does not depend on z and k and that (4.2.3) holds. Then (4.2.1) holds.

Proof *(i):* (4.2.1) $\Rightarrow$ (4.2.3): Integrating (4.2.1) from t to T we get

$$Y(T) - Y(t) = -\int_t^T g(s, Y(s), Z(s), K(s,\cdot))\mathrm{d}t + \int_t^T Z(s)\mathrm{d}B(s) + \int_t^T \int_{\mathbb{R}} K(s,\zeta)\tilde{N}(\mathrm{d}s, \mathrm{d}\zeta).$$

Taking conditional expectation and using that $t \to \int_0^t Z(s)\mathrm{d}B(s)$ and $t \to \int_t^T \int_{\mathbb{R}} K(s,\zeta)\tilde{N}(\mathrm{d}s, \mathrm{d}\zeta)$ are martingales, we get (4.2.3).

(ii): (4.2.3) $\Rightarrow$ (4.2.1): Assume (4.2.3) holds and that $g(s, y, z, k) = g(s, y)$ does not depend on z and k. Since $Y(T) = F$, we can write

$$\begin{aligned} Y(t) &= E\left[\int_0^T \{g(s, Y(s))\mathrm{d}t + F - \int_0^t g(s, Y(s))\mathrm{d}s\} \mid \mathcal{F}_t\right] \\ &= M(t) - \int_0^t g(s, Y(s))\mathrm{d}s, \end{aligned} \tag{4.2.4}$$

where $M(t)$ is the L^2-martingale

$$M(t) = E\left[\int_0^T \{g(s, Y(s))\mathrm{d}s + F\} \mid \mathcal{F}_t\right].$$

By the martingale representation theorem for Itô–Lévy processes (see e.g. [L]) there exists $Z(t)$ and $K(t,\zeta)$ such that

$$M(t) = M(0) + \int_0^t Z(s)\mathrm{d}B(s) + \int_0^t \int_{\mathbb{R}} K(s,\zeta)\tilde{N}(\mathrm{d}s, \mathrm{d}\zeta)\ ;\ 0 \le t \le T. \tag{4.2.5}$$

Substituting (4.2.5) into (4.2.4) and taking differentials, we get (4.2.1). □

Application to Recursive Utility

Motivated by Lemma 4.6 we now extend the definition of recursive utility given in Example 4.2 to the following: We call a process $c(t)$ a *consumption* process if $c(t)$ is predictable and $c(t) \ge 0$ for all t, a.s. The set of all consumption processes is denoted by $\mathcal{C}$.

Definition 4.7 Let $g(t, y, z, k, c) : [0, T] \times \mathbb{R} \times \mathbb{R} \times \mathcal{R} \times \mathcal{C} \to \mathbb{R}$ be a process satisfying the conditions of Theorem 4.5 for each given $c \in \mathcal{C}$. Suppose

$$(y, z, k, c) \to g(t, y, z, k, c) \text{ is concave for all } t. \tag{4.2.6}$$

Let $(Y_g^{(F)}, Z_g^{(F)}, K_g^{(F)})$ be the unique solution of the BSDE (4.2.1). Then we define

$$U_g(c) = Y_g^{(F)}(0) \tag{4.2.7}$$

to be the recursive utility of c with terminal payoff F.

4.3 Linear BSDEs

There is no solution formula for the general BSDE (4.2.1). However, in the *linear* case we get the following:

Theorem 4.8 *Let α, β, γ be bounded predictable processes, $F \in L^2(\mathcal{F}_T)$ and φ predictable with $E[\int_0^T \varphi^2(t)\mathrm{d}t] < \infty$. Assume $\gamma > -1$ a.s. Then the unique solution (Y, Z, K) of the linear BSDE*

$$\begin{cases} \mathrm{d}Y(t) = -\left[\varphi(t) + \alpha(t)Y(t) + \beta(t)Z(t) + \int_{\mathbb{R}} \gamma(t,\zeta)K(t,\zeta)\nu(\mathrm{d}\zeta)\right]\mathrm{d}t \\ \quad + Z(t)\mathrm{d}B(t) + \int_{\mathbb{R}} K(t,\zeta)\tilde{N}(\mathrm{d}t, \mathrm{d}\zeta)\ ;\ 0 \le t \le T \\ Y(T) = F \end{cases} \tag{4.3.1}$$

is given by

$$Y(t) = E\left[\left\{\frac{\Gamma(T)}{\Gamma(t)}F + \int_t^T \frac{\Gamma(s)}{\Gamma(t)}\varphi(s)\mathrm{d}s\right\} \mid \mathcal{F}_t\right]\ ;\ 0 \le t \le T, \tag{4.3.2}$$

where

$$\begin{cases} \mathrm{d}\Gamma(t) = \Gamma(t^-)\left[\alpha(t)\mathrm{d}t + \beta(t)\mathrm{d}B(t) + \int_{\mathbb{R}} \gamma(t,\zeta)\tilde{N}(\mathrm{d}t, \mathrm{d}\zeta)\right]\ ;\ t \ge 0 \\ \Gamma(0) = 1 \end{cases} \tag{4.3.3}$$

i.e.

$$\begin{aligned} \Gamma(t) = \exp\Bigg(&\int_0^t \beta(s)\mathrm{d}B(s) + \int_0^t \left\{\alpha(s) - \frac{1}{2}\beta^2(s)\right\}\mathrm{d}s \\ &+ \int_0^t \int_{\mathbb{R}} \ln(1+\gamma(s,\zeta))\tilde{N}(\mathrm{d}s, \mathrm{d}\zeta) \\ &+ \int_0^t \int_{\mathbb{R}} \{\ln(1+\gamma(s,\zeta)) - \gamma(s,\zeta)\}\nu(\mathrm{d}\zeta)\mathrm{d}s\Bigg). \end{aligned} \tag{4.3.4}$$

Proof (***Sketch***) By the Itô formula,

$$\begin{aligned}
&\mathrm{d}(\Gamma(t)Y(t)) = \Gamma(t^-)\mathrm{d}Y(t) + Y(t^-)\mathrm{d}\Gamma(t) + \mathrm{d}[\Gamma, Y](t) \\
&= \Gamma(t^-)\left[-\left(\varphi(t) + \alpha(t)Y(t) + \beta(t)Z(t) + \int_{\mathbb{R}} \gamma(t,\zeta)K(t,\zeta)\nu(\mathrm{d}\zeta)\right)\mathrm{d}t\right. \\
&\quad \left. + Z(t)\mathrm{d}B(t) + \int_{\mathbb{R}} K(t,\zeta)\tilde{N}(\mathrm{d}t,\mathrm{d}\zeta)\right] \\
&\quad + Y(t)\Gamma(t^-)\left[\alpha(t)\mathrm{d}t + \beta(t)\mathrm{d}B(t) + \int_{\mathbb{R}} \gamma(t,\zeta)\tilde{N}(\mathrm{d}t,\mathrm{d}\zeta)\right] \\
&\quad + Z(t)\Gamma(t)\beta(t)\mathrm{d}t + \int_{\mathbb{R}} K(t,\zeta)\Gamma(t^-)\gamma(t,\zeta)N(\mathrm{d}t,\mathrm{d}\zeta) \\
&= -\Gamma(t)\varphi(t)\mathrm{d}t + \Gamma(t)(Z(t) + \beta(t)Y(t))\mathrm{d}B(t) \\
&\quad + \int \Gamma(t^-)K(t,\zeta)(1+\gamma(t,\zeta))\tilde{N}(\mathrm{d}t,\mathrm{d}\zeta).
\end{aligned}$$

Hence $\Gamma(t)Y(t) + \int_0^t \Gamma(s)\varphi(s)\mathrm{d}s$ is a martingale and therefore

$$\Gamma(t)Y(t) + \int_0^t \Gamma(s)\varphi(s)\mathrm{d}s = E\left[\{\Gamma(T)Y(T) + \int_0^T \Gamma(s)\varphi(s)\mathrm{d}s\} \mid \mathcal{F}_t\right]$$

i.e.

$$\Gamma(t)Y(t) = E\left[\{\Gamma(T)F + \int_t^T \Gamma(s)\varphi(s)\mathrm{d}s\} \mid \mathcal{F}_t\right],$$

as claimed. □

Example 4.9 (***Replications portfolio problem***) Let 4.8 to solve the BSDE (4.1.6)–(4.1.3): In this case

$$\mathrm{d}\Gamma(t) = \Gamma(t)\left[r(t)\mathrm{d}t + \frac{\mu(t) - r(t)}{\sigma(t)}\mathrm{d}B(t)\right] \; ; \; \Gamma(0) = 1$$

i.e.

$$\Gamma(t) = \exp\left(\int_0^t \frac{\mu(s) - r(s)}{\sigma(s)}\mathrm{d}B(s) + \int_0^t \left\{r(s) - \frac{1}{2}\left(\frac{\mu(s) - r(s)}{\sigma(s)}\right)^2\right\}\mathrm{d}s\right)$$

and we get

$$Y(t) = \frac{1}{\Gamma(t)}E[F\Gamma(T)|\mathcal{F}_t].$$

Using Malliavin calculus we can write

$$Z(t) = D_{t^-}Y(t)\left(:= \lim_{s \to t^-} D_s Y(t)\right)$$

and this gives the replacing portfolio

$$\pi(t) = \frac{Z(t)}{Y(t)\sigma(t)} = \frac{D_t Y(t)}{Y(t)\sigma(t)}.$$

Here D_t denotes the Malliavin derivative at t (with respect to Brownian motion). See e.g. [DØP].

4.4 Comparison Theorems

Lemma 4.10 *Let α, β, γ, F be as in Theorem 4.8.*
Suppose $(Y(t), Z(t), K(t,\cdot))$ satisfies the linear backward stochastic inequality

$$\begin{cases} \mathrm{d}Y(t) = -h(t)\mathrm{d}t + Z(t)\mathrm{d}B(t) + \int_{\mathbb{R}} K(t,\zeta)\tilde{N}(\mathrm{d}t, \mathrm{d}\zeta)\ ;\ 0 \le t \le T \\ Y(T) \ge F, \end{cases} \tag{4.4.1}$$

where $h(t)$ is a given $\mathcal{F}_t$-adapted process such that

$$h(t) \ge \alpha(t)Y(t) + \beta(t)Z(t) + \int_{\mathbb{R}} \gamma(t,\zeta)K(t,\zeta)\nu(\mathrm{d}\zeta). \tag{4.4.2}$$

Then

$$Y(t) \ge E\left[\frac{\Gamma(T)F}{\Gamma(t)} \mid \mathcal{F}_t\right]\ ;\ 0 \le t \le T, \tag{4.4.3}$$

where $\Gamma(t)$ is given by (4.3.3)–(4.3.4).

Proof By the Itô formula we have

$$\begin{aligned} \mathrm{d}(\Gamma(t)Y(t)) &= \Gamma(t^-)\left[-h(t)\mathrm{d}t + Z(t)\mathrm{d}B(t) + \int_{\mathbb{R}} K(t,\zeta)\tilde{N}(\mathrm{d}t,\mathrm{d}\zeta)\right] \\ &\quad + Y(t^-)\Gamma(t^-)\left[\alpha(t)\mathrm{d}t + \beta(t)\mathrm{d}B(t) + \int_{\mathbb{R}} \gamma(t,\zeta)\tilde{N}(\mathrm{d}t,\mathrm{d}\zeta)\right] \\ &\quad + \Gamma(t)\beta(t)Z(t)\mathrm{d}t + \int_{\mathbb{R}} \Gamma(t^-)\gamma(t,\zeta)K(t,\zeta)N(\mathrm{d}t,\mathrm{d}\zeta) \\ &\le \Gamma(t)\left[-\alpha(t)Y(t) - \beta(t)Z(t) - \int_{\mathbb{R}} \gamma(t,\zeta)K(t,\zeta)\nu(\mathrm{d}\zeta)\right]\mathrm{d}t \end{aligned}$$

$$
\begin{aligned}
&+ \Gamma(t)Z(t)\mathrm{d}B(t) + \Gamma(t^-)\int_{\mathbb{R}} K(t,\zeta)\tilde{N}(\mathrm{d}t,\mathrm{d}\zeta)\\
&+ Y(t^-)\Gamma(t^-)\left[\alpha(t)\mathrm{d}t + \beta(t)\mathrm{d}B(t) + \int_{\mathbb{R}} \gamma(t,\zeta)\tilde{N}(\mathrm{d}t,\mathrm{d}\zeta)\right]\\
&+ \Gamma(t)\beta(t)Z(t)\mathrm{d}t + \int_{\mathbb{R}} \Gamma(t^-)\gamma(t,\zeta)K(t,\zeta)N(\mathrm{d}t,\mathrm{d}\zeta)\\
&= \mathrm{d}M(t),
\end{aligned}
$$

where

$$
M(t) := \int_0^t \Gamma(s)Z(s)\mathrm{d}B(s) + \int_0^t \int_{\mathbb{R}} \Gamma(s^-)\gamma(s,\zeta)K(s,\zeta)\tilde{N}(\mathrm{d}s,d\zeta)
$$

is a martingale. Hence

$$
\Gamma(T)Y(T) - \Gamma(t)Y(t) \le M(T) - M(t).
$$

Taking conditional expectation this gives

$$
\Gamma(t)Y(t) \ge E\left[\Gamma(T)F \mid \mathcal{F}_t\right].
$$

□

Corollary 4.11 *Let Y, Z, K be as in Lemma 4.10. Suppose that*

$$
F \ge 0 \ \textit{a.s.}
$$

Then

$$
Y(t) \ge 0 \ \textit{for a.a. } t, \omega.
$$

Proof Apply Lemma 4.10. □

In the following we assume that

$$
g_i(t, y, z, k, \omega) : [0, T] \times \mathbb{R} \times \mathbb{R} \times \mathcal{R} \times \Omega \to \mathbb{R}\ ;\ i = 1, 2
$$

are given $\mathbb{F}$-predictable processes satisfying (i)–(ii) in Theorem 4.5. We assume that $g_2(t, y, z, k, \omega)$ is Lipschitz continuous with respect to y, z, k, uniformly in t, ω. We also assume that there exists a bounded predictable process $\theta(t, \zeta)$ independent of y and z such that $\mathrm{d}t \otimes \mathrm{d}P \otimes \nu(\mathrm{d}u)$-a.s.,

$$
\theta(t, \zeta) \ge -1 \quad \text{and} \quad |\theta(t, \zeta)| \le \psi(\zeta), \tag{4.4.4}
$$

where $\psi \in L^2_\nu$, and such that

$$g_2(t, y, z, k_1(\cdot)) - g_2(t, y, z, k_2(\cdot)) \geq \int_{\mathbb{R}} \theta(t, \zeta)(k_1(\zeta) - k_2(\zeta))\nu(\mathrm{d}\zeta) \tag{4.4.5}$$

for all t, y, z.

We are now ready to state and prove a comparison theorem for BSDEs with jumps. For a stronger version see [QS].

Theorem 4.12 (***Comparison Theorem for BSDEs with Jumps***) *Suppose we have 2 process triples* (Y_1, Z_1, K_1) *and* (Y_2, Z_2, K_2)*, such that*

$$\begin{cases} \mathrm{d}Y_i(t) &= -g_i(t, Y_i(t), Z_i(t), K_i(t, \cdot))\mathrm{d}t + Z_i(t)\mathrm{d}B(t) \\ &\quad + \int_{\mathbb{R}} K_i(t, \zeta)\tilde{N}(\mathrm{d}t, \mathrm{d}\zeta)\ ;\ 0 \leq t \leq T \\ Y_i(T) &= F_i \end{cases} \tag{4.4.6}$$

for $i = 1, 2$*. where* $F_i \in L^2(\mathcal{F}_T)$*. Assume that*

$$g_1(t, Y_1(t), Z_1(t), K_1(t, \cdot)) \leq g_2(t, Y_1(t), Z_1(t), K_1(t, \cdot))\ ;\ t \in [0, T] \tag{4.4.7}$$

and

$$F_1 \leq F_2 \text{ a.s.} \tag{4.4.8}$$

Then

$$Y_1(t) \leq Y_2(t) \text{ for a.a. } (t, \omega) \in [0, T] \times \Omega. \tag{4.4.9}$$

Proof Put

$$\tilde{Y}(t) = Y_2(t) - Y_1(t),\ \tilde{Z}(t) = Z_2(t) - Z_1(t),\ \tilde{K}(t, \zeta) = K_2(t, \zeta) - K_1(t, \zeta).$$

Then

$$\begin{aligned} \mathrm{d}\tilde{Y}(t) &= -[g_2(t, Y_2(t), Z_2(t), K_2(t, \cdot)) - g_1(t, Y_1(t), Z_1(t), K_1(t, \cdot))]\mathrm{d}t \\ &\quad + \tilde{Z}(t)\mathrm{d}B(t) + \int_{\mathbb{R}} \tilde{K}(t, \zeta)\tilde{N}(\mathrm{d}t, \mathrm{d}\zeta)\ ;\ 0 \leq t \leq T. \end{aligned}$$

Note that

$$\begin{aligned} &g_2(t, Y_2(t), Z_2(t), K_2(t, \cdot)) - g_1(t, Y_1(t), Z_1(t), K_1(t, \cdot)) \\ &\quad = g_2(t, Y_2(t), Z_2(t), K_2(t, \cdot)) - g_2(t, Y_1(t), Z_2(t), K_2(t, \cdot)) \\ &\qquad + g_2(t, Y_1(t), Z_2(t), K_2(t, \cdot)) - g_2(t, Y_1(t), Z_1(t), K_2(t, \cdot)) \\ &\qquad + g_2(t, Y_1(t), Z_1(t), K_2(t, \cdot)) - g_2(t, Y_1(t), Z_1(t), K_1(t, \cdot)) \\ &\qquad + g_2(t, Y_1(t), Z_1(t), K_1(t, \cdot)) - g_1(t, Y_1(t), Z_1(t), K_1(t, \cdot)) \end{aligned}$$

$$= \varphi(t) + \alpha(t)\tilde{Y}(t) + \beta(t)\tilde{Z}(t) + \int_{\mathbb{R}} \theta(t,\zeta)\tilde{K}(t,\zeta)\nu(\mathrm{d}\zeta),$$

by (4.4.5), where

$$\varphi(t) := g_2(t, Y_1(t), Z_1(t), K_1(t,\cdot)) - g_1(t, Y_1(t), Z_1(t), K_1(t,\cdot)) \geq 0,$$

$$\alpha(t) := \frac{g_2(t, Y_2(t), Z_2(t), K_2(t,\cdot)) - g_2(t, Y_1(t), Z_2(t), K_2(t,\cdot))}{\tilde{Y}(t)} \chi_{\tilde{Y}(t)\neq 0}\tilde{Y}(t)$$

and

$$\beta(t) := \frac{g_2(t, Y_1(t), Z_2(t), K_2(t,\cdot)) - g_2(t, Y_1(t), Z_1(t), K_2(t,\cdot))}{\tilde{Z}(t)} \chi_{\tilde{Z}(t)\neq 0}\tilde{Z}(t).$$

Combining the above we get

$$\begin{cases} \mathrm{d}\tilde{Y}(t) &= -h(t)\mathrm{d}t + \tilde{Z}(t)\mathrm{d}B(t) + \int_{\mathbb{R}} \tilde{K}(t,\zeta)\tilde{N}(\mathrm{d}t, \mathrm{d}\zeta)\ ;\ 0 \leq t \leq T \\ \tilde{Y}(T) &= F_2 - F_1 \geq 0, \end{cases}$$

where

$$h(t) \geq \alpha(t)\tilde{Y}(t) + \beta(t)\tilde{Z}(t) + \int_{\mathbb{R}} \theta(t,\zeta)\tilde{K}(t,\zeta)\nu(\mathrm{d}\zeta).$$

By Corollary 4.11 it follows that $\tilde{Y}(t) \geq 0$ for all t, i.e. $Y_1(t) \leq Y_2(t)$ for all t. □

4.5 Convex Risk Measures and BSDEs

We now have the machinery we need to verify the connection between risk measures and BSDEs mentioned in Sect. 3.1.

Theorem 4.13 *Suppose $g(t,z,k) : [0,T] \times \mathbb{R} \times \mathcal{R} \to \mathbb{R}$ satisfies the conditions in Theorem 4.5, but now g does not depend on y, and is concave in z, k for all t. Assume g satisfies hypothesis (4.4.5). Define*

$$\rho_g(F) = -Y_g^{(F)}(0). \tag{4.5.1}$$

Then ρ_g is a convex risk measure.

Proof We must verify that ρ_g satisfies properties (i)–(iii) in Definition 4.4:

(i) (Convexity). Fix $\lambda \in (0,1)$ and let $F, G \in L^2(\mathcal{F}_T)$. We want to prove that

$$\rho_g(\lambda F + (1-\lambda)G) \leq \lambda\rho_g(F) + (1-\lambda)\rho_g(G)$$

i.e.

$$-Y^{(\lambda F+(1-\lambda)G)}(0) \leq \lambda(-Y^{(F)}(0)) + (1-\lambda)(-Y^{(G)}(0)).$$

Let $(\hat{Y}, \hat{Z}, \hat{K})$ be the solution of the BSDE

$$\begin{cases} \mathrm{d}\hat{Y}(t) &= -g(t, \hat{Z}(t), \hat{K}(t,\cdot))\mathrm{d}t + \hat{Z}(t)\mathrm{d}B(t) + \int_{\mathbb{R}} \hat{K}(t,\zeta)\tilde{N}(\mathrm{d}t, \mathrm{d}\zeta)\ ; \\ & \quad 0 \leq t \leq T \\ \hat{Y}(T) &= \lambda F + (1-\lambda)G \end{cases}$$

and put

$$\begin{aligned} \tilde{Y}(t) &= \lambda Y^{(F)}(t) + (1-\lambda)Y^{(G)}(t), \\ \tilde{Z}(t) &= \lambda Z^{(F)}(t) + (1-\lambda)Z^{(G)}(t), \\ \tilde{K}(t,\zeta) &= \lambda K^{(F)}(t,\zeta) + (1-\lambda)K^{(G)}(t,\zeta). \end{aligned}$$

Then

$$\begin{cases} \mathrm{d}\tilde{Y}(t) &= -[h(t) + g(t, \tilde{Z}(t), \tilde{K}(t,\cdot))]\mathrm{d}t \\ & \quad + \tilde{Z}(t)\mathrm{d}B(t) + \int_{\mathbb{R}} \tilde{K}(t,\zeta)\tilde{N}(\mathrm{d}t, \mathrm{d}\zeta)\ ;\ 0 \leq t \leq T \\ \tilde{Y}(T) &= \lambda F + (1-\lambda)G, \end{cases}$$

where

$$\begin{aligned} h(t) &= \lambda g(t, Z^{(F)}(t), K^{(F)}(t)) + (1-\lambda)g(t, Z^{(G)}(t), K^{(G)}(t,\cdot)) \\ &\quad - g(t, \tilde{Z}(t), \tilde{K}(t,\cdot)) \leq 0 \text{ since } g \text{ is concave.} \end{aligned}$$

By the comparison theorem (Theorem 4.12) we conclude that

$$\tilde{Y}(t) \leq \hat{Y}(t)\ ;\ 0 \leq t \leq T.$$

In particular, choosing $t = 0$ we get

$$\begin{aligned} \rho_g(\lambda F + (1-\lambda)G) &= -\hat{Y}(0) \leq -\tilde{Y}(0) = -\lambda Y^{(F)}(0) - (1-\lambda)Y^{(G)}(0) \\ &= \lambda\rho_g(F) + (1-\lambda)\rho_g(G). \end{aligned}$$

(ii) (Monotonicity) If $F_1 \leq F_2$, then $Y^{(F_1)}(t) \leq Y^{(F_2)}(t)$ by the comparison theorem. Hence

$$\rho_g(F_2) = -Y^{(F_2)}(0) \leq -Y^{(F_1)}(0) = \rho_g(F_1),$$

as required.

(iii) (Translation invariance) If $F \in L^2(\mathcal{F}_T, P)$ and $a \in \mathbb{R}$ is constant, then we check easily that $Y^{(F+a)}(t) = Y^{(F)}(t) + a$. Hence

$$\rho(F + a) = -Y^{(F+a)}(0) = -Y^{(F)}(0) - a = \rho_g(F) - a.$$

□

4.5.1 Dynamic Risk Measures

We now discuss an extension of the (static) risk measure ρ in Definition 4.4 to a *dynamic* risk measure ρ_t; $0 \le t \le T$.

Definition 4.14 A *dynamic risk measure* is a map ρ which to each bounded stopping time τ and each $\xi \in L^2(\mathcal{F}_\tau)$ assigns an adapted càdlàg process $(\rho_t(\xi, \tau))_{\{0 \le t \le \tau\}}$ which is non-increasing, translation invariant and *consistent*, in the sense that

$$\forall t \le S, \rho_t(\xi, \tau) = \rho_t(-\rho_S(\xi, \tau), S) \text{ a.s.} \tag{4.5.2}$$

for all stopping times $S \le \tau$.

Moreover we say that the risk measure satisfies

- the *zero-one law* property if
  $\rho_t(\mathbf{1}_A \xi, T) = \mathbf{1}_A \rho_t(\xi, T)$ *a.s* for $t \le T$, $A \in \mathcal{F}_t$, and $\xi \in L^2(\mathcal{F}_T)$.
- the *no arbitrage* property if
  $\left\{\xi^1 \ge \xi^2 \text{ a.s. and } \rho_t(\xi^1, \tau) = \rho_t(\xi^2, \tau) \text{ a.s. on some } A \in \mathcal{F}_t, t \le \tau\right\} \Longrightarrow \left\{\xi^1 = \xi^2 \text{ a.s. on } A\right\}$.

A natural way to construct dynamic risk measures is by means of BSDEs as follows:

Let g be a Lipschitz driver, which does not depend on y and such that $E\left[\int_0^T g^2(t, 0, 0)\mathrm{d}t\right] < \infty$. We assume that g satisfies (4.4.5)–(4.4.4) with $\theta(t, \zeta) > -1$. For a given stopping time $\tau \le T$ and $\xi \in L^2(\mathcal{F}_T)$, define the functional:

$$\rho_t^g(\xi, \tau) := -Y_g^{(\xi)}(t), \quad 0 \le t \le \tau, \tag{4.5.3}$$

where $Y_g^{(\xi)}$ denotes the solution of the BSDE with terminal condition ξ and terminal time τ. Then ρ^g defines a *dynamic risk measure* in the sense of Definition 4.14. To see this, we note that the consistency (4.5.2) follows from the flow property of BSDEs (see [QS]).

Moreover, the no-arbitrage property follows from the strict comparison theorem for BSDEs. We also note that if $g(t, 0, 0) = 0$, then the zero-one law holds. The dynamic risk measure is convex if g is concave.

It is natural to ask about the converse: When can a dynamic risk-measure be represented by a BSDE with jumps? The following proposition gives an answer.

Theorem 4.15 *Let ρ be a dynamic risk measure satisfying the zero-one law and the no arbitrage property. Moreover, suppose that ρ satisfies the following $\mathcal{E}^{C,C_1}$-*domination *property:*

There exists $C > 0$ and $-1 < C_1 \leq 0$ such that

$$\rho_t(\xi + \xi', T) - \rho_t(\xi, T) \geq -Y_t^{C,C_1}(\xi', T), \tag{4.5.4}$$

for any $\xi, \xi' \in L^2(\mathcal{F}_T)$, where $Y_t^{C,C_1}(\xi', T)$ is the solution of the BSDE associated with terminal condition ξ' and driver $f_{C,C_1}(t, \pi, \ell) := C|\pi| + C\int_{\mathbf{R}^*}(1 \wedge |u|)\ell^+(u)\nu(\mathrm{d}u) - C_1\int_{\mathbf{R}^*}(1 \wedge |u|)\ell^-(u)\nu(\mathrm{d}u)$. Then, there exists a Lipschitz driver $g(t, \pi, \ell)$ such that $\rho = \rho^g$, that is, ρ is the dynamic risk measure induced by a BSDE with jumps with driver $g(t, \pi, \ell)$.

For the proof, we refer to [Roy]. Additional properties of dynamic risk measures induced by BSDEs and dual representation in the convex case can be found in [QS].

For applications of convex risk measures and recursive utilities, see [Aa13] and Chaps. 5 and 6.

4.5.2 A Dual Representation of Convex Risk Measures

In Sect. 4.5 we saw that BSDEs can be used to define convex risk measures. With this representation the problem of risk minimization becomes a problem of stochastic control of FBSDEs, as we shall see in Sects. 5.4 and 6.2. There is another representation of convex risk measures based on convex duality methods. It goes as follows:

Theorem 4.16 (***Convex Duality Representation of Convex Risk Measures***) *Every convex risk measure $\rho : L^\infty(\mathcal{F}_T, P) \to \mathbb{R}$ is of the following form:*

$$\rho(X) = \sup_{Q \in \mathcal{P}} \{E_Q[-X] - \alpha(Q)\}\ ;\ X \in L^\infty(\mathcal{F}_T, P), \tag{4.5.5}$$

where $\mathcal{P}$ is a family of probability measures $Q \ll P$ and $\alpha : \mathcal{P} \to \mathbb{R}$ is a convex function, usually called a penalty function. See [FS1, FR].

For example, $\mathcal{P}$ could be the set $\mathcal{P}_\Theta$ of probability measures Q_θ defined by

$$\mathrm{d}Q_\theta(\omega) = M_\theta(T)\mathrm{d}P(\omega) \text{ on } \mathcal{F}_T, \tag{4.5.6}$$

where θ denotes the process $(\theta_0(t), \theta_1(t, \zeta))$ and $M_\theta(t)$ is the martingale defined by

$$\begin{cases} \mathrm{d}M_\theta(t) &= M_\theta(t^-)\left[\theta_0(t)\mathrm{d}B(t) + \displaystyle\int_{\mathbb{R}} \theta_1(t, \zeta)\tilde{N}(\mathrm{d}t, \mathrm{d}\zeta)\right]\ ;\ t \geq 0 \\ M_\theta(0) &= 1. \end{cases} \tag{4.5.7}$$

Here Θ is the set of admissible scenarios, defined by

$$\Theta = \{\theta(t) = (\theta_0, \theta_1(t,\zeta));\ \theta \text{ is } \mathcal{F}_t \text{ -predictable, } \theta_1(t,\zeta) > -1 \text{ and } E\left[\int_0^T \{\theta_0^2(t) + \int_{\mathbb{R}} \theta_1^2(t,\zeta)\nu(\mathrm{d}\zeta)\}\mathrm{d}t\right] < \infty\right\}. \tag{4.5.8}$$

A popular choice of the penalty function α is the *entropic penalty* α_e, defined by

$$\alpha_e(Q) = E\left[\frac{\mathrm{d}Q}{\mathrm{d}P}\log\frac{\mathrm{d}Q}{\mathrm{d}P}\right], \quad Q \in \mathcal{P}. \tag{4.5.9}$$

In information theory the quantity $E\left[\frac{\mathrm{d}Q}{\mathrm{d}P}\log\frac{\mathrm{d}Q}{\mathrm{d}P}\right]$ is known as the *entropy* of Q with respect to P and it is denoted by $H(Q \mid P)$. The corresponding convex risk measure in (4.5.5), with $\mathcal{P}$ and α_e as above, is called the *entropic risk measure* and denoted by ρ_e.

We will not give the proof of the representation (4.5.5) here, but we refer to [FS1, FR] for details. See also [FS2]. Note that if $\alpha = 0$ then the corresponding risk measure will be *sub-additive*, i.e.

$$\rho(X+Y) \le \rho(X) + \rho(Y) \tag{4.5.10}$$

and *positive homogeneous*, i.e.

$$\rho(cX) = c\rho(X) \text{ if } c > 0 \text{ is a constant.} \tag{4.5.11}$$

If (4.5.10) and (4.5.11) hold, then ρ is called a *coherent* risk measure. See [ADEH] for details.

The case when $\alpha = \alpha_e$ is the *entropic penalty case*. It corresponds to the case when the driver is

$$g(z) := -\frac{1}{2}z^2$$

in the BSDE representation of convex risk measures, as explained in Theorem 4.13. We will not prove this here, but we get an indirect confirmation of this connection by comparing the forthcoming Theorems 6.2 and 6.9 (based on (4.5.5) and $\alpha = \alpha_e$).

4.6 Exercises

Exercise 4.1 Use Theorem 4.8 to write the formula for the first component $Y(t)$ of the solution (Y, Z, K) of the following BSDEs:

(a)

$$\begin{cases} \mathrm{d}Y(t) &= \alpha Y(t)\mathrm{d}t + Z(t)\mathrm{d}B(t)\ ;\ 0 \le t \le T \\ Y(T) &= \exp(B(T)) \quad (\alpha \text{ constant}). \end{cases}$$

(b)

$$\begin{cases} \mathrm{d}Y(t) &= -Z(t)\mathrm{d}t + Z(t)\mathrm{d}B(t)\ ;\ 0 \le t \le T \\ Y(T) &= B^2(T). \end{cases}$$

(c)

$$\begin{cases} \mathrm{d}Y(t) &= -[B(t) + Y(t)]\mathrm{d}t + Z(t)\mathrm{d}B(t)\ ;\ 0 \le t \le T \\ Y(T) &= 0. \end{cases}$$

(d)

$$\begin{cases} \mathrm{d}Y(t) &= \int_{\mathbb{R}} (1+\zeta^2)^{-1} K(t,\zeta)\nu(\mathrm{d}\zeta)\mathrm{d}t \\ &\quad + \int_{\mathbb{R}} K(t,\zeta)\tilde{N}(\mathrm{d}t,\mathrm{d}\zeta)\ ;\ 0 \le t \le T \\ Y(T) &= \int_0^T \int_{\mathbb{R}} \zeta \tilde{N}(\mathrm{d}t,\mathrm{d}\zeta). \end{cases}$$

Exercise 4.2 Consider the financial market in Example 4.1, with $r = 0$. Find the replicating portfolio $\pi(t)$ for the T-claim $F = \exp(-B(T))$ by proceeding as suggested in this example, i.e. solving the BSDE (4.1.6) with $r = 0$.
Hint : Use Theorem 4.8 to write down the solution $Y(t)$. Then find $Z(t)$ from (4.1.6) and the replicating portfolio $\pi(t)$ from (4.1.5).

Exercise* 4.3 Consider a financial market of the form

$$\begin{cases} \mathrm{d}S_0(t) &= S_0(t)r(t)\mathrm{d}t\ ;\ S_0(0) = 1 \\ \mathrm{d}S_1(t) &= S_1(t^-)\left[\alpha(t)\mathrm{d}t + \beta(t)\mathrm{d}B(t) + \int_{\mathbb{R}} \gamma(t,\zeta)\tilde{N}(\mathrm{d}t,\mathrm{d}\zeta)\right]\ ;\ S_1(0) > 0, \end{cases} \tag{4.6.1}$$

where $r(t)$, $\alpha(t)$, $\beta(t)$ and $\gamma(t,\zeta)$ are given bounded predictable processes. Let $T > 0$ be a given constant. Suppose there exist bounded predictable processes $\theta_0(t)$ and $\theta_1(t,\eta) > -1$ such that

$$\alpha(t) - r(t) + \beta(t)\theta_0(t) + \int_{\mathbb{R}} \gamma(t,\zeta)\theta_1(t,\zeta)\nu(\mathrm{d}\zeta) = 0\ ;\ t \in [0,T]. \tag{4.6.2}$$

Define the measure Q_θ on $\mathcal{F}_r$ by

$$\frac{\mathrm{d}Q_0}{\mathrm{d}P} = G_\theta(T), \tag{4.6.3}$$

where

$$\begin{cases} \mathrm{d}G_\theta(t) &= G_\theta(t^-)\left[\theta_0(t)\mathrm{d}B(t) + \int_{\mathbb{R}} \theta_1(t,\zeta)\tilde{N}(\mathrm{d}t,\mathrm{d}\zeta)\right]\ ;\ 0 \le t \le T \\ G_\theta(0) &= 1. \end{cases} \tag{4.6.4}$$

(a) Explain why Q_θ is an equivalent martingale measure (EMM) for this market. In general this market will not be complete, but if F is a bounded T-claim we can define an arbitrage-free price $p_Q^F(\theta)$ at time $t = 0$ by

$$p_Q^F(0) = E_Q\left[\frac{1}{S_0(T)}F\right] = E_Q\left[\exp\left(-\int_0^T r(s)\mathrm{d}s\right)F\right]$$

(see e.g. Theorem 2.14). More generally, the price of F at time $t \in [0,T]$ can be defined by

$$Y(t) := p_Q^F(t) = E_Q\left[\exp\left(-\int_t^T r(s)\mathrm{d}s\right)F \mid \mathcal{F}_t\right]\ ;\ 0 \le t \le T. \tag{4.6.5}$$

(b) Show that there exist predictable processes $Z(t)$, $K(t,\zeta)$ such that (Y, Z, K) solves the linear BSDE

$$\begin{cases} \mathrm{d}Y(t) &= -\left\{-r(t)Y(t) + \theta_0(t)Z(t) + \int_{\mathbb{R}} \theta_1(t,\zeta)K(t,\zeta)\nu(\mathrm{d}\zeta)\right\}\mathrm{d}t \\ &\quad + Z(t)\mathrm{d}B(t) + \int_{\mathbb{R}} K(t,\zeta)\tilde{N}(\mathrm{d}t,\mathrm{d}\zeta)\ ;\ 0 \le t \le T \\ Y(T) &= F. \end{cases} \tag{4.6.6}$$

Chapter 5
Stochastic Control of Jump Diffusions



5.1 The Dynamic Programming Approach

Fix a domain $\mathcal{S} \subset \mathbb{R}^k$ (our solvency region) and let $Y(t) = Y^{(u)}(t)$ be a stochastic process of the form

$$\begin{aligned} \mathrm{d}Y(t) &= b(Y(t), u(t))\mathrm{d}t + \sigma(Y(t), u(t))\mathrm{d}B(t) \\ &\quad + \int_{\mathbb{R}^\ell} \gamma(Y(t^-), u(t^-), \zeta)\tilde{N}(\mathrm{d}t, \mathrm{d}\zeta), \quad Y(0) = y \in \mathbb{R}^k, \end{aligned} \tag{5.1.1}$$

where

$$b : \mathbb{R}^k \times U \to \mathbb{R}^k, \quad \sigma : \mathbb{R}^\ell \times U \to \mathbb{R}^{k\times m}, \quad \text{and} \quad \gamma : \mathbb{R}^k \times U \times \mathbb{R}^\ell \to \mathbb{R}^{k\times \ell}$$

are given functions, $U \subset \mathbb{R}^p$ is a given set. The process $u(t) = u(t, \omega) : [0, \infty) \times \Omega \to U$ is our *control process*, assumed to be càdlàg and adapted. We call $Y(t) = Y^{(u)}(t)$ a *controlled jump diffusion*.

We consider a performance criterion $J = J^{(u)}(y)$ of the form

$$J^{(u)}(y) = E^y\left[\int_0^{\tau_\mathcal{S}} f(Y(t), u(t))\mathrm{d}t + g(Y(\tau_\mathcal{S}))\cdot \mathcal{X}_{\{\tau_\mathcal{S}<\infty\}}\right], \tag{5.1.2}$$

where

$$\tau_\mathcal{S} = \inf\{t > 0; Y^{(u)}(t) \notin \mathcal{S}\} \quad \text{(the bankruptcy time)}$$

and $f : \mathcal{S} \times U \to \mathbb{R}$ and $g : \mathbb{R}^k \to \mathbb{R}$ are given functions.

We say that the control process u is *admissible* and write $u \in \mathcal{A}$ if (5.1.1) has a unique, strong solution $Y(t)$ for all $y \in \mathcal{S}$ and

$$E^y\left[\int_0^{\tau_\mathcal{S}} f^-(Y(t), u(t))\mathrm{d}t\right] < \infty,$$

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and the family $\{g^-(Y(\tau))\}_{\tau\in\mathcal{T}}$ is uniformly integrable, for all $y \in \mathcal{S}$.

The *stochastic control problem* is to find the *value function* $\Phi(y)$ and an *optimal control* $u^* \in \mathcal{A}$ defined by

$$\Phi(y) = \sup_{u\in\mathcal{A}} J^{(u)}(y) = J^{(u^*)}(y). \tag{5.1.3}$$

It turns out that – under mild conditions (see, e.g., [Ø1, Theorem 11.2.3]) – it suffices to consider *Markov* controls, i.e., controls $u(t)$ of the form

$$u(t) = u_0(Y(t^-))$$

for some function $u_0 : \mathbb{R}^k \to U$. Therefore, from now on we will only consider Markov controls and we will, with a slight abuse of notation, write $u(t) = u(Y(t^-))$.

Note that if $u = u(y)$ is a Markov control then $Y(t) = Y^{(u)}(t)$ is a Lévy diffusion with generator

$$\begin{aligned} A\phi(y) = A^u\phi(y) = &\sum_{i=1}^k b_i(y,u(y))\frac{\partial\phi}{\partial y_i}(y) + \frac{1}{2}\sum_{i,j=1}^k (\sigma\sigma^T)_{ij}(y,u(y))\cdot\frac{\partial^2\phi}{\partial y_i\partial y_j}(y)\\ &+\sum_{j=1}^{\ell}\int_{\mathbb{R}}\{\phi(y+\gamma^{(j)}(y,u(y),\zeta_j)) - \phi(y)\\ &-\nabla\phi(y)\cdot\gamma^{(j)}(y,u(y),z_j)\}\nu_j(\mathrm{d}\zeta_j). \end{aligned}$$

We now formulate a verification theorem for the optimal control problem (5.1.3), analogous to the classical Hamilton–Jacobi–Bellman (HJB) theorem for (continuous) Itô diffusions.

Theorem 5.1 (***HJB for Optimal Control of Jump Diffusions***)

(a) *Suppose* $\phi \in C^2(\mathcal{S}) \cap C(\bar{\mathcal{S}})$ *satisfies the following:*

(i) $A^v\phi(y) + f(y,v) \le 0$ *for all* $y \in \mathcal{S}$, $v \in U$.

(ii) $\phi(y) = g(y)$ *for all* $y \notin \mathcal{S}$.

(iii) $E^y\left[|\phi(Y(\tau))| + \int_0^{\tau_{\mathcal{S}}} |A\phi(Y(t))|\mathrm{d}t\right] < \infty$, *for all* $u \in \mathcal{A}$ *and all* $\tau \in \mathcal{T}$.

(iv) $\{\phi^-(Y(\tau))\}_{\tau\le\tau_{\mathcal{S}}}$ *is uniformly integrable for all* $u \in \mathcal{A}$ *and* $y \in \mathcal{S}$.

Then

$$\phi(y) \ge \Phi(y) \quad \text{for all } y \in \mathcal{S}. \tag{5.1.4}$$

(b) *Moreover, suppose that for each* $y \in \mathcal{S}$ *there exists a* $v = \hat{u}(y) \in U$ *such that*

(v) $A^{\hat{u}(y)}\phi(y) + f(y,\hat{u}(y)) = 0$

and

(vi) $\{\phi(Y^{(\hat{u})}(\tau))\}_{\tau \le \tau_S}$ *is uniformly integrable.*

Suppose $u^*(t) := \hat{u}(Y(t^-)) \in \mathcal{A}$. *Then* u^* *is an optimal control and*

$$\phi(y) = \Phi(y) = J^{(u^*)}(y) \quad \textit{for all } y \in \mathcal{S}. \tag{5.1.5}$$

Proof
(a) Let $u \in \mathcal{A}$. For $n = 1, 2, \ldots$ put $\tau_n = \min(n, \tau_S)$. Then by the Dynkin formula (Theorem 1.24) we have

$$E^y[\phi(Y(\tau_n))] = \phi(y) + E^y\left[\int_0^{\tau_n} A^u \phi(Y(t))\mathrm{d}t\right] \le \phi(y) - E\left[\int_0^{\tau_n} f(Y(t), u(t))\mathrm{d}t\right].$$

Hence,

$$\begin{aligned}\phi(y) &\ge \liminf_{n\to\infty} E^y\left[\int_0^{\tau_n} f(Y(t), u(t))\mathrm{d}t + \phi(Y(\tau_n))\right] \\ &\ge E^y\left[\int_0^{\tau_S} f(Y(t), u(t))\mathrm{d}t + g(Y(\tau_S)) \cdot \mathcal{X}_{\{\tau_S < \infty\}}\right] = J^{(u)}(y). \end{aligned} \tag{5.1.6}$$

Since $u \in \mathcal{A}$ was arbitrary we conclude that

$$\phi(y) \ge \Phi(y) \quad \text{for all } y \in \mathcal{S}. \tag{5.1.7}$$

(b) Now apply the above argument to $u(t) = \hat{u}(Y(t))$, where $\hat{u}$ is as in (v). Then we get *equality* in (5.1.6) and hence

$$\phi(y) = J^{(\hat{u})}(y) \le \Phi(y) \quad \text{for all} y \in \mathcal{S}. \tag{5.1.8}$$

Combining (5.1.7) and (5.1.8) we get (5.1.5). □

Example 5.2 (***Optimal Consumption and Portfolio in a Lévy Type Black–Scholes Market*** [Aa, FØS1])

Suppose we have a market with two possible investments:

(i) A *safe* investment (bond, bank account) with price dynamics

$$\mathrm{d}P_1(t) = r P_1(t)\mathrm{d}t, \quad P_1(0) = p_1 > 0.$$

(ii) A *risky* investment (stock) with price dynamics

$$\mathrm{d}P_2(t) = P_2(t^-)\left[\mu\,\mathrm{d}t + \sigma\,\mathrm{d}B(t) + \int_{-1}^{\infty} z\widetilde{N}(\mathrm{d}t, \mathrm{d}z)\right], \quad P_2(0) = p_2 > 0,$$

where $\mu > r > 0$, and $\sigma \in \mathbb{R}$ are constants. Assume that at any time t the investor can choose a consumption rate $c(t) \ge 0$ (adapted and càdlàg) and is also free to

transfer money from one investment to the other without transaction cost. Let $X_1(t)$ and $X_2(t)$ be the amounts of money invested in the bonds and the stocks, respectively. Let

$$\theta(t) = \frac{X_2(t)}{X_1(t) + X_2(t)}$$

be the fraction of the total wealth invested in stocks at time t. Define the *performance criterion* by

$$J^{(c,\theta)}(s, x_1, x_2) = E^{x_1,x_2}\left[\int_0^\infty \mathrm{e}^{-\delta(s+t)}\,\frac{c^\gamma(t)}{\gamma}\mathrm{d}t\right],$$

where $\delta > 0$, $\gamma \in (0, 1)$ are constants and E^{x_1,x_2} is the expectation w.r.t. the probability law P^{x_1,x_2} of $(X_1(t), X_2(t))$ when $X_1(0) = x_1$, $X_2(0) = x_2$. Call the control $u(t) = (c(t), \theta(t)) \in [0, \infty)\times[0, 1]$ *admissible* and write $u \in \mathcal{A}$ if the corresponding *total wealth*

$$W(t) = W^{(u)}(t) = X_1^{(u)}(t) + X_2^{(u)}(t)$$

is *nonnegative* for all $t \geq 0$.

The problem is to find $\Phi(s, x_1, x_2)$ and $u^* = (c^*, \theta^*) \in \mathcal{A}$ such that

$$\Phi(s, x_1, x_2) = \sup_{u\in\mathcal{A}} J^{(u)}(s, x_1, x_2) = J^{(u^*)}(s, x_1, x_2).$$

Case 1: $\nu = 0$

In this case the problem was solved by Merton [M]. He proved that if

$$\delta > \gamma\left[r + \frac{(\mu - r)^2}{2\sigma^2(1 - \gamma)}\right], \tag{5.1.9}$$

then the value function is

$$\Phi_0(s, x_1, x_2) = K_0\mathrm{e}^{-\delta s}(x_1 + x_2)^\gamma, \tag{5.1.10}$$

where

$$K_0 = \frac{1}{\gamma}\left[\frac{1}{1 - \gamma}\left(\delta - \gamma r - \frac{\gamma(\mu - r)^2}{2\sigma^2(1 - \gamma)}\right)\right]^{\gamma-1}. \tag{5.1.11}$$

Moreover, the optimal consumption rate $c_0^*(t)$ is given by

$$c_0^*(t) = (K_0\gamma)^{1/(\gamma-1)}(X_1(t) + X_2(t)) \tag{5.1.12}$$

and the optimal portfolio $\theta_0^*(t)$ is (the constant)

$$\theta_0^*(t) = \frac{\mu - r}{\sigma^2(1 - \gamma)} \quad \text{for all } t \in [0, \infty). \tag{5.1.13}$$

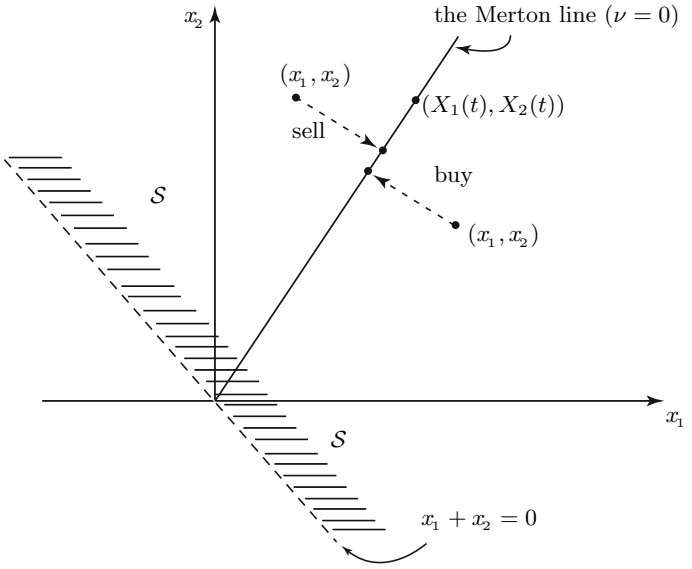


Fig. 5.1 The Merton line

In other words, it is optimal to keep the state $(X_1(t), X_2(t))$ on the line

$$x_2 = \frac{\theta_0^*}{1 - \theta_0^*} x_1 \tag{5.1.14}$$

in the (x_1, x_2)-plane at all times (the Merton line). See Fig. 5.1.

Case 2: $\nu \neq 0$

We now ask: How does the presence of jumps influence the optimal strategy?

As in [M] we reduce the dimension by introducing

$$W(t) = X_1(t) + X_2(t).$$

Then we see that

$$\begin{aligned} \mathrm{d}W(t) &= ([r(1 - \theta(t)) + \mu\theta(t)]W(t) - c(t))\,\mathrm{d}t + \sigma\theta(t)W(t)\mathrm{d}B(t) \\ &\quad + \theta(t)W(t^-)\int_{-1}^{\infty} \zeta\widetilde{N}(\mathrm{d}t, \mathrm{d}\zeta), \quad W(0) = x_1 + x_2 = w \geq 0. \end{aligned}$$

The generator $A^{(u)}$ of the controlled process

$$Y(t) = \begin{bmatrix} s + t \\ W(t) \end{bmatrix}; \quad t \geq 0, \quad Y(0) = y = \begin{bmatrix} s \\ w \end{bmatrix}$$

is

$$A^{(u)}\phi(y) = \frac{\partial\phi}{\partial s} + \big([r(1-\theta)+\mu\theta]w - c\big)\frac{\partial\phi}{\partial w} + \frac{1}{2}\sigma^2\theta^2w^2\frac{\partial^2\phi}{\partial w^2} + \int_{-1}^{\infty}\Big\{\phi(s, w+\theta w\zeta) - \phi(s, w) - \frac{\partial\phi}{\partial w}(s, w)\theta w\zeta\Big\}\nu(\mathrm{d}\zeta).$$

If we try

$$\phi(y) = \phi(s, w) = \mathrm{e}^{-\delta s}\psi(w)$$

we get

$$A^{(u)}\phi(y) = \mathrm{e}^{-\delta s}A_0^{(u)}\psi(w), \quad \text{where}$$
$$A_0^{(u)}\psi(w) = -\delta\psi(w) + ([r(1-\theta)+\mu\theta]\,w - c)\,\psi'(w) + \frac{1}{2}\sigma^2\theta^2w^2\psi''(w) + \int_{-1}^{\infty}\{\psi((1+\theta\zeta)w) - \psi(w) - \psi'(w)\theta w\zeta\}\nu(\mathrm{d}\zeta).$$

In particular, if we try

$$\psi(w) = Kw^{\gamma}$$

we get

$$A_0^{(u)}\psi(w) + f(w, u) = -\delta Kw^{\gamma} + \big([r(1-\theta)+\mu\theta]w - c\big)K\gamma w^{\gamma-1} + K\cdot\frac{1}{2}\sigma^2\theta^2w^2\gamma(\gamma-1)w^{\gamma-2} + Kw^{\gamma}\int_{-1}^{\infty}\{(1+\theta\zeta)^{\gamma} - 1 - \gamma\theta\zeta\}\nu(\mathrm{d}\zeta) + \frac{c^{\gamma}}{\gamma}. \quad (5.1.15)$$

Let $h(c, \theta)$ be the expression on the right-hand side. Then h is concave in (c, θ) and the maximum of h is attained at the critical points, i.e., when

$$\frac{\partial h}{\partial c} = -K\gamma w^{\gamma-1} + c^{\gamma-1} = 0 \quad (5.1.16)$$

and

$$\frac{\partial h}{\partial\theta} = (\mu - r)K\gamma w^{\gamma} + K\sigma^2\theta\gamma(\gamma-1)w^{\gamma} + Kw^{\gamma}\int_{-1}^{\infty}\{\gamma(1+\theta\zeta)^{\gamma-1}\zeta - \gamma\zeta\}\nu(\mathrm{d}\zeta) = 0. \quad (5.1.17)$$

From (5.1.16) we get

$$c = \hat{c} = \big(K\gamma\big)^{1/(\gamma-1)}w \quad (5.1.18)$$

and from (5.1.17) we get that $\theta = \hat{\theta}$ should solve the equation

$$\Lambda(\theta) := \mu - r - \sigma^2\theta(1-\gamma) - \int_{-1}^{\infty} \left\{1 - (1+\theta\zeta)^{\gamma-1}\right\}\zeta\nu(\mathrm{d}\zeta) = 0. \quad (5.1.19)$$

Since $\Lambda(0) = \mu - r > 0$ we see that if

$$\sigma^2(1-\gamma) + \int_{-1}^{\infty} \left\{1 - (1+\zeta)^{\gamma-1}\right\}\zeta\nu(\mathrm{d}\zeta) \geq \mu - r \quad (5.1.20)$$

then there exists an optimal $\theta = \hat{\theta} \in (0, 1]$.

With this choice of $c = \hat{c} = \left(K\gamma\right)^{1/(\gamma-1)} w$ and $\theta = \hat{\theta}$ (constant) we require that

$$\begin{aligned} &A_0^{(\hat{u})}\psi(w) + f(w, \hat{u}) = 0, \quad \text{i.e.,} \\ &\quad - \delta K + \left([r(1-\hat{\theta}) + \mu\hat{\theta}] - \left(K\gamma\right)^{1/(\gamma-1)}\right)K\gamma \\ &\quad + K\frac{1}{2}\sigma^2\hat{\theta}^2\gamma(\gamma-1) + K\int_{-1}^{\infty}\{(1+\hat{\theta}\zeta)^{\gamma} - 1 - \gamma\hat{\theta}\zeta\}\nu(\mathrm{d}\zeta) \\ &\quad + \left(K\gamma\right)^{\gamma/(\gamma-1)}\frac{1}{\gamma} = 0 \end{aligned}$$

or

$$\begin{aligned} &- \delta + \gamma[r(1-\hat{\theta}) + \mu\hat{\theta}] - (K\gamma)^{1/(\gamma-1)}\,\gamma \\ &-\frac{1}{2}\sigma^2\hat{\theta}^2(1-\gamma)\gamma + \int_{-1}^{\infty}\{(1+\hat{\theta}\zeta)^{\gamma} - 1 - \gamma\hat{\theta}\zeta\}\nu(\mathrm{d}\zeta) + K^{1/(\gamma-1)}\cdot\gamma^{\gamma/(\gamma-1)}\cdot\frac{1}{\gamma} \end{aligned}$$

or

$$\begin{aligned} \left(K\gamma\right)^{1/(\gamma-1)}[1-\gamma] = \delta - \gamma[r(1-\hat{\theta}) + \mu\hat{\theta}] + \frac{1}{2}\sigma^2\hat{\theta}^2(1-\gamma)\gamma \\ - \int_{-1}^{\infty}\{(1+\hat{\theta}\zeta)^{\gamma} - 1 - \gamma\hat{\theta}\zeta\}\nu(\mathrm{d}\zeta) \end{aligned}$$

or

$$\begin{aligned} K = \frac{1}{\gamma}\Bigg[\frac{1}{1-\gamma}\Bigg(\delta - \gamma\{r(1-\hat{\theta}) + \mu\hat{\theta}\} + \frac{1}{2}\sigma^2\hat{\theta}^2(1-\gamma)\gamma \\ - \int_{\mathbb{R}}\{(1+\hat{\theta}\zeta)^{\gamma} - 1 - \gamma\hat{\theta}\zeta\}\nu(\mathrm{d}\zeta)\Bigg)\Bigg]^{\gamma-1}. \end{aligned} \quad (5.1.21)$$

We now study condition (iii):

From (5.1.15) we see that

$$|A\phi(t,w)| \le e^{-\delta t} w^{\gamma} \left[\mathcal{C} + K \int_{\mathbb{R}} |(1+\theta\zeta)^{\gamma} - 1 - \gamma\theta\zeta| \nu(d\zeta) \right]$$

for some constant $\mathcal{C}$.

So (iii) holds if

$$E\left[\int_0^T \mathrm{e}^{-2\delta t} W^{2\gamma}(t)\mathrm{d}t\right] + \int_{\mathbb{R}} |(1+\theta\zeta)^{\gamma} - 1 - \gamma\theta\zeta| \nu(\mathrm{d}\zeta) < \infty. \tag{5.1.22}$$

We refer to [FØS1] for sufficient conditions on the parameters for (5.1.22) to hold. See also [FØS3].

We conclude that the value function is

$$\Phi(s,w) = \Phi(s,x_1,x_2) = \mathrm{e}^{-\delta s} K (x_1 + x_2)^{\gamma} \tag{5.1.23}$$

with optimal control $u^*(t) = (c^*(t), \theta^*(t))$, where $c^* = \hat{c} = (K\gamma)^{1/\gamma - 1}(x_1 + x_2)$ is given by (5.1.18) and $\theta^* = \hat{\theta}$ is given by (5.1.19), with K given by (5.1.21).

Finally we compare the solution in the jump case ($\nu \neq 0$) with Merton's solution in the no jump case ($\nu = 0$).

As before let Φ_0, c_0^*, and θ_0^* be the solution when there are no jumps ($\nu = 0$). Then it can be seen that

$$K < K_0 \quad \text{and hence} \quad \Phi(s,w) = \mathrm{e}^{-\delta s} K w^{\gamma} < \mathrm{e}^{-\delta s} K_0 w^{\gamma} = \Phi_0(s,w)$$
$$c^*(s,w) \ge c_0^*(s,w)$$
$$\theta^* \le \theta_0^*.$$

So with jumps it is optimal to place a *smaller* wealth fraction in the risky investment, consume *more* relative to the current wealth and the resulting value is *smaller* than in the no jump case. See Fig. 5.2.

For more details we refer to [FØS1, FØS3].

Remark 5.3 For more information and other applications of stochastic control of jump diffusions, see [GS, BKR1, BKR2, BKR3, BKR4, BKR5, BKR6, Ma] and the references therein.

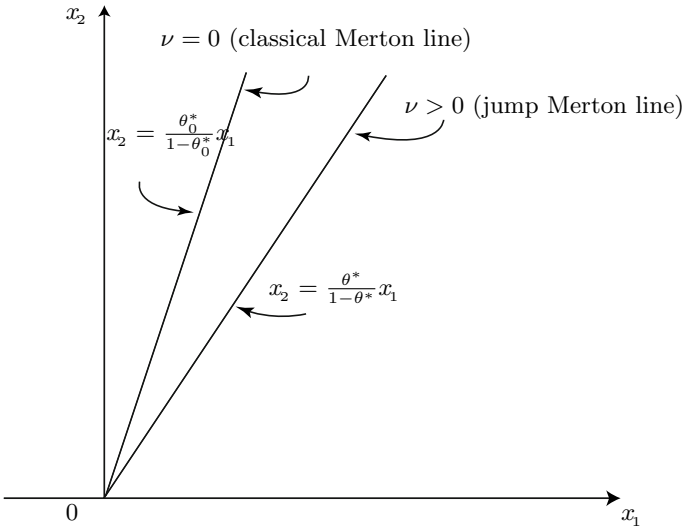


Fig. 5.2 The Merton line for $\nu = 0$ and $\nu > 0$

5.2 Stochastic Maximum Principles for Partial Information Control

Consider the following controlled stochastic differential equation:

$$\begin{cases} dX(t) &= b(t, X(t), u(t), \omega)dt + \sigma(t, X(t), u(t), \omega)dB(t) \\ &\quad + \int_{\mathbb{R}} \gamma(t, X(t), u(t), \omega, \zeta)\tilde{N}(dt, d\zeta) \ ; \ t \geq 0 \\ X(0) &= x \in \mathbb{R}. \end{cases} \tag{5.2.1}$$

For simplicity, we consider only the 1-dimensional case.

Let $\mathbb{G} := \{\mathcal{G}_t\}_{0 \leq t \leq T}$ be a given *subfiltration* of $\mathbb{F} := \{\mathcal{F}_t\}_{0 \leq t \leq T}$, i.e. $\mathcal{G}_t \subseteq \mathcal{F}_t$ for all t. We assume that $\mathbb{G}$ satisfies the usual conditions. We can interpret $\mathcal{G}_t$ as the information available to the controller at time t.

Let $\mathbb{U}$ be a given open convex subset of $\mathbb{R}$ and let $\mathcal{A}_{\mathbb{G}}$ be a given family of admissible controls, consisting of all $\mathbb{G}$-predictable processes $u = u(t)$ with values in $\mathbb{U}$.

The *performance functional* is given by

$$J(u) = E\left[\int_0^T f(t, X(t), u(t), \omega)dt + g(X(T), \omega)\right] \ ; \ u \in \mathcal{A}_{\mathbb{G}}. \tag{5.2.2}$$

Here $T > 0$ is a fixed (finite) constant. We want to find $u^* \in \mathcal{A}_{\mathbb{G}}$ such that

$$\sup_{u\in\mathcal{A}_{\mathbb{G}}} J(u) = J(u^*). \tag{5.2.3}$$

The setup is the same as in (5.1.1), except that now we allow the system to be non-Markovian, in the sense that we allow the coefficients to depend explicitly on ω, i.e. $b(t,x,u,\omega)$ is predictable for each $x\in\mathbb{R}^n$, $u\in U$, and similarly for σ and γ. Moreover, we have partial information constraints on the control.

We make the following assumptions:

$$f\in C^1 \text{ and } E\left[\int_0^T |\nabla f|^2(t)dt\right] < \infty, \tag{5.2.4}$$

$$b,\sigma,\gamma\in C^1 \text{ and } E\left[\int_0^T (|\nabla b|^2 + |\nabla\sigma|^2 + \|\nabla\gamma\|^2)(t)dt\right] < \infty, \tag{5.2.5}$$

where $\|\nabla\gamma(t,\cdot)\|^2 = \int_{\mathbb{R}} |\nabla\gamma|^2(t,\zeta)\nu(d\zeta)$,

$$g\in C^1 \text{ and } E\left[\int_0^T |\nabla g|^2(t)dt\right] < \infty, \tag{5.2.6}$$

for all $u\in\mathcal{A}_{\mathbb{G}}$. Let $\mathcal{R}$ denote the set of all functions $k:\mathbb{R}_0\to\mathbb{R}$.

The *Hamiltonian*

$$H : [0,T]\times\mathbb{R}\times\mathbb{U}\times\mathbb{R}\times\mathbb{R}\times\mathcal{R}\times\Omega\to\mathbb{R}$$

associated to the problem (5.2.3) is defined by

$$\begin{aligned} H(t,x,u,p,q,r,\omega) &= f(t,x,u,\omega) + b(t,x,u,\omega)p + \sigma(t,x,u,\omega)q \\ &\quad + \int_{\mathbb{R}} \gamma(t,x,u,\zeta,\omega)r(t,\zeta)\nu(d\zeta). \end{aligned} \tag{5.2.7}$$

Here p,q,r represent adjoint variables (see below).

For simplicity of notation the dependence on ω is suppressed in the following.

We assume that H is Fréchet differentiable (C^1) in the variables x,u. We let m denote Lebesgue measure on $[0,T]$. For $u\in\mathcal{A}_{\mathbb{G}}$ we let $X^u(t)$ be the associated solution of the system (5.2.1). We assume that for $u\in\mathcal{A}_{\mathbb{G}}$ this solution exists and is unique and satisfies

$$E\left[\int_0^T |X^u(t)|^2 dt\right] < \infty. \tag{5.2.8}$$

The associated BSDE system for the adjoint processes $(p(t),q(t),r(t,\cdot))$ is

$$\begin{cases} dp(t) &= -\dfrac{\partial H}{\partial x}(t)dt + q(t)dB(t) + \displaystyle\int_{\mathbb{R}} r(t,\zeta)\tilde{N}(dt,d\zeta)\ ;\ 0 \le t \le T \\ p(T) &= \varphi'(X(T)) + \lambda(T)h'(X(T)). \end{cases} \tag{5.2.9}$$

Here and in the following we are using the abbreviated notation

$$\frac{\partial H}{\partial x}(t) = \left[\frac{\partial}{\partial x}H(t,x,u(t))\right]_{x=X(t)} \text{ etc.}$$

5.2.1 A Sufficient Maximum Principle

We first formulate a sufficient maximum principle.

Theorem 5.4 (***Sufficient maximum principle***) *Let $\hat{u} \in \mathcal{A}_{\mathbb{G}}$ with corresponding solutions $\hat{X}(t), \hat{p}(t), \hat{q}(t), \hat{r}(t,\cdot)$ of Eqs. (5.2.1) and (5.2.9). Assume the following:*

$$\text{The function } x \to g(x) \text{ is concave} \tag{5.2.10}$$

$$\begin{aligned} &\text{(The Arrow condition). The function} \\ &\mathcal{H}(x) := \operatorname*{ess\,sup}_{v\in\mathbb{U}} E[H(t,x,v,\hat{p}(t),\hat{q}(t),\hat{r}(t,\cdot)) \mid \mathcal{G}_t] \\ &\text{is concave for all } t, \text{ a.s.} \end{aligned} \tag{5.2.11}$$

$$\begin{aligned} &\text{(The conditional maximum principle)} \\ &\operatorname*{ess\,sup}_{v\in\mathbb{U}} E[H(t,\hat{X}(t),v,\hat{p}(t),\hat{q}(t),\hat{r}(t,\cdot)) \mid \mathcal{G}_t] \\ &\quad = E[H(t,\hat{X}(t),\hat{u}(t),\hat{p}(t),\hat{q}(t),\hat{r}(t,\cdot)) \mid \mathcal{G}_t]\ ;\ t \in [0,T]. \end{aligned} \tag{5.2.12}$$

Then $\hat{u}$ is an optimal control for the problem (5.2.3).

Proof Define a sequence of stopping times $\tau_n\ ;\ n = 1, 2, \ldots,$ as follows

$$\begin{aligned} \tau_n = \inf\{t > 0\ ;\ &\int_0^t \hat{p}^2(s)\{(\sigma(s)-\hat{\sigma}(s))^2 + \int_{\mathbb{R}}(\gamma(s,\zeta)-\hat{\gamma}(s,\zeta))^2\nu(d\zeta)\}ds \\ &+ \int_0^t (X(s)-\hat{X}(s))^2\{\hat{q}(s)^2 + \int_{\mathbb{R}}\hat{r}(s,\zeta)^2\nu(d\zeta)\}ds \ge n\} \wedge T. \end{aligned} \tag{5.2.13}$$

Then note that $\tau_n \to T$ as $n \to \infty$ and

$$
\begin{aligned}
&E\left[\int_0^{\tau_n} \hat{p}(t)\left\{(\sigma(t)-\hat{\sigma}(t))dB(t)+\int_{\mathbb{R}}(\gamma(t,\zeta)-\hat{\gamma}(t,\zeta))\tilde{N}(dt,d\zeta)\right\}\right]\\
&\quad = E\left[\int_0^{\tau_n}(X(t)-\hat{X}(t))\left\{\hat{q}(t)dB(t)+\int_{\mathbb{R}}\hat{r}(t,\zeta)\tilde{N}(dt,d\zeta)\right\}\right]\\
&\quad = 0 \text{ for all } n. \qquad (5.2.14)
\end{aligned}
$$

Except for the introduction of these stopping times, the rest of the proof follows the proof in [FØS3]. See also the survey paper [ØS7]. For completeness we give the details:

Choose $u \in \mathcal{A}_{\mathbb{G}}$ and consider

$$J(u)-J(\hat{u}) = J_1 + J_2,$$

where

$$J_1 = E\left[\int_0^T \{f(t)-\hat{f}(t)\}dt\right],\ J_2 = E[g(X(T))-g(\hat{X}(T))],$$

where $f(t) = f(t, X(t), u(t))$, with $X(t) = X^u(t)$ etc.

By the definition of H we have

$$
\begin{aligned}
J_1 = E\Bigg[\int_0^T & H(t)-\hat{H}(t)-\hat{p}(t)(b(t)-\hat{b}(t))-\hat{q}(t)(\sigma(t)-\hat{\sigma}(t))\\
&-\int_{\mathbb{R}}\hat{r}(t,\zeta)(\gamma(t,\zeta)-\hat{\gamma}(t,\zeta))\nu(d\zeta)dt\Bigg]. \qquad (5.2.15)
\end{aligned}
$$

By concavity of g, (5.2.9), the Itô formula and (5.2.14),

$$
\begin{aligned}
J_2 &\le E[g'(\hat{X}(T))(X(T)-\hat{X}(T))]\\
&= E[\hat{p}(T)(X(T)-\hat{X}(T))]\\
&= \lim_{n\to\infty}\Bigg(E\left[\int_0^{\tau_n}\hat{p}(t^-)(dX(t)-d\hat{X}(t))+\int_0^{\tau_n}(X(t^-)-\hat{X}(t^-))d\hat{p}(t)\right.\\
&\quad\left.+\int_0^{\tau_n}\hat{q}(t)(\sigma(t)-\hat{\sigma}(t))dt+\int_0^{\tau_n}\int_{\mathbb{R}}\hat{r}(t,\zeta)(\gamma(t,\zeta)-\hat{\gamma}(t,\zeta))\nu(d\zeta)dt\right]\Bigg)\\
&= E\left[\int_0^T\hat{p}(t)(b(t)-\hat{b}(t))dt+\int_0^T(X(t)-\hat{X}(t))\left(-\frac{\partial\hat{H}}{\partial x}(t)\right)dt\right.\\
&\quad\left.+\int_0^T\hat{q}(t)(\sigma(t)-\hat{\sigma}(t))dt+\int_0^T\int_{\mathbb{R}}\hat{r}(t,\zeta)(\gamma(t,\zeta)-\hat{\gamma}(t,\zeta))\nu(d\zeta)dt\right].
\end{aligned}
$$

(5.2.16)

Adding (5.2.15), (5.2.16) we get

$$J(u) - J(\hat{u}) = J_1 + J_2$$
$$\leq E\left[\int_0^T \left\{H(t) - \hat{H}(t) - \frac{\partial \hat{H}}{\partial x}(X(t) - \hat{X}(t))\right\} dt\right]. \tag{5.2.17}$$

Using that $\hat{\mathcal{H}}$ is concave, we get by a separating hyperplane argument (see e.g. [R], Chap. 5, Sect. 23) that there exists a supergradient $a = (a_0, a_1, a_2, a_3(\cdot)) \in \mathbb{R}^3 \times \mathcal{R}$ for $\hat{\mathcal{H}}(x)$ at $x = \hat{X}(t)$, such that if we define

$$\Phi(x) := \hat{\mathcal{H}}(x) - \hat{\mathcal{H}}(\hat{X}(t)) - [a_0(x - \hat{X}(t))],$$

then

$$\Phi(x) \leq 0 \text{ for all } x.$$

On the other hand, since

$$\Phi(\hat{X}(t)) = 0$$

we get

$$\frac{\partial \hat{H}}{\partial x}(t) = \frac{\partial \hat{\mathcal{H}}}{\partial x}(\hat{X}(t)) = a_0.$$

If we combine this with (5.2.17) we obtain

$$J(u) - J(\hat{u}) \leq \hat{\mathcal{H}}(X(t)) - \hat{\mathcal{H}}(\hat{X}(t)) - \frac{\partial \hat{\mathcal{H}}}{\partial x}(\hat{X}(t))(X(t) - \hat{X}(t))$$
$$\leq 0, \text{ by concavity of } \mathcal{H}.$$

□

5.2.2 *A Necessary Maximum Principle*

We proceed to a *necessary* maximum principle. It stronger than the corresponding result in e.g. [ØS3], because of the weaker growth conditions.

We make the following assumptions:

A1. For all $t_0 \in [0, T]$ and all bounded $\mathcal{G}_{t_0}$-measurable random variables $\alpha(\omega)$, the control $\theta(t, \omega) := \chi_{[t_0,T]}(t)\alpha(\omega)$ belongs to $\mathcal{A}_{\mathbb{G}}$.

A2. For all $u, \beta_0 \in \mathcal{A}_{\mathbb{G}}$ with $\beta_0(t) \leq K < \infty$ for all t, define

$$\delta(t) := \frac{1}{2K}\text{dist}(u(t), \partial\mathbb{U}) \wedge 1 > 0$$

and put

$$\beta(t) := \delta(t)\beta_0(t). \tag{5.2.18}$$

Then the control

$$\tilde{u}(t) := u(t) + a\beta(t)\ ;\ t \in [0, T]$$

belongs to $\mathcal{A}_{\mathbb{G}}$ for all $a \in (-1, 1)$.

A3. For all β as in (5.2.18) the derivative process

$$x(t) := \frac{d}{da}X^{u+a\beta}(t)\ |_{a=0}$$

exists, belongs to $L^2(dm \times dP)$, and satisfies the equation

$$\begin{cases} dx(t) = \left\{\dfrac{\partial b}{\partial x}(t)x(t) + \dfrac{\partial b}{\partial u}(t)\beta(t)\right\}dt \\ \quad + \left\{\dfrac{\partial \sigma}{\partial x}(t)x(t) + \dfrac{\partial \sigma}{\partial u}(t)\beta(t)\right\}dB(t) \\ \quad + \displaystyle\int_{\mathbb{R}}\left\{\frac{\partial \gamma}{\partial x}(t,\zeta)x(t) + \frac{\partial \gamma}{\partial u}(t,\zeta)\beta(t)\right\}\tilde{N}(dt, d\zeta)\ ;\ t \in [0, T] \\ x(0) = 0. \end{cases} \tag{5.2.19}$$

Theorem 5.5 (***Necessary maximum principle***) *The following are equivalent:*

(i) $\dfrac{d}{da}J(u + a\beta)\ |_{a=0} = 0$ *for all bounded* $\beta \in \mathcal{A}_{\mathbb{G}}$ *of the form* (5.2.18).

(ii) $E\left[\dfrac{\partial H}{\partial u}(t) \mid \mathcal{G}_t\right] = 0$ *for all* $t \in [0, T]$.

Proof Define a sequence of stopping times $\tau_n\ ;\ n = 1, 2, \ldots$ as follows:

$$\tau_n = \inf\{t > 0\ ;\ \int_0^t p^2(s)\{\sigma^2(s) + \int_{\mathbb{R}}\gamma^2(s,\zeta)\nu(d\zeta)\}ds + \int_0^t x^2(s)\{q^2(s) + \int_{\mathbb{R}} r^2(s,\zeta)\nu(d\zeta)\}ds \geq n\} \wedge T.$$

Then it is clear that $\tau_n \to T$ as $n \to \infty$ and

$$E\left[\int_0^{\tau_n} p(t)\sigma(t)dB(t)\right] = E\left[\int_0^{\tau_n} p(t)\int_{\mathbb{R}}\gamma(t,\zeta)\tilde{N}(dt, d\zeta)\right]$$
$$= E\left[\int_0^{\tau_n} x(t)q(t)dB(t)\right] = E\left[\int_0^{\tau_n}\int_{\mathbb{R}} x(t)r(t,\zeta)\tilde{N}(dt, d\zeta)\right] = 0 \text{ for all } n. \tag{5.2.20}$$

We can write $\frac{d}{da}J(u+a\beta)\mid_{a=0}= I_1 + I_2$, where
$I_1 = \frac{d}{da}E[\int_0^T f(t, X^{u+a\beta}(t), Y^{u+a\beta}(t), Z^{u+a\beta}(t), K^{u+a\beta}(t), u(t)+a\beta(t))dt]_{a=0}$
and

$$I_2 = \frac{d}{da}E[g(X^{u+a\beta}(T))]_{a=0}.$$

By our assumptions on f, g we have

$$\begin{aligned} I_1 &= E\left[\int_0^T \left\{\frac{\partial f}{\partial x}(t)x(t) + \frac{\partial f}{\partial u}(t)\beta(t)\right\}dt\right] \\ I_2 &= E[\varphi'(X(T))x(T)] = E[p(T)x(T)]. \end{aligned} \tag{5.2.21}$$

By the Itô formula and (5.2.20)

$$\begin{aligned} I_2 &= E[p(T)x(T)] = \lim_{n\to\infty} E[p(\tau_n)x(\tau_n)] \\ &= \lim_{n\to\infty} E\left[\int_0^{\tau_n} p(t)dx(t) + \int_0^{\tau_n} x(t)dp(t) + \int_0^{\tau_n} d[p,x](t)\right] \\ &= \lim_{n\to\infty} E\left[\int_0^{\tau_n} p(t)\left\{\frac{\partial b}{\partial x}(t)x(t) + \frac{\partial b}{\partial u}(t)\beta(t)\right\}dt + \int_0^{\tau_n} x(t)\left(-\frac{\partial H}{\partial x}(t)\right)dt\right. \\ &\quad + \int_0^{\tau_n} q(t)\left\{\frac{\partial \sigma}{\partial x}(t)x(t) + \frac{\partial \sigma}{\partial u}(t)\beta(t)\right\}dt \\ &\quad \left. + \int_0^{\tau_n}\int_{\mathbb{R}} r(t,\zeta)\left\{\frac{\partial \gamma}{\partial x}(t,\zeta)x(t) + \frac{\partial \gamma}{\partial u}(t,\zeta)\beta(t)\right\}\nu(d\zeta)dt\right] \\ &= \lim_{n\to\infty} E\left[\int_0^{\tau_n}\left(x(t)\left\{\frac{\partial b}{\partial x}(t)p(t) + \frac{\partial \sigma}{\partial x}(t)q(t)\right\} + \int_{\mathbb{R}}\frac{\partial \gamma}{\partial x}(t,\zeta)r(t,\zeta)\nu(d\zeta)\right.\right. \\ &\quad \left.\left. -\frac{\partial H}{\partial x}(t)\right)dt\right] \\ &= \lim_{n\to\infty} E\left[\int_0^{\tau_n} x(t)\left\{-\frac{\partial f}{\partial x}(t)\right\}dt + \int_0^{\tau_n}\beta(t)\left\{\frac{\partial H}{\partial u}(t) - \frac{\partial f}{\partial u}(t)\right\}dt\right] \\ &= -I_1 + E\left[\int_0^T \frac{\partial H}{\partial u}(t)\beta(t)dt\right]. \end{aligned} \tag{5.2.22}$$

Summing (5.2.21) and (5.2.22) we get

$$\frac{d}{da}J(u+a\beta)\mid_{a=0}= I_1 + I_2 = E\left[\int_0^T \frac{\partial H}{\partial u}(t)\beta(t)dt\right].$$

We conclude that

$$\frac{d}{da}J(u+a\beta)\mid_{a=0}= 0$$

if and only if

$$E\left[\int_0^T \frac{\partial H}{\partial u}(t)\beta(t)dt\right] = 0\ ; \quad \text{for all bounded } \beta \in \mathcal{A}_{\mathbb{G}} \text{ of the form (5.2.18).}$$

In particular, applying this to $\beta(t) = \theta(t)$ as in A1, we get that this is again equivalent to

$$E\left[\frac{\partial H}{\partial u}(t) \mid \mathcal{G}_t\right] = 0 \text{ for all } t \in [0, T].$$

□

5.2.3 The Relation Between the Maximum Principle and Dynamic Programming

We mention briefly the relation to dynamic programming:

Define

$$J^{(u)}(s, x) = E\left[\int_0^{T-s} f(s + t, X^x(t), u(t))\mathrm{d}t + g(X^x(T - s))\right], \quad u \in \mathcal{A},$$

where $X^x(t)$ is the solution of (5.2.1) for $t \geq 0$ with initial value $X(0) = x$.

Then put

$$V(s, x) = \sup_{u \in \mathcal{A}} J^{(u)}(s, x). \tag{5.2.23}$$

Theorem 5.6 ([FØS3]) *Assume that $V(s, x) \in C^{1,3}(\mathbb{R} \times \mathbb{R}^n)$ and that there exists an optimal Markov control $u^*(t, x)$ for problem (5.1.3), with corresponding solution $X^*(t)$ of (5.1.1). Define*

$$\begin{aligned}
p_i(t) &= \frac{\partial V}{\partial x_i}(t, X^*(t)), \quad 1 \leq i \leq n, \\
q_{jk}(t) &= \sum_{i=1}^{n} \sigma_{ik}(t, X^*(t), u^*(t))\frac{\partial^2 V}{\partial x_i \partial x_j}(t, X^*(t)), \quad 1 \leq j \leq n,\ 1 \leq k \leq m, \\
r_{ik}(t, \zeta) &= \frac{\partial V}{\partial x_i}(t, X^*(t) + \gamma^{(k)}(t, X^*(t), u^*(t), \zeta)) - \frac{\partial V}{\partial x_i}(t, X^*(t)), \\
&\quad 1 \leq i \leq n,\ 1 \leq k \leq \ell.
\end{aligned}$$

Then $(p(t), q(t), r(t, \cdot))$ solves the adjoint equation (5.2.9).

For a proof see [FØS3].

5.2.4 Utility Maximization

We now give an example of how the maximum principle can be applied in finance. Consider a financial market where the unit price $S_0(t)$ of the risk free asset is

$$S_0(t) = 1 \; ; \; t \in [0, T] \tag{5.2.24}$$

and the unit price $S_1(t)$ of the risky asset is given by

$$dS_1(t) = S_1(t)[b_0(t)dt + \sigma_0(t)dB(t)] \; ; \; t \in [0, T] \tag{5.2.25}$$

where $b_0(t)$, $\sigma_0(t)$ are given $\mathbb{F}$-adapted processes.

Then the wealth process $X_\pi(t)$ associated to a (self-financing) portfolio $\pi(t)$ is given by

$$\begin{cases} dX_\pi(t) &= \pi(t)X(t^-)[b_0(t)dt + \sigma_0(t)dB(t)] \; ; \; t \geq 0 \\ X_\pi(0) &= x_0 > 0. \end{cases} \tag{5.2.26}$$

Let U be a given utility function. We want to find $\pi^* \in \mathcal{A}$ such that

$$E[U(X_{\pi^*}(T))] = \sup_{\pi \in \mathcal{A}} E[U(X_\pi(T))], \tag{5.2.27}$$

where $\mathcal{A}$ is the given family of admissible $\mathbb{F}$-adapted portfolios π with values in $\mathbb{R}$.

The Hamiltonian for this problem is

$$H(t, x, y, z, k, \pi, \lambda, p, q, r) = \pi x b_0 p + \pi x \sigma_0(t) q \tag{5.2.28}$$

and the adjoint equation is

$$\begin{cases} dp(t) &= -\pi(t)\{b_0(t)p(t) + \sigma_0(t)q(t)\}dt + q(t)dB(t) \; ; \; 0 \leq t \leq T \\ p(T) &= U'(X_\pi(T)). \end{cases} \tag{5.2.29}$$

Suppose $\hat{\pi} \in \mathcal{A}$ is an optimal portfolio for the problem (5.2.27) with corresponding solutions $\hat{X}$, $\hat{p}, \hat{q}, \hat{r}$ of (5.2.26) and (5.2.29). Then $\dfrac{\partial \hat{H}}{\partial \pi}(t) = 0$, which gives

$$b_0(t)\hat{p}(t) + \sigma_0(t)\hat{q}(t) = 0. \tag{5.2.30}$$

Hence

$$\hat{q}(t) = -\frac{b_0(t)}{\sigma_0(t)}\hat{p}(t)$$

and (5.2.29) becomes

$$\begin{cases} d\hat{p}(t) &= \hat{q}(t)dB(t) = -\dfrac{b_0(t)}{\sigma_0(t)}\hat{p}(t)dB(t)\ ;\ 0 \le t \le T \\ \hat{p}(T) &= U'(\hat{X}(T)). \end{cases} \tag{5.2.31}$$

Define

$$\theta_0(t) = -\frac{b_0(t)}{\sigma_0(t)}. \tag{5.2.32}$$

Then by (5.2.30)

$$b_0(t) + \sigma_0(t)\theta_0(t) = 0,$$

and the Girsanov theorem gives that the measure Q defined by

$$dQ = \Gamma(T)dP \text{ on } \mathcal{F}_T \tag{5.2.33}$$

is an equivalent local martingale measure, where $\Gamma(t) = \Gamma_{\theta_0}(t)$ is given by

$$d\Gamma(t) = \Gamma(t)\theta_0(t)dB(t)\ ;\ \Gamma(0) = 1. \tag{5.2.34}$$

Note that

$$\frac{d\hat{p}(t)}{\hat{p}(t)} = \frac{d\Gamma(t)}{\Gamma(t)}$$

so

$$\hat{p}(t) = E[U'(\hat{X}(T))]\Gamma(t).$$

By (5.2.32) and (5.2.34) we have

$$\Gamma(t) = \exp\left(-\int_0^t \frac{b_0(s)}{\sigma_0(s)}dB(s) - \frac{1}{2}\int_0^t \frac{b_0^2(s)}{\sigma_0^2(s)}ds\right)\ ;\ 0 \le t \le T. \tag{5.2.35}$$

Hence

$$\hat{p}(T) = U'(\hat{X}(T)) = E[U'(\hat{X}(T))]\Gamma(T),$$

i.e.

$$\hat{X}(T) = I(c\ \Gamma(T)), \tag{5.2.36}$$

where

$$I = (U')^{-1} \text{ and } c = E[U'(\hat{X}(T))].$$

It remains to find c. We can write (5.2.26) as

$$\begin{cases} d\hat{X}(t) &= \hat{\pi}(t)\hat{X}(t)[b_0(t)dt + \sigma_0(t)dB(t)]\ ;\ 0 \le t \le T \\ \hat{X}(T) &= I(c\Gamma(T)). \end{cases}$$

If we define

$$\hat{Z}(t) = \hat{\pi}(t)\hat{X}(t)\sigma_0(t) \tag{5.2.37}$$

this becomes a BSDE

$$\begin{cases} d\hat{X}(t) &= \dfrac{\hat{Z}(t)b_0(t)}{\sigma_0(t)}dt + \hat{Z}(t)db(t)\ ;\ 0 \le t \le T \\ \hat{X}(T) &= I(c\Gamma(T)). \end{cases} \tag{5.2.38}$$

The solution of this BSDE is

$$\hat{X}(t) = \frac{1}{\Gamma(t)}E[I(c\Gamma(T))\Gamma(T) \mid \mathcal{F}_t]. \tag{5.2.39}$$

In particular,

$$x = \hat{X}(0) = E[I(c\Gamma(T))\Gamma(T)]. \tag{5.2.40}$$

This is an equation which (implicitly) determines the value of c. When c is found, we have the optimal terminal wealth $\hat{X}(T)$ given by (5.2.36). Solving the resulting BSDE for $\hat{Z}(t)$, we get the corresponding optimal portfolio $\hat{\pi}(t)$ by (5.2.37). We have proved:

Theorem 5.7 *The optimal terminal wealth $\hat{X}(T) = X_{\pi^*}(T)$ for the portfolio optimization problem (5.2.27) is given by (5.2.36), where the constant $c > 0$ is found implicitly by Eq. (5.2.40).*

5.2.5 *Mean-Variance Portfolio Selection*

The following example is from [FØS3].

Consider a financial market with two investment possibilities, one risk free (e.g., a bond or bank account) and one risky (e.g., a stock), whose prices $S_0(t)$, $S_1(t)$ at time $t \in [0, T]$ are given by

$$\text{(bond)}\quad \mathrm{d}S_0(t) = \rho_t S_0(t)\mathrm{d}t, \quad S_0(0) = 1, \tag{5.2.41}$$

$$\text{(stock)}\quad \mathrm{d}S_1(t) = S_1(t^-)\left[\mu_t \mathrm{d}t + \sigma_t \mathrm{d}B(t) + \int_{\mathbb{R}} \gamma(t,z)\widetilde{N}(\mathrm{d}t, \mathrm{d}z)\right], \quad S_1(0) > 0, \tag{5.2.42}$$

where $\rho_t > 0$, μ_t, σ_t, and $\gamma(t,z) \ge -1$ are given bounded deterministic functions. We assume that the function

$$t \to \int_{\mathbb{R}} \gamma^2(t,z)\nu(\mathrm{d}z) \text{ is locally bounded.} \tag{5.2.43}$$

We may regard this market as a jump diffusion extension of the classical Black–Scholes market (see e.g. [Ø1]).

A *portfolio* in this market is a two-dimensional càdlàg and adapted process $\theta(t) = (\theta_0(t), \theta_1(t))$ giving the number of units of bonds and stocks, respectively, held at time t by an agent.

The corresponding *wealth* process $X(t) = X^{(\theta)}(t)$ is defined by

$$X(t) = \theta_0(t)S_0(t) + \theta_1(t)S_1(t), \quad t \in [0, T]. \tag{5.2.44}$$

The portfolio θ is called *self-financing* if

$$X(t) = X(0) + \int_0^t \theta_0(s)\mathrm{d}S_0(s) + \int_0^t \theta_1(s)\mathrm{d}S_1(s) \tag{5.2.45}$$

or, in shorthand notation,

$$\mathrm{d}X(t) = \theta_0(t)\mathrm{d}S_0(t) + \theta_1(t)\mathrm{d}S_1(t). \tag{5.2.46}$$

Alternatively, the portfolio can also be expressed in terms of the *amounts* $w_0(t)$ and $w_1(t)$ invested in the bond and stock, respectively. They are given by

$$w_i(t) = \theta_i(t)S_i(t), \quad i = 0, 1. \tag{5.2.47}$$

Now put

$$u(t) = w_1(t). \tag{5.2.48}$$

Then $w_0(t) = X(t) - u(t)$ and (5.2.46) gets the form

$$\mathrm{d}X(t) = [\rho_t X(t) + (\mu_t - \rho_t)u(t)]\mathrm{d}t + \sigma_t u(t)\mathrm{d}B(t) + u(t^-)\int_{\mathbb{R}} \gamma(t, z)\widetilde{N}(\mathrm{d}t, \mathrm{d}z). \tag{5.2.49}$$

We call $u(t)$ *admissible* and write $u(t) \in \mathcal{A}$ if (5.2.49) has a unique solution $X(t) = X^{(u)}(t)$ such that $E[(X^{(u)}(T))^2] < \infty$.

The *mean-variance portfolio selection problem* is to find $u(t)$ which minimizes

$$\mathrm{Var}[X(T)] := E\big[(X(T) - E[X(T)])^2\big]$$

under the condition that

$$E[X(T)] = A, \quad \text{a given constant.}$$

By the Lagrange multiplier method the problem can be reduced to minimizing, for a given constant $a \in \mathbb{R}$,

$$E[(X(T) - a)^2]$$

without constraints. To see this, consider

$$\begin{aligned} &E[(X(T)-A)^2-\lambda(E[X(T)]-A)] \\ &\quad = E\left[X^2(T)-2\left(A+\frac{\lambda}{2}\right)X(T)+A^2+\lambda A\right] \\ &\quad = E\left[\left(X(T)-\left(A+\frac{\lambda}{2}\right)\right)^2\right]-\frac{\lambda^2}{4}, \quad \text{where } \lambda\in\mathbb{R} \text{ is constant.} \end{aligned}$$

We will consider the equivalent problem

$$\sup_{u\in\mathcal{A}} E\left[-\frac{1}{2}(X^{(u)}(T)-a)^2\right].$$

In this case the Hamiltonian (5.2.7) gets the form

$$H(t,x,u,p,q,r)=\{\rho_t x+(\mu_t-\rho_t)u\}p+\sigma_t uq+u\int_{\mathbb{R}}\gamma(t,z)r(t,z)\nu(\mathrm{d}z).$$

Hence the adjoint equations (5.2.9) are

$$\begin{cases} \mathrm{d}p(t)=-\rho_t p(t)\mathrm{d}t+q(t)\mathrm{d}B(t)+\displaystyle\int_{\mathbb{R}} r(t^-,z)\widetilde{N}(\mathrm{d}t,\mathrm{d}z), \quad t<T \\ p(T)=-(X(T)-a). \end{cases} \tag{5.2.50}$$

We try a solution of the form

$$p(t)=\phi_t X(t)+\psi_t,$$

where ϕ_t,ψ_t are deterministic C^1 functions. Substituting into (5.2.50) and using (5.2.49) we get

$$\begin{aligned} \mathrm{d}p(t) &= \phi_t\Big[\{\rho_t X(t)+(\mu_t-\rho_t)u(t)\}\mathrm{d}t+\sigma_t u(t)\mathrm{d}B(t) \\ &\qquad +u(t^-)\int_{\mathbb{R}}\gamma(t,z)\widetilde{N}(\mathrm{d}t,\mathrm{d}z)\Big]+X(t)\phi_t'\,\mathrm{d}t+\psi_t'\,\mathrm{d}t \\ &= [\phi_t\rho_t X(t)+\phi_t(\mu_t-\rho_t)u(t)+X(t)\phi_t'+\psi_t']\mathrm{d}t \\ &\quad +\phi_t\sigma_t u(t)\mathrm{d}B(t)+\phi_t u(t^-)\int_{\mathbb{R}}\gamma(t,z)\widetilde{N}(\mathrm{d}t,\mathrm{d}z). \end{aligned} \tag{5.2.51}$$

Comparing with (5.2.50) we get

$$\phi_t\rho_t X(t)+\phi_t(\mu_t-\rho_t)u(t)+X(t)\phi_t'+\psi_t'=-\rho_t(\phi_t X(t)+\psi_t), \tag{5.2.52}$$

$$q(t) = \phi_t \sigma_t u(t), \tag{5.2.53}$$

$$r(t,z) = \phi_t u(t) \gamma(t,z). \tag{5.2.54}$$

Let $\hat{u} \in \mathcal{A}$ be a candidate for the optimal control with corresponding $\hat{X}$ and $\hat{p}, \hat{q}, \hat{r}$. Then

$$\begin{aligned} &H(t, \hat{X}(t), u, \hat{p}(t), \hat{q}(t), \hat{r}(t,\cdot)) \\ &\quad = \rho_t \hat{X}(t)\hat{p}(t) + u\left[(\mu_t - \rho_t)\hat{p}(t) + \sigma_t \hat{q}(t) + \int_{\mathbb{R}} \gamma(t,z)\hat{r}(t,z)\nu(\mathrm{d}z)\right]. \end{aligned}$$

Since this is a linear expression in u, it is natural to guess that the coefficient of u vanishes, i.e.,

$$(\mu_t - \rho_t)\hat{p}(t) + \sigma_t \hat{q}(t) + \int_{\mathbb{R}} \gamma(t,z)\hat{r}(t,z)\nu(\mathrm{d}z) = 0. \tag{5.2.55}$$

Using that by (5.2.53) and (5.2.54) we have

$$\hat{q}(t) = \phi_t \sigma_t \hat{u}(t), \quad \hat{r}(t,z) = \phi_t \hat{u}(t)\gamma(t,z)$$

we get from (5.2.55) that

$$\hat{u}(t) = \frac{(\rho_t - \mu_t)\hat{p}(t)}{\phi_t \Lambda_t} = \frac{(\rho_t - \mu_t)(\phi_t \hat{X}(t) + \psi_t)}{\phi_t \Lambda_t}, \tag{5.2.56}$$

where

$$\Lambda_t = \sigma_t^2 + \int_{\mathbb{R}} \gamma^2(t,z)\nu(\mathrm{d}z). \tag{5.2.57}$$

On the other hand, from (5.2.52) we have

$$\hat{u}(t) = \frac{(\phi_t \rho_t + \phi_t')\hat{X}(t) + \rho_t(\phi_t \hat{X}(t) + \psi_t) + \psi_t'}{\phi_t(\rho_t - \mu_t)}. \tag{5.2.58}$$

Combining (5.2.56) and (5.2.58) we get the equations

$$\begin{aligned} &(\rho_t - \mu_t)^2 \phi_t - [2\rho_t \phi_t + \phi_t']\Lambda_t = 0, \quad \phi_T = -1, \\ &(\rho_t - \mu_t)^2 \psi_t - [\rho_t \psi_t + \psi_t']\Lambda_t = 0, \quad \psi_T = a, \end{aligned}$$

which have the solutions

$$\phi_t = -\exp\left(\int_t^T \left\{\frac{(\rho_s - \mu_s)^2}{\Lambda_s} - 2\rho_s\right\}\mathrm{d}s\right), \quad 0 \le t \le T, \tag{5.2.59}$$

$$\psi_t = a \exp\left(\int_t^T \left\{ \frac{(\rho_s - \mu_s)^2}{\Lambda_s} - \rho_s \right\} \mathrm{d}s \right), \quad 0 \le t \le T. \tag{5.2.60}$$

With this choice of ϕ_t and ψ_t the processes

$$\hat{p}(t) := \phi_t \hat{X}(t) + \psi_t, \quad \hat{q}(t) := \phi_t \sigma_t \hat{u}(t), \quad \text{and} \quad \hat{r}(t,z) := \phi_t \hat{u}(t) \gamma(t,z)$$

solve the adjoint equation, and by (5.2.55) we see that all the conditions of the sufficient maximum principle (Theorem 5.4) are satisfied. We conclude that $\hat{u}(t)$ given by (5.2.56) is an optimal control. In feedback form the control can be written

$$\hat{u}(t,x) = \frac{(\rho_t - \mu_t)(\phi_t x + \psi_t)}{\phi_t \Lambda_t}. \tag{5.2.61}$$

5.3 The Maximum Principle with Infinite Horizon

In the previous section we have presented maximum principles for optimal control of an Itô–Lévy process with a performance functional defined within a finite time interval $[0, T]$, and we saw that these principles involved BSDEs in the adjoint processes $(p(t), q(t), r(t))$ with given terminal value $p(T)$ at time T. If the performance functional is over the infinite horizon $[0, \infty)$, however, it is not clear how to formulate the terminal value condition in the corresponding infinite horizon BSDE.

In this section we will study this situation. Our system is a controlled Itô–Lévy process of the form

$$\begin{cases} dX(t) = & b\,(t, X(t), u(t), \omega)\,dt + \sigma\,(t, X(t), u(t), \omega)\,dB(t) \\ & + \int_{\mathbb{R}_0} \theta\,(t, X(t), u(t), \zeta, \omega)\,\tilde{N}(dt, d\zeta); \quad t \in [0, \infty)\,, \\ X(0) = & x; \end{cases} \tag{5.3.1}$$

where $u(t)$ is the control process and

$$\begin{aligned} b &: [0,\infty) \times \mathbb{R} \times \mathbb{R} \times \mathbb{R} \times \mathcal{U} \times \Omega \to \mathbb{R}, \\ \sigma &: [0,\infty) \times \mathbb{R} \times \mathbb{R} \times \mathbb{R} \times \mathcal{U} \times \Omega \to \mathbb{R}, \\ \theta &: [0,\infty) \times \mathbb{R} \times \mathbb{R} \times \mathbb{R} \times \mathcal{U} \times \mathbb{R}_0 \times \Omega \to \mathbb{R}, \end{aligned}$$

are given functions such that for all t, $b(t,x,u)$, $\sigma(t,x,u)$ and $\theta(t,x,u,\zeta)$ are $\mathcal{F}_t$-predictable processes for all $x \in \mathbb{R}$ and $u \in \mathcal{U}$ and $\zeta \in \mathbb{R}_0$. We assume that b, σ, θ are C^1 (i.e. continuously differentiable) with respect to x, u, for all t, ζ and a.a. ω. Let $\mathbb{G} = \{\mathcal{G}_t\}_{t \ge 0}$ be a given subfiltration of $\mathbb{F}$, in the sense that $\mathcal{G}_t \subset \mathcal{F}_t$ for all t. We assume that $\mathbb{G}$ satisfies the usual conditions. The sigma-algebra $\mathcal{G}_t$ represents the information available to the controller at time t. Let $\mathcal{U}$ be a non-empty subset of $\mathbb{R}$. We let $\mathcal{A}_{\mathbb{G}}$ denote a given family of admissible $\mathbb{G}$-predictable control processes

u such that there exists a unique associated solution $X(t) = X^{(u)}(t)$ of (5.3.1) with $E[\int_0^T X^2(t)dt] < \infty$, for all $T < \infty$.

An element of $\mathcal{A}_{\mathbb{G}}$ is called an admissible control. The corresponding performance functional is

$$J(u) = E\left[\int_0^\infty f\,(t, X(t), u(t))\;dt\right]; u \in \mathcal{A}_{\mathbb{G}}, \tag{5.3.2}$$

where we assume that f is C^1 with respect to x, u for all t and a.a. ω and

$$E\left[\int_0^\infty \left\{|f\,(t, X(t), u(t))\;| + \left|\frac{\partial f}{\partial x}\,(t, X(t), u(t))\;\right|^2\right\} dt\right] < \infty. \tag{5.3.3}$$

An admissible control u^* in $\mathcal{A}_{\mathbb{G}}$ is called an optimal control if

$$\sup_{u \in \mathcal{A}_{\mathbb{G}}} J\,(u) = J(u^*). \tag{5.3.4}$$

A natural guess for the corresponding terminal condition in this case would be a zero limit condition for $p(t)$ as $t \to \infty$, but this turns out to be incorrect. In fact, [Ha] provides a counterexample. Thus some care is needed in the infinite horizon case. See also [MV, HØP].

For information about infinite horizon BSDEs, see [PeS, Pa3, Yi, BBP, Roy].

5.3.1 A Sufficient Maximum Principle

We establish here a sufficient maximum principle for optimal control in infinite horizon.

We first introduce the Hamiltonian and the adjoint equations for our problem. The Hamiltonian is defined in the same way as before by

$$\begin{aligned} H(t, x, u, p, q, r(\cdot)) &= f(t, x, u) + b(t, x, u)p + \sigma(t, x, u)q \\ &\quad + \int_{\mathbb{R}_0} \theta(t, x, u, \zeta)r(\zeta)\nu(d\zeta), \end{aligned} \tag{5.3.5}$$

where

$$H : [0, \infty) \times \mathbb{R} \times \mathcal{U} \times \mathbb{R} \times \mathbb{R} \times \mathcal{R} \times \Omega \to \mathbb{R}$$

and $\mathcal{R}$ is the set of functions $r\colon \mathbb{R}_0 \to \mathbb{R}$ such that the integral term in (5.3.5) converges and $\mathcal{U}$ is the set of possible control values.

We suppose that b, σ and θ are C^1 functions with respect to (x, u) and that

$$E\Big[\int_0^\infty \Big\{ \Big| \frac{\partial b}{\partial x}(t, X(t), u(t)) \Big|^2 + \Big| \frac{\partial \sigma}{\partial x}(t, X(t), u(t)) \Big|^2 + \int_{\mathbb{R}_0} \Big| \frac{\partial \theta}{\partial x}(t, X(t), u(t), \zeta) \Big|^2 \nu(d\zeta) \Big\} dt \Big] < \infty. \tag{5.3.6}$$

The adjoint processes $(p(t), q(t), r(t, \cdot))$, $t \in [0, \infty)$, are assumed to satisfy the equation

$$dp(t) = -\frac{\partial H}{\partial x}(t, X(t), u(t), p(t), q(t), r(t, \cdot))\, dt + q(t) dB(t) + \int_{\mathbb{R}_0} r(t, \zeta) \tilde{N}(dt, d\zeta); t \in [0, \infty). \tag{5.3.7}$$

Remark 5.8 Note that without further conditions there are infinitely many solutions (p, q, r) of this equation. We do not fix a priori a given solution of (5.3.7).

Theorem 5.9 (***Infinite Horizon Sufficient Maximum Principle***) *[HØP] Let $\hat{u} \in \mathcal{A}_{\mathbb{G}}$ with corresponding state processes $\hat{X}(t)$ and adjoint processes $\hat{p}(t)$, $\hat{q}(t)$ and $\hat{r}(t, \cdot)$ assumed to satisfy (5.3.7). Suppose that the following assertions hold:*

(i) (The transversality condition)

$$\overline{\lim_{T \to \infty}} E\Big[\hat{p}(T)(X(T) - \hat{X}(T)) \Big] \geq 0, \tag{5.3.8}$$

for all $u \in \mathcal{A}_{\mathcal{E}}$ with corresponding solution $X(t)$.

(ii) The function

$$(x, u) \to H(t, x, u, \hat{p}(t), \hat{q}(t), \hat{r}(t, \cdot)) \tag{5.3.9}$$

is concave for each $t \in [0, \infty)$ a.s.

(iii)

$$\max_{v \in \mathcal{U}} E\Big[H\Big(t, \hat{X}(t), v, \hat{p}(t), \hat{q}(t), \hat{r}(t, \cdot)\Big) \Big| \mathcal{G}_t \Big] = E\Big[H\Big(t, \hat{X}(t), \hat{u}(t), \hat{p}(t), \hat{q}(t), \hat{r}(t, \cdot)\Big) \Big| \mathcal{G}_t \Big], \tag{5.3.10}$$

for all $t \in [0, \infty)$ a.s.
Then $\hat{u}$ is an optimal control for the problem (5.3.4).

Proof Choose an arbitrary $u \in \mathcal{A}_{\mathbb{G}}$, and consider

$$J(u) - J(\hat{u}) = I_1, \tag{5.3.11}$$

where

$$I_1 = E\left[\int_0^\infty \left\{f\big(t, X(t), u(t)\big) - f\big(t, \hat{X}(t), \hat{u}(t)\big)\right\} dt\right]. \tag{5.3.12}$$

By introducing a sequence of stopping times τ_n as in the proof of Theorem 5.4, we see that we may assume that all the local martingales appearing in the following computations are martingales and therefore have expectations 0.

By the definition (5.3.5) of H and the concavity, we have

$$\begin{aligned} I_1 \le E\Bigg[\int_0^\infty \Bigg\{ & \frac{\partial H}{\partial x}(t)(X(t) - \hat{X}(t)) \\ & + \frac{\partial H}{\partial u}(t)(u(t) - \hat{u}(t)) - (b(t) - \hat{b}(t))\hat{p}(t) - (\sigma(t) - \hat{\sigma}(t))\hat{q}(t) \\ & - \int_{\mathbb{R}_0} (\theta(t,\zeta) - \hat{\theta}(t,\zeta))\hat{r}(t,\zeta)\nu(d\zeta)\Bigg\} dt\Bigg], \end{aligned} \tag{5.3.13}$$

where we have used the simplified notation

$$\frac{\partial H}{\partial x}(t) = \frac{\partial H}{\partial x}\left(t, \hat{X}(t), \hat{u}(t), \hat{p}(t), \hat{q}(t), \hat{r}(t,\cdot)\right),$$

and similarly for b and σ.

Applying the Itô formula to $\hat{p}(t)(X(t) - \hat{X}(t))$ we get, by (5.3.8) and (5.3.9),

$$\begin{aligned} 0 &\le \overline{\lim_{T\to\infty}} E[\ \hat{p}(T)(X(T) - \hat{X}(T))] \\ &= \overline{\lim_{T\to\infty}} E\Bigg[\Bigg(\int_0^T (b(t) - \hat{b}(t))\hat{p}(t)dt + \int_0^T (X(t) - \hat{X}(t))\frac{\partial H}{\partial x}(t)dt \\ &\quad + \int_0^T (\sigma(t) - \hat{\sigma}(t))\hat{q}(t)dt + \int_0^T \int_{\mathbb{R}_0} (\theta(t,\zeta) - \hat{\theta}(t,\zeta))\hat{r}(t,\zeta)\nu(d\zeta)dt\Bigg)\Bigg]. \end{aligned} \tag{5.3.14}$$

Subtracting and adding $\int_0^\infty \frac{\partial H}{\partial u}(t)(u(t) - \hat{u}(t))dt$ we conclude that

$$
\begin{aligned}
0 &\le \overline{\lim_{T\to\infty}} E\Big[\ \hat{p}(T)(X(T)-\hat{X}(T))\Big] \\
&= E\left[\left(\int_0^\infty (b(t)-\hat{b}(t))\hat{p}(t)dt\ -\int_0^\infty \frac{\partial H}{\partial x}(t)\,(X(t)-\hat{X}(t))dt\right.\right. \\
&+\int_0^\infty (\sigma(t)-\hat{\sigma}(t))\hat{q}(t)dt+\int_0^\infty\int_{\mathbb{R}_0}(\theta(t,\zeta)-\hat{\theta}(t,\zeta))\hat{r}(t,\zeta)\nu(d\zeta)dt \\
&\left.\left.-\int_0^\infty \frac{\partial H}{\partial u}(t)(u(t)-\hat{u}(t))dt+\int_0^\infty \frac{\partial H}{\partial u}(t)(u(t)-\hat{u}(t))dt\right)\right] \\
&\le -I_1+E\left[\int_0^\infty E\left[\frac{\partial H}{\partial u}(t)(u(t)-\hat{u}(t))\mid \mathcal{G}_t\right]dt\right].
\end{aligned}
$$

Hence

$$
I_1 \le E\left[\int_0^\infty E\left[\frac{\partial H}{\partial u}(t)\mid \mathcal{G}_t\right](u(t)-\hat{u}(t))dt\right]\le 0
$$

by (5.3.10).

Since $u\in\mathcal{A}_{\mathbb{G}}$ was arbitrary, this proves Theorem 5.9. □

Example 5.10 ([HØP]) Assume we are given the performance functional

$$
J(u)=E\left[\int_0^\infty e^{-\rho t}\frac{1}{\gamma}u^\gamma(t)dt\right], \tag{5.3.15}
$$

and the state equation

$$
\begin{cases} dX(t) &= [X(t)\mu-u(t)]\,dt+\sigma(t,X(t),u(t))dB(t);\ t\ge 0,\\ X(0) &= X_0>0, \end{cases} \tag{5.3.16}
$$

where $X_0>0$, $\gamma\in(0,1)$, $\rho>0$ and $\mu\in\mathbb{R}$ are given constants. We assume that

$$
\mu\gamma<\rho. \tag{5.3.17}
$$

Here $u(t)\ge 0$ is our control. It can be interpreted as the consumption rate from a cash flow $X(t)$. The performance $J(u)$ is the total expected discounted utility of the consumption. For u to be admissible we require that $E\big[X(t)\big]\ge 0$ for all $t\ge 0$.

In this case the Hamiltonian (5.3.5) takes the form

$$
H(t,x,u,p,q)=e^{-\rho t}\frac{1}{\gamma}u^\gamma+[x\mu-u]p+\sigma(t,x,u)q, \tag{5.3.18}
$$

so that we get the partial derivative

$$\frac{\partial H}{\partial u}(t,x,u,p,q) = e^{-\rho t}u^{\gamma-1} - p + \frac{\partial \sigma}{\partial u}(t,x,u)q.$$

Therefore, if $\frac{\partial H}{\partial u} = 0$ we get

$$p(t) = e^{-\rho t}u^{\gamma-1}(t) + \frac{\partial \sigma}{\partial u}(t,X(t),u(t))q(t). \tag{5.3.19}$$

We now see that the adjoint equation is given by:

$$dp(t) = -\Big[\mu p(t) + \frac{\partial \sigma}{\partial x}(t,X(t),u(t))q(t)\Big]dt + q(t)dB(t),$$

Now assume that

$$\sigma(t,x,u) = \sigma_0(t)x \tag{5.3.20}$$

for some bounded adapted process $\sigma_0(t)$. Let us try to choose $q = 0$. Then

$$dp(t) = -\mu p(t)dt,$$

which gives

$$p(t) = p(0)e^{-\mu t},$$

for some constant $p(0)$. Hence, by (5.3.19)

$$\hat{u}(t) = p(0)^{\frac{1}{\gamma-1}}e^{\frac{(\mu-\rho)t}{1-\gamma}}, \tag{5.3.21}$$

for all $t > 0$. Inserting $\hat{u}(t)$ into the dynamics of $\hat{X}(t)$, we get that

$$d\hat{X}(t) = \Big[\mu\hat{X}(t) - p^{\frac{1}{\gamma-1}}(0)e^{\frac{1}{\gamma-1}(\rho t-\mu t)}\Big]dt + \sigma_0(t)\hat{X}(t)dB(t).$$

So

$$\hat{X}(t) = \left[\hat{X}(0)\Gamma(t) - p^{\frac{1}{\gamma-1}}(0)\int_0^t \frac{\Gamma(t)}{\Gamma(s)}\exp\left(\frac{(\mu-\rho)s}{1-\gamma}\right)ds\right], \tag{5.3.22}$$

where

$$\Gamma(t) = \exp\left(\int_0^t \sigma_0(s)dB(s) + \mu t - \frac{1}{2}\int_0^t \sigma_0^2(s)ds\right). \tag{5.3.23}$$

Hence

$$E\left[\hat{X}(t)\right] = e^{\mu t}\left[\hat{X}(0) - p(0)^{\frac{1}{\gamma-1}}\int_0^t \exp\left(\frac{(\mu\gamma-\rho)s}{1-\gamma}\right)ds\right].$$

Therefore, to ensure that $E\left[\hat{X}(t)\right]$ is non-negative for all t, the maximal value we can have for $p(0)$ is

$$\hat{p}(0) = \left[\frac{\hat{X}(0)}{\int_0^\infty \exp\left(\frac{(\mu\gamma-\rho)s}{1-\gamma}\right)ds}\right]^{\gamma-1}. \tag{5.3.24}$$

With this choice of $\hat{p}(0)$, we see that $\overline{\lim}_{T\to\infty} E[\hat{p}(T)\hat{X}(T)] = 0$, so that we have

$$\overline{\lim}_{T\to\infty} E[\hat{p}(T)(X(T) - \hat{X}(T))] \geq 0.$$

This tells us that with $p(0) = \hat{p}(0)$ given by (5.3.24) the control $\hat{u}$ given by (5.3.21) is indeed an optimal control.

5.3.2 A Necessary Maximum Principle

In addition to the assumptions in the beginning of Sect. 5.3, we now assume the following.

(A_1) For all $u \in \mathcal{A}_{\mathbb{G}}$ and all $\beta \in \mathcal{A}_{\mathbb{G}}$ bounded, there exists an $\epsilon > 0$ such that

$$u + s\beta \in \mathcal{A}_{\mathbb{G}} \quad \text{for all } s \in (-\epsilon, \epsilon). \tag{5.3.25}$$

(A_2) For all t_0, h such that $0 \leq t_0 < t_0 + h \leq T$ and all bounded $\mathcal{G}_{t_0}$-measurable random variables α, the control process $\beta(t)$ defined by

$$\beta(t) = \alpha \mathbf{1}_{[t_0, t_0+h]}(t) \tag{5.3.26}$$

belongs to $\mathcal{A}_{\mathbb{G}}$.

(A_3) The derivative process

$$\xi(t) := \frac{d}{ds} X^{u+s\beta}(t) \mid_{s=0} \tag{5.3.27}$$

exists and belongs to $L^2(m \times P)$, where m denotes the Lebesgue measure on $\mathbb{R}$.

It follows from (A_3) that

$$d\xi(t) = \left\{\frac{\partial b}{\partial x}(t)\xi(t) + \frac{\partial b}{\partial u}(t)\beta(t)\right\}dt + \left\{\frac{\partial \sigma}{\partial x}(t)\xi(t) + \frac{\partial \sigma}{\partial u}(t)\beta(t)\right\}dB(t)$$
$$+ \int_{\mathbb{R}_0}\left\{\frac{\partial \theta}{\partial x}(t,\zeta)\xi(t) + \frac{\partial \theta}{\partial u}(t,\zeta)\beta(t)\right\}\tilde{N}(dt,d\zeta), \tag{5.3.28}$$

where, for simplicity of notation, we define

$$\frac{\partial b}{\partial x}(t) := \frac{\partial b}{\partial x}(t, X(t), u(t)).$$

Theorem 5.11 ([HØP] [***Infinite Horizon Necessary Maximum Principle***]) *Suppose that $\hat{u} \in \mathcal{A}_{\mathbb{G}}$ with corresponding solutions $\hat{X}(t)$ of (5.3.1) and $\hat{p}(t)$, $\hat{q}(t)$, and $\hat{r}(t,\cdot)$ of (5.3.7), and corresponding derivative process $\hat{\xi}(t)$ given by (5.3.28). Assume that*

$$\lim_{T\to\infty} E\left[\hat{p}(T)\hat{\xi}(T)\right] = 0. \tag{5.3.29}$$

Then the following assertions are equivalent:
(i) For all bounded $\beta \in \mathcal{A}_{\mathcal{E}}$,

$$\frac{d}{ds}J(\hat{u}+s\beta)\,|_{s=0} = 0.$$

(ii) For all $t \in [0,\infty)$,

$$E\left[\frac{\partial H}{\partial u}\left(t,\ \hat{X}(t), u, \hat{p}(t), \hat{q}(t), \hat{r}(t,\cdot)\right)\Big|\mathcal{G}_t\right]_{u=\hat{u}(t)} = 0 \ a.s.$$

Proof By introducing a sequence of stopping times τ_n as in the proof of Theorem 5.5, we see that we may assume that all the local martingales appearing in the following computations are martingales, and hence have expectation 0. Suppose that assertion (i) holds. Then

$$0 = \frac{d}{ds}J(\hat{u}+s\beta)\,|_{s=0} = \frac{d}{ds}E\Big[\int_0^\infty f(t, X^{\hat{u}+s\beta}(t), \hat{u}(t)+s\beta(t))dt\Big]\Big|_{s=0}$$
$$= E\left[\int_0^\infty \left\{\frac{\partial f}{\partial x}(t)\xi(t) + \frac{\partial f}{\partial u}(t)\beta(t)\right\}dt\right].$$

We know by the definition of H that

$$\frac{\partial f}{\partial x}(t) = \frac{\partial H}{\partial x}(t) - \frac{\partial b}{\partial x}(t)p(t) - \frac{\partial \sigma}{\partial x}(t)q(t) - \int_{\mathbb{R}_0} \frac{\partial \theta}{\partial x}(t,\zeta)r(t,\zeta)\nu(d\zeta)$$

and the same for $\frac{\partial f}{\partial u}(t)$.

Applying the Itô formula to $\hat{p}(t)\hat{\xi}(t)$, we obtain by (5.3.29) and (5.3.28)

$$\begin{aligned}
0 = \lim_{T\to\infty} E\left[\hat{p}(T)\xi(T)\right] &= E\Big[\int_0^\infty \hat{p}(t)\ \left\{\frac{\partial b}{\partial x}(t)\xi(t) + \frac{\partial b}{\partial u}(t)\beta(t)\right\} dt \\
&+ \int_0^\infty \xi(t)\left(-\frac{\partial H}{\partial x}(t)\right) dt + \int_0^\infty q(t)\left\{\frac{\partial \sigma}{\partial x}(t)\xi(t) + \frac{\partial \sigma}{\partial u}(t)\beta(t)\right\} dt \\
&+ \int_0^\infty \int_{\mathbb{R}_0} r(t,\zeta)\left\{\frac{\partial \theta}{\partial x}(t,\zeta)\xi(t) + \frac{\partial \theta}{\partial u}(t,\zeta)\beta(t)\right\}\nu(d\zeta)dt\Big] \\
&= -\frac{d}{ds}J(\hat{u}+s\beta)\mid_{s=0} + E\left[\int_0^\infty \frac{\partial H}{\partial u}(t)\beta(t)dt\right].
\end{aligned}$$

Therefore

$$E\left[\int_0^\infty \frac{\partial H}{\partial u}(t)\beta(t)dt\right] = \frac{d}{ds}J(\hat{u}+s\beta)\Big|_{s=0}. \tag{5.3.30}$$

Now apply this to

$$\beta(t) = \alpha 1_{[t_0,t_0+h]}(t)\ ,$$

where α is bounded and $\mathcal{G}_{t_0}$-measurable, $0 \le t_0 < t_o + h \le T$. Then if (i) holds we get

$$E\left[\int_{t_0}^{t_0+h} \frac{\partial H}{\partial u}(t)dt\ \alpha\right] = 0.$$

Differentiating with respect to h at 0, we have

$$E\left[\frac{\partial H}{\partial u}(t_0)\ \alpha\right] = 0.$$

This holds for all $\mathcal{G}_{t_0}$-measurable α and hence we obtain that

$$E\left[\frac{\partial H}{\partial u}(t_0)\mid \mathcal{G}_{t_0}\right] = 0.$$

This proves that assertion (i) implies (ii).

To complete the proof, we need to prove the converse implication. We obtain this by using that every bounded $\beta \in \mathcal{A}_{\mathbb{G}}$ can be approximated by linear combinations of controls β of the form (5.3.26). We omit the details. □

5.4 Optimal Control of FBSDEs by Means of Stochastic HJB Equations

In this section we study optimal stochastic control problems of general coupled systems of forward-backward stochastic differential equations with jumps.

Recall that in the classical theory of stochastic control of systems described by stochastic differential equations (SDEs) there are two main solution methods:

(i) Dynamic programming, which leads to a Hamilton–Jacobi–Bellman equation in the (unknown) value function.
(ii) The maximum principle, which involves the optimization of the Hamiltonian and an associated backward stochastic differential equation (BSDE) in the (unknown) adjoint processes, to obtain optimality conditions for the control.

Dynamic programming is a very efficient solution method for Markovian systems. The maximum principle, on the other hand, can also be applied in a non-Markovian setting, but leads to a coupled system of forward-backward SDEs (FBSDEs) with constraints, which is difficult to solve in general.

In view of this it is natural to ask if there is an extension of the HJB approach to non-Markovian systems. The answer is known to be yes, at least in some cases; we refer in particular to the papers [Pe, BM] and the references therein, where stochastic HJB equations associated to optimal control of non-Markovian SDEs driven by Brownian motions are studied. We also mention the paper [ETZ] where path-dependent stochastic control problems and associated path-dependent HJB equations are studied.

The purpose of the present section is to show that the answer is also yes in the general context of optimal control of non-Markovian systems of *forward-backward* stochastic differential equations (FBSDEs) driven by Itô–Lévy processes. By means of the Itô–Ventzell formula, we obtain a *stochastic* HJB equation for the value function of the control problem, which is a backward stochastic partial differential equation (BSPDE). In the case of Markovian optimal control of jump diffusions, this equation reduces to the classical HJB equation.

The relation between FBSDEs and BSPDEs has been known for several years. See in particular [MYZ] and the references therein, and [MPY] for the Markovian case, which leads to backward deterministic PDEs.

5.4.1 Optimal Control of FBSDEs

Consider the following controlled coupled FBSDE: The forward equation in $X(t)$ has the form

$$\begin{cases} dX(t) &= \alpha(t, X(t), Y(t), Z(t), K(t,\cdot), u(t, X(t)))dt \\ &\quad +\beta(t, X(t), Y(t), Z(t), K(t,\cdot), u(t, X(t)))dB(t) \\ &\quad + \int_{\mathbb{R}} \gamma(t, X(t), Y(t), Z(t), K(t,\cdot), u(t, X(t)), \zeta)\tilde{N}(dt, d\zeta)\ ;\ t \in [0, T] \\ X(0) &= x \in \mathbb{R}. \end{cases} \tag{5.4.1}$$

The backward equation in $Y(t)$, $Z(t)$, $K(t, \zeta)$ has the form

$$\begin{cases} dY(t) &= -g(t, X(t), Y(t), Z(t), K(t,\cdot), u(t, X(t)))dt + Z(t)dB(t) \\ &\quad + \int_{\mathbb{R}} K(t, \zeta)\tilde{N}(dt, d\zeta)\ ;\ t \in [0, T]; \\ Y(T) &= h(X(T)). \end{cases} \tag{5.4.2}$$

The given functions

$$\begin{aligned} \alpha(t, x, y, z, k, u, \omega) &: [0, T] \times \mathbb{R}^3 \times \mathcal{R} \times W \times \Omega \to \mathbb{R} \\ \beta(t, x, y, z, k, u, \omega) &: [0, T] \times \mathbb{R}^3 \times \mathcal{R} \times W \times \Omega \to \mathbb{R} \\ \gamma(t, x, y, z, k, u, \zeta, \omega) &: [0, T] \times \mathbb{R}^3 \times \mathcal{R} \times W \times \mathbb{R}_0 \times \Omega \to \mathbb{R} \\ g(t, x, y, z, k, u, \omega) &: [0, T] \times \mathbb{R}^3 \times \mathcal{R} \times W \times \Omega \to \mathbb{R} \end{aligned}$$

are assumed to be $\mathbb{F}$-predictable for each x, y, z, k, u.

Here, W is a given set of admissible control values $u(t, x, \omega)$, where $u(t) = u(t, X(t), \omega)$ is our control process. The function $h(x, \omega) : \mathbb{R} \times \Omega \to \mathbb{R}$ is assumed to be $\mathcal{F}_T$-measurable for each x and C^1 with respect to x for each ω.

We let $\mathcal{A}$ denote a given set of controls, contained in the set of predictable control processes $u(t)$ such that the FBSDE system (5.4.1)–(5.4.2) has a unique solution $X = X^u \in S^2$, $(Y = Y^u, Z = Z^u, K = K^u) \in S^2 \times H^2 \times H^2_\nu$ where S^2 is the set of real-valued càdlàg adapted processes ϕ with $E[\sup_{0\le t\le T} |\phi(t)|^2] < \infty$; H^2 is the set of real-valued predictable processes ϕ such that $E\left[\int_0^T \phi(t)^2 dt\right] < \infty$, and H^2_ν is the set of predictable processes ℓ such that $E\left[\int_0^T \int_{\mathbb{R}} |\ell(t, \zeta)|^2 \nu(d\zeta)\, dt\right] < \infty$.

We refer to [HP] for existence and uniqueness results on FBSDEs.

If $u \in \mathcal{A}$, we call u *admissible*. We want to find $\hat{u} \in \mathcal{A}$ such that

$$\sup_{u\in\mathcal{A}} Y^u(0) = Y^{\hat{u}}(0). \tag{5.4.3}$$

First we try to write the solution $Y(t)$ above in the form

$$Y(t) = y(t, X(t)) \tag{5.4.4}$$

for some random field $y(t,x) = y(t,x,\omega)$ which, together with $z(t,x)$ and $k(t,x,\zeta)$, satisfies a BSPDE of the form

$$\begin{cases} dy(t,x) &= A_u(y(\cdot), z(\cdot), k(\cdot))(t,x)dt + z(t,x)dB(t) \\ &\quad + \int_{\mathbb{R}} k(t,x,\zeta)\tilde{N}(dt,d\zeta)\ ;\ t \in [0,T] \\ y(T,x) &= h(x), \end{cases} \tag{5.4.5}$$

for some partial integro-differential operator A_u acting on x.

Remark 5.12 We interpret the Eq. (5.4.5) in the weak (variational) sense, which means that $y \in L^2([0,T]\times\mathbb{R}\times\Omega)$ and $y(\cdot,\cdot,\omega) \in C\big([0,T];\, L^2(\mathbb{R})\big) \cap L^2\big([0,T];\, V\big)$ for all ω, and for all $\phi \in C_0^\infty(\mathbb{R})\ ;\ t \geq 0$,

$$\begin{aligned} &< y(t,\cdot), \phi > = < y_0(\cdot), \phi > + \int_0^t < A_u y(s,\cdot), \phi > ds \\ &+ \int_0^t < z(s,\cdot), \phi > dB(s) + \int_0^t \int_{\mathbb{R}} < k(s,\cdot,\zeta), \phi > \tilde{N}(ds,d\zeta), \end{aligned} \tag{5.4.6}$$

for some $y_0(\cdot) \in V$, where $<,>$ denotes the dual pairing between the space V and its dual V^*, and $V := W_0^{1,2}(\mathbb{R})$ is the Sobolev space of order one with zero boundary condition at infinity. Note that with this framework the Itô calculus can be applied to (5.4.5). See [P, PR].

By the Itô–Ventzell formula (see [ØZ] and the references therein),

$$\begin{aligned} dY(t) &= A_u(y(\cdot), z(\cdot), k(\cdot))(t, X(t))dt + z(t, X(t))dB(t) \\ &\quad + \int_{\mathbb{R}} k(t, X(t), \zeta)\tilde{N}(dt, d\zeta) + y'(t, X(t))[\alpha(t)dt + \beta(t)dB(t)] \\ &\quad + \frac{1}{2}y''(t, X(t))\beta^2(t)dt \\ &\quad + \int_{\mathbb{R}} \{y(t, X(t) + \gamma(t,\zeta)) - y(t, X(t)) - y'(t, X(t))\gamma(t,\zeta)\}\nu(d\zeta)dt \\ &\quad + \int_{\mathbb{R}} \{y(t, X(t) + \gamma(t,\zeta)) - y(t, X(t))\}\tilde{N}(dt, d\zeta) + z'(t, X(t))\beta(t)dt \\ &\quad + \int_{\mathbb{R}} \{k(t, X(t) + \gamma(t,\zeta), \zeta) - k(t, X(t), \zeta)\}\nu(d\zeta)dt \\ &\quad + \int_{\mathbb{R}} k(t, X(t^-) + \gamma(t,\zeta), \zeta)\tilde{N}(dt, d\zeta), \end{aligned} \tag{5.4.7}$$

where $y'(t,x) = \dfrac{\partial y}{\partial x}(t,x)$ etc. and using the shorthand notation $\alpha(t) = \alpha(t, X(t), Y(t), Z(t), K(t,\cdot), u(t))$ etc. Rearranging the terms we see that

$$
\begin{aligned}
dY(t) &= [A_u(y(\cdot), z(\cdot), k(\cdot))(t, X(t)) + y'(t, X(t))\alpha(t) + \frac{1}{2}y''(t, X(t))\beta^2(t) \\
&+ \int_{\mathbb{R}} \{y(t, X(t) + \gamma(t, \zeta)) - y(t, X(t)) - y'(t, X(t))\gamma(t, \zeta)\}\nu(d\zeta) \\
&+ z'(t, X(t))\beta(t) + \int_{\mathbb{R}} \{k(t, X(t) + \gamma(t, \zeta), \zeta) - k(t, X(t), \zeta)\}\nu(d\zeta)]dt \\
&+ [z(t, X(t)) + y'(t, X(t))\beta(t)]dB(t) \\
&+ \int_{\mathbb{R}} \{y(t, X(t) + \gamma(t, \zeta)) - y(t, X(t)) + k(t, X(t) + \gamma(t, \zeta), \zeta)\}\tilde{N}(dt, d\zeta).
\end{aligned} \tag{5.4.8}
$$

Comparing (5.4.8) with the original Eq. (5.4.2) for Y we deduce the following result

Theorem 5.13 *Suppose that* $(y(t, x), z(t, x), k(t, x, \cdot))$ *satisfies the BSPDE*

$$
dy(t, x) = -A_u(t, x)dt + z(t, x)dB(t) + \int_{\mathbb{R}} k(t, x, \zeta)\tilde{N}(dt, d\zeta); \quad y(T, x) = h(x), \tag{5.4.9}
$$

where

$$
\begin{aligned}
A_u(t, x) &= A_u(y(\cdot), z(\cdot), k(\cdot))(t, x) \\
&:= g(t, x, y(t, x), z(t, x) + y'(t, x)\beta(t), \\
&\qquad y(t, x + \gamma(t, \cdot)) - y(t, x) + k(t, x + \gamma(t, \cdot), \cdot), u(t, x)) \\
&+ y'(t, x)\alpha(t) + \frac{1}{2}y''(t, x)\beta^2(t) + z'(t, x)\beta(t) \\
&+ \int_{\mathbb{R}} \{y(t, x + \gamma(t, \zeta)) - y(t, x) - y'(t, x)\gamma(t, \zeta)\}\nu(d\zeta) \\
&\quad + \int_{\mathbb{R}} \{k(t, x + \gamma(t, \zeta), \zeta) - k(t, x, \zeta)\}\nu(d\zeta).
\end{aligned} \tag{5.4.10}
$$

Then $(Y(t), Z(t), K(t, \zeta))$*, given by*

$$
Y(t) := y(t, X(t)), \tag{5.4.11}
$$

$$
Z(t) := z(t, X(t)) + y'(t, X(t))\beta(t), \tag{5.4.12}
$$

$$
K(t, \zeta) := y(t, X(t) + \gamma(t, \zeta)) - y(t, X(t)) + k(t, X(t) + \gamma(t, \zeta), \zeta), \tag{5.4.13}
$$

is a solution of the FBSDE system (5.4.1)–(5.4.2)*.*

Definition 5.14 We say that the BSPDE (5.4.9) satisfies the comparison principle with respect to u if for all $u_1, u_2 \in \mathcal{A}$ and all $\mathcal{F}_T$-measurable random variables h_1, h_2 with corresponding solutions (y_i, z_i, k_i), $i = 1, 2$, of (5.4.9) such that

$$A_{u_1}(t,x) \le A_{u_2}(t,x) \text{ for all } t, x \in [0,T] \times \mathbb{R}$$

$$\text{and} \quad h_1(x) \le h_2(x) \text{ for all } x \in \mathbb{R},$$

we have

$$y_1(t,x) \le y_2(t,x) \text{ for all } t, x \in [0,T] \times \mathbb{R}.$$

Sufficient conditions for the validity of comparison principles for BSPDEs with jumps is still an open question in this setting. For related results see [ØSZ2]. However, in the Brownian case, sufficient conditions for the validity of comparison principles for BSPDEs of the type (5.4.10) are given in Theorem 7.1 in [MYZ]. Using this result we get

Theorem 5.15 *Assume that the following hold:*

- $N = K = 0$, *i.e. there are no jumps*
- *The coefficients* α, β, *and* g *are* $\mathbb{F}$*-progressively measurable for each fixed* (x, y, z) *and* h *is* $\mathcal{F}_T$*-measurable for each fixed* x
- α, β, g, h *are uniformly Lipschitz-continuous in* (x, y, z)
- α *and* β *are bounded and* $E[\int_0^T g^2(t,0,0,0)dt + h^2(0)] < \infty$
- α *and* β *do not depend on* z.

Then the comparison principle holds for the BSPDE (5.4.9).

From the above we deduce the following result, which may be regarded as a *stochastic HJB equation* for optimal control of possibly non-Markovian FBSDEs.

Theorem 5.16 (***Stochastic HJB equation***) *Suppose the comparison principle holds for the BSPDE* (5.4.9). *Moreover, suppose that for all* t, x, ω *there exists a maximizer* $u = \hat{u}(t,x) = \hat{u}(y, y', y'', z, z', k)(t,x,\omega)$ *of the function* $u \to A_u(t,x)$. *Suppose the system* (5.4.9) *with* $u = \hat{u}$ *has a unique solution* $(\hat{y}(t,x), \hat{z}(t,x), \hat{k}(t,x,\cdot))$ *and that* $\hat{u}(t, X(t)) \in \mathcal{A}$. *Then* $\hat{u}(t, X(t))$ *is an optimal control for the problem* (5.4.3), *with optimal value*

$$\sup_{u \in \mathcal{A}} Y^u(0) = Y^{\hat{u}}(0) = \hat{y}(0,x). \tag{5.4.14}$$

Note that in this general non-Markovian setting the classical *value function* from the dynamic programming is replaced by the solution $\hat{y}(t,x)$ of the BSPDE (5.4.9) for $u = \hat{u}$.

5.4.2 Applications in Mathematical Finance

We now illustrate Theorem 5.16 by looking at some examples.

Example 5.17 (***Utility Maximization***) First we consider the Merton problem, the solution of which is well known in the Markovian case with deterministic coefficients. Here we consider the general non-Markovian case, when the coefficients are stochastic processes. Consider a financial market consisting of a risk free investment, with unit price $S_0(t) := 1\ ;\ t \in [0, T]$ and a risky investment, with unit price

$$dS_1(t) = S_1(t)[b(t)dt + \sigma(t)dB(t)]\ ;\ t \in [0, T]. \tag{5.4.15}$$

Here $b(t) = b(t, \omega)$ and $\sigma(t) = \sigma(t, \omega) > 0$ are given adapted processes. Let $u(t, X(t))$ be a *portfolio*, representing the *amount* invested in the risky asset at time t. If u is self-financing, then the corresponding wealth $X(t)$ at time t is given by the stochastic differential equation

$$dX(t) = dX_x^u(t) = u(t, X(t))[b(t)dt + \sigma(t)dB(t)]\ , t \in [0, T]\ ;\ X(0) = x > 0. \tag{5.4.16}$$

Let $(Y(t), Z(t)) = (Y_x^u(t), Z_x^u(t))$ be the solution of the BSDE

$$dY(t) = Z(t)dB(t)\ ,\ t \in [0, T]\ ;\ Y(T) = U(X(T)), \tag{5.4.17}$$

where $U(X) = U(X, \omega)$ is a given utility function, possibly random. Then

$$Y_x^u(0) = E[U(X_x^u(T))].$$

Therefore, the classical portfolio optimization problem of Merton is to find $\hat{u} \in \mathcal{A}$ such that

$$\sup_{u \in \mathcal{A}} Y_x^u(0) = Y_x^{\hat{u}}(0). \tag{5.4.18}$$

In the following we assume that

$$\sup_{u \in \mathcal{A}} Y_x^u(0) < \infty. \tag{5.4.19}$$

In this general non-Markovian setting with stochastic coefficients $b(t) = b(t, \omega)$ and $\sigma(t) = \sigma(t, \omega) > 0$, an explicit expression for the optimal portfolio $\hat{u}$ is not known. Let us apply the theory from the previous sections to study this problem. In this case we get, from (5.4.10),

$$A_u(t, x) = y'(t, x)ub(t) + \frac{1}{2}y''(t, x)u^2\sigma^2(t, x) + z'(t, x)u\sigma(t), \tag{5.4.20}$$

which is maximal when

$$u = \hat{u}(t,x) = -\frac{y'(t,x)b(t) + z'(t,x)\sigma(t)}{y''(t,x)\sigma^2(t)}. \tag{5.4.21}$$

Substituting this into $A_{\hat{u}}(t,x)$ we obtain

$$A_{\hat{u}}(t,x) = -\frac{(y'(t,x)b(t) + z'(t,x)\sigma(t))^2}{2y''(t,x)\sigma^2(t)}. \tag{5.4.22}$$

Hence the BSPDE for $y(t,x), z(t,x)$ gets the form

$$\begin{cases} dy(t,x) &= \dfrac{(y'(t,x)b(t) + z'(t,x)\sigma(t))^2}{2y''(t,x)\sigma^2(t)}dt + z(t,x)dB(t)\ ;\ t \in [0,T] \\ y(T,x) &= U(x). \end{cases} \tag{5.4.23}$$

We have proved:

Proposition 5.18 *Suppose there exists a solution $(y(t,x), z(t,x))$ of the BSPDE (5.4.23) with $y''(t,x) < 0$. Suppose that $\hat{u}$ defined in (5.4.21) is admissible. Then $\hat{u}$ is optimal for problem (5.4.18) and*

$$y(0,x) = \sup_{u \in \mathcal{A}} Y_x^u(0) = Y_x^{\hat{u}}(0). \tag{5.4.24}$$

Note that if b, σ and U are *deterministic*, we can choose $z(t,x) = 0$ in (5.4.23) and this leads to the following (deterministic) PDE for $y(t,x)$:

$$\frac{\partial y}{\partial t}(t,x) - \frac{y'(t,x)^2 b^2(t)}{2y''(t,x)\sigma^2(t)} = 0\ ;\ t \in [0,T]\ ;\ y(T,x) = U(x). \tag{5.4.25}$$

This is the classical Merton PDE for the value function, usually obtained by dynamic programming and the HJB equation. Hence we may regard (5.4.21)–(5.4.23) as a generalization of the Merton equation (5.4.25) to the non-Markovian case with stochastic $b(t)$, $\sigma(t)$ and $U(x)$. The Markovian case corresponds to the special case when $z(t,x) = 0$ in the BSDE (5.4.23). Therefore $\hat{y}(s,x)$ is a stochastic generalization of the value function

$$\varphi(s,x) := \sup_{u \in \mathcal{A}} E[U(X_{s,x}^u(T))], \tag{5.4.26}$$

where

$$dX_{s,x}^u(t) = u(t)[b(t)dt + \sigma(t)dB(t)]\ ;\ t \geq s\ ;\ X_{s,x}^u(s) = x. \tag{5.4.27}$$

Let us compare with the use of the classical HJB:

$$\begin{cases} \dfrac{\partial \varphi}{\partial s}(s,x) + \max\limits_{v} \left\{ \dfrac{1}{2} v^2 \sigma_0^2(s) \varphi''(s,x) + v b_0(s) \varphi'(s,x) \right\} & = 0 \ ; \ s < T \\ \varphi(T,x) & = U(x). \end{cases} \tag{5.4.28}$$

The maximum is attained at

$$v = \hat{u}(s,x) = -\frac{b_0(s)\varphi'(s,x)}{\varphi''(s,x)\sigma_0^2(s)}. \tag{5.4.29}$$

Substituted into (5.4.28) this gives the HJB equation

$$\frac{\partial \varphi}{\partial s}(s,x) - \frac{\varphi'(s,x)^2 b_0^2(s)}{\varphi''(s,x)\sigma_0^2(s)} = 0, \tag{5.4.30}$$

which is identical to (5.4.25).

Example 5.19 (***Risk Minimizing Portfolios***) Now suppose $X(t) = X_x^u(t)$ is as in (5.4.16), while $(Y(t), Z(t)) = (Y_x^u(t), Z_x^u(t))$ is given by the BSDE

$$dY(t) = -(-\frac{1}{2}Z^2(t))dt + Z(t)dB(t) \ ; \ Y(T) = X(T). \tag{5.4.31}$$

Note that the driver $g(z) := -\frac{1}{2}z^2$ is concave. We want to minimize the *risk* of the terminal financial standing $X(T)$, denoted by $\rho(X(T))$. If we interpret the risk in the sense of the *convex risk measure* defined in terms of the BSDE (5.4.31) we have

$$\rho(X(T)) = -Y(0).$$

The risk minimization problem is to find $\hat{u} \in \mathcal{A}$ such that

$$\inf_{u \in \mathcal{A}} -Y_x^u(0) = -Y_x^{\hat{u}}(0), \tag{5.4.32}$$

where $Y_x^u(t)$ is given by (5.4.31). By changing sign we can consider the supremum problem instead. In this case we get

$$\begin{aligned} A_u(t,x) = &-\frac{1}{2}(z(t,x) + y'(t,x)u\sigma(t))^2 + y'(t,x)ub(t) \\ &+ \frac{1}{2}y''(t,x)u^2\sigma^2(t) + z'(t,x)u\sigma(t), \end{aligned} \tag{5.4.33}$$

which is maximal when $u = \hat{u}(t,x)$ satisfies

$$\hat{u}(t,x) = -\frac{z(t,x)y'(t,x)\sigma(t) - y'(t,x)b(t) - z'(t,x)\sigma(t)}{((y'(t,x))^2 - y''(t,x))\sigma^2(t)}. \tag{5.4.34}$$

This gives

$$A_{\hat{u}}(t,x) = -\frac{1}{2}\hat{z}^2(t,x) + \frac{(\hat{z}(t,x)\hat{y}'(t,x)\sigma(t) - \hat{y}'(t,x)b(t) - \hat{z}'(t,x)\sigma(t))^2}{2((\hat{y}'(t,x))^2 - \hat{y}''(t,x))\sigma^2(t)}, \tag{5.4.35}$$

and hence $(\hat{y}(t,x), \hat{z}(t,x))$ solves the BSPDE

$$d\hat{y}(t,x) = -A_{\hat{u}}(t,x)dt + \hat{z}(t,x)dB(t),\ 0 \le t \le T\ ;\ \hat{y}(T,x) = x. \tag{5.4.36}$$

We have proved:

Proposition 5.20 *Suppose there exists a solution $(\hat{y}(t,x), \hat{z}(t,x))$ of the BSPDE (5.4.36). Suppose $\hat{u}$ defined by (5.4.34) belongs to $\mathcal{A}$. Then $\hat{u}$ is optimal for the risk minimizing problem (5.4.32), and the minimal risk is*

$$\inf_{u\in\mathcal{A}} -Y_x^u(0) = -Y_x^{\hat{u}}(0) = -\hat{y}(0,x). \tag{5.4.37}$$

Next we look at the special case when $b(t)$ and $\sigma(t)$ are *deterministic*. Let us try to choose $\hat{z}(t,x) = 0$ in (5.4.36). Then this BSPDE reduces to the PDE

$$\begin{cases} \dfrac{\partial\hat{y}(t,x)}{\partial t} = -\dfrac{(\hat{y}'(t,x)b(t))^2}{2((\hat{y}'(t,x))^2 - \hat{y}''(t,x))\sigma^2(t)}\ ;\ 0 \le t \le T \\ \hat{y}(T,x) = x. \end{cases} \tag{5.4.38}$$

We try a solution of the form

$$\hat{y}(t,x) = x + a(t), \tag{5.4.39}$$

where $a(t)$ is deterministic. Substituted into (5.4.38) this gives

$$a'(t) = -\frac{1}{2}\left(\frac{b(t)}{\sigma(t)}\right)^2,\ 0 \le t \le T\ ;\ a(T) = 0, \tag{5.4.40}$$

which gives

$$a(t) = \int_t^T \frac{1}{2}\left(\frac{b(s)}{\sigma(s)}\right)^2 ds\ ;\ 0 \le t \le T.$$

With this choice of $a(t)$, (5.4.38) is satisfied and the minimal risk is

$$\rho_{\min}(X(T)) = -Y^{(\hat{u})}(0) = -\hat{y}(0,x) = -x - \int_0^T \frac{1}{2}\left(\frac{b(s)}{\sigma(s)}\right)^2 ds. \tag{5.4.41}$$

Hence by (5.4.34) the optimal (risk minimizing) portfolio is

$$\hat{u}(t, X(t)) = \frac{b(t)}{\sigma^2(t)}. \tag{5.4.42}$$

Remark 5.21 Note that (5.4.41) can be interpreted by means of *entropy* as follows: Recall that in general the *entropy* of a measure Q with respect to the measure P is defined by

$$H(Q \mid P) := E\left[\frac{dQ}{dP} \ln \frac{dQ}{dP}\right].$$

Define

$$\Gamma(t) = \exp\left(-\int_0^t \frac{b(s)}{\sigma(s)} dB(s) - \frac{1}{2}\int_0^t \left(\frac{b(s)}{\sigma(s)}\right)^2 ds\right). \tag{5.4.43}$$

By the Itô formula we have

$$\begin{aligned} d(\Gamma(t)\ln\Gamma(t)) &= \Gamma(t)\left[-\frac{b(t)}{\sigma(t)} dB(t) - \frac{1}{2}\left(\frac{b(t)}{\sigma(t)}\right)^2 dt\right] \\ &\quad + (\ln\Gamma(t))\Gamma(t)\left(-\frac{b(t)}{\sigma(t)} dB(t)\right) + \Gamma(t)\left(-\frac{b(t)}{\sigma(t)}\right)\left(-\frac{b(t)}{\sigma(t)}\right) dt. \end{aligned}$$

Hence, if we define the measure $Q_\Gamma(\omega)$ by

$$dQ_\Gamma(\omega) := \Gamma(T) dP(\omega) \tag{5.4.44}$$

we get

$$\begin{aligned} E\left[\frac{dQ_\Gamma}{dP} \ln \frac{dQ_\Gamma}{dP}\right] &= E[\Gamma(T)\ln\Gamma(T)] \\ &= E\left[\int_0^T \Gamma(t)\frac{1}{2}\left(\frac{b(t)}{\sigma(t)}\right)^2 dt\right] = \frac{1}{2}\int_0^T \left(\frac{b(t)}{\sigma(t)}\right)^2 dt, \end{aligned}$$

which proves that (5.4.41) can be written

$$\rho_{\min}(X(T)) = -x - H(Q_\Gamma \mid P). \tag{5.4.45}$$

Note that Q_Γ is the *unique equivalent martingale measure* for the market (5.4.15), when deterministic coefficients are considered.

Thus we have proved that if the coefficients $b(t)$ and $\sigma(t)$ in (5.4.15) are deterministic and if the portfolio $\hat{u}(t, X(t)) := \frac{b(t)}{\sigma^2(t)}$ is admissible, then $\hat{u}$ is a risk minimizing portfolio for the problem (5.4.32) and the minimal risk is equal to minus the initial wealth x minus the entropy of the equivalent martingale measure.

5.5 Optimal Control of Stochastic Delay Equations

It is a remarkable feature of the stochastic maximum principle that it often also applies, with appropriate modifications, to optimal control problems of non-standard stochastic systems. We will see examples of such situations in the next chapter on mean-field games and in Chap. 13 on optimal control of stochastic partial differential equations. In this section we present another example, namely the optimal control of systems with *delay*. More precisely, as in [ØSZ3] we consider a controlled stochastic delay equation of the form

$$dX(t) = b(t, X(t), Y(t), A(t), u(t), \omega)dt + \sigma(t, X(t), Y(t), A(t), u(t), \omega)dB(t) + \int_{\mathbb{R}} \theta(t, X(t), Y(t), A(t), u(t), z, \omega)\tilde{N}(dt, dz)\ ;\ t \in [0, T] \tag{5.5.1}$$

$$X(t) = x_0(t)\ ;\ t \in [-\delta, 0], \tag{5.5.2}$$

where

$$Y(t) = X(t - \delta), \quad A(t) = \int_{t-\delta}^{t} e^{-\rho(t-r)} X(r)dr, \tag{5.5.3}$$

and $\delta > 0$, $\rho \geq 0$ and $T > 0$ are given constants. Here

$$b : [0, T] \times \mathbb{R} \times \mathbb{R} \times \mathbb{R} \times \mathcal{U} \times \Omega \to \mathbb{R}$$
$$\sigma : [0, T] \times \mathbb{R} \times \mathbb{R} \times \mathbb{R} \times \mathcal{U} \times \Omega \to \mathbb{R}$$

and

$$\theta : [0, T] \times \mathbb{R} \times \mathbb{R} \times \mathbb{R} \times \mathcal{U} \times \mathbb{R}_0 \times \Omega \to \mathbb{R}$$

are given functions such that, for all t, $b(t, x, y, a, u, \cdot)$, $\sigma(t, x, y, a, u, \cdot)$ and $\theta(t, x, y, a, u, z, \cdot)$ are $\mathcal{F}_t$-measurable for all $x \in \mathbb{R}$, $y \in \mathbb{R}$, $a \in \mathbb{R}$, $u \in \mathcal{U}$ and $z \in \mathbb{R}_0 := \mathbb{R}\backslash\{0\}$. The function $x_0(t)$ is assumed to be continuous and deterministic.

Let $\mathcal{E}_t \subseteq \mathcal{F}_t\ ;\ t \in [0, T]$ be a given subfiltration of $\{\mathcal{F}_t\}_{t\in[0,T]}$, representing the information available to the controller who decides the value of $u(t)$ at time t. For example, we could have $\mathcal{E}_t = \mathcal{F}_{(t-c)^+}$ for some given $c > 0$. Let $\mathcal{U} \subset \mathbb{R}$ be a given set of admissible control values $u(t)\ ;\ t \in [0, T]$ and let $\mathcal{A}_\mathcal{E}$ be a given family of admissible control processes $u(\cdot)$, included in the set of càdlàg, $\mathcal{E}$-predictable and $\mathcal{U}$-valued processes $u(t)\ ;\ t \in [0, T]$ such that (5.5.1)–(5.5.2) has a unique solution $X(\cdot) \in L^2(\lambda \times P)$, where λ denotes the Lebesgue measure on $[0, T]$.

The performance functional is assumed to have the form

$$J(u) = E\left[\int_0^T f(t, X(t), Y(t), A(t), u(t), \omega)dt + g(X(T), \omega)\right]\ ;\ u \in \mathcal{A}_\mathcal{E}, \tag{5.5.4}$$

where $f = f(t, x, y, a, u, \omega) : [0, T] \times \mathbb{R} \times \mathbb{R} \times \mathbb{R} \times \mathcal{U} \times \Omega \to \mathbb{R}$ and $g = g(x, \omega) : \mathbb{R} \times \Omega \to \mathbb{R}$ are given $\mathcal{C}^1$ functions w.r.t. (x, y, a, u) such that

$$E\Big[\int_0^T \Big\{|f(t, X(t), A(t), u(t))| + \Big|\frac{\partial f}{\partial x_i}(t, X(t), Y(t), A(t), u(t))\Big|^2\Big\}dt + |g(X(T))| + |g'(X(T))|^2\Big] < \infty \text{ for } x_i = x, y, a \text{ and } u.$$

Here, and in the following, we suppress the ω, for notational simplicity. The problem we consider in this section is the following:

Find $\Phi(x_0)$ and $u^* \in \mathcal{A}_\mathcal{E}$ such that

$$\Phi(x_0) := \sup_{u \in \mathcal{A}_\mathcal{E}} J(u) = J(u^*). \tag{5.5.5}$$

Any control $u^* \in \mathcal{A}_\mathcal{E}$ satisfying (5.5.5) is called an *optimal control*.

Variants of this problem have been studied in several papers. Stochastic control of delay systems is a challenging research area, because delay systems have, in general, an infinite-dimensional nature. Hence, the natural general approach to them is infinite-dimensional. For this kind of approach in the context of control problems we refer to [C-M, Fe, GoMa, GoMaSa] in the stochastic Brownian case.

Nevertheless, in some cases systems with delay can be reduced to finite-dimensional systems, in the sense that the information we need from their dynamics can be represented by a finite-dimensional variable evolving in terms of itself. In such a context, the crucial point is to understand when this finite-dimensional reduction of the problem is possible and/or to find conditions ensuring it. There are some papers dealing with this subject in the stochastic Brownian case: We refer to [KoSh, EØS, La1, LR, ØS8]. The paper [Dav] represents an extension of [ØS8] to the case when the equation is driven by a Lévy noise.

We also mention the paper [EH], where certain control problems of stochastic functional differential equations are studied by means of the Girsanov transformation. This approach, however, does not work if there is a delay in the noise components.

Our approach in this section is different from all the above. Note that the presence of the terms $Y(t)$ and $A(t)$ in (5.5.1) makes the problem non-Markovian and we cannot use a (finite-dimensional) dynamic programming approach. However, we will show that it is possible to obtain a maximum principle for the problem. To this end, we define the following appropriately modified *Hamiltonian*

$$H : [0, T] \times \mathbb{R} \times \mathbb{R} \times \mathbb{R} \times \mathcal{U} \times \mathbb{R} \times \mathbb{R} \times \mathcal{R} \times \Omega \to \mathbb{R}$$

by

$$H(t,x,y,a,u,p,q,r(\cdot),\omega) = H(t,x,y,a,u,p,q,r(\cdot)) = f(t,x,y,a,u) + b(t,x,y,a,u)p + \sigma(t,x,y,a,u)q + \int_{\mathbb{R}_0} \theta(t,x,y,a,u,z)r(z)\nu(dz); \tag{5.5.6}$$

where $\mathcal{R}$ is the set of functions $r : \mathbb{R}_0 \to \mathbb{R}$ such that the last term in (5.5.6) converges. We assume that b, σ and θ are $\mathcal{C}^1$ functions with respect to (x, y, a, u) and that

$$E\left[\int_0^T \left\{ \left| \frac{\partial b}{\partial x_i}(t, X(t), Y(t), A(t), u(t)) \right|^2 + \left| \frac{\partial \sigma}{\partial x_i}(t, X(t), Y(t), A(t), u(t)) \right|^2 + \int_{\mathbb{R}_0} \left| \frac{\partial \theta}{\partial x_i}(t, X(t), Y(t), A(t), u(t), z) \right|^2 \nu(dz) \right\} dt \right] < \infty \tag{5.5.7}$$

for $x_i = x, y, a$ and u.

Associated to H we define the adjoint processes $p(t), q(t), r(t,z); t \in [0,T], z \in \mathbb{R}_0$, by the following backward stochastic differential equation (BSDE):

$$\begin{cases} dp(t) &= E[\mu(t) \mid \mathcal{F}_t]dt + q(t)dB(t) + \int_{\mathbb{R}_0} r(t,z)\tilde{N}(dt,dz) \ ; \ t \in [0,T] \\ p(T) &= g'(X(T)), \end{cases} \tag{5.5.8}$$

where

$$\begin{aligned} \mu(t) = &-\frac{\partial H}{\partial x}(t, X(t), Y(t), A(t), u(t), p(t), q(t), r(t,\cdot)) \\ &- \frac{\partial H}{\partial y}(t+\delta, X(t+\delta), Y(t+\delta), A(t+\delta), \\ &u(t+\delta), p(t+\delta), q(t+\delta), r(t+\delta,\cdot))\chi_{[0,T-\delta]}(t) \\ &- e^{\rho t}\Big(\int_t^{t+\delta} \frac{\partial H}{\partial a}(s, X(s), Y(s), A(s), \\ &\quad u(s), p(s), q(s), r(s,\cdot))e^{-\rho s}\chi_{[0,T]}(s)ds\Big). \end{aligned} \tag{5.5.9}$$

Note that this BSDE is *anticipative*, or *time-advanced*, in the sense that the driver $\mu(t)$ contains future values of $X(s), u(s), p(s), q(s), r(s,\cdot) \ ; \ s \le t+\delta$.

In the case when there are no jumps and no integral term in (5.5.9), anticipative BSDEs (ABSDEs for short) have been studied by [PY], who prove the existence and uniqueness of such equations under certain conditions. They also relate a class of linear ABSDEs to a class of linear stochastic delay control problems with no delay

in the noise coefficients. Thus, in this section we extend this relation to general nonlinear control problems and general nonlinear ABSDEs by means of the maximum principle, and throughout the discussion we also include the possibility of delays in all the noise coefficients, as well as the possibility of jumps.

5.5.1 A Sufficient Maximum Principle

We now establish a maximum principle of sufficient type, i.e. we show that – under some assumptions – maximizing the Hamiltonian leads to an optimal control.

Theorem 5.22 (***Sufficient maximum principle*** [ØSZ3]) *Let $\hat{u} \in \mathcal{A}_{\mathcal{E}}$ with corresponding state processes $\hat{X}(t)$, $\hat{Y}(t)$, $\hat{A}(t)$ and adjoint processes $\hat{p}(t)$, $\hat{q}(t)$, $\hat{r}(t,z)$, assumed to satisfy the ABSDE (5.5.8)–(5.5.9). Suppose the following hold:*

(i) *The functions $x \to g(x)$ and*

$$(x, y, a, u) \to H(t, x, y, a, u, \hat{p}(t), \hat{q}(t), \hat{r}(t, \cdot)) \tag{5.5.10}$$

are concave, for each $t \in [0, T]$, a.s.

(ii)

$$\max_{v \in \mathcal{U}} E\left[H(t, \hat{X}(t), \hat{X}(t-\delta), \hat{A}(t), v, \hat{p}(t), \hat{q}(t), \hat{r}(t, \cdot)) \mid \mathcal{E}_t\right]$$
$$= E\left[H(t, \hat{X}(t), \hat{X}(t-\delta), \hat{A}(t), \hat{u}(t), \hat{p}(t), \hat{q}(t), \hat{r}(t, \cdot)) \mid \mathcal{E}_t\right] \tag{5.5.11}$$

for all $t \in [0, T]$, a.s.

Then $\hat{u}(t)$ is an optimal control for the problem (5.5.5).

Proof As in the proof of Theorem 5.4, by considering an increasing sequence of stopping times converging to T we may assume that all the local martingale terms in the following are martingales, and hence have expectation 0.

Choose $u \in \mathcal{A}_{\mathcal{E}}$ and consider

$$J(u) - J(\hat{u}) = I_1 + I_2, \tag{5.5.12}$$

where

$$I_1 = E\Big[\int_0^T \{f(t, X(t), Y(t), A(t), u(t)) - f(t, \hat{X}(t), \hat{Y}(t), \hat{A}(t), \hat{u}(t))\} dt\Big], \tag{5.5.13}$$

$$I_2 = E[g(X(T)) - g(\hat{X}(T))]. \tag{5.5.14}$$

By the definition of H and concavity of H we have

$$\begin{aligned}
I_1 &= E\Bigg[\int_0^T \{H(t, X(t), Y(t), A(t), u(t), \hat{p}(t), \hat{q}(t), \hat{r}(t,\cdot)) \\
&\quad - H(t, \hat{X}(t), \hat{Y}(t), \hat{A}(t), \hat{u}(t), \hat{p}(t), \hat{q}(t), \hat{r}(t,\cdot)) \\
&\quad - (b(t, X(t), Y(t), A(t), u(t)) - b(t, \hat{X}(t), \hat{Y}(t), \hat{A}(t), \hat{u}(t)))\hat{p}(t) \\
&\quad - (\sigma(t, X(t), Y(t), A(t), u(t)) - \sigma(t, \hat{X}(t), \hat{Y}(t), \hat{A}(t), \hat{u}(t)))\hat{q}(t) \\
&\quad - \int_{\mathbb{R}} (\theta(t, X(t), Y(t), A(t), u(t), z) \\
&\qquad\qquad -\theta(t, \hat{X}(t), \hat{Y}(t), \hat{A}(t), \hat{u}(t), z))\hat{r}(t,z)\nu(dz)\}dt\Bigg] \\
&\leq E\Bigg[\int_0^T \{\frac{\partial \hat{H}}{\partial x}(t)(X(t) - \hat{X}(t)) + \frac{\partial \hat{H}}{\partial y}(t)(Y(t) - \hat{Y}(t)) \\
&\quad + \frac{\partial \hat{H}}{\partial a}(t)(A(t) - \hat{A}(t)) + \frac{\partial \hat{H}}{\partial u}(t)(u(t) - \hat{u}(t)) \\
&\quad - (b(t) - \hat{b}(t))\hat{p}(t) - (\sigma(t) - \hat{\sigma}(t))\hat{q}(t) \\
&\quad - \int_{\mathbb{R}} (\theta(t,z) - \hat{\theta}(t,z))\hat{r}(t,z)\nu(dz)\}dt\Bigg], \qquad (5.5.15)
\end{aligned}$$

where we have used the abbreviated notation

$$\begin{aligned}
\frac{\partial \hat{H}}{\partial x}(t) &= \frac{\partial H}{\partial x}(t, \hat{X}(t), \hat{Y}(t), \hat{A}(t), \hat{u}(t), \hat{p}(t), \hat{q}(t), \hat{r}(t,\cdot)), \\
b(t) &= b(t, X(t), Y(t), A(t), u(t)), \\
\hat{b}(t) &= b(t, \hat{X}(t), \hat{Y}(t), \hat{A}(t), \hat{u}(t) \text{ etc.}
\end{aligned}$$

Since g is concave we have

$$\begin{aligned}
I_2 &\leq E[g'(\hat{X}(T))(X(T) - \hat{X}(T))] = E[\hat{p}(T)(X(T) - \hat{X}(T))] \\
&= E\Bigg[\int_0^T \hat{p}(t)(dX(t) - d\hat{X}(t)) + \int_0^T (X(t) - \hat{X}(t))d\hat{p}(t) \\
&\quad + \int_0^T (\sigma(t) - \hat{\sigma}(t))\hat{q}(t)dt + \int_0^T \int_{\mathbb{R}} (\theta(t,z) - \hat{\theta}(t,z))\hat{r}(t,z)\nu(dz)dt\Bigg] \\
&= E\Bigg[\int_0^T (b(t) - \hat{b}(t))\hat{p}(t)dt + \int_0^T (X(t) - \hat{X}(t))\mu(t)dt \\
&\quad + \int_0^T (\sigma(t) - \hat{\sigma}(t))\hat{q}(t)dt + \int_0^T \int_{\mathbb{R}} (\theta(t,z) - \hat{\theta}(t,z))\hat{r}(t,z)\nu(dz)dt\Bigg]. \qquad (5.5.16)
\end{aligned}$$

Combining (5.5.12)–(5.5.16) we get, using that $X(t) = \hat{X}(t) = x_0(t)$ for all $t \in [-\delta, 0]$,

$$\begin{aligned} J(u) - J(\hat{u}) &\leq E\Big[\int_0^T \Big\{\frac{\partial \hat{H}}{\partial x}(t)(X(t) - \hat{X}(t)) + \frac{\partial \hat{H}}{\partial y}(t)(Y(t) - \hat{Y}(t)) \\ &\quad + \frac{\partial \hat{H}}{\partial a}(t)(A(t) - \hat{A}(t)) + \frac{\partial \hat{H}}{\partial u}(t)(u(t) - \hat{u}(t)) + \mu(t)(X(t) - \hat{X}(t))\Big\}dt\Big] \\ &= E\Big[\int_\delta^{T+\delta} \Big\{\frac{\partial \hat{H}}{\partial x}(t-\delta) + \frac{\partial \hat{H}}{\partial y}(t)\chi_{[0,T]}(t) + \mu(t-\delta)\Big\}(Y(t) - \hat{Y}(t))dt \\ &\quad + \int_0^T \frac{\partial \hat{H}}{\partial a}(t)(A(t) - \hat{A}(t))dt + \int_0^T \frac{\partial \hat{H}}{\partial u}(t)(u(t) - \hat{u}(t))dt\Big]. \end{aligned} \tag{5.5.17}$$

Using integration by parts and substituting $r = t - \delta$, we get

$$\begin{aligned} &\int_0^T \frac{\partial \hat{H}}{\partial a}(s)(A(s) - \hat{A}(s))ds \\ &= \int_0^T \frac{\partial \hat{H}}{\partial a}(s) \int_{s-\delta}^s e^{-\rho(s-r)}(X(r) - \hat{X}(r))drds \\ &= \int_0^T \left(\int_r^{r+\delta} \frac{\partial \hat{H}}{\partial a}(s)e^{-\rho s}\chi_{[0,T]}(s)ds\right) e^{\rho r}(X(r) - \hat{X}(r))dr \\ &= \int_\delta^{T+\delta} \left(\int_{t-\delta}^t \frac{\partial \hat{H}}{\partial a}(s)e^{-\rho s}\chi_{[0,T]}(s)ds\right) e^{\rho(t-s)}(X(t-\delta) - \hat{X}(t-\delta))dt. \end{aligned} \tag{5.5.18}$$

Combining this with (5.5.17) and using (5.5.9) we obtain

$$\begin{aligned} J(u) - J(\hat{u}) &\leq E\left[\int_\delta^{T+\delta} \left\{\frac{\partial \hat{H}}{\partial x}(t-\delta) + \frac{\partial \hat{H}}{\partial y}(t)\chi_{[0,T]}(t)\right. \right. \\ &\quad \left. + \left(\int_{t-\delta}^t \frac{\partial \hat{H}}{\partial a}(s)e^{-\rho s}\chi_{[0,T]}(s)ds\right) e^{\rho(t-\delta)} + \mu(t-\delta)\right\}(Y(t) - \hat{Y}(t))dt \\ &\quad \left. + \int_0^T \frac{\partial \hat{H}}{\partial u}(t)(u(t) - \hat{u}(t))dt\right] \\ &= E\left[\int_0^T \frac{\partial \hat{H}}{\partial u}(t)(u(t) - \hat{u}(t))dt\right] \\ &= E\left[\int_0^T E\left[\frac{\partial \hat{H}}{\partial u}(t)(u(t) - \hat{u}(t)) \mid \mathcal{E}_t\right]dt\right] \end{aligned}$$

$$= E\left[\int_0^T E\left[\frac{\partial \hat{H}}{\partial u}(t) \mid \mathcal{E}_t\right](u(t) - \hat{u}(t))dt\right] \leq 0.$$

The last inequality holds because for each $t \in [0, T]$, $v = \hat{u}(t)$ maximizes $E[H(t, \hat{X}(t), \hat{Y}(t), \hat{A}(t), v, \hat{p}(t), \hat{q}(t), \hat{r}(t, \cdot) \mid \mathcal{E}_t]$.

This proves that $\hat{u}$ is an optimal control. □

5.5.2 A Necessary Maximum Principle

A drawback with the sufficient maximum principle in Sect. 5.5.1 is the condition of concavity, which does not always hold in applications. In this section we will prove a result going in the other direction. More precisely, we will prove the equivalence between being a directional critical point for $J(u)$ and a critical point for the conditional Hamiltonian. To this end, we need to make the following assumptions:

A 1 For all $u \in \mathcal{A}_{\mathcal{E}}$ and all bounded $\beta \in \mathcal{A}_{\mathcal{E}}$ there exists an $\varepsilon > 0$ such that

$$u + s\beta \in \mathcal{A}_{\mathcal{E}} \text{ for all } s \in (-\varepsilon, \varepsilon).$$

A 2 For all $t_0 \in [0, T]$ and all bounded $\mathcal{E}_{t_0}$-measurable random variables α the control process $\beta(t)$ defined by

$$\beta(t) = \alpha\chi_{[t_0,T]}(t)\ ;\ t \in [0, T] \tag{5.5.19}$$

belongs to $\mathcal{A}_{\mathcal{E}}$. For all bounded $\beta \in \mathcal{A}_{\mathcal{E}}$ the derivative process

$$\xi(t) := \frac{d}{ds}X^{u+s\beta}(t)\ |_{s=0} \tag{5.5.20}$$

exists and belongs to $L^2(\lambda \times P)$.

It follows from (5.5.1) that

$$\begin{aligned}
&d\xi(t) = \\
&\left\{\frac{\partial b}{\partial x}(t)\xi(t) + \frac{\partial b}{\partial y}(t)\xi(t-\delta) + \frac{\partial b}{\partial a}(t)\int_{t-\delta}^{t} e^{-\rho(t-r)}\xi(r)dr + \frac{\partial b}{\partial u}(t)\beta(t)\right\}dt \\
&+\left\{\frac{\partial \sigma}{\partial x}(t)\xi(t) + \frac{\partial \sigma}{\partial y}(t)\xi(t-\delta)\right. \\
&\quad\left.+\frac{\partial \sigma}{\partial a}(t)\int_{t-\delta}^{t} e^{-\rho(t-r)}\xi(r)dr + \frac{\partial \sigma}{\partial u}(t)\beta(t)\right\}dB(t) \\
&+\int_{\mathbb{R}_0}\left\{\frac{\partial \theta}{\partial x}(t,z)\xi(t) + \frac{\partial \theta}{\partial y}(t,z)\xi(t-\delta)\right.
\end{aligned}$$

$$+\frac{\partial\theta}{\partial a}(t)\int_{t-\delta}^{t} e^{-\rho(t-r)}\xi(r)dr + \frac{\partial\theta}{\partial u}(t)\beta(t)\Big\}\tilde{N}(dt,dz), \tag{5.5.21}$$

where for simplicity of notation we have put

$$\frac{\partial b}{\partial x}(t) = \frac{\partial b}{\partial x}(t, X(t), X(t-\delta), A(t), u(t)) \text{ etc } \ldots$$

and we have used that

$$\frac{d}{ds}Y^{u+s\beta}(t)\mid_{s=0} = \frac{d}{ds}X^{u+s\beta}(t-\delta)\mid_{s=0} = \xi(t-\delta) \tag{5.5.22}$$

and

$$\begin{aligned}\frac{d}{ds}A^{u+s\beta}(t)\mid_{s=0} &= \frac{d}{ds}\left(\int_{t-\delta}^{t} e^{-\rho(t-r)}X^{u+s\beta}(r)dr\right)\mid_{s=0}\\ &= \int_{t-\delta}^{t} e^{-\rho(t-r)}\frac{d}{ds}X^{u+s\beta}(r)\mid_{s=0} dt = \int_{t-\delta}^{t} e^{-\rho(t-r)}\xi(r)dr.\end{aligned} \tag{5.5.23}$$

Note that

$$\xi(t) = 0 \text{ for } t \in [-\delta, 0]. \tag{5.5.24}$$

Theorem 5.23 (***Necessary maximum principle*** [ØSZ3]) *Suppose $\hat{u} \in \mathcal{A}_{\mathcal{E}}$ with corresponding solutions $\hat{X}(t)$ of (5.5.1)–(5.5.2) and $\hat{p}(t)$, $\hat{q}(t)$, $\hat{r}(t,z)$ of (5.5.7)–(5.5.8) and corresponding derivative process $\hat{\xi}(t)$ given by (5.5.20).*

Then the following are equivalent:

(i) $\frac{d}{ds}J(\hat{u}+s\beta)\mid_{s=0} = 0$ *for all bounded $\beta \in \mathcal{A}_{\mathcal{E}}$.*

(ii) $E\left[\frac{\partial H}{\partial u}(t, \hat{X}(t), \hat{Y}(t), \hat{A}(t), u, \hat{p}(t), \hat{q}(t), \hat{r}(t,\cdot)) \mid \mathcal{E}_t\right]_{u=\hat{u}(t)} = 0$ *a.s. for all $t \in [0,T]$.*

Proof For simplicity of notation we write $\hat{u} = u$, $\hat{X} = X$, $\hat{p} = p$, $\hat{q} = q$ and $\hat{r} = r$ in the following. Suppose **(i)** holds. Then

$$\begin{aligned}0 &= \frac{d}{ds}J(u+s\beta)\mid_{s=0}\\ &= \frac{d}{ds}E[\int_0^T f(t, X^{u+s\beta}(t), Y^{u+s\beta}(t), A^{u+s\beta}(t), u(t)+s\beta(t))dt\\ &\quad + g(X^{u+s\beta}(T))]\mid_{s=0}\\ &= E\Big[\int_0^T \Big\{\frac{\partial f}{\partial x}(t)\xi(t) + \frac{\partial f}{\partial y}(t)\xi(t-\delta)\end{aligned}$$

$$
\begin{aligned}
&+\frac{\partial f}{\partial a}(t)\int_{t-\delta}^{t} e^{-\rho(t-r)}\xi(r)dt+\frac{\partial f}{\partial u}(t)\beta(t)\Big\}\,dt+g'(X(T))\xi(T)\Big]\\
&=E\Big[\int_0^T\Big\{\frac{\partial H}{\partial x}(t)-\frac{\partial b}{\partial x}(t)p(t)-\frac{\partial \sigma}{\partial x}(t)q(t)\\
&-\int_{\mathbb{R}}\frac{\partial \theta}{\partial x}(t,z)r(t,z)\nu(dz)\Big\}\xi(t)dt\\
&+\int_0^T\Big\{\frac{\partial H}{\partial y}(t)-\frac{\partial b}{\partial y}(t)p(t)-\frac{\partial \sigma}{\partial y}(t)q(t)\\
&-\int_{\mathbb{R}}\frac{\partial \theta}{\partial y}(t,z)r(t,z)\nu(dz)\Big\}\xi(t-\delta)dt\\
&+\int_0^T\Big\{\frac{\partial H}{\partial a}(t)-\frac{\partial b}{\partial a}(t)p(t)-\frac{\partial \sigma}{\partial a}(t)q(t)\\
&-\int_{\mathbb{R}}\frac{\partial \theta}{\partial a}(t,z)r(t,z)\nu(dz)\Big\}\Big(\int_{t-\delta}^{t}e^{-\rho(t-r)}\xi(r)dr\Big)\,dt\\
&+\int_0^T\frac{\partial f}{\partial u}(t)\beta(t)dt+g'(X(T))\xi(T)\Big].
\end{aligned}
\tag{5.5.25}
$$

By (5.5.21),

$$
\begin{aligned}
&E[g'(X(T))\xi(T)]=E[p(T)\xi(T)]\\
&=E[\int_0^T p(t)d\xi(t)+\int_0^T\xi(t)dp(t)+\int_0^T q(t)\Big\{\frac{\partial \sigma}{\partial x}(t)\xi(t)+\frac{\partial \sigma}{\partial y}(t)\xi(t-\delta)\\
&+\frac{\partial \sigma}{\partial a}(t)\int_{t-\delta}^{t}e^{-\rho(t-r)}\xi(r)dr+\frac{\partial \sigma}{\partial u}(t)\beta(t)\Big\}\,dt\\
&+\int_0^T\int_{\mathbb{R}} r(t,z)\Big\{\frac{\partial \theta}{\partial x}(t,z)\xi(t)+\frac{\partial \theta}{\partial y}(t,z)\xi(t-\delta)\\
&+\frac{\partial \theta}{\partial a}(t,z)\int_{t-\delta}^{t}e^{-\rho(t-r)}\xi(r)dr\ +\frac{\partial \theta}{\partial u}(t)\beta(t)\Big\}\nu(dz)dt\Big]\\
&=E\Big[\int_0^T p(t)\Big\{\frac{\partial b}{\partial x}(t)\xi(t)+\frac{\partial b}{\partial y}(t)\xi(t-\delta)+\frac{\partial b}{\partial a}(t)\int_{t-\delta}^{t}e^{-\rho(t-r)}\xi(r)dr\\
&+\frac{\partial b}{\partial u}(t)\beta(t)\Big\}\,dt+\int_0^T\xi(t)\mu(t)dt\\
&+\int_0^T q(t)\Big\{\frac{\partial \sigma}{\partial x}(t)\xi(t)+\frac{\partial \sigma}{\partial y}(t)\xi(t-\delta)+\frac{\partial \sigma}{\partial a}(t)\int_{t-\delta}^{t}e^{-\rho(t-r)}\xi(r)dr\\
&+\frac{\partial \sigma}{\partial u}(t)\beta(t)\Big\}\,dt\\
&+\int_0^T\int_{\mathbb{R}} r(t,z)\Big\{\frac{\partial \theta}{\partial x}(t,z)\xi(t)+\frac{\partial \theta}{\partial y}(t,z)\xi(t-\delta)
\end{aligned}
$$

$$+\frac{\partial\theta}{\partial a}(t,z)\int_{t-\delta}^{t}e^{-\rho(t-r)}\xi(r)dr+\frac{\partial\theta}{\partial u}(t,z)\beta(t)\Big\}\nu(dz)dt\Big]. \tag{5.5.26}$$

Combining (5.5.25) and (5.2.7) we get

$$\begin{aligned}
0&=E\Big[\int_0^T\xi(t)\Big\{\frac{\partial H}{\partial x}(t)+\mu(t)\Big\}dt+\int_0^T\xi(t-\delta)\frac{\partial H}{\partial y}(t)dt\\
&\quad+\int_0^T\Big(\int_{t-\delta}^{t}e^{-\rho(t-r)}\xi(r)dr\Big)\frac{\partial H}{\partial a}(t)dt+\int_0^T\frac{\partial H}{\partial u}(t)\beta(t)dt\Big]\\
&=E\Big[\int_0^T\xi(t)\Big\{\frac{\partial H}{\partial x}(t)-\frac{\partial H}{\partial x}(t)-\frac{\partial H}{\partial y}(t+\delta)\chi_{[0,T-\delta]}(t)\\
&\quad-e^{\rho t}\Big(\int_t^{t+\delta}\frac{\partial H}{\partial a}(s)e^{-\rho s}\chi_{[0,T]}(s)ds\Big)\Big\}dt+\int_0^T\xi(t-\delta)\frac{\partial H}{\partial y}(t)dt\\
&\quad+\int_0^T\Big(\int_{s-\delta}^{s}e^{-\rho(s-t)}\xi(t)dt\Big)\frac{\partial H}{\partial a}(s)ds+\int_0^T\frac{\partial H}{\partial u}(t)\beta(t)dt\Big]\\
&=E\Big[\int_0^T\xi(t)\Big\{-\frac{\partial H}{\partial y}(t+\delta)\chi_{[0,T-\delta]}(t)\\
&\quad-e^{\rho t}\Big(\int_t^{t+\delta}\frac{\partial H}{\partial a}(s)e^{-\rho s}\chi_{[0,T]}(s)ds\Big)\Big\}dt+\int_0^T\xi(t-\delta)\frac{\partial H}{\partial y}(t)dt\\
&\quad+e^{\rho t}\int_0^T\Big(\int_t^{t+\delta}\frac{\partial H}{\partial a}(s)e^{-\rho s}\chi_{[0,T]}(s)ds\Big)\xi(t)dt+\int_0^T\frac{\partial H}{\partial u}(t)\beta(t)dt\Big]\\
&=E\Big[\int_0^T\frac{\partial H}{\partial u}(t)\beta(t)dt\Big],
\end{aligned} \tag{5.5.27}$$

where we have again used integration by parts. If we apply (5.2.8) to

$$\beta(t)=\alpha(\omega)\chi_{[s,T]}(t),$$

where $\alpha(\omega)$ bounded and $\mathcal{E}_{t_0}$-measurable, $s\geq t_0$, we get

$$E\Big[\int_s^T\frac{\partial H}{\partial u}(t)dt\ \alpha\Big]=0.$$

Differentiating with respect to s we obtain

$$E\Big[\frac{\partial H}{\partial u}(s)\alpha\Big]=0.$$

Since this holds for all $s\geq t_0$ and all α we conclude that

$$E\left[\frac{\partial H}{\partial u}(t_0) \mid \mathcal{E}_{t_0}\right] = 0.$$

This shows that **(i)** $\Rightarrow$ **(ii)**.

Conversely, since every bounded $\beta \in \mathcal{A}_{\mathcal{E}}$ can be approximated by linear combinations of controls β of the form (5.5.20), we can prove that **(ii)** $\Rightarrow$ **(i)** by reversing the above argument. □

5.5.3 *Time-Advanced BSDEs with Jumps*

We now study time-advanced backward stochastic differential equations driven both by Brownian motion $B(t)$ and compensated Poisson random measures $\tilde{N}(dt, dz)$.

Framework

Given a positive constant δ, denote by $D([0, \delta], \mathbb{R})$ the space of all càdlàg paths from $[0, \delta]$ into $\mathbb{R}$. For a path $X(\cdot) : \mathbb{R}_+ \to \mathbb{R}$, X_t will denote the function defined by $X_t(s) = X(t + s)$ for $s \in [0, \delta]$. Put $\mathcal{H} = L^2(\nu)$. Consider the L^2 spaces $V_1 := L^2([0, \delta], ds)$ and $V_2 := L^2([0, \delta] \to \mathcal{H}, ds)$. Let

$$F : \mathbb{R}_+ \times \mathbb{R} \times \mathbb{R} \times V_1 \times \mathbb{R} \times \mathbb{R} \times V_1 \times \mathcal{H} \times \mathcal{H} \times V_2 \times \Omega \to \mathbb{R}$$

be a predictable function. Introduce the following Lipschitz condition: There exists a constant C such that

$$\begin{aligned} &|F(t, p_1, p_2, p, q_1, q_2, q, r_1, r_2, r, \omega) - F(t, \bar{p}_1, \bar{p}_2, \bar{p}, \bar{q}_1, \bar{q}_2, \bar{q}, \bar{r}_1, \bar{r}_2, \bar{r}, \omega)| \\ &\quad \le C(|p_1 - \bar{p}_1| + |p_2 - \bar{p}_2| + |p - \bar{p}|_{V_1} + |q_1 - \bar{q}_1| + |q_2 - \bar{q}_2| + |q - \bar{q}|_{V_1} \\ &\quad + |r_1 - \bar{r}_1|_{\mathcal{H}} + |r_2 - \bar{r}_2|_{\mathcal{H}} + |r - \bar{r}|_{V_2}. \end{aligned} \tag{5.5.28}$$

First Existence and Uniqueness Theorem (Special Case)

We first consider the following time-advanced backward stochastic differential equation in the unknown $\mathbb{F}$-predictable processes $(p(t), q(t), r(t, z))$:

$$\begin{aligned} dp(t) = E[F\big(&t, p(t), p(t+\delta)\chi_{[0,T-\delta]}(t), p_t\chi_{[0,T-\delta]}(t), q(t), q(t+\delta)\chi_{[0,T-\delta]}(t), \\ &q_t\chi_{[0,T-\delta]}(t), r(t), r(t+\delta)\chi_{[0,T-\delta]}(t), r_t\chi_{[0,T-\delta]}(t)\big) \mid \mathcal{F}_t]dt \\ &+ q(t)dB(t) + \int_{\mathbb{R}} r(t, z)\tilde{N}(dt, dz)\ ;\ t \in [0, T] \end{aligned} \tag{5.5.29}$$

$$p(T) = G, \tag{5.5.30}$$

where G is a given $\mathcal{F}_t$-measurable random variable.

Note that the time-advanced BSDE (5.5.8)–(5.5.9) for the adjoint processes of the Hamiltonian is of this form.

For this type of time-advanced BSDEs we have the following result:

Theorem 5.24 ([ØSZ3]) *Assume that* $E[G^2] < \infty$ *and that condition* (5.5.28) *is satisfied. Then the BSDE* (5.5.29)–(5.5.30) *has a unique solution* $p(t), q(t), r(t,z))$ *such that*

$$E\left[\int_0^T \left\{p^2(t) + q^2(t) + \int_{\mathbb{R}} r^2(t,z)\nu(dz)\right\} dt\right] < \infty. \tag{5.5.31}$$

Moreover, the solution can be found by inductively solving a sequence of BSDEs backwards as follows:

Step 0:

In the interval $[T-\delta, T]$ we let $p(t), q(t)$ and $r(t,z)$ be defined as the solution of the classical BSDE

$$\begin{aligned} dp(t) &= F\,(t, p(t), 0, 0, q(t), 0, 0, r(t,z), 0, 0)\,dt \\ &\quad + q(t)dB(t) + \int_{\mathbb{R}} r(t,z)\tilde{N}(dt,dz)\ ;\ t \in [T-\delta, T], \end{aligned} \tag{5.5.32}$$

$$p(T) = G. \tag{5.5.33}$$

Step k ; $k \geq 1$
If the values of $(p(t), q(t), r(t,z))$ have been found for $t \in [T-k\delta, T-(k-1)\delta]$, then if $t \in [T-(k+1)\delta, T-k\delta]$ the values of $p(t+\delta)$, $p_t, q(t+\delta), q_t, r(t+\delta, z)$ and r_t are known and hence the BSDE

$$\begin{aligned} dp(t) &= E[F\,(t, p(t), p(t+\delta), p_t, q(t), q(t+\delta), q_t, r(t), r(t+\delta), r_t)\,|\mathcal{F}_t]dt \\ &+ q(t)dB(t) + \int_{\mathbb{R}} r(t,z)\tilde{N}(dt,dz)\ ;\ t \in [T-(k+1)\delta, T-k\delta] \end{aligned} \tag{5.5.34}$$

$$p(T-k\delta) = \text{ the value found in Step } k-1 \tag{5.5.35}$$

has a unique solution in $[T-(k+1)\delta, T-k\delta]$.

We proceed like this until k is such that $T-(k+1)\delta \leq 0 < T-k\delta$ and then we solve the corresponding BSDE on the interval $[0, T-k\delta]$.

Proof The proof follows directly from the above inductive procedure. The estimate (5.5.31) is a consequence of known estimates for classical BSDEs (Pardoux–Peng). □

Second Existence and Uniqueness Theorem (General Case)

Next, consider the following backward stochastic differential equation in the unknown $\mathbb{F}$-predictable processes $(p(t), q(t), r(t,x))$:

$$\begin{aligned} dp(t) &= E[F(t, p(t), p(t+\delta), p_t, q(t), q(t+\delta), q_t, r(t), r(t+\delta), r_t)|\mathcal{F}_t]dt \\ &\quad + q(t)dB_t + \int_{\mathbb{R}} r(t,z)\tilde{N}(dt,dz),\ ;\ t \in [0,T], \end{aligned} \tag{5.5.36}$$

$$p(t) = G(t), \qquad t \in [T, T+\delta], \tag{5.5.37}$$

where G is a given $\mathcal{F}_T$-measurable stochastic process.

Theorem 5.25 ([ØSZ3]) *Assume* $E[\sup_{T\le t\le T+\delta} |G(t)|^2] < \infty$ *and that the condition* (5.5.28) *is satisfied. Then the backward stochastic differential equation* (5.5.36) *admits a unique solution* $(p(t), q(t), r(t,z))$ *such that*

$$E\left[\int_0^T \{p^2(t) + q^2(t) + \int_{\mathbb{R}} r^2(t,z)\nu(dz)\}dt\right] < \infty.$$

Proof *Step 1. Assume* F *is independent of* p_1, p_2 *and* p.

Set $q^0(t) := 0, r^0(t,x) = 0$. For $n \ge 1$, define $(p^n(t), q^n(t), r^n(t,x))$ to be the unique solution to the following backward stochastic differential equation:

$$\begin{aligned} dp^n(t) &= E[F(t, q^{n-1}(t), q^{n-1}(t+\delta), q_t^{n-1}, r^{n-1}(t,\cdot), r^{n-1}(t+\delta,\cdot), \\ &\qquad r_t^{n-1}(\cdot))|\mathcal{F}_t]dt + q^n(t)dB_t + r^n(t,z)\tilde{N}(dt,dz), \quad t \in [0,T] \\ p^n(t) &= G(t) \qquad t \in [T, T+\delta]. \end{aligned} \tag{5.5.38}$$

As pointed out in Chap. 4 the existence and uniqueness of the solution of above BSDE is well known. We then show that $(p^n(t), q^n(t), r^n(t,x))$ forms a Cauchy sequence.

Step 2. General case

Let $p^0(t) = 0$. For $n \ge 1$, define $(p^n(t), q^n(t), r^n(t,z))$ to be the unique solution to the following BSDE:

$$\begin{aligned} dp^n(t) &= E[F(t, p^{n-1}(t), p^{n-1}(t+\delta), p_t^{n-1}, q^n(t), q^n(t+\delta), q_t^n, r^n(t,\cdot), \\ &\qquad r^n(t+\delta,\cdot), r_t^n(\cdot))|\mathcal{F}_t]dt + q^n(t)dB_t + r^n(t,z)\tilde{N}(dt,dz), \\ p^n(t) &= G(t); \qquad t \in [T, T+\delta]. \end{aligned} \tag{5.5.39}$$

The existence of $(p^n(t), q^n(t), r^n(t,z))$ is proved in Step 1.

By similar arguments as above, we deduce after some computations that

$$E\left[\int_0^T |p^{n+1}(s) - p^n(s)|^2 ds\right] \le \frac{e^{CNT}T^n}{n!}.$$

By this inequality and a similar argument as in Step 1, it can be shown that $(p^n(t), q^n(t), r^n(t,z))$ converges to some limit $(p(t), q(t), r(t,z))$, which is the unique solution of equation (5.5.36). □

Theorem 5.26 ([ØSZ3]) *Assume* $E\left[\sup_{T\le t\le T+\delta} |G(t)|^{2\alpha}\right] < \infty$ *for some* $\alpha > 1$ *and that the following condition holds:*

$$
\begin{aligned}
&|F(t, p_1, p_2, p, q_1, q_2, q, r_1, r_2, r) - F(t, \bar{p}_1, \bar{p}_2, \bar{p}, \bar{q}_1, \bar{q}_2, \bar{q}, \bar{r}_1, \bar{r}_2, \bar{r})| \\
&\quad \le C(|p_1 - \bar{p}_1| + |p_2 - \bar{p}_2| + \sup_{0\le s\le\delta} |p(s) - \bar{p}(s)| + |q_1 - \bar{q}_1| \\
&\quad + |q_2 - \bar{q}_2| + |q - \bar{q}|_{V_1} + |r_1 - \bar{r}_1|_{\mathcal{H}} + |r_2 - \bar{r}_2|_{\mathcal{H}} + |r - \bar{r}|_{V_2}. \qquad (5.5.40)
\end{aligned}
$$

Then the BSDE (5.5.36) *admits a unique solution* $(p(t), q(t), r(t, z))$ *such that*

$$
E\left[\sup_{0\le t\le T} |p(t)|^{2\alpha} + \int_0^T \{q^2(t) + \int_{\mathbb{R}} r^2(t, z)\nu(dz)\}dt\right] < \infty.
$$

Proof *Step 1. Assume* F *is independent of* p_1, p_2 *and* p.

In this case the condition above reduces to assumption (5.5.28). By Step 1 in the proof of Theorem 5.25, there is a unique solution $(p(t), q(t), r(t, z))$ to Eq. (5.5.36).

Step 2. General case.

Let $p^0(t) = 0$. For $n \ge 1$, define $(p^n(t), q^n(t), r^n(t, z))$ to be the unique solution to the following BSDE:

$$
\begin{aligned}
dp^n(t) =& E[F(t, p^{n-1}(t), p^{n-1}(t+\delta), p_t^{n-1}, q^n(t), q^n(t+\delta), q_t^n, r^n(t, \cdot), \\
&\quad r^n(t+\delta, \cdot), r_t^n(\cdot))|\mathcal{F}_t]dt + q^n(t)dB_t + \int_{\mathbb{R}} r^n(t, z)\tilde{N}(dt, dz), \\
p^n(t) =& G(t), \qquad t \in [T, T+\delta]. \qquad (5.5.41)
\end{aligned}
$$

By Step 1, $(p^n(t), q^n(t), r^n(t, z))$ exists. We then proceed to show that $(p^n(t), q^n(t), r^n(t, z))$ forms a Cauchy sequence.

After some computations we end up with the estimate

$$
E\left[\int_0^T \sup_{t\le s\le T} |p^{n+1}(s) - p^n(s)|^{2\alpha} ds\right] \le \frac{e^{CNT}T^n}{n!}.
$$

Using this inequality and a similar argument as in Step 1, we can show that $(p^n(t), q^n(t), r^n(t, z))$ converges to some limit $(p(t), q(t), r(t, z))$, which is the unique solution of equation (5.5.36). □

Finally we give a result when the coefficient f is independent of z and r.

Theorem 5.27 ([ØSZ3]) *Assume that* $E\left[\sup_{T\le t\le T+\delta} |G(t)|^2\right] < \infty$ *and that* F *satisfies*

$$
|F(t, y_1, y_2, p) - F(t, \bar{y}_1, \bar{y}_2, \bar{p})| \le C(|y_1 - \bar{y}_1| + |y_2 - \bar{y}_2| + \sup_{0\le s\le\delta} |p(s) - \bar{p}(s)|). \qquad (5.5.42)
$$

Then the backward stochastic differential equation (5.5.36) *admits a unique solution.*

5.5.4 Example: Optimal Consumption from a Cash Flow with Delay

Let $\alpha(t)$, $\beta(t)$ and $\gamma(t,z)$ be given bounded adapted processes, $\alpha(t)$ deterministic. Assume that $\int_{\mathbb{R}} \gamma^2(t,z)\nu(dz) < \infty$. Consider a cash flow $X^0(t)$ with the dynamics, for $t \in [0,T]$,

$$dX^0(t) = X^0(t-\delta)\left[\alpha(t)dt + \beta(t)dB(t) + \int_{\mathbb{R}} \gamma(t,z)\tilde{N}(dt,dz)\right]; t \in [0,T] \tag{5.5.43}$$

and such that

$$X^0(t) = x_0(t) > 0\ ;\ t \in [-\delta, 0], \tag{5.5.44}$$

where $x_0(t)$ is a given bounded deterministic function.

Suppose that at time $t \in [0,T]$ we consume at the rate $c(t) \geq 0$, a càdlàg adapted process. Then the dynamics of the corresponding net cash flow $X(t) = X^c(t)$ is

$$\begin{aligned} dX(t) &= [X(t-\delta)\alpha(t) - c(t)]dt + X(t-\delta)\beta(t)dB(t) \\ &\quad + X(t-\delta)\int_{\mathbb{R}} \gamma(t,z)\tilde{N}(dt,dz)\ ;\ t \in [0,T] \\ X(t) &= x_0(t)\ ;\ t \in [-\delta, 0]. \end{aligned} \tag{5.5.45}$$

Let $U_1(t,c,\omega) : [0,T] \times \mathbb{R}^+ \times \Omega \to \mathbb{R}$ be a given stochastic utility function satisfying the following conditions

$$\begin{aligned} t &\to U_1(t,c,\omega) \text{ is } \mathcal{F}_t\text{-adapted for each } c \geq 0, \\ c &\to U_1(t,c,\omega) \text{ is } \mathcal{C}^1,\ \frac{\partial U_1}{\partial c}(t,c,\omega) > 0, \\ c &\to \frac{\partial U_1}{\partial c}(t,c,\omega) \text{ is strictly decreasing}, \\ \lim_{c\to\infty} &\frac{\partial U_1}{\partial c}(t,c,\omega) = 0 \text{ for all } t,\omega \in [0,T] \times \Omega. \end{aligned} \tag{5.5.46}$$

Put $v_0(t,\omega) = \dfrac{\partial U_1}{\partial c}(t,0,\omega)$ and define

$$I(t,v,\omega) = \begin{cases} 0 & \text{if } v \geq v_0(t,\omega), \\ \left(\dfrac{\partial U_1}{\partial c}(t,\cdot,\omega)\right)^{-1}(v) & \text{if } 0 \leq v < v_0(t,\omega). \end{cases} \tag{5.5.47}$$

Suppose we want to find a consumption rate $\hat{c}(t)$ such that

$$J(\hat{c}) = \sup\{J(c)\ ;\ c \in \mathcal{A}\}, \tag{5.5.48}$$

where

$$J(c) = E\left[\int_0^T U_1(t, c(t), \omega)dt + kX(T)\right].$$

Here $k > 0$ is constant and $\mathcal{A}$ is the family of all càdlàg, $\mathcal{F}_t$-adapted processes $c(t) \geq 0$ such that $E[|X(T)|] < \infty$.

In this case the Hamiltonian given by (5.5.6) gets the form

$$H(t, x, y, a, u, p, q, r(\cdot), \omega) = U_1(t, c, \omega) + (\alpha(t)y - c)p + y\beta(t)q + y\int_{\mathbb{R}} \gamma(t, z)r(z)\nu(dz). \tag{5.5.49}$$

Maximizing H with respect to c gives the following first-order condition for an optimal $\hat{c}(t)$:

$$\frac{\partial U_1}{\partial c}(t, \hat{c}(t), \omega) = p(t). \tag{5.5.50}$$

The time-advanced BSDE for $p(t), q(t), r(t, z)$ is, by (5.5.8)–(5.5.9),

$$\begin{aligned} dp(t) = &-E[\{\alpha(t)p(t+\delta) + \beta(t)q(t+\delta) \\ &+ \int_{\mathbb{R}} \gamma(t, z)r(t+\delta, z)\nu(dz)\Big\}\chi_{[0,T-\delta]}(t)|\mathcal{F}_t]dt \\ &+ q(t)dB(t) + \int_{\mathbb{R}} r(t, z)\tilde{N}(dt, dz)\ ;\ t \in [0, T] \end{aligned} \tag{5.5.51}$$

$$p(T) = k. \tag{5.5.52}$$

Since k is deterministic, we can choose $q = r = 0$ and (5.5.51)–(5.5.52) becomes

$$dp(t) = -\alpha(t)p(t+\delta)\chi_{[0,T-\delta]}(t)dt\ ;\ t < T, \tag{5.5.53}$$

$$p(t) = k \quad \text{for}\, t \in [T-\delta, T+\delta]. \tag{5.5.54}$$

To solve this we introduce

$$h(t) := p(T-t)\ ;\ t \in [-\delta, T].$$

Then

$$\begin{aligned} dh(t) &= -dp(T-t) = \alpha(T-t)p(T-t+\delta)dt \\ &= \alpha(T-t)p(T-(t-\delta))dt = \alpha(T-t)h(t-\delta)dt \end{aligned} \tag{5.5.55}$$

for $t \in [0, T]$, and

$$h(t) = p(T-t) = k \text{ for } t \in [-\delta, 0]. \tag{5.5.56}$$

This determines $h(t)$ inductively on each interval $[j\delta, (j+1)\delta]$; $j = 1, 2, \ldots$, as follows: If $h(s)$ is known on $[(j-1)\delta, j\delta]$, then for $t \in [j\delta, (j+1)\delta]$ we have

$$h(t) = h(j\delta) + \int_{j\delta}^{t} h'(s)ds = h(j\delta) + \int_{j\delta}^{t} \alpha(T-s)h(s-\delta)ds. \; ; \; t \in [j\delta, (j+1)\delta]. \tag{5.5.57}$$

We have proved:

Proposition 5.28 (***Optimal consumption rate in a stochastic system with delay***) *[ØSZ3] The optimal consumption rate $\hat{c}_\delta(t)$ for the problem (5.5.45), (5.5.48) is given by*

$$\hat{c}_\delta(t) = I(t, h_\delta(T-t), \omega), \tag{5.5.58}$$

where $h_\delta(\cdot) = h(\cdot)$ is determined by (5.5.56)–(5.5.57).

Remark 5.29 Assume that $\alpha(t) = \alpha > 0$ for all $t \in [0, T]$. Then we see by induction on (5.5.57) that

$$0 \le \delta_1 < \delta_2 \Rightarrow h_{\delta_1}(t) > h_{\delta_2(t)} \text{ for all } t \in (0, T]$$

and hence, perhaps surprisingly,

$$0 \le \delta_1 < \delta_2 \Rightarrow \hat{c}_{\delta_1}(t) < \hat{c}_{\delta_2}(t) \text{ for all } t \in [0, T).$$

Thus the optimal consumption rate *increases* if the delay increases. The explanation for this may be that the delay postpones the negative effect on the growth of the cash flow caused by the consumption.

5.6 Exercises

Exercise* 5.1 Suppose the wealth $X(t) = X^{(u)}(t)$ of a person with consumption rate $u(t) \ge 0$ satisfies the following Lévy type mean reverting Ornstein–Uhlenbeck SDE

$$\begin{aligned} &\mathrm{d}X(t) = (\mu - \rho X(t) - u(t))\mathrm{d}t + \sigma\,\mathrm{d}B(t) + \theta\int_{\mathbb{R}} z\widetilde{N}(\mathrm{d}t, \mathrm{d}z), \quad t > 0, \\ &X(0) = x > 0. \end{aligned}$$

Fix $T > 0$ and define

$$J^{(u)}(s,x) = E^{s,x}\Big[\int_0^{T-s} \mathrm{e}^{-\delta(s+t)}\frac{u^\gamma(t)}{\gamma}\mathrm{d}t + \lambda X(T-s)\Big].$$

Use dynamic programming to find the value function $\Phi(s,x)$ and the optimal consumption rate (control) $u^*(t)$ such that

$$\Phi(s,x) = \sup_{u(\cdot)} J^{(u)}(s,x) = J^{(u^*)}(s,x).$$

In the above $\mu, \rho, \sigma, \theta, T, \delta > 0, \gamma \in (0,1)$, and $\lambda > 0$ are constants.

Exercise* 5.2 Solve the problem of Exercise 5.1 by using the stochastic maximum principle.

Exercise* 5.3 Define

$$\mathrm{d}X^{(u)}(t) = \mathrm{d}X(t) = \begin{bmatrix}\mathrm{d}X_1(t)\\ \mathrm{d}X_2(t)\end{bmatrix} = \begin{bmatrix}u(t,\omega)\int_{\mathbb{R}} z\widetilde{N}(\mathrm{d}t,\mathrm{d}z)\\ \int_{\mathbb{R}} z^2\widetilde{N}(\mathrm{d}t,\mathrm{d}z)\end{bmatrix} \in \mathbb{R}^2$$

and, for fixed $T > 0$ (deterministic),

$$J(u) = E\big[-(X_1(T) - X_2(T))^2\big].$$

Use the stochastic maximum principle to find u^* such that

$$J(u^*) = \sup_u J(u).$$

Interpretation. Put $F(\omega) = \int_{\mathbb{R}} z^2\widetilde{N}(T,\mathrm{d}z)$. We may regard F as a given T-claim in the normalized market with the two investment possibilities bond and stock, whose prices are

(bond) $\mathrm{d}S_0(t) = 0, \quad S_0(0) = 1,$
(stock) $\mathrm{d}S_1(t) = \int_{\mathbb{R}} z\widetilde{N}(\mathrm{d}t,\mathrm{d}z),$ a Lévy martingale.

Then $-J(u)$ is the variance of the difference between $F = X_2(T)$ and the wealth $X_1(T)$ generated by a self-financing portfolio $u(t,\omega)$. See [BDLØP] for more information on minimal variance hedging in markets driven by Lévy martingales.

Exercise* 5.4 Solve the stochastic control problem

$$\Phi_1(s,x) = \inf_{u(t)\in\mathbb{R}} E^{s,x}\Big[\int_0^\infty \mathrm{e}^{-\rho(s+t)}(X^2(t) + \theta u^2(t))\mathrm{d}t\Big],$$

where

$$\mathrm{d}X(t) = u(t)\mathrm{d}t + \sigma \mathrm{d}B(t) + \int_{\mathbb{R}} z\tilde{N}(\mathrm{d}t, \mathrm{d}z),\ X(0) = x,$$

where $\rho > 0$, $\theta > 0$, and $\sigma > 0$ are constants.

The interpretation of this problem is that we want to push the process $X(t)$ as close as possible to 0 by using a minimum of energy, its rate being measured by $\theta u^2(t)$.

[*Hint: Try* $\varphi(s, x) = \mathrm{e}^{-\rho s}(ax^2 + b)$ *for some constants* a, b.]

Exercise* 5.5 (***The Stochastic Linear Regulator Problem***) Solve the stochastic control problem

$$\Phi_0(x) = \inf_u E^x \left[\int_0^T (X^2(t) + \theta u(t)^2)\mathrm{d}t + \lambda X^2(T)\right],$$

where

$$\mathrm{d}X(t) = u(t)\mathrm{d}t + \sigma \mathrm{d}B(t) + \int_{\mathbb{R}} z\tilde{N}(\mathrm{d}t, \mathrm{d}z),\ X(0) = x$$

and

$$T > 0 \text{ is a constant.}$$

(a) By using dynamic programming (Theorem 5.1).
(b) By using the stochastic maximum principle (Theorem 5.4).

Exercise* 5.6 Solve the stochastic control problem

$$\Phi(s, x) = \sup_{c(t)\geq 0} E^{s,x}\left[\int_0^{\tau_0} \mathrm{e}^{-\delta(s+t)} \ln c(t)\mathrm{d}t\right],$$

where the supremum is taken over all $\mathcal{F}_t$-adapted processes $c(t) \geq 0$ and

$$\tau_0 = \inf\{t > 0;\ X(t) \leq 0\},$$

where

$$\mathrm{d}X(t) = X(t^-)\left[\mu\,\mathrm{d}t + \sigma\,\mathrm{d}B(t) + \theta\int_{\mathbb{R}} z\tilde{N}(\mathrm{d}t, \mathrm{d}z)\right] - c(t)\mathrm{d}t, \quad X(0) = x > 0,$$

where $\delta > 0$, μ, σ, and θ are constants, and

$$\theta z > -1 \text{ for a.a. } z \text{ w.r.t. } \nu.$$

We may interpret $c(t)$ as the consumption rate, $X(t)$ as the corresponding wealth, and τ_0 as the bankruptcy time. Thus Φ represents the maximal expected total discounted logarithmic utility of the consumption up to bankruptcy time.

[*Hint: Try* $\varphi(s,x) = \mathrm{e}^{-\delta s}(a \ln x + b)$ *as a candidate for* $\Phi(s,x)$, *where* a *and* b *are suitable constants.*]

Exercise 5.7 Use the stochastic maximum principle (Theorem 5.4) to solve the problem

$$\sup_{c(t)\geq 0} E\left[\int_0^T \mathrm{e}^{-\delta t} \ln c(t)\mathrm{d}t + \lambda \mathrm{e}^{-\delta T} \ln X(T)\right],$$

where

$$\mathrm{d}X(t) = X(t^-)\left[\mu\,\mathrm{d}t + \sigma\,\mathrm{d}B(t) + \theta\int_{\mathbb{R}} z\tilde{N}(\mathrm{d}t,\mathrm{d}z)\right] - c(t)\mathrm{d}t, \quad X(0) > 0.$$

Here $\delta > 0$, $\lambda > 0$, μ, σ, and θ are constants, and

$$\theta z > -1 \text{ for a.a. } z(\nu)$$

(see Exercise 5.6 for an interpretation of this problem).
[*Hint: Try* $p(t) = a\mathrm{e}^{-\mu t}X^{-1}(t)$ *and* $c(t) = e^{-\delta t}p^{-1}(t) = e^{(\mu-\delta)t}X(t)/a$, *for some constant* $a > 0$.]

Exercise 5.8 (***Optimal Portfolio Problem***) Suppose a financial market has two investment possibilities: a risk free asset with unit price

$$S_0(t) = 1 \text{ for all } t,$$

and a risky asset, with unit price $S(t)$ given by the SDE

$$\begin{cases} dS(t) = S(t^-)\left[\alpha(t)dt + \beta(t)dB(t) + \displaystyle\int_{\mathbb{R}} \gamma(t,\zeta)\tilde{N}(dt,d\zeta)\right]; \quad t \in [0,T] \\ S(0) > 0, \end{cases} \tag{5.6.1}$$

where $\alpha(t), \beta(t), \gamma(t,\zeta) > -1$ are given bounded deterministic functions.

Suppose we introduce a portfolio $\pi(t)$ in this market, representing the fraction of the total wealth $X(t)$ invested in the risky asset at time t. Then the corresponding equation for the wealth process $X(t) = X^{\pi}(t)$ will be

$$\begin{cases} dX(t) = \pi(t)X(t^-)[\alpha(t)dt + \beta(t)dB(t) + \displaystyle\int_{\mathbb{R}} \gamma(t,\zeta)\tilde{N}(dt,d\zeta)]; \quad t \in [0,T] \\ X(0) = x_0 > 0. \end{cases} \tag{5.6.2}$$

Let $U : (0,\infty) \mapsto [-\infty,\infty)$ be a given utility function, assumed to be C^1 and concave. We want to find the portfolio $\pi^*(t)$ which maximizes the expected utility of the terminal wealth, i.e. such that

$$\sup_{\pi} J_1(\pi) = J_1(\pi^*), \tag{5.6.3}$$

where $J_1(\pi)$ is the *performance functional* defined by

$$J_1(\pi) = E[U(X^{\pi}(T))]. \tag{5.6.4}$$

1. Use dynamic programming and the HJB equation to solve this problem in the following cases (discuss possible admissible conditions for the control):

   (a) (Logarithmic utility) $N = 0$ and $U(x) = \ln(x)$
   Hint: Try a value function of the form $\varphi(s, x) = h(s) + \ln(x)$, *for some function* h.

   (b) (Power utility) $N = 0$ and $U(x) = \frac{1}{\rho}x^{\rho}$, where $\rho \in (-\infty, 1)$ is a constant
   Hint: Try a value function of the form $\varphi(s, x) = k(s) \times \frac{1}{\rho}x^{\rho}$, *for some function* k.

   (c) (Exponential utility) $N = 0$ and $U(x) = \exp(-\alpha x)$, where $\alpha > 0$ is a constant

   (d) (Jump diffusion case) $N \neq 0$ and $U(x) = \ln(x)$.
2. Use now the maximum principle to solve the optimal portfolio problem for the various cases given above.
   Hint to case (d): In the adjoint BSDE for (p, q, r), *try to put* $q(t) = p(t)\theta_0(t)$ *and* $r(t, \zeta) = p(t^-)\theta_1(t, \zeta)$ *for some processes* θ_0, θ_1.

Exercise 5.9 (***Optimal Consumption Problem***) Suppose we are given a cash flow $X^0(t)$ of the form

$$\begin{cases} dX^0(t) = X^0(t^-)\left[\alpha(t)dt + \beta(t)dB(t) + \int_{\mathbb{R}} \gamma(t, \zeta)\tilde{N}(dt, d\zeta)\right]; \quad t \in [0, T] \\ X^0(0) = x_0 > 0, \end{cases} \tag{5.6.5}$$

with $\alpha(t), \beta(t), \gamma(t, \zeta)$ as in Exercise 5.8. If we introduce a *relative consumption rate* $c(t)$ from this wealth process, the corresponding wealth $X(t) = X^c(t)$ gets the dynamics

$$\begin{aligned} dX(t) &= X(t^-)\left[(\alpha(t) - c(t))dt + \beta(t)dB(t) + \int_{\mathbb{R}} \gamma(t, \zeta)\tilde{N}(dt, d\zeta)\right]; \ t \in [0, T] \\ X(0) &= x_0 > 0. \end{aligned}$$

We want to find the relative consumption rate $c^*(t)$ which maximizes the total expected utility from consumption up to time T, i.e. we want to find $c^*(t)$ such that

$$\sup_{c} J_2(c) = J_2(c^*), \tag{5.6.6}$$

where $J_2(c)$ is the performance functional defined by

$$J_2(c) = E\Big[\int_0^T U(c(t)X^c(t))dt + \lambda U(X^c(T))\Big], \tag{5.6.7}$$

where $\lambda > 0$ is a given constant.

1. Use dynamic programming and the HJB equation to solve this problem in the following cases:

   (a) (Logarithmic utility) $N = 0$ and $U(x) = \ln(x)$
   Hint: Try a value function of the form $\varphi(s, x) = f(s)\ln(x) + g(s)$, *for some functions* f, g.

   (b) (Power utility) $N = 0$ and $U(x) = \frac{1}{\rho}x^\rho$, where $\rho \in (-\infty, 1)$.

   (c) (Exponential utility) $N = 0$ and $U(x) = \exp(-\alpha x)$, where $\alpha > 0$.

   (d) (Jump diffusion case) $N \neq 0$ and $U(x) = \ln(x)$.
2. Now use the maximum principle to solve the optimal consumption problem in the various cases given above.
   Hint to case (a): In the adjoint BSDE for (p, q) *try to put* $q(t) = p(t)\theta_0(t)$ *for some process* θ_0.

Exercise 5.10 (***Infinite Horizon Consumption Problem***) We consider an infinite horizon problem as in Example 5.9. Solve this problem with the performance functional (5.3.15) replaced by

$$J(u) = E\left[\int_0^\infty e^{-\rho t}\ln u(t)dt\right]$$

and with $\sigma(t, x, u) = \sigma_0(t)X$.

Chapter 6
Stochastic Differential Games



6.1 Stochastic Differential (Markov) Games, HJB–Isaacs Equations

In this section we present the dynamic programming approach to stochastic differential games. We only present the case for zero sum games. For the extension to non-zero sum games, we refer to [MØ].

Suppose the state $Y(t) = Y_y^{(u)}(t) \in \mathbb{R}^k$ is described by a controlled Markovian SDE of the form

$$\begin{cases} \mathrm{d}Y(t) &= b(Y(t), u(t))\mathrm{d}t + \sigma(Y(t), u(t))\mathrm{d}B(t) \\ &\quad + \int_{\mathbb{R}^l} \gamma(Y(t), u(t), \zeta)\tilde{N}(\mathrm{d}t, \mathrm{d}\zeta); t \geq 0 \\ Y(0) &= y \in \mathbb{R}^k. \end{cases} \tag{6.1.1}$$

Here $u = (u_1, u_2)$ is a control process with values in $V_1 \times V_2 \subset \mathbb{R}^{n_1} \times \mathbb{R}^{n_2}$. The control u_i is chosen by player number i, and $\mathcal{A}_i$ is the set of admissible control processes for player number i; $i = 1, 2$. We put $\mathcal{A} = \mathcal{A}_1 \times \mathcal{A}_2$. Let

$$f : \mathbb{R}^k \times V_1 \times V_2 \to \mathbb{R}$$

be a given profit rate and let

$$g : \mathbb{R}^k \to \mathbb{R}$$

be a given bequest function. We assume that

$$E^y\left[\int_0^{\tau_S} |f(Y(t), u(t))|\mathrm{d}t + |g(Y(\tau_S))|\right] < \infty$$

for all y, where $E^y[\varphi(Y(t))]$ means $E[\varphi(Y_y^{(u)}(t))]$.

Here

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$$\tau_{\mathcal{S}} := \inf\{t > 0 \,;\; Y_y^{(u)}(t) \notin \mathcal{S}\}, \tag{6.1.2}$$

with $\mathcal{S} \in \mathbb{R}^k$ being a given *solvency region*. We may interpret $\tau_{\mathcal{S}}$ as the *bankruptcy time*.

Let $\mathcal{A}_1, \mathcal{A}_2$ be given families of admissible control processes u_1, u_2 for player number 1 and 2, respectively

Problem 6.1 (***Zero-sum stochastic differential games***)
Find $\varphi(y)$ *and* $u_i^* \in \mathcal{A}_{\epsilon_i}$; $i = 1, 2$ *such that*

$$\Phi(y) = \sup_{u_1 \in \mathcal{A}_1} \left(\inf_{u_2 \in \mathcal{A}_2} J^u(y) \right) = J^{u_1^*, u_2^*}(y), \tag{6.1.3}$$

where

$$J^u(y) = J^{u_1,u_2}(y) = E\left[\int_0^{\tau_S} f(Y(s), u(s))\mathrm{d}s + g(Y(\tau_S))\chi_{\tau_S < \infty} \right] \tag{6.1.4}$$

is the performance functional.

As in classical Markovian stochastic control we can restrict ourselves to consider Markov controls (feedback controls), i.e. we assume that

$$u_i(t) = \tilde{u}_i(Y(t))$$

for some deterministic function $\tilde{u}_i : \mathbb{R}^k \to \mathbb{R}$; $i = 1, 2$. As is customary we do not distinguish notationally between u_i and $\tilde{u}_i$.

When the control $u = (u_1, u_2) \in \mathcal{A}_1 \times \mathcal{A}_2$ is Markovian, the corresponding controlled process $Y^{(u)}(t)$ will be a (Markovian) jump diffusion with generator A^u given by

$$\begin{aligned} A^u\varphi(y) &= \sum_{i=1}^k b_i(y, u(y)) \frac{\partial \varphi}{\partial y_i} + \frac{1}{2} \sum_{i,j=1}^k (\sigma\sigma^T)_{ij}(y, u(y)) \frac{\partial^2 \varphi}{\partial y_i \partial y_j}(y) \\ &+ \sum_{j=1}^{\ell} \int_{\mathbb{R}^k} \{\varphi(y + \gamma^{(j)}(y, u(y), \zeta)) - \varphi(y) - \nabla\varphi(y)\gamma^{(j)}(y, u(y), \zeta)\}\nu_j(d\zeta) \end{aligned} \tag{6.1.5}$$

for all smooth functions $\varphi : \mathbb{R}^k \to \mathbb{R}$ with compact support, where $\gamma^{(j)}$ denotes column number j in the $k \times \ell$ matrix $\gamma = [\gamma_{ij}]_{\substack{1 \le i \le k \\ 1 \le j \le \ell}}$.

We now state the main result of this section.

Theorem 6.1 ([MØ]) (***The Hamilton–Jacobi–Bellman–Isaacs (HJBI) equation***)
Suppose there exists a function $\varphi \in C^2(\mathcal{S}) \times C(\mathbb{R}^k)$ *and a Markov control* $\hat{u} = (\hat{u}_1, \hat{u}_2) \in \mathcal{A}_1 \times \mathcal{A}_2$ *such that*

(i) $A^{\hat{u}_1,v_2}\varphi(y) + f(y,\hat{u}_1(y),v_2) \geq 0$ *for all* $v_2 \in V_2,\ y \in \mathcal{S}$,
(ii) $A^{v_1,\hat{u}_2}\varphi(y) + f(y,v_1,\hat{u}_2(y)) \leq 0$ *for all* $v_1 \in V_1;\ y \in \mathcal{S}$,
(iii) $A^{\hat{u}_1,\hat{u}_2}\varphi(y) + f(y,\hat{u}_1(y),\hat{u}_2(y)) = 0$ *for all* $y \in \mathcal{S}$,
(iv) $\varphi(y) = g(y)$ *for all* $y \notin \mathcal{S}$,
(v) The family $\{\varphi(Y^{(u)}(\tau))\}_{\tau\in\mathcal{T}}$ *is uniformly integrable, for all* $y \in \mathcal{S}, u \in \mathcal{A}_1 \times \mathcal{A}_2$. *Here* $\mathcal{T}$ *denotes the set of all* $\mathcal{F}$*-stopping times* $\tau \leq \tau_{\mathcal{S}}$.

Then

$$\begin{aligned}\varphi(y) = \Phi(y) &:= \sup_{u_1\in\mathcal{A}_1}\left(\inf_{u_2\in\mathcal{A}_2} J^u(y)\right)\\ &= \inf_{u_2\in\mathcal{A}_2}\left(\sup_{u_1\in\mathcal{A}_1} J^u(y)\right) = \sup_{u_1\in\mathcal{A}_1} J^{u_1,\hat{u}_2}(y)\\ &= \inf_{u_2\in\mathcal{A}_2} J^{\hat{u}_1,u_2}(y) = J^{\hat{u}_1,\hat{u}_2}(y) \qquad (6.1.6)\end{aligned}$$

and

$$u^* := (\hat{u}_1,\hat{u}_2) \text{ is an optimal control.} \qquad (6.1.7)$$

Proof Choose $u \in \mathcal{A}$. Then by the Dynkin formula for jump diffusions (see Theorem 1.24) we have

$$E^y[\varphi(Y^u(\tau_{\mathcal{S}}^{(N)}))] = \varphi(y) + E^y\left[\int_0^{\tau_{\mathcal{S}}^{(N)}} A^u\varphi(Y^u(t))\mathrm{d}t\right],$$

where

$$\tau_{\mathcal{S}}^{(N)} = \tau_{\mathcal{S}} \wedge N \wedge \inf\{t > 0\ ;\ |Y(t)| \geq N\}\ ;\ N = 1,2,\ldots$$

Hence, by (i) we get, with $u_i(s) = u_i(Y(s))$,

$$E^y[\varphi(Y^{\hat{u}_1,u_2}(\tau_{\mathcal{S}}^{(N)}))] \geq \varphi(y) - E^y\left[\int_0^{\tau_{\mathcal{S}}^{(N)}} f(Y^{\hat{u}_1,u_2}(s),u_1(s),\hat{u}_2(s))\mathrm{d}s\right].$$

Therefore,

$$\begin{aligned}\varphi(y) &\leq E^y\left[\int_0^{\tau_{\mathcal{S}}^{(N)}} f(Y^{\hat{u}_1,u_2}(s),\hat{u}_1(s),u_2(s))\mathrm{d}s + \varphi(Y^{\hat{u}_1,u_2}(\tau_{\mathcal{S}}^{(N)}))\right]\\ &\to J^{\hat{u}_1,u_2}(y) \text{ as } N \to \infty. \qquad (6.1.8)\end{aligned}$$

Here we have used condition (v) and the fact that $Y(\cdot)$ is quasi-left continuous (i.e. left continuous at increasing stopping times; see [JS], Propositions I.2.26 and I.3.27). Since $u_2 \in \mathcal{A}_2$ was arbitrary, we deduce from (6.1.8) that

$$\varphi(y) \leq \inf_{u_2 \in \mathcal{A}_2} J^{\hat{u}_1, u_2}(y). \tag{6.1.9}$$

It follows that

$$\varphi(y) \leq \sup_{u_1 \in \mathcal{A}_1} \left(\inf_{u_2 \in \mathcal{A}_2} J^{u_1, u_2}(y) \right) = \Phi(y). \tag{6.1.10}$$

Similarly, applying (ii) we get

$$\varphi(y) \geq J^{u_1, \hat{u}_2}(y) \text{ for all } u_1 \in \mathcal{A}_1 \tag{6.1.11}$$

and therefore

$$\varphi(y) \geq \sup_{u_1 \in \mathcal{A}_1} J^{u_1, \hat{u}_2}(y) \geq \inf_{u_2 \leq \mathcal{A}_2} \left(\sup_{u_1 \in \mathcal{A}_1} J^{u_1, u_2}(y) \right). \tag{6.1.12}$$

In the same way, applying (iii) we get

$$\varphi(y) = J^{\hat{u}_1, \hat{u}_2}(y). \tag{6.1.13}$$

By combining (6.1.9)–(6.1.13) we obtain

$$\begin{aligned} \varphi(y) = \inf_{u_2 \in \mathcal{A}_2} J^{\hat{u}_1, u_2}(y) &\leq \sup_{u_1 \in \mathcal{A}_1} \left(\inf_{u_2 \in \mathcal{A}_2} J^{u_1, u_2}(y) \right) = \Phi(y) \\ &\leq \inf_{u_2 \in \mathcal{A}_2} \left(\sup_{u_1 \in \mathcal{A}_1} J^{u_1, u_2}(y) \right) \leq \sup_{u_1 \in \mathcal{A}_1} J^{u_1, \hat{u}_2}(y) \leq \varphi(y). \end{aligned}$$

We conclude that $\varphi(y) = \Phi(y)$, that (6.1.6) holds and that $u^* := (\hat{u}_1, \hat{u}_2)$ is optimal. □

6.1.1 *Entropic Risk Minimization Example*

We now return to the problem of risk minimization in a financial market, but now the risk β is represented by (4.5.5). If $X_\beta(t)$ is the wealth processes corresponding to a portfolio β, then the risk minimization problem becomes

$$\inf_{\beta \in \mathcal{A}} \left(\sup_{Q \in \mathcal{P}} \{E_Q[-X_\beta(T)] - \alpha(Q)\} \right). \tag{6.1.14}$$

This is a min-max problem, which in our setting is an example of a *stochastic differential game*:

Suppose the financial market is as in (4.1.1), but with $r = 0$ and with jumps added, i.e.

$$\begin{cases} S_0(t) = 1 \text{ for all } t \\ \mathrm{d}S_1(t) = S_1(t^-)\left[\mu(t)\mathrm{d}t + \sigma(t)\mathrm{d}B(t) + \int_{\mathbb{R}} \gamma(t,\zeta)\tilde{N}(\mathrm{d}t,\mathrm{d}\zeta)\right]; t \geq 0 \\ S_1(0) > 0. \end{cases} \tag{6.1.15}$$

If $\beta(t)$ is a self-financing portfolio representing the number of units of the risky asset (with unit price $S_1(t)$) held at time t, the corresponding wealth process $X(t) = X_\beta(t)$ is given by

$$\begin{aligned} \mathrm{d}X(t) &= \beta(t)\mathrm{d}S_1(t) \\ &= w(t)\left[\mu(t)\mathrm{d}t + \sigma(t)\mathrm{d}B(t) + \int_{\mathbb{R}} \gamma(t,\zeta)\tilde{N}(\mathrm{d}t,\mathrm{d}\zeta)\right] \\ &= \mathrm{d}X^w(t), \end{aligned} \tag{6.1.16}$$

where $w(t) := \beta(t)S_1(t^-)$ is the *amount* held in the risky asset at time t.

If we use the representation (4.5.5) of the risk measure $\rho = \rho_e$ corresponding to the family $\mathcal{P}_\Theta$ of measures Q given in (4.5.6)–(4.5.8) and the entropic penalty α_e given by (4.5.9), the risk minimizing portfolio problem (6.1.14) can be written

$$\inf_{w\in\mathcal{W}}\left(\sup_{\theta\in\Theta}\left\{E\left[-\frac{\mathrm{d}Q}{\mathrm{d}P}X^w(T) - \frac{\mathrm{d}Q}{\mathrm{d}P}\log\left(\frac{\mathrm{d}Q}{\mathrm{d}P}\right)\right]\right\}\right), \tag{6.1.17}$$

where $\mathcal{W}$ is the family of admissible portfolios w. To put this problem into the setting above, we represent Q by $\frac{\mathrm{d}Q}{\mathrm{d}P} = M^\theta(T)$ and we put

$$\begin{aligned} \mathrm{d}Y(t) = \mathrm{d}Y^{\theta,w}(t) = \begin{bmatrix} \mathrm{d}t \\ \mathrm{d}X^w(t) \\ \mathrm{d}M^\theta(t) \end{bmatrix} &= \begin{bmatrix} 1 \\ w(t)\mu(t) \\ 0 \end{bmatrix}\mathrm{d}t + \begin{bmatrix} 0 \\ w(t)\sigma(t) \\ M^\theta(t)\theta_0(t) \end{bmatrix}\mathrm{d}B(t) \\ &\quad + \int_{\mathbb{R}} \begin{bmatrix} 0 \\ w(t)\gamma(t,\zeta) \\ M^\theta(t^-)\theta_1(t_1,\zeta) \end{bmatrix}\tilde{N}(\mathrm{d}t,\mathrm{d}\zeta), \end{aligned} \tag{6.1.18}$$

with initial value

$$Y^{\theta,w}(0) = y = \begin{bmatrix} s \\ x \\ m \end{bmatrix} \; ; \; s \in [0,T], x > 0, m > 0. \tag{6.1.19}$$

In this case the solvency region is $\mathcal{S} = [0,T] \times \mathbb{R}_+ \times \mathbb{R}_+$ and the performance functional is

$$J^{\theta,w}(s,x,m) = E^{s,x,m}[-M^{\theta}(T)X^{w}(T) - M^{\theta}(T)\log M^{\theta}(T)]. \tag{6.1.20}$$

Assume from now on that

$$\mu(t),\ \sigma(t) \text{ and } \gamma(t,\zeta) \text{ are deterministic.} \tag{6.1.21}$$

Then $Y^{\theta,w}(t)$ becomes a controlled jump diffusion, and the risk minimization problem (6.1.14) is the following special case of Problem 6.1:

Problem 6.2 (***Entropic Risk Minimization***)
Find $w^* \in \mathcal{W}$, $\theta^* \in \Theta$ and $\Phi(y)$ such that

$$\Phi(y) = \inf_{w\in\mathcal{W}} \sup_{\theta\in\Theta} J^{\theta,w}(y) = J^{\theta^*,w^*}(y)\ ;\ \ y \in \mathcal{S}. \tag{6.1.22}$$

By (6.1.18) we see that the generator $A^{\theta,w}$ is given by (see (6.1.5))

$$\begin{aligned} A^{\theta,w}\varphi(s,x,m) &= \frac{\partial\varphi}{\partial s}(s,x,m) + w\mu(s)\frac{\partial\varphi}{\partial x}(s,x,m) \\ &+ \frac{1}{2}w^2\sigma^2(s)\frac{\partial^2\varphi}{\partial x^2}(s,x,m) + \frac{1}{2}m^2\theta_0^2\frac{\partial^2\varphi}{\partial m^2}(s,x,m) + w\theta_0 m\sigma(s)\frac{\partial^2\varphi}{\partial x\partial m}(s,x,m) \\ &+ \int_{\mathbb{R}} \Big\{\varphi(s,x+w\gamma(s,\zeta),m+m\theta_1(\zeta)) - \varphi(s,x,m) \\ &- \frac{\partial\varphi}{\partial x}(s,x,m)w\gamma(s,\zeta) - \frac{\partial\varphi}{\partial m}(s,x,m)m\theta_1(\zeta)\Big\}\nu(\mathrm{d}\zeta). \end{aligned} \tag{6.1.23}$$

Comparing with the general formulation in (6.1.4), we see that in this case

$$f = 0 \text{ and } g(y) = g(x,m) = -mx - m\log(m).$$

Therefore, according to Theorem 6.1, we should try to find a function $\varphi(s,x,m) \in C^2(\mathcal{S}) \cap C(\mathbb{R}^3)$ and control values $\theta = \hat{\theta}(y)$, $w = \hat{w}(y)$ such that

$$\inf_{w\in\mathbb{R}} \sup_{\theta\in\mathbb{R}^2} A^{\theta,w}\varphi(y) = A^{\hat{\theta},\hat{w}}\varphi(y)\ ;\ \ y \in \mathcal{S} \tag{6.1.24}$$

and

$$\lim_{t\to T^-} \varphi(s,x,m) = -xm - m\log(m). \tag{6.1.25}$$

Let us try a function of the form

$$\varphi(s,x,m) = -xm - m\log(m) + \kappa(s)m, \tag{6.1.26}$$

where κ is a deterministic function, $\kappa(T) = 0$. Then by (6.1.23)

$$
\begin{aligned}
A^{\theta,w}\varphi(s,x,m) &= \kappa'(s)m + m\mu(t)(-m) + \frac{1}{2}m^2\theta_0^2\left(-\frac{1}{m}\right) + w\theta_0 m\sigma(t)(-1)\\
&+ \int_{\mathbb{R}}\{-(x + w\gamma(s,\zeta))(m + m\theta_1(\zeta)) + xm - (m + m\theta_1(\zeta))\log(m + m\theta_1(\zeta))\\
&+ m\log m + \kappa(s)(m + m\theta_1(\zeta)) - \kappa(s)m + mw\gamma(s,\zeta)\\
&- m\theta_1(\zeta)(-x - 1 - \log m + \kappa(s))\}\nu(\mathrm{d}\zeta)\\
&= m\left[\kappa'(s) - w\mu(t) - \frac{1}{2}\theta_0^2 - w\theta_0\sigma(t)\right.\\
&\left.+ \int_{\mathbb{R}}\theta_1(\zeta)\{1 - \log(1 + \theta_1(\zeta)) - w\gamma(s,\zeta)\}\nu(\mathrm{d}\zeta)\right]. \qquad (6.1.27)
\end{aligned}
$$

Maximizing $A^{\theta,w}\varphi(y)$ with respect to $\theta = (\theta_0, \theta_1)$ and minimizing with respect to w gives the following first-order equations

$$\hat{\theta}_0(s) + \hat{w}(s)\sigma(s) = 0, \qquad (6.1.28)$$

$$1 - \log(1 + \hat{\theta}_1(s,\zeta)) - \hat{w}(s)\gamma(s,\zeta) - \frac{\hat{\theta}_1(s,\zeta)}{1 + \hat{\theta}_1(s,\zeta)} = 0, \qquad (6.1.29)$$

$$\mu(s) + \hat{\theta}_0(s)\sigma(s) - \int_{\mathbb{R}}\hat{\theta}_1(s,\zeta)\gamma(s,\zeta)\nu(\mathrm{d}\zeta) = 0. \qquad (6.1.30)$$

These are 3 equations in the 3 unknown candidates $\hat{\theta}_0, \hat{\theta}_1$ and $\hat{w}$ for the optimal control for the game in Problem 6.2. To get an explicit solution, let us now assume that

$$N = 0 \text{ and } \gamma = \theta_1 = 0. \qquad (6.1.31)$$

Then (6.1.28)–(6.1.30) gives

$$\hat{\theta}_0(s) = -\frac{\mu(s)}{\sigma(s)}, \hat{w}(s) = \frac{\mu(s)}{\sigma^2(s)}. \qquad (6.1.32)$$

Substituted into (6.1.27) we get by (iii) Theorem 6.1

$$A^{\hat{\theta},\hat{w}}\varphi(s,x,w) = m\left[\kappa'(s) - \frac{1}{2}\left(\frac{\mu(s)}{\sigma(s)}\right)^2\right] = 0.$$

Combining this with (iv) Theorem 6.1 we obtain

$$\kappa(s) = -\int_s^T \frac{1}{2}\left(\frac{\mu(t)}{\sigma(t)}\right)^2 \mathrm{d}t. \qquad (6.1.33)$$

Now all the conditions of Theorem 6.1 are satisfied, and we get:

Theorem 6.2 (***Entropic Risk Minimization***) *Assume that* (6.1.21) *and* (6.1.31) *hold. Then the solution of Problem 6.2 is*

$$\Phi(s,x,m) = -xm - m\log m - \int_s^T \frac{1}{2}\left(\frac{\mu(t)}{\sigma(t)}\right)^2 \mathrm{d}t \tag{6.1.34}$$

and the optimal controls are

$$\hat{\theta}_0(s) = -\frac{\mu(s)}{\sigma(s)} \text{ and } \hat{w}(s) = \frac{\mu(s)}{\sigma^2(s)}\ ;\ s \in [0,T]. \tag{6.1.35}$$

In particular, choosing the initial values $s = 0$ and $m = 1$ we get

$$\Phi(0,x,1) = -x - \int_0^T \frac{1}{2}\left(\frac{\mu(t)}{\sigma(t)}\right)^2 \mathrm{d}t. \tag{6.1.36}$$

This agrees with what we obtained in Proposition 5.20 using the stochastic HJB equation for optimal control of FBSDEs. See also Theorem 6.9.

Remark 6.3 Theorem 6.2 could also have been obtained by using a stochastic HJB approach to the optimal control of forward-backward SDEs. This was presented in Sect. 5.4.

6.2 Stochastic Maximum Principles

We first recall some basic concepts and results from Banach space theory. Let V be an open subset of a Banach space $\mathcal{X}$ with norm $\|\cdot\|$ and let $F : V \to \mathbb{R}$.

(i) We say that F has a directional derivative (or Gâteaux derivative) at $x \in X$ in the direction $y \in \mathcal{X}$ if

$$D_y F(x) := \lim_{\varepsilon \to 0} \frac{1}{\varepsilon}(F(x+\varepsilon y) - F(x))$$

exists.

(ii) We say that F is Fréchet differentiable at $x \in V$ if there exists a linear map

$$L : \mathcal{X} \to \mathbb{R}$$

such that

$$\lim_{\substack{h \to 0 \\ h \in \mathcal{X}}} \frac{1}{\|h\|} |F(x+h) - F(x) - L(h)| = 0.$$

In this case we call L the *gradient* (or Fréchet derivative) of F at x and we write

$$L = \nabla_x F.$$

(iii) If F is Fréchet differentiable, then F has a directional derivative in all directions $y \in \mathcal{X}$ and

$$D_y F(x) = \nabla_x F(y).$$

6.2.1 General (Non-zero) Stochastic Differential Games

In this section, we formulate and prove a sufficient and a necessary maximum principle for general stochastic differential games (not necessarily zero-sum games) of forward-backward SDEs.

Consider a controlled *forward* SDE of the form

$$\begin{aligned} \mathrm{d}X(t) = \mathrm{d}X^u(t) &= b(t, X(t), u(t), \omega)\mathrm{d}t + \sigma(t, X(t), u(t), \omega)\mathrm{d}B(t) \\ &+ \int_{\mathbb{R}} \gamma(t, X(t^-), u(t), \zeta, \omega)\tilde{N}(\mathrm{d}t, \mathrm{d}\zeta)\ ;\quad 0 \le t \le T\ ;\ X(0) = x \in \mathbb{R}. \end{aligned} \tag{6.2.1}$$

Here $u = (u_1, u_2)$, where $u_i(t)$ is the control of player i ; $i = 1, 2$. We assume that we are given two subfiltrations

$$\mathcal{E}_t^{(i)} \subseteq \mathcal{F}_t\ ;\ t \in [0, T], \tag{6.2.2}$$

representing the information available to player i at time t ; $i = 1, 2$. We let $\mathcal{A}_i$ denote a given set of admissible control processes for player i, contained in the set of $\mathcal{E}_t^{(i)}$-predictable processes; $i = 1, 2$, with values in $A_i \subset \mathbb{R}^d$, $d \ge 1$. Define $\mathbb{U} := A_1 \times A_2$ and $\mathcal{U} := \mathcal{A}_1 \times \mathcal{A}_2$. We assume that $b(\cdot, x, u, \omega)$, $\sigma(\cdot, , x, u, \omega)$, $\gamma(\cdot, , x, u, \zeta, \omega)$ are given predictable processes for each x in $\mathbb{R}$, u in $\mathcal{U}$, ζ in $\mathbb{R}_0 := \mathbb{R}\backslash\{0\}$ such that (6.2.1) has a unique solution for each u in $\mathcal{U}$.

We consider the associated controlled *backward* SDEs (i.e. BSDEs) in the unknowns $(Y_i^u(t), Z_i^u(t), K_i^u(t, \zeta)) = (Y_i(t), Z_i(t), K_i(t, \zeta))$ of the form

$$\begin{aligned} \mathrm{d}Y_i(t) &= -g_i(t, X(t), Y_i(t), Z_i(t), K_i(t, \cdot), u(t), \omega)dt \\ &+ Z_i(t)\mathrm{d}B(t) + \int_{\mathbb{R}} K_i(t, \zeta)\tilde{N}(\mathrm{d}t, \mathrm{d}\zeta)\ ;\ 0 \le t \le T \\ Y_i(T) &= h_i(X(T), \omega)\ ;\ i = 1, 2. \end{aligned} \tag{6.2.3}$$

Here $g_i(\cdot, x, y, z, k, u, \omega)$ are given predictable processes for each x in $\mathbb{R}$, y in $\mathbb{R}$, z in $\mathbb{R}$, k in $\mathcal{R}^{\mathbb{R}_0}$, u in $\mathbb{U}$, and where $h_i(x, \omega)$ is $\mathcal{F}_T$-measurable for each given x in $\mathbb{R}$, such that the BSDEs (6.2.3) have unique solutions for each u in $\mathbb{U}$.

Let $f_i(t, x, u) : [0, T] \times \mathbb{R} \times \mathbb{U} \to \mathbb{R}$, $\varphi_i(x) : \mathbb{R} \to \mathbb{R}$ and $\psi_i(x) : \mathbb{R} \to \mathbb{R}$ be given profit rates, bequest functions and "risk evaluations" respectively, of player

$i\ ;\ i = 1, 2$. Define for $i = 1, 2$

$$J_i(u) := E\left[\int_0^T f_i(t, X^u(t), u(t), \omega)\mathrm{d}t + \varphi_i(X^u(T), \omega) + \psi_i(Y_i^u(0))\right], \quad (6.2.4)$$

provided the integrals and expectations exist. We call $J_i(u)$ the *performance functional* of player $i\ ;\ i = 1, 2$. We assume that $b, \sigma, \gamma, g_i, h_i, f_i, \varphi_i$ and ψ_i are C^1 with respect to x, y, z, u and that

$$\psi_i'(x) \geq 0 \quad \text{for all } x; \quad i = 1, 2. \quad (6.2.5)$$

A *Nash equilibrium* for the FBSDE game (6.2.1)–(6.2.4) is a pair $(\hat{u}_1, \hat{u}_2) \in \mathcal{A}_1 \times \mathcal{A}_2$ such that

$$J_1(u_1, \hat{u}_2) \leq J_1(\hat{u}_1, \hat{u}_2) \text{ for all } u_1 \in \mathcal{A}_1 \quad (6.2.6)$$

and

$$J_2(\hat{u}_1, u_2) \leq J_2(\hat{u}_1, \hat{u}_2) \text{ for all } u_2 \in \mathcal{A}_2. \quad (6.2.7)$$

Heuristically, this means that player i has no incentive to deviate from the control $\hat{u}_i$, as long as player j ($j \neq i$) does not deviate from $\hat{u}_j\ ;\ i = 1, 2$. Therefore a Nash equilibrium is in some cases a likely outcome of a game.

Suppose there exists a Nash equilibrium $(\hat{u}_1, \hat{u}_2)$. We now present a method to find it, based on the maximum principle for stochastic control.

Define the *Hamiltonians*

$$H_i(t, x, y, z, k, u_1, u_2, \lambda, p, q, r)\ :\ [0, T] \times \mathbb{R}^3 \times \mathcal{R} \times \mathbb{U} \times \mathbb{R}^3 \times \mathcal{R} \to \mathbb{R}$$

of this game by

$$H_i(t, x, y, z, k, u_1, u_2, \lambda, p, q, r) := f_i(t, x, u_1, u_2) + \lambda g_i(t, x, y, z, k, u_1, u_2) + pb(t, x, u_1, u_2) + q\sigma(t, x, u_1, u_2) + \int_{\mathbb{R}} r(\zeta)\gamma(t, x, u_1, u_2, \zeta)\nu(\mathrm{d}\zeta)\ ;\ i = 1, 2, \quad (6.2.8)$$

where $\mathcal{R}$ is the set of functions from $\mathbb{R}_0$ into $\mathbb{R}$ such that the integral in (6.2.8) converges.

We assume that H_i is Fréchet differentiable ($\mathcal{C}^1$) in the variables x, y, z, k, u i=1,2, and that $\nabla_k H_i(t, \zeta)$ as a random measure is absolutely continuous with respect to $\nu\ ;\ i = 1, 2$. We also assume that H_i and its derivatives with respect to u_1 and u_2 are integrable with respect to $P; i = 1, 2$.

In the following, we use the shorthand notation

$$\frac{\partial H_i}{\partial y}(t)$$
$$= \frac{\partial H_i}{\partial y}(t, X(t), Y_i(t), Z_i(t), K_i(t,\cdot), u_1(t), u_2(t), \lambda_i(t), p_i(t), q_i(t), r_i(t,\cdot))$$

and similarly for the other partial derivatives of H_i.

To these Hamiltonians we associate a system of FBSDEs in the adjoint processes $\lambda_i(t)$, $p_i(t)$, $q_i(t)$ and $r_i(t,\zeta)$ as follows:

(i) Forward SDE in $\lambda_i(t)$:

$$\begin{aligned} \mathrm{d}\lambda_i(t) &= \frac{\partial H_i}{\partial y}(t)\mathrm{d}t + \frac{\partial H_i}{\partial z}(t)\mathrm{d}B(t) + \int_{\mathbb{R}} \frac{d}{\mathrm{d}\nu}\nabla_k H_i(t,\zeta)\tilde{N}(\mathrm{d}t, \mathrm{d}\zeta) \\ &= \lambda_i(t)\left[\frac{\partial g_i}{\partial y}(t)\mathrm{d}t + \frac{\partial g_i}{\partial z}(t)\mathrm{d}B(t) + \int_{\mathbb{R}} \frac{\mathrm{d}}{\mathrm{d}\nu}\nabla_k g_i(t,\zeta)\tilde{N}(\mathrm{d}t,\mathrm{d}\zeta)\right]; 0 \le t \le T \\ \lambda_i(0) &= \psi_i'(Y_i(0))\left(= \frac{\mathrm{d}\psi_i}{\mathrm{d}y}(Y_i(0))\right) \ge 0 \end{aligned} \tag{6.2.9}$$

where $\frac{\mathrm{d}}{\mathrm{d}\nu}\nabla_k g_i(t,\zeta)$ is the Radon–Nikodym derivative of $\nabla_k g_i(t,\zeta)$ with respect to $\nu(\zeta)$.

(ii) Backward SDE in $p_i(t), q_i(t), r_i(t,\zeta)$:

$$\begin{aligned} \mathrm{d}p_i(t) &= -\frac{\partial H_i}{\partial x}(t)\mathrm{d}t + q_i(t)\mathrm{d}B(t) + \int_{\mathbb{R}} r_i(t,\zeta)\tilde{N}(\mathrm{d}t,\mathrm{d}\zeta)\ ;\ 0 \le t \le T \\ p_i(T) &= \varphi_i'(X(T)) + h_i'(X(T))\lambda_i(T). \end{aligned} \tag{6.2.10}$$

In the following we assume that

$$\frac{\mathrm{d}}{\mathrm{d}\nu}\Delta_k g_i(t,\zeta) > 1 \text{ a.s. for all } t, \zeta \text{ and } i = 1, 2. \tag{6.2.11}$$

This ensures that

$$\lambda_i(t) \ge 0 \text{ for all } t\ ;\ i = 1, 2. \tag{6.2.12}$$

Theorem 6.4 (***Sufficient Maximum Principle for FBSDE Games***) *Let* $(\hat{u}_1, \hat{u}_2) \in \mathcal{A}_1 \times \mathcal{A}_2$ *with corresponding solutions* $\hat{X}(t), \hat{Y}_i(t), \hat{Z}_i(t), \hat{K}_i(t), \hat{\lambda}_i(t), \hat{p}_i(t), \hat{q}_i(t), \hat{r}_i(t,\zeta)$ *of Eqs.* (6.2.1), (6.2.3), (6.2.9) *and* (6.2.10) *for* $i = 1, 2$. *Suppose that the following hold:*

- *(Concavity I) The functions* $x \to h_i(x), x \to \varphi_i(x), x \to \psi_i(x)$ *are concave,* $i =$ *1,2.*
- *(The conditional maximum principle)*

$$\begin{aligned} ess\ sup_{v\in A_1}\{E[H_1(t,\hat{X}(t),\hat{Y}_1(t),\hat{Z}_1(t),\hat{K}_1(t,\cdot),v,\hat{u}_2(t), \\ \hat{\lambda}_1(t),\hat{p}_1(t),\hat{q}_1(t),\hat{r}_1(t,\cdot))\mid \mathcal{E}_t^{(1)}]\} \\ = E[H_1(t,\hat{X}(t),\hat{Y}_1(t),\hat{Z}_1(t),\hat{K}_1(t,\cdot),\hat{u}_1(t),\hat{u}_2(t), \\ \hat{\lambda}_1(t),\hat{p}_1(t),\hat{q}_1(t),\hat{r}_1(t,\cdot))\mid \mathcal{E}_t^{(1)}] \end{aligned} \tag{6.2.13}$$

$$\begin{aligned} ess\ sup_{v\in A_2}\{E[H_2(t,\hat{X}(t),\hat{Y}_2(t),\hat{Z}_2(t),\hat{K}_2(t,\cdot),u_1(t),v, \\ \hat{\lambda}_2(t),\hat{p}_2(t),\hat{q}_2(t),\hat{r}_2(t,\cdot))\mid \mathcal{E}_t^{(2)}]\} \\ = E[H_2(t,\hat{X}(t),\hat{Y}_2(t),\hat{Z}_2(t),\hat{K}_2(t,\cdot),\hat{u}_1(t),\hat{u}_2(t), \\ \hat{\lambda}_2(t),\hat{p}_2(t),\hat{q}_2(t),\hat{r}_2(t,\cdot))\mid \mathcal{E}_t^{(2)}]. \end{aligned} \tag{6.2.15}$$

and similarly

- *(Concavity II) (The Arrow conditions). The functions*

$$\begin{aligned} \hat{h}_1(x,y,z,k) := ess\ sup_{v_1\in A_1} E[H_1(t,x,y,z,k,v_1,\hat{u}_2(t), \\ \hat{\lambda}_1(t),\hat{p}_1(t),\hat{q}_1(t),\hat{r}_1(t,\cdot))\mid \mathcal{E}_t^{(1)}] \end{aligned}$$

and

$$\begin{aligned} \hat{h}_2(x,y,z,k) := ess\ sup_{v_2\in A_2} E[H_2(t,x,y,z,k,\hat{u}_1(t),v_2, \\ \hat{\lambda}_2(t),\hat{p}_2(t),\hat{q}_2(t),\hat{r}_2(t,\cdot))\mid \mathcal{E}_t^{(2)}] \end{aligned}$$

are concave for all t, a.s.
Then $\hat{u}(t)=(\hat{u}_1(t),\hat{u}_2(t))$ is a Nash equilibrium for (6.2.1)–(6.2.4).

Above and in the proof in Sect. 6.2.3, we have used the following shorthand notation:

If $i=1$, then $X(t)=X^{(u_1,\hat{u}_2)}(t)$ and $Y_1(t)=Y_1^{(u_1,\hat{u}_2)}(t)$ are the processes corresponding to the control $u(t)=(u_1(t),\hat{u}_2(t))$, while $\hat{X}(t)=X^{\hat{u}}(t)$ and $\hat{Y}_1(t)=Y_1^{\hat{u}}(t)$ are those corresponding to the control $\hat{u}(t)=(\hat{u}_1(t),\hat{u}_2(t))$. An analogous notation is used for $i=2$. Moreover, we put

$$\frac{\partial \hat{H}_i}{\partial x}(t)=\frac{\partial H_i}{\partial x}(t,\hat{X}(t),\hat{Y}_i(t),\hat{Z}_i(t),\hat{K}_i(t,\cdot),\hat{u}(t),\hat{\lambda}_i(t),\hat{p}_i(t),\hat{q}_i(t),\hat{r}_i(t,\cdot))$$

and similarly with $\dfrac{\partial \hat{H}_i}{\partial z}(t)$ and $\nabla_k \hat{H}_i(t,\zeta)$, $i=1,2$.

Proof See Sect. 6.2.3.

It is also of interest to prove a version of the maximum principle which does not require the concavity conditions. One such version is the following *necessary*

maximum principle (Theorem 6.5), which makes the following assumptions:

- For all $t_0 \in [0,T]$ and all bounded $\mathcal{E}_t^{(i)}$-measurable random variables $\alpha_i(\omega)$, the control $\beta_i(t) := \chi_{(t_0,T)}(t)\alpha_i(\omega)$ belongs to $\mathcal{A}_i$; $i = 1,2$. (6.2.16)
- For all $u_i, \beta_i \in \mathcal{A}_i$ with β_i bounded there exists $\delta_i > 0$ such that the control $\tilde{u}_i(t) := u_i(t) + s\beta_i(t)$; $t \in [0,T]$ belongs to $\mathcal{A}_i$ for all $s \in (-\delta_i, \delta_i)$; $i = 1,2$. (6.2.17)
- The following derivative processes exist and belong to $L^2([0,T]\times\Omega)$: (6.2.18)

$$x_1(t) = \frac{\mathrm{d}}{\mathrm{d}s}X^{(u_1+s\beta_1,u_2)}(t)\mid_{s=0}\ ; \quad y_1(t) = \frac{\mathrm{d}}{\mathrm{d}s}Y_1^{(u_1+s\beta_1,u_2)}(t)\mid_{s=0}\ ;$$
$$z_1(t) = \frac{\mathrm{d}}{\mathrm{d}s}Z_1^{(u_1+s\beta_1,u_2)}(t)\mid_{s=0}\ ; \quad k_1(t,\zeta) = \frac{\mathrm{d}}{\mathrm{d}s}K_1^{(u_1+s\beta_1,u_2)}(t)\mid_{s=0}$$
$$\text{and, similarly } x_2(t) = \frac{\mathrm{d}}{\mathrm{d}s}X^{(u_1,u_2+s\beta_2)}(t)\mid_{s=0} \text{ etc.}$$

Note that since $X^u(0) = x$ for all u we have $x_i(0) = 0$ for $i = 1,2$.

In the following we write

$$\frac{\partial b}{\partial x}(t) \text{ for } \frac{\partial b}{\partial x}(t, X(t), u(t)) \text{ etc.}$$

By (6.2.1) and (6.2.3) we have, using the estimates in [AØ12],

$$\begin{aligned}
\mathrm{d}x_1(t) =& \\
&\left\{\frac{\partial b}{\partial x}(t)x_1(t) + \frac{\partial b}{\partial u_1}(t)\beta_1(t)\right\}\mathrm{d}t + \left\{\frac{\partial \sigma}{\partial x}(t)x_1(t) + \frac{\partial \sigma}{\partial u_1}(t)\beta_1(t)\right\}\mathrm{d}B(t)\\
&+ \int_{\mathbb{R}}\left\{\frac{\partial \gamma}{\partial x}(t,\zeta)x_1(t) + \frac{\partial \gamma}{\partial u_1}(t,\zeta)\beta_1(t)\right\}\tilde{N}(\mathrm{d}t,\mathrm{d}\zeta), \qquad (6.2.19)\\
\mathrm{d}y_1(t) =& -\left\{\frac{\partial g_1}{\partial x}(t)x_1(t) + \frac{\partial g_1}{\partial y}(t)y_1(t) + \frac{\partial g_1}{\partial z}(t)z_1(t)\right.\\
&\left. + \int_{\mathbb{R}}\nabla_k g_1(t,\zeta)k_1(t,\zeta)\nu(\mathrm{d}\zeta) + \frac{\partial g_1}{\partial u_1}(t)\beta_1(t)\right\}\mathrm{d}t\\
&+ z_i(t)\mathrm{d}B(t) + \int_{\mathbb{R}}k_1(t,\zeta)\tilde{N}(\mathrm{d}t,\mathrm{d}\zeta)\ ;\ 0 \le t \le T,\\
y_1(T) =& \ h_1'(X^{(u_1,u_2)}(T))x_1(T), \qquad (6.2.20)
\end{aligned}$$

and similarly for $\mathrm{d}x_2(t), \mathrm{d}y_2(t)$.

We are now ready to state a necessary maximum principle, which is an extension of Theorem 3.1 in [AØ12] and Theorem 2.2 in [ØS3]. See also Theorem 3.2 in [ØS7]. In the sequel, $\frac{\partial H}{\partial v}$ means $\nabla_v H$.

Theorem 6.5 (***Necessary Maximum Principle***) *Suppose $u \in \mathcal{A}$ with corresponding solutions $X(t)$, $Y_i(t)$, $Z_i(t)$, $K_i(t,\zeta)$, $\lambda_i(t)$, $p_i(t)$, $q_i(t)$, $r_i(t,\zeta)$ of Eqs. (6.2.1), (6.2.3), (6.2.9) and (6.2.10). Suppose (6.2.16)–(6.2.18) hold.*

Then the following are equivalent:

(i)

$$\frac{\mathrm{d}}{\mathrm{d}s} J_1(u_1 + s\beta_1, u_2) \mid_{s=0} = \frac{\mathrm{d}}{\mathrm{d}s} J_2(u_1, u_2 + s\beta_2) \mid_{s=0} = 0$$

for all bounded $\beta_1 \in \mathcal{A}_1$, $\beta_2 \in \mathcal{A}_2$.

(ii)

$$\begin{aligned} & E\left[\frac{\partial}{\partial v_1} H_1(t, X(t), Y_1(t), Z_1(t), K_1(t,\cdot), v_1, u_2(t), \lambda_1(t),\right. \\ & \left. p_1(t) q_1(t), r_1(t,\cdot)) \mid \mathcal{E}_t^{(1)}\right]_{v_1 = u_1(t)} \\ & = E\left[\frac{\partial}{\partial v_2} H_2(t, X(t), Y_2(t), Z_2(t), K_2(t,\cdot), u_1(t), v_2, \lambda_2(t),\right. \\ & \left. p_2(t), q_2(t), r_2(t,\cdot)) \mid \mathcal{E}_t^{(2)}\right]_{v_2 = u_2(t)} \\ & = 0. \end{aligned}$$

Proof See Sect. 6.2.3.

6.2.2 The Zero-Sum Game Case

In the *zero-sum case* we have

$$J_1(u_1, u_2) + J_2(u_1, u_2) = 0. \tag{6.2.21}$$

Then the Nash equilibrium $(\hat{u}_1, \hat{u}_2) \in \mathcal{A}_1 \times \mathcal{A}_2$ satisfying (6.2.6)–(6.2.7) becomes a *saddle point* for $J(u_1, u_2) := J_1(u_1, u_2)$. To see this, note that (6.2.6)–(6.2.7) imply that

$$J_1(u_1, \hat{u}_2) \le J_1(\hat{u}_1, \hat{u}_2) = -J_2(\hat{u}_1, \hat{u}_2) \le -J_2(\hat{u}_1, u_2)$$

and hence

$$J(u_1, \hat{u}_2) \le J(\hat{u}_1, \hat{u}_2) \le J(\hat{u}_1, u_2) \text{ for all } u_1, u_2.$$

From this we deduce that

$$\begin{aligned}\inf_{u_2\in\mathcal{A}_2}\sup_{u_1\in\mathcal{A}_1} J(u_1,u_2) &\le \sup_{u_1\in\mathcal{A}_1} J(u_1,\hat{u}_2) \le J(\hat{u}_1,\hat{u}_2)\\ &\le \inf_{u_2\in\mathcal{A}_2} J(\hat{u}_1,u_2) \le \sup_{u_1\in\mathcal{A}_1}\inf_{u_2\in\mathcal{A}_2} J(u_1,u_2).\end{aligned} \tag{6.2.22}$$

Since we always have inf sup $\ge$ sup inf, we conclude that

$$\begin{aligned}\inf_{u_2\in\mathcal{A}_2}\sup_{u_1\in\mathcal{A}_1} J(u_1,u_2) &= \sup_{u_1\in\mathcal{A}_1} J(u_1,\hat{u}_2) = J(\hat{u}_1,\hat{u}_2)\\ &= \inf_{u_2\in\mathcal{A}_2} J(\hat{u}_1,u_2) = \sup_{u_1\in\mathcal{A}_1}\inf_{u_2\in\mathcal{A}_2} J(u_1,u_2),\end{aligned} \tag{6.2.23}$$

i.e. $(\hat{u}_1,\hat{u}_2)\in\mathcal{A}_1\times\mathcal{A}_2$ is a *saddle point* for $J(u_1,u_2)$.

In this case, only one Hamiltonian is needed, and only one set of adjoint equations. Indeed, let $g_1 = g_2 =: g$, $h_1 = h_2 =: h$, $f_1 = -f_2 := f$, $\varphi_1 = -\varphi_2 := \varphi$, and $\psi_1 = -\psi_2 := \psi$. Then

$$\begin{aligned}H_1(t,x,y,z,k,u_1,u_2,\lambda,p,q,r) &= f(t,x,u_1,u_2)+\lambda g(t,x,y,z,k,u_1,u_2)\\ &+ pb(t,x,u_1,u_2)+q\sigma(t,x,u_1,u_2)+\int_{\mathbb{R}} r(\zeta)\gamma(t,x,u_1,u_2,\zeta)\nu(\mathrm{d}\zeta)\end{aligned} \tag{6.2.24}$$

and

$$\begin{aligned}H_2(t,x,y,z,k,u_1,u_2,\lambda,p,q,r) &= -f(t,x,u_1,u_2)+\lambda g(t,x,y,z,k,u_1,u_2)\\ &+ pb(t,x,u_1,u_2)+q\sigma(t,x,u_1,u_2)+\int_{\mathbb{R}} r(\zeta)\gamma(t,x,u_1,u_2,\zeta)\nu(\mathrm{d}\zeta).\end{aligned} \tag{6.2.25}$$

For $u=(u_1,u_2)\in\mathcal{A}_1\times\mathcal{A}_2$, let $X^u(t)$ be defined by (6.2.1), and $(Y^u(t), Z^u(t), K^u(t,\zeta))$ be defined by (6.2.3) with $g_i = g$, and $h_i = h$. The adjoint processes λ_i, $i=1,2$, satisfy

$$\begin{cases}\mathrm{d}\lambda_i(t) &= \lambda_i(t)\Big[\dfrac{\partial g}{\partial y}(t,X(t),Y(t),Z(t),K(t,\cdot),u_1(t),u_2(t))\mathrm{d}t\\ &\quad +\dfrac{\partial g}{\partial z}(t,X(t),Y(t),Z(t),K(t,\cdot),u_1(t),u_2(t))\mathrm{d}B(t)\\ &\quad +\displaystyle\int_{\mathbb{R}}\frac{\mathrm{d}\nabla_k g(t,\zeta)}{\mathrm{d}\nu(\zeta)}\tilde{N}(\mathrm{d}t,\mathrm{d}\zeta)\Big]\ ;\ 0\le t\le T\\ \lambda_i(0) &= \psi_i'(Y(0))\ge 0.\end{cases} \tag{6.2.26}$$

Recall that by (6.2.11) we have $\lambda_i(t)\ge 0$ for all t,ζ ; $i=1,2$.

It follows that

$$\lambda_2(t) = -\lambda_1(t);\quad t\in[0,T]. \tag{6.2.27}$$

The adjoint processes for p_i, q_i and r_i $i = 1, 2$ become by (6.2.10)

$$\begin{cases} \mathrm{d}p_1(t) &= -\left[\dfrac{\partial f}{\partial x}(t) + \lambda_1(t)\dfrac{\partial g}{\partial x}(t) + p_1(t)\dfrac{\partial b}{\partial x}(t) + q_1(t)\dfrac{\partial \sigma}{\partial x}(t)\right. \\ &\quad \left. + \displaystyle\int_{\mathbb{R}} r_1(t,\zeta)\frac{\partial \gamma}{\partial x}(t,\zeta)\nu(\mathrm{d}\zeta)\right]\mathrm{d}t \\ &\quad + q_1(t)\mathrm{d}B(t) + \displaystyle\int_{\mathbb{R}} r_1(t,\zeta)\tilde{N}(\mathrm{d}t,\mathrm{d}\zeta)\ ;\ 0 \le t \le T \\ p_1(T) &= \varphi_1'(X(T)) + h_1'(X(T))\lambda_1(T) \end{cases} \tag{6.2.28}$$

and, by (6.2.10) and (6.2.27),

$$\begin{cases} \mathrm{d}p_2(t) &= -\left[-\dfrac{\partial f}{\partial x}(t) - \lambda_1(t)\dfrac{\partial g}{\partial x}(t) + p_2(t)\dfrac{\partial b}{\partial x}(t) + q_2(t)\dfrac{\partial \sigma}{\partial x}(t)\right. \\ &\quad \left. + \displaystyle\int_{\mathbb{R}} r_2(t,\zeta)\frac{\partial \gamma}{\partial x}(t,\zeta)\nu(\mathrm{d}\zeta)\right]\mathrm{d}t \\ &\quad + q_2(t)\mathrm{d}B(t) + \displaystyle\int_{\mathbb{R}} r_2(t,\zeta)\tilde{N}(\mathrm{d}t,\mathrm{d}\zeta)\ ;\ 0 \le t \le T \\ p_2(T) &= -\varphi_1'(X(T)) - h_1'(X(T))\lambda_1(T). \end{cases}$$

Thus, we see that

$$(p_2(t), q_2(t), r_2(t)) = -(p_1(t), q_1(t), r_1(t)); \quad t \in [0, T]. \tag{6.2.29}$$

Consequently,

$$\begin{aligned} &- H_2(t, X(t), Y(t), Z(t), K(t,\cdot), u_1(t), u_2(t), \lambda_2(t), p_2(t), q_2(t), r_2(t,\cdot)) \\ &= H_1(t, X(t), Y(t), Z(t), K(t,\cdot), u_1(t), u_2(t), \lambda_1(t), p_1(t), q_1(t), r_1(t,\cdot)). \end{aligned}$$

We thus conclude that in the zero-sum game case, we only need one Hamiltonian and one quadruple of controlled adjoint processes. In the following, we set:

$$J(u_1, u_2) := E\left[\int_0^T f(t, X^u(t), u(t))\mathrm{d}t + \varphi(X^{(u)}(T)) + \psi(Y^u(0))\right], \tag{6.2.30}$$

$H := H_1$ as defined in (6.2.24), and $(\lambda(t), p(t), q(t), r(t,\cdot)) := (\lambda_1(t), p_1(t), q_1(t), r_1(t,\cdot))$ as defined in (6.2.26), (6.2.28).

We can now state the necessary and the sufficient maximum principles for the zero-sum game:

Theorem 6.6 (***Necessary Maximum Principle for Zero-Sum Games***) *Assume the conditions of Theorem 6.5 hold. Then the following are equivalent:*

(i)

$$\frac{\mathrm{d}}{\mathrm{d}s}J(u_1 + s\beta_1, u_2)\,|_{s=0} = \frac{\mathrm{d}}{\mathrm{d}s}J(u_1, u_2 + s\beta_2)\,|_{s=0} = 0 \tag{6.2.31}$$

for all bounded $\beta_1 \in \mathcal{A}_1, \beta_2 \in \mathcal{A}_2$.

(ii)

$$\begin{aligned} & E\Big[\frac{\partial}{\partial v_1} H(t, X(t), Y(t), Z(t), K(t,\cdot), v_1, u_2(t), \lambda(t), p(t), \\ & \quad q(t), r(t,\cdot)) \mid \mathcal{E}_t^{(1)}\Big]_{v_1=u_1(t)} \\ & = E\Big[\frac{\partial}{\partial v_2} H(t, X(t), Y(t), Z(t), K(t,\cdot), u_1(t), v_2, \lambda(t), p(t), \\ & \quad q(t), r(t,\cdot)) \mid \mathcal{E}_t^{(2)}\Big]_{v_2=u_2(t)} \\ & = 0. \end{aligned} \tag{6.2.32}$$

Proof The proof is similar to that of Theorem 6.5 and is omitted.

Corollary 6.7 *Let* $u = (u_1, u_2) \in \mathcal{A}_1 \times \mathcal{A}_2$ *be a Nash equilibrium (saddle point) for the zero-sum game in Theorem 6.6. Then* (6.2.32) *holds.*

Proof This follows from Theorem 6.6 by noting that if $u = (u_1, u_2)$ is a Nash equilibrium, then (6.2.31) holds by (6.2.23).

Similarly we get

Theorem 6.8 *(***Sufficient Maximum Principle for Zero-Sum Forward-Backward Games***) Let* $(\hat{u}_1, \hat{u}_2) \in \mathcal{A}_1 \times \mathcal{A}_2$, *with corresponding solutions* $\hat{X}(t), \hat{Y}(t), \hat{Z}(t), \hat{K}(t), \hat{\lambda}(t), \hat{p}(t), \hat{q}(t), \hat{r}(t,\zeta)$. *Suppose the following hold*

- *The functions* $x \to \varphi(x)$ *and* $x \to \psi(x)$ *are affine, and the function* $x \to h(x)$ *is concave.*
- *(The conditional maximum principle)*

$$\begin{aligned} & ess\ sup_{v_1 \in A_1} E[H(t, \hat{X}(t), \hat{Y}(t), \hat{Z}(t), \hat{K}(t,\cdot), v_1, \hat{u}_2(t), \hat{\lambda}(t), \hat{p}(t), \hat{q}(t), \hat{r}(t,\cdot)) \mid \mathcal{E}_t^{(1)}] \\ & \quad = E[H(t, \hat{X}(t), \hat{Y}(t), \hat{Z}(t), \hat{K}(t,\cdot), \hat{u}_1(t), \hat{u}_2(t), \hat{\lambda}(t), \hat{p}(t), \hat{q}(t), \hat{r}(t,\cdot)) \mid \mathcal{E}_t^{(1)}] \end{aligned} \tag{6.2.33}$$

and

$$\begin{aligned} & \text{ess inf}_{v_2 \in A_2} E[H(t, \hat{X}(t), \hat{Y}(t), \hat{Z}(t), \hat{K}(t,\cdot), \hat{u}_1(t), v_2, \hat{\lambda}(t), \hat{p}(t), \hat{q}(t), \hat{r}(t,\cdot)) \mid \mathcal{E}_t^{(2)}] \\ & \quad = E[H(t, \hat{X}(t), \hat{Y}(t), \hat{Z}(t), \hat{K}(t,\cdot), \hat{u}_1(t), \hat{u}_2(t), \hat{\lambda}(t), \hat{p}(t), \hat{q}(t), \hat{r}(t,\cdot)) \mid \mathcal{E}_t^{(2)}]. \end{aligned} \tag{6.2.34}$$

- *(The Arrow conditions) The function*

$$\begin{aligned} & \hat{h}(x, y, z, k) := \\ & \qquad ess\ sup_{v_1 \in A_1} E[H(t, x, y, z, k, v_1, \hat{u}_2(t), \hat{\lambda}(t), \hat{p}(t), \hat{q}(t), \hat{r}(t,\cdot)) \mid \mathcal{E}_t^{(1)}] \end{aligned}$$

is concave, and the function

$$\check{h}(x, y, z, k) := \operatorname{ess\,inf}_{v_2 \in A_2} E[H(t, x, y, z, k, \hat{u}_1(t), v_2, \hat{\lambda}(t), \hat{p}(t), \hat{q}(t), \hat{r}(t, \cdot)) \mid \mathcal{E}_t^{(2)}]$$

is convex, for all $t \in [0, T]$, *a.s.*

Then $\hat{u}(t) = (\hat{u}_1(t), \hat{u}_2(t))$ *is a saddle point for* $J(u_1, u_2)$.

Proof The proof is similar to that of Theorem 6.4 and is omitted.

6.2.3 Proofs of the Stochastic Maximum Principles

Proof of Theorem 6.4 (***Sufficient maximum principle***)
By introducing a sequence of stopping times τ_n as in the proof of Theorem 5.4, we see that we may assume that all the local martingales appearing in the following computations are martingales and therefore have expectation 0.

We first prove that

$$J_1(u_1, \hat{u}_2) \leq J_1(\hat{u}_1, \hat{u}_2) \text{ for all } u_1 \in \mathcal{A}_1.$$

To this end, fix $u_1 \in \mathcal{A}_1$ and consider

$$\Delta := J_1(u_1, \hat{u}_2) - J_1(\hat{u}_1, \hat{u}_2) = I_1 + I_2 + I_3, \tag{6.2.35}$$

where

$$I_1 = E\left[\int_0^T \{f_1(t, X(t), u(t)) - f_1(t, \hat{X}(t), \hat{u}(t))\}\mathrm{d}t\right], \tag{6.2.36}$$

$$I_2 = E[\varphi_1(X(T)) - \varphi_1(\hat{X}(T))], \tag{6.2.37}$$

$$I_3 = E[\psi_1(Y_1(0)) - \psi_1(\hat{Y}_1(0))]. \tag{6.2.38}$$

By (6.2.8) we have

$$I_1 = E\left[\int_0^T \Big\{H_1(t) - \hat{H}_1(t) - \hat{\lambda}_1(t)(g_1(t) - \hat{g}_1(t)) - \hat{p}_1(t)(b(t) - \hat{b}(t)) - \hat{q}_1(t)(\sigma(t) - \hat{\sigma}(t)) - \int_{\mathbb{R}} \hat{r}_1(t, \zeta)(\gamma(t, \zeta) - \hat{\gamma}(t, \zeta))\nu(\mathrm{d}\zeta)\Big\}\,\mathrm{d}t\right]. \tag{6.2.39}$$

By concavity of φ_1, (6.2.10) and the Itô formula,

$$
\begin{aligned}
I_2 &\le E[\varphi_1'(\hat{X}(T))(X(T)-\hat{X}(T))] = E[\hat{p}_1(T)(X(T)-\hat{X}(T))] \\
&\quad - E[\hat{\lambda}_1(T)h_1'(\hat{X}(T))(X(T)-\hat{X}(T))] \\
&= E\left[\int_0^T \hat{p}_1(t^-)(\mathrm{d}X(t)-\mathrm{d}\hat{X}(t)) + \int_0^T (X(t^-)-\hat{X}(t^-))\mathrm{d}\hat{p}_1(t)\right. \\
&\quad \left. + \int_0^T \hat{q}_1(t)(\sigma(t)-\hat{\sigma}(t))\mathrm{d}t + \int_0^T\int_{\mathbb{R}} \hat{r}_1(t,\zeta)(\gamma(t,\zeta)-\hat{\gamma}(t,\zeta))\nu(\mathrm{d}\zeta)\mathrm{d}t\right] \\
&\quad - E[\hat{\lambda}_1(T)h_1'(\hat{X}(T))(X(T)-\hat{X}(T))] \\
&= E\left[\int_0^T \hat{p}_1(t)(b(t)-\hat{b}(t))\mathrm{d}t + \int_0^T (X(t)-\hat{X}(t))\left(-\frac{\partial \hat{H}_1}{\partial x}(t)\right)\mathrm{d}t\right. \\
&\quad \left. + \int_0^T \hat{q}_1(t)(\sigma(t)-\hat{\sigma}(t))\mathrm{d}t \quad + \int_0^T\int_{\mathbb{R}} \hat{r}_1(t,\zeta)(\gamma(t,\zeta)-\hat{\gamma}(t,\zeta))\nu(\mathrm{d}\zeta)\mathrm{d}t\right] \\
&\quad - E[\hat{\lambda}_1(T)h_1'(\hat{X}(T))(X(T)-\hat{X}(T))]. \qquad (6.2.40)
\end{aligned}
$$

By concavity of ψ_1, (6.2.5), (6.2.9), and concavity of φ_1 :

$$
\begin{aligned}
I_3 &= E[\psi_1(Y_1(0)) - \psi_1(\hat{Y}_1(0))] \le E[\psi_1'(\hat{Y}_1(0))(Y_1(0)-\hat{Y}_1(0))] \\
&\quad = E[\hat{\lambda}_1(0)(Y_1(0)-\hat{Y}_1(0))] \\
&= E[(Y_1(T)-\hat{Y}_1(T))\hat{\lambda}_1(T)] - E\left[\int_0^T (Y_1(t^-)-\hat{Y}_1(t^-))\mathrm{d}\hat{\lambda}_1(t)\right. \\
&\quad + \int_0^T \hat{\lambda}_1(t^-)(\mathrm{d}Y_1(t)-\mathrm{d}\hat{Y}_1(t)) + \int_0^T \frac{\partial \hat{H}_1}{\partial z}(t)(Z_1(t)-\hat{Z}_1(t))\mathrm{d}t \\
&\quad \left. + \int_0^T\int_{\mathbb{R}} \nabla_k \hat{H}_1(t,\zeta)(K_1(t,\zeta)-\hat{K}_1(t,\zeta))\nu(\mathrm{d}\zeta)\mathrm{d}t\right] \\
&= E[(h_1(X(T))-h_1(\hat{X}(T)))\hat{\lambda}_1(T)] - E\left[\int_0^T \frac{\partial \hat{H}_1}{\partial y}(t)(Y_1(t)-\hat{Y}_1(t))\mathrm{d}t\right. \\
&\quad + \int_0^T \hat{\lambda}_1(t)(-g_1(t)+\hat{g}_1(t))\mathrm{d}t + \int_0^T \frac{\partial \hat{H}_1}{\partial z}(t)(Z_1(t)-\hat{Z}_1(t))\mathrm{d}t \\
&\quad \left. + \int_0^T\int_{\mathbb{R}} \nabla_k \hat{H}_1(t,\zeta)(K_1(t,\zeta)-\hat{K}_1(t,\zeta))\nu(\mathrm{d}\zeta)\mathrm{d}t\right] \\
&\le E[\hat{\lambda}_1(T)h_1'(\hat{X}(T))(X(T)-\hat{X}(T))] - E\left[\int_0^T \frac{\partial \hat{H}_1}{\partial y}(t)(Y_1(t)-\hat{Y}_1(t))\mathrm{d}t\right. \\
&\quad + \int_0^T \hat{\lambda}_1(t)(-g_1(t)+\hat{g}_1(t))\mathrm{d}t + \int_0^T \frac{\partial \hat{H}_1}{\partial z}(t)(Z_1(t)-\hat{Z}_1(t))\mathrm{d}t \\
&\quad \left. + \int_0^T\int_{\mathbb{R}} \nabla_k \hat{H}_1(t,\zeta)(K_1(t,\zeta)-\hat{K}_1(t,\zeta))\nu(\mathrm{d}\zeta)\mathrm{d}t\right], \qquad (6.2.41)
\end{aligned}
$$

where we have used $\hat{\lambda}_1(T) \geq 0$ (see (6.2.11)). Adding (6.2.39)–(6.2.41) we get

$$\begin{aligned}\Delta &= I_1 + I_2 + I_3 \\ &\leq E\Bigg[\int_0^T \Big\{H_1(t) - \hat{H}_1(t) - \frac{\partial \hat{H}_1}{\partial x}(t)(X(t) - \hat{X}(t)) \\ &- \frac{\partial H_1}{\partial y}(t)(Y_1(t) - \hat{Y}_1(t)) - \frac{\partial \hat{H}_1}{\partial z}(t)(Z_1(t) - \hat{Z}_1(t)) \\ &- \int_{\mathbb{R}} \nabla_k \hat{H}_1(t,\zeta)(K_1(t,\zeta) - \hat{K}_1(t,\zeta))\nu(\mathrm{d}\zeta)\Big\}\mathrm{d}t\Bigg]. \qquad (6.2.42)\end{aligned}$$

Since $\hat{h}_1(x, y, z, k)$ is concave, it follows by a standard separating hyperplane argument (see e.g. [SeSy], Chap. 5, Sect. 23) that there exists a supergradient $a = (a_0, a_1, a_2, a_3(\cdot)) \in \mathbb{R}^3 \times \mathcal{R}$ for $\hat{h}_1(x, y, z, k)$ at $x = \hat{X}(t)$, $y = \hat{Y}_1(t)$, $z = \hat{Z}_1(t^-)$ and $k = \hat{K}_1(t^-, \cdot)$ such that if we define

$$\begin{aligned}\varphi_1(x, y, z, k) &:= \hat{h}_1(x, y, z, k) - \hat{h}_1(\hat{X}(t^-), \hat{Y}_1(t^-), \hat{Z}_1(t^-), \hat{K}_1(t, \cdot)) \\ &- \Big[a_0(x - \hat{X}(t)) + a_1(y - \hat{Y}_1(t)) + a_2(z - \hat{Z}_1(t)) \\ &+ \int_{\mathbb{R}} a_3(\zeta)(k(\zeta) - \hat{K}(t,\zeta))\nu(\mathrm{d}\zeta)\Big]\end{aligned}$$

then

$$\varphi_1(x, y, z, k) \leq 0 \text{ for all } x, y, z, k.$$

On the other hand we clearly have

$$\varphi_1(\hat{X}(t), \hat{Y}(t), \hat{Z}(t), \hat{K}_1(t, \cdot)) = 0.$$

It follows that

$$\begin{aligned}\frac{\partial \hat{H}_1}{\partial x}(t) &= \frac{\partial \hat{h}_1}{\partial x}(\hat{X}(t), \hat{Y}_1(t), \hat{Z}_1(t), \hat{K}_1(t, \cdot)) = a_0, \\ \frac{\partial \hat{H}_1}{\partial y}(t) &= \frac{\partial \hat{h}_1}{\partial y}(\hat{X}(t), \hat{Y}_1(t), \hat{Z}_1(t), \hat{K}_1(t, \cdot)) = a_1, \\ \frac{\partial \hat{H}_1}{\partial z}(t) &= \frac{\partial \hat{h}_1}{\partial z}(\hat{X}(t), \hat{Y}_1(t), \hat{Z}_1(t), \hat{K}_1(t, \cdot)) = a_2, \\ \nabla_k \hat{H}_1(t,\zeta) &= \nabla_k \hat{h}_1(\hat{X}(t), \hat{Y}_1(t), \hat{Z}_1(t), \hat{K}_1(t, \cdot)) = a_3.\end{aligned}$$

Combining this with (6.2.42) we get

$$
\begin{aligned}
\Delta &\le \hat{h}_1(X(t), Y_1(t), Z_1(t), K_1(t,\cdot)) \\
&\quad - \hat{h}_1(\hat{X}(t), \hat{Y}_1(t), \hat{Z}_1(t), \hat{K}_1(t,\cdot)) \\
&\quad - \frac{\partial \hat{h}_1}{\partial x}(\hat{X}(t), \hat{Y}_1(t), \hat{Z}_1(t), \hat{K}_1(t,\cdot))(X(t) - \hat{X}(t)) \\
&\quad - \frac{\partial \hat{h}_1}{\partial y}(\hat{X}(t), \hat{Y}_1(t), \hat{Z}_1(t), \hat{K}_1(t,\cdot))(Y_1(t) - \hat{Y}_1(t)) \\
&\quad - \frac{\partial \hat{h}_1}{\partial z}(\hat{X}(t), \hat{Y}_1(t), \hat{Z}_1(t), \hat{K}_1(t,\cdot))(Z_1(t) - \hat{Z}_1(t)) \\
&\quad - \int_{\mathbb{R}} \nabla_k \hat{h}_1(\hat{X}(t), \hat{Y}_1(t), \hat{Z}_1(t), \hat{K}_1(t,\cdot))(K_1(t,\zeta) - \hat{K}_1(t,\zeta))\nu(\mathrm{d}\zeta) \\
&\quad \le 0 \text{ since } \hat{h}_1 \text{ is concave.}
\end{aligned}
$$

Hence

$$J_1(u_1, \hat{u}_2) \le J_1(\hat{u}_1, \hat{u}_2) \quad \text{for all } u_1 \in \mathcal{A}_1.$$

The inequality

$$J_2(\hat{u}_1, u_2) \le J_2(\hat{u}_1, \hat{u}_2) \text{ for all } u_2 \in \mathcal{A}_2$$

is proved similarly. This completes the proof of Theorem 6.4. □

Proof of Theorem 6.5 (***Necessary maximum principle***)
By introducing a sequence of stopping times τ_n as in the proof of Theorem 5.5, we see that we may assume that all the local martingales appearing in the following computations are martingales and therefore have expectation 0. Consider

$$
\begin{aligned}
D_1 &:= \frac{\mathrm{d}}{\mathrm{d}s} J_1(u_1 + s\beta_1, u_2) \mid_{s=0} \\
&= E\left[\int_0^T \left\{\frac{\partial f_1}{\partial x}(t)x_1(t) + \frac{\partial f_1}{\partial u_1}(t)\beta_1(t)\right\} \mathrm{d}t \right. \\
&\quad \left. + \varphi_1'(X^{(u_1,u_2)}(T))x_1(T) + \psi_1'(Y_1(0))y_1(0)\right]. \qquad (6.2.43)
\end{aligned}
$$

By (6.2.10) and the Itô formula,

$$
\begin{aligned}
&E[\varphi_1'(X^{(u_1,u_2)}(T))x_1(T)] \\
&\quad = E[p_1(T)x_1(T)] - E[h_1'(X^{(u_1,u_2)}(T))\lambda_1(T)] \\
&\quad = E\left[\int_0^T \left\{p_1(t^-)\mathrm{d}x_1(t) + x_1(t^-)\mathrm{d}p_1(t)\right.\right. \\
&\qquad + q_1(t)\left[\frac{\partial \sigma}{\partial x}(t)x_1(t) + \frac{\partial \sigma}{\partial u_1}(t)\beta_1(t)\right] \\
&\qquad \left.\left. + \int_{\mathbb{R}} r_1(t,\zeta)\left[\frac{\partial \gamma}{\partial x}(t,\zeta)x_1(t) + \frac{\partial \gamma}{\partial u_1}(t,\zeta)\beta_1(t,\zeta)\right]\nu(\mathrm{d}\zeta)\right\}\mathrm{d}t\right]
\end{aligned}
$$

$$
\begin{aligned}
&- E[h_1'(X^{(u_1,u_2)}(T))\lambda_1(T)] \\
&= E\left[\int_0^T \left\{ p_1(t)\left[\frac{\partial b}{\partial x}(t)x_1(t) + \frac{\partial b}{\partial u_1}(t)\beta_1(t)\right]\right.\right. \\
&\quad + x_1(t)\left(-\frac{\partial H_1}{\partial x}(t)\right) + q_1(t)\left[\frac{\partial \sigma}{\partial x}(t)x_1(t) + \frac{\partial \sigma}{\partial u_1}(t)\beta_1(t)\right] \\
&\quad \left.\left. + \int_{\mathbb{R}} r_1(t,\zeta)\left[\frac{\partial \gamma}{\partial x}(t,\zeta)x_1(t) + \frac{\partial \gamma}{\partial u_1}(t,\zeta)\beta_1(t,\zeta)\right]\nu(\mathrm{d}\zeta)\right\}\mathrm{d}t\right] \\
&\quad - E[h_1'(X^{(u_1,u_2)}(T))\lambda_1((T)]. \qquad (6.2.44)
\end{aligned}
$$

By (6.2.9) and the Itô formula

$$
\begin{aligned}
&E[\psi_1'(Y_1(0))y_1(0)] = E[\lambda_1(0)y_1(0)] \\
&= E[\lambda_1(T)y_1(T)] - E\left[\int_0^T \lambda_1(t^-)\mathrm{d}y_1(t) + y_1(t^-)\mathrm{d}\lambda_1(t)\right. \\
&\quad \left. + \frac{\partial H_1}{\partial z}(t)z_1(t)\mathrm{d}t + \int_{\mathbb{R}} \nabla_k H_1(t,\zeta)k_1(t,\zeta)\nu(\mathrm{d}\zeta)\mathrm{d}t\right] \\
&= E[\lambda_1(T)h_1'(X^{(u_1,u_2)}(T))] \\
&\quad - E\left[\int_0^T \left\{\lambda_1(t)\left[-\frac{\partial g_1}{\partial x}(t)x_1(t) - \frac{\partial g_1}{\partial y}(t)y_1(t) - \frac{\partial g_1}{\partial z}(t)z_1(t)\right.\right.\right. \\
&\quad \left. - \int_{\mathbb{R}} \nabla_k g_1(t,\zeta)k_1(t,\zeta)\nu(\mathrm{d}\zeta) - \frac{\partial g_1}{\partial u_1}(t)\beta_1(t)\right] \\
&\quad \left.\left. + \frac{\partial H_1}{\partial y}(t)y_1(t) + \frac{\partial H_1}{\partial z}(t)z_1(t) + \int_{\mathbb{R}} \nabla_k H_1(t,\zeta)k_1(t,\zeta)\nu(\mathrm{d}\zeta)\right\}\mathrm{d}t\right]. \qquad (6.2.45)
\end{aligned}
$$

Adding (6.2.44) and (6.2.45) we get, by (6.2.43),

$$
\begin{aligned}
D_1 = E\left[\int_0^T \left\{\left[\frac{\partial f_1}{\partial x}(t) + p_1(t)\frac{\partial b}{\partial x}(t) + q_1(t)\frac{\partial \sigma}{\partial x}(t)\right.\right.\right. \\
+ \int_{\mathbb{R}} r_1(t,\zeta)\frac{\partial \gamma}{\partial x}(t,\zeta)\nu(\mathrm{d}\zeta) - \frac{\partial H_1}{\partial x}(t) + \lambda_1(t)\frac{\partial g_1}{\partial x}(t)\Big]x_1(t) \\
+ \left[-\frac{\partial H_1}{\partial y}(t) + \lambda_1(t)\frac{\partial g_1}{\partial y}(t)\right]y_1(t) + \left[-\frac{\partial H_1}{\partial z}(t) + \lambda_1(t)\frac{\partial g_1}{\partial z}(t)\right]z_1(t) \\
+ \int_{\mathbb{R}} [-\nabla_k H_1(t,\zeta) + \lambda_1(t)\nabla_k g_1(t,\zeta)]\,k_1(t,\zeta)\nu(\mathrm{d}\zeta) \\
+ \left[\frac{\partial f_1}{\partial u_1}(t) + p_1(t)\frac{\partial b}{\partial u_1}(t) + q_1(t)\frac{\partial \sigma}{\partial u_1}(t)\right. \\
\left.\left.\left. + \int_{\mathbb{R}} r_1(t,\zeta)\frac{\partial \gamma}{\partial u_1}(t,\zeta)\nu(\mathrm{d}\zeta) + \frac{\partial g_1}{\partial u_1}(t)\right]\beta_1(t)\right\}\mathrm{d}t\right]
\end{aligned}
$$

$$= E\left[\int_0^T \frac{\partial H_1}{\partial u_1}(t)\beta_1(t)\mathrm{d}t\right]$$
$$= E\left[\int_0^T E\left[\frac{\partial H_1}{\partial u_1}(t)\beta_1(t) \mid \mathcal{E}_t^{(1)}\right]\mathrm{d}t\right]. \tag{6.2.46}$$

If $D_1 = 0$ for all bounded $\beta_1 \in \mathcal{A}_1$, then this holds in particular for β_1 of the form in (6.2.16), i.e.

$$\beta_1(t) = \chi_{(t_0,T]}(t)\alpha_1(\omega),$$

where $\alpha_1(\omega)$ is bounded and $\mathcal{E}_{t_0}^{(1)}$-measurable. Hence

$$E\left[\int_{t_0}^T E\left[\frac{\partial H_1}{\partial u_1}(t) \mid \mathcal{E}_t^{(1)}\right]\alpha_1\mathrm{d}t\right] = 0.$$

Differentiating with respect to t_0 we get

$$E\left[\frac{\partial H_1}{\partial u_1}(t_0)\alpha_1\right] = 0 \text{ for a.a. } t_0.$$

Since this holds for all bounded $\mathcal{E}_{t_0}^{(1)}$-measurable random variables α_1 we conclude that

$$E\left[\frac{\partial H_1}{\partial u_1}(t) \mid \mathcal{E}_t^{(1)}\right] = 0 \text{ for a.a. } t \in [0,T].$$

A similar argument gives that

$$E\left[\frac{\partial H_2}{\partial u_2}(t) \mid \mathcal{E}_t^{(2)}\right] = 0$$

provided that

$$D_2 := \frac{\mathrm{d}}{\mathrm{d}s} J_2(u_1, u_2 + s\beta_2) \mid_{s=0} = 0 \text{ for all bounded } \beta_2 \in \mathcal{A}_2.$$

This shows that (i) $\Rightarrow$ (ii). The argument above can be reversed, to give that (ii) $\Rightarrow$ (i). We omit the details. □

6.2.4 *Risk Minimization by FBSDE Games*

We illustrate the general results above by again studying the risk minimization problem.

The starting point is the same as in Sect. 5.2.4, with a wealth process $X_\pi(t)$ associated to a (self-financing) portfolio $\pi(t)$ given by

$$\begin{cases} \mathrm{d}X_\pi(t) &= \pi(t)X(t^-)[b_0(t)\mathrm{d}t + \sigma_0(t)\mathrm{d}B(t)]\ ;\ t \geq 0 \\ X_\pi(0) &= x_0 > 0. \end{cases} \tag{6.2.47}$$

This time we want to *minimize the risk* $\rho(X_\pi(T))$ of the terminal value $X_\pi(T)$, defined by

$$\rho(X_\pi(T)) = -Y_\pi(0), \tag{6.2.48}$$

where

$$\begin{cases} \mathrm{d}Y_\pi(t) &= -g(Z(t))\mathrm{d}t + Z(t)\mathrm{d}B(t)\ ;\ t \in [0,T] \\ Y_\pi(T) &= X_\pi(T), \end{cases} \tag{6.2.49}$$

for some given concave function g. Thus we want to find $\hat{\pi} \in \mathcal{A}$ and $\rho(X_{\hat{\pi}}(T)) := -Y_{\hat{\pi}}(0)$ such that

$$\inf_{\pi\in\mathcal{A}}\{-Y_\pi(0)\} = -Y_{\hat{\pi}}(0). \tag{6.2.50}$$

In this case the Hamiltonian becomes

$$H(t,x,y,z,k,\pi,\lambda,p,q,r) = \pi x b_0(t)p + \pi x\sigma_0(t)q + \lambda g(z). \tag{6.2.51}$$

The adjoint equations are (6.2.10)–(6.2.9)

$$\begin{cases} \mathrm{d}p(t) &= -\{\pi(t)b_0(t)p(t) + \pi(t)\sigma_0(t)q(t)\}\mathrm{d}t + q(t)\mathrm{d}B(t) \\ p(T) &= \lambda(T) \end{cases} \tag{6.2.52}$$

and

$$\begin{cases} \mathrm{d}\lambda(t) &= \lambda(t)g'(Z(t))\mathrm{d}B(t) \\ \lambda(0) &= 1 \end{cases} \tag{6.2.53}$$

i.e.

$$\lambda(t) = \exp\left(\int_0^t g'(Z(s))\mathrm{d}B(s) - \frac{1}{2}g'(Z(s))^2\mathrm{d}s\right). \tag{6.2.54}$$

If $\hat{\pi}$ is optimal, then

$$b_0(t)\hat{p}(t) + \sigma_0(t)\hat{q}(t) = 0. \tag{6.2.55}$$

This gives

$$\begin{cases} \mathrm{d}\hat{p}(t) &= \hat{q}(t)\mathrm{d}B(t) = -\dfrac{b_0(t)}{\sigma_0(t)}\hat{p}(t)\mathrm{d}B(t)\ ;\ 0 \leq t \leq T \\ \hat{p}(T) &= \hat{\lambda}(T). \end{cases} \tag{6.2.56}$$

Comparing with (6.2.53) we see that the solution $(\hat{p}, \hat{q})$ of the BSDE (6.2.56) is

$$\hat{p}(t) = \hat{\lambda}(t), \hat{q}(t) = \hat{\lambda}(t)g'(\hat{Z}(t)). \tag{6.2.57}$$

Hence by (6.2.55)

$$g'(\hat{Z}(t)) = -\frac{b_0(t)}{\sigma_0(t)}. \tag{6.2.58}$$

If, for example,

$$g(z) = -\frac{1}{2}z^2 \tag{6.2.59}$$

then (6.2.58) gives

$$\hat{Z}(t) = \frac{b_0(t)}{\sigma_0(t)}.$$

Substituted into (6.2.47) this gives, using (6.2.48) (with $\Gamma(t)$ as in (5.2.35)),

$$\begin{aligned}\hat{X}(T) = \hat{Y}(T) &= \hat{Y}(0) + \int_0^T \frac{1}{2}\left(\frac{b_0(s)}{\sigma_0(s)}\right)^2 \mathrm{d}s + \int_0^T \frac{b_0(s)}{\sigma_0(s)}\mathrm{d}B(s) \\ &= \hat{Y}(0) - \ln \Gamma(T).\end{aligned} \tag{6.2.60}$$

We take expectation w.r.t. the martingale measure Q defined by

$$\mathrm{d}Q(\omega) = \Gamma(T)\mathrm{d}P(\omega) \tag{6.2.61}$$

and get

$$-\hat{Y}(0) = -x - E_Q[\ln \Gamma(T)] = -x - E\left[\frac{\mathrm{d}Q}{\mathrm{d}P}\ln\frac{\mathrm{d}Q}{\mathrm{d}P}\right]. \tag{6.2.62}$$

Note that $H(Q \mid P) := E\left[\frac{\mathrm{d}Q}{\mathrm{d}P}\ln\frac{\mathrm{d}Q}{\mathrm{d}P}\right]$ is the *entropy* of Q with respect to P.

Now that the optimal value $\hat{Y}(0)$ has been found, we can use (6.2.60) to find the corresponding optimal terminal wealth $\hat{X}(T)$, and from there the optimal portfolio as we did in Sect. 5.2.4. We have proved:

Theorem 6.9 *Suppose (6.2.59) holds. Then the minimal risk $-Y_{\hat{\pi}}(0) = -\hat{Y}(0)$ of problem (6.2.50) is given by (6.2.62), where $\mathrm{d}Q = \Gamma(T)\mathrm{d}P$ is the unique equivalent martingale measure for the market of Sect. 5.2.4.*

6.3 Mean-Field Games

Mean-field SDEs are stochastic differential equations where the coefficients depend not only on the current value of the solution but also on its probability law. The motivations of such equations, also called McKean–Vlasov equations, go back to M. Kac and H.P. McKean, with more recent fundamental contributions from J.M. Lasry and P.L. Lions. We refer to [CD1] and [CD2] and the references therein for more information about the background for mean-field SDEs.

Mean-field SDEs are not Markovian, and classical dynamic programming and HJB equations cannot be used to study optimal control problems for such equations. However, it is possible to apply a suitably modified maximum principle approach. The first paper on this was [AD]. Subsequently a number of extensions have appeared, e.g. to optimal control of coupled systems of mean-field FBSDEs, singular mean-field control, mean-field stochastic partial differential equations, mean-field games, mean-field systems with memory and mean-field BSDEs. We refer to e.g. [CD1, CD2, HØS, AØ1, AØ2, DØS, AHØ] and the references therein. Our presentation is mainly based on [HØS].

6.3.1 Two Motivating Examples

(a) Optimal Harvesting from a Mean-Field System

Suppose we model the density $X^0(t) = X_t^0$ of an unharvested population at time t by an equation of the form

$$\begin{aligned} \mathrm{d}X_t^0 &= E[X_t^0]a(t)\mathrm{d}t + X_{t^-}^0\left[\sigma(t)\mathrm{d}B(t) + \int_{\mathbb{R}} \gamma(t,\zeta)\tilde{N}(\mathrm{d}t,\mathrm{d}\zeta)\right]; \quad t \in [0,T] \\ X_{0^-}^0 &= x > 0. \end{aligned} \tag{6.3.1}$$

We assume that $a(t), \sigma(t)$, and $\gamma(t,\zeta)$ are given predictable processes. We may regard (6.3.1) as a limit as $n \to \infty$ of a large population interacting system of the form

$$\mathrm{d}x^{i,n}(t) = \left[\frac{1}{n}\sum_{j=1}^{n} x^{j,n}(t)\right] a(t)\mathrm{d}t + x^{i,n}(t)\left[\sigma(t)\mathrm{d}B^i(t) + \int_{\mathbb{R}} \gamma(t,\zeta)\tilde{N}(\mathrm{d}t,\mathrm{d}\zeta)\right]. \tag{6.3.2}$$

Here we have divided the lake into a grid of size n, and $x^{i,n}(t)$ represents the population density in grid box i.

Thus the mean-field term $E[X(t)]$ represents an approximation to the weighted average $\frac{1}{n}\sum_{j=1}^{n} x^{j,n}(t)$ for large n. Now suppose we introduce harvesting of the population. The size of the harvested population $X(t) = X^u(t)$ at time t can then be

modeled by a mean-field control stochastic differential equation of the form

$$\begin{aligned} \mathrm{d}X(t) &= E[X(t)]a(t)\mathrm{d}t + X(t^-)\left[\sigma(t)\mathrm{d}B(t) + \int_{\mathbb{R}} \gamma(t,\zeta)\tilde{N}(\mathrm{d}t,\mathrm{d}\zeta)\right] \\ &\quad - X(t)u(t)\mathrm{d}t; \quad t \in [0,T] \\ X(0) &= x > 0, \end{aligned} \tag{6.3.3}$$

where $u \geq 0$ is a predictable process, representing the relative harvesting rate.

The *performance functional* is assumed to be of the form

$$J(u) = E\left[\int_o^T \{h_0 X(t)u(t) - \frac{1}{2}\rho\, X^2(t)\, u^2(t)\}\mathrm{d}t + K X(T)\right], \tag{6.3.4}$$

where $h_0 > 0$ is a fixed given unit price and $K = K(\omega) > 0$ is a given salvage price, assumed to be $\mathcal{F}_T$-measurable. The term $\frac{1}{2}\rho\, u^2(t)$ represents a quadratic cost rate for the harvesting effort $u(t)$ and $\rho > 0$ is a given constant.

The problem is to find u^* such that

$$J(u^*) = \sup_u J(u). \tag{6.3.5}$$

This is an example of a *mean-field control problem*. We will return to this problem in Sect. 6.22.

(b) Optimal Irreversible Investments Under Model Uncertainty

Let $V(t) = V_0 + \int_0^t u(s)\mathrm{d}s$ denote the production capacity of a production plant and let $D(t)$ denote the demand rate at time t. At any time t the production capacity can be increased by the rate $u(t)$ at the price $\lambda_0(t, D(t))$ per capacity unit, where $u(t) \geq 0$ is a predictable process (our control). The total expected net profit of the production is assumed to be

$$\begin{aligned} J(u,\theta) = & E^{Q_\theta}\left[\int_0^T a(t, E[\varphi(D(t))])\min[D(t), V(t)]dt + g(D(T))\right. \\ & \left. - \int_0^T \lambda_0(t, D(t))u(t)\mathrm{d}t\right], \end{aligned} \tag{6.3.6}$$

where $g(D(T))$ is some salvage value of the closed-down production plant, φ is a given real function and $a(t)$ is the unit sales price of the production. Here $\{Q^\theta\}_{\theta\in\Theta}$ is a family of probability measures representing the model uncertainty. We let $\mathcal{A}_{\mathbb{E}}$

denote the set of right-continuous, non-decreasing $\mathcal{E}_t$-adapted processes $u(t) \geq 0$, where $\mathbb{E} = \{\mathcal{E}_t\}_{t\geq 0}$ with $\mathcal{E}_t \subseteq \mathcal{F}_t$ is a given subfiltration. Thus $\mathcal{E}_t$ represents the information available to the investor at time t. We assume that the demand $D(t)$ is given by a jump diffusion of the form

$$\begin{cases} \mathrm{d}D(t) &= D(t^-)\left[\alpha(t,\omega)\mathrm{d}t + \beta(t,\omega)\mathrm{d}B(t) + \int_{\mathbb{R}} \eta(t,\omega,\zeta)\tilde{N}(\mathrm{d}t,\mathrm{d}\zeta)\right], \ 0\leq t\leq T \\ D(0) &> 0, \end{cases} \tag{6.3.7}$$

where $\eta(t,\omega,\zeta) \geq -1$ for all t,ω,ζ.

We want to maximize the expected total net profit under the worst possible scenario, i.e. we want to find $(u^*,\theta^*) \in \mathcal{A}_{\mathbb{E}} \times \Theta$ such that

$$\sup_{u\in\mathcal{A}_{\mathbb{E}}} \inf_{\theta\in\Theta} J(u,\theta) = \inf_{\theta\in\Theta} \sup_{u\in\mathcal{A}_{\mathbb{E}}} J(u,\theta) = J(u^*,\theta^*). \tag{6.3.8}$$

This is an example of a (partial information) *mean-field control game* of a jump diffusion. Note that the system is non-Markovian, both because of the mean-field term and the partial information constraint.

6.3.2 General Mean-Field Non-zero Sum Games

In the following, we study general mean-field control games and prove maximum principles for the problem to find the following types of equilibria:

(i) Nash equilibria for non-zero sum games.
(ii) Saddle points (as in (6.3.8)) for zero sum games.

Recall (see the beginning of Sect. 6.2) that if $F : L^2(P) \to \mathbb{R}$ is Fréchet differentiable, then its Fréchet derivative at $X \in L^2(P)$ denoted by $\nabla_X F$ is a bounded linear functional on $L^2(P)$. Hence, by the Riesz representation theorem, we can identify $\nabla_X F$ with a random variable $\tilde{\nabla}_X F$ in $L^2(P)$ and the action of $\nabla_X F$ on $Y \in L^2(P)$ is given by $< \nabla_X F, Y >= E\left[\left(\tilde{\nabla}_X F\right) Y\right]$. In the following we do not distinguish between $\nabla_X F$ and $\tilde{\nabla}_X F$.

For example, if $F(X) = E[\varphi(X)]$; $X \in L^2(P)$, where φ is a real C^1- function such that $\varphi(X) \in L^2(P)$ and $\frac{\partial\varphi}{\partial x}(X) \in L^2(P)$, then $\nabla_X F = \frac{\partial\varphi}{\partial x}(X)$ and $\nabla_X F(Y) = < \frac{\partial\varphi}{\partial x}(X), Y >= E[\frac{\partial\varphi}{\partial x}(X)Y]$ for $Y \in L^2(P)$.

Put $u = (u_1, u_2)$, where u_i represents the control of player i ; $i = 1, 2$.

Consider the system with state process $X(t) = X^u(t)$ of the form

$$\begin{aligned} \mathrm{d}X(t) &= b(t, X(t), Y(t), u(t), \omega)\mathrm{d}t + \sigma(t, X(t), Y(t), u(t), \omega)\mathrm{d}B(t) \\ &\quad + \int_{\mathbb{R}} \gamma(t, X(t), Y(t), u(t), \zeta, \omega)\tilde{N}(\mathrm{d}t, \mathrm{d}\zeta), \end{aligned} \tag{6.3.9}$$

where

$$Y(t) = F(X(t, \cdot)), \tag{6.3.10}$$

and F is a Fréchet differentiable operator on $L^2(P)$.

The *performance functional* for player i is assumed to be of the form

$$J_i(u) = E\left[\int_0^T f_i(t, X(t), Y(t), u(t), \omega)\mathrm{d}t + g_i(X(T), Y(T), \omega)\right]; \quad i = 1, 2. \tag{6.3.11}$$

We may interpret the functions f_i as profit rates and g_i as bequest or salvage value functions.

We want to find a *Nash equilibrium* for this game, i.e. find $u_1^* \in \mathcal{A}^{(1)}$ and $u_2^* \in \mathcal{A}^{(2)}$ such that

$$\sup_{u_1 \in \mathcal{A}^{(1)}} J_1(u_1, u_2^*) = J_1(u_1^*, u_2^*) \tag{6.3.12}$$

and

$$\sup_{u_2 \in \mathcal{A}^{(2)}} J_2(u_1^*, u_2) = J_2(u_1^*, u_2^*). \tag{6.3.13}$$

Here $\mathcal{A}^{(i)}$ is a given family of $\mathbb{E}^{(i)}$-predictable processes such that the corresponding state equation has a unique solution X such that $\omega \to X(t, \omega) \in L^2(P)$ for all t. We let $A^{(i)}$ denote the set of possible values of $u_i(t); t \in [0, T]$ when $u_i \in \mathcal{A}^{(i)}; i = 1, 2$.

6.3.3 A Sufficient Maximum Principle

In this section we prove a sufficient maximum principle for the mean-field control games described above.

We first consider the general non-zero sum game.
To this end, define two *Hamiltonians* $H_i; i = 1, 2$, as follows:

$$\begin{aligned} &H_i(t, x, y, u_1, u_2, p_i, q_i, r_i) \\ &\quad = f_i(t, x, y, u) + b(t, x, y, u)p_i + \sigma(t, x, y, u)q_i + \int_{\mathbb{R}} \gamma(t, x, y, u, \zeta)r_i(\zeta)\nu(d\zeta). \end{aligned} \tag{6.3.14}$$

We assume that for $i = 1, 2$, $H = H_i$ is Fréchet differentiable (C^1) in the variables x, y, z, k, ξ, u and that the Fréchet derivative $\nabla_k H$ of H with respect to $k \in \mathcal{R}$ as a random measure is absolutely continuous with respect to ν, with Radon–Nikodym derivative $\frac{\mathrm{d}\nabla_k H}{\mathrm{d}\nu}$. Thus, if $\langle \nabla_k H, h\rangle$ denotes the action of the linear operator $\nabla_k H$ on the function $h \in \mathcal{R}$ we have

$$\langle \nabla_k H, h\rangle = \int_{\mathbb{R}} h(\zeta)\mathrm{d}\nabla_k H(\zeta) = \int_{\mathbb{R}} h(\zeta)\frac{\mathrm{d}\nabla_k H(\zeta)}{\mathrm{d}\nu(\zeta)}\mathrm{d}\nu(\zeta). \tag{6.3.15}$$

In the rest of the book we will assume that

$$\left\| \frac{\mathrm{d}\nabla_k \hat{H}_i(t,\cdot)}{\mathrm{d}\nu} \right\| < \infty \text{ for all } t \in [0,T],\ i = 1,2, \tag{6.3.16}$$

where here, and in the following, we are using the abbreviated notation

$$H_i(t) = H_i(t, X(t), Y(t), u(t), p_i(t), q_i(t), r_i(t,\cdot))$$

and similarly for

$$\hat{H}_i(t) = H_i(t, \hat{X}(t), \hat{Y}(t), \hat{u}(t), \hat{p}_i(t), \hat{q}_i(t), \hat{r}_i(t,\cdot))$$

and related processes.

The BSDE for the adjoint processes p_i, q_i, r_i is

$$\begin{cases} \mathrm{d}p_i(t) &= -\dfrac{\partial H_i}{\partial x}(t, X(t), Y(t), u(t), p_i(t), q_i(t), r_i(t))\mathrm{d}t \\ &\quad -\dfrac{\partial H_i}{\partial y}(t, X(t), Y(t), u(t), p_i(t), q_i(t), r_i(t))\nabla_{X(t)}F)\mathrm{d}t \\ &\quad +q_i(t)\mathrm{d}B(t) + \displaystyle\int_{\mathbb{R}} r_i(t,\zeta)\tilde{N}(\mathrm{d}t, \mathrm{d}\zeta) \\ p_i(T) &= \dfrac{\partial g_i}{\partial x}(X(T), Y(T)) + \dfrac{\partial g_i}{\partial y}(X(T), Y(T))\nabla_{X(T)}F; \quad i = 1,2. \end{cases} \tag{6.3.17}$$

Theorem 6.10 (***Sufficient maximum principle***) *Let $\hat{u}_1 \in \mathcal{A}^{(1)}$, $\hat{u}_2 \in \mathcal{A}^{(2)}$ with corresponding solutions $\hat{X}, \hat{p}_i, \hat{q}_i, \hat{r}_i$ of (6.3.9) and (6.3.17). Assume the following:*

- *The maps*

$$X, u_1 \to H_1(t, X, F(X), u_1, \hat{u}_2(t), \hat{p}_1(t), \hat{q}_1(t), \hat{r}_1(t)), \tag{6.3.18}$$

$$X, u_2 \to H_2(t, X, F(X), \hat{u}_1(t), u_2, \hat{p}_2(t), \hat{q}_2(t), \hat{r}_2(t)), \tag{6.3.19}$$

and

$$X \to g_i(X, F(X)) \tag{6.3.20}$$

are concave for all t; $i = 1, 2$.

- *(The conditional maximum properties)*

$$\begin{aligned} &ess\ sup_{u_1} E[H_1(t, \hat{X}(t), \hat{Y}(t), u_1, \hat{u}_2(t), \hat{p}_1(t), \hat{q}_1(t), \hat{r}_1(t, \cdot)) \mid \mathcal{E}_t^{(1)}] \\ &\quad = E[H_1(t, \hat{X}(t), \hat{Y}(t), \hat{u}_1(t), \hat{u}_2(t), \hat{p}_1(t), \hat{q}_1(t), \hat{r}_1(t, \cdot)) \mid \mathcal{E}_t^{(1)}] \end{aligned} \tag{6.3.21}$$

and

$$\begin{aligned} &ess\ sup_{u_2} E[H_2(t, \hat{X}(t), \hat{Y}(t), \hat{u}_1(t), u_2, \hat{p}_2(t), \hat{q}_2(t), \hat{r}_2(t, \cdot)) \mid \mathcal{E}_t^{(2)}] \\ &\quad = E[H_2(t, \hat{X}(t), \hat{Y}(t), \hat{u}_1(t), \hat{u}_2(t), \hat{p}_2(t), \hat{q}_2(t), \hat{r}_2(t, \cdot)) \mid \mathcal{E}_t^{(2)}]. \end{aligned} \tag{6.3.22}$$

Then $(\hat{u}_1, \hat{u}_2)$ *is a Nash equilibrium, in the sense that* (6.3.12) *and* (6.3.13) *hold with* $u_i^* := \hat{u}_i$; $i = 1, 2$.

Proof By introducing an increasing sequence of stopping times converging to T, we see that we may assume that all local martingales appearing in the proof below are martingales, and hence have expectation 0. See the proof of Theorem 5.4.
We first study the stochastic control problem (6.3.12):
For simplicity of notation, in the following we put $X(t) = X^{u_1,\hat{u}_2}(t), Y(t) = Y^{u_1,\hat{u}_2}(t)$ and $\hat{X}(t) = X^{\hat{u}_1,\hat{u}_2}(t), \hat{Y}(t) = Y^{\hat{u}_1,\hat{u}_2}(t)$.

Consider $J_1(u_1, \hat{u}_2) - J_1(\hat{u}_1, \hat{u}_2) = I_1 + I_2$, where

$$I_1 = E\left[\int_0^T \{f_1(t, X(t), Y(t), u_1(t), \hat{u}_2(t)) - f_1(t, \hat{X}(t), \hat{Y}(t), \hat{u}_1(t), \hat{u}_2(t))\}\mathrm{d}t\right],$$

$$I_2 = E[g_1(X(T), Y(T)) - g_1(\hat{X}(T), \hat{Y}(T))].$$

By the definition of H_1 we have

$$\begin{aligned} I_1 = E\Bigg[\int_0^T \Big\{ &H_1(t, X_t, Y_t, u_1(t), \hat{u}_2(t), \hat{p}_1(t), \hat{q}_1(t), \hat{r}_1(t)) \\ &- H_1(t, \hat{X}_t, \hat{Y}_t, \hat{u}(t), \hat{p}_1(t), \hat{q}_1(t), \hat{r}_1(t)) \\ &-(b - \hat{b})\hat{p}_1 - (\sigma - \hat{\sigma})\hat{q}_1 - \int (\gamma - \hat{\gamma})\hat{r}_1(\zeta)\nu(\mathrm{d}\zeta)\Big\}\mathrm{d}t\Bigg]. \end{aligned} \tag{6.3.23}$$

By concavity of g_1 and the Itô formula we have

$$
\begin{aligned}
I_2 &\le E\Big[\frac{\partial g_1}{\partial x}(\hat{X}(T),\hat{Y}(T))(X(T)-\hat{X}(T)) \\
&\quad +\frac{\partial g_1}{\partial y}(\hat{X}(T),\hat{Y}(T))\nabla_{\hat{X}(T)}F(X(T)-\hat{X}(T))\Big] \\
&=< \hat{p}_1(T), X(T)-\hat{X}(T) > \\
&= E\Big[\int_0^T \hat{p}_1(t)\mathrm{d}\widetilde{X}(t)+\int_0^T \widetilde{X}(t)\mathrm{d}\hat{p}_1(t) \\
&\quad +\int_0^T \hat{q}_1(t)\widetilde{\sigma}(t)\mathrm{d}t+\int_0^T\int_{\mathbb{R}} \hat{r}_1(t,\zeta)\tilde{\gamma}(t,\zeta)\nu(\mathrm{d}\zeta)\mathrm{d}t\Big], \qquad (6.3.24)
\end{aligned}
$$

where we have put

$$
\widetilde{X}(t)=X(t)-\hat{X}(t),\quad \widetilde{\sigma}(t)=\sigma(t)-\hat{\sigma}(t). \qquad (6.3.25)
$$

Note that

$$
E\Big[\int_0^T \hat{p}_1(t)\mathrm{d}\widetilde{X}(t)\Big]=E\Big[\int_0^T \hat{p}_1(t)(b-\hat{b})\mathrm{d}t\Big] \qquad (6.3.26)
$$

and that

$$
\begin{aligned}
&E\Big[\int_0^T \widetilde{X}(t)\mathrm{d}\hat{p}_1(t)\Big] \\
&\quad = E\Big[\int_0^T \widetilde{X}(t)\Big[-\frac{\partial H_1}{\partial x}(t,\hat{X}(t),\hat{Y}(t),\hat{u}(t),\hat{p}_1(t),\hat{q}_1(t),\hat{r}_1(t)) \\
&\quad -\frac{\partial H_1}{\partial y}(t,\hat{X}(t),\hat{Y}(t),\hat{u}(t),\hat{p}_1(t),\hat{q}_1(t),\hat{r}_1(t))\nabla_{\hat{X}(t)}F)\Big]\mathrm{d}t\Big]. \qquad (6.3.27)
\end{aligned}
$$

Combining the above we get

$$
\begin{aligned}
&J_1(u_1,\hat{u}_2)-J_1(\hat{u}_1,\hat{u}_2) \\
&\le E\Big[\int_0^T\{H_1(t)-\hat{H}_1(t)-\frac{\partial \hat{H}_1}{\partial x}(X-\hat{X})-\frac{\partial \hat{H}_1}{\partial y}\nabla_{\hat{X}(t)}F(X-\hat{X})\}\mathrm{d}t\Big] \\
&= E\Big[\int_0^T E[H_1(t)-\hat{H}_1(t)-\frac{\partial \hat{H}_1}{\partial x}(X-\hat{X})-\frac{\partial \hat{H}_1}{\partial y}\nabla_{\hat{X}(t)}F(X-\hat{X})\mid \mathcal{E}_t^{(1)}]\mathrm{d}t\Big], \qquad (6.3.28)
\end{aligned}
$$

where $\hat{H}_1(t)$ means H_1 evaluated at $(t,\hat{X}(t),\hat{Y}(t),\hat{u}(t),\hat{p}_1(t),\hat{q}_1(t),\hat{r}_1(t))$, and $H_1(t)$ is H_1 evaluated at $(t,X(t),Y(t),u_1(t),\hat{u}_2(t),\hat{p}_1(t),\hat{q}_1(t),\hat{r}_1(t))$.

Note that by concavity of H_1 we have

$$H_1(t, X, F(X), u_1, \hat{u}_2, \hat{p}_1, \hat{q}_1, \hat{r}_1) - H_1(t, \hat{X}, F(\hat{X}), \hat{u}_1, \hat{u}_2, \hat{p}_1, \hat{q}_1, \hat{r}_1)$$
$$\leq \frac{\partial \hat{H}_1}{\partial x}(\hat{X})(X - \hat{X}) + \frac{\partial \hat{H}_1}{\partial y}(\hat{X})\nabla_{\hat{X}} F(X - \hat{X}) + \frac{\partial \hat{H}_1}{\partial u_1}(\hat{u})(u_1 - \hat{u}_1). \quad (6.3.29)$$

Therefore, to obtain that $J_1(u_1, \hat{u}_2) - J_1(\hat{u}_1, \hat{u}_2) \leq 0$, it suffices that

$$E\left[\frac{\partial \hat{H}_1}{\partial u_1}(\hat{u}) \mid \mathcal{E}_t^{(1)}\right](u_1 - \hat{u}_1) \leq 0 \quad (6.3.30)$$

for all u_1.

The inequality (6.3.30) holds by our assumption (6.3.21).
The difference

$$J_2(\hat{u}_1, u_2) - J_2(\hat{u}_1, \hat{u}_2)$$

is handled similarly. □

6.3.4 A Necessary Maximum Principle

In Theorem 6.10 we proved a verification theorem, stating that if a given control $\hat{u}$ satisfies certain conditions, then it is indeed optimal for the mean-field control game. We now establish a partial converse, implying that if a control $\hat{u}$ is optimal for the mean-field control game, then it is a conditional saddle point for the Hamiltonian.

To achieve this, we start with the setup of Theorem 6.5 as follows:

Assume that (6.2.16) and (6.2.17) hold, and assume the analogue of (6.2.18)–(6.2.20), namely that the following derivative processes exit and belong to $L^2([0, T] \times \Omega)$:

$$x_1(t) = \frac{\mathrm{d}}{\mathrm{d}s} X^{(u_1 + s\beta_1, u_2)}(t) \mid_{s=0},$$
$$x_2(t) = \frac{\mathrm{d}}{\mathrm{d}s} X^{(u_1, u_2 + s\beta_2)}(t) \mid_{s=0},$$

and that they satisfy the SDEs

$$\mathrm{d}x_i = \left\{\frac{\partial b}{\partial x}(t)x_i(t) + \left\langle \frac{\partial b}{\partial y}(t)\nabla_{X(t)} F, x_i(t)\right\rangle + \frac{\partial b}{\partial u_i}(t)\beta_i(t)\right\}\mathrm{d}t$$
$$+ \left\{\frac{\partial \sigma}{\partial x}(t)x_i(t) + \left\langle \frac{\partial \sigma}{\partial y}(t)\nabla_{X(t)} F, x_i(t)\right\rangle + \frac{\partial \sigma}{\partial u_i}(t)\beta_i(t)\right\}\mathrm{d}B(t)$$

$$+\int_{\mathbb{R}}\left\{\frac{\partial\gamma}{\partial x}(t)x_i(t)+\left\langle\frac{\partial\gamma}{\partial y}(t)\nabla_{X(t)}F,x_i(t)\right\rangle+\frac{\partial\gamma}{\partial u_i}(t)\beta_i(t)\right\}\tilde{N}(\mathrm{d}t,\mathrm{d}\zeta)$$

for all u_i, β_i as in (6.2.17); $i=1,2$. Note that, by the chain rule, if $h\in\mathcal{C}^1(\mathbb{R})$,

$$\begin{aligned}\frac{\mathrm{d}}{\mathrm{d}s}h\left(Y^{(u_1+s\beta_1,u_2)}(t)\right)\mid_{s=0}&=\frac{\mathrm{d}}{\mathrm{d}s}h\left(F\left(X^{(u_1+s\beta_1,u_2)}(t)\right)\right)\mid_{s=0}\\&=\left\langle h'(Y^u(t))\nabla_{X^u(t)}F,x_1(t)\right\rangle\end{aligned}$$

and similarly

$$\frac{\mathrm{d}}{\mathrm{d}s}h\left(Y^{(u_1,u_2+s\beta_2)}(t)\right)\mid_{s=0}=\left\langle h'(Y^u(t))\nabla_{X^u(t)}F,x_2(t)\right\rangle.$$

Under these assumptions we can prove the following:

Theorem 6.11 (***Necessary maximum principle for mean-field games***)
Suppose $\hat{u}_1\in\mathcal{A}^{(1)}$ and $\hat{u}_2\in\mathcal{A}^{(2)}$ constitute a Nash equilibrium for the game, i.e. they satisfy (6.3.12) *and* (6.3.13). *Then*

$$E\left[\frac{\partial H_1}{\partial u_1}(t,\hat{X}(t),\hat{Y}(t),u_1,\hat{u}_2(t),\hat{p}_1(t),\hat{q}_1(t),\hat{r}_1(t,\cdot))_{u_1=\hat{u}_1(t)}\mid\mathcal{E}_t^{(1)}\right]=0 \tag{6.3.31}$$

and

$$E\left[\frac{\partial H_2}{\partial u_2}(t,\hat{X}(t),\hat{Y}(t),\hat{u}_1(t),u_2,\hat{p}_2(t),\hat{q}_2(t),\hat{r}_2(t,\cdot))_{u_2=\hat{u}_2(t)}\mid\mathcal{E}_t^{(2)}\right]=0. \tag{6.3.32}$$

Proof Let u_i, β_i be as in (6.2.17); $i=1,2$. First, consider

$$\begin{aligned}D_1^{\beta_i}&:=\frac{\mathrm{d}}{\mathrm{d}s}J_i(u_1+s\beta_1,u_2)\mid_{s=0}\\&=E\left[\int_0^T\left\{\frac{\partial f_1}{\partial x}(t)x_1(t)+\frac{\partial f_1}{\partial y}(t)\nabla_{X(t)}Fx_1(t)+\frac{\partial f_1}{\partial u}(t)\beta_1(t)\right\}\mathrm{d}t\right.\\&\left.+\frac{\partial g_1}{\partial x}(X^u(T),Y^u(T))x_1(T)+\frac{\partial g_1}{\partial y}(X^u(T),Y^u(T))\nabla_{X(T)}Fx_1(T)\right].\end{aligned}$$

By (6.3.17) we have

$$\begin{aligned}&E\left[\frac{\partial g_1}{\partial x}(X(T),Y(T))x_1(T)+\frac{\partial g_1}{\partial y}(X(T),Y(T))\nabla_{X(T)}Fx_1(T)\right]\\&=E[p_1(T)x_1(T)]\\&=E\left[\int_0^T p_1(t^-)\mathrm{d}x_1(t)+x_1(t^-)\mathrm{d}p_1(t)\right]\end{aligned}$$

$$+ q_1(t)\left\{\frac{\partial\sigma}{\partial x}(t)x_1(t) + \frac{\partial\sigma}{\partial y}(t)\nabla_{X(t)}Fx_1(t) + \frac{\partial\sigma}{\partial u_1}(t)\beta_1(t)\right\}\mathrm{d}t$$
$$+ \int_{\mathbb{R}} r_1(t,\zeta)\left\{\frac{\partial\gamma}{\partial x}(t)x_1(t) + \frac{\partial\gamma}{\partial y}(t)\nabla_{X(t)}Fx_1(t) + \frac{\partial\gamma}{\partial u_1}(t)\beta_1(t)\right\}\nu(\mathrm{d}\zeta)\mathrm{d}t.$$

Hence, by (6.3.14) and (6.3.17),

$$\begin{aligned}
D_1^{\beta_1} &= E\left[\int_0^T \left(\frac{\partial f_1}{\partial x}(t) + \frac{\partial f_1}{\partial y}(t)\nabla_{X(t)}F + p_1(t)\left(\frac{\partial b}{\partial x}(t) + \frac{\partial b}{\partial y}(t)\nabla_{X(t)}F\right)\right.\right.\\
&\quad + q_1(t)\left(\frac{\partial\sigma}{\partial x}(t) + \frac{\partial\sigma}{\partial y}(t)\nabla_{X(t)}F\right)\\
&\quad + \int_{\mathbb{R}} r_1(t,\zeta)\left(\frac{\partial\gamma}{\partial x}(t,\zeta) + \frac{\partial\gamma}{\partial y}(t,\zeta)\nabla_{X(t)}F\right)\nu(\mathrm{d}\zeta)\\
&\quad \left.\left. - \frac{\partial H_1}{\partial x}(t) - \frac{\partial H_1}{\partial y}(t)\nabla_{X(t)}F\right)x_1(t)\mathrm{d}t\right]\\
&\quad + E\left[\int_0^T \left\{\frac{\partial f_1}{\partial u_1}(t) + p_1(t)\frac{\partial b}{\partial u_1}(t) + q_1(t)\frac{\partial\sigma}{\partial u_1}(t)\right.\right.\\
&\quad \left.\left. + \int_{\mathbb{R}} r_1(t,\zeta)\frac{\partial\gamma}{\partial u_1}(t,\zeta)\nu(\mathrm{d}\zeta)\right\}\beta_1(t)\mathrm{d}t\right]\\
&= E\left[\int_0^T \frac{\partial H_1}{\partial u_1}(t)\beta_1(t)\mathrm{d}t\right] = E\left[\int_0^T E\left[\frac{\partial H_1}{\partial u_1}(t)\beta_1(t) \mid \mathcal{E}_t^{1)}\right]\mathrm{d}t\right]\\
&= E\left[\int_0^T \beta_1(t)E\left[\frac{\partial H_1}{\partial u_1}(t) \mid \mathcal{E}_t^{(1)}\right]\mathrm{d}t\right].
\end{aligned}$$

Thus we see that if $D_1^{\beta_1} = 0$ for all bounded $\mathcal{E}_t^{(1)}$-adapted $\beta_1(t)$ we obtain that

$$E\left[\frac{\partial H_1}{\partial u_1}(t) \mid \mathcal{E}_t^{(1)}\right] = 0 \text{ for all } t \in [0,T].$$

The proof for $D_2^{\beta_2} := \frac{\mathrm{d}}{\mathrm{d}s}J_2(u_1, u_2 + s\beta_2)$ is similar. □

Remark 6.12 Note that the argument in the above proof can be reversed, to give the following result of independent interest:

Corollary 6.13 *Under the assumptions of the previous theorem the following are equivalent, for each* $i = 1, 2$*:*

(i) $D_i^{\beta_i} = 0$ *for all bounded* $\beta_i \in \mathcal{A}_i$,
(ii) $E\left[\frac{\partial H_i}{\partial u_i}(t) \mid \mathcal{E}_t^{(i)}\right] = 0$ *for all* $t \in [0,T]$.

□

6.3.5 Application to Model Uncertainty Control

General Theory

We represent *model uncertainty* by a family of probability measures $Q = Q^\theta$ equivalent to P, with the Radon–Nikodym derivative on $\mathcal{F}_t$ given by

$$\frac{\mathrm{d}(Q \mid \mathcal{F}_t)}{\mathrm{d}(P \mid \mathcal{F}_t)} = G^\theta(t), \tag{6.3.33}$$

where, for $0 \le t \le T$, $G^\theta(t)$ is a martingale of the form

$$\mathrm{d}G^\theta(t) = G^\theta(t^-)\left[\theta_0(t)\mathrm{d}B(t) + \int_{\mathbb{R}} \theta_1(t,\zeta)\tilde{N}(\mathrm{d}t,\mathrm{d}\zeta)\right];\quad G^\theta(0) = 1. \tag{6.3.34}$$

Here $\theta = (\theta_0, \theta_1)$ may be regarded as a *scenario control*. Let $\mathcal{A}_1 := \mathcal{A}_{\mathbb{G}}$ denote a given family of admissible singular controls ξ and let $\mathcal{A}_2 := \Theta$ denote a given set of admissible scenario controls θ such that

$$E\left[\int_0^T \{|\theta_0^2(t)| + \int_{\mathbb{R}} \theta_1^2(t,\zeta)\nu(\mathrm{d}\zeta)\}\mathrm{d}t\right] < \infty \tag{6.3.35}$$

and $\theta_1(t,\zeta) \ge -1 + \epsilon$ for some $\epsilon > 0$.

Now assume that $X_1(t) = X^u(t)$ is a controlled mean-field Itô–Lévy process of the form

$$\begin{aligned}\mathrm{d}X_1(t) &= b_1(t, X_1(t), Y_1(t), \omega)\mathrm{d}t + \sigma_1(t, X_1(t), Y_1(t), \omega)\mathrm{d}B(t)\\ &\quad + \int_{\mathbb{R}} \gamma_1(t, X_1(t), Y_1(t), \zeta, \omega)\tilde{N}(\mathrm{d}t, \mathrm{d}\zeta),\end{aligned} \tag{6.3.36}$$

where

$$Y_1(t) = F(X_1(t,\cdot)), \tag{6.3.37}$$

and F is a Fréchet differentiable operator on $L^2(P)$.

As before, let $\mathbb{E}^{(1)} = \{\mathbb{E}_t^{(1)}\}_{0\le t\le T}$ and $\mathbb{E}^{(2)} = \{\mathcal{E}_t^{(2)}\}_{0\le t\le T}$ be given subfiltrations of $\mathbb{F} = \{\mathcal{F}_t\}_{0\le t\le T}$, representing the information available to the controllers at time t. It is required that $\xi \in \mathcal{A}_1$ be $\mathbb{E}^{(1)}$-predictable, and $\theta \in \mathcal{A}_2$ be $\mathbb{E}^{(2)}$-predictable. We set $w = (u, \theta)$ and consider the stochastic differential game to find $(\hat{u}, \hat{\theta}) \in \mathcal{A}_1 \times \mathcal{A}_2$ such that

$$\sup_{u\in\mathcal{A}_1}\inf_{\theta\in\mathcal{A}_2} E_{Q^\theta}[j(u,\theta)] = E_{Q^{\hat{\theta}}}[j(\hat{u},\hat{\theta})] = \inf_{\theta\in\mathcal{A}_2}\sup_{u\in\mathcal{A}_1} E_{Q^\theta}[j(u,\theta)], \tag{6.3.38}$$

where

$$j(u,\theta) = \int_0^T \{f_1(t, X(t), Y(t), u(t), \omega) + \lambda(\theta(t))\}\mathrm{d}t + g_1(X(T), Y(T), \omega). \tag{6.3.39}$$

The term $E_{Q^\theta}[\int_0^T \lambda(\theta(t))\mathrm{d}t]$ can be seen as a penalty term, penalizing the difference between Q^θ and the original probability measure P.

Note that since $G^\theta(t)$ is a martingale we have

$$E_{Q^\theta}[j(u,\theta)] = E\Big[G^\theta(T)g_1(X(T), Y(T)) + \int_0^T G^\theta(t)\{f_1(t, X(t), Y(t), u(t))) + \lambda(\theta(t))\}\mathrm{d}t\Big]. \tag{6.3.40}$$

We see that this is a mean-field stochastic differential game of the type discussed in Sect. 6.3.2, with a two-dimensional state space

$$X(t) := (X_1(t), X_2(t)) := (X^u(t), G^\theta(t)) \tag{6.3.41}$$

and with

$$\begin{aligned} f(t, X(t), Y(t), u, \theta)) &:= G^\theta(t)\{f_1(t, X_1(t), Y_1(t), u(t)) + \lambda(\theta(t))\} \\ &= X_2(t)\{f_1(t, X_1(t), Y_1(t), u(t)) + \lambda(\theta(t))\} \end{aligned} \tag{6.3.42}$$

and

$$g(X(T), Y(T)) := G^\theta(T)g_1(X_1(T), Y_1(T)) = X_2(T)g_1(X_1(T), Y_1(T)). \tag{6.3.43}$$

Accordingly, using the result from Sect. 6.3.2, we get the following Hamiltonian for the game (6.3.38):

$$\begin{aligned} H(t, x_1, x_2, y_1, u, \theta, p, q, r) &= x_2\{f_1(t, x_1, y_1, u) + \lambda(\theta)\} \\ &+ b_1(t, x_1, y_1)p_1 + \sigma(t, x_1, y_1)q_1 + x_2\theta_0 q_2 \\ &+ \int_{\mathbb{R}} \gamma_1(t, x_1, y_1(t), \zeta)r_1(\zeta)\nu(\mathrm{d}\zeta) + \int_{\mathbb{R}} x_2\theta_1(\zeta)r_2(\zeta)\nu(\mathrm{d}\zeta) \end{aligned} \tag{6.3.44}$$

and the corresponding mean-field BSDEs for the adjoint processes become

$$\begin{cases} \mathrm{d}p_1(t) & = -\dfrac{\partial H}{\partial x_1}(t, X(t), Y(t), w(t), p(t), q(t), r(t))\mathrm{d}t \\ & \quad -\dfrac{\partial H}{\partial y_1}(t, X(t), Y(t), w(t), p(t), q(t), r(t))\nabla_{X_1(t)}F)\mathrm{d}t \\ & \quad +q_1(t)\mathrm{d}B(t) + \displaystyle\int_{\mathbb{R}} r_1(t,\zeta)\tilde{N}(\mathrm{d}t, \mathrm{d}\zeta) \\ p_1(T) & = X_2(T)\dfrac{\partial g_1}{\partial x_1}(X_1(T), Y_1(T)) + E\left[\dfrac{\partial g_1}{\partial y_1}(X_1(T), Y_1(T))\right]\nabla_{X_1(T)}F \end{cases} \tag{6.3.45}$$

and

$$\begin{cases} \mathrm{d}p_2(t) & = -\{f_1(t, X_1(t), Y_1(t), u(t)) + \lambda(\theta(t)) + \theta_0(t)q_2(t) \\ & \quad + \displaystyle\int_{\mathbb{R}} \theta_1(t,\zeta)r_2(t,\zeta)\nu(\mathrm{d}\zeta)\}\mathrm{d}t \\ & \quad +q_2(t)\mathrm{d}B(t) + \displaystyle\int_{\mathbb{R}} r_2(t,\zeta)\tilde{N}(\mathrm{d}t, \mathrm{d}\zeta) \\ p_2(T) & = g_1(X_1(T), Y_1(T)). \end{cases}$$

Minimizing the Hamiltonian with respect to $\theta = (\theta_0, \theta_1)$ gives the following first-order conditions:

$$\frac{\partial \lambda}{\partial \theta_0}(t) = -E[q_2(t) \mid \mathcal{E}_t^{(2)}] \tag{6.3.46}$$

and

$$\nabla_{\theta_1}\lambda(\theta)(t)(\cdot) = -E\left[\int_{\mathbb{R}} r_2(t,\zeta)(\cdot) \mid \mathcal{E}_t^{(2)}\right]\nu(\mathrm{d}\zeta) \tag{6.3.47}$$

(as linear operators).

A Special Case

For simplicity, consider the special case with

$$\mathcal{E}_t^{(i)} = \mathcal{F}_t, \quad N = 0. \tag{6.3.48}$$

Then, writing $X_1(t) = X^u(t)$, $Y_1(t) = Y^u(t)$ and $X_2(t) = G^\theta(t)$ and $f_1 = f, g_1 = g, b_1 = b, \sigma_1 = \sigma$, and $\theta_0 = \theta$, the controlled system gets the form

$$\mathrm{d}X^u(t) = b(t, X^u(t), Y^u(t))\mathrm{d}t + \sigma(t, X^u(t), Y^u(t))\mathrm{d}B(t); \; X^u(0) = x \tag{6.3.49}$$

$$\mathrm{d}G^\theta(t) = G^\theta(t)\theta(t)\mathrm{d}B(t); \quad G^\theta(0) = 1. \tag{6.3.50}$$

The performance functional becomes

$$E_{Q^\theta}[j(u,\theta)] = E\Big[G^\theta(T)g(X^u(T), Y^u(T)) + \int_0^T G^\theta(t)\{f(t, X^u(t), Y^u(t), u(t)) + \lambda(\theta(t))\}\mathrm{d}t\Big], \tag{6.3.51}$$

the Hamiltonian becomes

$$\begin{aligned} &H(t,x,g,y,u,\theta,p,q) \\ &\quad = g\{f_1(t,x,y,u) + \lambda(\theta)\} + b_1(t,x,y)p_1 + \sigma(t,x,y)q_1 + g\theta q_2 \end{aligned} \tag{6.3.52}$$

and the corresponding mean-field BSDEs for the adjoint processes become

$$\begin{cases} \mathrm{d}p_1(t) &= -\dfrac{\partial H}{\partial x}(t, X^u(t), Y^u(t), u(t), p(t), q(t))\mathrm{d}t \\ &\quad -\dfrac{\partial H}{\partial y}(t, X^u(t), Y^u(t), u(t), p(t), q(t))\nabla_{X^u(t)}F)\mathrm{d}t \\ &\quad +q_1(t)\mathrm{d}B(t) \\ p_1(T) &= G^\theta(T)\dfrac{\partial g}{\partial x}(X^u(T), Y^u(T)) + E\left[\dfrac{\partial g}{\partial y}(X^u(T), Y^u(T))\right]\nabla_{X^u(T)}F \end{cases} \tag{6.3.53}$$

and

$$\begin{cases} \mathrm{d}p_2(t) &= -\{f(t, X^u(t), Y^u(t), u) + \lambda(\theta(t)) + \theta(t)q_2(t)\}\mathrm{d}t + q_2(t)\mathrm{d}B(t) \\ p_2(T) &= g_1(X^u(T), Y^u(T)). \end{cases} \tag{6.3.54}$$

Then the first-order condition for a minimum of the Hamiltonian with respect to θ reduces to

$$\rho'(\theta)(t) = -q_2(t). \tag{6.3.55}$$

In general it is difficult to solve such a coupled system of forward-backward mean-field SDEs. However, in some cases a possible solution procedure can be described, as in the next example.

Example 6.14 (***Optimal harvesting under uncertainty***)

Now we consider a model uncertainty version of the optimal harvesting problem in Sect. 6.3.1(a). For simplicity we put $K = 1$ and $N = 0$. Thus we have the following mean-field forward system, with $X(t) = X^u(t)$,

$$\begin{cases} \mathrm{d}X(t) &= E[X(t)]a(t)\mathrm{d}t + X(t)\sigma(t)\mathrm{d}B(t) - X(t)u(t)\mathrm{d}t \\ X(0) &= x > 0 \end{cases} \tag{6.3.56}$$

and

$$\mathrm{d}G^\theta(t) = G^\theta(t)\theta(t)\mathrm{d}B(t)\ ; \quad G^\theta(0) = 1, \tag{6.3.57}$$

with performance functional

$$J(u,\theta) = E[\int_0^T G^\theta(t)\{h_0 X(t)u(t) - \frac{1}{2}X^2(t)u^2(t) + \lambda\theta(t)\}\mathrm{d}t + G^\theta(T)X(T)]. \tag{6.3.58}$$

Hence the Hamiltonian is

$$\begin{aligned} &H(t,x,g,y,u,\theta,p,q) \\ &= g\{h_0 xu - \frac{1}{2}\rho x^2u^2 - \lambda(\theta)\} + ya(t)p_1 + x\sigma(t)q_1 + g\theta q_2. \end{aligned} \tag{6.3.59}$$

Minimizing the Hamiltonian with respect to θ gives the first-order equation

$$\rho'(\theta)(t) = -q_2(t). \tag{6.3.60}$$

The corresponding reflected backward system is

$$\begin{cases} \mathrm{d}p_1(t) &= -[gh_0u - \rho xu^2 + a(t)p_1(t) + \sigma(t)q_1(t)]\mathrm{d}t + q_1(t)\mathrm{d}B(t) \\ p_1(T) &= G^\theta(T) \end{cases} \tag{6.3.61}$$

and

$$\begin{cases} \mathrm{d}p_2(t) = &-[h_0X(t)u(t) - \frac{1}{2}\rho X^2(t)u^2(t) - \lambda\theta(t) \\ &+\theta(t)q_2(t)]\mathrm{d}t + q_2(t)\mathrm{d}B(t) \\ p_2(T) &= X(T). \end{cases} \tag{6.3.62}$$

Then we get the following result:

Proposition 6.15 *Suppose there exists a solution* $\hat{X}(t) := X^{\hat{u}}(t)$, $\hat{G}(t) := G^{\hat{\theta}}(t)$, $\hat{p}_1(t)$, $\hat{q}_1(t)$, $\hat{p}_2(t)$, $\hat{q}_2(t)$, $\hat{u}(t)$, $\hat{\theta}(t)$ *of the coupled system of mean-field forward-backward stochastic differential equations consisting of the forward equations* (6.3.56), (6.3.57) *and the backward equations* (6.3.61), (6.3.62), *and satisfying the constraint* (6.3.60).
Then $\hat{u}(t)$ *is the optimal harvesting strategy and* $\hat{\theta}(t)$ *is the optimal scenario parameter for the model uncertainty harvesting problem* (6.3.38).

Example 6.16 (***Optimal insider consumption under model uncertainty***) Suppose we have a cash flow with consumption, modeled by the process $X(t) = X^{c,\mu}(t)$ defined by:

$$\begin{cases} \mathrm{d}X(t) = &(\alpha(t) + \mu(t) - c(t))X(t)\mathrm{d}t + \beta(t)X(t)\mathrm{d}B(t) \\ &+\int_{\mathbb{R}}\gamma(t,\zeta)X(t)\tilde{N}(\mathrm{d}t,\mathrm{d}\zeta) \\ X(0) = &x > 0. \end{cases}$$

Here $\alpha(t)$, $\beta(t)$, $\gamma(t,\zeta)$ are given coefficients, while $c(t) > 0$ is the relative consumption rate chosen by the consumer (player number 1) and $\mu(t)$ is a perturbation

of the drift term, representing the model uncertainty chosen by the environment (player number 2). Define the performance functional by

$$J(c,\mu)=E\left[\int_0^T\{\log(c(t)X(t))+\frac{1}{2}\mu^2(t)\}\mathrm{d}t+\theta\log X(T)\right],\tag{6.3.63}$$

where $\theta=\theta(\omega)>0$ is a given $\mathcal{F}_T$-measurable random variable, and $\frac{1}{2}\mu^2(t)$ represents a penalty rate, penalizing μ for being away from 0. We assume that c and μ are $\mathbb{F}$-adapted.

We want to find $c^*\in\mathcal{A}_1$ and $\mu^*\in\mathcal{A}_2$ such that

$$\sup_{c\in\mathcal{A}_1}\inf_{\mu\in\mathcal{A}_2}J(c,\mu)=J(c^*,\mu^*).\tag{6.3.64}$$

The Hamiltonian for this problem is

$$\begin{aligned}&H(t,x,y,c,\mu,p,q,r)\\&=\left\{\log(cx)+\frac{1}{2}\mu^2\right\}+(\alpha(t)+\mu-c)xp+\beta(t)xq+x\int_{\mathbb{R}}\gamma(t,\zeta)r(\zeta)\mathrm{d}\nu(\zeta)\end{aligned}\tag{6.3.65}$$

and the BSDE for the adjoint processes p,q,r is

$$\begin{cases}\mathrm{d}p(t) &= -[\frac{1}{X(t)}+(\alpha(t)+\mu(t)-c(t))p(t)\\ &\quad+\beta(t)q(t)+\int_{\mathbb{R}}\gamma(t,\zeta)r(\zeta)\mathrm{d}\nu(\zeta)]\mathrm{d}t\\ &\quad+q(t)\mathrm{d}B(t)+\int_{\mathbb{R}}r(t)\tilde{N}(\mathrm{d}t,\mathrm{d}\zeta);\ 0\le t\le T\\ p(T) &= \frac{\theta}{X(T)}.\end{cases}\tag{6.3.66}$$

Define

$$h(t)=p(t)X(t).\tag{6.3.67}$$

Then by the Itô formula we get

$$\begin{aligned}\mathrm{d}h(t)=X(t)\Big[&-\frac{1}{X(t)}-(\alpha(t)+\mu(t)-c(t))p(t)-\beta(t)q(t)\\&-\int_{\mathbb{R}}\gamma(t,y,\zeta)r(t,\zeta)\mathrm{d}\nu(\zeta)\Big]\mathrm{d}t\\&+p(t)(\alpha(t)+\mu(t)-c(t))X(t)\mathrm{d}t+p(t)\beta(t)X(t)\mathrm{d}B(t)+X(t)q(t)\mathrm{d}B(t)\\&+q(t)\beta(t)X(t)\mathrm{d}t\\&+\int_{\mathbb{R}}[(X(t)+\gamma(t,\zeta)X(t))(p(t)+r(t,\zeta))-p(t)X(t)-p(t)\gamma(t,\zeta)X(t)\\&-X(t)r(t,\zeta]\mathrm{d}\nu(\zeta)\mathrm{d}t\end{aligned}$$

$$+\int_{\mathbb{R}}[(X(t)+\gamma(t,\zeta)X(t))(p(t)+r(t,\zeta))-p(t)X(t)]\tilde{N}(\mathrm{d}t,\mathrm{d}\zeta)$$
$$=\mathrm{d}F(t)+h(t)\beta(t)\mathrm{d}B(t)+h(t)\int_{\mathbb{R}}\gamma(t,\zeta)\tilde{N}(\mathrm{d}t,\mathrm{d}\zeta)), \tag{6.3.68}$$

where

$$\mathrm{d}F(t)=-\mathrm{d}t+X(t)q(t)\mathrm{d}B(t)+X(t)\int_{\mathbb{R}}r(t,\zeta)(1+\gamma(t,\zeta))\tilde{N}(\mathrm{d}t,\mathrm{d}\zeta). \tag{6.3.69}$$

To simplify this, we define the process $k(t)$ by the equation

$$\mathrm{d}k(t)=k(t)\Big[b(t)\mathrm{d}B(t)+\int_{\mathbb{R}}\lambda(t,\zeta)\tilde{N}(\mathrm{d}t,\mathrm{d}\zeta)\Big] \tag{6.3.70}$$

for suitable processes b, λ (to be determined).

Then again by the Itô formula we get

$$\begin{aligned}\mathrm{d}(h(t)k(t))&=h(t)k(t)\Big[b(t)\mathrm{d}B(t)+\int_{\mathbb{R}}\lambda(t,\zeta)\tilde{N}(\mathrm{d}t,\mathrm{d}\zeta)\Big]\\&+k(t)\Big[\mathrm{d}F(t)+h(t)\beta(t)\mathrm{d}B(t)+h(t)\int_{\mathbb{R}}\gamma(t,\zeta)\tilde{N}(\mathrm{d}t,\mathrm{d}\zeta)\Big]\\&+(h(t)\beta(t)+X(t)q(t))k(t)b(t)\mathrm{d}t\\&+\int_{\mathbb{R}}\Big(h(t)\gamma(t,\zeta)+X(t)r(t,\zeta)(1+\gamma(t,\zeta)\Big)k(t)\lambda(t,\zeta)\tilde{N}(\mathrm{d}t,\mathrm{d}\zeta)\\&+\int_{\mathbb{R}}\Big(h(t)\gamma(t,\zeta)+X(t)r(t,\zeta)(1+\gamma(t,\zeta)\Big)k(t)\lambda(t,\zeta)\mathrm{d}\nu(\zeta)\mathrm{d}t.\end{aligned} \tag{6.3.71}$$

Define

$$u(t):=h(t)k(t). \tag{6.3.72}$$

Then the equation above can be written

$$\begin{aligned}\mathrm{d}u(t)&=u(t)\Big[\int_{\mathbb{R}}\gamma(t,\zeta)\lambda(t,\zeta)\mathrm{d}\nu(\zeta)\mathrm{d}t\\&+\{\beta(t)+b(t)\}\mathrm{d}B(t)+\beta(t)b(t)\mathrm{d}t+\int_{\mathbb{R}}\{\lambda(t,\zeta)+\gamma(t,\zeta)+\lambda(t,\zeta)\\&\gamma(t,\zeta)\}\tilde{N}(\mathrm{d}t,\mathrm{d}\zeta)\Big]k(t)\Big[\mathrm{d}F(t)+X(t)q(t)b(t)\mathrm{d}t\end{aligned}$$

$$+\int_{\mathbb{R}} X(t)r(t,\zeta)\lambda(t,\zeta)(1+\gamma(t,\zeta))\mathrm{d}\nu(\zeta)\mathrm{d}t$$
$$+\int_{\mathbb{R}} X(t)r(t,\zeta)\lambda(t,\zeta)(1+\gamma(t,\zeta))\tilde{N}(\mathrm{d}t,\mathrm{d}\zeta)\Big]. \tag{6.3.73}$$

Choose

$$\begin{aligned} b(t) &:= -\beta(t), \\ \lambda(t,\zeta) &:= -\frac{\gamma(t,\zeta)}{1+\gamma(t,\zeta)}. \end{aligned} \tag{6.3.74}$$

Then from (6.3.70) we get

$$\begin{aligned} k(t) = \exp\Big(&\int_0^t -\beta(s)\mathrm{d}B(s) - \frac{1}{2}\int_0^t \beta^2(s)\mathrm{d}s - \int_0^t\int_{\mathbb{R}} \ln(1+\gamma(s,\zeta))\tilde{N}(\mathrm{d}s,\mathrm{d}\zeta) \\ &+ \int_0^t\int_{\mathbb{R}} \{\frac{\gamma(s,\zeta)}{1+\gamma(s,\zeta)} - \ln(1+\gamma(s,\zeta))\}\nu(\mathrm{d}\zeta)\mathrm{d}s\Big), \end{aligned} \tag{6.3.75}$$

and (6.3.73) reduces to

$$\begin{aligned} \mathrm{d}u(t) &= f(t)\mathrm{d}t + k(t)X(t)q(t)\mathrm{d}B(t) \\ &\quad + \int_{\mathbb{R}} \{X(t)r(t,\zeta)(1+\gamma(t,\zeta))[k(t)+k(t)\lambda(t,\zeta)]\}\tilde{N}(\mathrm{d}t,\mathrm{d}\zeta), \end{aligned} \tag{6.3.76}$$

where

$$\begin{aligned} f(t) &= -k(t) + u(t)\left[\int_{\mathbb{R}} \gamma(t,\zeta)\lambda(t,\zeta)\mathrm{d}\nu(\zeta) + \beta(t)b(t)\right] \\ &\quad + k(t)X(t)q(t)b(t) + k(t)\int_{\mathbb{R}} X(t)r(t,\zeta)\lambda(t,\zeta)(1+\gamma(t,\zeta))\mathrm{d}\nu(\zeta). \end{aligned} \tag{6.3.77}$$

Now define

$$\begin{aligned} v(t) &:= k(t)X(t)q(t), \\ w(t) &:= k(t)X(t)r(t,\zeta). \end{aligned} \tag{6.3.78}$$

Then from (6.3.71) and (6.3.74) we get the following BSDE in the unknowns u, v, w:

$$\begin{aligned} \mathrm{d}u(t) = \Big(&-k(t) - u(t)[\int_{\mathbb{R}} \frac{\gamma^2(t,\zeta)}{1+\gamma(t,\zeta)}\mathrm{d}\nu(\zeta) + \beta^2(t)] \\ &- \beta(t)v(t) - \int_{\mathbb{R}} \gamma(t,\zeta)w(t,\zeta)\mathrm{d}\nu(\zeta)\Big)\mathrm{d}t \end{aligned} \tag{6.3.79}$$

$$+ v(t)\mathrm{d}B(t) + \int_{\mathbb{R}} w(t,,\zeta)\tilde{N}(\mathrm{d}t,\mathrm{d}\zeta); \quad 0 \le t \le T$$
$$u(T) = \theta k(T).$$

This is a linear BSDE which has a unique solution $u(t) = p(t)X(t)k(t)$, $v(t)$, $w(t,\zeta)$, where $u(t)$ is given explicitly by (see Theorem 4.8)

$$u(t) = \frac{1}{\Gamma(t)} E\left[\Gamma(T)\theta k(T) + \int_t^T \Gamma(s)k(s)\mathrm{d}s|\mathcal{F}_t\right], \tag{6.3.80}$$

where

$$\Gamma(t) = \exp\Big(\int_0^t \beta(s)\mathrm{d}B(s) + \frac{1}{2}\int_0^t \beta^2(s)\mathrm{d}s + \int_0^t\int_{\mathbb{R}} \ln(1+\gamma(s,\zeta))\tilde{N}(\mathrm{d}s,\mathrm{d}\zeta)$$
$$+ \int_0^t\int_{\mathbb{R}} \Big\{\ln(1+\gamma(s,\zeta)) - \frac{\gamma(s,\zeta)}{1+\gamma(s,\zeta)}\Big\}\nu(\mathrm{d}\zeta)\mathrm{d}s\Big). \tag{6.3.81}$$

Combining (6.3.75) and (6.3.81) we see that

$$\Gamma(t)k(t) = 1. \tag{6.3.82}$$

Substituted into (6.3.80) this gives

$$u(t) = k(t)\Big(E[\theta|\mathcal{F}_t] + T - t\Big). \tag{6.3.83}$$

In particular, we get

$$p(t)X(t) = \frac{u(t)}{k(t)} = E[\theta|\mathcal{F}_t] + T - t. \tag{6.3.84}$$

Maximizing H with respect to c gives the first-order equation

$$\frac{1}{c(t)} - X(t)p(t) = 0, \tag{6.3.85}$$

i.e., by (6.3.84),

$$c(t) = \hat{c}(t) = \frac{1}{X(t)p(t)} = \frac{1}{E[\theta|\mathcal{F}_t] + T - t}. \tag{6.3.86}$$

Minimizing H with respect to μ gives the first-order equation

$$\mu(t) + X(t)p(t) = 0, \tag{6.3.87}$$

i.e., by (6.3.84),

$$\mu(t) = \hat{\mu}(t) = -X(t)p(t) = -\Big(E[\theta|\mathcal{F}_t] + T - t\Big). \tag{6.3.88}$$

We can now verify that $\hat{c}$, $\hat{\mu}$ satisfies all the conditions of the sufficient maximum principle, and hence we conclude the following:

Proposition 6.17 (***Optimal Consumption for an Insider under Model Uncertainty***) *The solution* (c^*, μ^*) *of the stochastic differential game* (6.3.64) *is given by*

$$c^*(t) = \frac{1}{E[\theta|\mathcal{F}_t] + T - t}; \quad 0 \le t \le T \tag{6.3.89}$$

and

$$\mu^*(t) = -\Big(E[\theta|\mathcal{F}_t] + T - t\Big); \quad 0 \le t \le T. \tag{6.3.90}$$

An interesting, and perhaps surprising, consequence of this result is the following

Corollary 6.18 *Suppose* θ *is a deterministic constant. Then* $c^*(t)$ *and* $\mu^*(t)$ *are also deterministic. In fact, we have*

$$c^*(t) = \frac{1}{\theta + T - t}; \quad 0 \le t \le T \tag{6.3.91}$$

and

$$\mu^*(t) = -\Big(\theta + T - t\Big); \quad 0 \le t \le T. \tag{6.3.92}$$

Remark 6.19 Note that this last result states that if θ is deterministic, then the two players do not need any information about the system to find the optimal respective controls.

6.3.6 The Zero-Sum Game Case

In the *zero-sum case* we have

$$J_1(u_1, u_2) + J_2(u_1, u_2) = 0. \tag{6.3.93}$$

Then the Nash equilibrium $(\hat{u}_1, \hat{u}_2) \in \mathcal{A}_1 \times \mathcal{A}_2$ satisfying (6.3.12)–(6.3.13) becomes a *saddle point* for

$$J(u_1, u_2) := J_1(u_1, u_2). \tag{6.3.94}$$

To see this, note that (6.3.12)–(6.3.13) imply that

$$J_1(u_1, \hat{u}_2) \leq J_1(\hat{u}_1, \hat{u}_2) = -J_2(\hat{u}_1, \hat{u}_2) \leq -J_2(\hat{u}_1, u_2)$$

and hence

$$J(u_1, \hat{u}_2) \leq J(\hat{u}_1, \hat{u}_2) \leq J(\hat{u}_1, u_2) \text{ for all } u_1, u_2.$$

From this we deduce that

$$\begin{aligned} \inf_{u_2 \in \mathcal{A}_2} \sup_{u_1 \in \mathcal{A}_1} J(u_1, u_2) &\leq \sup_{u_1 \in \mathcal{A}_1} J(u_1, \hat{u}_2) \leq J(\hat{u}_1, \hat{u}_2) \\ &\leq \inf_{u_2 \in \mathcal{A}_2} J(\hat{u}_1, u_2) \leq \sup_{u_1 \in \mathcal{A}_1} \inf_{u_2 \in \mathcal{A}_2} J(u_1, u_2). \end{aligned} \tag{6.3.95}$$

Since we always have inf sup $\geq$ sup inf, we conclude that

$$\begin{aligned} \inf_{u_2 \in \mathcal{A}_2} \sup_{u_1 \in \mathcal{A}_1} J(u_1, u_2) &= \sup_{u_1 \in \mathcal{A}_1} J(u_1, \hat{u}_2) = J(\hat{u}_1, \hat{u}_2) \\ &= \inf_{u_2 \in \mathcal{A}_2} J(\hat{u}_1, u_2) = \sup_{u_1 \in \mathcal{A}_1} \inf_{u_2 \in \mathcal{A}_2} J(u_1, u_2), \end{aligned} \tag{6.3.96}$$

i.e. $(\hat{u}_1, \hat{u}_2) \in \mathcal{A}_1 \times \mathcal{A}_2$ is a *saddle point* for $J(u_1, u_2)$.

Thus we want to find $(u_1^*, u_2^*) \in \mathcal{A}_1 \times \mathcal{A}_2$ such that

$$\sup_{u_1 \in \mathcal{A}_1} \left\{ \inf_{u_2 \in \mathcal{A}_2} J(u_1, u_2) \right\} = \inf_{u_2 \in \mathcal{A}_2} \left\{ \sup_{u_1 \in \mathcal{A}_1} J(u_1, u_2) \right\} = J(u_1^*, u_2^*), \tag{6.3.97}$$

where

$$J(u_1, u_2) = E\left[\int_0^T f(t, X(t), Y(t), u(t), \omega) \mathrm{d}t + g(X(T), Y(T), \omega) \right]. \tag{6.3.98}$$

In this case we see that only one Hamiltonian H is needed, namely

$$\begin{aligned} &H(t, x, y, u_1, u_2, p, q, r) \\ &\quad = f(t, x, y, u) + b(t, x, y, u)p + \sigma(t, x, y, u)q + \int_{\mathbb{R}} \gamma(t, x, y, u, \zeta) r(\zeta) \nu(\mathrm{d}\zeta), \end{aligned} \tag{6.3.99}$$

and we have put $g_i = g$; $i = 1, 2$ and $f_1 = f = -f_2$.

Moreover, there is only one triple (p, q, r) of adjoint processes, given by the BSDE

$$\begin{cases} \mathrm{d}p(t) &= -\dfrac{\partial H}{\partial x}(t, X(t), Y(t), u(t), p(t), q(t), r(t))\mathrm{d}t \\ &\quad -\dfrac{\partial H}{\partial y}(t, X(t), Y(t), u(t), p(t), q(t), r(t))\nabla_{X(t)}F)\mathrm{d}t \\ &\quad +q(t)\mathrm{d}B(t) + \displaystyle\int_{\mathbb{R}} r(t,\zeta)\tilde{N}(\mathrm{d}t, \mathrm{d}\zeta) \\ p(T) &= \dfrac{\partial g}{\partial x}(X(T), Y(T)) + E\Big[\dfrac{\partial g}{\partial y}(X(T), Y(T))\Big]\nabla_{X(T)}F. \end{cases} \tag{6.3.100}$$

We can now state the corresponding sufficient maximum principle for the zero-sum game:

Theorem 6.20 (***Sufficient maximum principle for zero-sum mean-field games***) *Let* $(\hat{u}_1, \hat{u}_2) \in \mathcal{A}_1 \times \mathcal{A}_2$, *with corresponding solutions* $\hat{X}(t), \hat{Y}(t), \hat{p}(t), \hat{q}(t), \hat{r}(t,\zeta)$.

Suppose the following hold

- *The function*

$$X, u_1 \to H(t, X, F(X), u_1, \hat{u}_2(t), \hat{p}(t), \hat{q}(t), \hat{r}(t)) \tag{6.3.101}$$

is concave for all t, the function

$$X, u_2 \to H(t, X, F(X), \hat{u}_1(t), u_2, \hat{p}(t), \hat{q}(t), \hat{r}(t)) \tag{6.3.102}$$

is convex for all t, and the function

$$X \to g(X, F(X)) \tag{6.3.103}$$

is affine.

- *(The conditional maximum property)*

$$\begin{aligned} &ess\ sup_{v_1 \in A_1} E[H(t, \hat{X}(t), \hat{Y}(t), v_1, \hat{u}_2(t), \hat{p}(t), \hat{q}(t), \hat{r}(t,\cdot)) \mid \mathcal{E}_t^{(1)}] \\ &\quad = E[H(t, \hat{X}(t), \hat{Y}(t), \hat{u}_1(t), \hat{u}_2(t), \hat{p}(t), \hat{q}(t), \hat{r}(t,\cdot)) \mid \mathcal{E}_t^{(1)}] \end{aligned} \tag{6.3.104}$$

and

$$\begin{aligned} &\text{ess inf}_{v_2 \in A_2} E[H(t, \hat{X}(t), \hat{Y}(t), \hat{u}_1(t), v_2, \hat{p}(t), \hat{q}(t), \hat{r}(t,\cdot)) \mid \mathcal{E}_t^{(2)}] \\ &\quad = E[H(t, \hat{X}(t), \hat{Y}(t), \hat{u}_1(t), \hat{u}_2(t), \hat{p}(t), \hat{q}(t), \hat{r}(t,\cdot)) \mid \mathcal{E}_t^{(2)}]. \end{aligned} \tag{6.3.105}$$

Then $\hat{u}(t) = (\hat{u}_1(t), \hat{u}_2(t))$ *is a saddle point for* $J(u_1, u_2)$.

Proof The proof is similar to (and simpler than) the proof of Theorem 6.10 and is omitted. □

6.3.7 The Single Player Case

If there is only one player, then $u = u_1$ and the problem (6.3.12) reduces to the problem of maximizing the performance functional

$$J(u) = E\left[\int_0^T f(t, X(t), Y(t), u(t), \omega)\mathrm{d}t + g(X(T), Y(T), \omega)\right]. \quad (6.3.106)$$

The corresponding Hamiltonian reduces to

$$\begin{aligned} &H(t, x, y, u, p, q, r) \\ &\quad = f(t, x, y, u) + b(t, x, y, u)p + \sigma(t, x, y, u)q + \int_{\mathbb{R}} \gamma(t, x, y, u, \zeta)r(\zeta)\nu(\mathrm{d}\zeta) \end{aligned} \quad (6.3.107)$$

and the associated mean-field BSDE for the adjoint processes becomes

$$\begin{cases} \mathrm{d}p(t) &= -\dfrac{\partial H}{\partial x}(t, X(t), Y(t), u(t), p(t), q(t), r(t))\mathrm{d}t \\ &\quad -\dfrac{\partial H}{\partial y}(t, X(t), Y(t), u(t), p(t), q(t), r(t))\nabla_{X(t)}F)\mathrm{d}t \\ &\quad +q(t)\mathrm{d}B(t) + \displaystyle\int_{\mathbb{R}} r(t, \zeta)\tilde{N}(\mathrm{d}t, \mathrm{d}\zeta) \\ p(T) &= \dfrac{\partial g}{\partial x}(X(T), Y(T)) + E\left[\dfrac{\partial g}{\partial y}(X(T), Y(T))\right]\nabla_{X(T)}F. \end{cases} \quad (6.3.108)$$

The corresponding sufficient maximum principle can therefore be stated as follows:

Theorem 6.21 (***Sufficient maximum principle for mean-field control***) *Let $\hat{u} \in \mathcal{A}$, with corresponding solutions $\hat{X}(t), \hat{Y}(t), \hat{p}(t), \hat{q}(t), \hat{r}(t, \zeta)$.*

Suppose the following hold

- *The function*

$$X, u \to H(t, X, F(X), u, \hat{p}(t), \hat{q}(t), \hat{r}(t)) \quad (6.3.109)$$

is concave for all t.

- *(The conditional maximum property)*

$$\begin{aligned} &ess\ sup_{v\in A} E[H(t, \hat{X}(t), \hat{Y}(t), v, \hat{p}(t), \hat{q}(t), \hat{r}(t, \cdot)) \mid \mathcal{E}_t] \\ &\quad = E[H(t, \hat{X}(t), \hat{Y}(t), \hat{u}(t), \hat{p}(t), \hat{q}(t), \hat{r}(t, \cdot)) \mid \mathcal{E}_t]. \end{aligned} \quad (6.3.110)$$

Then $\hat{u}(t)$ is an optimal control for $J(u)$.

Proof The proof is similar to the proof of Theorem 6.10 and is omitted.

Similarly, we obtain a single player version of the necessary maximum principle by reducing Theorem 6.11 in the same way.

Example 6.22 (***Return to the optimal harvesting problem***)
To illustrate our result, we apply it to the optimal harvesting problem (6.3.3), (6.3.4) in Sect. 6.3.1(a): For simplicity we assume that N=0. Then the Hamiltonian (6.3.107) gets the form

$$H(t,x,y,u,p,q) = h_0 xu - \frac{1}{2}\rho x^2 u^2 + [ya(t) - xu]p + x\sigma(t)q. \tag{6.3.111}$$

The corresponding BSDE (6.3.108) reduces to

$$\begin{cases} \mathrm{d}p(t) = & -[h_0 u(t) - \rho x(t)u^2(t) + (a(t) - u(t))p(t) + \sigma(t)q(t)]\mathrm{d}t \\ & +q(t)\mathrm{d}B(t) \\ p(T) = & K. \end{cases} \tag{6.3.112}$$

The first-order equation for a maximum of the Hamiltonian as a function of u is

$$\frac{\partial H}{\partial u} = h_0 x(t) - \rho x^2(t)u(t) - x(t)p(t) = 0,$$

i.e.

$$u(t) = \hat{u}(t) = \frac{h_0 - p(t)}{\rho x(t)}. \tag{6.3.113}$$

If we substitute this into (6.3.112) we get the BSDE

$$\begin{cases} \mathrm{d}p(t) &= -[(a(t) - u(t))p(t) + \sigma(t)q(t)]\mathrm{d}t + q(t)\mathrm{d}B(t) \\ p(T) &= K. \end{cases} \tag{6.3.114}$$

This is a linear BSDE with solution (see Theorem 4.8)

$$p(t) = \frac{1}{\Gamma(t)} E[K\Gamma(T) \mid \mathcal{F}_t]; \quad 0 \le t \le T,$$

where

$$\mathrm{d}\Gamma(t) = \Gamma(t)[a(t)\mathrm{d}t + \sigma(t)\mathrm{d}B(t)]; \quad \Gamma(0) = 1.$$

If, for example, $a(t)$ and K are deterministic, then we can choose $q(t) = 0$ in (6.3.114) and get

$$p(t) = K \exp\left(\int_t^T a(s)\mathrm{d}s\right).$$

Therefore, if $a \geq 0$ and

$$h_0 \geq K \exp\left(\int_0^T a(s)\mathrm{d}s\right),$$

then we get $\hat{u}(t) \geq 0$.

Moreover, if

$$x \geq \int_0^T \frac{h_0 - p(s)}{\rho} \exp\left(-\int_a^s a(r)\mathrm{d}r\right)\mathrm{d}s,$$

then we can see that the population $\hat{X}(t) = X^{\hat{u}}(t)$ corresponding to the control $\hat{u}$ is non-negative. We conclude that the relative harvesting rate $\hat{u}$ given by (6.3.113) is admissible and optimal under the above assumptions. We leave the details to the reader. (Exercise 6.4).

6.4 Exercises

Exercise* 6.1 (***Optimal portfolio under model uncertainty***)
Suppose we have a financial market of the form

$$\begin{aligned}
\mathrm{d}S_0(t) &= S_0(t)r(t)\mathrm{d}t \ ; \ S_0(0) = 1, \\
\mathrm{d}S_1(t) &= S_1(t^-)\left[\alpha(t)\mathrm{d}t + \beta(t)\mathrm{d}B(t) + \int_{\mathbb{R}} \gamma(t,\zeta)\tilde{N}(\mathrm{d}t,\mathrm{d}\zeta)\right]; \ S_1(0) > 0,
\end{aligned}$$

where r, β and $\gamma > -1$ are given deterministic functions. Assume that the mean rate of return of the stock, $\alpha(t)$, is not given a priori, but represents a model uncertainty chosen by the environment.

The wealth process $V^{(\alpha,\pi)}(t)$ corresponding to α and to a self-financing portfolio π is given by

$$\begin{aligned}
\mathrm{d}V^{(\alpha,\pi)}(t) &= V^{(\alpha,\pi)}(t^-)\left[\{(1-\pi(t))r(t) + \pi(t)\alpha(t)\}\mathrm{d}t\right. \\
&\qquad \left. + \pi(t)\beta(t) + \pi(t)\int_{\mathbb{R}} \zeta\tilde{N}(\mathrm{d}t,\mathrm{d}\zeta)\right] \ ; \ S \leq t \leq T \\
V^{(\alpha\pi)}(0) &= x > 0.
\end{aligned}$$

If U is a given strictly increasing, C^2 utility function on $[0,\infty]$, then the expected utility of the terminal wealth is

$$J(\alpha,\pi)(s,x) = E\left[U\left(V^{(\alpha,\pi)}(T)\right)\right].$$

The trader is convinced that "the world is against her" and therefore she wants to maximize $J^{(\alpha,\pi)}(s,x)$ in the worst possible case regarding α ("worst case scenario").

This leads to the zero-sum game to find (α^*, π^*) and $\Phi(s, x)$ (the value function) such that

$$\Phi(s,x) = \sup_{\pi \in \mathcal{A}_\varepsilon} \left(\inf_{\alpha \in \mathcal{A}_1} J^{(\alpha,\pi)} s, x) \right) = \inf_{\alpha \in \mathcal{A}_1} \left(\sup_{\pi \in \mathcal{A}_2} J^{(\alpha,\pi)}(s,x) \right),$$

where $\mathcal{A}_1$ and $\mathcal{A}_2$ denote the usual sets of admissible π and α, respectively.

To solve this problem, we first put it into the framework of Theorem 4.16 by introducing $Y = (Y_0, Y_1)$, where

$$\mathrm{d}Y_0(t) = \mathrm{d}t \ ; \ Y_0(0) = y_0 = s \in \mathbb{R}$$

and

$$\begin{aligned} \mathrm{d}Y_1(t) = \mathrm{d}V^{(\alpha,\pi)}(t) = Y_1(t^-)\Big[&\{(1-\pi(t))r(t) + \pi(t)\mathrm{d}(t)\}\mathrm{d}t \\ &+ \pi(t)\beta(t)\mathrm{d}B(t) + \pi(t)\int_{\mathbb{R}} \gamma(t,\zeta)\tilde{N}(\mathrm{d}t, \mathrm{d}\zeta)\Big] \\ Y_1(0) = y_1 = x > 0. \end{aligned}$$

(i) Write down the generator $A^{\alpha,\pi}$ of Y.
(ii) Write down the HJBI equation for the value function Φ.
(iii) To find the value function Φ, we guess that it has the form

$$\hat{\varphi}(s,x) = U\left(x \exp\left(\int_s^T r(t)\mathrm{d}t \right) \right) \ ; \ (s,x) \in (-\infty, T) \times (0, \infty).$$

Compute $A^{\alpha,\pi}\hat{\varphi}(s,x)$.
(iv) Find a saddle point $(\hat{\alpha}, \hat{\pi})$ for the functions

$$(\alpha, \pi) \to A^{\alpha,\pi}\hat{\varphi}(s,x)$$

by requiring that

$$\frac{\partial^2}{\partial\pi\partial\alpha} A^{\alpha,\pi}\hat{\varphi}(s,x)_{\alpha=\hat{\alpha},\pi=\hat{\pi}} = 0.$$

(v) Verify that $(\hat{\alpha}, \pi)$ found in (iv) satisfies all the conditions of Theorem 4.16, and conclude that $(\hat{\alpha}, \pi) = (\alpha^*, \pi^*)$ is indeed a Nash equilibrium and that

$$\Phi(s,x) = \hat{\varphi}(s,x) = U\left(x \exp\left(\int_s^T r(t)\mathrm{d}t \right) \right) \ ; \ (s,x) \in (-\infty, T] \times (0, \infty).$$

Exercise* 6.2
Consider the following jump extension of Example 5.17, with the FBSDE in (X^π, Y^π) given by

$$\begin{cases} \mathrm{d}X^{\pi}(t) &= \pi(t)X^{\pi}(t^{-})\left[b(t)\mathrm{d}t + \sigma(t)\mathrm{d}B(t) + \int_{\mathbb{R}} \gamma(t,\zeta)\tilde{N}(\mathrm{d}t,\mathrm{d}\zeta)\right] \\ X^{\pi}(0) &= x > 0 \end{cases}$$

and

$$\begin{cases} \mathrm{d}Y^{\pi}(t) &= Z^{\pi}(t)\mathrm{d}B(t) + \int_{\mathbb{R}} K^{\pi}(t,\zeta)\tilde{N}(\mathrm{d}t,\mathrm{d}\zeta) \\ Y^{\pi}(T) &= \ln X^{\pi}(T). \end{cases}$$

Here π denotes the *fraction* of the wealth $X^{\pi}(t)$ invested in the risky asset. Assuming that the conditions of Theorem 5.16 hold, use this theorem to find $\pi^{*} \in \mathcal{A}$ and $Y^{\pi *}(0)$ such that

$$\sup_{\pi\in\mathcal{A}} E[\ln(X^{\pi}(T))] = E[\ln(X^{\pi *}(T))],$$

i.e.

$$\sup_{\pi\in\mathcal{A}} Y^{\pi}(0) = Y^{\pi *}(0) = y(0,x).$$

We assume that the coefficients $b(t)$, $\sigma(t)$ and $\gamma(t,\zeta) > -1$ are deterministic and that $X^{\pi}(t) > 0$ for all $t \in [0,T]$ and all $\pi \in \mathcal{A}$.

Exercise* 6.3 (***Optimal insider portfolio under model uncertainty***)
Consider a financial market with two investment possibilities:

(i) A risk free investment possibility with unit price $S_0(t) = 1$ for all $t \in [0,T]$
(ii) A risky investment, where the unit price $S(t)$ is modeled by the SDE

$$\mathrm{d}S(t) = S(t)[(\alpha(t) + \mu(t))\mathrm{d}t + \beta(t)\mathrm{d}B(t)];\ S(0) > 0. \tag{6.4.1}$$

Here $\alpha(t)$, $\beta(t)$ are given $\mathbb{F}$-adapted coefficients, while $\mu(t)$ is a perturbation of the drift term, representing the model uncertainty chosen by the environment (player number 2).

Suppose the wealth process $X(t) = X^{\pi,\mu}(t)$ associated to an insider portfolio $\pi(t)$ (representing the fraction of the wealth invested in the risky asset) is given by:

$$\begin{cases} \mathrm{d}X(t) = \pi(t)X(t)[(\alpha(t) + \mu(t))\mathrm{d}t + \beta(t)]\mathrm{d}B(t) \\ X(0) = x > 0. \end{cases} \tag{6.4.2}$$

Define the performance functional by

$$J(\pi,\mu) = E\left[\int_0^T \frac{1}{2}\mu^2(t)\mathrm{d}t + \theta \log X(T)\right], \tag{6.4.3}$$

where $\theta > 0$ is a given (deterministic) constant, and $\frac{1}{2}\mu^2(t)$ represents a penalty rate, penalizing μ for being away from 0. We assume that π and μ are $\mathbb{F}$-adapted.

Let $\mathcal{A}_1, \mathcal{A}_2$ be given families of admissible controls for π and μ, respectively. For π, μ to be admissible we require the usual integrability and adaptedness conditions. Moreover, we require that the corresponding $X^{\pi,\mu}(t) > 0$ for all t. We want to find $\pi^* \in \mathcal{A}_1$ and $\mu^* \in \mathcal{A}_2$ such that

$$\sup_{\pi\in\mathcal{A}_1}\inf_{\mu\in\mathcal{A}_2} J(\pi,\mu) = \inf_{\mu\in\mathcal{A}_2}\sup_{\pi\in\mathcal{A}_1} J(\pi,\mu) = J(\pi^*,\mu^*). \tag{6.4.4}$$

(i) Write down the Hamiltonian $H(t, x, \pi, \mu, p, q)$ for this problem.

(ii) Write down the BSDE for the adjoint processes p, q.

(iii) Show that the first-order equation for maximizing H with respect to π is

$$X(t)[(\alpha(t) + \mu(t))p(t) + \beta(t)q(t)] = 0. \tag{6.4.5}$$

(iv) Combine this with (ii) above to obtain that

$$\begin{cases} \mathrm{d}p(t) = -\frac{\alpha(t)+\mu(t)}{\beta(t)}p(t)\mathrm{d}B(t) \\ p(T) = \frac{\theta}{X(T)}. \end{cases} \tag{6.4.6}$$

(v) Define

$$h(t) = p(t)X(t). \tag{6.4.7}$$

Use the Itô formula to show that

$$\begin{cases} \mathrm{d}h(t) &= (\pi(t)\beta(t) - \frac{\alpha(t)+\mu(t)}{\beta(t)})h(t)\mathrm{d}B(t) \\ h(T) &= p(T)X(T) = \theta. \end{cases} \tag{6.4.8}$$

(vi) Explain why this BSDE has the unique solution

$$h(t) = \theta; \quad 0 \le t \le T \tag{6.4.9}$$

and deduce the following expression for our candidate $\hat{\pi}(t)$ for the optimal portfolio

$$\hat{\pi}(t) = \frac{\alpha(t) + \hat{\mu}(t)}{\beta^2(t)}, \tag{6.4.10}$$

where $\hat{\mu}(t)$ is the corresponding candidate for the optimal perturbation.

(vii) Write the first-order equation for the optimal $\hat{\mu}$ obtained by minimizing H with respect to μ.

(viii) Combine this with the result in (vi) to conclude the following:

Proposition 6.23 *(***Optimal Portfolio under Model Uncertainty***) The saddle point* $(\pi^*(t), \mu^*(t))$ *of the stochastic differential game* (6.4.2) *is given by*

$$\pi^*(t) = \frac{\alpha(t)}{\beta^2(t) + \theta}, \tag{6.4.11}$$

and

$$\mu^*(t) = -\frac{\alpha(t)\theta}{\beta^2(t) + \theta}. \tag{6.4.12}$$

Exercise* 6.4 (***Optimal Mean-Field Control***)
Complete the details of Example 6.22.

Chapter 7
Combined Optimal Stopping and Stochastic Control of Jump Diffusions



7.1 Introduction

In this chapter we discuss combined optimal stopping and stochastic control problems and their associated Hamilton–Jacobi–Bellman (HJB) variational inequalities. This is a subject which deserves to be better known because of its many applications. A thorough treatment of such problems (but without the associated HJB variational inequalities) can be found in Krylov [K].

This chapter may also serve as a brief review of the theory of optimal stopping and their variational inequalities on one hand, and the theory of stochastic control and their HJB equations on the other. An introduction to these topics separately can be found in [Ø1].

As an illustration of how combined optimal stopping and stochastic control problems may appear in economics, let us consider the following example which is an extension of Exercise 3.2.

Example 7.1 (***An Optimal Resource Extraction Control and Stopping Problem***) Suppose the price $P_t = P(t)$ of one unit of a resource (e.g., gas or oil) at time t is varying like a geometric Lévy process, i.e.,

$$\mathrm{d}P(t) = P(t^-)\left(\alpha\,\mathrm{d}t + \beta\,\mathrm{d}B(t) + \gamma\int_{\mathbb{R}} z\tilde{N}(\mathrm{d}t,\mathrm{d}z)\right), \quad P_0 = p \geq 0, \tag{7.1.1}$$

where $\alpha, \beta \neq 0, \gamma$ are constants and $\gamma z \geq 0$ a.s. ν.

Let Q_t denote the amount of remaining resources at time t. If we extract the resources at the "intensity" $u_t = u_t(\omega) \in [0, m]$ at time t, then the dynamics of Q_t is

$$\mathrm{d}Q_t = -u_t Q_t \mathrm{d}t, \quad Q_0 = q \geq 0. \tag{7.1.2}$$

(m is a constant giving the maximal intensity.)

We assume as before that our *control* $u_t(\omega)$ is adapted to the filtration $\mathbb{F} = \{\mathcal{F}_t\}_{t\geq 0}$. If the running cost is given by $K_0 + K_1 u_t$ (with $K_0, K_1 \geq 0$ constants) as long as

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the field is open and if we decide to stop the extraction for good at time $\tau(\omega) \geq 0$ let us assume that the expected total discounted profit is

$$J^{(u,\tau)}(s,p,q) = E^{(p,q)}\left[\int_0^\tau \mathrm{e}^{-\rho(s+t)}(u_t(P_tQ_t - K_1) - K_0)\mathrm{d}t + \mathrm{e}^{-\rho(s+\tau)}(\theta P_\tau Q_\tau - a)\right], \tag{7.1.3}$$

where $\rho > 0$ is the discounting exponent and $\theta > 0$, $a \geq 0$ are constants. Thus $\mathrm{e}^{-\rho(s+t)}\big(u_t(P_tQ_t - K_1) - K_0\big)$ gives the discounted net profit rate when the field is in operation, while $\mathrm{e}^{-\rho(s+\tau)}(\theta P_\tau Q_\tau - a)$ gives the discounted net value of the remaining resources at time τ. (We may interpret $a \geq 0$ as a transaction cost.) We assume that the closing time τ is a *stopping time* with respect to the filtration $\{\mathcal{F}_t\}_{t\geq 0}$, i.e., that

$$\{\omega;\, \tau(\omega) \leq t\} \in \mathcal{F}_t \quad \text{for all } t.$$

Thus both the extraction intensity u_t and the decision whether to close before or at time t must be based on the information $\mathbb{F}_t$ only, not on any future information.

The problem is to find the *value function* $\Phi(s,p,q)$ and the *optimal control* $u_t^* \in [0,m]$ and the *optimal stopping time* τ^* such that

$$\Phi(s,p,q) = \sup_{u_t,\tau} J^{(u,\tau)}(s,p,q) = J^{(u^*,\tau^*)}(s,p,q). \tag{7.1.4}$$

This problem is an example of a combined optimal stopping and stochastic control problem. It is a modification of a problem discussed in [BØ1, DZ].

We will return to this and other examples after presenting a general theory for problems of this type.

7.2 A General Mathematical Formulation

Consider a controlled stochastic system of the same type as in Chap. 5, where the state $Y^{(u)}(t) = Y(t) \in \mathbb{R}^k$ at time t is given by

$$\begin{aligned} \mathrm{d}Y(t) &= b(Y(t),u(t))\mathrm{d}t + \sigma(Y(t),u(t))\mathrm{d}B(t) + \int_{\mathbb{R}^k} \gamma(Y(t^-),u(t^-),z)\tilde{N}(\mathrm{d}t,\mathrm{d}z), \\ Y(0) &= y \in \mathbb{R}^k. \end{aligned} \tag{7.2.1}$$

Here $b : \mathbb{R}^k \times U \to \mathbb{R}^k$, $\sigma : \mathbb{R}^k \times U \to \mathbb{R}^{k\times m}$, and $\gamma : \mathbb{R}^k \times U \times \mathbb{R}^k \to \mathbb{R}^{k\times \ell}$ are given continuous functions and $u(t) = u(t,\omega)$ is our *control*, assumed to be $\mathbb{F}_t$-adapted and with values in a given closed, convex set $U \subset \mathbb{R}^p$.

Associated to a control $u = u(t,\omega)$ and an $\mathbb{F}$-stopping time $\tau = \tau(\omega)$ belonging to a given set $\mathcal{T}$ of *admissible* stopping times, we assume there is a *performance*

criterion of the form

$$J^{(u,\tau)}(y) = E^y\left[\int_0^\tau f(Y(t), u(t))\mathrm{d}t + g(Y(\tau))\chi_{\{\tau<\infty\}}\right], \tag{7.2.2}$$

where $f : \mathbb{R}^k \times U \to \mathbb{R}$ (the profit rate) and $g : \mathbb{R}^k \to \mathbb{R}$ (the bequest function) are given functions.

We assume that we are given a set $\mathcal{U} = \mathcal{U}(y)$ of admissible controls u which is contained in the set of controls u such that a unique strong solution $Y(t) = Y^{(u)}(t)$ of (7.2.1) exists and the following, (7.2.3) and (7.2.4), hold:

$$-\ E^y\left[\int_0^{\tau_{\mathcal{S}}} |f(Y(t), u(t))|\mathrm{d}t\right] < \infty, \quad \text{for all } y \in \mathcal{S} \tag{7.2.3}$$

where $\tau_{\mathcal{S}} = \tau_{\mathcal{S}}(y, u) = \inf\{t > 0; Y^{(u)}(t) \notin \mathcal{S}\}$.

– The family $\{g^-(Y^{(u)}(\tau)); \tau \in \mathcal{T}\}$ is uniformly P^y-integrable for all $y \in \mathcal{S}$, where $g^-(y) = \max(0, -g(y))$. (7.2.4)

We interpret $g(Y(\tau(\omega)))$ as 0 if $\tau(\omega) = \infty$. Here, and in the following, E^y denotes expectation with respect to P when $Y(0) = y$ and $\mathcal{S} \subset \mathbb{R}^k$ is a fixed Borel set such that

$$\mathcal{S} \subset \overline{\mathcal{S}^0}.$$

We can think of $\mathcal{S}$ as the "universe" or "solvency set" of our system, in the sense that we are only interested in the system up to time T, which may be interpreted as the time of bankruptcy.

We now consider the following *combined optimal stopping and control* problem.

Let $\mathcal{T}$ be the set of $\mathbb{F}_t$-stopping times $\tau \leq \tau_{\mathcal{S}}$. Find $\Phi(y)$ and $u^* \in \mathcal{U}$, $\tau^* \in \mathcal{T}$ such that

$$\Phi(y) = \sup\{J^{(u,\tau)}(y); u \in \mathcal{U}, \tau \in \mathcal{T}\} = J^{(u^*,\tau^*)}(y). \tag{7.2.5}$$

We will prove a verification theorem for this problem. The theorem can be regarded as a combination of the variational inequalities for optimal stopping (Theorem 3.2) and the HJB equation for stochastic control (Theorem 5.1).

We say that the control u is *Markov* or *Markovian* if it has the form

$$u(t) = u_0(Y(t))$$

for some function $u_0 : \bar{\mathcal{S}} \to U$. If this is the case we usually do not distinguish notationally between u and u_0 and write (with abuse of notation)

$$u(t) = u(Y(t)).$$

If $u \in \mathcal{U}$ is Markovian then $Y^{(u)}(t)$ is a Markov process whose generator coincides on $C_0^2(\mathbb{R}^k)$ with the differential operator $L = L^u$ defined for $y \in \mathbb{R}^k$ by

$$L^u\psi(y) = \sum_{i=1}^{k} b_i(y,u(y))\frac{\partial\psi}{\partial y_i} + \frac{1}{2}\sum_{i,j=1}^{k}(\sigma\sigma^T)_{ij}(y,u(y))\frac{\partial^2\psi}{\partial y_i\partial y_j}$$
$$+\sum_{j=1}^{\ell}\int_{\mathbb{R}}\{\psi(y+\gamma^{(j)}(y,u(y),z_j)) - \psi(y)$$
$$-\nabla\psi(y).\gamma^{(j)}(y,u(y),z_j)\}\nu_j(\mathrm{d}z_j) \tag{7.2.6}$$

for all functions $\psi : \mathbb{R}^k \to \mathbb{R}$ which are twice differentiable at y.

Typically the value function Φ will be C^2 outside the boundary ∂D of the continuation region D (see (ii) below) and it will satisfy an HJB equation in D and an HJB inequality outside $\bar{D}$. Across ∂D the function Φ will not be C^2, but it will usually be C^1, and this feature is often referred to as the "high contact" – or "smooth fit" – principle. This is the background for the verification theorem given below (Theorem 7.2). Note, however, that there are cases when Φ is not even C^1 at ∂D. To handle such cases one can use a verification theorem based on the viscosity solution concept. See Chap. 12 and in particular Sect. 12.2.

Theorem 7.2 (***HJB-Variational Inequalities for Optimal Stopping and Control***)

(a) Suppose we can find a function $\varphi : \bar{\mathcal{S}} \to \mathbb{R}$ such that

(i) $\varphi \in C^1(\mathcal{S}^0) \cap C(\bar{\mathcal{S}})$

(ii) $\varphi \geq g$ *on* $\mathcal{S}^0$

Define

$$D = \{y \in \mathcal{S};\ \varphi(y) > g(y)\} \quad (\textit{the continuation region}).$$

Suppose $Y^{(u)}(t)$ spends 0 time on ∂D a.s., i.e.,

(iii) $E^y\left[\int_0^{\tau_{\mathcal{S}}} \mathcal{X}_{\partial D}(Y^{(u)}(t))\mathrm{d}t\right] = 0$ *for all* $y \in \mathcal{S}$, $u \in \mathcal{U}$

and suppose that

(iv) ∂D *is a Lipschitz surface*

(v) $\varphi \in C^2(\mathcal{S}^0 \setminus \partial D)$ *and the second-order derivatives of* φ *are locally bounded near* ∂D

(vi) $L^v\varphi(y) + f(y,v) \leq 0$ *on* $\mathcal{S}^0 \setminus \partial D$ *for all* $v \in U$

(vii) $\varphi(y) = g(y)$ *for all* $y \notin \mathcal{S}$.

(viii) $E^y\left[|\varphi(Y^{(u)}(\tau))| + \int_0^{\tau_{\mathcal{S}}} |L^u\varphi(Y^{(u)}(t))|\mathrm{d}t\right] < \infty$ *for all* $u \in \mathcal{U}$, $\tau \in \mathcal{T}$.

Then

$$\varphi(y) \geq \Phi(y) \quad \textit{for all } y \in \mathcal{S}.$$

(b) Suppose, in addition to (i)–(viii) above, that

(ix) for each $y \in D$ there exists $\hat{u}(y) \in U$ such that

$$L^{\hat{u}(y)}\varphi(y) + f(y, \hat{u}(y)) = 0,$$

(x) $\tau_D := \inf\{t > 0; Y^{(\hat{u})}(t) \notin D\} < \infty$ *a.s. for all* $y \in \mathcal{S}$,

and

(xi) the family $\{\varphi(Y^{(\hat{u})}(\tau)); \tau \in \mathcal{T}\}$ *is uniformly integrable with respect to* P^y *for all* $y \in D$.

Suppose $\hat{u} \in \mathcal{U}$. *Then*

$$\varphi(y) = \Phi(y) \quad \text{for all } y \in \mathcal{S}.$$

Moreover, $u^* := \hat{u}$ *and* $\tau^* := \tau_D$ *are optimal control and stopping times, respectively.*

Proof The proof is a synthesis of the proofs of Theorems 3.2 and 5.1. For completeness we give some details:
(a) By Theorem 3.1 we may assume that $\varphi \in C^2(\mathcal{S}^0) \cap C(\bar{\mathcal{S}})$. Choose $u \in \mathcal{U}$ and put $Y(t) = Y^{(u)}(t)$. Let $\tau \leq \tau_{\mathcal{S}}$ be a stopping time. Then by Dynkin's formula (Theorem 1.24) we have, for $m = 1, 2, \ldots$,

$$E^y\big[\varphi(Y(\tau \wedge m))\big] = \varphi(y) + E^y\left[\int_0^{\tau \wedge m} L^u\varphi(Y(t))\mathrm{d}t\right]. \tag{7.2.7}$$

Hence by (vii) and the Fatou lemma

$$\begin{aligned}\varphi(y) &= \varliminf_{m\to\infty} E^y\left[\int_0^{\tau\wedge m} -L^u\varphi(Y(t))\mathrm{d}t + \varphi(Y(\tau \wedge m))\right] \\ &\geq E^y\left[\int_0^{\tau} -L^u\varphi(Y(t))\mathrm{d}t + g(Y(\tau))\chi_{\{\tau<\infty\}}\right].\end{aligned} \tag{7.2.8}$$

Here we have used that $Y(\cdot)$ is quasi-left continuous (i.e. left continuous at increasing stopping times; see [JS], Propositions I.2.26 and I.3.27).

If we now use (vi) we can conclude that

$$\varphi(y) \geq E^y\left[\int_0^{\tau} f(Y(t), u(t))\mathrm{d}t + g(Y(\tau))\chi_{\{\tau<\infty\}}\right] = J^{u,\tau}(y). \tag{7.2.9}$$

Since $u \in \mathcal{U}$ and $\tau \leq \tau_{\mathcal{S}}$ were arbitrary we conclude that

$$\varphi(y) \geq \sup_{u,\tau} J^{(u,\tau)}(y) = \Phi(y), \tag{7.2.10}$$

which proves (a).

To prove the opposite inequality, assume that (ix)–(xi) hold. Choose a point $y \in D$. Then apply the argument above to the Markovian control $\hat{u}(t) := \hat{u}(Y(t))$ and the stopping time $\hat{\tau} = \tau_D = \inf\{t > 0; \hat{Y}(t) \notin D\}$, where $\hat{Y}(t) = Y^{(\hat{u})}(t)$. We get

$$E^y\big[\varphi(\hat{Y}(\hat{\tau}))\big] = \varphi(y) + E^y\left[\int_0^{\hat{\tau}} L^{\hat{u}}\varphi(\hat{Y}(t))\mathrm{d}t\right], \tag{7.2.11}$$

which implies that

$$\varphi(y) = E^y\left[\int_0^{\hat{\tau}} f(\hat{Y}(t), \hat{u}(t))\mathrm{d}t + g(\hat{Y}(\hat{\tau}))\right] = J^{(\hat{u},\hat{\tau})}(y).$$

Combined with (a) this shows that $\varphi(y) = \Phi(y)$ and that $(\hat{u}, \hat{\tau})$ is optimal if $y \in D$.

Finally, if $y \notin D$ then $\varphi(y) = g(y) \le \Phi(y)$ and hence by (a) we have $\varphi(y) = \Phi(y)$ also if $y \notin D$. In this case $\tau^* = 0$ is an optimal stopping time. □

Remark 7.3 If we neglect all the technical conditions of Theorem 7.2 and concentrate on conditions (ii), (vi), and (ix), then we can write the conditions of Theorem 7.2 in the following condensed form:

$$\max\left(\sup_{v\in U}\{L^v\phi(y) + f(y, v)\}, g(y) - \phi(y)\right) = 0, \quad y \in \mathcal{S}^0. \tag{7.2.12}$$

Since this is a combination of the HJB equation of stochastic control and the variational inequality (VI) of optimal stopping, we call (7.2.12) an HJBVI.

One can prove that, under some conditions, the value function ϕ is indeed a solution of (7.2.12) in the weak sense of *viscosity*. See Chap. 12 for a discussion of this concept.

Remark 7.4 Note that the problem (7.2.5) contains the general optimal stopping problem as a special case.

More precisely, if the functions b, σ, and f do not depend on u, then the problem reduces to the optimal stopping problem

$$\Phi(y) = \sup_{\tau\in\mathcal{T}} E^y\left[\int_0^{\tau} f(Y(t))\mathrm{d}t + g(Y(\tau))\chi_{\{\tau<\infty\}}\right]$$

discussed in Chap. 3 and the HJBVI (7.2.12) becomes the VI

$$\max(L\Phi(y) + f(y), g(y) - \Phi(y)) = 0, \quad y \in \mathcal{S}^0. \tag{7.2.13}$$

The problem (7.2.5) is also closely related to the general *stochastic control problem* discussed in Chap. 5. In such a problem the stopping time τ is fixed to $\tau = \tau_{\mathcal{S}}$ and hence the problem is to find $\Phi(y)$ and $u^* \in \mathcal{U}$ such that

$$\Phi(y) = \sup_{u \in \mathcal{U}} J^{(u)}(y) = J^{(u^*)}(y), \tag{7.2.14}$$

where

$$J^{(u)}(y) = E^y \left[\int_0^{\tau_S} f(Y(t), u(t)) \mathrm{d}t + g(Y(\tau_S)) \chi_{\{\tau_S < \infty\}} \right].$$

7.3 Applications

To illustrate Theorem 7.2 let us apply it to the problem of Example 7.1.

In this case the generator L^u of the time–space state process

$$\mathrm{d}Y(t) = (\mathrm{d}t, \mathrm{d}P_t, \mathrm{d}Q_t), \quad Y(0) = (s, p, q) \in [0, \infty)^3$$

is given by (see Theorem 1.22)

$$L^u \psi(s, p, q) = \frac{\partial \psi}{\partial s} + \alpha p \frac{\partial \psi}{\partial p} - uq \frac{\partial \psi}{\partial q} + \frac{1}{2} \beta^2 p^2 \frac{\partial^2 \psi}{\partial p^2} + \int_{\mathbb{R}} \left\{ \psi(s, p + \gamma p z, q) - \psi(s, p, q) - \frac{\partial \psi}{\partial p}(s, p, q).\gamma z p \right\} \nu(\mathrm{d}z).$$

In view of Theorem 7.2 we are looking for a subset D of $\mathcal{S} = [0, \infty)^3$ and a function $\varphi(s, p, q) : \mathcal{S} \to \mathbb{R}$ such that

$$\varphi(s, p, q) = \mathrm{e}^{-\rho s}(\theta pq - a) \quad \text{for all } (s, p, q) \notin D, \tag{7.3.1}$$

$$\varphi(s, p, q) \geq \mathrm{e}^{-\rho s}(\theta pq - a) \quad \text{for all } (s, p, q) \in \mathcal{S}, \tag{7.3.2}$$

$$L^v \varphi(s, p, q) + \mathrm{e}^{-\rho s}(v(pq - K_1) - K_0) \leq 0 \quad \text{for all } (s, p, q) \in \mathcal{S}^0 \setminus \bar{D}$$

$$\text{and all } v \in [0, m], \tag{7.3.3}$$

$$\sup_{v \in [0,m]} \left\{ L^v \varphi(s, p, q) + \mathrm{e}^{-\rho s}(v(pq - K_1) - K_0) \right\} = 0 \quad \text{for all } (s, p, q) \in D. \tag{7.3.4}$$

Let us try a function φ of the form

$$\varphi(s, p, q) = \mathrm{e}^{-\rho s} F(w), \quad \text{where } w = pq \tag{7.3.5}$$

and a continuation region D of the form

$$D = \{(s, p, q);\ pq > w_0\} \quad \text{for some } w_0 > 0.$$

Then (7.3.1)–(7.3.4) get the form

$$F(w) = \theta w - a \quad \text{for all } w \le w_0, \tag{7.3.6}$$

$$F(w) > \theta w - a \quad \text{for all } w > w_0, \tag{7.3.7}$$

$$\begin{aligned} &-\rho F(w) + (\alpha - v) w F'(w) + \frac{1}{2}\beta^2 w^2 F''(w) \\ &+ \int_{\mathbb{R}} \{F(w + \gamma z w) - F(w) - F'(w)\gamma z w\}\nu(\mathrm{d}z) + v(w - K_1) - K_0 \le 0 \\ &\quad \text{for all } w < w_0,\ v \in [0, m], \end{aligned} \tag{7.3.8}$$

$$\begin{aligned} \sup_{v\in[0,m]} \Big\{ &-\rho F(w) + (\alpha - v) w F'(w) + \frac{1}{2}\beta^2 w^2 F''(w) + \int_{\mathbb{R}} \{F(w + \gamma z w) - F(w) \\ &- F'(w)\gamma z w\}\nu(\mathrm{d}z) + v(w - K_1) - K_0 \Big\} = 0 \quad \text{for all } w > w_0. \end{aligned} \tag{7.3.9}$$

From (7.3.9) and (xi) of Theorem 7.2 we get the following candidate $\hat{u}$ for the optimal control:

$$\begin{aligned} v = \hat{u}(w) &= \operatorname*{Argmax}_{v\in[0,m]} \Big\{ v(w(1 - F'(w)) - K_1) \Big\} \\ &= \begin{cases} m & \text{if } F'(w) < 1 - (K_1/w) \\ 0 & \text{if } F'(w) > 1 - (K_1/w) \end{cases}. \end{aligned} \tag{7.3.10}$$

Let $F_m(w)$ be the solution of (7.3.9) with $v = m$, i.e., the solution of

$$\begin{aligned} &-\rho F_m(w) + (\alpha - m) w F_m'(w) + \frac{1}{2}\beta^2 w^2 F_m''(w) \\ &+ \int_{\mathbb{R}} \{F(w + \gamma z w) - F(w) - F'(w)\gamma z w\}\nu(\mathrm{d}z) = K_0 + m K_1 - m w. \end{aligned} \tag{7.3.11}$$

A solution of (7.3.11) is

$$F_m(w) = C_1 w^{\lambda_1} + C_2 w^{\lambda_2} + \frac{m w}{\rho + m - \alpha} - \frac{K_0 + m K_1}{\rho}, \tag{7.3.12}$$

where C_1, C_2 are constants and $\lambda_1 > 0$, $\lambda_2 < 0$ are roots of the equation

$$h(\lambda) = 0 \tag{7.3.13}$$

with

$$h(\lambda) = -\rho + (\alpha - m)\lambda + \frac{1}{2}\beta^2\lambda(\lambda - 1) + \int_{\mathbb{R}} \{(1 + \gamma z)^\lambda - 1 - \lambda\gamma z\}\nu(\mathrm{d}z). \tag{7.3.14}$$

(Note that $h(0) = -\rho < 0$ and $\lim_{|\lambda|\to\infty} h(\lambda) = \infty$.) The solution will depend on the relation between the parameters involved and we will not give a complete discussion, but only consider some special cases.

Case 1
Let us assume that

$$\alpha \le \rho, \quad K_1 = a = 0, \quad \text{and} \quad 0 < \theta < \frac{m}{\rho + m - \alpha}. \tag{7.3.15}$$

It is easy to see that

$$\lambda_1 > 1 \iff \rho + m > \alpha. \tag{7.3.16}$$

Let us try (guess) that

$$C_1 = 0 \tag{7.3.17}$$

and that the continuation region $D = \{(s, p, q);\ pq > w_0\}$ is such that (see (7.3.10))

$$F_m'(w) < 1 \quad \text{for all} \quad w > w_0. \tag{7.3.18}$$

The intuitive motivation for trying this is the belief that it is optimal to use the maximal extraction intensity m all the time until closure, at least if θ is small enough.

These guesses lead to the following candidate for the value function $F(w)$:

$$F(w) = \begin{cases} \theta w & \text{if } 0 \le w \le w_0 \\ F_m(w) = C_2 w^{\lambda_2} + \dfrac{mw}{\rho + m - \alpha} - \dfrac{K_0}{\rho} & \text{if } w > w_0. \end{cases} \tag{7.3.19}$$

We now use continuity and differentiability at $w = w_0$ to determine w_0 and C_2:

$$\text{(Continuity)} \quad C_2 w_0^{\lambda_2} + \frac{m w_0}{\rho + m - \alpha} - \frac{K_0}{\rho} = \theta w_0, \tag{7.3.20}$$

$$\text{(Differentiability)} \quad C_2 \lambda_2 w_0^{\lambda_2 - 1} + \frac{m}{\rho + m - \alpha} = \theta. \tag{7.3.21}$$

Easy calculations show that the unique solution of (7.3.20) and (7.3.21) is

$$w_0 = \frac{(-\lambda_2) K_0 (\rho + m - \alpha)}{(1 - \lambda_2)\rho[m - \theta(\rho + m - \alpha)]} \quad (> 0 \ \text{ by (7.3.15)}) \tag{7.3.22}$$

and

$$C_2 = \frac{[m - \theta(\rho + m - \alpha)] w_0^{1-\lambda_2}}{(-\lambda_2)(\rho + m - \alpha)} \quad (> 0 \ \text{ by (7.3.15)}). \tag{7.3.23}$$

It remains to verify that with these values of w_0 and C_2 the set $D = \{(s, p, q);\ pq > w_0\}$ and the function $F(w)$ given by (7.3.19) satisfies (7.3.6)–(7.3.9), as well as all the other conditions of Theorem 7.2.

To verify (7.3.6) we have to check that (7.3.18) holds, i.e., that

$$F'_m(w) = C_2\lambda_2 w^{\lambda_2 - 1} + \frac{m}{\rho + m - \alpha} < 1 \quad \text{for all} \quad w > w_0.$$

Since $\lambda_2 < 0$ and we have assumed $\alpha \leq \rho$ (in (7.3.15)) this is clear. So (7.3.9) holds. If we substitute $F(w) = \theta w$ in (7.3.8) we get

$$-\rho\theta w + (\alpha - m)w\theta + mw - K_0 = w[m - \theta(\rho + m - \alpha)] - K_0.$$

We know that this is 0 for $w = w_0$ by (7.3.20) and (7.3.21). Hence it is less than 0 for $w < w_0$. So (7.3.8) holds. Condition (7.3.6) holds by definition of D and F. Finally, since $F(w_0) = \theta w_0$, $F'(w_0) = \theta$, and

$$F''(w) = F''_m(w) = C_2\lambda_2(\lambda_2 - 1)w^{\lambda_2 - 2} > 0$$

we must have $F(w) > \theta w$ for $w > w_0$. Hence (7.3.7) holds. Similarly one can verify all the other conditions of Theorem 7.2. We have proved.

Theorem 7.5 *Suppose* (7.3.15) *holds. Then the optimal strategy* (u^*, τ^*) *for problems* (7.1.3) *and* (7.1.4) *is*

$$u^* = m, \quad \tau^* = \inf\{t > 0;\ P_t Q_t \leq w_0\}, \tag{7.3.24}$$

where w_0 *is given by* (7.3.22). *The corresponding value function is* $\Phi(s, p, q) = e^{-\rho s} F(p \cdot q)$, *where* F *is given by* (7.3.19) *with* $\lambda_2 < 0$ *as in* (7.3.11) *and* $C_2 > 0$ *as in* (7.3.23).

For other values of the parameters it might be optimal not to produce at all but just wait for the best closing/sellout time. For example, we mention without proof the following cases (see Exercise 7.2):

Case 2
Assume that

$$\theta = 1 \quad \text{and} \quad \rho \leq \alpha. \tag{7.3.25}$$

Then $u^* = 0$ and

$$\Phi = \infty.$$

Case 3
Assume that

$$\theta = 1, \quad \rho > \alpha, \quad \text{and} \quad K_0 < \rho a < K_0 + \rho K_1. \tag{7.3.26}$$

Then

$$u^* = 0 \quad \text{and} \quad \tau^* = \inf\{t > 0;\ P_t Q_t \geq w_1\},$$

for some $w_1 > 0$.

7.4 Exercises

Exercise* 7.1 (a) Solve the following stochastic control problem

$$\Phi(s,x) = \sup_{u(t)\geq 0} J^{(u)}(s,x) = J^{(u^*)}(s,x),$$

where

$$J^{(u)}(s,x) = E^x\left[\int_0^{\tau_{\mathcal{S}}} \mathrm{e}^{-\delta(s+t)}\frac{u^\gamma(t)}{\gamma}\mathrm{d}t\right].$$

Here

$$\tau_{\mathcal{S}} = \tau_{\mathcal{S}}(\omega) = \inf\{t > 0;\, X(t) \leq 0\} \quad \text{(the time of bankruptcy)}$$

and

$$\mathrm{d}X(t) = (\mu X(t) - u(t))\mathrm{d}t + \sigma X(t)\mathrm{d}B(t) + \theta X(t^-)\int_{\mathbb{R}} z\tilde{N}(\mathrm{d}t, \mathrm{d}z), \quad X_0 = x > 0$$

with $\gamma \in (0,1)$, $\delta > 0$, $\mu, \sigma \neq 0$, θ constants, $\theta z > -1$ a.s. ν. The interpretation of this is the following. $X(t)$ represents the total wealth at time t, $u(t) = u(t,\omega) \geq 0$ represents the chosen consumption rate (the control). We want to find the consumption rate $u^*(t)$ which maximizes the expected total discounted utility of the consumption up to the time of bankruptcy, $\tau_{\mathcal{S}}$.

[Hint: Try a value function of the form

$$\phi(s,x) = K\mathrm{e}^{-\delta s}x^\gamma$$

for a suitable value of the constant K.]

(b) Consider the following combined stochastic control and optimal stopping problem

$$\Phi(s,x) = \sup_{u,\tau} J^{(u,\tau)}(s,x) = J^{(u^*,\tau^*)}(s,x),$$

where

$$J^{(u,\tau)}(s,x) = E^x\left[\int_0^{\tau} \mathrm{e}^{-\delta(s+t)}\frac{u^\gamma(t)}{\gamma}\mathrm{d}t + \lambda\mathrm{e}^{-\delta(s+\tau)}X^\gamma(\tau)\right]$$

with $X(t)$ as in (a), $\lambda > 0$ a given constant.

Now the supremum is taken over all $\mathbb{F}$-adapted controls $u(t) \geq 0$ and all $\mathbb{F}$-stopping times $\tau \leq \tau_{\mathcal{S}}$.

Let K be the constant found in (a). Show that:

1. If $\lambda \geq K$ then it is optimal to stop immediately.
2. If $\lambda < K$ then it is never optimal to stop.

Exercise 7.2 (a) Verify the statements in Cases 2 and 3 at the end of Sect. 7.3.

(b) What happens in the cases

Case 4: $\theta = 1$, $\rho > \alpha$, and $\rho a \leq K_0$?

Case 5: $\theta = 1$, $\rho > \alpha$, and $K_0 + \rho K_1 \leq \rho a$?

Exercise 7.3 (***A Stochastic Linear Regulator Problem with Optimal Stopping***) Consider the stochastic linear regulator problem in Exercise 5.5, with the additional option of stopping, i.e., solve the problem

$$\Phi(s,x) = \inf\left\{J^{(u,\tau)}(s,x);\, u \in \mathcal{U}, \tau \in \mathcal{T}\right\},$$

where

$$J^{(u,\tau)}(s,x) = E^{s,x}\left[\int_0^\tau \mathrm{e}^{-\rho(s+t)}\left(X^2(t) + \theta u^2(t)\right)\mathrm{d}t + \lambda \mathrm{e}^{-\rho(s+\tau)}X^2(\tau)\chi_{\{\tau<\infty\}}\right],$$

($\rho > 0, \theta > 0, \lambda > 0$ constants), where the state process is

$$Y(t) = \begin{bmatrix} s+t \\ X(t) \end{bmatrix}; \quad t \geq 0, \quad Y(0) = \begin{bmatrix} s \\ x \end{bmatrix} = y \in \mathbb{R}^2$$

with

$$\mathrm{d}X(t) = \mathrm{d}X^{(u)}(t) = u(t)\mathrm{d}t + \sigma\,\mathrm{d}B(t) + \int_{\mathbb{R}} z\widetilde{N}(\mathrm{d}t, \mathrm{d}z), \quad X(0) = x.$$

(a) Explain why

$$\Phi(s,x) \leq \lambda \mathrm{e}^{-\rho s}x^2 \quad \text{for all } s, x.$$

(b) Prove that

$$\Phi(s,x) \leq \mathrm{e}^{-\rho s}\left(x^2 + \sigma^2 + \int_{\mathbb{R}} z^2 \nu(\mathrm{d}z)\right) \quad \text{for all } s, x.$$

[Hint: What happens if we choose $\tau = \infty$?]

(c) Show that if

$$0 < \lambda \leq \frac{1}{2}\left(-\rho\theta + \sqrt{\rho^2\theta^2 + 4\theta}\right)$$

then

$$\Phi(s, x) = \lambda e^{-\rho s} x^2$$

and it is optimal to stop immediately.
[Hint: Apply the (minimum version of) Theorem 7.2a to the function $\Phi(s, x) = g(s, x) = \lambda e^{-\rho s} x^2$.]

Chapter 8
Singular Control for Jump Diffusions



8.1 An Illustrating Example

We illustrate singular control problems by the following example, studied in [FØS2].

Example 8.1 (***Optimal Consumption and Portfolio Under Proportional Transaction Costs***) Consider again a financial market of the forms (5.2.41) and (5.2.42), where we have two investment possibilities:

(i) A bank account/bond where the value/price $P_1(t)$ at time t grows with interest rate r, i.e.,

$$\mathrm{d}P_1(t) = rP_1(t)\mathrm{d}t, \quad P_1(0) = 1.$$

(ii) A stock, with price $P_2(t)$ satisfying the equation

$$\mathrm{d}P_2(t) = P_2(t^-)\left[\alpha\,\mathrm{d}t + \beta\,\mathrm{d}B(t) + \int_{\mathbb{R}} z\widetilde{N}(\mathrm{d}t, \mathrm{d}z)\right], \quad P_2(0) = p_2 > 0.$$

Here $r > 0$, α and $\beta > 0$ are given constants, and we assume that $z > -1$ a.s. ν.

Assume that at any time t the investor can choose an adapted and càdlàg consumption rate process $c(t) = c(t, \omega) \geq 0$, taken from the bank account. Moreover, the investor can at any time transfer money from one investment to another with a transaction cost which is proportional to the size of the transaction. Let $X_1(t)$ and $X_2(t)$ denote the amounts of money invested in the bank and in the stocks, respectively. Then the evolution equations for $X_1(t)$ and $X_2(t)$ are

$$\begin{aligned}\mathrm{d}X_1(t) = \mathrm{d}X_1^{x_1,c,\xi}(t) &= (rX_1(t) - c(t))\mathrm{d}t - (1+\lambda)\mathrm{d}\xi_1(t)\\ &\quad + (1-\mu)\mathrm{d}\xi_2(t), \quad X_1(0^-) = x_1 \in \mathbb{R},\end{aligned}$$

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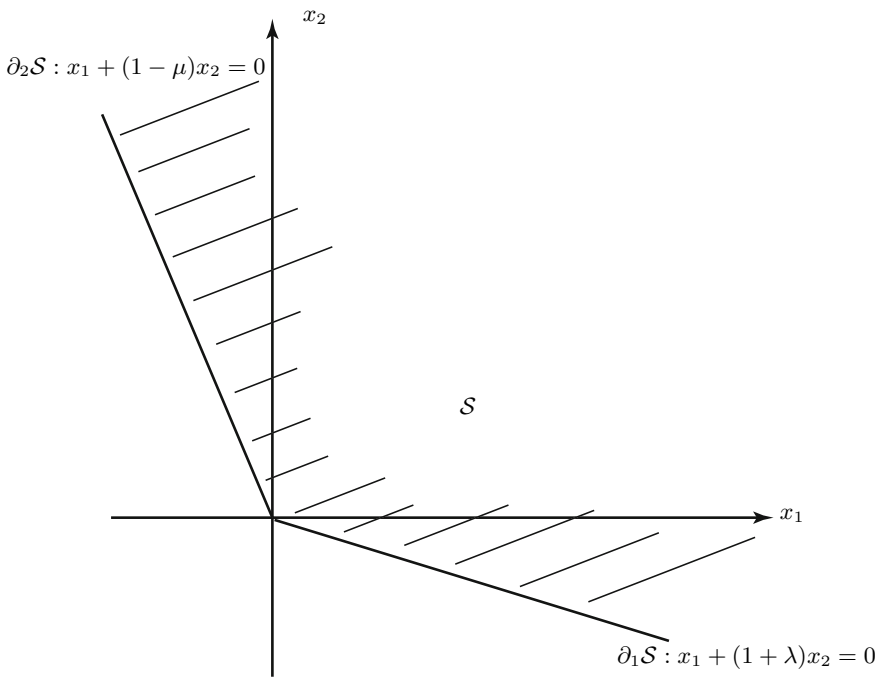


Fig. 8.1 The solvency region $\mathcal{S}$

$$\mathrm{d}X_2(t) = \mathrm{d}X_2^{x,c,\xi}(t) = X_2(t^-)\left[\alpha\,\mathrm{d}t + \beta\,\mathrm{d}B(t) + \int_{\mathbb{R}} z\widetilde{N}(\mathrm{d}t, \mathrm{d}z)\right] + \mathrm{d}\xi_1(t) - \mathrm{d}\xi_2(t), \quad X_2(0^-) = x_2 \in \mathbb{R}.$$

Here $\xi = (\xi_1, \xi_2)$, where $\xi_1(t)$ and $\xi_2(t)$ represent cumulative purchase and sale, respectively, of stocks up to time t. The constants $\lambda \geq 0$, $\mu \in [0, 1]$ represent the constants of proportionality of the transaction costs.

Define the *solvency region* $\mathcal{S}$ to be the set of states (x_1, x_2) such that the net wealth is nonnegative, i.e., (see Fig. 8.1)

$$\mathcal{S} = \{(x_1, x_2) \in \mathbb{R}^2,\ x_1 + (1+\lambda)x_2 \geq 0, \quad \text{and} \quad x_1 + (1-\mu)x_2 \geq 0\}. \tag{8.1.1}$$

We define the set $\mathcal{A}$ of *admissible* controls as the set of adapted consumption–investment policies (c, ξ) such that $c = c(t) \geq 0$ is càdlàg and $\xi = (\xi_1, \xi_2)$ where each $\xi_i(t)$, $i = 1, 2$ is right-continuous, nondecreasing, $\xi(0^-) = 0$.

Define

$$\tau_{\mathcal{S}} = \inf\{t > 0;\ (X_1(t), X_2(t)) \notin \mathcal{S}\}.$$

The performance criterion is defined by

$$J^{c,\xi}(s,x_1,x_2) = E^{s,x_1,x_2}\left[\int_0^{\tau_S} \mathrm{e}^{-\delta(s+t)}\,\frac{c^\gamma(t)}{\gamma}\mathrm{d}t\right], \tag{8.1.2}$$

where $\delta > 0$, $\gamma \in (0,1)$ are constants. We seek $(c^*,\xi^*) \in \mathcal{A}$ and $\Phi(s,x_1,x_2)$ such that

$$\Phi(s,x_1,x_2) = \sup_{(c,\xi)\in\mathcal{A}} J^{c,\xi}(s,x_1,x_2) = J^{c^*,\xi^*}(s,x_1,x_2). \tag{8.1.3}$$

This is an example of a *singular* stochastic control problem. It is called singular because the investment control measure $\mathrm{d}\xi(t)$ is allowed to be singular with respect to Lebesgue measure $\mathrm{d}t$. In fact, as we shall see, the optimal control measure $\mathrm{d}\xi^*(t)$ turns out to be singular.

We now give a general theory of singular control of jump diffusions and return to the above example afterward.

8.2 A General Formulation

Let $\kappa = [\kappa_{ij}] : \mathbb{R}^k \to \mathbb{R}^{k\times p}$ and $\theta = [\theta_i] : \mathbb{R}^k \to \in \mathbb{R}^p$ be given continuous functions. Suppose the state $Y(t) = Y^{u,\xi}(t) \in \mathbb{R}^k$ is described by the equation

$$\begin{aligned}\mathrm{d}Y(t) &= b(Y(t),u(t))\mathrm{d}t + \sigma(Y(t),u(t))\mathrm{d}B(t)\\ &\quad + \int_{\mathbb{R}^\ell} \gamma(Y(t^-),u(t^-),z)\widetilde{N}(\mathrm{d}t,\mathrm{d}z) + \kappa(Y(t^-))\mathrm{d}\xi(t), \quad Y(0^-) = y \in \mathbb{R}^k.\end{aligned} \tag{8.2.1}$$

Here $\xi(t) \in \mathbb{R}^p$ is an adapted càdlàg finite variation process with increasing components and $\xi(0^-) = 0$. Since $\mathrm{d}\xi(t)$ may be singular with respect to Lebesgue measure $\mathrm{d}t$, we call ξ our *singular control* or our *intervention control*. The process $u(t)$ is an adapted càdlàg process with values in a given set U ($u(t)\mathrm{d}t$ is our absolutely continuous control). Suppose we are given a performance functional $J^{u,\xi}(y)$ of the form

$$\begin{aligned}J^{u,\xi}(y) = E^y\Bigg[&\int_0^{\tau_S} f(Y(t),u(t))\mathrm{d}t + g(Y(\tau_S))\cdot\mathcal{X}_{\{\tau_S<\infty\}}\\ &+ \int_0^{\tau_S} \theta^T(Y(t^-))\mathrm{d}\xi(t)\Bigg],\end{aligned}$$

where $f : \mathbb{R}^k \times U \to \mathbb{R}$, $g : \mathbb{R}^k \to \mathbb{R}$ and $\theta : \mathbb{R}^k \to \mathbb{R}^p$ are given continuous functions and

$\tau_{\mathcal{S}} = \inf\{t > 0;\, Y^{u,\xi}(t) \notin \mathcal{S}\} \leq \infty$ is the time of bankruptcy,

where $\mathcal{S} \subset \mathbb{R}^k$ is a given *solvency set*, assumed to satisfy $\mathcal{S} \subset \overline{\mathcal{S}^0}$.

Let $\mathcal{A}$ be a given family of *admissible controls* (u, ξ), contained in the set of (u, ξ) such that a unique strong solution $Y(t)$ of (8.2.1) exists and

$$E^y\left[\int_0^{\tau_{\mathcal{S}}} |f(Y(t), u(t))|\mathrm{d}t + |g(Y(\tau_{\mathcal{S}}))| \cdot \mathcal{X}_{\{\tau_{\mathcal{S}} < \infty\}} + \int_0^{\tau_{\mathcal{S}}} \sum_{j=1}^{p} |\theta_j(Y(t^-))|\mathrm{d}\xi_j(t)\right] < \infty.$$

The problem is to find the value function $\Phi(y)$ and an optimal control $(u^*, \xi^*) \in \mathcal{A}$ such that

$$\Phi(y) = \sup_{(u,\xi)\in\mathcal{A}} J^{u,\xi}(y) = J^{u^*,\xi^*}(y).$$

Note that if we apply a Markov control $u(t) = u(Y(t)) \in U$ and $\mathrm{d}\xi = 0$, then $Y(t) = Y^{u,0}(t)$ has the generator A^u given by

$$A^u\phi(y) = \sum_{i=1}^{k} b_i(y, u(y))\frac{\partial\phi}{\partial y_i} + \frac{1}{2}\sum_{i,j=1}^{k} (\sigma\sigma^T)_{ij}(y, u(y))\frac{\partial^2\phi}{\partial y_i \partial y_j} + \sum_{j=1}^{\ell}\int_{\mathbb{R}} \left\{\phi(y + \gamma^{(j)}(y, u(y), z_j)\right) - \phi(y) - \nabla\phi(y)^T\gamma^{(j)}(y, u(y), z_j)\Big\}\nu_j(\mathrm{d}z_j).$$

In the following we let A^v denote this same integrodifferential operator obtained by choosing $u(y) = v$ (constant) $\in U$.

Note that we distinguish between the jumps of $Y^{u,\xi}(t)$ caused by the jump of $N(t, z)$, denoted by $\Delta_N Y(t)$, and the jump caused by the singular control ξ, denoted by $\Delta_\xi Y(t)$. Thus

$$\Delta_N Y(t) = \int_{\mathbb{R}^k} \gamma(Y(t^-), u(t^-), z)\widetilde{N}(\{t\}, \mathrm{d}z), \quad \text{while} \quad \Delta_\xi Y(t) = \kappa(Y(t^-))\Delta\xi(t), \tag{8.2.2}$$

where

$$\widetilde{N}(\{t\}, \mathrm{d}z) = \widetilde{N}(t, \mathrm{d}z) - \widetilde{N}(t^-, \mathrm{d}z).$$

We let $t_1, t_2, \ldots$ denote the jumping times of $\xi(t)$ and we let

$$\Delta_\xi\phi(Y(t_n)) = \phi(Y(t_n)) - \phi(Y(t_n^-) + \Delta_N Y(t_n)) \tag{8.2.3}$$

be the increase in ϕ due to the jump $\Delta\xi(t) = \xi(t) - \xi(t^-)$ of $\xi(t)$ at $t = t_n$.

We give now a verification theorem for singular control problems.

Theorem 8.2 (***Integrovariational Inequalities for Singular Control***)

(a) *Suppose there exists a function* $\phi \in C^2(\mathcal{S}^0) \cap C(\mathbb{R}^k)$ *such that*

(i) $A^v\phi(y) + f(y, v) \le 0$ *for all (constant)* $v \in U$ *and* $y \in \mathcal{S}$.

(ii) $$\sum_{i=1}^{k} \kappa_{ij}(y)\frac{\partial\phi}{\partial y_i}(y) + \theta_j(y) \le 0 \text{ for all } y \in \mathcal{S}, \quad j = 1, \dots, p.$$

(iii) $$E^y\Bigg[\int_0^{\tau_{\mathcal{S}}} \Big\{|\sigma^T(Y(t), u(t))\nabla\phi(Y(t))|^2 + \sum_{k=1}^{\ell}\int_{\mathbb{R}} |\phi(Y(t) + \gamma^{(k)}(Y(t), u(t), z)) - \phi(Y(t))|^2\nu_k(\mathrm{d}z)\Big\}\mathrm{d}t\Bigg] < \infty$$

for all $(u, \xi) \in \mathcal{A}$.

(iv) $\phi(y) = g(y)$ *for all* $y \notin \mathcal{S}$.

(v) $\{\phi^-(Y(\tau))\}_{\tau \le \tau_{\mathcal{S}}}$ *is uniformly integrable for all* $(u, \xi) \in \mathcal{A}$, $y \in \mathcal{S}$.

Then

$$\phi(y) \ge \Phi(y) \quad \text{for all } y \in \mathcal{S}. \tag{8.2.4}$$

(b) *Define the* nonintervention region D *by*

$$D = \left\{y \in \mathcal{S};\ \max_{1\le j\le p}\left\{\sum_{i=1}^{k}\kappa_{ij}(y)\frac{\partial\phi}{\partial y_i}(y) + \theta_j(y)\right\} < 0\right\}. \tag{8.2.5}$$

Suppose, in addition to (i)–(v) *above, that for all* $y \in \overline{D}$ *there exists* $v = \hat{u}(y)$ *such that*

(vi) $A^{\hat{u}(y)}\phi(y) + f(y, \hat{u}(y)) = 0$.

Moreover, suppose there exists $\hat{\xi}$ *such that* $(\hat{u}, \hat{\xi}) \in \mathcal{A}$ *and*

(vii) $Y^{\hat{u},\hat{\xi}}(t) \in \overline{D}$ *for all* t.

(viii) $$\sum_{j=1}^{p}\left\{\sum_{i=1}^{k}\kappa_{ij}(Y(t^-))\frac{\partial\phi}{\partial y_i}(Y(t^-)) + \theta_j(Y(t^-))\right\}\mathrm{d}\hat{\xi}_j^{(c)} = 0$$

for all t, $1 \le j \le p$,

where $\xi^{(c)}(t)$ *is the continuous part of* $\xi(t)$, *i.e., the process obtained by removing the jumps of* $\xi(t)$.

(ix) $$\Delta_{\hat{\xi}}\phi(Y(t_n)) + \sum_{j=1}^{p}\theta_j(Y(t_n^-))\Delta\hat{\xi}_j(t_n) = 0 \quad \text{for all jumping times } t_n \text{ of } \hat{\xi}(t)$$

and

(x) $$\lim_{R\to\infty} E^y\left[\phi\left(Y^{\hat{u},\hat{\xi}}(T_R)\right)\right] = E^y\left[g\left(Y^{\hat{u},\hat{\xi}}(\tau_{\mathcal{S}})\right)\cdot\mathcal{X}_{\{\tau_{\mathcal{S}}<\infty\}}\right],$$

where $T_R = \min(\tau_{\mathcal{S}}, R)$ *for* $R < \infty$.

Then

$$\phi(y) = \Phi(y)$$

and

$$(\hat{u}, \hat{\xi}) \quad \textit{is an optimal control.}$$

Proof (a) Choose $(u, \xi) \in \mathcal{A}$. Then by the Itô formula for the semimartingale $Y(t) = Y^{u,\xi}(t)$ we have (see [P, Theorem II.33] and Theorem 1.24)

$$E^y[\phi(Y(T_R))] = \phi(y) + E^y\Bigg[\int_0^{T_R} A^u\phi(Y(t))\mathrm{d}t + \int_0^{T_R}\sum_{i=1}^{k}\frac{\partial\phi}{\partial y_i}(Y(t^-))\sum_{j=1}^{p}\kappa_{ij}(Y(t^-))\mathrm{d}\xi_j^{(c)}(t) + \sum_{0<t_n\le T_R}\Delta_\xi\phi(Y(t_n))\Bigg]. \quad (8.2.6)$$

By the mean value theorem we have

$$\begin{aligned}\Delta_\xi\phi(Y(t_n)) &= \nabla\phi(\hat{Y}^{(n)})^T\Delta_\xi Y(t_n)\\ &= \sum_{i=1}^{k}\sum_{j=1}^{p}\frac{\partial\phi}{\partial y_i}(\hat{Y}^{(n)})\kappa_{ij}(Y(t_n^-))\Delta\xi_j(t_n),\end{aligned} \quad (8.2.7)$$

where $\hat{Y}^{(n)}$ is some point on the straight line between $Y(t_n)$ and $Y(t_n^-) + \Delta_N Y(t_n)$ (see (8.2.3)).

Hence, by (i) and (ii) and (8.2.6), (8.2.7),

$$\begin{aligned}\phi(y) &\ge E^y\Bigg[\int_0^{T_R} f(Y(t), u(t))\mathrm{d}t + \phi(Y(T_R))\\ &\quad - \sum_{j=1}^{p}\sum_{i=1}^{k}\Bigg\{\int_0^{T_R}\frac{\partial\phi}{\partial y_i}(Y(t^-))\kappa_{ij}\mathrm{d}\xi_j^{(c)}(t)\\ &\quad + \sum_{0<t_n\le T_R}\frac{\partial\phi}{\partial y_i}(\hat{Y}^{(n)})\kappa_{ij}(Y(t_n^-))\Delta\xi_j(t_n)\Bigg\}\Bigg]\\ &\ge E^y\left[\int_0^{T_R} f(Y(t), u(t))\mathrm{d}t + \phi(Y(T_R)) + \sum_{j=1}^{p}\int_0^{T_R}\theta_j(Y(t^-))\mathrm{d}\xi_j(t)\right].\end{aligned}$$

Letting $R \to \infty$ and applying (iv), (v) and quasi-left continuity (see the proof of Theorem 3.2 we obtain from (8.2.7) that

$$\phi(y) \ge J^{u,\xi}(y).$$

Since $(u, \xi) \in \mathcal{A}$ was arbitrary, this proves (8.2.4).

(b) Now apply the above argument to $(\hat{u}, \hat{\xi}) \in \mathcal{A}$, as given by (vi)–(xi). Then we get *equality* everywhere in (a) and we end up with

$$\phi(y) = J^{\hat{u},\hat{\xi}}(y)$$

and the proof is complete. □

The optimal singular control $\hat{\xi}$ described in Theorem 8.2b can be characterized as a *local time* at ∂D of the jump diffusion $Y(t)$ reflected back into $\overline{D}$ with respect to the directions given by the rows of the matrix κ.

Definition 8.3 Let $D \subset \mathbb{R}^k$ be an open set and let $b : \mathbb{R}^k \to \mathbb{R}^k$, $\sigma : \mathbb{R}^k \to \mathbb{R}^{k\times m}$, $\gamma : \mathbb{R}^k \times \mathbb{R}^\ell \to \mathbb{R}^{k\times \ell}$, and $\kappa : \mathbb{R}^k \to \mathbb{R}^{k\times k}$ be given continuous functions. The *Skorohod problem* for reflected jump diffusions into $\overline{D}$ (with respect to the direction matrix κ) is to find a pair $(Y(t), \xi(t))$ of càdlàg and adapted processes such that the following conditions are satisfied:

$$\begin{aligned} \mathrm{d}Y(t) &= b(Y(t))\mathrm{d}t + \sigma(Y(t))\mathrm{d}B(t) + \int_{\mathbb{R}^\ell} \gamma(Y(t^-), z)\widetilde{N}(\mathrm{d}t, \mathrm{d}z) \\ &\quad + \kappa(Y(t^-))\mathrm{d}\xi(t), \quad Y(0^-) = y \in \mathbb{R}^k, \qquad (8.2.8) \\ Y(t) &\in \overline{D} \quad \text{for all } t \geq 0, \qquad (8.2.9) \\ \xi(t) &\in \mathbb{R}^k \textit{has finite variation and } \mathrm{d}\,\xi(t) = 0 \textit{if} Y(t) \in D. \qquad (8.2.10) \end{aligned}$$

The process $\xi(t)$ is called a *local time* at the boundary ∂D for the reflected process $Y(t)$.

The following result is an extension to jump diffusions of a well-known result for (continuous) Itô diffusions. For related results see [CEM].

Theorem 8.4 *Suppose ∂D is C^1, i.e., ∂D is locally the graph of a C^1-function. Suppose that $b(\cdot)$, $\sigma(\cdot)$, $\gamma(\cdot, z)$, and $\kappa(\cdot)$ are Lipschitz continuous with at most linear growth (uniformly in $z \in \mathbb{R}^\ell$). Then there exists a solution $(Y(t), \xi(t))$ to the Skorohod problem (8.2.8)–(8.2.10).*

Sketch of Proof We adapt the approach of Anderson and Orey [AØ] to the jump diffusion case. (See also the presentation in [Fre].) Let us first assume that D is the half-space

$$D = \{(y_1, \ldots, y_k) \in \mathbb{R}^k, \quad y_1 > 0\} \qquad (8.2.11)$$

and that the rows $\kappa_1, \ldots, \kappa_k$ of the matrix κ are

$$\kappa_1 = (1, 0, \ldots, 0), \kappa_i \in \mathbb{R}^k \text{arbitrary constants}, \ i = 2, \ldots, k. \qquad (8.2.12)$$

Let $\mathcal{C}$ denote the set of all càdlàg functions $f : [0, \infty) \to \mathbb{R}^k$.

Define the map $\Gamma : \mathcal{C} \to \mathcal{C}$ by

$$\Gamma f(t) = \left(f_1(t) + \sup_{0\le s\le t} f_1^-(s), f_2(t), \dots, f_k(t) \right), \quad f = (f_1, \dots, f_k) \in \mathcal{C} \tag{8.2.13}$$

and define the map $\Lambda : \mathcal{C} \to \mathcal{C}$ by

$$\Lambda(f) = (\Lambda(f)_1, 0, \dots, 0),$$

where

$$\Lambda(f)_1(t) = \Gamma(f)_1(t) - f_1(t) = \sup_{0\le s\le t} f_1^-(s). \tag{8.2.14}$$

Note that

$$\sup_{0\le s\le t} |\Gamma(f)(s) - \Gamma(g)(s)| \le 2 \sup_{0\le s\le t} |f(s) - g(s)|$$

and

$$|\Gamma(f)(s) - \Gamma(f)(t)| \le |f(s) - f(t)|, \quad f, g \in \mathcal{C}.$$

Hence we can define $Z(t)$ to be the càdlàg adapted solution of the stochastic differential equation

$$\begin{aligned} \mathrm{d}Z(t) &= b(\Gamma(Z)(t))\mathrm{d}t + \sigma(\Gamma(Z)(t))\mathrm{d}B(t) + \int_{\mathbb{R}^\ell} \gamma(\Gamma(Z)(t^-), z)\widetilde{N}(\mathrm{d}t, \mathrm{d}z) \\ Z(0) &= y. \end{aligned} \tag{8.2.15}$$

Now define

$$Y(t) := \Gamma(Z)(t) \quad \text{and} \quad \xi(t) := \Lambda(Z)(t). \tag{8.2.16}$$

Then $(Y(t), \xi(t))$ satisfies (8.2.8). This is because

$$\begin{aligned} \mathrm{d}Y_1(t) - \kappa_1(Y(t))\mathrm{d}\xi(t) &= \mathrm{d}Y_1(t) - \mathrm{d}\xi_1(t) \\ &= \mathrm{d}(\Gamma(Z)_1(t) - \Lambda_1(Z)(t)) \\ &= \mathrm{d}(\Gamma(Z)_1(t) - \Gamma(Z)_1(t) + Z_1(t)) = \mathrm{d}Z_1(t) \\ &= b_1(Y(t))\mathrm{d}t + \sigma_1(Y(t))\mathrm{d}B(t) \\ &\quad + \int_{\mathbb{R}} \gamma_1(Y(t^-), z)\widetilde{N}(\mathrm{d}t, \mathrm{d}z) \end{aligned}$$

and, trivially,

$$\mathrm{d}Y_i(t) - \kappa_i(Y(t))\mathrm{d}\xi(t) = \mathrm{d}Y_i(t) = \mathrm{d}Z_i(t)$$

for $i = 2, \dots, k$. It follows from the definition of Γ that

$$Y(t) = (\Gamma(Z)_1(t), Z_2(t), \dots, Z_k(t)) \in \overline{D}$$

since

$$\Gamma(Z)_1(t) = Z_1(t) + \sup_{0\le s\le t} Z_1^-(s) \ge 0 \quad \text{for all } t.$$

Moreover, by (8.2.14) we see that $\xi_1(t) = \Lambda(Z)_1(t)$ is nondecreasing and hence $\xi(t)$ has finite variation. Moreover, if $Y(t) = \Gamma(Z)(t) \in D$ then

$$\Gamma(Z)_1(t) = Z_1(t) + \sup_{0\le s\le t} Z_1^-(s) > 0$$

and hence

$$Z_1(t) > \min\left(\inf_{0\le s\le t} Z(s), 0\right).$$

Therefore either $Z_1(t) > 0$ or $Z_1(t) > \inf_{0\le s\le t} Z(s)$. In either case the process $\xi_1(t) = \sup_{0\le s\le t} Z_1^-(s)$ is constant in an interval $[t, t+\epsilon)$ for some $\epsilon > 0$, by right continuity of $Z(t)$. Therefore $\mathrm{d}\xi(t) = 0$ if $Y(t) \in D$, as claimed. Hence $(Y(t), \xi(t))$ defined by (8.2.16) solves the Skorohod equation (8.2.8)–(8.2.10) in the case when D and κ are given by (8.2.11) and (8.2.12).

The proof in the general case follows by mapping neighborhoods of $x \in \partial D$ into neighborhoods of the boundary point 0 of a half-space and applying the argument above. □

8.3 Application to Portfolio Optimization with Transaction Costs

We now apply this theorem to Example 8.1. In this case our state process is

$$\begin{aligned}\mathrm{d}Y(t) = \begin{bmatrix}\mathrm{d}t\\ \mathrm{d}X_1(t)\\ \mathrm{d}X_2(t)\end{bmatrix} &= \begin{bmatrix}1\\ rX_1(t) - c(t)\\ \alpha X_2(t)\end{bmatrix}\mathrm{d}t + \begin{bmatrix}0\\ 0\\ \beta X_2(t)\end{bmatrix}\mathrm{d}B(t)\\ &+ \begin{bmatrix}0\\ 0\\ X_2(t^-)\int_{\mathbb{R}} z\widetilde{N}(\mathrm{d}t, \mathrm{d}z)\end{bmatrix} + \begin{bmatrix}0 & 0\\ -(1+\lambda) & (1-\mu)\\ 1 & -1\end{bmatrix}\begin{bmatrix}\mathrm{d}\xi_1(t)\\ \\ \mathrm{d}\xi_2(t)\end{bmatrix}.\end{aligned}$$

The generator of $Y(t)$ when there are no interventions is

$$\begin{aligned}A^c\phi(y) = \frac{\partial\phi}{\partial s} + (rx_1 - c)\frac{\partial\phi}{\partial x_1} + \alpha x_2\frac{\partial\phi}{\partial x_2} + \frac{1}{2}\beta^2 x_2^2\frac{\partial^2\phi}{\partial x_2^2}\\ + \int_{\mathbb{R}}\left\{\phi(s, x_1, x_2 + x_2 z) - \phi(s, x_1, x_2) - x_2 z\frac{\partial\phi}{\partial x_2}(s, x_1, x_2)\right\}\nu(\mathrm{d}z).\end{aligned}$$

Or, if

$$\phi(s, x_1, x_2) = \mathrm{e}^{-\delta s}\psi(x_1, x_2)$$

we have

$$A^c\phi(s, x_1, x_2) = \mathrm{e}^{-\delta s}A_0^c\psi(x_1, x_2),$$

where

$$\begin{aligned} A_0^c\psi(x_1, x_2) =& -\delta\psi + (rx_1 - c)\frac{\partial\psi}{\partial x_1} + \alpha x_2\frac{\partial\psi}{\partial x_2} + \frac{1}{2}\beta^2 x_2^2\frac{\partial^2\psi}{\partial x_2^2} \\ &+ \int_{\mathbb{R}}\left\{\psi(x_1, x_2 + x_2 z) - \psi(x_1, x_2) - x_2 z\frac{\partial\psi}{\partial x_2}(x_1, x_2)\right\}\nu(\mathrm{d}z). \end{aligned}$$

Here

$$\theta = g = 0, \quad u(t) = c(t), \quad f(y, u) = f(s, x_1, x_2, c) = \mathrm{e}^{-\delta s}\frac{c^\gamma}{\gamma}.$$

Condition (ii) of Theorem 8.2 gets the form

$$-(1 + \lambda)\frac{\partial\psi}{\partial x_1} + \frac{\partial\psi}{\partial x_2} \le 0$$

and

$$(1 - \mu)\frac{\partial\psi}{\partial x_1} - \frac{\partial\psi}{\partial x_2} \le 0.$$

The nonintervention region D in (8.2.5) therefore becomes

$$D = \left\{(s, x_1, x_2) \in \mathcal{S};\ -(1 + \lambda)\frac{\partial\psi}{\partial x_1} + \frac{\partial\psi}{\partial x_2} < 0 \text{ and } (1 - \mu)\frac{\partial\psi}{\partial x_1} - \frac{\partial\psi}{\partial x_2} < 0\right\}.$$

Conditions (i) and (vi) become, respectively,

$$\begin{aligned} A_0^c\psi(x_1, x_2) + \frac{c^\gamma}{\gamma} &\le 0 \quad \text{for all } c \ge 0, \\ A_0^{\hat{c}}\psi(x_1, x_2) + \frac{\hat{c}^\gamma}{\gamma} &= 0 \quad \text{on } D. \end{aligned}$$

Using these integrovariational inequalities together with the remaining conditions of Theorem 8.2 it is possible to prove the following about the optimal consumption rate $c^*(t)$ and the optimal portfolio $\xi^*(t)$.

Theorem 8.5 ([FØS2]) *Suppose*

$$\delta > \gamma\alpha - \frac{1}{2}\beta^2\gamma(1-\gamma) - \gamma\|\nu\| + \int_{-1}^{\infty}\{(1+z)^\gamma - 1\}\nu(\mathrm{d}z),$$

where

$$\|\nu\| = \nu((-1,\infty)) < \infty.$$

Then $\Phi \in C^2(\mathcal{S}^0) \cap C(\overline{\mathcal{S}})$ *and*

$$c^*(x_1, x_2) = \left(\frac{\partial \Phi}{\partial x_1}\right)^{1/(\gamma-1)}$$

is optimal and there exist $\hat{\theta}_1, \hat{\theta}_2 \in [0, \pi/2]$ *with* $\hat{\theta}_1 < \hat{\theta}_2$ *such that*

$$D = \{re^{i\theta};\, \hat{\theta}_1 < \theta < \hat{\theta}_2\} \quad (i = \sqrt{-1})$$

is the non-intervention region and the optimal intervention strategy (portfolio) $\xi^*(t)$ *is the* local time *of the process* $(X_1(t), X_2(t))$ *reflected back into* D *in the direction parallel to* $\partial_1\mathcal{S}$ *at* $\theta = \hat{\theta}_1$ *and in the direction parallel to* $\partial_2\mathcal{S}$ *at* $\theta = \hat{\theta}_2$. *See Fig. 8.2.*

For proofs and more details we refer to [FØS2].

Remark 8.6 For other applications of singular control theory of jump diffusions see [Ma].

8.4 Exercises

Exercise* 8.1 (***Optimal Dividend Policy Under Proportional Transaction Costs***) Suppose the cash flow $X(t) = X^{(\xi)}(t)$ of a firm at time t is given by (with $\alpha, \sigma, \beta, \lambda > 0$ constants)

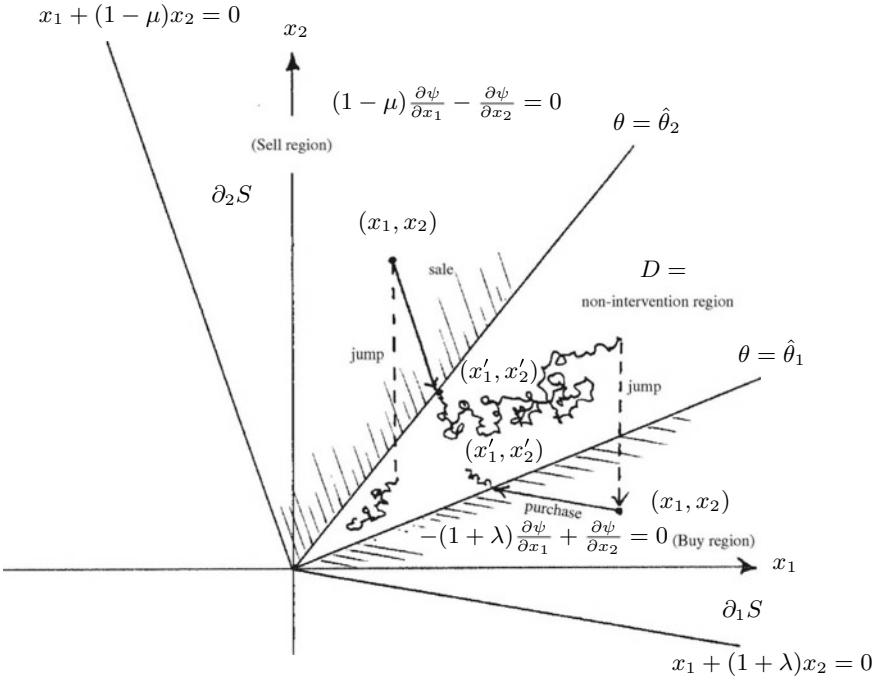
$$\mathrm{d}X(t) = \alpha\,\mathrm{d}t + \sigma\,\mathrm{d}B(t) + \beta\int_{\mathbb{R}} z\widetilde{N}(\mathrm{d}t, \mathrm{d}z) - (1+\lambda)\mathrm{d}\xi(t), \quad X(0^-) = x > 0,$$

where $\xi(t)$ is an increasing, adapted càdlàg process representing the total dividend taken out up to time t (our control process). We assume that $\beta z \le 0$ for a.a. z (ν). Let

$$\tau_{\mathcal{S}} = \inf\{t > 0;\, X^{(\xi)}(t) \le 0\}$$

be the time of bankruptcy of the firm and let

$$J^{\xi}(s,x) = E^{s,x}\left[\int_0^{\tau_{\mathcal{S}}} e^{-\rho(s+t)}\mathrm{d}\xi(t)\right], \quad \rho > 0 \text{ constant}$$



$\ell =$ purchase amount, $m =$ sale amount (at transaction)
$x_1' = x_1 - (1+\lambda)\ell + (1-\mu)m$ (new value of x_1 after transaction)
$x_2' = x_2 + \ell - m$ (new value of x_2 after transaction)

Fig. 8.2 The optimal portfolio $\xi^*(t)$

be the expected total discounted amount taken out up to bankruptcy time.

Find $\Phi(s,x)$ and a dividend policy ξ^* such that

$$\Phi(s,x) = \sup_{\xi} J^{(\xi)}(s,x) = J^{\xi^*}(s,x).$$

Exercise* 8.2 Let $\Phi(s,x_1,x_2)$ be the value function of the optimal consumption problem (8.1.2) with proportional transaction costs and let $\Phi_0(s,x_1,x_2) = K\mathrm{e}^{-\delta s}(x_1+x_2)^\gamma$ be the corresponding value function when there are no transaction costs, i.e., $\mu = \lambda = 0$ (Example 5.2). Use Theorem 8.2a to prove that

$$\Phi(s,x_1,x_2) \le K\mathrm{e}^{-\delta s}(x_1+x_2)^\gamma.$$

Exercise 8.3 (***Optimal Harvesting***)
Suppose the size $X(t)$ at time t of a certain fish population is modeled by a geometric Lévy process, i.e.,

$$\mathrm{d}X(t) = \mathrm{d}X^{(\xi)}(t) = X(t^-)\left[\mu\,\mathrm{d}t + \int_{\mathbb{R}} z\widetilde{N}(\mathrm{d}t, \mathrm{d}z)\right] - \mathrm{d}\xi(t), \quad t > 0,$$
$$X(0^-) = x > 0,$$

where $\mu > 0$ is a constant, $z > -1$ a.s. $\nu(\mathrm{d}z)$ and $\xi(t)$ is an increasing adapted process giving the amount harvested from the population from time 0 up to time t. We assume that $\xi(t)$ is right-continuous. Consider the *optimal harvesting problem*

$$\Phi(s, x) = \sup_{\xi} J^{(\xi)}(s, x),$$

where

$$J^{(\xi)}(s, x) = E^{s,x}\left[\int_0^{\tau_s} \theta \mathrm{e}^{-\rho(s+t)}\mathrm{d}\xi(t)\right],$$

with $\theta > 0$, $\rho > 0$ constants and

$$\tau_{\mathcal{S}} = \inf\{t > 0;\, X^{(\xi)}(t) \leq 0\} \quad \text{(extinction time)}.$$

If we interpret θ as the price per unit harvested, then $J^{(\xi)}(s, x)$ represents the expected total discounted value of the harvested amount up to extinction time.

(a) Write down the integrovariational inequalities (i), (ii), (vi), and (ix) of Theorem 8.2 in this case, with the state process

$$Y(t) = \begin{bmatrix} s + t \\ X(t) \end{bmatrix}, \quad t \geq 0, \quad Y(0) = y = \begin{bmatrix} s \\ x \end{bmatrix} \in \mathbb{R}^+ \times \mathbb{R}^+.$$

(b) Suppose $\mu \leq \rho$.
Show that in this case it is optimal to harvest all the population immediately, i.e., it is optimal to choose the harvesting strategy $\hat{\xi}$ defined by

$$\hat{\xi}(t) = x \text{ for all } t \geq 0$$

(sometimes called the "take the money and run" strategy).
This gives the value function

$$\Phi(s, x) = \theta \mathrm{e}^{-\rho s} x.$$

(c) Suppose $\mu > \rho$.
Show that in this case $\Phi(s, x) = \infty$.

Chapter 9
Impulse Control of Jump Diffusions



9.1 A General Formulation and a Verification Theorem

Suppose that—if there are no interventions—the state $Y(t) \in \mathbb{R}^k$ of the system we consider is a jump diffusion of the form

$$\mathrm{d}Y(t) = b(Y(t))\mathrm{d}t + \sigma(Y(t))\mathrm{d}B(t) + \int_{\mathbb{R}^\ell} \gamma(Y(t^-), z)\widetilde{N}(\mathrm{d}t, \mathrm{d}z), \qquad (9.1.1)$$

$$Y(0) = y \in \mathbb{R}^k,$$

where $b : \mathbb{R}^k \to \mathbb{R}^k$, $\sigma : \mathbb{R}^k \to \mathbb{R}^{k\times m}$, and $\gamma : \mathbb{R}^k \times \mathbb{R}^\ell \to \mathbb{R}^{k\times \ell}$ are given functions satisfying the conditions for the existence and uniqueness of a solution $Y(t)$ (see Theorem 1.19).

The generator A of $Y(t)$ is

$$A\phi(y) = \sum_{i=1}^{k} b_i(y)\frac{\partial \phi}{\partial y_i} + \frac{1}{2}\sum_{i,j=1}^{k} (\sigma\sigma^T)_{ij}(y)\frac{\partial^2 \phi}{\partial y_i \partial y_j}$$
$$+ \sum_{j=1}^{\ell} \int_{\mathbb{R}} \{\phi(y + \gamma^{(j)}(y, z_j)) - \phi(y) - \nabla\phi(y) \cdot \gamma^{(j)}(y, z_j)\}\nu_j(\mathrm{d}z_j).$$

Now suppose that at any time t and any state y we are free to intervene and give the system an impulse $\zeta \in \mathcal{Z} \subset \mathbb{R}^p$, where $\mathcal{Z}$ is a given set (the set of admissible impulse values). Suppose the result of giving the impulse ζ when the state is y is that the state jumps immediately from $y = Y(t^-)$ to $Y(t) = \Gamma(y, \zeta) \in \mathbb{R}^k$, where $\Gamma : \mathbb{R}^k \times \mathcal{Z} \to \mathbb{R}^k$ is a given function.

An *impulse control* for this system is a double (possibly finite) sequence

$$v = (\tau_1, \tau_2, \ldots, \tau_j, \ldots; \zeta_1, \zeta_2, \ldots, \zeta_j, \ldots)_{j\le M}, \quad M \le \infty,$$

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where $0 \leq \tau_1 \leq \tau_2 \leq \cdots$ are $\mathbb{F}_t$-stopping times (the *intervention times*) and $\zeta_1, \zeta_2, \ldots$ are the corresponding *impulses* at these times. We assume that ζ_j is $\mathbb{F}_{\tau_j}$-measurable for all j.

If $v = (\tau_1, \tau_2, \ldots; \zeta_1, \zeta_2, \ldots)$ is an impulse control, the corresponding state process $Y^{(v)}(t)$ is defined by

$$Y^{(v)}(0^-) = y \quad \text{and} \quad Y^{(v)}(t) = Y(t), \quad 0 < t \leq \tau_1, \tag{9.1.2}$$

$$Y^{(v)}(\tau_j) = \Gamma(\check{Y}^{(v)}(\tau_j^-), \zeta_j), \quad j = 1, 2, \ldots \tag{9.1.3}$$

(If $\tau_1 = 0$ we put $Y^{(v)}(\tau_1) = \Gamma(Y^{(v)}(0^-), \zeta_1) = \Gamma(y, \zeta_1)$.)

$$\begin{aligned} \mathrm{d}Y^{(v)}(t) &= b(Y^{(v)}(t))\mathrm{d}t + \sigma(Y^{(v)}(t))\mathrm{d}B(t) \\ &\quad + \int_{\mathbb{R}^\ell} \gamma(Y^{(v)}(t), z)\widetilde{N}(\mathrm{d}t, \mathrm{d}z) \quad \text{for } \tau_j < t < \tau_{j+1} \wedge \tau^*. \end{aligned} \tag{9.1.4}$$

If $\tau_{j+1} = \tau_j$, then $Y^{(v)}(t)$ jumps from $\check{Y}^{(v)}(\tau_j^-)$ to $\Gamma(\Gamma(\check{Y}^{(v)}(\tau_j^-), \zeta_j), \zeta_{j+1})$,

where

$$\check{Y}^{(v)}(\tau_j^-) = Y^{(v)}(\tau_j^-) + \Delta_N Y(\tau_j), \tag{9.1.5}$$

$\Delta_N Y^{(v)}(t)$ is as in (8.2.2) the jump of $Y^{(v)}$ stemming from the jump of the random measure $N(t, \cdot)$ only and

$$\tau^* = \tau^*(\omega) = \lim_{R \to \infty} (\inf\{t > 0; |Y^{(v)}(t)| \geq R\}) \leq \infty \tag{9.1.6}$$

is the explosion time of $Y^{(v)}(t)$. Note that here we must distinguish between the (possible) jump of $Y^{(v)}(\tau_j)$ stemming from the random measure N, denoted by $\Delta_N Y^{(v)}(\tau_j)$ and the jump caused by the intervention v, given by

$$\Delta_v Y^{(v)}(\tau_j) := \Gamma(\check{Y}^{(v)}(\tau_j^-), \zeta) - \check{Y}^{(v)}(\tau_j^-). \tag{9.1.7}$$

Let $\mathcal{S} \subset \mathbb{R}^k$ be a fixed open set (the solvency region). Define

$$\tau_{\mathcal{S}} = \inf\{t \in (0, \tau^*); Y^{(v)}(t) \notin \mathcal{S}\}. \tag{9.1.8}$$

Suppose we are given a continuous *profit function* $f : \mathcal{S} \to \mathbb{R}$ and a continuous *bequest function* $g : \mathbb{R}^k \to \mathbb{R}$. Moreover, suppose the profit/utility of making an intervention with impulse $\zeta \in \mathcal{Z}$ when the state is y is $K(y, \zeta)$, where $K : \mathcal{S} \times \mathcal{Z} \to \mathbb{R}$ is a given continuous function.

We assume we are given a set $\mathcal{V}$ of *admissible impulse controls* which is included in the set of $v = (\tau_1, \tau_2, \ldots; \zeta_1, \zeta_2, \ldots)$ such that a unique solution $Y^{(v)}$ of (9.1.2)–(9.1.4) exists and

$$\tau^* = \infty \quad \text{a.s.} \tag{9.1.9}$$

and (if $M = \infty$)

$$\lim_{j\to\infty} \tau_j = \tau_{\mathcal{S}}. \tag{9.1.10}$$

We also assume that

$$E^y\left[\int_0^{\tau_{\mathcal{S}}} \left|f(Y^{(v)}(t))\right| \mathrm{d}t\right] < \infty \quad \text{for all } y \in \mathbb{R}^k,\ v \in \mathcal{V}, \tag{9.1.11}$$

$$E\left[g^-(Y^{(v)}(\tau_{\mathcal{S}}))\mathcal{X}_{\{\tau_{\mathcal{S}}<\infty\}}\right] < \infty \quad \text{for all } y \in \mathbb{R}^k,\ v \in \mathcal{V}, \tag{9.1.12}$$

and

$$E\left[\sum_{\tau_j \le \tau_{\mathcal{S}}} K^-(\check{Y}^{(v)}(\tau_j^-), \zeta_j)\right] < \infty \quad \text{for all } y \in \mathbb{R}^k,\ v \in \mathcal{V}. \tag{9.1.13}$$

Now define the performance criterion

$$J^{(v)}(y) = E^y\Bigg[\int_0^{\tau_{\mathcal{S}}} f(Y^{(v)}(t))\mathrm{d}t + g(Y^{(v)}(\tau_{\mathcal{S}}))\mathcal{X}_{\{\tau_{\mathcal{S}}<\infty\}} + \sum_{\tau_j \le \tau_{\mathcal{S}}} K(\check{Y}^{(v)}(\tau_j^-), \zeta_j)\Bigg].$$

The *impulse control* problem is the following.

Find $\Phi(y)$ and $v^* \in \mathcal{V}$ such that

$$\Phi(y) = \sup\{J^{(v)}(y);\ v \in \mathcal{V}\} = J^{(v^*)}(y). \tag{9.1.14}$$

The following concept is crucial.

Definition 9.1 Let $\mathcal{H}$ be the space of all measurable functions $h : \mathcal{S} \to \mathbb{R}$. The *intervention operator* $\mathcal{M} : \mathcal{H} \to \mathcal{H}$ is defined by

$$\mathcal{M}h(y) = \sup\{h(\Gamma(y,\zeta)) + K(y,\zeta);\ \zeta \in \mathcal{Z} \quad \text{and} \quad \Gamma(y,\zeta) \in \mathcal{S}\}. \tag{9.1.15}$$

As in Chap. 3 we put

$$\mathcal{T} = \{\tau;\ \tau \text{ stopping time},\ 0 \le \tau \le \tau_{\mathcal{S}} \text{ a.s.}\}.$$

We can now state the main result of this chapter, a verification theorem for impulse control problems.

Theorem 9.2 (***Quasi-Integrovariational Inequalities for Impulse Control***)

(a) Suppose we can find $\phi : \bar{\mathcal{S}} \to \mathbb{R}$ *such that*

(i) $\phi \in C^1(\mathcal{S}) \cap C(\bar{\mathcal{S}})$.

(ii) $\phi \geq \mathcal{M}\phi$ *on* $\mathcal{S}$.
Define

$$D = \{y \in \mathcal{S};\ \phi(y) > \mathcal{M}\phi(y)\} \quad \textit{(the continuation region).}$$

Assume

(iii) $E^y\left[\int_0^{\tau_{\mathcal{S}}} \mathcal{X}_{\partial D}(Y^{(v)}(t))\mathrm{d}t\right] = 0$ *for all* $y \in \mathcal{S}$, $v \in \mathcal{V}$.

(iv) ∂D *is a Lipschitz surface.*

(v) $\phi \in C^2(\mathcal{S} \setminus \partial D)$ *with locally bounded derivatives near* ∂D.

(vi) $A\phi + f \leq 0$ *on* $\mathcal{S} \setminus \partial D$.

(vii) $\phi(y) = g(y)$ *for all* $y \notin \mathcal{S}$.

(viii) $\{\phi^-(Y^{(v)}(\tau));\ \tau \in \mathcal{T}\}$ *is uniformly integrable, for all* $y \in \mathcal{S}$, $v \in \mathcal{V}$.

(ix) $E^y\left[|\phi(Y^{(v)}(\tau))| + \int_0^{\tau_{\mathcal{S}}} |A\phi(Y^{(v)}(t))|\mathrm{d}t\right] < \infty$ *for all* $\tau \in \mathcal{T}$, $v \in \mathcal{V}$, $y \in \mathcal{S}$.
Then

$$\phi(y) \geq \Phi(y) \quad \textit{for all } y \in \mathcal{S}. \tag{9.1.16}$$

(b) Suppose in addition that

(x) $A\phi + f = 0$ *in* D.

(xi) $\hat{\zeta}(y) \in \mathrm{Argmax}\{\phi(\Gamma(y,\cdot)) + K(y,\cdot)\} \in \mathcal{Z}$ *exists for all* $y \in \mathcal{S}$ *and* $\hat{\zeta}(\cdot)$ *is a Borel measurable selection.*

Put $\hat{\tau}_0 = 0$ *and define* $\hat{v} = (\hat{\tau}_1, \hat{\tau}_2, \ldots; \hat{\zeta}_1, \hat{\zeta}_2, \ldots)$ *inductively by* $\hat{\tau}_{j+1} = \inf\{t > \hat{\tau}_j;\ Y^{(\hat{v}_j)}(t) \notin D\} \wedge \tau_{\mathcal{S}}$ *and* $\hat{\zeta}_{j+1} = \hat{\zeta}(Y^{(\hat{v}_j)}(\hat{\tau}_{j+1}^-))$ *if* $\hat{\tau}_{j+1} < \tau_{\mathcal{S}}$, *where* $Y^{(\hat{v}_j)}$ *is the result of applying* $\hat{v}_j := (\hat{\tau}_1, \ldots, \hat{\tau}_j; \hat{\zeta}_1, \ldots, \hat{\zeta}_j)$ *to* Y.
Suppose

(xii) $\hat{v} \in \mathcal{V}$ *and* $\{\phi(Y^{(\hat{v})}(\tau));\ \tau \in \mathcal{T}\}$ *is uniformly integrable.*
Then

$$\phi(y) = \Phi(y) \quad \textit{and } \hat{v} \textit{ is an optimal impulse control.} \tag{9.1.17}$$

Sketch of Proof (a) By Theorem 3.1 and (iii)–(v), we may assume that $\phi \in C^2(\mathcal{S}) \cap C(\bar{\mathcal{S}})$. Choose $v = (\tau_1, \tau_2, \ldots; \zeta_1, \zeta_2, \ldots) \in \mathcal{V}$ and set $\tau_0 = 0$. By another approximation argument we may assume that we can apply the Dynkin formula to the stopping times τ_j. Then for $j = 0, 1, 2, \ldots$, with $Y = Y^{(v)}$

$$E^y[\phi(Y(\tau_j))] - E^y[\phi(\check{Y}(\tau_{j+1}^-))] = -E^y\left[\int_{\tau_j}^{\tau_{j+1}} A\phi(Y(t))\mathrm{d}t\right], \tag{9.1.18}$$

where $\check{Y}(\tau_{j+1}^-) = Y(\tau_{j+1}^-) + \Delta_N Y(\tau_{j+1})$, as before. Summing this from $j = 0$ to $j = m$ we get

$$\phi(y) + \sum_{j=1}^{m} E^y[\phi(Y(\tau_j)) - \phi(\check{Y}(\tau_j^-))] - E^y[\phi(\check{Y}(\tau_{m+1}^-))]$$
$$= -E^y\left[\int_0^{\tau_{m+1}} A\phi(Y(t))\mathrm{d}t\right] \geq E^y\left[\int_0^{\tau_{m+1}} f(Y(t))\mathrm{d}t\right]. \tag{9.1.19}$$

Now

$$\phi(Y(\tau_j)) = \phi(\Gamma(\check{Y}(\tau_j^-), \zeta_j))$$
$$\leq \mathcal{M}\phi(\check{Y}(\tau_j^-)) - K(\check{Y}(\tau_j^-), \zeta_j) \quad \text{if } \tau_j < \tau_{\mathcal{S}} \text{ by (9.1.15)}$$

and

$$\phi(Y(\tau_j)) = \phi(\check{Y}(\tau_j^-)) \quad \text{if } \tau_j = \tau_{\mathcal{S}} \text{ by } (vii).$$

Therefore

$$\mathcal{M}\phi(\check{Y}(\tau_j^-)) - \phi(\check{Y}(\tau_j^-)) \geq \phi(Y(\tau_j)) - \phi(\check{Y}(\tau_j^-)) + K(\check{Y}(\tau_j^-), \zeta_j)$$

and

$$\phi(y) + \sum_{j=1}^{m} E^y[\{\mathcal{M}\phi(\check{Y}(\tau_j^-)) - \phi(\check{Y}(\tau_j^-))\} \cdot \mathcal{X}_{\{\tau_j < \tau_{\mathcal{S}}\}}]$$
$$\geq E^y\left[\int_0^{\tau_{m+1}} f(Y(t))\mathrm{d}t + \phi(\check{Y}(\tau_{m+1}^-)) + \sum_{j=1}^{m} K(\check{Y}(\tau_j^-), \zeta_j)\right].$$

Letting $m \to M$ and using quasi-left continuity of $Y(\cdot)$ (see the proof of Theorem 3.2) we get

$$\phi(y) \geq E^y\left[\int_0^{\tau_{\mathcal{S}}} f(Y(t))\mathrm{d}t + g(Y(\tau_{\mathcal{S}}))\mathcal{X}_{\{\tau_{\mathcal{S}} < \infty\}} + \sum_{j=1}^{M} K(\check{Y}(\tau_j^-), \zeta_j)\right] = J^{(v)}(y). \tag{9.1.20}$$

Hence $\phi(y) \geq \Phi(y)$.

(b) Next assume (x)–(xii) also hold. Apply the above argument to $\hat{v} = (\hat{\tau}_1, \hat{\tau}_2, \ldots; \hat{\zeta}_1, \hat{\zeta}_2, \ldots)$. Then by (x) we get *equality* in (9.1.19) and by our choice of $\zeta_j = \hat{\zeta}_j$ we have *equality* in (9.1.20). Hence

$$\phi(y) = J^{(\hat{v})}(y),$$

which combined with (a) completes the proof. □

Remark 9.3 In the case of a pure diffusion process, the same verification theorem holds, just skip condition (ix).

9.2 Examples

Example 9.4 (***Optimal Stream of Dividends Under Transaction Costs***) This example is an extension to the jump diffusion case of a problem studied in [J-PS]. Suppose that if we make no interventions the amount $X(t)$ available (cash flow) is given by

$$\mathrm{d}X(t) = \mu\,\mathrm{d}t + \sigma\,\mathrm{d}B(t) + \theta\int_{\mathbb{R}} z\widetilde{N}(\mathrm{d}t,\mathrm{d}z), \quad X(0) = x > 0, \tag{9.2.1}$$

where $\mu, \sigma > 0$, $\theta \geq 0$ are constants and we assume that $z \leq 0$ a.s. ν. Suppose that at any time t we are free to take out an amount $\zeta > 0$ from $X(t)$ by applying the transaction cost

$$k(\zeta) = c + \lambda\zeta, \tag{9.2.2}$$

where $c > 0$, $\lambda \geq 0$ are constants. The constant c is called the *fixed* part and the quantity $\lambda\zeta$ is called the *proportional* part, respectively, of the transaction cost. The resulting cash flow $X^{(v)}(t)$ is given by

$$X^{(v)}(t) = X(t) \quad \text{if } 0 \leq t < \tau_1, \tag{9.2.3}$$

$$X^{(v)}(\tau_j) = \Gamma(X^{(v)}(\tau_j^-) + \Delta_N X(\tau_j), \zeta_j) = \check{X}^{(v)}(\tau_j^-) - (1+\lambda)\zeta_j - c, \tag{9.2.4}$$

and

$$\mathrm{d}X^{(v)}(t) = \mu\,\mathrm{d}t + \sigma\,\mathrm{d}B(t) + \theta\int_{\mathbb{R}} z\widetilde{N}(\mathrm{d}t,\mathrm{d}z) \text{ if } \tau_j \leq t < \tau_{j+1}. \tag{9.2.5}$$

Put

$$\tau_{\mathcal{S}} = \inf\{t > 0;\, X^{(v)}(t) \leq 0\} \quad \text{(time of bankruptcy)} \tag{9.2.6}$$

and

$$J^{(v)}(s,x) = E^{s,x}\left[\sum_{\tau_j \leq \tau_{\mathcal{S}}} \mathrm{e}^{-\rho(s+\tau_j)}\zeta_j\right], \tag{9.2.7}$$

where $\rho > 0$ is constant (the discounting exponent).

We seek $\Phi(s,x)$ and $v^* = (\tau_1^*, \tau_2^*, \ldots; \zeta_1^*, \zeta_2^*, \ldots) \in \mathcal{V}$ such that

$$\Phi(s,x) = \sup_{v\in\mathcal{V}} J^{(v)}(s,x) = J^{(v^*)}(s,x), \tag{9.2.8}$$

where $\mathcal{V}$ is the set of impulse controls s.t. $X^{(v)}(t) \geq 0$ for all $t \leq \tau_{\mathcal{S}}$. This is a problem of the type (9.1.14), with

$$Y^{(v)}(t) = \begin{bmatrix} s+t \\ X^{(v)}(t) \end{bmatrix}, \quad t \geq 0, \quad Y^{(v)}(0^-) = \begin{bmatrix} s \\ x \end{bmatrix} = y,$$
$$\Gamma(y,\zeta) = \Gamma(s,x,\zeta) = \begin{bmatrix} s \\ x - c - (1+\lambda)\zeta \end{bmatrix},$$
$$K(y,\zeta) = K(s,x,\zeta) = \mathrm{e}^{-\rho s}\zeta, \quad f = g = 0,$$

and

$$\mathcal{S} = \{(s,x); x > 0\}.$$

As a candidate for the value function Φ we try

$$\phi(s,x) = \mathrm{e}^{-\rho s}\psi(x). \tag{9.2.9}$$

Then

$$\mathcal{M}\psi(x) = \sup\left\{\psi(x - c - (1+\lambda)\zeta) + \zeta,\ 0 < \zeta < \frac{x-c}{1+\lambda}\right\}.$$

We now guess that the continuation region has the form

$$D = \{(s,x); 0 < x < x^*\} \quad \text{for some } x^* > 0. \tag{9.2.10}$$

Then (x) of Theorem 9.2 gives

$$-\rho\psi(x) + \mu\psi'(x) + \frac{1}{2}\sigma^2\psi''(x) + \int_{\mathbb{R}}\{\psi(x+\theta z) - \psi(x) - \psi'(x)\theta z\}\nu(\mathrm{d}z) = 0.$$

To solve this equation we try a function of the form

$$\psi(x) = \mathrm{e}^{rx}$$

for some constant $r \in \mathbb{R}$. Then r must solve the equation

$$h(r) := -\rho + \mu r + \frac{1}{2}\sigma^2 r^2 + \int_{\mathbb{R}}\{\mathrm{e}^{r\theta z} - 1 - r\theta z\}\nu(\mathrm{d}z) = 0. \tag{9.2.11}$$

Since $h(0) = -\rho < 0$ and $\lim_{|r|\to\infty} h(r) = \infty$, we see that there exist two solutions r_1, r_2 of $h(r) = 0$ such that

$$r_2 < 0 < r_1.$$

Moreover, since $\mathrm{e}^{r\theta z} - 1 - r\theta z \geq 0$ for all r, z we have

$$|r_2| > r_1.$$

With such a choice of r_1, r_2 we try

$$\psi(x) = A_1 e^{r_1 x} + A_2 e^{r_2 x}, \quad A_i \text{ constants.}$$

Since

$$\psi(0) = 0 \quad \text{we have} \quad A_1 + A_2 = 0$$

so we write $A_1 = A = -A_2 > 0$ and

$$\psi(x) = A\left(e^{r_1 x} - e^{r_2 x}\right), \quad 0 < x < x^*.$$

Define

$$\psi_0(x) = A\left(e^{r_1 x} - e^{r_2 x}\right) \quad \text{for } all x > 0. \tag{9.2.12}$$

To study $\mathcal{M}\psi$ we first consider

$$g(\zeta) := \psi_0(x - c - (1+\lambda)\zeta) + \zeta, \quad \zeta > 0.$$

The first-order condition for a maximum point $\hat{\zeta} = \hat{\zeta}(x)$ for $g(\zeta)$ is that

$$\psi_0'(x - c - (1+\lambda)\hat{\zeta}) = \frac{1}{1+\lambda}.$$

Now

$$\psi_0'(x) > 0 \quad \text{for all } x \text{ and}$$
$$\psi_0''(x) < 0 \quad \text{iff} \quad x < \tilde{x} := \frac{2(\ln|r_2| - \ln r_1)}{r_1 - r_2}.$$

Therefore the equation $\psi_0'(x) = 1/(1+\lambda)$ has exactly two solutions $x = \underline{x}$ and $x = \bar{x}$ where

$$0 < \underline{x} < \tilde{x} < \bar{x}$$

(provided that $\psi_0'(\tilde{x}) < 1/(1+\lambda) < \psi_0'(0)$). See Fig. 9.1.

Choose

$$x^* = \bar{x} \quad \text{and put} \quad \hat{x} = \underline{x}. \tag{9.2.13}$$

If we require that $\psi(x) = \mathcal{M}\psi_0(x)$ for $x \geq x^*$ we get

$$\psi(x) = \psi_0(\hat{x}) + \hat{\zeta}(x) \quad \text{for} \quad x \geq x^*,$$

where

$$x - c - (1+\lambda)\hat{\zeta}(x) = \hat{x},$$

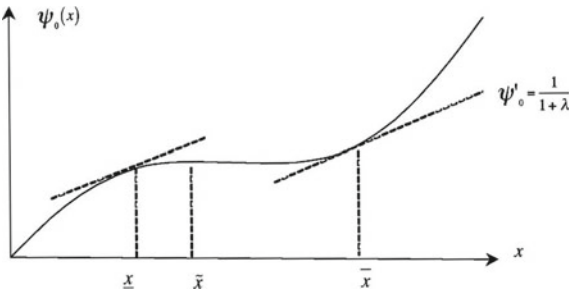


Fig. 9.1 The function $\psi_0(x)$

i.e.,

$$\hat{\zeta}(x) = \frac{x - \hat{x} - c}{1 + \lambda} \quad \text{for} \quad x \geq x^*. \tag{9.2.14}$$

Hence we propose that ψ has the form

$$\psi(x) = \begin{cases} \psi_0(x) = A(e^{r_1 x} - e^{r_2 x}), & 0 < x < x^*, \\ \psi_0(\hat{x}) + \dfrac{x - \hat{x} - c}{1 + \lambda}, & x \geq x^*. \end{cases} \tag{9.2.15}$$

Now choose A such that ψ is continuous at $x = x^*$. This gives

$$A = (1 + \lambda)^{-1}\left[e^{r_1 x^*} - e^{r_2 x^*} - e^{r_1 \hat{x}} + e^{r_2 \hat{x}}\right]^{-1}(x^* - \hat{x} - c). \tag{9.2.16}$$

By our choice of x^* we then have that ψ is also differentiable at $x = x^*$.

We can now check that, with these values of x^*, $\hat{x}$, and A, our choice of $\phi(s, x) = e^{-\rho s}\psi(x)$ satisfies all the requirements of Theorem 9.2, provided that some conditions on the parameters are satisfied. We leave this verification to the reader.

Thus the solution of the impulse control problem (9.2.8) can be described as follows. As long as $X(t) < x^*$ we do nothing. If $X(t)$ reaches the value x^* then immediately we make an intervention to bring $X(t)$ down to the value $\hat{x}$. See Fig. 9.2.

Example 9.5 As another illustration of how to apply Theorem 9.2 we consider the following example, which is a jump diffusion version of the example in [Ø2] studied in connection with questions involving vanishing fixed costs. Variations of this problem have been studied by many authors (see, e.g., [HST, J-P, MuØ, ØS, ØUZ, V]). One possible economic interpretation is that the given process represents the exchange rate of a given currency and the impulses represent the interventions taken in order to keep the exchange rate in a given "target zone." See, e.g., [J-P, MØ].

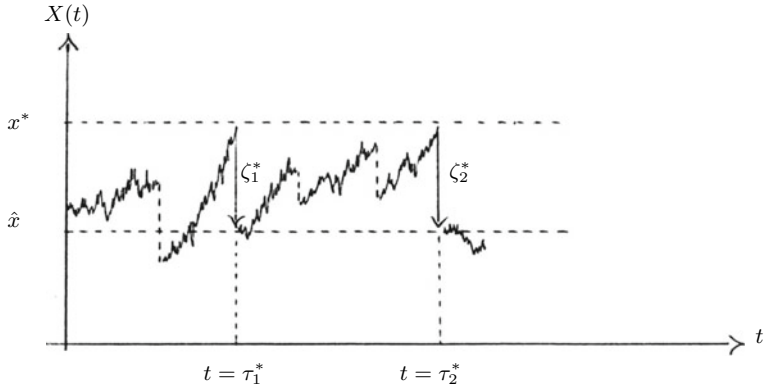


Fig. 9.2 The optimal impulse control of Example 9.4

Suppose that without interventions the system has the form

$$Y(t) = \begin{bmatrix} s+t \\ X(t) \end{bmatrix} \in \mathbb{R}^2, \quad Y(0) = y = (s, x), \tag{9.2.17}$$

where $X(t) = x + B(t) + \int_0^t \int_{\mathbb{R}} z\tilde{N}(\mathrm{d}s, \mathrm{d}z)$ and $B(0) = 0$. We assume that $z \leq 0$ a.s. z. Suppose that we are only allowed to give the system impulses ζ with values in $\mathcal{Z} := (0, \infty)$ and that if we apply an impulse control $v = (\tau_1, \tau_2, \ldots; \zeta_1, \zeta_2, \ldots)$ to $Y(t)$ it gets the form

$$Y^{(v)}(t) = \begin{bmatrix} s+t \\ X(t) - \sum\limits_{\tau_k \leq t} \zeta_k \end{bmatrix} = \begin{bmatrix} s+t \\ X^{(v)}(t) \end{bmatrix}. \tag{9.2.18}$$

Suppose that the *cost rate* $f(t, \xi)$ if $X^{(v)}(t) = \xi$ at time t is given by

$$f(t, \xi) = \mathrm{e}^{-\rho t}\xi^2, \tag{9.2.19}$$

where $\rho > 0$ is constant. In an effort to reduce the cost one can apply the impulse control v in order to reduce the value of $X^{(v)}(t)$. However, suppose the cost of an intervention of size $\zeta > 0$ at time t is

$$K(t, \xi, \zeta) = K(\zeta) = c + \lambda\zeta, \tag{9.2.20}$$

where $c > 0$ and $\lambda \geq 0$ are constants. Then the expected total discounted cost associated to a given impulse control is

$$J^{(v)}(s,x) = E^x\left[\int_0^\infty \mathrm{e}^{-\rho(s+t)}(X^{(v)}(t))^2\mathrm{d}t + \sum_{k=1}^{N}\mathrm{e}^{-\rho(s+\tau_k)}(c+\lambda\zeta_k)\right]. \quad (9.2.21)$$

We seek $\Phi(s,x)$ and $v^* = (\tau_1^*, \tau_2^*, \ldots; \zeta_1^*, \zeta_2^*, \ldots)$ such that

$$\Phi(s,x) = \inf_v J^{(v)}(s,x) = J^{(v^*)}(s,x). \quad (9.2.22)$$

This is an impulse control problem of the type described above, except that it is a minimum problem rather than a maximum problem. Theorem 9.2 still applies, with the corresponding changes.

Note that it is not optimal to move $X(t)$ downward if $X(t)$ is already below 0. Hence we may restrict ourselves to consider impulse controls $v = (\tau_1, \tau_2, \ldots; \zeta_1, \zeta_2, \ldots)$ such that

$$\sum_{j=1}^{\tau_k}\zeta_j \le X(\tau_k) \quad \text{for all } k. \quad (9.2.23)$$

We let $\mathcal{V}$ denote the set of such impulse controls.

We guess that the optimal strategy is to wait until the level of $X(t)$ reaches an (unknown) value $x^* > 0$. At this time, τ_1, we intervene and give $X(t)$ an impulse ζ_1, which brings it down to a lower value $\hat{x} > 0$. Then we do nothing until the next time, τ_2, that $X(t)$ reaches the level x^*, etc. This suggests that the continuation region D in Theorem 9.2 has the form

$$D = \{(s,x),\ x < x^*\}. \quad (9.2.24)$$

See Fig. 9.3.

Let us try a value function φ of the form

$$\varphi(s,x) = \mathrm{e}^{-\rho s}\psi(x), \quad (9.2.25)$$

where ψ remains to be determined.

Condition (x) of Theorem 9.2 gives that for $x < x^*$ we should have

$$A\varphi + f = \mathrm{e}^{-\rho s}\left(-\rho\psi(x) + \frac{1}{2}\psi''(x) + \int_{\mathbb{R}}\{\psi(x+z) - \psi(x) - z\psi'(x)\}\nu(\mathrm{d}z)\right) + \mathrm{e}^{-\rho s}x^2 = 0.$$

So for $x < x^*$ we let ψ be a solution $h(x)$ of the equation

$$\int_{\mathbb{R}}\{h(x+z) - h(x) - zh'(x)\}\nu(\mathrm{d}z) + \frac{1}{2}h''(x) - \rho h(x) + x^2 = 0. \quad (9.2.26)$$

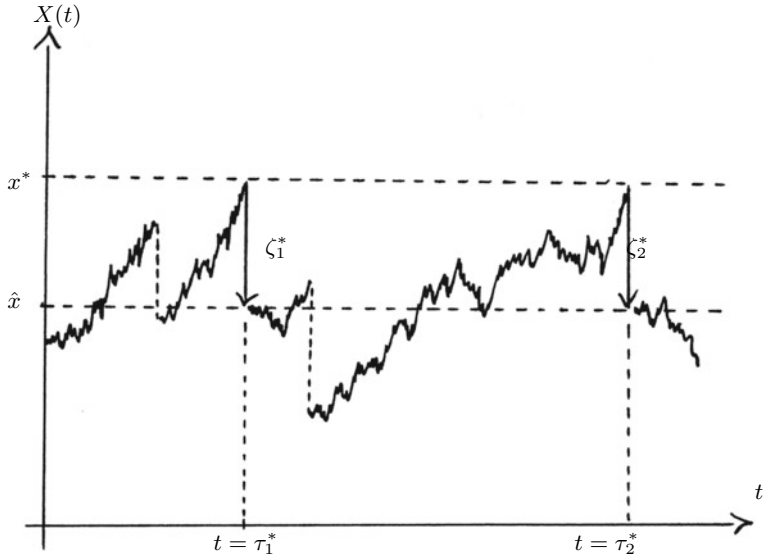


Fig. 9.3 The optimal impulse control of Example 9.5

We see that any function $h(x)$ of the form

$$h(x) = C_1 e^{r_1 x} + C_2 e^{r_2 x} + \frac{1}{\rho}x^2 + \frac{1 + \int_{\mathbb{R}} z^2 \nu(\mathrm{d}z)}{\rho^2}, \tag{9.2.27}$$

where C_1, C_2 are arbitrary constants, is a solution of (9.2.26), provided that $r_1 > 0, r_2 < 0$ are roots of the equation

$$K(r) := \int_{\mathbb{R}} \{e^{rz} - 1 - rz\}\nu(\mathrm{d}z) + \frac{1}{2}r^2 - \rho = 0.$$

Note that if we make no interventions at all, then the cost is

$$\begin{aligned} J^{(v)}(s,x) &= e^{-\rho s} E^x \left[\int_0^\infty e^{-\rho t} (X(t))^2 \mathrm{d}t \right] \\ &= e^{-\rho s} \int_0^\infty e^{-\rho t}(x^2 + tb)\mathrm{d}t = e^{-\rho s}\left(\frac{1}{\rho}x^2 + \frac{b}{\rho^2}\right), \end{aligned} \tag{9.2.28}$$

where $b = 1 + \int_{\mathbb{R}} z^2 \nu(\mathrm{d}z)$. Hence we must have

$$0 \le \psi(x) \le \frac{1}{\rho}x^2 + \frac{b}{\rho^2} \quad \text{for all } x. \tag{9.2.29}$$

Comparing this with (9.2.27) we see that we must have $C_2 = 0$. Hence $C_1 \leq 0$. So we put

$$\psi(x) = \psi_0(x) := \frac{1}{\rho}x^2 + \frac{b}{\rho^2} - ae^{r_1 x} \quad \text{for } x \leq x^*, \tag{9.2.30}$$

where $a = -C_1$ remains to be determined.

We guess that $a > 0$.

To determine a we first find ψ for $x > x^*$ and then require ψ to be C^1 at $x = x^*$.

By (ii) and (9.2.24) we know that for $x > x^*$ we have

$$\psi(x) = \mathcal{M}\psi(x) := \inf\{\psi(x - \zeta) + c + \lambda\zeta, \zeta > 0\}. \tag{9.2.31}$$

The first-order condition for a minimum $\hat{\zeta} = \hat{\zeta}(x)$ of the function

$$G(\zeta) := \psi(x - \zeta) + c + \lambda\zeta, \quad \zeta > 0$$

is

$$\psi'(x - \hat{\zeta}) = \lambda.$$

Suppose there is a unique point $\hat{x} \in (0, x^*)$ such that

$$\psi'(\hat{x}) = \lambda. \tag{9.2.32}$$

Then

$$\hat{x} = x - \hat{\zeta}(x), \text{ i.e., } \hat{\zeta}(x) = x - \hat{x}$$

and from (9.2.31) we deduce that

$$\psi(x) = \psi_0(\hat{x}) + c + \lambda(x - \hat{x}) \quad \text{for } x \geq x^*.$$

In particular,

$$\psi'(x^*) = \lambda \tag{9.2.33}$$

and

$$\psi(x^*) = \psi_0(\hat{x}) + c + \lambda(x^* - \hat{x}). \tag{9.2.34}$$

To summarize we put

$$\psi(x) = \begin{cases} \dfrac{1}{\rho}x^2 + \dfrac{b}{\rho^2} - ae^{r_1 x}, & \text{for } x \leq x^*, \\ \psi_0(\hat{x}) + c + \lambda(x - \hat{x}), & \text{for } x > x^*, \end{cases} \tag{9.2.35}$$

where $\hat{x}$, x^*, and a are determined by (9.2.32)–(9.2.34), i.e.,

$$ar_1 e^{r_1\hat{x}} = \frac{2}{\rho}\hat{x} - \lambda \quad \text{(i.e., } \psi'(\hat{x}) = \lambda\text{)}, \tag{9.2.36}$$

$$ar_1 e^{r_1 x^*} = \frac{2}{\rho}x^* - \lambda \quad \text{(i.e., } \psi'(x^*) = \lambda\text{)}, \tag{9.2.37}$$

$$ae^{r_1 x^*} - ae^{r_1\hat{x}} = \frac{1}{\rho}((x^*)^2 - (\hat{x})^2) - c - \lambda(x^* - \hat{x}). \tag{9.2.38}$$

One can now prove (see [Ø2, Theorem 2.5]).

For each $c > 0$ there exists $a = a^*(c) > 0$, $\hat{x} = \hat{x}(c) > 0$, and $x^* = x^*(c) > \hat{x}$ such that (9.2.36)–(9.2.38) hold. With this choice of a, $\hat{x}$, and x^*, the function $\varphi(s,x) = e^{-\rho s}\psi(x)$ with ψ given by (9.2.35) coincides with the value function Φ defined in (9.2.22). Moreover, the optimal impulse control $v^* = (\tau_1^*, \tau_2^*, \ldots; \zeta_1^*, \zeta_2^*, \ldots)$ is to do nothing while $X(t) < x^*$, then move $X(t)$ from x^* down to $\hat{x}$ (i.e., apply $\zeta_1^* = x^* - \hat{x}$) at the first time τ_1^* when $X(t)$ reaches a value $\geq x^*$, then wait until the next time, τ_2^*, $X(t)$ again reaches the value x^*, etc.

Remark 9.6 In [Ø2] this result is used to study how the value function $\Phi(s,x) = \Phi_c(s,x)$ depends on the fixed part $c > 0$ of the intervention cost. It is proved that the function

$$c \to \Phi_c(s,x)$$

is continuous but not differentiable at $c = 0$. In fact, we have

$$\frac{\partial}{\partial c}\Phi_c(s,x) \to \infty \text{ as } c \to 0^+.$$

Subsequently this high c-sensitivity of the value function for c close to 0 was proved for other processes as well. See [ØUZ].

Remark 9.7 For applications of impulse control theory in inventory control see, e.g., [S1, S2] and the references therein.

9.3 Exercises

Exercise* 9.1 Solve the impulse control problem

$$\Phi(s,x) = \inf_v J^{(v)}(s,x) = J^{(v^*)}(s,x),$$

where

$$J^{(v)}(s,x) = E\left[\int_0^\infty e^{-\rho(s+t)}(X^{(v)}(t))^2 dt + \sum_{k=1}^N e^{-\rho(s+\tau_k)}(c + \lambda|\zeta_k|)\right].$$

The inf is taken over all impulse controls $v = (\tau_1, \tau_2, \ldots; \zeta_1, \zeta_2, \ldots)$ with $\zeta_i \in \mathbb{R}$ and the corresponding process $X^{(v)}(t)$ is given by

$$\begin{aligned} \mathrm{d}X^{(v)}(t) &= \mathrm{d}B(t) + \int_{\mathbb{R}} \theta(X^{(v)}(t), z)\tilde{N}(\mathrm{d}t, \mathrm{d}z), \quad \tau_k < t < \tau_{k+1}, \\ X^{(v)}(\tau_{k+1}) &= \check{X}^{(v)}(\tau_{k+1}^-) + \zeta_{k+1}, \quad k = 0, 1, 2, \ldots, \end{aligned}$$

where $B(0) = 0$, $x \in \mathbb{R}$, and we assume that there exists $\xi > 0$ such that

$$\theta(x, z) = 0 \text{ for a.a. } z \text{ if } |x| < \xi, \tag{9.3.1}$$

$$\theta(x, z) \geq 0 \text{ if } x \geq \xi \text{ and } \theta(x, z) \leq 0 \text{ if } x \leq -\xi \text{ for a.a. } z, \tag{9.3.2}$$

Exercise* 9.2 (***Optimal Stream of Dividends with Transaction Costs from a Geometric Lévy Process***) For $v = (\tau_1, \tau_2, \ldots; \zeta_1, \zeta_2, \ldots)$ with $\zeta_i \in \mathbb{R}_+$ define $X^{(v)}(t)$ by

$$\begin{aligned} \mathrm{d}X^{(v)}(t) &= \mu X^{(v)}(t)\mathrm{d}t + \sigma X^{(v)}(t)\mathrm{d}B(t) \\ &\quad + \theta X^{(v)}(t^-)\int_{\mathbb{R}} z\tilde{N}(\mathrm{d}s, \mathrm{d}z), \quad \tau_i \leq t \leq \tau_{i+1}, \\ X^{(v)}(\tau_{i+1}) &= \check{X}^{(v)}(\tau_{i+1}^-) - (1 + \lambda)\zeta_{i+1} - c, \quad i = 0, 1, 2, \ldots, \\ X^{(v)}(0^-) &= x > 0, \end{aligned}$$

where $\mu, \sigma \neq 0, \theta, \lambda \geq 0$, and $c > 0$ are constants (see (9.1.5)), $-1 \leq \theta z \leq 0$ a.s. ν.

Find Φ and v^* such that

$$\Phi(s, x) = \sup_v J^{(v)}(s, x) = J^{(v^*)}(s, x).$$

Here

$$J^{(v)}(s, x) = E^x\left[\sum_{\tau_k < \tau_{\mathcal{S}}} \mathrm{e}^{-\rho(s+\tau_k)}\zeta_k\right] \qquad (\rho > 0 \text{ constant})$$

is the expected discounted total dividend up to time $\tau_{\mathcal{S}}$, where

$$\tau_{\mathcal{S}} = \tau_{\mathcal{S}}(\omega) = \inf\{t > 0;\, X^{(v)}(t) \leq 0\}$$

is the time of bankruptcy. (See also Exercise 10.2.)

Exercise* 9.3 (***Optimal Forest Management*** (***Inspired by Y. Willassen*** [W])) Suppose the biomass of a forest at time t is given by

$$X(t) = x + \mu t + \sigma B(t) + \theta\int_{\mathbb{R}} z\widetilde{N}(t, \mathrm{d}z),$$

where $\mu > 0$, $\sigma > 0$, $\theta > 0$ are constants and we assume that $z \leq 0$ a.s. ν. At times $0 \leq \tau_1 < \tau_2 < \cdots$ we decide to cut down the forest and replant it, with the cost

$$c + \lambda \check{X}(\tau_k^-), \quad \text{with} \quad \check{X}(\tau_k^-) = X(\tau_k^-) + \Delta_N X(\tau_k),$$

where $c > 0$ and $\lambda \in [0, 1)$ are constants and $\Delta_N X(t)$ is the (possible) jump in X at t coming from the jump in $N(t, \cdot)$ only, not from the intervention.

Find the sequence of stopping times $v = (\tau_1, \tau_2, \ldots)$ which maximizes the expected total discounted net profit $J^{(v)}(s, x)$ given by

$$J^{(v)}(s, x) = E^x \left[\sum_{k=1}^{\infty} \mathrm{e}^{-\rho(s+\tau_k)} (\check{X}(\tau_k^-) - c - \lambda \check{X}(\tau_k^-)) \right],$$

where $\rho > 0$ is a given discounting exponent.

Chapter 10
Approximating Impulse Control by Iterated Optimal Stopping



10.1 Iterative Scheme

In general it is not possible to reduce impulse control to optimal stopping, because the choice of the first intervention time τ_1 and the first impulse ζ_1 will influence the next and so on. However, if we allow only (up to) a fixed finite number n of interventions, then the corresponding impulse control problem can be solved by solving iteratively n optimal stopping problems. Moreover, if we restrict the number of interventions to (at most) n in a given impulse control problem, then the value function of this restricted problem will converge to the value function of the original problem as $n \to \infty$. Thus it is possible to reduce a given impulse control problem to a sequence of iterated optimal stopping problems. This is useful both for theoretical purposes and numerical applications.

We now make this more precise.

Using the notation of Chap. 9 consider the impulse control problem

$$\Phi(y) = \sup\{J^{(v)}(y);\, v \in \mathcal{V}\} = J^{(v^*)}(y), \quad y \in \mathcal{S}, \tag{10.1.1}$$

where, with $\tau_{\mathcal{S}} = \tau_{\mathcal{S}}^{(v)} = \inf\{t > 0;\, Y^{(v)}(t) \notin \mathcal{S}\}$,

$$J^{(v)}(y) = E^y\left[\int_0^{\tau_{\mathcal{S}}} f(Y^{(v)}(t))\mathrm{d}t + g(Y^{(v)}(\tau_{\mathcal{S}}))\chi_{\{\tau_{\mathcal{S}}<\infty\}} + \sum_{\tau_j<\tau_{\mathcal{S}}} K(\check{Y}^{(v)}(\tau_j^-), \zeta_j)\right]. \tag{10.1.2}$$

Here $\mathcal{V}$ denotes the set of admissible controls $v = (\tau_1, \tau_2, \ldots; \zeta_1, \zeta_2, \ldots)$. See (9.1.9)–(9.1.13).

For $n = 1, 2, \ldots$ let $\mathcal{V}_n$ denote the set of all $v \in \mathcal{V}$ such that $v = (\tau_1, \tau_2, \ldots, \tau_k; \zeta_1, \zeta_2, \ldots, \zeta_k)$ with $k \le n$. In other words, $\mathcal{V}_n$ is the set of all admissible controls *with at most n interventions*. Then

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$$\mathcal{V}_n \subseteq \mathcal{V}_{n+1} \subseteq \mathcal{V} \quad \text{for all } n. \tag{10.1.3}$$

Define

$$\Phi_n(y) = \sup\{J^{(v)}(y);\, v \in \mathcal{V}_n\}, \quad n = 1, 2, \ldots \tag{10.1.4}$$

Then $\Phi_n(y) \leq \Phi_{n+1}(y) \leq \Phi(y)$ because $\mathcal{V}_n \subseteq \mathcal{V}_{n+1} \subseteq \mathcal{V}$. Moreover, we have:

Lemma 10.1 *Suppose $g \geq 0$. Then*

$$\lim_{n\to\infty} \Phi_n(y) = \Phi(y) \quad \textit{for all } y \in \mathcal{S}.$$

Proof We have already seen that

$$\lim_{n\to\infty} \Phi_n(y) \leq \Phi(y).$$

To get the opposite inequality let us first assume $\Phi(y) < \infty$. Then for each $\varepsilon > 0$ there exists $v = (\tau_1, \tau_2, \ldots; \zeta_1, \zeta_2, \ldots) \in \mathcal{V}$ such that

$$\begin{aligned} J^{(v)}(y) = E^y\Bigg[\int_0^{\tau_{\mathcal{S}}^{(v)}} f(Y^{(v)}(t))\mathrm{d}t + g(Y^{(v)}(\tau_{\mathcal{S}}^{(v)}))\chi_{\{\tau_{\mathcal{S}}^{(v)}<\infty\}} \\ + \sum_{\tau_j < \tau_{\mathcal{S}}^{(v)}} K(\check{Y}^{(v)}(\tau_j^-), \zeta_j)\Bigg] \geq \Phi(y) - \varepsilon. \end{aligned} \tag{10.1.5}$$

For $n = 1, 2, \ldots$ define $v_n = (\tau_1, \tau_2, \ldots, \tau_n, \tau_{\mathcal{S}}; \zeta_1, \zeta_2, \ldots, \zeta_n)$, i.e., v_n is obtained by truncating the v sequence after n steps. Then

$$Y^{(v_n)}(t) = Y^{(v)}(t) \quad \text{for all } t \leq \tau_n. \tag{10.1.6}$$

Since $\tau_j \to \tau_{\mathcal{S}}$ a.s. when $j \to \infty$, we get by assumptions (9.1.11) and (9.1.13) that there exists n such that

$$E^y\left[\int_{\tau_n}^{\tau_{\mathcal{S}}^{(v)}} \left|f(Y^{(v)}(t))\right| \mathrm{d}t + \int_{\tau_n}^{\tau_{\mathcal{S}}^{(v_n)}} \left|f(Y^{(v_n)}(t))\right| \mathrm{d}t\right] < \varepsilon \tag{10.1.7}$$

and

$$E^y\Bigg[\sum_{\substack{j>n \\ \tau_j < \tau_{\mathcal{S}}^{(v)}}} K^-(\check{Y}^{(v)}(\tau_j^-), \zeta_j)\Bigg] < \varepsilon. \tag{10.1.8}$$

Moreover, by (10.1.6) we have

$$\chi_{\{\tau_{\mathcal{S}}^{(v)}<\infty\}} \leq \liminf_{n\to\infty} \chi_{\{\tau_{\mathcal{S}}^{(v_n)}<\infty\}}. \tag{10.1.9}$$

Combining (10.1.6)–(10.1.9) we get

$$\begin{aligned}\liminf_{n\to\infty} J^{(v_n)}(y) &= \liminf_{n\to\infty}\left\{E^y\left[\int_0^{\tau_n} + \int_{\tau_n}^{\tau_{\mathcal{S}}^{(v_n)}} f(Y^{(v_n)}(t))\mathrm{d}t\right]\right.\\ &\quad\left. + E^y\left[g(Y^{(v_n)}(\tau_{\mathcal{S}}^{(v_n)}))\chi_{\{\tau_{\mathcal{S}}^{(v_n)}<\infty\}}\right] + E^y\left[\sum_{j=1}^{n} K(\check{Y}^{(v_n)}(\tau_j^-),\zeta_j)\right]\right\}\\ &\geq J^{(v)}(y) - 2\varepsilon + \liminf_{n\to\infty} E^y\left[g(Y^{(v_n)}(\tau_{\mathcal{S}}^{(v_n)}))\chi_{\{\tau_{\mathcal{S}}^{(v_n)}<\infty\}}\right.\\ &\quad\left. - g(Y^{(v)}(\tau_{\mathcal{S}}^{(v)}))\chi_{\{\tau_{\mathcal{S}}^{(v)}<\infty\}}\right] \geq J^{(v)}(y) - 2\varepsilon\\ &\quad + E^y\left[\liminf_{n\to\infty}(g(Y^{(v_n)}(\tau_{\mathcal{S}}^{(v_n)})) - g(Y^{(v)}(\tau_{\mathcal{S}}^{(v)})))\chi_{\{\tau_{\mathcal{S}}^{(v_n)}<\infty\}}\right]\\ &= J^{(v)}(y) - 2\varepsilon.\end{aligned}$$

Hence by (10.1.5)

$$\liminf_{n\to\infty} \Phi_n(y) \geq \liminf_{n\to\infty} J^{(v_n)}(y) \geq \Phi(y) - 3\varepsilon.$$

Since $\varepsilon > 0$ was arbitrary this proves Lemma 10.1 in the case when $\Phi(y) < \infty$. If $\Phi(y) = \infty$ the proof is similar, except that now we use that for each $M < \infty$ there exists $v \in \mathcal{V}$ such that $J^{(v)}(y) \geq M$. Choosing v_n as before and using (10.1.6)–(10.1.9) with $\varepsilon = 1$ we get $J^{(v_n)}(y) \geq M - 2$. Since M was arbitrary this shows that

$$\lim_{n\to\infty} \Phi_n(y) \geq \lim_{n\to\infty} J^{(v_n)}(y) = \infty.$$

□

Let

$$\mathcal{M}h(y) = \sup_{\zeta\in\mathcal{Z}}\{h(\Gamma(y,\zeta)) + K(y,\zeta)\}, \quad h \in \mathcal{H},\ y \in \mathbb{R}^k \tag{10.1.10}$$

be the intervention operator (Definition 9.1). The iterative procedure is the following.

Let $Y(t) = Y^{(0)}(t)$ be the process (9.1.1) without interventions. Define

$$\varphi_0(y) = E^y\left[\int_0^{\tau_{\mathcal{S}}} f(Y(t))\mathrm{d}t + g(Y(\tau_{\mathcal{S}}))\chi_{\{\tau_{\mathcal{S}}<\infty\}}\right] \tag{10.1.11}$$

and inductively, for $j = 1, 2, \ldots, n$,

$$\varphi_j(y) = \sup_{\tau\in\mathcal{T}} E^y\left[\int_0^{\tau} f(Y(t))\mathrm{d}t + \mathcal{M}\varphi_{j-1}(Y(\tau))\chi_{\{\tau_{\mathcal{S}}<\infty\}}\right], \tag{10.1.12}$$

where, as before $\mathcal{T}$ denotes the set of stopping times $\tau \leq \tau_{\mathcal{S}}$, with

$$\tau_{\mathcal{S}} = \inf\{t > 0; Y(t) \notin \mathcal{S}\}.$$

Let $\mathcal{P}(\mathbb{R}^k)$ denote the set of functions $h : \mathbb{R}^k \to \mathbb{R}$ of at most *polynomial growth*, i.e., with the property that there exists constants C and m (depending on h) such that

$$|h(y)| \leq C(1 + |y|^m) \quad \text{for all } y \in \mathbb{R}^k. \tag{10.1.13}$$

The main result of this chapter is the following.

Theorem 10.2 *Suppose*

$$f, g \text{ and } \mathcal{M}\varphi_{j-1} \in \mathcal{P}(\mathbb{R}^k) \quad \text{for } j = 1, 2, \ldots, n. \tag{10.1.14}$$

Then

$$\varphi_n = \Phi_n.$$

To prove this we need a *dynamic programming principle* (or *Bellman* principle). This principle is due to Krylov [K, Theorems 9 and 11, p. 134] when there are no jumps. The proof of the dynamic programming principle for jump processes can be found in Ishikawa [Ish, Sect. 4].

Lemma 10.3 (***Dynamic programming principle (I)***) *Suppose $G \in \mathcal{P}(\mathbb{R}^k)$. Define*

$$\psi(y) = \sup_{\tau \in \mathcal{T}} E^y\left[\int_0^{\tau} f(Y(s))\mathrm{d}s + G(Y(\tau))\chi_{\{\tau_{\mathcal{S}} < \infty\}}\right]. \tag{10.1.15}$$

(a) *Then for all finite stopping times β we have*

$$\psi(y) = \sup_{\tau \in \mathcal{T}} E^y\left[\int_0^{\tau \wedge \beta} f(Y(t))\mathrm{d}t + G(Y(\tau))\chi_{\{\tau \leq \beta\}} + \psi(Y(\beta))\chi_{\{\tau > \beta\}}\right]. \tag{10.1.16}$$

(b) *For $\varepsilon > 0$ define*

$$D^{(\varepsilon)} = \{y \in \mathcal{S}; \psi(y) > G(y) + \varepsilon\}$$

and put

$$\tau^{(\varepsilon)} = \inf\{t > 0; Y(t) \notin D^{(\varepsilon)}\}.$$

Then if β is a finite stopping time such that $\beta \leq \tau^{(\varepsilon)}$ for some $\varepsilon > 0$ we have

$$\psi(y) = E^y\left[\int_0^{\beta} f(Y(t))\mathrm{d}t + \psi(Y(\beta))\right]. \tag{10.1.17}$$

Later we will also need the following stochastic control version, a proof of which can be found in [Ph1]. See also [Ph2], p. 40–41.

Lemma 10.4 (***Dynamic programming principle (II)***) *Let $Y(t) = U^{(u)}(t)$ be a controlled jump diffusion as in (5.1.1). Define*

$$\varphi(y) = \sup_{u \in \mathcal{A}} E^y \left[\int_0^{\tau_s} f(Y(t))dt + g(Y(\tau_s))\chi_{\{\tau_s < \infty\}} \right]. \tag{10.1.18}$$

Then

$$\varphi(y) = \sup_{u \in \mathcal{A}} E^y \left[\int_0^{\tau} f(Y(t))dt + \varphi(Y(\tau))\chi_{\{\tau_s < \infty\}} \right] \tag{10.1.19}$$

for all stopping times $\tau \leq \tau_s$.

Corollary 10.5 (a) *For each given $\tau \in \mathcal{T}$ the process*

$$U(t) := \int_0^{t \wedge \tau} f(Y(s))\mathrm{d}s + \psi(Y(t \wedge \tau)), \quad t \geq 0$$

is a supermartingale. *In particular, if τ_1, τ_2 are finite stopping times, $\tau_1 \leq \tau_2 \leq \tau_S$, then*

$$E^y[\psi(Y(\tau_1))] \geq E^y \left[\int_{\tau_1}^{\tau_2} f(Y(s))\mathrm{d}s + \psi(Y(\tau_2)) \right].$$

(b) *For $\varepsilon > 0$ let $\tau^{(\varepsilon)}$ be as in Lemma 10.3a and let β_1, β_2 be finite stopping times, $\beta_1 \leq \beta_2 \leq \tau^{(\varepsilon)}$. Then*

$$E^y[\psi(Y(\beta_1))] = E^y \left[\int_{\beta_1}^{\beta_2} f(Y(t))\mathrm{d}t + \psi(Y(\beta_2)) \right].$$

Proof of Corollary 10.5 (a) Define

$$R(t) = \left(Y(t \wedge \tau), \int_0^{t \wedge \tau} f(Y(r))\mathrm{d}r \right) \in \mathbb{R}^{k+1}$$

and put

$$H(y, u) = \psi(y) + u, \quad (y, u) \in \mathbb{R}^k \times \mathbb{R}.$$

Then if $t > s$ we have by the Markov property

$$E^y[U(t)|\mathcal{F}_s] = E^y[H(R(t))|\mathcal{F}_s] = E^{R(s)}[H(R(t-s))].$$

Then by (10.1.16) applied to $\beta = (t - s) \wedge \tau$ and $(y, u) = R(s)$

$$
\begin{aligned}
E^{R(s)}[H(R(t-s))] &= E^y\left[\psi(Y((t-s)\wedge\tau)) + u + \int_0^{(t-s)\wedge\tau} f(Y(r))\mathrm{d}r\right] \\
&= E^y\left[\psi(Y(\beta)) + u + \int_0^{\beta} f(Y(r))\mathrm{d}r\right] \\
&\le u + \sup_{\tau\ge\beta} E^y\left[\int_0^{\tau\wedge\beta} f(Y(r))\mathrm{d}r + \psi(Y(\tau\wedge\beta))\right] \\
&\le u + \psi(y) = H(R(s)) = U(s).
\end{aligned}
$$

Hence $E^y[U(t)|\mathcal{F}_s] \le U(s)$ for all $t > s$. This proves that $U(t)$ is a supermartingale. The second statement follows from Doob's optional sampling theorem. (b) By Lemma 10.3b we have

$$
\begin{aligned}
\psi(y) &= E^y\left[\int_0^{\beta_1} f(Y(t))\mathrm{d}t + \psi(Y(\beta_1))\right] \\
&\quad + E^y\left[\int_{\beta_1}^{\beta_2} f(Y(t))\mathrm{d}t + \psi(Y(\beta_2)) - \psi(Y(\beta_1))\right] \\
&= \psi(y) + E^y\left[\int_{\beta_1}^{\beta_2} f(Y(t))\mathrm{d}t + \psi(Y(\beta_2)) - \psi(Y(\beta_1))\right],
\end{aligned}
$$

from which (b) follows. □

Proof of Theorem 10.2 Choose $v_n = (\tau_1, \tau_2, \ldots, \tau_n; \zeta_1, \zeta_2, \ldots, \zeta_n)$ with $\tau_n \le \tau_{\mathcal{S}}$ and set $\tau_{n+1} = \tau_{\mathcal{S}}$. Let $Y(t) = Y^{(v_n)}(t)$. Then by Corollary 10.5a we have

$$
E^y[\varphi_{n-j}(Y(\tau_j))] \ge E^y\left[\int_{\tau_j}^{\tau_{j+1}} f(Y(t))\mathrm{d}t + \varphi_{n-j}(\check{Y}(\tau_{j+1}^-))\right]. \tag{10.1.20}
$$

Choosing $\tau = 0$ in (10.1.12) we obtain that

$$
\varphi_{n-j} \ge \mathcal{M}\varphi_{n-j-1} \quad \text{if } n-j \ge 1. \tag{10.1.21}
$$

Moreover, from the definition of $\mathcal{M}$ it follows that

$$
\mathcal{M}\varphi_{n-j-1}(\check{Y}(\tau_{j+1}^-)) \ge \varphi_{n-j-1}(Y(\tau_{j+1})) + K(\check{Y}(\tau_{j+1}^-), \zeta_{j+1}). \tag{10.1.22}
$$

Combining (10.1.20)–(10.1.22) we get

$$
\begin{aligned}
&E^y[\varphi_{n-j}(Y(\tau_j))] \\
&\ge E^y\left[\int_{\tau_j}^{\tau_{j+1}} f(Y(t))\mathrm{d}t + \varphi_{n-j-1}(Y(\tau_{j+1})) + K(\check{Y}(\tau_{j+1}^-), \zeta_{j+1})\right]
\end{aligned} \tag{10.1.23}
$$

for $j = 0, 1, \ldots, n-1$, where we have put $\tau_0 = 0$. Summing (10.1.23) from $j = 0$ to $j = n-1$ we get

$$\sum_{j=0}^{n-1} E^y[\varphi_{n-j}(Y(\tau_j)) - \varphi_{n-j-1}(Y(\tau_{j+1}))]$$
$$\geq E^y\left[\int_0^{\tau_n} f(Y(t))\mathrm{d}t + \sum_{j=1}^{n} K(\check{Y}(\tau_j^-), \zeta_j)\right]$$

or

$$E^y[\varphi_n(Y(0)) - \varphi_0(Y(\tau_n))] \geq E^y\left[\int_0^{\tau_n} f(Y(t))\mathrm{d}t + \sum_{j=1}^{n} K(\check{Y}(\tau_j^-), \zeta_j)\right]. \tag{10.1.24}$$

Now

$$E^y[\varphi_n(Y(0))] = \varphi_n(y) \tag{10.1.25}$$

and by the strong Markov property

$$E^y[\varphi_0(Y(\tau_n))] = E^y\left[E^{Y(\tau_n)}\left[\int_0^{\tau_S} f(Y(t))\mathrm{d}t + g(Y(\tau_S))\chi_{\{\tau_S<\infty\}}\right]\right]$$
$$= E^y\left[\int_{\tau_n}^{\tau_S} f(Y(t))\mathrm{d}t + g(Y(\tau_S))\chi_{\{\tau_S<\infty\}}\right]. \tag{10.1.26}$$

Combining (10.1.24)–(10.1.26) we obtain

$$\varphi_n(y) \geq E^y\left[\int_0^{\tau_S} f(Y(t))\mathrm{d}t + g(Y(\tau_S))\chi_{\{\tau_S<\infty\}} + \sum_{j=1}^{n} K(\check{Y}(\tau_j^-), \zeta_j)\right]$$
$$= J^{(v)}(y). \tag{10.1.27}$$

Since $v \in \mathcal{V}_n$ was arbitrary we conclude that

$$\varphi_n(y) \geq \Phi_n(y). \tag{10.1.28}$$

To get the opposite inequality choose $\varepsilon > 0$ and define an increasing sequence of stopping times $0 = \hat{\tau}_0 < \hat{\tau}_1 < \cdots < \hat{\tau}_n$ as follows.

For $j = 1, 2, \ldots, n$ let

$$D_j^{(\varepsilon)} = \{y;\, \varphi_j(y) > \mathcal{M}\varphi_{j-1}(y) + \varepsilon\}. \tag{10.1.29}$$

Define

$$\hat{\tau}_1 = \inf\{t > 0; Y^{(0)}(t) \notin D_n^{(\varepsilon)}\}, \tag{10.1.30}$$

where $Y^{(0)}(t) = Y(t)$ is the process without interventions. Then choose $\hat{\zeta}_1 = \bar{\zeta}_1(Y(\hat{\tau}_1^-))$, where $\bar{\zeta}_1 = \bar{\zeta}_1(y) \in \mathcal{Z}$ is ε-optimal for φ_{n-1}, in the sense that

$$\mathcal{M}\varphi_{n-1}(y) \le \varphi_{n-1}(\Gamma(y, \bar{\zeta}_1)) + K(y, \bar{\zeta}_1) + \varepsilon. \tag{10.1.31}$$

Inductively, if $0 = \hat{\tau}_0, \ldots, \hat{\tau}_j; \hat{\zeta}_1, \ldots, \hat{\zeta}_j$ have been chosen, where $j \le n-1$, we let $Y^{(j)}(t)$ be the process obtained by applying $\hat{v}_j = (\hat{\tau}_1, \ldots, \hat{\tau}_j; \hat{\zeta}_1, \ldots, \hat{\zeta}_j)$ to $Y(t)$. Define

$$\hat{\tau}_{j+1} = \inf\{t > \hat{\tau}_j; Y^{(j)}(t) \notin D_{n-j}^{(\varepsilon)}\} \tag{10.1.32}$$

and choose $\hat{\zeta}_{j+1} = \bar{\zeta}_{j+1}(Y^{(j)}(\hat{\tau}_{j+1}^-))$, where $\bar{\zeta}_{j+1} = \bar{\zeta}_{j+1}(y) \in \mathcal{Z}$ is ε-optimal for φ_{n-j-1}, in the sense that

$$\mathcal{M}\varphi_{n-j-1}(y) \le \varphi_{n-j-1}(\Gamma(y, \bar{\zeta}_{j+1})) + K(y, \bar{\zeta}_{j+1}) + \varepsilon. \tag{10.1.33}$$

Finally define

$$\hat{v} = (\hat{\tau}_1, \ldots, \hat{\tau}_n; \hat{\zeta}_1, \ldots, \hat{\zeta}_n) \in \mathcal{V}_n.$$

Now apply the argument (10.1.20)–(10.1.27) to $\hat{v}$.
By Corollary 10.5b we have

$$E^y[\varphi_{n-j}(Y(\hat{\tau}_j))] \le E^y\left[\int_{\hat{\tau}_j}^{\hat{\tau}_{j+1}} f(Y(t))\mathrm{d}t + \varphi_{n-j}(\check{Y}(\hat{\tau}_{j+1}^-))\right]. \tag{10.1.34}$$

Since $\check{Y}(\hat{\tau}_{j+1}^-) \notin D_{n-j}^{(\varepsilon)}$ we deduce that

$$\varphi_{n-j}(\check{Y}(\hat{\tau}_{j+1}^-)) \le \mathcal{M}\varphi_{n-j-1}(\check{Y}(\hat{\tau}_{j+1}^-)) + \varepsilon \tag{10.1.35}$$

and by (10.1.33), we get, with $Y = Y^{(\hat{v})}$,

$$\mathcal{M}\varphi_{n-j-1}(\check{Y}(\hat{\tau}_{j+1}^-)) \le \varphi_{n-j-1}(Y(\hat{\tau}_{j+1})) + K(\check{Y}(\hat{\tau}_{j+1}^-), \hat{\zeta}_{j+1}) + \varepsilon. \tag{10.1.36}$$

Combining (10.1.34)–(10.1.36) we obtain

$$\begin{aligned} &E^y[\varphi_{n-j}(Y(\hat{\tau}))] \\ &\quad \le E^y\left[\int_{\hat{\tau}_j}^{\hat{\tau}_{j+1}} f(Y(t))\mathrm{d}t + \varphi_{n-j-1}(Y(\hat{\tau}_{j+1})) + K(\check{Y}(\hat{\tau}_{j+1}^-), \hat{\zeta}_{j+1})\right] + 2\varepsilon. \end{aligned} \tag{10.1.37}$$

Now sum (10.1.37) from $j = 0$ to $j = n - 1$. The result is

$$\varphi_n(y) \le E^y \left[\int_0^{\hat{\tau}_n} f(Y(t))\mathrm{d}t + \varphi_0(Y(\hat{\tau}_n)) + \sum_{j=1}^{n} K(\check{Y}(\hat{\tau}_j^-), \hat{\zeta}_j) \right] + 2n\varepsilon.$$

Therefore by (10.1.26),

$$\begin{aligned}\varphi_n(y) &\le E^y \left[\int_0^{\hat{\tau}_{\mathcal{S}}} f(Y(t))\mathrm{d}t + g(Y(\hat{\tau}_{\mathcal{S}}))\chi_{\{\hat{\tau}_{\mathcal{S}}<\infty\}} + \sum_{j=1}^{n} K(\check{Y}(\hat{\tau}_j^-), \hat{\zeta}_j) \right] + 2n\varepsilon \\ &= J^{(\hat{v})}(y) + 2n\varepsilon.\end{aligned}$$

Since ε was arbitrary we deduce that

$$\varphi_n(y) \le \sup\{J^{(v)}(y);\, v \in \mathcal{V}_n\} = \Phi_n(y).$$

Combined with (10.1.28) this proves Theorem 10.2. □

Remark 10.6 Note that the proof of Theorem 10.2 actually also gives a $2n\varepsilon$-optimal impulse control $\hat{v} = (\hat{\tau}_1, \dots, \hat{\tau}_n; \hat{\zeta}_1, \dots, \hat{\zeta}_n)$. It is defined inductively by (10.1.29)–(10.1.33).

In particular, if it is possible to choose $\hat{\zeta}_j = \zeta_j^*$ to be optimal (i.e., (10.1.33) holds with $\varepsilon = 0$), then $v^* = (\hat{\tau}_1, \dots, \hat{\tau}_n; \zeta_1^*, \dots, \zeta_n^*)$ will be an *optimal impulse* control for Φ_n given by the following procedure.

For $j = 1, 2, \dots, n$ let

$$D_j = \{y;\, \varphi_j(y) > \mathcal{M}\varphi_{j-1}(y)\}. \tag{10.1.38}$$

Define

$$\hat{\tau}_1 = \inf\{t > 0;\, Y^{(0)}(t) \notin D_n\} \tag{10.1.39}$$

and

$$\hat{\zeta}_1 = \bar{\zeta}_1(Y^{(0)}(\hat{\tau}_1^-)), \tag{10.1.40}$$

where $\bar{\zeta}_1 = \bar{\zeta}_1(y)$ is a Borel measurable function such that

$$\mathcal{M}\varphi_{n-1}(y) = \varphi_{n-1}(\Gamma(y, \bar{\zeta}_1)) + K(y, \bar{\zeta}_1), \quad y \in \mathcal{S}. \tag{10.1.41}$$

Then if $(\hat{\tau}_1, \dots, \hat{\tau}_j; \hat{\zeta}_1, \dots, \hat{\zeta}_j)$ is defined, put

$$\hat{\tau}_{j+1} = \inf\{t > \hat{\tau}_j;\, Y^{(j)}(t) \notin D_{n-j}\} \tag{10.1.42}$$

and

$$\hat{\zeta}_{j+1} = \bar{\zeta}_{j+1}(Y^{(j)}(\hat{\tau}_{j+1}^{-})), \tag{10.1.43}$$

where $\bar{\zeta}_{j+1} = \bar{\zeta}_{j+1}(y)$ is a Borel measurable function such that

$$\mathcal{M}\varphi_{n-(j+1)}(y) = \varphi_{n-(j+1)}(\Gamma(y, \bar{\zeta}_{j+1})) + K(y, \bar{\zeta}_{j+1}), \quad y \in \mathcal{S},\ j+1 \le n. \tag{10.1.44}$$

As before $Y^{(j)}(t)$ denotes the result of applying the impulse control $\hat{v}_j = (\hat{\tau}_1, \ldots, \hat{\tau}_j; \hat{\zeta}_1, \ldots, \hat{\zeta}_j)$ to Y.

Corollary 10.7 *Assume that $g \ge 0$ and that $f, g, \mathcal{M}\varphi_n \in \mathcal{P}(\mathbb{R}^k)$ for $n = 0, 1, 2, \ldots$, where φ_n is as defined in (13.2.5) and (10.1.12). Then*

$$\varphi_n(y) \uparrow \Phi(y) \text{ as } n \to \infty, \quad \text{for all } y.$$

Corollary 10.8 *Suppose $g \ge 0$ and $f, g, \mathcal{M}\varphi_n \in \mathcal{P}(\mathbb{R}^k)$ for $n = 0, 1, 2, \ldots$ Then Φ is a solution of the following nonlinear optimal stopping problem*

$$\Phi(y) = \sup_{\tau \in \mathcal{T}} E^y \left[\int_0^\tau f(Y(t))\mathrm{d}t + \mathcal{M}\Phi(Y(\tau))\chi_{\{\tau<\infty\}} \right]. \tag{10.1.45}$$

Proof Let $\{\varphi_n\}_{n=0}^{\infty}$ be as in (13.2.5) and (10.1.12). Then $\varphi_n \uparrow \Phi$ as $n \to \infty$, by Corollary 10.7. Therefore

$$\varphi_n(y) \le \sup_{\tau \in \mathcal{T}} E^y \left[\int_0^\tau f(Y(t))\mathrm{d}t + \mathcal{M}\Phi(Y(\tau))\chi_{\{\tau<\infty\}} \right]$$

for all n and hence

$$\Phi(y) \le \sup_{\tau \in \mathcal{T}} E^y \left[\int_0^T f(Y(t))\mathrm{d}t + \mathcal{M}\Phi(Y(\tau))\chi_{\{\tau<\infty\}} \right].$$

To get the opposite inequality, choose $\varepsilon > 0$ and let $\hat{\tau} \in \mathcal{T}$ be a stopping time such that

$$E^y \left[\int_0^{\hat{\tau}} f(Y(t))\mathrm{d}t + \mathcal{M}\Phi(Y(\hat{\tau}))\chi_{\{\hat{\tau}<\infty\}} \right] \ge \sup_{\tau \in \mathcal{T}} E^y \left[\int_0^\tau f(Y(t))\mathrm{d}t + \mathcal{M}\Phi(Y(\tau))\chi_{\{\tau<\infty\}} \right] - \varepsilon.$$

Then by monotone convergence

$$\begin{aligned}\Phi(y) &= \lim_{n\to\infty} \varphi_n(y) \geq \lim_{n\to\infty} E^y\left[\int_0^{\hat{\tau}} f(Y(t))\mathrm{d}t + \mathcal{M}\varphi_{n-1}(Y(\hat{\tau}))\chi_{\{\hat{\tau}<\infty\}}\right] \\ &= E^y\left[\int_0^{\hat{\tau}} f(Y(t))\mathrm{d}t + \mathcal{M}\Phi(Y(\hat{\tau}))\chi_{\{\hat{\tau}<\infty\}}\right] \\ &\geq \sup_{\tau\in\mathcal{T}} E^y\left[\int_0^{\tau} f(Y(t))\mathrm{d}t + \mathcal{M}\Phi(Y(\tau))\chi_{\{\tau<\infty\}}\right] - \varepsilon.\end{aligned} \tag{10.1.46}$$

Combining the results above we get the following general picture of the optimal impulse control $(\hat{\tau}_1, \ldots, \hat{\tau}_n; \hat{\zeta}_1, \ldots, \hat{\zeta}_n)$ for Φ_n (see (13.2.5), (10.1.12), and (10.1.38)–(10.1.44)). Define

$$D = \{y;\, \Phi(y) > \mathcal{M}\Phi(y)\} \tag{10.1.47}$$

and

$$D_n = \{y;\, \varphi_n(y) > \mathcal{M}\varphi_{n-1}(y)\}, \quad n = 1, 2, \ldots \tag{10.1.48}$$

Make the first intervention the first time $t = \hat{\tau}_1$ that $Y(t) \notin D_n$. Then give the system the impulse $\hat{\zeta}_1$ according to (10.1.40). Now we have only $n-1$ interventions left, so we wait until $Y(t)$ exits from the set D_{n-1} before making the next intervention, and so on. The last intervention time $\hat{\tau}_n$ is the first time after $\hat{\tau}_{n-1}$ that $Y(t) \notin D_1$. See Fig. 10.1.

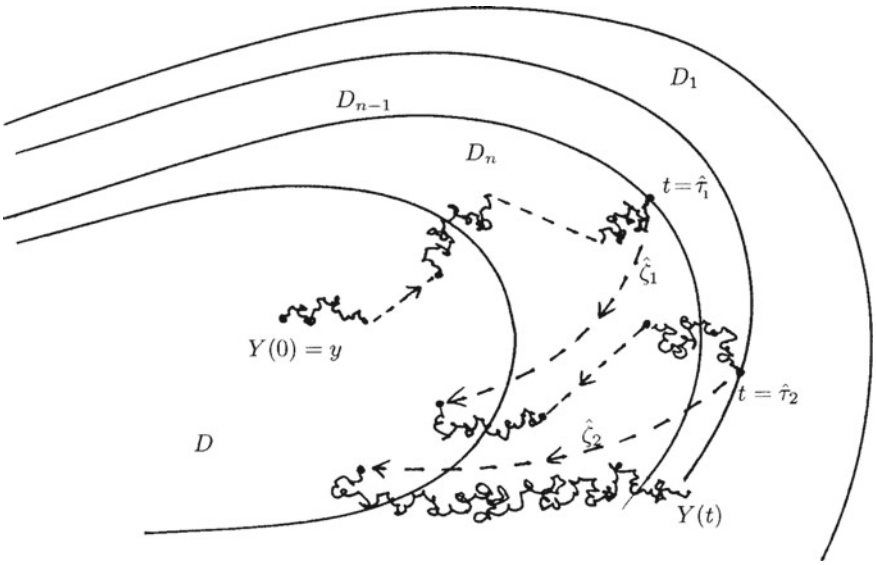


Fig. 10.1 Optimal impulse control for Φ_n

10.2 Examples

Example 10.9 Consider the impulse control problems

$$\Phi_n(s,x) = \inf_{v\in\mathcal{V}_n} J^{(v)}(s,x) = J^{(v_n^*)}(s,x), \quad n = 1,2,\dots, \tag{10.2.1}$$

$$\Phi(s,x) = \inf_{v\in\mathcal{V}} J^{(v)}(s,x) = J^{(v^*)}(s,x), \tag{10.2.2}$$

where

$$J^{(v)}(s,x) = E^x\left[\int_0^\infty \mathrm{e}^{-\rho(s+t)}(X^{(v)}(t))^2\mathrm{d}t + c\sum_j \mathrm{e}^{-\rho(s+\tau_j)}\right],$$

$c > 0$ and $\rho > 0$ are constants, and

$$X^{(v)}(t) = x + B(t) + \int_0^t\int_{\mathbb{R}} z\tilde{N}(\mathrm{d}s,\mathrm{d}z) - \sum_j \zeta_j\chi_{\{\tau_j\le t\}}, \quad B(0) = 0,$$

when

$$v = (\tau_1,\tau_2,\dots,\tau_n;\zeta_1,\zeta_2,\dots,\zeta_n) \in \mathcal{V}_n$$

or

$$v = (\tau_1,\tau_2,\dots;\zeta_1,\zeta_2,\dots) \in \mathcal{V}, \quad \zeta_j > 0.$$

Put

$$X(t) = x + B(t) + \int_0^t\int_{\mathbb{R}} z\tilde{N}(\mathrm{d}s,\mathrm{d}z), \quad t \ge 0.$$

This is a finite number of interventions version of Example 9.5, except that we have for simplicity put $\lambda = 0$. We assume that $z \le 0$ a.s. with respect to ν.

First, let us find $\Psi_n(x) := \Phi_n(0,x)$ by using the iterative procedure (13.2.25) and (10.1.12).

In this case we get from (13.2.25) (see (9.2.29))

$$\Psi_0(x) = E^x\left[\int_0^\infty \mathrm{e}^{-\rho t}\left(x + B(t) + \int_0^t\int_{\mathbb{R}} z\tilde{N}(\mathrm{d}s,\mathrm{d}z)\right)^2\mathrm{d}t\right] = \frac{1}{\rho}x^2 + \frac{b}{\rho^2}, \tag{10.2.3}$$

where $b = 1 + \int_{\mathbb{R}} z^2\nu(\mathrm{d}z)$. Hence

$$\mathcal{M}\Psi_0(x) = \inf\{\Psi_0(x-\zeta) + c;\zeta > 0\} = \Psi_0(x\wedge 0) + c = \frac{1}{\rho}(x\wedge 0)^2 + \frac{b}{\rho^2} + c. \tag{10.2.4}$$

Therefore, by (10.1.12)

$$\Psi_1(x) = \inf_{\tau \geq 0} E^x \left[\int_0^\tau \mathrm{e}^{-\rho t} X^2(t) \mathrm{d}t + \mathrm{e}^{-\rho\tau} \left(\frac{1}{\rho} (X(\tau) \wedge 0)^2 + \frac{b}{\rho^2} + c \right) \right]. \tag{10.2.5}$$

To solve this optimal stopping problem we consider the three basic associated variational inequalities

$$-\rho\psi_1(x) + \frac{1}{2}\psi_1''(x) + \int_{\mathbb{R}} \{\psi_1(x+z) - \psi_1(x) - z\psi_1'(x)\}\nu(\mathrm{d}z) + x^2 \geq 0 \quad \text{for all } x, \tag{10.2.6}$$

$$\psi_1(x) \leq \frac{1}{\rho}(x \wedge 0)^2 + \frac{b}{\rho^2} + c \quad \text{for all } x, \tag{10.2.7}$$

$$-\rho\psi_1(x) + \frac{1}{2}\psi_1''(x) + \int_{\mathbb{R}} \{\psi_1(x+z) - \psi_1(x) - z\psi_1'(x)\}\nu(\mathrm{d}z) + x^2 = 0$$

$$\text{on } D_1 := \left\{ x; \psi_1(x) < \frac{1}{\rho}(x \wedge 0)^2 + \frac{b}{\rho^2} + c \right\}.$$

If we guess that D_1 has the form

$$D_1 = (-\infty, x_1^*) \quad \text{for some } x_1^* > 0$$

then we are led to the following candidate for ψ_1 (see Example 9.5)

$$\psi_1(x) = \begin{cases} \dfrac{1}{\rho}x^2 + \dfrac{b}{\rho^2} - a_1 \exp(rx), & x < x_1^*, \\ \dfrac{b}{\rho^2} + c, & x \geq x_1^*, \end{cases} \tag{10.2.8}$$

where a_1 is a constant to determine and $r = r_1 > 0$ is a root of the equation

$$K(r) := \int_{\mathbb{R}} \{\mathrm{e}^{rz} - 1 - rz\}\nu(\mathrm{d}z) + \frac{1}{2}r^2 - \rho = 0.$$

See Fig. 10.2.

If we require ψ_1 to be continuous and differentiable at $x = x_1^*$, we get the following two equations to determine x_1^* and a_1:

$$\frac{1}{\rho}(x_1^*)^2 + \frac{b}{\rho^2} - a_1 \exp(rx_1^*) = \frac{b}{\rho^2} + c, \tag{10.2.9}$$

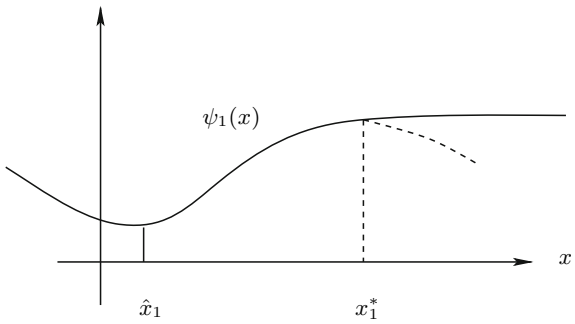


Fig. 10.2 The graph of $\psi_1(x)$

$$\frac{2}{\rho}x_1^* - a_1 r \exp(r x_1^*) = 0. \tag{10.2.10}$$

Combining these two equations we get

$$x_1^* = \frac{1}{r}\left(1 + \sqrt{1 + c\rho r^2}\right) \tag{10.2.11}$$

and

$$a_1 = \frac{2x_1^*}{\rho r}\exp(-r x_1^*). \tag{10.2.12}$$

If we choose these values of x_1^* and a_1, we can now verify that, under some additional assumptions on ν, the candidate ψ_1 given by (10.2.8) satisfies all the conditions of the verification theorem and we conclude that

$$\psi_1 = \Psi_1.$$

We now repeat the procedure to find Ψ_2.
First note that

$$\begin{aligned}\mathcal{M}\Psi_1(x) &= \inf\{\Psi_1(x-\zeta) + c; \zeta > 0\} = \Psi_1(x \wedge \hat{x}_1) + c \\ &= \frac{1}{\rho}(x \wedge \hat{x}_1)^2 + \frac{b}{\rho^2} + c - a_1 \exp(r(x \wedge \hat{x}_1)),\end{aligned} \tag{10.2.13}$$

where

$$\hat{x}_1 \in \mathrm{Argmin}\Psi_1(x).$$

Hence

$$\Psi_2(x) = \inf_{\tau \geq 0} E^x\left[\int_0^\tau \mathrm{e}^{-\rho t} X^2(t)\mathrm{d}t\right.$$

$$+ \mathrm{e}^{-\rho\tau}\left(\frac{1}{\rho}(X(\tau)\wedge\hat{x}_1)^2+\frac{b}{\rho^2}+c-a_1\exp(r(X(\tau)\wedge\hat{x}_1))\right)\Bigg].$$

The same procedure as above leads us to the candidate

$$\psi_2(x)=\begin{cases}\dfrac{1}{\rho}x^2+\dfrac{b}{\rho^2}-a_2\exp(rx), & x<x_2^*,\\ \dfrac{1}{\rho}\hat{x}_1^2+\dfrac{b}{\rho^2}+c-a_1\exp(r\hat{x}_1), & x\geq x_2^*,\end{cases} \tag{10.2.14}$$

where x_2^* and a_2 are given by

$$x_2^*=\frac{1}{r}\left(1+\sqrt{1+r^2(\hat{x}_1^2+\rho(c-a_1\exp(r\hat{x}_1)))}\right), \tag{10.2.15}$$

$$a_2=\frac{2x_2^*}{\rho r}\exp(-rx_2^*). \tag{10.2.16}$$

If we choose these values of x_2^* and $a_2>0$ we can verify that the candidate ψ_2 given by (10.2.14) coincides with Ψ_2. Note that $x_2^*<x_1^*$ and $a_2>a_1$.

Continuing this inductively we get a sequence of functions ψ_n of the form

$$\psi_n(x)=\begin{cases}\dfrac{1}{\rho}x^2+\dfrac{b}{\rho^2}-a_n\exp(rx), & x<x_n^*,\\ \dfrac{1}{\rho}\hat{x}_{n-1}^2+\dfrac{b}{\rho^2}+c-a_{n-1}\exp(r\hat{x}_{n-1}), & x\geq x_n^*,\end{cases} \tag{10.2.17}$$

where x_n^* and a_n are given by

$$x_n^*=\frac{1}{r}\left(1+\sqrt{1+r^2(\hat{x}_{n-1}^2+\rho(c-a_{n-1}\exp(r\hat{x}_{n-1})))}\right) \tag{10.2.18}$$

with

$$\hat{x}_{n-1}\in\operatorname{Argmin}\Psi_{n-1}(x),$$

and

$$a_n=\frac{2x_n^*}{\rho r}\exp(-rx_n^*). \tag{10.2.19}$$

We find that $x_n^*<x_{n-1}^*$ and $a_n>a_{n-1}$. Using the verification theorem we conclude that $\psi_n=\Psi_n$. In the limiting case when there is no bound on the number of interventions the corresponding value function will be

$$\Psi(x) = \begin{cases} \frac{1}{\rho}x^2 + \frac{b}{\rho^2} - a\exp(rx), & x < x^*, \\ \frac{1}{\rho}\hat{x}^2 + \frac{b}{\rho^2} + c - a\exp(r\hat{x}), & x \geq x^*, \end{cases} \tag{10.2.20}$$

where x^* and a solve the coupled system of equations

$$x^* = \frac{1}{r}\Big(1+\sqrt{1 + r^2(\hat{x}^2 + \rho(c - a\exp(r\hat{x})))}\Big), \tag{10.2.21}$$

$$a = \frac{2x^*}{\rho r}\exp(-rx^*), \tag{10.2.22}$$

where

$$\hat{x} \in \mathrm{Argmin}\Psi(x).$$

It is easy to see that this system has a unique solution $x^* > 0$, $a > 0$, and that $x^* < x_n^*$ and $a > a_n$ for all n. The situation in the case $n = 3$ is shown in Fig. 10.3. Note that with only three interventions allowed the optimal strategy is first to wait until the first time τ_1^* when $X(t) \geq x_3^*$, then move $X(t)$ down to $\hat{x}_2$, next wait until the first time $\tau_2^* > \tau_1^*$ when $X(t) \geq x_2^*$, then move $X(t)$ down to $\hat{x}_1$ and finally wait until the first time $t = \tau_3^* > \tau_2^*$ when $X(t) \geq x_1^*$ before making the last intervention, which is moving $X(t)$ down to $\hat{x}_0 = 0$.

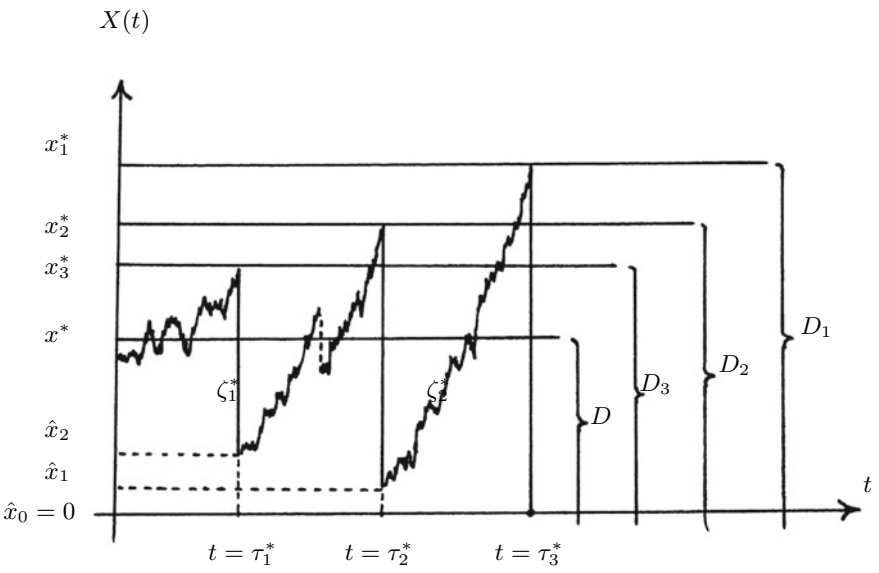


Fig. 10.3 The optimal impulse control for Ψ_3 (Example 10.9)

10.3 Exercises

Exercise* 10.1 (***Optimal Forest Management Revisited***) Using the notation of Exercise 9.3 let

$$\Phi(x) = \sup\left\{J^{(v)}(x);\, v = (\tau_1, \tau_2, \ldots)\right\}$$

be the value function when there are no restrictions on the number of interventions. For $n = 1, 2, \ldots$ let

$$\Phi_n(x) = \sup\left\{J^{(v)}(x);\, v = (\tau_1, \tau_2, \ldots, \tau_k);\, k \le n\right\}$$

be the value function when up to n interventions are allowed. Use Theorem 10.2 to find Φ_1 and Φ_2.

Exercise* 10.2 (***Optimal Stream of Dividends with Transaction Costs from a Geometric Lévy Process***) This is an addition to Exercise 9.2. Suppose that we at times $0 \le \tau_1 \le \tau_2 \le \cdots$ decide to take out dividends of sizes $\zeta_1, \zeta_2, \ldots \in (0, \infty)$ from an economic quantity growing like a geometric Lévy process. If we let $X^{(v)}(t)$ denote the size at time t of this quantity when the dividend policy $v = (\tau_1, \tau_2, \ldots; \zeta_1, \zeta_2, \ldots)$ is applied, we assume that $X^{(v)}(t)$ is described by (see (9.2.4))

$$\begin{aligned}
&\mathrm{d}X^{(v)}(t) = \\
&\qquad \mu X^{(v)}(t)\mathrm{d}t + \sigma X^{(v)}(t)\mathrm{d}B(t) + \theta X^{(v)}(t^-)\int_{\mathbb{R}} z\tilde{N}(\mathrm{d}t, \mathrm{d}z), \quad \tau_i \le t < \tau_{i+1}, \\
&X^{(v)}(\tau_{i+1}) = \breve{X}^{(v)}(\tau_{i+1}^-) - (1+\lambda)\zeta_{i+1} - c, \quad i = 0, 1, 2, \ldots \quad \text{(see (9.1.5))},
\end{aligned}$$

where $\mu, \sigma \ne 0$, $\theta, \lambda \ge 0$, and $c > 0$ are constants, $-1 \le \theta z \le 0$, a.s. (ν). Let

$$\Phi(x) = \sup\left\{J^{(v)}(x);\, v = (\tau_1, \tau_2, \ldots; \zeta_1, \zeta_2, \ldots)\right\}$$

and

$$\Phi_n(x) = \sup\left\{J^{(v)}(x);\, v = (\tau_1, \tau_2, \ldots, \tau_k; \zeta_1, \zeta_2, \ldots, \zeta_k);\, k \le n\right\}$$

be the value function with no restrictions on the number of interventions and with at most n interventions, respectively, where

$$J^{(v)}(x) = E^x\left[\sum_{\tau_k < \tau_{\mathcal{S}}} \mathrm{e}^{-\rho\tau_k}\zeta_k\right] \quad (\rho > 0 \text{ constant})$$

is the expected total discounted dividend and

$$\tau_{\mathcal{S}} = \inf\left\{t > 0;\, X^{(v)}(t) \le 0\right\}$$

is the time of bankruptcy. Show that

$$\Phi(x) = \Phi_n(x) = \Phi_1(x) \quad \text{for all } n.$$

Thus in this case we achieve the optimal result with just one intervention.

Chapter 11
Combined Stochastic Control and Impulse Control of Jump Diffusions



11.1 A Verification Theorem

Consider the general situation in Chap. 9, except that now we assume that we, in addition, are free at any state $y \in \mathbb{R}^k$ to choose a Markov control $u(y) \in U$, where U is a given closed convex set in $\mathbb{R}^p$. Let $\mathcal{U}$ be a given family of such Markov controls u. If, as before, $v = (\tau_1, \tau_2, \ldots; \zeta_1, \zeta_2, \ldots) \in \mathcal{V}$ denotes a given impulse control we call $w := (u, v)$ a *combined control*. If $w = (u, v)$ is applied, we assume that the corresponding state $Y(t) = Y^{(w)}(t)$ at time t is given by (see (5.1.1) and (9.1.2)–(9.1.5))

$$\begin{aligned} Y(0^-) &= y, \\ \mathrm{d}Y(t) &= b(Y(t), u(t))\mathrm{d}t + \sigma(Y(t), u(t))\mathrm{d}B(t) \\ &\quad + \int_{\mathbb{R}^\ell} \gamma(Y(t^-), u(t^-), z)\tilde{N}(\mathrm{d}t, \mathrm{d}z), \quad \tau_j < t < \tau_{j+1}, \end{aligned} \tag{11.1.1}$$

$$Y(\tau_{j+1}) = \Gamma(\breve{Y}(\tau_{j+1}^-), \zeta_{j+1}), \quad j = 0, 1, 2, \ldots \tag{11.1.2}$$

where $u(t) = u(Y(t))$ and $b : \mathbb{R}^k \times U \to \mathbb{R}^k$, $\sigma : \mathbb{R}^k \times U \to \mathbb{R}^{k\times m}$, and $\gamma : \mathbb{R}^k \times U \times \mathbb{R}^\ell \to \mathbb{R}^{k\times\ell}$ are given continuous functions, $\tau_0 = 0$.

As before we let our "universe" $\mathcal{S}$ be a fixed Borel set in $\mathbb{R}^k$ such that $\mathcal{S} \subset \overline{\mathcal{S}^0}$ and we define

$$\tau^* = \lim_{R\to\infty} \inf\{t > 0; |Y^{(w)}(t)| \geq R\} \leq \infty \tag{11.1.3}$$

and

$$\tau_{\mathcal{S}} = \inf\{t \in (0, \tau^*(\omega)); Y^{(w)}(t, \omega) \notin \mathcal{S}\}. \tag{11.1.4}$$

If $Y^{(w)}(t, \omega) \in \mathcal{S}$ for all $t < \tau^*$ we put $\tau_{\mathcal{S}} = \tau^*$.

We assume that we are given a set $\mathcal{W} = \mathcal{U} \times \mathcal{V}$ of *admissible* combined controls $w = (u, v)$ which is included in the set of combined controls $w = (u, v)$ such that a unique strong solution $Y^{(w)}(t)$ of (11.1.1) and (11.1.2) exists and

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$$\tau^* = \infty \quad \text{and} \quad \lim_{j\to\infty} \tau_j = \tau_{\mathcal{S}} \text{ a.s. } Q^y \text{ for all } y \in \mathbb{R}^k. \tag{11.1.5}$$

Define the *performance* or *total expected profit/utility* of $w = (u, v) \in \mathcal{W}$, $v = (\tau_1, \tau_2, \ldots; \zeta_1, \zeta_2, \ldots)$, by

$$J^{(w)}(y) = E^y\Big[\int_0^{\tau_S} f(Y^{(w)}(t), u(t))\mathrm{d}t + g(Y^{(w)}(\tau_S))\mathcal{X}_{\{\tau_S<\infty\}} + \sum_{\tau_j\le\tau_S} K(\check{Y}^{(w)}(\tau_j^-), \zeta_j)\Big], \tag{11.1.6}$$

where $f : \mathcal{S} \times U \to \mathbb{R}$, $g : \mathbb{R}^k \to \mathbb{R}$, and $K : \bar{\mathcal{S}} \times \mathcal{Z} \to \mathbb{R}$ are given functions satisfying the conditions similar to (9.1.11)–(9.1.13).

The *combined stochastic control and impulse control problem* is the following.

Find the value function $\Phi(y)$ and an optimal control $w^* = (u^*, v^*) \in \mathcal{W}$ such that

$$\Phi(y) = \sup\{J^{(w)}(y);\, w \in \mathcal{W}\} = J^{(w^*)}(y). \tag{11.1.7}$$

We now state a verification theorem for this problem. It is a combination of the HJB equation of control theory and the QVI for impulse control.

Define

$$L^\alpha h(y) = \sum_{i=1}^k b_i(y,\alpha)\frac{\partial h}{\partial y_i} + \frac{1}{2}\sum_{i,j=1}^k (\sigma\sigma^T)_{ij}(y,\alpha)\frac{\partial^2 h}{\partial y_i \partial y_j} + \int_{\mathbb{R}} \sum_{j=1}^{\ell} \{h(y + \gamma^{(j)}(y,\alpha,z)) - h(y) - \nabla h(y).\gamma^{(j)}(y,\alpha,z)\}\nu_j(\mathrm{d}z_j) \tag{11.1.8}$$

for each $\alpha \in U$ and for each twice differentiable function h. This is the generator of $Y^{(w)}(t)$ if we apply the constant control α and no (impulse) interventions.

As in Chap. 9 we let

$$\mathcal{M}h(y) = \sup\{h(\Gamma(y,\zeta)) + K(y,\zeta);\, \zeta \in \mathcal{Z} \quad \text{and} \quad \Gamma(y,\zeta) \in \mathcal{S}\} \tag{11.1.9}$$

be the intervention operator.

Then the verification theorem is the following (compare with Theorems 5.1 and 9.2).

***Theorem 11.1* (*HJBQVI Verification Theorem for Combined Stochastic and Impulse Control*)**

(a) *Suppose we can find a function* $\phi : \bar{\mathcal{S}} \to \mathbb{R}$ *such that*

(i) $\phi \in C^1(\mathcal{S}^0) \cap C(\bar{\mathcal{S}})$,
(ii) $\phi \ge \mathcal{M}\phi$ *on* $\mathcal{S}$.

Define

$$D = \{y \in \mathcal{S};\, \phi(y) > \mathcal{M}\phi(y)\} \quad \textit{(the continuation region)}. \tag{11.1.10}$$

Suppose that $Y^{(w)}(t)$ spends 0 time on ∂D a.s., i.e.,

(iii) $E^y\left[\int_0^{\tau_{\mathcal{S}}} \mathcal{X}_{\partial D}(Y^{(w)}(t))\mathrm{d}t\right] = 0$ *for all* $y \in \mathcal{S}$, $w \in \mathcal{W}$

and suppose that

(iv) ∂D *is a Lipschitz surface,*

(v) $\phi \in C^2(S^0 \setminus \partial D)$ *and the second-order derivatives of* ϕ *are locally bounded near* ∂D,

(vi) $L^{\alpha}\phi(y) + f(y, \alpha) \le 0$ *for all* $\alpha \in U$, $y \in S^0 \setminus \partial D$,

(vii) $\phi(y) = g(y)$ *fort all* $y \notin \mathcal{S}$,

(viii) *the family* $\{\phi^-(Y^{(w)}(\tau));\, \tau \in \mathcal{T}\}$ *is uniformly* Q^y*-integrable for all* $y \in \mathcal{S}$, $w \in \mathcal{W}$,

(ix) $E^y\left[|\phi(Y(\tau))| + \int_0^{\tau_{\mathcal{S}}} |A\phi(Y(t))|\mathrm{d}t\right] < \infty$ *for all* $\tau \in \mathcal{T}$, $w \in \mathcal{W}$, $y \in \mathcal{S}$.

Then

$$\phi(y) \ge \Phi(y) \quad \textit{for all} \quad y \in \mathcal{S}.$$

(b) *Suppose in addition that*

(x) *there exists a function* $\hat{u} : \mathcal{S} \to \mathbb{R}$ *such that*

$$L^{\hat{u}(y)}\phi(y) + f(y, \hat{u}(y)) = 0 \quad \textit{for all} \quad y \in D$$

and

(xi)

$$\bar{\zeta}(y) \in \textit{argmax}\{\phi(\Gamma(y, \cdot)) + K(y, \cdot)\}$$

exists for all $y \in \mathcal{S}$ and $\bar{\zeta}(\cdot)$ is a Borel measurable selection.
Define an impulse control $\hat{v} = (\hat{\tau}_1, \hat{\tau}_2, \ldots; \hat{\zeta}_1, \hat{\zeta}_2, \ldots)$ *as follows.*
Put $\hat{\tau}_0 = 0$ *and inductively*

$$\hat{\tau}_{k+1} = \inf\{t > \hat{\tau}_k;\, Y^{(k)}(t) \notin D\},$$
$$\hat{\zeta}_{k+1} = \bar{\zeta}(Y^{(k)}(\hat{\tau}_{k+1}^-)), \quad k = 0, 1, \ldots,$$

where $Y^{(k)}(t)$ *is the result of applying the combined control*

$$\widehat{w}_k := (\hat{u}, (\hat{\tau}_1, \ldots, \hat{\tau}_k; \hat{\zeta}_1, \ldots, \hat{\zeta})).$$

Put $\widehat{w} = (\hat{u}, \hat{v})$. *Suppose* $\widehat{w} \in \mathcal{W}$ *and that*

(xii) $\{\phi(Y^{(\widehat{w})}(\tau)); \tau \in \mathcal{T}, \tau \leq \tau_D\}$ *is* Q^y*-uniformly integrable for all* $y \in \mathcal{S}$. *Then*

$$\phi(y) = \Phi(y) \quad \textit{for all} \quad y \in \mathcal{S}$$

and

$$\widehat{w} \in \mathcal{W} \quad \textit{is an optimal combined control.}$$

Proof The proof is similar to the proof of Theorem 9.2 and is omitted. □

11.2 Examples

Example 11.2 (***Optimal Combined Control of the Exchange Rate***) This example is a simplification of a model studied in [MuØ].

Suppose a government has two means of influencing the foreign exchange rate of its own currency:

1. At all times the government can choose the domestic interest rate r.
2. At times selected by the government it can intervene in the foreign exchange market by buying or selling large amounts of foreign currency.

Let $r(t)$ denote the interest rate chosen and let $\tau_1, \tau_2, \ldots$ be the (stopping) times when it is decided to intervene, with corresponding amounts $\zeta_1, \zeta_2, \ldots$ If $\zeta > 0$ the government buys foreign currency, if $\zeta < 0$ it sells. Let $v = (\tau_1, \tau_2, \ldots; \zeta_1, \zeta_2, \ldots)$ be corresponding impulse control.

If the combined control $w = (r, v)$ is applied, we assume that the corresponding exchange rate $X(t)$ (measured in the number of domestic monetary units it takes to buy one average foreign monetary unit) is given by

$$X(t) = x - \int_0^t F(r(s) - \bar{r}(s))\mathrm{d}s + \sigma B(t) + \sum_{j:\tau_j \leq t} \gamma(\zeta_j), \quad t \geq 0, \tag{11.2.1}$$

where $\sigma > 0$ is a constant, $\bar{r}(s)$ is the (average) foreign interest rate, and $F : \mathbb{R} \to \mathbb{R}$ and $\gamma : \mathbb{R} \to \mathbb{R}$ are known functions which give the effects on the exchange rate by the interest rate differential $r(s) - \bar{r}(s)$ and the amount ζ_j, respectively.

The total expected cost of applying the combined control $w = (r, v)$ is assumed to be of the form

$$J^{(w)}(s, x) = E^x\left[\int_s^T \mathrm{e}^{-\rho t}\{M(X(t) - m) + N(r(t) - \bar{r}(t))\}\mathrm{d}t + \sum_{j;\tau_j \leq T} L(\zeta_j)\mathrm{e}^{-\rho\tau_j}\right], \tag{11.2.2}$$

where $M(X(t) - m)$ and $N(r(t) - \bar{r}(t))$ give the costs incurred by the difference $X(t) - m$ between $X(t)$ and an optimal value m, and by the difference $r(t) - \bar{r}(t)$ between the domestic and the average foreign interest rate $\bar{r}(t)$, respectively. The

cost of buying/selling the amount ζ_j is $L(\zeta_j)$ and $\rho > 0$ is a constant discounting exponent.

The problem is to find $\Phi(s,x)$ and $w^* = (r^*, v^*)$ such that

$$\Phi(s,x) = \inf_w J^{(w)}(s,x) = J^{(w^*)}(s,x). \tag{11.2.3}$$

Since this is a minimum problem, the corresponding HJBQVIs of Theorem 11.1 are changed to minima also and they get the form

$$\min\left(\inf_{r\in\mathbb{R}}\left\{e^{-\rho s}\big(M(x-m)+N(r-\bar{r}(s))\big)+\frac{\partial\phi}{\partial s}-F(r-\bar{r}(s))\frac{\partial\phi}{\partial x}+\frac{1}{2}\sigma^2\frac{\partial^2\phi}{\partial x^2}\right\}, \mathcal{M}\phi(s,x)-\phi(s,x)\right)=0, \tag{11.2.4}$$

where

$$\mathcal{M}\phi(s,x) = \inf_{\zeta\in\mathbb{R}}\{\phi(s,x+\gamma(\zeta))+e^{-\rho s}L(\zeta)\}. \tag{11.2.5}$$

In general this is difficult to solve for ϕ, even for simple choices of the functions M, N, and F. A detailed discussion on a special case can be found in [MuØ].

Example 11.3 (***Optimal Consumption and Portfolio with Both Fixed and Proportional Transaction Costs (1)***) This application is studied in [ØS].

Suppose there are two investment possibilities, say a bank account and a stock. Let $X_1(t)$ and $X_2(t)$ denote the amount of money invested in these two assets, respectively, at time t. In the absence of consumption and transactions suppose that

$$dX_1(t) = rX_1(t)dt \tag{11.2.6}$$

and

$$dX_2(t) = X_2(t^-)\left(\mu\,dt+\sigma\,dB(t)+\theta\int_{\mathbb{R}} z\tilde{N}(dt,dz)\right), \tag{11.2.7}$$

where r, μ, θ, and $\sigma \neq 0$ are constants and

$$\mu > r > 0, \quad \theta z \geq -1 \text{ a.s. } \nu. \tag{11.2.8}$$

Suppose that at any time t the investor is free to choose a *consumption rate* $u(t) \geq 0$. This consumption is automatically drawn from the bank account holding with no extra costs. In addition the investor may at any time transfer money from the bank to the stock and conversely. Suppose that such a transaction of size ζ incurs a transaction cost given by

$$c + \lambda|\zeta|, \tag{11.2.9}$$

where $c > 0$ and $\lambda \in [0, 1)$ are constants. (If $\zeta > 0$ we buy stocks and if $\zeta < 0$ we sell stocks.) Thus the control of the investor consists of a combination of a stochastic control $u(t)$ and an impulse control $v = (\tau_1, \tau_2, \ldots; \zeta_1, \zeta_2, \ldots)$, where $\tau_1, \tau_2, \ldots$ are the chosen transaction times and $\zeta_1, \zeta_2, \ldots$ are corresponding transaction amounts.

If such a combined control $w = (u, v)$ is applied, the corresponding system $(X_1(t), X_2(t)) = (X_1^{(w)}(t), X_2^{(w)}(t))$ gets the form

$$\mathrm{d}X_1(t) = (rX_1(t) - u(t))\mathrm{d}t, \quad \tau_i < t < \tau_{i+1}, \tag{11.2.10}$$

$$\mathrm{d}X_2(t) = X_2(t^-)\left(\mu\,\mathrm{d}t + \sigma\,\mathrm{d}B(t) + \theta\int_{\mathbb{R}} z\tilde{N}(\mathrm{d}t, \mathrm{d}z)\right), \quad \tau_i < t < \tau_{i+1}, \tag{11.2.11}$$

$$X_1(\tau_{i+1}) = X_1(\tau_{i+1}^-) - \zeta_{i+1} - c - \lambda|\zeta_{i+1}|, \tag{11.2.12}$$

$$X_2(\tau_{i+1}) = \breve{X}_2(\tau_{i+1}^-) + \zeta_{i+1}. \tag{11.2.13}$$

If we do not allow any negative amounts held in the bank account or in the stock, the *solvency region* $\mathcal{S}$ is given by

$$\mathcal{S} = [0, \infty) \times [0, \infty). \tag{11.2.14}$$

We call $w = (u, v)$ *admissible* if $X^{(w)}(t) := (X_1^{(w)}(t), X_2^{(w)}(t))$ exists for all t. The set of admissible controls is denoted by $\mathcal{W}$. Let

$$\tau_{\mathcal{S}} = \inf\{t > 0;\, X^{(w)}(t) \notin \mathcal{S}\}$$

be the bankruptcy time. The investor's objective is to maximize

$$J^{(w)}(y) = E^y\left[\int_0^{\tau_{\mathcal{S}}} \mathrm{e}^{-\rho(s+t)}\frac{u^\gamma(t)}{\gamma}\mathrm{d}t\right], \tag{11.2.15}$$

where $\rho > 0$ and $\gamma \in (0, 1)$ are constants and E^y with $y = (s, x_1, x_2)$ denotes the expectation when $X_1(0^-) = x_1 \geq 0$ and $X_2(0^-) = x_2 \geq 0$. Thus we seek the value function $\Phi(y)$ and an optimal control $w^* = (u^*, v^*) \in \mathcal{W}$ such that

$$\Phi(y) = \sup_{w \in \mathcal{W}} J^{(w)}(y) = J^{(w^*)}(y). \tag{11.2.16}$$

This problem may be regarded as a generalization of optimal consumption and portfolio problems studied by Merton [M] and Davis and Norman [DN] (see also Shreve and Soner [SS]). Merton [M] considers the case with no jumps and no-transaction costs ($c = \lambda = \theta = 0$). The problem then reduces to an ordinary stochastic control problem and it is optimal to keep the positions $(X_1(t), X_2(t))$ on the line $y = (\pi^*/(1 - \pi^*))x$ in the (x, y)-plane at all times (the Merton line), where $\pi^* = (\mu - r)/(1 - \gamma)\sigma^2$ (see Example 5.2).

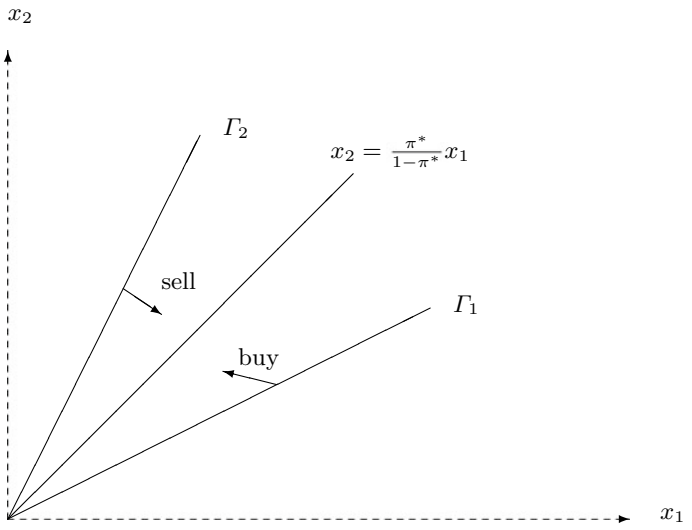


Fig. 11.1 The no-transaction cone (no fixed cost: $c = 0$), $\theta = 0$

Davis and Norman [DN] and Shreve and Soner [SS] consider the case when the cost is proportional ($\lambda > 0$), with no fixed component ($c = 0$) and no jumps ($\theta = 0$). In this case the problem can be formulated as a singular stochastic control problem and under some conditions it is proved that there exists a *no-transaction cone* NT bounded by two straight lines Γ_1 and Γ_2 such that it is optimal to make no transactions if $(X_1(t), X_2(t)) \in NT$ and make transactions corresponding to local time at $\partial(NT)$, resulting in reflections back to NT every time $(X_1(t), X_2(t)) \in \partial(NT)$. See Fig. 11.1. These results have subsequently been extended to jump diffusion markets by [FØS2] (see Example 8.1).

In the general combined control case numerical results indicate (see [CØS]) that the optimal control $w^* = (u^*, v^*)$ has the following form.

There exist two pairs of lines, $(\Gamma_1, \hat{\Gamma}_1)$ and $(\Gamma_2, \hat{\Gamma}_2)$ from the origin such that the following is optimal. Make no transactions (only consume at the rate $u^*(t)$) while $(X_1(t), X_2(t))$ belongs to the region D bounded by the outer curves Γ_1, Γ_2, and if $(X_1(t), X_2(t))$ hits $\partial D = \Gamma_1 \cup \Gamma_2$ then sell or buy so as to bring $(X_1(t), X_2(t))$ to the curve $\hat{\Gamma}_1$ or $\hat{\Gamma}_2$. See Fig. 11.2.

Note that if we *sell* stocks ($\zeta < 0$) then $(X_1(t), X_2(t)) = (x_1, x_2)$ moves to a point $(x_1', x_2') = (x_1'(\zeta), x_2'(\zeta))$ on the line

$$x_1' + (1 - \lambda)x_2' = x_1 + (1 - \lambda)x_2 - c, \tag{11.2.17}$$

i.e., (x_1', x_2') lies on the straight line through (x_1, x_2) with slope $-1/(1 - \lambda)$.

Similarly, if we *buy* stocks ($\zeta > 0$) then the new position $(x_1', x_2') = (x_1'(\zeta), x_2'(\zeta))$ is on the line

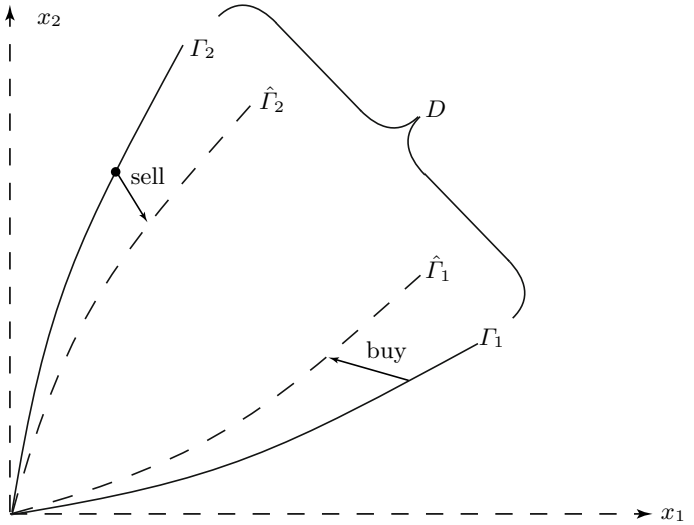


Fig. 11.2 The no-transaction region D ($c > 0$)

$$x_1' + (1+\lambda)x_2' = x_1 + (1+\lambda)x_2 - c, \tag{11.2.18}$$

i.e., (x_1', x_2') lies on the straight line through (x_1, x_2) with slope $-1/(1+\lambda)$. If there are no interventions then the process

$$Y(t) = \begin{bmatrix} s+t \\ X_1(t) \\ X_2(t) \end{bmatrix} \tag{11.2.19}$$

has the generator

$$\begin{aligned} L^u\phi(s,x_1,x_2) &= \frac{\partial\phi}{\partial s} + (rx_1 - u)\frac{\partial\phi}{\partial x_1} + \mu x_2\frac{\partial\phi}{\partial x_2} + \frac{1}{2}\sigma^2 x_2^2\frac{\partial^2\phi}{\partial x_2^2} \\ &\quad + \int_{\mathbb{R}}\Big\{\phi(s,x_1,x_2+x_2\theta z) - \phi(s,x_1,x_2) \\ &\quad - x_2\theta z\frac{\partial\phi}{\partial x_2}(s,x_1,x_2)\Big\}\nu(\mathrm{d}z). \end{aligned} \tag{11.2.20}$$

Therefore, if we put $\phi(s,x_1,x_2) = \mathrm{e}^{-\rho s}\psi(x_1,x_2)$ the corresponding HJBQVI is

$$\max\Bigg(\sup_{u\geq 0}\Bigg\{\frac{u^\gamma}{\gamma}-\rho\psi(x_1,x_2)+(rx_1-u)\frac{\partial\psi}{\partial x_1}+\mu x_2\frac{\partial\psi}{\partial x_2}+\frac{1}{2}\sigma^2x_2^2\frac{\partial^2\psi}{\partial x_2^2}$$
$$+\int_{\mathbb{R}}\Big\{\psi(x_1,x_2+x_2\theta z)-\psi(x_1,x_2)-x_2\theta z\frac{\partial\psi}{\partial x_2}(x_1,x_2)\Big\}\nu(\mathrm{d}z)\Bigg\}$$
$$\psi(x_1,x_2)-\mathcal{M}\psi(x_1,x_2)\Bigg)=0\quad\text{for all }(x_1,x_2)\in\mathcal{S},\tag{11.2.21}$$

where (see (11.2.17) and (11.2.18))

$$\mathcal{M}\psi(x_1,x_2)=\sup\{\psi(x_1'(\zeta),x_2'(\zeta));\,\zeta\in\mathbb{R}\setminus\{0\},\,(x_1'(\zeta),x_2'(\zeta))\in\mathcal{S}\}.\tag{11.2.22}$$

See Example 12.12 for a further discussion of this.

11.3 Iterative Methods

In Chap. 10 we saw that an impulse control problem can be regarded as a limit of iterated optimal stopping. A similar result holds for combined control problems. More precisely, a combined stochastic control and impulse control problem can be regarded as a limit of iterated combined stochastic control and optimal stopping problems.

We now describe this in more detail. The presentation is similar to the approach in Chap. 10.

For $n=1,2,\ldots$ let $\mathcal{W}_n$ denote the set of all admissible combined controls $w=(u,v)\in\mathcal{W}$ with $v\in\mathcal{V}_n$, where $\mathcal{V}_n$ is the set of impulse controls $v=(\tau_1,\ldots,\tau_k;\zeta_1,\zeta_2,\ldots,\zeta_k)$ with *at most n interventions* (i.e., $k\leq n$). Then

$$\mathcal{W}_n\subseteq\mathcal{W}_{n+1}\subseteq\mathcal{W}\quad\text{for all }n.\tag{11.3.1}$$

Define, with $J^{(w)}(y)$ as in (11.1.6),

$$\Phi_n(y)=\sup\big\{J^{(w)}(y);\,w\in\mathcal{W}_n\big\},\quad n=1,2,\ldots\tag{11.3.2}$$

Then

$$\Phi_n(y)\leq\Phi_{n+1}(y)\leq\Phi(y)\quad\text{because }\mathcal{W}_n\subseteq\mathcal{W}_{n+1}\subseteq\mathcal{W}.$$

Moreover, we have:

Lemma 11.4 *Suppose $g\geq 0$. Then*

$$\lim_{n\to\infty}\Phi_n(y)=\Phi(y)\quad\textit{for all }y\in\mathcal{S}.$$

Proof The proof is similar to the proof of Lemma 10.1 and is omitted. □

The iterative procedure is the following.

Let $Y(t) = Y^{(u,0)}(t)$ be the process in (11.1.1) obtained by using the control u and no interventions. Define

$$\phi_0(y) = \sup_{u \in \mathcal{U}} E^y\left[\int_0^{\tau_S} f(Y(t), u(t))\mathrm{d}t + g(Y(\tau_S))\chi_{\{\tau_S<\infty\}}\right] \tag{11.3.3}$$

and inductively, for $j = 1, 2, \ldots, n$,

$$\phi_j(y) = \sup_{u \in \mathcal{U}, \tau \in \mathcal{T}} E^y\left[\int_0^{\tau} f(Y(t), u(t))\mathrm{d}t + \mathcal{M}\phi_{j-1}(Y(\tau))\chi_{\{\tau<\infty\}}\right]. \tag{11.3.4}$$

As in (10.1.13) we let $\mathcal{P}(\mathbb{R}^k)$ denote the set of functions $h : \mathbb{R}^k \to \mathbb{R}$ with at most polynomial growth. Then we have as in Chap. 10:

Theorem 11.5 *Suppose*

$$f, g \text{ and } \mathcal{M}\phi_{j-1} \in \mathcal{P}(\mathbb{R}^k) \tag{11.3.5}$$

for $j = 1, 2, \ldots, n$. *Then*

$$\phi_n = \Phi_n.$$

Proof The proof is basically the same as the proof of Theorem 10.2 and is left to the reader. □

Similarly we obtain combined control versions of the rest of the results of Chap. 10, with obvious modifications. We omit the details.

11.4 Exercises

Exercise* 11.1 Let $\Phi(s, x_1, x_2)$ be the value function of the optimal consumption problem (11.2.16) with fixed and proportional transaction costs and let $\Phi_0(s, x_1, x_2) = K\mathrm{e}^{-\rho s}(x_1 + x_2)^\gamma$ be the corresponding value function in the case when there are no-transaction costs, i.e., $c = \lambda = 0$ (see Example 5.2). Use Theorem 11.1a to prove that

$$\Phi(s, x_1, x_2) \le K\mathrm{e}^{-\rho s}(x_1 + x_2)^\gamma.$$

Exercise* 11.2 (***A Combined Impulse Linear Regulator Problem***) (Compare with Exercises 5.5 and 7.3.)

Suppose the state process is

$$Y(t) = \begin{bmatrix} s+t \\ X(t) \end{bmatrix},\ t \ge 0, \quad Y(0) = \begin{bmatrix} s \\ x \end{bmatrix} = y \in \mathbb{R}^2,$$

with

$$\mathrm{d}X(t) = \mathrm{d}X^{(w)}(t) \text{given by}$$

$$\begin{aligned}\mathrm{d}X(t) &= u(t)\mathrm{d}t + \sigma\,\mathrm{d}B(t) + \int_{\mathbb{R}} z\tilde{N}(\mathrm{d}t, \mathrm{d}z), \quad \tau_i < t < \tau_{i+1},\\ X(\tau_{i+1}) &= X(\tau_{i+1}^-) + \Delta_N X(\tau_{i+1}) + \zeta_{i+1},\end{aligned}$$

where $w = (u, v)$ is a combined control, $v = (\tau_1, \tau_2, \ldots; \zeta_1, \zeta_2, \ldots)$ with $\tau_i \leq T$ (constant), σ constant, and $\zeta_{i+1} \in \mathbb{R}\backslash\{0\}$.

Write down and study the HJBQVI for the combined control problem

$$\Phi(y) = \inf_{w \in \mathcal{W}} J^{(w)}(y),$$

where

$$J^{(w)}(y) = E^{(s,x)}\left[\int_0^{T-s} (X^2(t) + u^2(t))\mathrm{d}t + \sum_{\tau_i \leq T-s} c\right],$$

and $c > 0$ is a given constant.

This models the situation where one is trying to keep $X(t)$ close to 0 with a minimum of cost of the two controls, represented by the rate $u^2(t)$ and the intervention cost c.

Chapter 12
Viscosity Solutions



The main results of Chaps. 3–9 and 11 and are all *verification theorems*. Any function ϕ which satisfies the given requirements is necessarily the value function Φ of the corresponding problem. These requirements are made as weak as possible in order to include as many cases as possible. For example, except for the singular control case, we do not require the function ϕ to be C^2 everywhere (only outside ∂D), because except for that case, Φ will usually not be C^2 everywhere. On the other hand, all the above-mentioned verification theorems require ϕ to be C^1 everywhere, because this is often the case for Φ. This C^1 assumption on Φ is usually called the "high contact" – or "smooth fit" – principle. As we have seen in the examples and exercises this principle is very convenient, because it provides us with extra information needed to find Φ and the continuation region D.

However, it is important to know that in general Φ need not be C^1. In fact, it need not even be continuous everywhere. Nevertheless, it turns out that Φ does satisfy the corresponding verification theorems, provided that we interpret these equations in an appropriate weak sense. More precisely, they should be interpreted in the sense of *viscosity solutions*. This weak solution concept was first introduced by Crandall and Lions to handle the HJB equations of stochastic control and later extended by them and others to more general equations. See [CIL, FS, BCa] and the references therein.

In case the equation in question is a *linear* partial differential operator, the viscosity solution is the same as the well-known distribution solution. See [Is2]. However, the nice feature of the viscosity solution is that it also applies to the nonlinear equations appearing in control theory.

We now proceed to define viscosity solutions. We will do this in two steps.

First we consider the viscosity solutions of the variational inequalities appearing in the optimal stopping problems of Chap. 3. Then we proceed to discuss more general equations.

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12.1 Viscosity Solutions of Variational Inequalities

Consider the optimal stopping problem of Chap. 3. The state $Y(t)$ is given by

$$\mathrm{d}Y(t) = b(Y(t))\mathrm{d}t + \sigma(Y(t))\mathrm{d}B(t) + \int_{\mathbb{R}^\ell} \gamma(Y(t^-), z)\bar{N}(\mathrm{d}t, \mathrm{d}z), \quad Y(0) = y \in \mathbb{R}^k \tag{12.1.1}$$

and the performance criterion is

$$J^\tau(y) = E^y\left[\int_0^\tau f(Y(t))\mathrm{d}t + g(Y(\tau))\chi_{\{\tau<\infty\}}\right], \quad \tau \in \mathcal{T}, \tag{12.1.2}$$

with f and g as in (3.1.2) and (3.1.3). The associated variational inequality of the optimal stopping problem

$$\Phi(y) = \sup\{J^\tau(y); \tau \in \mathcal{T}\} \tag{12.1.3}$$

is (see (7.2.13))

$$\max(L\phi(y) + f(y), g(y) - \phi(y)) = 0, \quad y \in \mathcal{S}^0. \tag{12.1.4}$$

In addition we have the boundary requirement

$$\phi(y) = g(y), \quad y \in \partial\mathcal{S} \tag{12.1.5}$$

(see Theorem 3.2).

We will prove that, under some conditions, the function $\phi = \Phi$ is the unique viscosity solution of (12.1.4) and (12.1.5). First we give the definition of a viscosity solution of such equations.

Definition 12.1 Let $\phi \in C(\bar{\mathcal{S}})$.

(i) We say that ϕ is a *viscosity subsolution* of (12.1.4) and (12.1.5) if (12.1.5) holds and for all $h \in C^2(\mathbb{R}^k)$ and all $y_0 \in \mathcal{S}^0$ such that

$$h \geq \phi \quad \text{on } \mathcal{S} \quad \text{and} \quad h(y_0) = \phi(y_0)$$

we have

$$\max(Lh(y_0) + f(y_0), g(y_0) - \phi(y_0)) \geq 0. \tag{12.1.6}$$

(ii) ϕ is a *viscosity supersolution* of (12.1.4) and (12.1.5) if (12.1.5) holds and for all $h \in C^2(\mathbb{R}^k)$ and all $y_0 \in \mathcal{S}$ such that

$$h \leq \phi \quad \text{on } \mathcal{S} \quad \text{and} \quad h(y_0) = \phi(y_0)$$

we have

$$\max(Lh(y_0) + f(y_0), g(y_0) - \phi(y_0)) \leq 0. \tag{12.1.7}$$

(iii) ϕ is a *viscosity solution* of (12.1.4) and (12.1.5) if ϕ is both a viscosity subsolution and a viscosity supersolution of (12.1.4) and (12.1.5).

Theorem 12.2 ([ØR]) *Assume that the set $\partial\mathcal{S}$ is* regular *for the process $Y(t)$, i.e.,*

$$\tau_{\mathcal{S}^0} = \tau_{\mathcal{S}^0}(y) := \inf\{t > 0; Y(t) \notin \mathcal{S}^0\} = 0 \quad \text{a.s. } Q^y \text{ for all } y \in \partial\mathcal{S}. \tag{12.1.8}$$

Moreover, assume that the value function Φ of the optimal stopping problem (12.1.3) *is continuous on $\bar{\mathcal{S}}$. Then Φ is a viscosity solution of* (12.1.4) *and* (12.1.5).

Proof First note that (12.1.5) follows directly from (12.1.8). So it remains to consider (12.1.4).

We first prove that Φ is a subsolution. To this end suppose $h \in C^2(\mathbb{R}^k)$ and $y_0 \in \mathcal{S}$ with $h \geq \Phi$ on $\mathcal{S}$ and $h(y_0) = \Phi(y_0)$. As before let

$$D = \{y \in \mathcal{S}; \Phi(y) > g(y)\}. \tag{12.1.9}$$

Then if $y_0 \notin D$ we have $\Phi(y_0) = g(y_0)$ and hence (12.1.6) holds trivially.

Next, assume $y_0 \in D$. Then by the dynamic programming principle (Lemma 10.3b) we have

$$\Phi(y_0) = E^{y_0}\left[\int_0^\tau f(Y(t))\mathrm{d}t + \Phi(Y(\tau))\right] \tag{12.1.10}$$

for all bounded stopping times $\tau \leq \tau_D = \inf\{t > 0; Y(t) \notin D\}$. Choose $h_m \in C_0^2(\mathbb{R}^k)$ such that $h_m \to h$ and $Lh_m \to Lh$ pointwise dominatedly on $\mathbb{R}^k$. Then by combining (12.1.10) with the Dynkin formula we get

$$\begin{aligned}
\Phi(y_0) &= E^{y_0}\left[\int_0^\tau f(Y(t))\mathrm{d}t + \Phi(Y(\tau))\right] \\
&\leq E^{y_0}\left[\int_0^\tau f(Y(t))\mathrm{d}t + h(Y(\tau))\right] \\
&= \lim_{m\to\infty} E^{y_0}\left[\int_0^\tau f(Y(t))\mathrm{d}t + h_m(Y(\tau))\right] \\
&= h(y_0) + \lim_{m\to\infty} E^{y_0}\left[\int_0^\tau (Lh_m(Y(t)) + f(Y(t)))\mathrm{d}t\right].
\end{aligned}$$

Hence

$$\lim_{m\to\infty} E^{y_0}\left[\int_0^\tau (Lh_m(Y(t)) + f(Y(t)))\mathrm{d}t\right] \geq 0.$$

In particular, if we choose $\tau = \beta_j := \inf\{t > 0; |Y(t) - y_0| \geq 1/j\} \wedge 1/j \wedge \tau_D$, we get

$$E^{y_0}\left[\int_0^{\beta_j} (Lh(Y(t)) + f(Y(t)))\mathrm{d}t\right]$$
$$= \lim_{m\to\infty} E^{y_0}\left[\int_0^{\beta_j} (Lh_m(Y(t)) + f(Y(t)))\mathrm{d}t\right] \geq 0. \tag{12.1.11}$$

If we divide (12.1.11) by $E^{y_0}[\beta_j]$ and let $j \to \infty$ we get, by right continuity,

$$Lh(y_0) + f(y_0) \geq 0.$$

Hence (12.1.6) holds and we have proved that Φ is a viscosity subsolution.

Finally we show that Φ is a viscosity supersolution. So we assume that $h \in C^2(\mathbb{R}^k)$ and $y_0 \in \mathcal{S}$ are such that $h \leq \Phi$ on $\mathcal{S}$ and $h(y_0) = \Phi(y_0)$. Then by the dynamic programming principle (Lemma 10.3a) we have

$$\Phi(y_0) \geq E^{y_0}\left[\int_0^{\tau} f(Y(t))\mathrm{d}t + \Phi(Y(\tau))\right] \tag{12.1.12}$$

for all stopping times $\tau \leq \tau_{\mathcal{S}^0}$. Hence, by the Dynkin formula, with h_m and β_j as above,

$$\begin{aligned}
\Phi(y_0) &\geq E^{y_0}\left[\int_0^{\beta_j} f(Y(t))\mathrm{d}t + \phi(Y(\tau))\right] \\
&\geq \lim_{m\to\infty} E^{y_0}\left[\int_0^{\beta_j} f(Y(t))\mathrm{d}t + h(Y(\tau))\right] \\
&= h(y_0) + \lim_{m\to\infty} E^{y_0}\left[\int_0^{\beta_j} (Lh(Y(t))\mathrm{d}t + f(Y(t)))\mathrm{d}t\right] \\
&= h(y_0) + E^{y_0}\left[\int_0^{\beta_j} (Lh(Y(t))\mathrm{d}t + f(Y(t)))\mathrm{d}t\right] \quad \text{for all } j.
\end{aligned}$$

Hence

$$E^{y_0}\left[\int_0^{\beta_j} (Lh(Y(t)) + f(Y(t)))\mathrm{d}t\right] \leq 0$$

and by dividing by $E^{y_0}[\beta_j]$ and letting $j \to \infty$ we get, by right continuity,

$$Lh(y_0) + f(y_0) \leq 0.$$

Hence (12.1.7) holds and we have proved that Φ is also a viscosity supersolution. □

12.1.1 Uniqueness

One important application of the viscosity solution concept is that it can be used as a verification method. In order to verify that a given function ϕ is indeed the value function Φ it suffices to verify that the function is a viscosity solution of the corresponding variational inequality. For this method to work, however, it is necessary that we know that Φ is the *unique* viscosity solution. Therefore the question of uniqueness is crucial.

In general we need not have uniqueness. The following simple example illustrates this.

Example 12.3 Let $Y(t) = B(t) \in \mathbb{R}$ and choose $f = 0$, $\mathcal{S} = \mathbb{R}$, and

$$g(y) = \frac{y^2}{1+y^2}, \quad y \in \mathbb{R}. \tag{12.1.13}$$

Then the value function Φ of the optimal stopping problem

$$\Phi(y) = \sup_{\tau \in \mathcal{T}} E^y[g(B(\tau))] \tag{12.1.14}$$

is easily seen to be $\Phi(y) \equiv 1$. The corresponding VI is

$$\max\left(\frac{1}{2}\phi''(y), g(y) - \phi(y)\right) = 0 \tag{12.1.15}$$

and this equation is trivially satisfied by all constant functions

$$\phi(y) \equiv a$$

for any $a \geq 1$.

Theorem 12.4 (***Uniqueness***) *Suppose that*

$$\tau_{\mathcal{S}^0} < \infty \quad a.s. \quad P^y \text{ for all } \ y \in \mathcal{S}^0. \tag{12.1.16}$$

Let $\phi \in C(\bar{\mathcal{S}})$ be a viscosity solution of (12.1.4) *and* (12.1.5) *with the property that*

$$\begin{aligned} &\text{the family } \{\phi(Y(\tau)); \tau \text{ stopping time, } \tau \leq \tau_{\mathcal{S}^0}\} \\ &\quad \text{is } P^y\text{-uniformly integrable, for all } y \in \mathcal{S}^0. \end{aligned} \tag{12.1.17}$$

Then

$$\phi(y) = \Phi(y) \quad \text{for all } \ y \in \bar{\mathcal{S}}.$$

Proof We refer the reader to [ØR] for the proof in the case where there are no jumps. □

12.2 The Value Function is Not Always C^1

Example 12.5 We now give an example of an optimal stopping problem where the value function Φ is not C^1 everywhere. In this case Theorem 6.10 cannot be used to find Φ. However, we can use Theorem 12.4. The example is taken from [ØR]:

Define

$$k(x) = \begin{cases} 1 & \text{for } x \leq 0 \\ 1 - cx & \text{for } 0 < x < a, \\ 1 - ca & \text{for } x \geq a \end{cases} \tag{12.2.1}$$

where c and a are constants to be specified more closely later. Consider the optimal stopping problem

$$\Phi(s, x) = \sup_{\tau \in \mathcal{T}} E^{(s,x)} \left[\mathrm{e}^{-\rho(s+\tau)} k(B(\tau)) \right], \tag{12.2.2}$$

where $B(t)$ is one-dimensional Brownian motion, $B(0) = x \in \mathbb{R} = \mathcal{S}$, and $\rho > 0$ is a constant. The corresponding variational inequality is (see (12.1.4))

$$\max \left(\frac{\partial \phi}{\partial s} + \frac{1}{2} \frac{\partial^2 \phi}{\partial x^2}, \mathrm{e}^{-\rho s} k(x) - \phi(s, x) \right) = 0. \tag{12.2.3}$$

If we try a solution of the form

$$\phi(s, x) = \mathrm{e}^{-\rho s} \psi(x) \tag{12.2.4}$$

for some function ψ, then (12.2.3) becomes

$$\max \left(-\rho \psi(x) + \frac{1}{2} \psi''(x), k(x) - \psi(x) \right) = 0. \tag{12.2.5}$$

Let us guess that the continuation region D has the form

$$D = \{(s, x); 0 < x < x_1\} \tag{12.2.6}$$

for some $x_1 > a$. Then (12.2.5) can be split into the three equations

$$\begin{aligned} -\rho \psi(x) + \frac{1}{2} \psi''(x) &= 0 \quad ; \quad 0 < x < x_1 \\ \psi(x) &= 1 \quad ; \quad x \leq 0 \\ \psi(x) &= 1 - ca \quad ; \quad x \geq x_1. \end{aligned} \tag{12.2.7}$$

The general solution of (12.2.7) is

$$\psi(x) = C_1 \mathrm{e}^{\sqrt{2\rho}\,x} + C_2 \mathrm{e}^{-\sqrt{2\rho}\,x}, \quad 0 < x < x_1,$$

where C_1, C_2 are arbitrary constants. If we require ψ to be continuous at $x = 0$ and at $x = x_1$ we get the two equations

$$C_1 + C_2 = 1, \tag{12.2.8}$$

$$C_1 \mathrm{e}^{\sqrt{2\rho}\,x_1} + C_2 \mathrm{e}^{-\sqrt{2\rho}\,x_1} = 1 - ca \tag{12.2.9}$$

in the three unknowns C_1, C_2, and x_1. If we also guess that ψ will be C^1 at $x = x_1$ we get the third equation

$$C_1 \sqrt{2\rho}\, \mathrm{e}^{\sqrt{2\rho}\,x_1} - C_2 \sqrt{2\rho}\, \mathrm{e}^{-\sqrt{2\rho}\,x_1} = 0. \tag{12.2.10}$$

If we assume that

$$ca < 1 \quad \text{and} \quad \sqrt{2\rho} < \frac{1}{a} \ln\left(\frac{1 - ca}{1 - \sqrt{ca(2 - ca)}} \right) \tag{12.2.11}$$

then the three equations (12.2.8), (12.2.9), and (12.2.10) have the unique solution

$$C_1 = \frac{1}{2}\left(1 - \sqrt{ca(2 - ca)}\right) > 0, \quad C_2 = 1 - C_1 > 0 \tag{12.2.12}$$

and

$$x_1 = \frac{1}{\sqrt{2\rho}} \ln\left(\frac{1 - ca}{2C_1} \right) > a. \tag{12.2.13}$$

With these values of C_1, C_2, and x_1 we put

$$\psi(x) = \begin{cases} 1 & \text{if } x \le 0 \\ C_1 \mathrm{e}^{\sqrt{2\rho}\,x} + C_2 \mathrm{e}^{-\sqrt{2\rho}\,x} & \text{if } 0 < x < x_1. \\ 1 - ca & \text{if } x_1 \le x \end{cases} \tag{12.2.14}$$

See Fig. 12.1.

We claim that ψ is a viscosity solution of (12.2.5).

(i) First we verify that ψ is a viscosity *sub*solution: let $h \in C^2(\mathbb{R})$, $h \ge \psi$, and $h(x_0) = \psi(x_0)$. Then if $x_0 \le 0$ or $x_0 \ge x_1$ we have $k(x_0) - \psi(x_0) = 0$. And if $0 < x_0 < x_1$ then $h - \psi$ is C^2 at $x = x_0$ and has a local minimum at x_0 so

$$h''(x_0) - \psi''(x_0) \ge 0.$$

Therefore

$$-\rho h(x_0) + \frac{1}{2} h''(x_0) \ge -\rho \psi(x_0) + \frac{1}{2} \psi''(x_0) = 0.$$

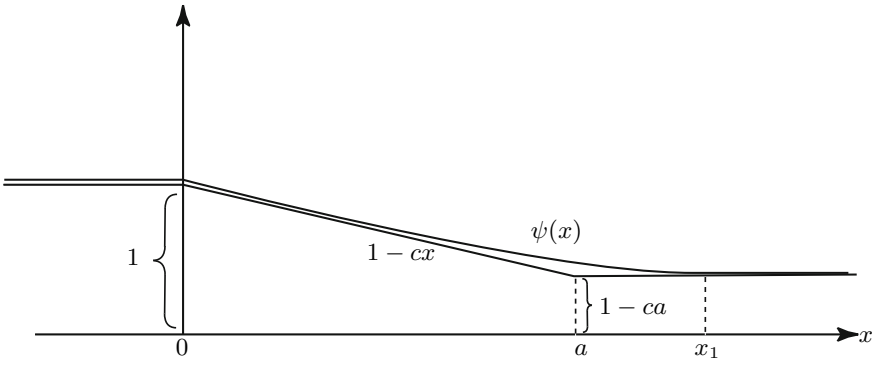


Fig. 12.1 The function ψ

This proves that

$$\max\left(-\rho h(x_0) + \frac{1}{2}h''(x_0), k(x_0) - \psi(x_0)\right) \geq 0,$$

so ψ is a viscosity subsolution of (12.2.5).

(ii) Second, we prove that ψ is a viscosity *super*solution. So let $h \in C^2(\mathbb{R})$, $h \leq \psi$, and $h(x_0) = \psi(x_0)$. Note that we always have

$$k(x_0) - \psi(x_0) \leq 0$$

so in order to prove that

$$\max\left(-\rho h(x_0) + \frac{1}{2}h''(x_0), k(x_0) - \psi(x_0)\right) \leq 0$$

it suffices to prove that

$$-\rho h(x_0) + \frac{1}{2}h''(x_0) \leq 0.$$

At any point x_0 where ψ is C^2 this follows in the same way as in (i) above. So it remains only to consider the two cases $x_0 = 0$ and $x_0 = x_1$. If $x_0 = 0$ then no such h exists, so the conclusion trivially holds. If $x_0 = x_1$ then the function $h - \psi$ has a local maximum at $x = x_0$ and it is C^2 to the left of x_0 so

$$\lim_{x \to x_0^-} h''(x) - \psi''(x) \leq 0,$$

i.e., $h''(x_0) - \psi''(x_0^-) \le 0$.
This gives

$$-\rho h(x_0) + \frac{1}{2}h''(x_0) \le -\rho\psi(x_0) + \frac{1}{2}\psi''(x_0^-) = 0,$$

and the proof is complete.
We have proved.
Suppose (12.2.11) holds. Then the value function $\Phi(s,x)$ of problem (12.2.2) is given by

$$\Phi(s,x) = \mathrm{e}^{-\rho s}\psi(x)$$

with ψ as in (12.2.14), C_1, C_2, and x_1 as in (12.2.12) and (12.2.13). Note in particular that $\psi(x)$ is not C^1 at $x = 0$.

12.3 Viscosity Solutions of HJBQVI

We now turn to the general combined stochastic control and impulse control problem from Chap. 11. Thus the state $Y(t) = Y^{(w)}(t)$ is

$$\begin{aligned} Y(0^-) &= y \in \mathbb{R}^k, \\ \mathrm{d}Y(t) &= b(Y(t), u(t))\mathrm{d}t + \sigma(Y(t), u(t))\mathrm{d}B(t) \\ &\quad + \int_{\mathbb{R}^\ell} \gamma(Y(t^-), u(t^-), z)\tilde{N}(\mathrm{d}t, \mathrm{d}z), \quad \tau_i < t < \tau_{i+1}, \\ Y(\tau_{i+1}) &= \Gamma(\check{Y}(\tau_{i+1}^-), \zeta_{i+1}), \quad i = 0, 1, 2, \ldots, \end{aligned} \tag{12.3.1}$$

where $w = (u, v) \in \mathcal{W}$, $u \in \mathcal{U}$, $v = (\tau_1, \tau_2, \ldots; \zeta_1, \zeta_2, \ldots) \in \mathcal{V}$.
The performance is given by

$$J^{(w)}(y) = E^y\left[\int_0^{\tau_\mathcal{S}} f(Y(t), u(t))\mathrm{d}t + g(Y(\tau_\mathcal{S}))\chi_{\{\tau_\mathcal{S}<\infty\}} + \sum_j K(\check{Y}(\tau_j^-), \zeta_j)\right] \tag{12.3.2}$$

and we want to find the value function Φ defined by

$$\Phi(y) = \sup_{w\in\mathcal{W}} J^{(w)}(y). \tag{12.3.3}$$

To simplify the presentation we will from now on assume that

$$\mathcal{S} \text{ is an open set, i.e., } \mathcal{S} = \mathcal{S}^0, \tag{12.3.4}$$

and that $\partial\mathcal{S}$ is *regular* for $Y^{(w)}(t)$ for all $w \in \mathcal{W}$, i.e.,

$$\tau_{\mathcal{S}} = \tau_{\mathcal{S}}(y) = \inf\{t > 0;\, Y^{(w)}(t) \notin \mathcal{S}\} = 0 \quad \text{for all } \ y \in \partial\mathcal{S} \ \text{ and all } \ w \in \mathcal{W}. \tag{12.3.5}$$

These conditions (12.3.4) and (12.3.5) exclude cases where Φ also satisfies certain HJBQVIs on $\partial\mathcal{S}$ (see, e.g., [ØS]), but it is often easy to see how to extend the results to such situations.

We need to make the following two assumptions on the set of admissible controls W:

(1) If $w = (u, v)$, with $v = (\tau_1, \tau_2, \ldots; \zeta_1, \zeta_2, \ldots)$, belongs to W and $\hat{\zeta}$ is any point in Z, then $w := (u, \hat{v})$ belongs to W also, when $\hat{v} := (0, \tau_1, \tau_2, \ldots; \zeta, \zeta_1, \zeta_2, \ldots)$.

(2) If α is any constant in U, then the combined control $w := (\alpha, 0)$ (no interventions, just the constant control α) belongs to W.

Theorem 11.1 associates Φ to the HJBQVI

$$\max\left(\sup_{\alpha\in U}\{L^{\alpha}\Phi(y) + f(y,\alpha)\}, \mathcal{M}\Phi(y) - \Phi(y)\right) = 0, \quad y \in \mathcal{S} \tag{12.3.6}$$

with boundary values

$$\Phi(y) = g(y), \quad y \in \partial\mathcal{S}, \tag{12.3.7}$$

where

$$\begin{aligned} L^{\alpha}\Phi(y) = \sum_{i=1}^{k} b_i(y,\alpha)\frac{\partial\Phi}{\partial y_i} + \frac{1}{2}\sum_{i,j=1}^{k}\left(\sigma\sigma^T\right)_{ij}(y,\alpha)\frac{\partial^2\Phi}{\partial y_i \partial y_j} \\ + \sum_{j=1}^{\ell}\int_{\mathbb{R}}\Big\{\Phi\left(y + \gamma^{(j)}(y,\alpha,z_j)\right) - \Phi(y) \\ - \nabla\Phi(y)\cdot\gamma^{(j)}(y,\alpha,z_j)\Big\}\nu_j(\mathrm{d}z_j) \end{aligned} \tag{12.3.8}$$

and

$$\mathcal{M}\Phi(y) = \sup\left\{\Phi(\Gamma(y,\zeta)) + K(y,\zeta);\, \zeta \in \mathcal{Z},\, \Gamma(y,\zeta) \in \mathcal{S}\right\}. \tag{12.3.9}$$

Unfortunately, as we have seen already for optimal stopping problems, the value function Φ need not be C^1 everywhere – in general not even continuous! So (12.3.6) is not well defined, if we interpret the equation in the usual sense. However, it turns out that if we interpret (12.3.6) in the *weak sense of viscosity* then Φ does indeed solve the equation. In fact, under some assumptions Φ is the *unique* viscosity solution of (12.3.6) and (12.3.7) (see Theorem 12.11). This result is an important supplement to Theorem 11.1.

We now define the concept of viscosity solutions of general HJBQVIs of type (12.3.6) and (12.3.7).

Definition 12.6 Let $\varphi \in C(\bar{\mathcal{S}})$.

(i) We say that φ is a *viscosity subsolution* of

$$\max\left(\sup_{\alpha\in U}\{L^{\alpha}\varphi(y)+f(y,\alpha)\},\ \mathcal{M}\varphi(y)-\varphi(y)\right)=0,\quad y\in\mathcal{S},\tag{12.3.10}$$

$$\varphi(y)=g(y),\quad y\in\partial\mathcal{S}\tag{12.3.11}$$

if (12.3.11) holds and for every $h\in C^2(\mathbb{R}^k)$ and every $y_0\in\mathcal{S}$ such that $h\geq\varphi$ on $\mathcal{S}$ and $h(y_0)=\varphi(y_0)$ we have

$$\max\left(\sup_{\alpha\in U}\{L^{\alpha}h(y_0)+f(y_0,\alpha)\},\ \mathcal{M}\varphi(y_0)-\varphi(y_0)\right)\geq 0.\tag{12.3.12}$$

(ii) We say that φ is a *viscosity supersolution* of (12.3.10) and (12.3.11) if (12.3.11) holds and for every $h\in C^2(\mathbb{R}^k)$ and every $y_0\in\mathcal{S}$ such that $h\leq\varphi$ on $\mathcal{S}$ and $h(y_0)=\varphi(y_0)$ we have

$$\max\left(\sup_{\alpha\in U}\{L^{\alpha}h(y_0)+f(y_0,\alpha)\},\ \mathcal{M}\varphi(y_0)-\varphi(y_0)\right)\leq 0.\tag{12.3.13}$$

(iii) We say that φ is a *viscosity solution* of (12.3.10) and (12.3.11) if φ is both a viscosity subsolution and a viscosity supersolution of (12.3.10) and (12.3.11).

Lemma 12.7 *Let Φ be as in (12.3.3). Then $\Phi(y)\geq\mathcal{M}\Phi(y)$ for all $y\in\mathcal{S}$.*

Proof Suppose there exists $y\in\mathcal{S}$ with

$$\Phi(y)<\mathcal{M}\Phi(y),$$

i.e.,

$$\Phi(y)<\sup_{\zeta\in\mathcal{Z}}\{\Phi(\Gamma(y,\zeta))+K(y,\zeta)\}.$$

Then there exist $\varepsilon>0$ and $\hat{\zeta}\in\mathcal{Z}$ such that, with $\hat{y}=\Gamma(y,\hat{\zeta})$,

$$\Phi(y)<\Phi(\hat{y})+K(y,\hat{\zeta})-2\varepsilon.$$

Let $w=(u,v)$, with $v=(\tau_1,\tau_2,\ldots;\zeta_1,\zeta_2,\ldots)$ be ε-optimal for Φ at $\hat{y}$, in the sense that

$$J^{(w)}(\hat{y})>\Phi(\hat{y})-\varepsilon.$$

Define $\hat{w}:=(u,\hat{v})$, where $\hat{v}=(0,\tau_1,\tau_2,\ldots;\hat{\zeta},\zeta_1,\zeta_2,\ldots)$. Then, with $\tau_0=0$ and $\zeta_0=\hat{\zeta}$,

$$\Phi(y) \geq J^{(\hat{w})}(y) = E^y\Bigg[\int_0^{\tau_S} f(Y(t), u(t))\mathrm{d}t + g(Y(\tau_S))\chi_{\{\tau_S<\infty\}} + \sum_{i=0}^{\infty} K(\check{Y}(\tau_i^-), \zeta_i)\Bigg] = K(y, \hat{\zeta}) + J^{(w)}(\hat{y}).$$

Combining the above we get

$$\begin{aligned}\Phi(\hat{y}) + K(y, \hat{\zeta}) &< J^{(w)}(\hat{y}) + K(y, \hat{\zeta}) + \varepsilon \\ &\leq \Phi(y) + \varepsilon \\ &< \Phi(\hat{y}) + K(y, \hat{\zeta}) - \varepsilon,\end{aligned}$$

a contradiction. □

Our first main result in this section is the following.

Theorem 12.8 *The value function*

$$\Phi(y) = \sup_{w \in \mathcal{W}} J^{(w)}(y)$$

of the combined stochastic control and impulse control problem (12.3.3) *is a viscosity solution of* (12.3.6) *and* (12.3.7).

Proof By (12.3.2) and (12.3.5) we see that Φ satisfies (12.3.7).
(a) We first prove that Φ is a viscosity subsolution. To this end, choose $h \in C^2(\mathbb{R}^k)$ and $y_0 \in \mathcal{S}$ such that $h \geq \Phi$ on $\mathcal{S}$ and $h(y_0) = \Phi(y_0)$. We must prove that

$$\max\left(\sup_{\alpha \in U}\{L^\alpha h(y_0) + f(y_0, \alpha)\},\ \mathcal{M}\Phi(y_0) - \Phi(y_0)\right) \geq 0. \tag{12.3.14}$$

If $\mathcal{M}\Phi(y_0) \geq \Phi(y_0)$ then (12.3.14) holds trivially, so we may assume that

$$\mathcal{M}\Phi(y_0) < \Phi(y_0). \tag{12.3.15}$$

Choose $\epsilon > 0$ and let $w = (u, v) \in \mathcal{W}$, with $v = (\tau_1, \tau_2, \ldots; \zeta_1, \zeta_2, \ldots) \in \mathcal{V}$, be an ϵ-optimal portfolio, i.e.,

$$\Phi(y_0) < J^{(w)}(y_0) + \epsilon.$$

Since τ_1 is a stopping time we know that $\{\omega; \tau_1(\omega) = 0\}$ is $\mathcal{F}_0$-measurable and hence either

$$\tau_1(\omega) = 0 \text{ a.s.} \quad \text{or} \quad \tau_1(\omega) > 0 \text{ a.s.} \tag{12.3.16}$$

If $\tau_1 = 0$ a.s. then $Y^{(w)}$ makes an immediate jump from y_0 to the point $y' = \Gamma(y_0, \zeta_1) \in \mathcal{S}$ and hence

$$\Phi(y_0) - \epsilon \le J^{(w')}(y') + K(y_0, \zeta_1) \le \Phi(y') + K(y_0, \zeta_1) \le \mathcal{M}\Phi(y_0),$$

where $w' = (\tau_2, \tau_3, \ldots; \zeta_2, \zeta_3, \ldots)$.

This is a contradiction if $\epsilon < \Phi(y_0) - \mathcal{M}\Phi(y_0)$. This proves that (12.3.15) implies that it is impossible to have $\tau_1 = 0$ a.s.

So by (12.3.16), we can now assume that $\tau_1 > 0$ a.s. Choose $\rho > 0$ and define

$$\tau := \tau_1 \wedge \rho.$$

By the dynamic programming principle (see Lemma 10.3) we have: for each $\varepsilon > 0$, there exists a control $u = u_\rho$ such that

$$\Phi(y_0) \le E^{y_0}\left[\int_0^\tau f(Y(t), u(t))\mathrm{d}t + \Phi(\check{Y}(\tau^-))\right] + \rho^2, \tag{12.3.17}$$

where, as before, $\check{Y}(\tau^-) = Y(\tau^-) + \Delta_N Y(\tau)$.

Choose $h_m \in C_0^2(\mathbb{R}^k)$ such that $h_m \to h$ and $L^u h_m \to L^u h$ pointwise dominatedly as $m \to \infty$. Then by (12.3.17) and the Dynkin formula we have, using that $\Phi \le h$,

$$\begin{aligned}
\Phi(y_0) &\le E^{y_0}\left[\int_0^\tau f(Y(t), u(t))\mathrm{d}t + h(\check{Y}(\tau^-))\right] + \varepsilon \\
&\le \liminf_{m\to\infty} E^{y_0}\left[\int_0^\tau f(Y(t), u(t))\mathrm{d}t + h_m(\check{Y}(\tau^-))\right] + \varepsilon \\
&= h(y_0) + \liminf_{m\to\infty} E^{y_0}\left[\int_0^\tau (L^u h_m(Y(t)) + f(Y(t), u(t)))\mathrm{d}t\right] + \varepsilon \\
&= h(y_0) + E^{y_0}\left[\int_0^\tau (L^u h(Y(t)) + f(Y(t), u(t)))\mathrm{d}t\right] + \varepsilon. \qquad (12.3.18)
\end{aligned}$$

Using that $h(y_0) = \Phi(y_0)$, we obtain

$$E^{y_0}\left[\int_0^\tau \{L^u h(Y(t)) + f(Y(t), u(t))\}\mathrm{d}t\right] \ge -\rho^2.$$

Dividing by $E^{y_0}[\tau]$ and letting $\rho \to 0$ we get

$$L^{\alpha_0} h(y_0) + f(y_0, \alpha_0) \ge 0,$$

where

$$\alpha_0 = \limsup_{\rho\to 0} \lim_{s\to 0^+} u(s).$$

Since ε is arbitrary, this proves (12.3.14) and hence that Φ is a viscosity subsolution.

(b) Next we prove that Φ is a viscosity supersolution. So we choose $h \in C^2(\mathbb{R}^k)$ and $y_0 \in \mathcal{S}$ such that $h \leq \Phi$ on $\mathcal{S}$ and $h(y_0) = \Phi(y_0)$. We must prove that

$$\max\left(\sup_{\alpha\in U}\{L^\alpha h(y_0) + f(y_0,\alpha)\}, \mathcal{M}\Phi(y_0) - \Phi(y_0)\right) \leq 0. \tag{12.3.19}$$

Since $\Phi \geq \mathcal{M}\Phi$ always (Lemma 12.7) it suffices to prove that

$$L^\alpha h(y_0) + f(y_0,\alpha) \leq 0 \text{ for all } \alpha \in U.$$

To this end, fix $\alpha \in U$ and let $w_\alpha = (\alpha, 0)$, i.e., w_α is the combined control $(u_\alpha, v_\alpha) \in \mathcal{W}$ where $u_\alpha = \alpha$ (constant) and $v_\alpha = 0$ (no interventions). Then by the dynamic programming principle and the Dynkin formula we have, with $Y(t) = Y^{(w_\alpha)}(t)$, $\tau = \tau_{\mathcal{S}} \wedge \rho$, and h_m as in (a),

$$\begin{aligned}
\Phi(y_0) &\geq E^{y_0}\left[\int_0^\tau f(Y(s),\alpha)\mathrm{d}s + \Phi(\check{Y}(\tau^-))\right]\\
&\geq E^{y_0}\left[\int_0^\tau f(Y(s),\alpha)\mathrm{d}s + h(\check{Y}(\tau^-))\right]\\
&= h(y_0) + \lim_{m\to\infty} E^{y_0}\left[\int_0^\tau \{L^\alpha h_m(Y(t)) + f(Y(t),\alpha)\}\mathrm{d}t\right]\\
&= h(y_0) + E^{y_0}\left[\int_0^\tau \{L^\alpha h(Y(t)) + f(Y(t),\alpha)\}\mathrm{d}t\right].
\end{aligned}$$

Hence

$$E\left[\int_0^\tau \{L^\alpha h(Y(t)) + f(Y(t),\alpha)\}\mathrm{d}t\right] \leq 0.$$

Dividing by $E[\tau]$ and letting $\rho \to 0$ we get (12.3.19).

This completes the proof of Theorem 12.8. □

Next we turn to the question of *uniqueness* of viscosity solutions of (12.3.10) and (12.3.11). Many types of uniqueness results can be found in the literature. See the references in the end of this section.

Here we give a proof in the case when the process $Y(t)$ has no jumps, i.e. when $N(\cdot,\cdot) = \nu(\cdot) = 0$. The method we use is a generalization of the method in [ØS, Theorem 3.8].

First we introduce some convenient notation:

Define $\Lambda : \mathbb{R}^{k\times k} \times \mathbb{R}^k \times \mathbb{R}^{\mathcal{S}} \times \mathbb{R}^k \to \mathbb{R}$ by

$$\Lambda(R, r, \varphi, y) := \sup_{\alpha\in U}\left\{\sum_{i=1}^k b_i(y,\alpha)r_i + \frac{1}{2}\sum_{i,j=1}^k (\sigma\sigma^T)_{ij}(y,\alpha)R_{ij}\right.$$

$$+\sum_{j=1}^{\ell}\int_{\mathbb{R}}\Big\{\varphi(y+\gamma^{(j)}(y,\alpha,z_j))-\varphi(y)$$

$$-r\cdot\gamma^{(j)}(y,\alpha,z_j)\Big\}\nu_j(\mathrm{d}z_j)+f(y,\alpha)\Big\} \tag{12.3.20}$$

for $R=[R_{ij}]\in\mathbb{R}^{k\times k}$, $r=(r_i,\dots,r_k)\in\mathbb{R}^k$, $\varphi:\mathcal{S}\to\mathbb{R}$, $y\in\mathbb{R}^k$, and define $F:\mathbb{R}^{k\times k}\times\mathbb{R}^k\times\mathbb{R}^{\mathcal{S}}\times\mathbb{R}^k\to\mathbb{R}$ by

$$F(R,r,\varphi,y)=\max\{\Lambda(R,r,\varphi,y),\mathcal{M}\varphi(y)-\varphi(y)\}. \tag{12.3.21}$$

Note that if $\varphi\in C^2(\mathbb{R}^k)$ then

$$\Lambda(D^2\varphi,D\varphi,\varphi,y)=\sup_{\alpha\in U}\{L^{\alpha}\varphi(y)+f(y,\alpha)\},$$

where

$$D^2\varphi(y)=\left[\frac{\partial^2\varphi}{\partial y_i\partial y_j}\right](y)\text{ and }D\varphi(y)=\left[\frac{\partial\varphi}{\partial y_i}\right](y).$$

We recall the concepts of "superjets" $J_{\mathcal{S}}^{2,+}$, $J_{\mathcal{S}}^{2,-}$ and $\bar{J}_{\mathcal{S}}^{2,+}$, $\bar{J}_{\mathcal{S}}^{2,-}$ (see [CIL, Sect. 2]):

$$J_{\mathcal{S}}^{2,+}\varphi(y):=\Big\{(R,r)\in\mathbb{R}^{k\times k}\times\mathbb{R}^k\ ;$$

$$\limsup_{\substack{\eta\to y\\ \eta\in\mathcal{S}}}[\varphi(\eta)-\varphi(y)-(\eta-y)^Tr-\frac{1}{2}(\eta-y)^TR(\eta-y)]\cdot|\eta-y|^{-2}\le 0\Big\},$$

$$\bar{J}_{\mathcal{S}}^{2,+}\varphi(y):=\Big\{(R,r)\in\mathbb{R}^{k\times k}\times\mathbb{R}^k;\ \text{ for all } n \text{ there exists}$$
$$(R^{(n)},r^{(n)},y^{(n)})\in\mathbb{R}^{k\times k}\times\mathbb{R}^k\times\mathcal{S}\text{ such that }(R^{(n)},r^{(n)})\in J_{\mathcal{S}}^{2,+}\varphi(y^{(n)})\text{ and}$$
$$(R^{(n)},r^{(n)},\varphi(y^{(n)}),y^{(n)})\to(R,r,\varphi(y),y)\text{ as }n\to\infty\Big\}$$

and

$$J_{\mathcal{S}}^{2,-}\varphi=-J_{\mathcal{S}}^{2,+}(-\varphi),\quad \bar{J}_{\mathcal{S}}^{2,-}\varphi=-\bar{J}_{\mathcal{S}}^{2,+}(-\varphi).$$

In terms of these superjets one can give an equivalent definition of viscosity solutions as follows.

Theorem 12.9 ([CIL, Sect. 2])

(i) A function $\varphi\in C(\mathcal{S})$ is a viscosity subsolution of (12.3.10) *and* (12.3.11) *if and only if* (12.3.11) *holds and*

$$\max(\Lambda(R,r,\varphi,y),\mathcal{M}\varphi(y)-\varphi(y))\ge 0\ \textit{for all}\ (R,r)\in\bar{J}_{\mathcal{S}}^{2,+}\varphi(y),\ \ y\in\mathcal{S}.$$

(ii) A function $\varphi \in C(\mathcal{S})$ is a viscosity supersolution of (12.3.10) *and* (12.3.11) *if and only if* (12.3.11) *holds and*

$$\max(\Lambda(R, r, \varphi, y), \mathcal{M}\varphi(y) - \varphi(y)) \leq 0 \text{ for all } (R, r) \in \bar{J}_{\mathcal{S}}^{2,-}\varphi(y), \quad y \in \mathcal{S}.$$

We have now ready for the second main theorem of this section.

Theorem 12.10 (***Comparison Theorem***) *Assume that*

$$N(\cdot, \cdot) = 0. \tag{12.3.22}$$

Suppose that there exists a positive function $\beta \in C^2(\bar{\mathcal{S}})$ which satisfies the strict quasivariational inequality

$$\max\left(\sup_{\alpha \in U}\{L^{\alpha}\beta(y)\}, \sup_{\zeta \in \mathcal{Z}} \beta(\Gamma(y, \zeta)) - \beta(y)\right) \leq -\delta(y) < 0, \quad y \in \mathcal{S}, \tag{12.3.23}$$

where $\delta(y) > 0$ is bounded away from 0 on compact subsets of $\mathcal{S}$.

Let u be a viscosity subsolution and v be a viscosity supersolution of (12.3.10) *and* (12.3.11) *and suppose that*

$$\lim_{|y| \to \infty} \left\{ \frac{u^+(y)}{\beta(y)} + \frac{v^-(y)}{\beta(y)} \right\} = 0. \tag{12.3.24}$$

Then

$$u(y) \leq v(y) \quad \text{for all } y \in \mathcal{S}.$$

Proof (***Sketch***) We argue by contradiction. Suppose that

$$\sup_{y \in \mathcal{S}}\{u(y) - v(y)\} > 0.$$

Then by (12.3.24) there exists $\epsilon > 0$ such that if we put

$$v_{\epsilon}(y) := v(y) + \epsilon\beta(y), \quad y \in \mathcal{S}$$

then

$$M := \sup_{y \in \mathcal{S}}\{u(y) - v_{\epsilon}(y)\} > 0.$$

For $n = 1, 2, \ldots$ and $(x, y) \in \mathcal{S} \times \mathcal{S}$ define

$$H_n(x, y) := u(x) - v(y) - \frac{n}{2}|x - y|^2 - \frac{\epsilon}{2}(\beta(x) + \beta(y))$$

and

$$M_n := \sup_{(x,y)\in\mathcal{S}\times\mathcal{S}} H_n(x, y).$$

Then by (12.3.24) we have

$$0 < M_n < \infty \quad \text{for all } n,$$

and there exists $(x^{(n)}, y^{(n)}) \in \mathcal{S} \times \mathcal{S}$ such that

$$M_n = H_n(x^{(n)}, y^{(n)}).$$

Then by Lemma 3.1 in [CIL] the following holds:

$$\lim_{n\to\infty} n|x^{(n)} - y^{(n)}|^2 = 0$$

and

$$\lim_{n\to\infty} M_n = u(\hat{y}) - v_\epsilon(\hat{y}) = \sup_{y\in\mathcal{S}}\{u(y) - v_\epsilon(y)\} = M,$$

for any limit point $\hat{y}$ of $\{y^{(n)}\}_{n=1}^{\infty}$.

Since v is a supersolution of (12.3.10), (12.3.11), and (12.3.23) holds, we see that v_ϵ is a *strict* supersolution of (12.3.10), in the sense that $\varphi = v_\epsilon$ satisfies (12.3.13) in the following strict form:

$$\max\left(\sup_{\alpha\in U}\{L^\alpha h(y_0) + f(y_0, \alpha)\},\ \mathcal{M}v_\epsilon(y_0) - v_\epsilon(y_0)\right) \le -\delta(y_0),$$

with $\delta(\cdot)$ as in (12.3.23).

By [CIL, Theorem 3.2], there exist $k \times k$ matrices $P^{(n)}$, $Q^{(n)}$ such that, if we put

$$p^{(n)} = q^{(n)} = n(x^{(n)} - y^{(n)})$$

then

$$(P^{(n)}, p^{(n)}) \in \bar{J}^{2,+}u(x^{(n)}) \quad \text{and} \quad (Q^{(n)}, q^{(n)}) \in \bar{J}^{2,-}v_\epsilon(y^{(n)})$$

and

$$\begin{bmatrix} P^{(n)} & 0 \\ 0 & -Q^{(n)} \end{bmatrix} \le 3n \begin{bmatrix} I & -I \\ -I & I \end{bmatrix},$$

in the sense that

$$\xi^T P^{(n)}\xi - \eta^T Q^{(n)}\eta \le 3n|\xi - \eta|^2 \quad \text{for all } \xi, \eta \in \mathbb{R}^k. \tag{12.3.25}$$

Since u is a subsolution we have, by Theorem 12.9,

$$\max\left(\Lambda(P^{(n)}, p^{(n)}, u, x^{(n)}), \mathcal{M}u(x^{(n)}) - u(x^{(n)})\right) \geq 0 \tag{12.3.26}$$

and since v_ϵ is a supersolution we have

$$\max\left(\Lambda(Q^{(n)}, q^{(n)}, v_\epsilon, y^{(n)}), \mathcal{M}v_\epsilon(y^{(n)}) - v_\epsilon(y^{(n)})\right) \leq 0. \tag{12.3.27}$$

By (12.3.25) we get

$$\begin{aligned}
&\Lambda(P^{(n)}, p^{(n)}, u, x^{(n)}) - \Lambda(Q^{(n)}, q^{(n)}, v_\epsilon, y^{(n)}) \\
&\leq \sup_{\alpha\in U}\left\{\sum_{i=1}^{k}(b_i(x^{(n)}, \alpha) - b_i(y^{(n)}, \alpha))(p_i^{(n)} - q_i^{(n)})\right. \\
&\qquad \left. + \frac{1}{2}\sum_{i,j=1}^{k}\left[(\sigma\sigma^T)_{ij}(x^{(n)}, \alpha) - (\sigma\sigma^T)_{ij}(y^{(n)}, \alpha)\right](P_{ij}^{(n)} - Q_{ij}^{(n)})\right\} \\
&\leq 0.
\end{aligned}$$

Therefore, by (12.3.27),

$$\Lambda(P^{(n)}, p^{(n)}, u, x^{(n)}) \leq \Lambda(Q^{(n)}, q^{(n)}, v_\epsilon, y^{(n)}) \leq 0$$

and hence, by (12.3.26),

$$\mathcal{M}u(x^{(n)}) - u(x^{(n)}) \geq 0. \tag{12.3.28}$$

On the other hand, since v_ϵ is a strict supersolution we have

$$\mathcal{M}v_\epsilon(y^{(n)}) - v_\epsilon(y^{(n)}) < -\delta \quad \text{for all } n, \tag{12.3.29}$$

for some constant $\delta > 0$.

Combining the above we get

$$M_n < u(x^{(n)}) - v_\epsilon(y^{(n)}) < \mathcal{M}u(x^{(n)}) - \mathcal{M}v_\epsilon(y^{(n)}) - \delta$$

and hence

$$\begin{aligned}
M = \lim_{n\to\infty} M_n &\leq \lim_{n\to\infty}(\mathcal{M}u(x^{(n)}) - \mathcal{M}v_\epsilon(y^{(n)}) - \delta) \\
&\leq \mathcal{M}u(\hat{y}) - \mathcal{M}v_\epsilon(\hat{y}) - \delta \\
&= \sup_{\zeta\in\mathcal{Z}}\{u(\Gamma(\hat{y}, \zeta)) + K(\hat{y}, \zeta)\} - \sup_{\zeta\in\mathcal{Z}}\{v_\epsilon(\Gamma(\hat{y}, \zeta)) + K(\hat{y}, \zeta)\} - \delta \\
&\leq \sup_{\zeta\in\mathcal{Z}}\{u(\Gamma(\hat{y}, \zeta)) - v_\epsilon(\Gamma(\hat{y}, \zeta))\} - \delta \leq M - \delta.
\end{aligned}$$

This contradiction proves Theorem 12.10. □

Theorem 12.11 (***Uniqueness of Viscosity Solutions***)
Suppose that the process $Y(t)$ has no jumps, i.e.,

$$N(\cdot,\cdot) = 0$$

and let $\beta \in C^2(\bar{\mathcal{S}})$ be as in Theorem 12.10. Then there is at most one viscosity solution φ of (12.3.10) and (12.3.11) with the property that

$$\lim_{|y|\to\infty} \frac{|\varphi(y)|}{\beta(y)} = 0. \tag{12.3.30}$$

Proof Let φ_1, φ_2 be two viscosity solutions satisfying (12.3.30). If we apply Theorem 12.10 to $u = \varphi_1$ and $v = \varphi_2$ we get

$$\varphi_1 \le \varphi_2.$$

If we apply Theorem 12.10 to $u = \varphi_2$ and $v = \varphi_1$ we get

$$\varphi_2 \le \varphi_1.$$

Hence $\varphi_1 = \varphi_2$. □

Example 12.12 (***Optimal Consumption and Portfolio with Both Fixed and Proportional Transaction Costs (2)***)

Let us return to Example 11.3. In this case (12.3.10) takes the form (11.2.21) and (11.2.22) in $\mathcal{S}^0$. For simplicity we assume Dirichlet boundary conditions, e.g., $\psi = 0$, on $\partial\mathcal{S}$. Fix $\gamma' \in (\gamma, 1)$ such that (see (5.1.9))

$$\rho > \gamma' \left[r + \frac{(\mu - r)^2}{2\sigma^2(1-\gamma)} \right]$$

and define

$$\beta(x_1, x_2) = (x_1 + x_2)^{\gamma'}. \tag{12.3.31}$$

Then with $\mathcal{M}$ as in (11.2.22) we have

$$(\mathcal{M}\beta - \beta)(x_1, x_2) \le (x_1 + x_2)^{\gamma'} \left[\left(1 - \frac{c}{x_1 + x_2} \right)^{\gamma'} - 1 \right] < 0. \tag{12.3.32}$$

Moreover, with

$$L^u\psi(x_1,x_2) := -\rho\psi(x_1,x_2) + (rx_1 - u)\frac{\partial\psi}{\partial x_1}(x_1,x_2) + \mu x_2\frac{\partial\psi}{\partial x_2}(x_1,x_2) + \frac{1}{2}\sigma^2 x_2^2\frac{\partial^2\psi}{\partial x_2^2}(x_1,x_2), \quad \psi \in C^2(\mathbb{R}^2) \tag{12.3.33}$$

we get

$$\max_{u\geq 0} L^u\beta(x_1,x_2) < 0, \tag{12.3.34}$$

and in both (12.3.32) and (12.3.34) the strict inequality is uniform on compact subsets of $\mathcal{S}^0$. The proofs of these inequalities are left as an exercise (Exercise 12.3).

We conclude that the function β in (12.3.31) satisfies the conditions (12.3.23) of Theorem 12.10. Thus by Theorem 12.11 we have in this example uniqueness of viscosity solutions φ satisfying the growth condition

$$\lim_{|(x_1,x_2)|\to\infty} (x_1+x_2)^{-\gamma'}|\varphi(x_1,x_2)| = 0. \tag{12.3.35}$$

For other results regarding uniqueness of viscosity solutions of equations associated to impulse control, stochastic control and optimal stopping for jump diffusions, we refer to [Am, AKL, BKR2, CIL, Is1, Is2, Ish, MS, AT, Ph1, JK1, FS, BCa, BCe] and the references therein.

12.4 Numerical Analysis of HJBQVI

In this section we give some insights in the numerical solution of HJBQVI. We refer, e.g., to [LST] for details on the finite difference approximations and the description of the algorithms to solve dynamic programming equations. Here we focus on the main problem which arises in the case of quasivariational inequalities, i.e., the presence of a nonexpansive operator due to the intervention operator.

12.4.1 Finite Difference Approximation

We want to solve the following HJBQVI numerically

$$\max\left(\sup_{\alpha\in U}\{L^\alpha\Phi(x) + f(x,\alpha)\}, \mathcal{M}\Phi(x) - \Phi(x)\right) = 0, \quad x \in \mathcal{S} \tag{12.4.1}$$

with boundary values

$$\Phi(x) = g(x), \quad x \in \partial\mathcal{S}, \tag{12.4.2}$$

where

$$L^{\alpha}\Phi(x) = -\rho\Phi + \sum_{i=1}^{k} b_i(x,\alpha)\frac{\partial\Phi}{\partial x_i} + \frac{1}{2}\sum_{i,j=1}^{k} a_{ij}(x,\alpha)\frac{\partial^2\Phi}{\partial x_i \partial x_j} \tag{12.4.3}$$

and

$$\mathcal{M}\Phi(x) = \sup\left\{\Phi(\Gamma(x,\zeta)) + K(x,\zeta);\, \zeta \in \mathcal{Z},\, \Gamma(x,\zeta) \in \mathcal{S}\right\}. \tag{12.4.4}$$

We have denoted here $a_{ij} := \left(\sigma\sigma^T\right)_{ij}$. We shall also write $K^{\zeta}(x)$ for $K(x,\zeta)$.

We assume that $\mathcal{S}$ is bounded, otherwise a change of variable or a localization procedure has to be performed in order to reduce to a bounded domain. Moreover we assume for simplicity that $\mathcal{S}$ is a box, i.e., a cartesian product of bounded intervals in $\mathbb{R}^k$. We can also handle Neumann type boundary conditions without additional difficulty.

We discretize (12.4.1) by using a finite difference approximation. Let δ_i denote the finite difference step in each coordinate direction and set $\delta = (\delta_1, \ldots, \delta_k)$. Denote by e_i the unit vector in the ith coordinate direction, and consider the grid $\mathcal{S}_\delta = \mathcal{S} \bigcap \prod_{i=1}^{k}(\delta_i\mathbb{Z})$. Set $\partial\mathcal{S}_\delta = \partial\mathcal{S} \bigcap \prod_{i=1}^{k}(\delta_i\mathbb{Z})$. We use the following approximations:

$$\frac{\partial\Phi}{\partial x_i}(x) \sim \frac{\Phi(x+\delta_i e_i) - \Phi(x-\delta_i e_i)}{2\delta_i} \equiv \partial_i^{\delta_i}\Phi(x) \tag{12.4.5}$$

or (see (12.4.16))

$$\frac{\partial\Phi}{\partial x_i}(x) \sim \begin{cases} \dfrac{\Phi(x+\delta_i e_i) - \Phi(x)}{\delta_i} \equiv \partial_i^{\delta_i +}\Phi(x) \text{ if } b_i(x) \geq 0, \\ \dfrac{\Phi(x) - \Phi(x-\delta_i e_i)}{\delta_i} \equiv \partial_i^{\delta_i -}\Phi(x) \text{ if } b_i(x) \leq 0. \end{cases} \tag{12.4.6}$$

$$\frac{\partial^2\Phi}{\partial x_i^2}(x) \sim \frac{\Phi(x+\delta_i e_i) - 2\Phi(x) + \Phi(x-\delta_i e_i)}{\delta_i^2} \equiv \partial_{ii}^{\delta_i}\Phi(x). \tag{12.4.7}$$

If $a_{ij}(x) \geq 0$, $i \neq j$, then

$$\begin{aligned} \frac{\partial^2\Phi}{\partial x_i \partial x_j}(x) &\sim \frac{2\Phi(x) + \Phi(x+\delta_i e_i + \delta_j e_j) + \Phi(x-\delta_i e_i - \delta_j e_j)}{2\delta_i\delta_j} \\ &\quad - \left[\frac{\Phi(x+\delta_i e_i) + \Phi(x-\delta_i e_i) + \Phi(x+\delta_j e_j) + \Phi(x-\delta_j e_j)}{2\delta_i\delta_j}\right] \\ &\equiv \partial_{ij}^{\delta_i\delta_j +}\Phi(x). \end{aligned} \tag{12.4.8}$$

If $a_{ij}(x) < 0$, $i \neq j$, then

$$\begin{aligned}\frac{\partial^2 \Phi}{\partial x_i \partial x_j}(x) &\sim -\frac{[2\Phi(x) + \Phi(x + \delta_i e_i - \delta_j e_j) + \Phi(x - \delta_i e_i + \delta_j e_j)]}{2\delta_i \delta_j} \\ &+ \frac{\Phi(x + \delta_i e_i) + \Phi(x - \delta_i e_i) + \Phi(x + \delta_j e_j) + \Phi(x - \delta_j e_j)}{2\delta_i \delta_j} \\ &\equiv \partial_{ij}^{\delta_i \delta_j -} \Phi(x). \end{aligned} \tag{12.4.9}$$

These approximations can be justified when the function Φ is smooth by Taylor expansions. Using approximations (12.4.5), (12.4.7)–(12.4.9), we obtain the following approximation problem:

$$\max\left(\sup_{\alpha \in U}\{L_\delta^\alpha \Phi_\delta(x) + f(x,\alpha)\}, \mathcal{M}_\delta \Phi_\delta(x) - \Phi_\delta(x)\right) = 0 \quad \text{for all } x \in \mathcal{S}_\delta,$$

$$\Phi_\delta(x) = g(x) \quad \text{for all } x \in \partial \mathcal{S}_\delta, \tag{12.4.10}$$

where

$$\begin{aligned} L_\delta^\alpha \Phi(x) = \Phi(x) &\left\{ \sum_{i=1}^k \frac{-a_{ii}(x,\alpha)}{\delta_i^2} + \sum_{j \neq i} \frac{|a_{ij}(x,\alpha)|}{2\delta_i \delta_j} - \rho \right\} \\ &+ \frac{1}{2} \sum_{i, \kappa = \pm 1} \Phi(x + \kappa \delta_i e_i) \left\{ \frac{a_{ii}(x,\alpha)}{\delta_i^2} - \sum_{j, j \neq i} \frac{|a_{ij}(x,\alpha)|}{\delta_i \delta_j} + \kappa \frac{b_i(x,\alpha)}{\delta_i} \right\} \\ &+ \frac{1}{2} \sum_{i \neq j, \kappa = \pm 1, \lambda = \pm 1} \Phi(x + \kappa e_i \delta_i + \lambda e_j \delta_j) \frac{a_{ij}(x,\alpha)^{[\kappa\lambda]}}{\delta_i \delta_j} \end{aligned} \tag{12.4.11}$$

and

$$\mathcal{M}_\delta \Phi_\delta(x) = \sup\left\{\Phi(\Gamma(x,\zeta)) + K(x,\zeta); \zeta \in \mathcal{Z}_\delta(x)\right\} \tag{12.4.12}$$

with

$$\mathcal{Z}_\delta(x) = \left\{\zeta \in \mathcal{Z}, \Gamma(x,\zeta) \in \mathcal{S}_\delta\right\}. \tag{12.4.13}$$

We have used here the notation

$$a_{ij}^{[\kappa\lambda]}(x,\alpha) = \begin{cases} a_{ij}^+(x,\alpha) \equiv \max(0, a_{ij}(x,\alpha)) & \text{if } \kappa\lambda = 1, \\ a_{ij}^-(x,\alpha) \equiv -\min(0, a_{ij}(x,\alpha)) & \text{if } \kappa\lambda = -1. \end{cases}$$

In (12.4.10), Φ_δ denotes an approximation of Φ at the grid points. This approximation is consistent and stable if the following condition holds: (see [LST] for a proof)

$$|b_i(x,\alpha)| \le \frac{a_{ii}(x,\alpha)}{\delta_i} - \sum_{j\neq i} \frac{|a_{ij}(x,\alpha)|}{\delta_j} \quad \text{for all } \alpha \text{ in } U,\ x \text{ in } \mathcal{S}_\delta,\ i = 1,\dots,k. \tag{12.4.14}$$

In this case ϕ_δ converges to the viscosity solution of (12.4.1) when the step δ goes to 0. This can be proved by using techniques introduced by Barles and Souganidis [BaS], provided a comparison theorem holds for viscosity sub- and supersolutions of the continuous-time problem.

If (12.4.14) does not hold but only the following weaker condition

$$0 \le \frac{a_{ii}(x,\alpha)}{\delta_i} - \sum_{j\neq i} \frac{|a_{ij}(x,\alpha)|}{\delta_j} \quad \text{for all } \alpha \text{ in } U,\ x \text{ in } \mathcal{S}_\delta,\ i = 1\dots k. \tag{12.4.15}$$

is satisfied, then it can be shown that we can also obtain a stable approximation (but of lower order) by using the one-sided approximations (12.4.6) for the approximation of the gradient instead of the centered difference (12.4.5). Instead of (12.4.11), the operator L_δ^α is then equal to

$$\begin{aligned} L_\delta^\alpha \Phi(x) = \Phi(x) &\left\{ \sum_{i=1}^{k} \frac{-a_{ii}(x,\alpha)}{\delta_i^2} + \sum_{j\neq i} \frac{|a_{ij}(x,\alpha)|}{2\delta_i\delta_j} - \frac{|b_i(x,\alpha)|}{\delta_i} - \rho \right\} \\ &+ \frac{1}{2} \sum_{i,\kappa=\pm 1} \Phi(x+\kappa\delta_i e_i) \left\{ \frac{a_{ii}(x,\alpha)}{\delta_i^2} - \sum_{j,j\neq i} \frac{|a_{ij}(x,\alpha)|}{\delta_i\delta_j} + \frac{b_i(x,\alpha)^{[\kappa]}}{\delta_i} \right\} \\ &+ \frac{1}{2} \sum_{i\neq j,\kappa=\pm 1,\lambda=\pm 1} \Phi(x+\kappa e_i\delta_i + \lambda e_j\delta_j) \frac{a_{ij}(x,\alpha)^{[\kappa\lambda]}}{\delta_i\delta_j}. \end{aligned} \tag{12.4.16}$$

By replacing the values of the function Φ_δ by their known values on the boundary, we obtain the following equation in $\mathcal{S}_\delta$:

$$\max\left(\sup_{\alpha\in U}\{\bar{L}_\delta^\alpha \Phi_\delta(x) + f_\delta(x,\alpha)\}, \mathcal{M}_\delta\Phi_\delta(x) - \Phi_\delta(x) \right) = 0, \quad x \in \mathcal{S}_\delta, \tag{12.4.17}$$

where $\bar{L}_\delta^\alpha$ is a square $N_\delta \times N_\delta$ matrix, obtained by retrieving the first and last column from L_δ^α, $N_\delta = \mathrm{Card}(\mathcal{S}_\delta)$, i.e., the number of points of the grid, and $f_\delta(x,\alpha)$ (which will also be denoted by $f_\delta^\alpha(x)$) takes into account the boundary values.

12.4.2 A Policy Iteration Algorithm for HJBQVI

When the stability conditions (12.4.14) or (12.4.15) hold, then the matrix $\bar{L}_\delta^\alpha$ is diagonally dominant, i.e.,

$$(\bar{L}_\delta^\alpha)_{ij} \geq 0 \text{ for } i \neq j \quad \text{and} \quad \sum_{j=1}^{N_\delta} (\bar{L}_\delta^\alpha)_{ij} \leq -\rho < 0 \quad \text{for all } i = 1, \ldots, N_\delta.$$

Now let h be a positive number such that

$$h \leq \min_i \frac{1}{|(\bar{L}_\delta^\alpha)_{ii} + \rho|} \tag{12.4.18}$$

and let I_δ denote the $N_\delta \times N_\delta$ identity matrix. It is easy to check that the matrix

$$P_\delta^\alpha := I_\delta + h(\bar{L}_\delta^\alpha + \rho I_\delta)$$

is sub-Markovian, i.e., $(P_\delta^\alpha)_{ij} \geq 0$ for all i, j and $\sum_{j=1}^{N_\delta} (P_\delta^\alpha)_{ij} \leq 1$ for all i. Consequently (12.4.17) can be rewritten as

$$\max\left(\sup_{\alpha\in U}\left\{\frac{1}{h}\Big(P_\delta^\alpha \Phi_\delta(x) - (1+\rho h)\Phi_\delta(x)\Big) + f_\delta^\alpha(x)\right\}, \mathcal{M}_\delta \Phi_\delta(x) - \Phi_\delta(x)\right) = 0, \tag{12.4.19}$$

which is equivalent to

$$\Phi_\delta(x) = \max\left(\sup_{\alpha\in U} \mathcal{L}_\delta^\alpha \Phi_\delta(x), \sup_{\zeta\in\mathcal{Z}_\delta(x)} B^\zeta \Phi_\delta(x)\right), \tag{12.4.20}$$

where

$$\mathcal{L}_\delta^\alpha \Phi(x) := \frac{P_\delta^\alpha \Phi(x) + h f_\delta^\alpha(x)}{1+\rho h}, \tag{12.4.21}$$

$$B^\zeta \Phi(x) := \Phi(\Gamma(x,\zeta)) + K^\zeta(x). \tag{12.4.22}$$

Let $\mathcal{P}(\mathcal{S}_\delta)$ denote the set of all subsets of $\mathcal{S}_\delta$ and for (T, α, ζ) in $\mathcal{P}(\mathcal{S}_\delta) \times U \times \mathcal{Z}_\delta$, denote by $\mathcal{O}_{T,\alpha,\zeta}$ the operator:

$$\mathcal{O}_{T,\alpha,\zeta} v(x) := \begin{cases} \mathcal{L}_\delta^\alpha v(x) & \text{if } x \in \mathcal{S}_\delta \backslash T, \\ B^\zeta v(x) & \text{if } x \in T. \end{cases} \tag{12.4.23}$$

Problem (12.4.20) is equivalent to the fixed point problem

$$\Phi_\delta(x) = \sup_{T\in\mathcal{P}(\mathcal{S}_\delta), \alpha\in U, \zeta\in\mathcal{Z}_\delta} \mathcal{O}_{T,\alpha,\zeta} \Phi_\delta(x).$$

We define $\mathbf{T}_{\text{ad}}$ as

$$\mathbf{T}_{\text{ad}} := \mathcal{P}(\mathcal{S}_\delta) \backslash \mathcal{S}_\delta$$

and restrict ourselves to the following problem

$$\Phi_\delta(x) = \sup_{T\in\mathbf{T}_{\rm ad},\alpha\in U,\zeta\in\mathcal{Z}_\delta} \mathcal{O}_{T,w,z}\Phi_\delta(x) =: \mathcal{O}\Phi_\delta(x). \tag{12.4.24}$$

In other words, it is not admissible to make interventions at all points of $\mathcal{S}_\delta$ (i.e., the continuation region is never the empty set). We can always assume that we order the points of the grid in such a way that it is not admissible to intervene at $x_1 \in \mathcal{S}_\delta$.

The operator $\mathcal{L}_\delta^\alpha$ is contractive (because $\|P\|_\infty \le 1$ and $rh > 0$) and satisfies the discrete maximum principle, i.e.,

$$\mathcal{L}_\delta^\alpha v_1 - \mathcal{L}_\delta^\alpha v_2 \le v_1 - v_2 \Rightarrow v_1 - v_2 \ge 0. \tag{12.4.25}$$

(If v is a function from S_δ into $\mathbb{R}$, $v \ge 0$ means $v(x) \ge 0$ for all $x \in \mathcal{S}_\delta$.)

The operator B^ζ is nonexpansive and we need some additional hypothesis in order to be able to use a policy iteration algorithm for computing a solution of (12.4.21). We assume

There exists an integer function $\sigma : \{1, 2, \dots, N_\delta\} \times \mathcal{Z}_\delta \to \{1, 2, \dots, N_\delta\}$ such that for all $\zeta \in \mathcal{Z}_\delta$ and all $i = 1, \dots, N_\delta$

$$\Gamma(x_i, \zeta) = x_{\sigma(i,\zeta)} \ \text{ with } \sigma(i, \zeta) < i. \tag{12.4.26}$$

The operator B_ζ defined in (12.4.22) can be rewritten as

$$B^\zeta v = \mathbf{B}^\zeta v + K^\zeta,$$

where $(\mathbf{B}^\zeta, \zeta \in \mathcal{Z}_\delta)$ is a family of $N_\delta \times N_\delta$ Markovian matrices (except for the first row) defined by: $\mathbf{B}^z_{i,j} = 1$ if $j = \sigma(i, z)$ and $i \ne 1$, and 0 elsewhere.

Let $\zeta(\cdot)$ be a feedback Markovian control from $\mathcal{S}_\delta$ into $\mathcal{Z}_\delta$, and define the function $\bar\sigma$ on $\mathcal{S}_\delta$ by $\bar\sigma(x) := \sigma(x, \zeta(x))$. Condition (12.4.26) implies that the pth composition of $\bar\sigma$ starting in $T \in \mathbf{T}_{\rm ad}$ will end up in $\mathcal{S}_\delta \backslash T$ after a finite number of iterations.

We can now consider the following *Howard* or *policy iteration* algorithm to solve problem (12.4.20) in the finite set $\mathcal{S}_\delta$. It consists of constructing two sequences of feedback Markovian policies $\{(T_k, \alpha_k, \zeta_k), k \in \mathbb{N}\}$ and functions $\{v_k, k \in \mathbb{N}\}$ as follows. Let v_0 be a given initial function in $\mathcal{S}_\delta$. For $k \ge 0$ we do the following iterations:

- (step $2k$) Given v_k, compute a feedback Markovian admissible policy $(T_{k+1}, \alpha_{k+1}, \zeta_{k+1})$ such that

$$(T_{k+1}, \alpha_{k+1}, \zeta_{k+1}) \in \operatorname*{Argmax}_{T,\alpha,\zeta} \{\mathcal{O}_{T,\alpha,\zeta} v_k\}. \tag{12.4.27}$$

In other words

$$\alpha_{k+1}(x) \in \operatorname*{Argmax}_{\alpha\in U} \mathcal{L}_\delta^\alpha v_k(x); \quad \text{for all } x \text{ in } \mathcal{S}_\delta,$$
$$\zeta_{k+1}(x) \in \operatorname*{Argmax}_{\beta\in\mathcal{Z}_\delta} B_\delta^\zeta v_k(x); \quad \text{for all } x \text{ in } \mathcal{S}_\delta,$$
$$T_{k+1} = \{x \in \mathcal{S}_\delta, \mathcal{L}_\delta^{\alpha_{k+1}(x)} v_k(x) > B_\delta^{\zeta_{k+1}(x)} v_k(x)\}.$$

- (step $2k+1$) Compute v_{k+1} as the solution of

$$v_{k+1} = \mathcal{O}_{T_{k+1},\alpha_{k+1},\zeta_{k+1}} v_{k+1}. \tag{12.4.28}$$

Set $k \leftarrow k+1$ and go to step $2k$.

It can be proved that if (12.4.15), (12.4.18), and (12.4.26) hold, then the sequence $\{v_k\}$ converges to the solution Φ_δ of (12.4.20) and the sequence $\{(T_k, \alpha_k, \zeta_k)\}$ converges to the optimal feedback Markovian strategy. See [CMS] for a proof and [BT] for similar problems. For more information on the Howard algorithm, we refer to [Pu, LST]. For complements on numerical methods for HJB equations we refer, e.g., to [KD, LST].

Example 12.13 (***Optimal Consumption and Portfolio with Both Fixed and Proportional Transaction Costs (3)***) We go back to Example 12.12. We want to solve (11.2.21) numerically. We assume now that $\mathcal{S} = (0,l) \times (0,l)$ with $l > 0$, and that the following boundary conditions hold:

$$\psi(0,x_2) = \psi(x_1,0) = 0,$$
$$\frac{\partial\psi}{\partial x_1}(l,x_2) = \frac{\partial\psi}{\partial x_2}(x_1,l) = 0 \quad \text{for all } (x_1,x_2) \text{ in } (0,l)\times(0,l).$$

Moreover we assume that the consumption is bounded by $u_{\max} > 0$ so that $U = [0, u_{\max}]$. Let $\delta > 0$ be a positive step and let $\mathcal{S}_\delta = \{(i\delta, j\delta), i, j \in \{1,\dots,N\}\}$ be the finite difference grid (we suppose that $N = l/\delta$ is an integer). We denote by ψ_δ the approximation of ψ on the grid. We approximate the operator L^u defined in (12.3.33) by the following finite difference operator on $\mathcal{S}_\delta$:

$$L_\delta^u\psi := -\rho\psi + rx_1\partial_1^{\delta+}\psi + \mu x_2\partial_2^{\delta+}\psi - u\partial_1^{\delta-}\psi + \frac{1}{2}\sigma^2x_2^2\partial_{22}^{\delta_2+}\psi$$

and set the following boundary values:

$$\psi_\delta(0,x_2) = \psi_\delta(x_1,0) = 0,$$
$$\psi_\delta(l-\delta,x_2) = \psi_\delta(l,x_2),$$
$$\psi_\delta(x_1,l-\delta) = \psi_\delta(x_1,l).$$

We then obtain a stable approximation. Take now

$$h \leq \frac{rx_1}{\delta} + \frac{\mu x_2}{\delta} + \left(\frac{\sigma x_2}{\delta}\right)^2 + \frac{u_{\max}}{\delta}.$$

We obtain a problem of the form (12.4.20). In order to be able to apply the Howard algorithm described above, it remains to check that (12.4.26) holds. This is indeed the case since a finite number of transactions brings the state to the continuation region. The details are left as an exercise.

This problem is solved in [CØS] by using another numerical method based on the iterative methods of Chap. 10.

12.5 Exercises

Exercise* 12.1 Let $k > 0$ be a constant and define

$$G(x) = \begin{cases} K|x| & \text{for } -\frac{1}{K} \leq x \leq \frac{1}{K}, \\ 1 & \text{for } |x| > \frac{1}{K}. \end{cases}$$

Solve the optimal stopping problem

$$\Phi(s,x) = \sup_{\tau \geq 0} E^x\left[\mathrm{e}^{-\rho(s+\tau)} G(B(\tau))\right],$$

where $B(t)$ is a one-dimensional Brownian motion starting at $x \in \mathbb{R}$. Distinguish between the two cases

(a) $K \leq \sqrt{2\rho}/z$, where $z > 0$ is the unique positive solution of the equation

$$\operatorname{tgh}(z) = \frac{1}{z},$$

and

$$\operatorname{tgh}(z) = \frac{\mathrm{e}^z - \mathrm{e}^{-z}}{\mathrm{e}^z + \mathrm{e}^{-z}}.$$

(b) $K > \sqrt{2\rho}/z$.

Exercise* 12.2 Assume that the state $X(t) = X^{(w)}(t)$ at time t obtained by using a combined control $w = (u, v)$, where $u = u(t,\omega) \in \mathbb{R}$ and $v = (\tau_1, \tau_2, \ldots ; \zeta_1, \zeta_2, \ldots)$ with $\zeta_i \in \mathbb{R}$ given by

$$\begin{aligned} \mathrm{d}X(t) &= u(t)\mathrm{d}t + dB(t) + \int_{\mathbb{R}} z\tilde{N}(\mathrm{d}t, \mathrm{d}z), \quad \tau_i \leq t < \tau_{i+1}, \\ X(\tau_{i+1}) &= X(\tau_{i+1}^-) + \Delta_N X(\tau_{i+1}) + \zeta_{i+1}, \quad X(0^-) = x \in \mathbb{R}. \end{aligned}$$

Assume that the cost of applying such a control is

$$J^{(w)}(s,x) = E^x\left[\int_0^\infty \mathrm{e}^{-\rho(s+t)}(X^{(w)}(t)^2 + \theta u(t)^2)\mathrm{d}t + c\sum_i \mathrm{e}^{-\rho(s+\tau_i)}\right],$$

where ρ, θ, and c are positive constants. Consider the problem to find $\Phi(s,x)$ and $w^* = (u^*, v^*)$ such that

$$\Phi(s,x) = \inf_w J^{(w)}(s,x) = J^{(w^*)}(s,x). \tag{12.5.1}$$

Let

$$\Phi_1(s,x) = \inf_u J^{(u,0)}(s,x)$$

be the value function if we do not allow any impulse control (i.e., $v = 0$) and let

$$\Phi_2(s,x) = \inf_v J^{(0,v)}(s,x)$$

be the value function if u is fixed equal to 0, and only impulse controls are allowed. (See Exercises 5.4 and 9.1, respectively.)

Prove that for $i = 1, 2$, there exists $(s,x) \in \mathbb{R} \times \mathbb{R}$ such that

$$\Phi(s,x) < \Phi_i(s,x).$$

In other words, no matter how the positive parameter values ρ, θ, and c are chosen it is never optimal for the problem (12.5.1) to choose $u = 0$ or $v = 0$ (compare with Exercise 11.2).

[Hint: Use Theorem 12.8].

Exercise 12.3 Prove the inequalities (12.3.32) and (12.3.34) and verify that the inequalities hold uniformly on compact subsets of $\mathcal{S}^0$.

Chapter 13
Optimal Control of Stochastic Partial Differential Equations and Partial (Noisy) Observation Control



13.1 A Motivating Example

Example 13.1 Suppose the density $Y(t,x)$ of a fish population at time $t \in [0,T]$ and at the point $x \in D \subset \mathbb{R}^n$ (where D is a given open set) is modeled by a stochastic *partial* differential equation (SPDE for short) of the form

$$\begin{aligned} \mathrm{d}Y(t,x) &= \left[\frac{1}{2}\Delta Y(t,x) + \alpha Y(t,x) - u(t,x)\right]\mathrm{d}t \\ &\quad + \beta Y(t,x)\mathrm{d}B(t) + Y(t^-,x)\int_{\mathbb{R}} \zeta \tilde{N}(\mathrm{d}t,\mathrm{d}\zeta);\ (t,x) \in (0,T)\times D, \end{aligned} \tag{13.1.1}$$

where we assume that $\zeta \geq -1+\varepsilon$ a.s. $\nu(\mathrm{d}\zeta)$ for some constant $\varepsilon > 0$. The boundary conditions are:

$$Y(0,x) = \xi(x);\ x \in D \tag{13.1.2}$$

$$Y(t,x) = \eta(t,x);\ (t,x) \in [0,T)\times \partial D. \tag{13.1.3}$$

Here $\mathrm{d}Y(t,x) = d_t Y(t,x)$ is the differential with respect to t, and

$$\Delta = \Delta_x = \sum_{i=1}^{n} \frac{\partial^2}{\partial x_i^2}$$

is the Laplacian operator acting on the variable x. We assume that α and β are constants and $\xi(x)$ and $\eta(t,x)$ are given deterministic functions. The process $u(t,x) \geq 0$ is our *control*, representing the harvesting rate density at (t,x). Equation (13.1.1) is an example of a *reaction-diffusion*. With $u = 0$ and without the Δ-term, the equation reduces to a geometric Lévy equation describing the growth with respect to t. The Δ-term models the *diffusion in space* of the population.

Let $\mathcal{A}$ be a family of *admissible controls*, contained in the set of all $\mathbb{F}$-adapted processes $u(t,x,\omega)$ such that (13.1.1)–(13.1.3) has a unique solution $Y(t,x)$.

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Suppose the total expected utility from the harvesting rate $u(\cdot)$ and the corresponding terminal density $Y(T,x)$ is given by

$$J(u) = E\left[\int_0^T \left(\int_D \frac{u^\gamma(t,x)}{\gamma}\mathrm{d}x\right)\mathrm{d}t + \rho\int_D Y(T,x)\mathrm{d}x\right], \tag{13.1.4}$$

where $\gamma \in (0,1)$ and $\rho > 0$ are constants.

We want to find $u^* \in \mathcal{A}$ such that

$$\sup_{u\in\mathcal{A}} J(u) = J(u^*). \tag{13.1.5}$$

Such a control u^* is called an *optimal control*.

This is an example of a stochastic control problem for *random jump fields*, i.e., random fields which are solutions of stochastic partial differential equations driven by Brownian motions and Poisson random measures.

How do we solve problem (13.1.5)? It is possible to use a dynamic programming approach and formulate an infinite-dimensional HJB equation for the value function (see [Mort]) but this HJB equation is difficult to use. Therefore, we will instead formulate a maximum principle for such problems (Theorem 13.2). This principle can be used to solve such stochastic control problems in some cases. We will illustrate this by solving (13.1.5) by this method.

Our presentation is an extension of the presentation given in the previous edition of the book. It is based on the recent paper [DØ4]. See also [Ø1, ØPZ].

13.2 The Maximum Principle

We first give a general formulation of the stochastic control problem we consider. Suppose that the state $Y(t,x) = Y^{(u)}(t,x)$ at (t,x) is described by a stochastic partial differential equation of the form

$$\begin{aligned}\mathrm{d}Y(t,x) &= \left[L_{u(t,x)}Y(t,x) + b(t,x,Y(t,x),u(t,x))\right]\mathrm{d}t\\ &\quad + \sigma(t,x,Y(t,x),u(t,x))\mathrm{d}B(t)\\ &\quad + \int_{\mathbb{R}^k}\theta(t,x,Y(t,x),u(t,x),\zeta)\tilde{N}(\mathrm{d}t,\mathrm{d}\zeta);\ (t,x)\in(0,T)\times D\end{aligned} \tag{13.2.1}$$

with boundary conditions

$$Y(0,x) = \xi(x);\ x\in D \tag{13.2.2}$$

$$Y(t,x) = \eta(t,x);\ (t,x)\in[0,T]\times\partial D \tag{13.2.3}$$

The operator L_u is a linear integro-differential operator acting on x, with parameter u, and the expression $L_{u(t,x)}Y(t,x)$ means $L_u Y(t,x)|_{u=u(t,x)}$.

The Eq. (13.2.1) for Y is interpreted in the weak (variational) sense, i.e., $Y(t,\cdot)$ satisfies the equation

$$\begin{aligned}(Y(t,\cdot),\phi)_{L^2(D)} &= (a,\phi)_{L^2(D)} + \int_0^t (Y(s,\cdot), L_u^*\phi)_{L^2(D)}\mathrm{d}s \\ &+ \int_0^t (b(s,Y(s,\cdot),\cdot),\phi)_{L^2(D)}\mathrm{d}s + \int_0^t (\sigma(s,Y(s,\cdot),\cdot),\phi)_{L^2(D)}\mathrm{d}B(s) \\ &+ \int_0^t \int_{\mathbb{R}^k} \gamma(s,Y(s,\cdot),\zeta,\cdot),\phi)_{L^2(D)}\tilde{N}(\mathrm{d}s,\mathrm{d}\zeta),\end{aligned} \tag{13.2.4}$$

for all smooth functions ϕ with compact support in D. Here

$$(\psi,\phi)_{L^2(D)} = \int_D \psi(x)\phi(x)\mathrm{d}x \tag{13.2.5}$$

is the L^2 inner product on D and L_u^* is the adjoint of the operator L_u, in the sense that

$$(L_u\psi,\phi)_{L^2(D)} = (\psi, L_u^*\phi)_{L^2(D)} \tag{13.2.6}$$

for all smooth L^2 functions ψ, ϕ with compact support in D. It is known that the Itô formula can be applied to such SPDEs. See [Pa1, PrR].

The coefficients $b : [0,T]\times D\times\mathbb{R}^n\times V \to \mathbb{R}$ and $\sigma : [0,T]\times D\times\mathbb{R}^n\times V \to \mathbb{R}$ and $\theta : [0,T]\times D\times\mathbb{R}^n\times V\times\mathbb{R}^k \to \mathbb{R}$ are given functions and $V \subseteq \mathbb{R}^\ell$ is a given closed set of admissible control values.

Let $f : [0,T]\times D\times\mathbb{R}^n\times V \to \mathbb{R}$ and $g : D\times\mathbb{R}^n \to \mathbb{R}$ be a given *profit rate* function and *bequest rate* function, respectively. Let $\mathcal{A}$ be a given family of *admissible* controls, contained in the set of $\mathbb{F}_t$-adapted right-continuous stochastic processes $u(t,x) \in U$ such that (13.2.1)–(13.2.3) has a unique solution $Y(t,x)$ and such that

$$E\left[\int_0^T \left(\int_D \left|f(t,x,Y(t,x),u(t,x))\right|\mathrm{d}x\right)\mathrm{d}t + \int_D \left|g(x,Y(T,x))\right|\mathrm{d}x\right] < \infty. \tag{13.2.7}$$

If $u \in \mathcal{A}$ we define its *performance functional* $J(u)$ by

$$J(u) = E\left[\int_0^T \left(\int_D f(t,x,Y(t,x),u(t,x))\mathrm{d}x\right)\mathrm{d}t + \int_D g(x,Y(T,x))\mathrm{d}x\right]. \tag{13.2.8}$$

Problem 13.1 Find $u^* \in \mathcal{A}$ such that

$$\sup_{u\in\mathcal{A}} J(u) = J(u^*). \tag{13.2.9}$$

Such a process u^* is called an it optimal control (if it exists). The number

$$J^* := \sup_{u \in \mathcal{A}} J(u) \tag{13.2.10}$$

is called the *value* of this problem.

We now state the *maximum principle* for this problem. Define the *Hamiltonian* $H : [0, T] \times D \times \mathbb{R}^n \times \mathcal{D} \times V \times \mathbb{R} \times \mathbb{R} \times \mathcal{R} \times \Omega \to \mathbb{R}$ by

$$\begin{aligned} &H(t, x, y, \varphi, u, p, q, r) = H(t, x, y, \varphi, u, p, q, r, \omega) \\ &= f(t, x, y, u) + [L_u(\varphi) + b(t, x, y, u)]p \\ &+ \sigma(t, x, y, u)q + \int_{\mathbb{R}^k} \gamma(t, x, y, u, \zeta) r(\zeta) \nu(\mathrm{d}\zeta). \end{aligned} \tag{13.2.11}$$

Here $\mathcal{D}$ denotes the domain of definition for the operator L_u, while $\mathcal{R}$ denotes the set of all functions $r(\cdot) : \mathbb{R} \to \mathbb{R}$ such that the last integral above converges. We assume that $\mathcal{D}$ is a Banach space. The quantities p, q, $r(\cdot)$ are called the *adjoint variables*. We define the *adjoint processes* $p(t, x)$, $q(t, x)$, $r(t, x, \zeta)$ as the solution of the backward stochastic partial differential equation (BSPDE)

$$\begin{cases} \mathrm{d}p(t, x) = -[L^*_{u(t,x)} p(t, x) + \dfrac{\partial H}{\partial y}(t, x)]\mathrm{d}t + q(t, x)\mathrm{d}B(t) \\ \quad + \displaystyle\int_{\mathbb{R}} r(t, x, \zeta)\tilde{N}(\mathrm{d}t, \mathrm{d}\zeta); \ (t, x) \in (0, T) \times D \\ p(T, x) = \dfrac{\partial g}{\partial y}(x, Y(T, x)); \quad x \in D \\ p(t, x) = 0; \quad (t, x) \in [0, T] \times \partial D, \end{cases} \tag{13.2.12}$$

where

$$\frac{\partial H}{\partial y}(t, x) = \frac{\partial H}{\partial y}(t, x, y, Y(t, .), u(t, x), p(t, x), q(t, x), r(t, x, .))|_{y=Y(t,x)}. \tag{13.2.13}$$

Note that for fixed t, u, p, q, r we can regard

$$\varphi \mapsto h(\varphi)(x) := H(t, x, \varphi(x), \varphi, u, p, q, r) \tag{13.2.14}$$

as a map from $\mathcal{D}$ into $\mathbb{R}$. The Fréchet derivative at φ of this map is the linear operator $\nabla_\varphi h$ on $\mathcal{D}$ given by

$$\langle \nabla_\varphi h, \psi \rangle = \nabla_\varphi h(\psi) = L_u(\psi)(x)p + \psi(x)\frac{\partial H}{\partial y}(t, x, y, \varphi, u, p, q, r)|_{y=\varphi(x)}; \ \psi \in \mathcal{D}. \tag{13.2.15}$$

For simplicity of notation, if there is no risk of confusion, we will denote h by H from now on.

The following result is an extension to control-dependent integro-differential operators taken from [DØ4]. For earlier, related results see [Ø1, ØPZ].

Theorem 13.2 (***A Sufficient Maximum Principle for Random Jump Fields***) *[DØ4] Let $\hat{u} \in \mathcal{A}$ with associated solution $\hat{Y}(t,x), \hat{p}(t,x), \hat{q}(t,x), \hat{r}(t,x,\zeta)$ of (13.2.1) and (13.2.12). Assume that the following hold:*

$$y \to g(x,y) \text{ is concave for all } x \tag{13.2.16}$$

$$(\varphi, u) \to H(t,x,\varphi(x),\varphi,u,\widehat{p}(t,x,z),\widehat{q}(t,x,z),\hat{r}(t,x,z,\zeta)) \tag{13.2.17}$$

$$\text{is concave for all } t,x,\zeta \tag{13.2.18}$$

$$\begin{aligned}&\sup_{w\in\mathbb{U}} H\big(t,x,\widehat{Y}(t,x),\widehat{Y}(t,\cdot)(x),w,\widehat{p}(t,x),\widehat{q}(t,x),\hat{r}(t,x,\zeta)\big)\\ &\quad = H\big(t,x,\widehat{Y}(t,x),\widehat{Y}(t,\cdot,)(x),\widehat{u}(t,x),\widehat{p}(t,x),\widehat{q}(t,x),\hat{r}(t,x,\zeta)\big) \text{ for all } t,x,\zeta.\end{aligned} \tag{13.2.19}$$

Then $\widehat{u}(\cdot,\cdot)$ is an optimal insider control for Problem 13.1.

Proof By considering an increasing sequence of stopping times τ_n converging to T, we may assume that all local integrals appearing in the computations below are martingales and hence have expectation 0. See [ØS2]. We omit the details.
Choose arbitrary $u(.,.) \in \mathcal{A}$, and let the corresponding solution of (13.2.1) and (13.2.12) be $Y(t,x)$ and $p(t,x)$, $q(t,x)$, $r(t,x,\zeta)$, *respectively*. For simplicity of notation we write $f = f(t,x,Y(t,x),u(t,x))$, $\widehat{f} = f(t,x,\widehat{Y}(t,x),\widehat{u}(t,x))$ and similarly with $b, \widehat{b}, \sigma, \widehat{\sigma}$ and so on.
Moreover put

$$\hat{H}(t,x) = H(t,x,\widehat{Y}(t,x),\widehat{Y}(t,\cdot)(x),\widehat{u}(t,x),\widehat{p}(t,x),\widehat{q}(t,x),\widehat{r}(t,x,.)) \tag{13.2.20}$$

and

$$H(t,x) = H(t,x,Y(t,x),Y(t,\cdot)(x),u(t,x),\widehat{p}(t,x),\widehat{q}(t,x),\widehat{r}(t,x,.)) \tag{13.2.21}$$

In the following we write $\widetilde{f} = f - \widehat{f}$, $\widetilde{b} = b - \widehat{b}$, $\widetilde{Y} = Y - \widehat{Y}$.
Consider

$$J(u) - J(\widehat{u}) = I_1 + I_2,$$

where

$$I_1 = E\left[\int_0^T \left(\int_D \{f(t,x) - \widehat{f}(t,x)\}\mathrm{d}x\right)\mathrm{d}t\right], \quad I_2 = E\left[\int_D \{g(x) - \hat{g}(x)\}\mathrm{d}x\right]. \tag{13.2.22}$$

By the definition of H we have

$$I_1 = E\Big[\int_0^T \int_D \{H(t,x) - \widehat{H}(t,x) - \widehat{p}(t,x)[L_u Y(t,x) - L_{\hat{u}}\widehat{Y}(t,x) + \widetilde{b}(t,x)] - \widehat{q}(t,x)\widetilde{\sigma}(t,x) - \int_{\mathbb{R}} \hat{r}(t,x,\zeta)\tilde{\gamma}(t,x,\zeta)\nu(\mathrm{d}\zeta)\}\mathrm{d}x\mathrm{d}t\Big]. \tag{13.2.23}$$

Since g is concave with respect to y we have

$$(g(x, Y(T,x)) - g(x, \hat{Y}(T,x))) \leq \frac{\partial g}{\partial y}(x, \hat{Y}(T,x))(Y(T,x) - \hat{Y}(T,x)), \tag{13.2.24}$$

and hence

$$\begin{aligned} I_2 &\leq E\Big[\int_D \frac{\partial g}{\partial y}(x, \widehat{Y}(T,x))\tilde{Y}(T,x)\mathrm{d}x\Big] = E\Big[\int_D \widehat{p}(T,x)\widetilde{Y}(T,x)\mathrm{d}x\Big] \qquad (13.2.25)\\ &= E\Big[\int_D \Big(\int_0^T \widehat{p}(t,x)d\widetilde{Y}(t,x) + \int_0^T \widetilde{Y}(t,x)d\widehat{p}(t,x) + \int_0^T \mathrm{d}[\hat{p}, \tilde{Y}]_t\Big)\mathrm{d}x\Big]\\ &= E\Big[\int_D \int_0^T \{\widehat{p}(t,x)[L_u Y(t,x) - L_{\hat{u}}\widehat{Y}(t,x) + \widetilde{b}(t,x)] - \widetilde{Y}(t,x)[L^*_{\hat{u}}\widehat{p}(t,x) + \frac{\partial \widehat{H}(t,x)}{\partial y}\Big] + \widetilde{\sigma}(t,x)\widehat{q}(t,x) + \int_{\mathbb{R}} \tilde{\gamma}(t,x,\zeta)\hat{r}(t,x,\zeta)\nu(\mathrm{d}\zeta)\}\mathrm{d}t\mathrm{d}x\Big]. \end{aligned}$$

where

$$\frac{\partial \widehat{H}(t,x)}{\partial y} = \frac{\partial H}{\partial y}(t,x,y,\widehat{Y}(t,\cdot)(x), \hat{u}(t,x), \hat{p}(t,x), \hat{q}(t,x), \hat{r}(t,x,.))\,|_{y=\widehat{Y}(t,x)}\,. \tag{13.2.26}$$

By a slight extension of (13.2.6) we get

$$\int_D \widetilde{Y}(t,x)L^*_{\hat{u}}\widehat{p}(t,x)\mathrm{d}x = \int_D \widehat{p}(t,x)L_{\hat{u}}\widetilde{Y}(t,x)\mathrm{d}x. \tag{13.2.27}$$

Therefore, adding (13.2.23)–(13.2.25) and using (13.2.27) we get,

$$J(u) - J(\widehat{u}) \leq E\Big[\int_D \Big(\int_0^T \Big\{H(t,x) - \hat{H}(t,x) - \Big[\hat{p}(t,x)L_{\hat{u}}(\tilde{Y})(t,x) + \tilde{Y}(t,x)\frac{\partial \hat{H}(t,x)}{\partial y}\Big]\Big\}\mathrm{d}t\Big)\mathrm{d}x\Big]. \tag{13.2.28}$$

Hence

$$J(u) - J(\widehat{u}) \leq E\left[\int_D \left(\int_0^T \{H(t,x) - \hat{H}(t,x) - \nabla_{\hat{Y}}\widehat{H}(\tilde{Y})(t,x)\}\mathrm{d}t\right)\mathrm{d}x\right] \tag{13.2.29}$$

where

$$\nabla_{\hat{Y}}\widehat{H}(\tilde{Y}) = \nabla_{\varphi}\widehat{H}(\tilde{Y})|_{\varphi=\hat{Y}} \tag{13.2.30}$$

By the concavity assumption of H in (φ, u) we have:

$$H(t,x) - \hat{H}(t,x) \leq \nabla_{\hat{Y}}\widehat{H}(Y - \hat{Y})(t,x) + \frac{\partial\widehat{H}}{\partial u}(t,x)(u(t,x) - \hat{u}(t,x)), \tag{13.2.31}$$

and the maximum condition implies that

$$\frac{\partial\widehat{H}}{\partial u}(t,x)(u(t,x) - \hat{u}(t,x)) \leq 0. \tag{13.2.32}$$

Hence by (13.2.29) we get $J(u) \leq J(\hat{u})$. Since $u \in \mathcal{A}$ was arbitrary, this shows that $\hat{u}$ is optimal. □

In many cases the Hamiltonian

$$h(\varphi, u) := H(t, x, \varphi(x), \varphi, u, \hat{p}(t,x), \hat{q}(t,x), \hat{r}(t,x,\cdot))$$

is not concave in both variables φ, u.

In such cases it might be useful to replace the concavity condition in (φ, u) (see (13.2.18)) by a weaker condition, called the *Arrow condition*, which is the following:

The Arrow condition. *For each fixed t, x the function*

$$\hat{h}(\varphi) := \max_{v \in V} H(t, x, \varphi(x), \varphi, v, \hat{p}(t,x), \hat{q}(t,x), \hat{r}(t,x,\cdot)) \tag{13.2.33}$$

exists and is a concave function of φ.

We then get the following extension of Theorem 13.2:

Corollary 13.3 (***Strengthened Maximum Principle***) *Let $\hat{u}(t,x)$, $\hat{Y}(t,x)$, $\hat{p}(t,x)$, $\hat{q}(t,x)$ and $\hat{r}(t,x,\cdot)$ be as in Theorem 13.2. Assume that $g(x,y)$ is concave in y for each x and that the Arrow condition (13.2.33) and the maximum condition (13.2.19) hold. Then $\hat{u}(t,x)$ is an optimal control for the stochastic control problem (13.2.9).*

Proof We proceed as in the proof of Theorem 13.2, up to and including (13.2.30). Then to obtain

$$H - \hat{H} - \nabla_{\hat{Y}}\widehat{H}(\tilde{Y}) \leq 0 \tag{13.2.34}$$

we note that

$$H - \hat{H} - \nabla_{\tilde{Y}}\widehat{H}(\tilde{Y}) = h(\varphi, u) - h(\hat{\varphi}, \hat{u}) - \langle \nabla_{\varphi} h, \tilde{Y} \rangle. \tag{13.2.35}$$

This is ≤ 0 by the same argument as in the deterministic case. See [SeSy], Theorem 5, p. 107–108. For completeness we give the details:

Note that by the maximum condition (13.2.19) we have

$$h(\hat{\varphi}, \hat{u}) = \hat{h}(\hat{\varphi}). \tag{13.2.36}$$

Moreover, by definition of $\hat{h}$,

$$h(\varphi, u) \leq \hat{h}(\varphi) \text{ for all } \varphi, u. \tag{13.2.37}$$

Therefore, subtracting (13.2.36) from (13.2.37) gives

$$h(\varphi, u) - h(\hat{\varphi}, \hat{u}) \leq \hat{h}(\varphi) - \hat{h}(\hat{\varphi}) \quad \text{for all } \varphi, u. \tag{13.2.38}$$

Accordingly, to prove (13.2.34) it suffices to prove that (see (13.2.35))

$$\hat{h}(\varphi) - \hat{h}(\hat{\varphi}) - \langle \nabla_{\hat{\varphi}} h, \tilde{Y} \rangle \leq 0. \tag{13.2.39}$$

To this end, note that since the function $\hat{h}(\varphi)$ is concave it follows by a standard separating hyperplane argument (see e.g., [SeSy], Chap. 5, Sect. 23) that there exists a *supergradient* a for $\hat{h}(\varphi)$ at $\varphi = \hat{\varphi}$, i.e.,

$$\hat{h}(\varphi) - \hat{h}(\hat{\varphi}) \leq \langle a, \varphi - \hat{\varphi} \rangle \quad \text{for all } \varphi. \tag{13.2.40}$$

Define

$$F(\varphi) = h(\varphi, \hat{u}) - h(\hat{\varphi}, \hat{u}) - \langle a, \varphi - \hat{\varphi} \rangle.$$

Then by (13.2.38) and (13.2.40) we have

$$F(\varphi) \leq 0 \quad \text{for all } \varphi.$$

Moreover, by definition of F we have

$$F(\hat{\varphi}) = 0.$$

Therefore

$$\nabla_{\hat{\varphi}} F = a.$$

Combining this with (13.2.40), we obtain (13.2.39) and the proof is complete. □

13.2.1 Return to Example 13.1

As an illustration of the maximum principle let us apply it to solve the problem in Example 13.1. In this case the Hamiltonian is

$$H(t,x,y,u,p,q,r) = \frac{u^\gamma}{\gamma} + (\alpha y - u)p + \beta y q + y\int_{\mathbb{R}} r(\zeta)\zeta\nu(\mathrm{d}\zeta) \tag{13.2.41}$$

which is clearly concave in (y,u). The adjoint equation is

$$\begin{aligned}\mathrm{d}p(t,x) = &-\left[\frac{1}{2}\Delta p(t,x) + \alpha p(t,x) + \beta q(t,x) + \int_{\mathbb{R}} r(t,x,\zeta)\zeta\nu(\mathrm{d}\zeta)\right]\mathrm{d}t\\ &+ q(t,x)\mathrm{d}B(t) + \int_{\mathbb{R}} r(t,x,\zeta)\tilde{N}(\mathrm{d}t,\mathrm{d}\zeta);\ (t,x)\in(0,T)\times D\end{aligned} \tag{13.2.42}$$

$$p(T,x) = \rho;\ x\in D, \tag{13.2.43}$$

$$p(t,x) = 0\ ;\ (t,x)\in(0,T)\times\partial D. \tag{13.2.44}$$

Since the coefficients α, β and the boundary value ρ are all deterministic, we see that we can choose

$$q(t,x) = r(t,x,z) = 0$$

and solve the resulting deterministic equation

$$\frac{\partial}{\partial t}p(t,x) = -\frac{1}{2}\Delta p(t,x) - \alpha p(t,x)\ ;\ (t,x)\in(0,T)\times D \tag{13.2.45}$$

(together with (13.2.43)–(13.2.44)) for a deterministic solution $p(t,x)$.

This is a classical boundary value problem, and it is well known that the solution can be expressed as follows (Fig. 13.1):

$$p(t,x) = \rho\mathrm{e}^{\alpha(T-t)}P[W^x(s)\in D \quad \text{for all } s\in[t,T]], \tag{13.2.46}$$

where $W^x(\cdot)$ denotes an auxiliary n-dimensional Brownian motion starting at $x\in\mathbb{R}^n$ with probability law P. (See e.g., [KS, Chap. 4], or [Ø1, Chap. 9]).

The function

$$u \to H(t,x,y,u,p,q,r) = \frac{u^\gamma}{\gamma} + (\alpha y - u)p + \beta y q + y\int_{\mathbb{R}} r(\zeta)\zeta\nu(\mathrm{d}\zeta)$$

is maximal when

$$u = \hat{u}(t,x) = (p(t,x))^{1/(\gamma-1)}, \tag{13.2.47}$$

where $p(t,x)$ is given by (13.2.46).

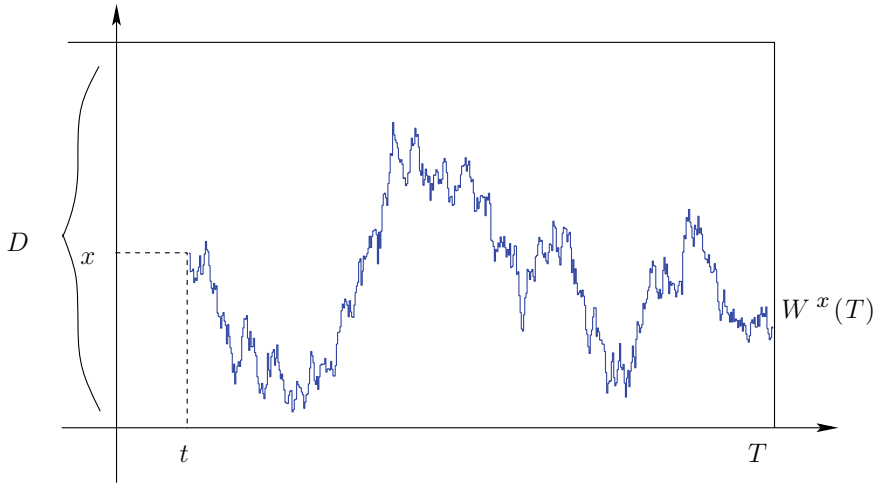


Fig. 13.1 Interpretation of the function $p(t, x)$

With this choice of $\hat{u}(t, x)$ we see that all the conditions of Theorem 13.2 are satisfied and we conclude that $\hat{u}(t, x)$ is an optimal harvesting rate for Example 13.1.

Example 13.4 ([ØPZ]) The solution $\hat{u}(t, x)$ of Example 13.1 is a bit degenerate, in the sense that it is deterministic and hence independent of the history of the population density $Y(t, x)$. The mathematical reason for this is the deterministic parameters of the adjoint equation, including the terminal condition $p(T, x) = \rho$. Therefore, let us consider a more general situation, where the performance functional $J(u)$ of (13.1.4) is replaced by

$$J(u) = E\left[\int_0^T \int_{\mathbb{R}} \frac{u^\gamma(t, x)}{\gamma} \mathrm{d}x\mathrm{d}t + \int_{\mathbb{R}} g(x, Y(T, x))\mathrm{d}x\right], \tag{13.2.48}$$

where $g : \mathbb{R}^2 \to \mathbb{R}$ is a given C^1-function. The Hamiltonian remains the same as in Example 13.1, and hence the candidate $\hat{u}(t, x)$ for the optimal control has the same form as in (13.2.47), i.e.,

$$\hat{u}(t, x) = (p(t, x))^{\frac{1}{\gamma-1}}. \tag{13.2.49}$$

The difference is that now we have to work harder to find $p(t, x)$. The backward SPDE is now

$$\begin{aligned} \mathrm{d}p(t, x) = &-\left[\alpha p(t, x) + \beta q(t, x) + \int_{\mathbb{R}} r(t, x, \zeta)\zeta\nu(\mathrm{d}\zeta) + \frac{1}{2}\Delta p(t, x)\right]\mathrm{d}t \\ &+ q(t, x)\mathrm{d}B(t) + \int_{\mathbb{R}} r(t, x, \zeta)\tilde{N}(\mathrm{d}t, \mathrm{d}\zeta);\ (t, x) \in [0, T] \times \mathbb{R}, \end{aligned} \tag{13.2.50}$$

$$p(T,x) = F(x);\ x \in \mathbb{R}. \tag{13.2.51}$$

$$\lim_{|x|\to\infty} p(t,x) = 0;\ t \in [0,T], \tag{13.2.52}$$

where

$$F(x) = F(x,\omega) = \frac{\partial g}{\partial y}(x, Y(T,x));\ x \in \mathbb{R}. \tag{13.2.53}$$

To solve this equation we proceed as follows:
Put

$$\tilde{p}(t,x) = \mathrm{e}^{\alpha t} p(t,x). \tag{13.2.54}$$

This transforms (13.2.50)–(13.2.52) into

$$\begin{aligned} \mathrm{d}\tilde{p}(t,x) = &-\beta \mathrm{e}^{\alpha t} q(t,x)\mathrm{d}t - \frac{1}{2}\Delta\tilde{p}(t,x)\mathrm{d}t \\ &+ \mathrm{e}^{\alpha t}\int_{\mathbb{R}} r(t,x,\zeta)\zeta\nu(\mathrm{d}\zeta)\mathrm{d}t + \mathrm{e}^{\alpha t} q(t,x)\mathrm{d}B(t) \\ &+ \mathrm{e}^{\alpha t}\int_{\mathbb{R}} r(t,x,\zeta)\tilde{N}(\mathrm{d}t,\mathrm{d}\zeta);\ t < T, \end{aligned} \tag{13.2.55}$$

$$\tilde{p}(T,x) = \mathrm{e}^{\alpha T} F(x), \tag{13.2.56}$$

$$\lim_{|x|\to\infty} \tilde{p}(t,x) = 0. \tag{13.2.57}$$

Define the probability measure P_0 on $\mathbb{F}_T$ by

$$\mathrm{d}P_0(\omega) = Z(T)\mathrm{d}P(\omega),$$

where

$$\begin{aligned} Z(t) = \exp\Bigg[\beta B(t) - \frac{1}{2}\beta^2 t + \int_0^t\int_{\mathbb{R}} \ln(1+\zeta)\tilde{N}(\mathrm{d}s,\mathrm{d}\zeta) \\ + \int_0^t\int_{\mathbb{R}} \{\ln(1+\zeta) - \zeta\}\nu(\mathrm{d}\zeta)\mathrm{d}s\Bigg];\ 0 \le t \le T. \end{aligned} \tag{13.2.58}$$

Then by the Girsanov theorem (use Theorem 1.33 with $u = -\beta$ and $\theta(t,\zeta) = -\zeta$) the process $B_0(t) := B(t) - \beta t$ is a Brownian motion with respect to P_0 and the random measure $\tilde{N}_0(\mathrm{d}t,\mathrm{d}\zeta) := \tilde{N}(\mathrm{d}t,\mathrm{d}\zeta) - \zeta\nu(\mathrm{d}\zeta)\mathrm{d}t$ is a compensated Poisson random measure with respect to P_0.

In terms of $\mathrm{d}B_0(t)$ and $\tilde{N}_0(\mathrm{d}t, \mathrm{d}\zeta)$ (13.2.55) gets the form

$$\begin{aligned}\mathrm{d}\tilde{p}(t,x) = &-\frac{1}{2}\Delta\tilde{p}(t,x)\mathrm{d}t + \mathrm{e}^{\alpha t}q(t,x)\mathrm{d}B_0(t)\\ &+\mathrm{e}^{\alpha t}\int_{\mathbb{R}} r(t,x,\zeta)\tilde{N}_0(\mathrm{d}t,\mathrm{d}\zeta).\end{aligned} \tag{13.2.59}$$

Suppose

$$E_0\left[\int_{\mathbb{R}} F^2(x)\mathrm{d}x\right] < \infty, \tag{13.2.60}$$

where E_0 denotes the expectation with respect to P_0. Then by the Itô representation theorem (see e.g., [I]), there exists for a.a., $x \in \mathbb{R}$ a unique pair of adapted processes $(\varphi(t,x), \psi(t,x,\zeta))$ such that

$$E_0\left[\int_0^T \varphi^2(t,x)\mathrm{d}t + \int_0^T\int_{\mathbb{R}} \psi^2(t,x,\zeta)\nu(\mathrm{d}\zeta)\mathrm{d}t\right] < \infty$$

and

$$\mathrm{e}^{\alpha T}F(x) = h(x) + \int_0^T \varphi(t,x)\mathrm{d}B_0(t) + \int_0^T\int_{\mathbb{R}} \psi(t,x,\zeta)\tilde{N}_0(\mathrm{d}t,\mathrm{d}\zeta), \tag{13.2.61}$$

where

$$h(x) = E_0\left[\mathrm{e}^{\alpha T}F(x)\right].$$

Let Q_t be the *heat operator*, defined by

$$(Q_t f)(x) = (2\pi t)^{-1/2}\int_{\mathbb{R}} f(y)\exp\left(-\frac{(x-y)^2}{2t}\right)\mathrm{d}y; f \in \mathcal{D}$$

where $\mathcal{D}$ is the set of functions $f : \mathbb{R} \to \mathbb{R}$ such that the integral exists. Define

$$\begin{aligned}\hat{p}(t,x) &:= Q_{T-t}\left(\int_0^t \varphi(s,\cdot)\mathrm{d}B_0(s) + \int_0^t\int_{\mathbb{R}} \psi(s,\cdot,\zeta)\tilde{N}_0(\mathrm{d}s,\mathrm{d}\zeta) + h(\cdot)\right)(x)\\ &= \int_0^t Q_{T-t}\varphi(s,\cdot)(x)\mathrm{d}B_0(s) + \int_0^t\int_{\mathbb{R}} Q_{T-t}\psi(s,\cdot,\zeta)(x)\tilde{N}_0(\mathrm{d}s,\mathrm{d}\zeta)\\ &\quad + Q_{T-t}h(x).\end{aligned} \tag{13.2.62}$$

Then, since

$$\frac{\mathrm{d}}{\mathrm{d}t}Q_t f = \frac{1}{2}\Delta(Q_t f),$$

we see that

$$\begin{aligned}
\mathrm{d}\hat{p}(t,x)x &= Q_{T-t}\varphi(t,\cdot)(x)\mathrm{d}B_0(t) + \left[\int_0^t \left(-\frac{1}{2}\Delta(Q_{T-t}\varphi(s,\cdot))(x)\right)\mathrm{d}B_0(s)\right]\mathrm{d}t \\
&\quad + \int_{\mathbb{R}} Q_{T-t}\psi(t,\cdot,\zeta)(x)\tilde{N}_0(\mathrm{d}s,\mathrm{d}\zeta) \\
&\quad + \left[\int_0^t \int_{\mathbb{R}} -\frac{1}{2}\Delta(Q_{T-t}\psi(s,\cdot,\zeta))(x)\tilde{N}_0(\mathrm{d}s,\mathrm{d}\zeta)\right]\mathrm{d}t \\
&\quad - \frac{1}{2}\Delta(Q_{T-t}h(x))\mathrm{d}t \\
&= -\frac{1}{2}\Delta\left(\int_0^t Q_{T-t}\varphi(s,\cdot)(x)\mathrm{d}B_0(s)\right. \\
&\quad \left. + \int_0^t \int_{\mathbb{R}} Q_{T-t}\psi(s,\cdot\zeta)(x)\tilde{N}_0(\mathrm{d}s,\mathrm{d}\zeta) + Q_{T-t}h(x)\right)\mathrm{d}t \\
&\quad + Q_{T-t}\varphi(t,\cdot)(x)\mathrm{d}B_0(t) + \int_{\mathbb{R}} Q_{T-t}\psi(t,\cdot,\zeta)(x)\tilde{N}_0(\mathrm{d}t,\mathrm{d}\zeta) \\
&= -\frac{1}{2}\Delta\hat{p}(t,x)\mathrm{d}t + Q_{T-t}\varphi(t,\cdot)(x)\mathrm{d}B_0(t) \\
&\quad + \int_{\mathbb{R}} Q_{T-t}\psi(t,\cdot,\zeta)(x)\tilde{N}_0(\mathrm{d}t,\mathrm{d}\zeta).
\end{aligned} \tag{13.2.63}$$

Comparing with (13.2.59) we see that the triple $(\tilde{p}, q, r)$ given by

$$\tilde{p}(t,x) := \hat{p}(t,x), \tag{13.2.64}$$

$$q(t,x) := \mathrm{e}^{-\alpha t}Q_{T-t}\varphi(t,\cdot)(x), \tag{13.2.65}$$

$$r(t,x,\zeta) := \mathrm{e}^{-\alpha t}Q_{T-t}\psi(t,\cdot,\zeta)(x) \tag{13.2.66}$$

solves the backward SPDE 13.2.59, and hence it solves (13.2.55), together with the terminal values (13.2.56) and (13.2.57).

We have proved:

Theorem 13.5 *Assume that* (13.2.60) *holds Then the optimal control of the problem* (13.1.1)–(13.1.3), (13.1.5), *with performance functional* $J(u)$ *as in* (13.2.48), *satisfies*

$$\hat{u}(t,x) = (p(t,x))^{1/(\gamma-1)},$$

where

$$p(t,x) = \mathrm{e}^{-\alpha t}\hat{p}(t,x)$$

with $\hat{p}(t,x)$ *defined by* (13.2.62) *and* (13.2.61).

13.3 A Necessary Maximum Principle

We proceed to establish a corresponding necessary maximum principle. For this, we do not need concavity conditions, but instead we need the following assumptions about the set of admissible control processes:

- For all $t_0 \in [0, T]$ and all bounded $\mathcal{F}_{t_0}$-measurable random variables $\alpha(x, \omega)$, the control $\theta(t, x, \omega) := \mathbf{1}_{[t_0,T]}(t)\alpha(x, \omega)$ belongs to $\mathcal{A}$.
- For all $u, \beta_0 \in \mathcal{A}$ with $\beta_0(t, x) \leq K < \infty$ for all t, x define

$$\delta(t, x) = \frac{1}{2K}\text{dist}(u(t, x), \partial V) \wedge 1 > 0 \tag{13.3.1}$$

and put

$$\beta(t, x) = \delta(t, x)\beta_0(t, x). \tag{13.3.2}$$

Then the control

$$\widetilde{u}(t, x) = u(t, x) + a\beta(t, x); \quad t \in [0, T]$$

belongs to $\mathcal{A}$ for all $a \in (-1, 1)$.

- For all β as in (13.3.2) the derivative process

$$\eta(t, x) := \frac{d}{\mathrm{d}a} Y^{u+a\beta}(t, x)|_{a=0} \tag{13.3.3}$$

exists, and belongs to $\mathbf{L}^2(\lambda \times \mathbf{P})$ and

$$\begin{aligned}
\mathrm{d}\eta(t, x) = &\left[\frac{\mathrm{d}L}{\mathrm{d}u}(Y)(t, x)\beta(t, x) + L_u\eta(t, x) + \frac{\partial b}{\partial y}(t, x)\eta(t, x)\right.\\
&\left.+\frac{\partial b}{\partial u}(t, x)\beta(t, x)\right]\mathrm{d}t\\
&+\left[\frac{\partial \sigma}{\partial y}(t, x)\eta(t, x) + \frac{\partial \sigma}{\partial u}(t, x)\beta(t, x)\right]dB(t)\\
&+\int_{\mathbb{R}}\left[\frac{\partial \gamma}{\partial y}(t, x, \zeta)\eta(t, x) + \frac{\partial \gamma}{\partial u}(t, x, \zeta)\beta(t, x)\right]\tilde{N}(\mathrm{d}t, \mathrm{d}\zeta);\\
&\qquad (t, x) \in [0, T] \times D,\\
\eta(0, x) = &\frac{d}{\mathrm{d}a} Y^{u+a\beta}(0, x)|_{a=0} = 0,\\
\eta(t, x) = &\,0;\ (t, x) \in [0, T] \times \partial D.
\end{aligned} \tag{13.3.4}$$

Theorem 13.6 (***Necessary maximum principle***) *[DØ4] Let $\hat{u} \in \mathcal{A}$. Then the following are equivalent:*

1. $\frac{\mathrm{d}}{\mathrm{d}a} J(\hat{u} + a\beta)|_{a=0} = 0$ *for all bounded* $\beta \in \mathcal{A}$ *of the form* (13.3.2).
2. $\frac{\partial H}{\partial u}(t, x)_{u=\hat{u}} = 0$ *for all* $(t, x) \in [0, T] \times D$.

Proof For simplicity of notation we write u instead of $\hat{u}$ in the following.
By considering an increasing sequence of stopping times τ_n converging to T, we may assume that all local integrals appearing in the computations below are martingales and have expectation 0. See [ØS2]. We omit the details.
We can write

$$\frac{\mathrm{d}}{\mathrm{d}a} J((u + a\beta))|_{a=0} = I_1 + I_2$$

where

$$I_1 = \frac{\mathrm{d}}{\mathrm{d}a}\left[\int_D \int_0^T f(t, x, Y^{u+a\beta}(t, x), u(t, x) + a\beta(t, x)\mathrm{d}t\mathrm{d}x\right] \|_{a=0}$$

and

$$I_2 = \frac{\mathrm{d}}{\mathrm{d}a} E\left[\int_D g(x, Y^{u+a\beta}(T, x))\mathrm{d}x\right] |_{a=0}.$$

By our assumptions on f and g and by (13.3.3) we have

$$I_1 = E\left[\int_D \int_0^T \left\{\frac{\partial f}{\partial y}(t, x)\eta(t, x) + \frac{\partial f}{\partial u}(t, x)\beta(t, x)\right\} \mathrm{d}t\mathrm{d}x\right], \tag{13.3.5}$$

$$I_2 = E\left[\int_D \frac{\partial g}{\partial y}(x, Y(T, x))\eta(T, x)\mathrm{d}x\right] = \mathbb{E}\left[\int_D p(T, x)\eta(T, x)\mathrm{d}x\right]. \tag{13.3.6}$$

By the Itô formula

$$\begin{aligned} I_2 &= E\left[\int_D p(T, x)\eta(T, x)\mathrm{d}x\right] = \mathbb{E}\left[\int_D \int_0^T p(t, x)\mathrm{d}\eta(t, x)\mathrm{d}x\right. \\ &\quad \left. + \int_D \int_0^T \eta(t, x)\mathrm{d}p(t, x)\mathrm{d}x + \int_D \int_0^T \mathrm{d}[\eta, p](t, x)\mathrm{d}x\right] \\ &= E\left[\int_D \int_0^T p(t, x)\left\{\frac{\mathrm{d}L}{\mathrm{d}u}(Y)(t, x)\beta(t, x) + L_u\eta(t, x)\right.\right. \\ &\quad \left.\left. + \frac{\partial b}{\partial y}(t, x)\eta(t, x) + \frac{\partial b}{\partial u}(t, x)\beta(t, x)\right\} \mathrm{d}t\mathrm{d}x\right. \\ &\quad \left. + \int_D \int_0^T p(t, x)\left\{\frac{\partial \sigma}{\partial y}(t, x)\eta(t, x) + \frac{\partial \sigma}{\partial u}(t, x)\beta(t, x)\right\} \mathrm{d}B(t)\right. \end{aligned}$$

$$
\begin{aligned}
&+\int_D\int_0^T\int_{\mathbb{R}} p(t,x)\left\{\frac{\partial\gamma}{\partial y}(t,x,\zeta)\eta(t,x)+\frac{\partial\gamma}{\partial u}(t,x,\zeta)\beta(t,x)\right\}\tilde{N}(\mathrm{d}t,\mathrm{d}\zeta)\mathrm{d}x\\
&-\int_D\int_0^T \eta(t,x)\left[L_u^*p(t,x)+\frac{\partial H}{\partial y}(t,x)\right]\mathrm{d}t\mathrm{d}x\\
&+\int_D\int_0^T \eta(t,x)q(t,x)\mathrm{d}B(t)\mathrm{d}x+\int_D\int_0^T\int_{\mathbb{R}}\eta(t,x)r(t,x,\zeta)\tilde{N}(\mathrm{d}t,\mathrm{d}\zeta)\mathrm{d}x\\
&+\int_D\int_0^T q(t,x)\left\{\frac{\partial\sigma}{\partial y}(t,x)\eta(t,x)+\frac{\partial\sigma}{\partial u}(t,x)\beta(t,x)\right\}\mathrm{d}t\mathrm{d}x\\
&+\int_D\int_0^T\int_{\mathbb{R}}\left\{\frac{\partial\gamma}{\partial y}(t,x,\zeta)\eta(t,x)+\frac{\partial\gamma}{\partial u}(t,x,\zeta)\beta(t,x)\right\}r(t,x,\zeta)\nu(\zeta)\mathrm{d}t\mathrm{d}x\Bigg]\\
&=E\Bigg[\int_D\Bigg(\int_0^T\left\{p(t,x)\left(\frac{\mathrm{d}L}{\mathrm{d}u}(Y)(t,x)\beta(t,x)+L_u\eta(t,x)\right)\right\}\mathrm{d}t\\
&+\int_0^T\eta(t,x)\left\{p(t,x)\frac{\partial b}{\partial y}(t,x)+q(t,x)\frac{\partial\sigma}{\partial y}(t,x)-L_u^*p(t,x)-\frac{\partial H}{\partial y}(t,x)\right.\\
&\left.+\int_{\mathbb{R}}\frac{\partial\gamma}{\partial y}(t,x,\zeta)r(t,x,\zeta)\nu(\mathrm{d}\zeta)\right\}\mathrm{d}t\\
&+\int_0^T\beta(t,x)\left\{p(t,x)\frac{\partial b}{\partial u}(t,x)+q(t,x)\frac{\partial\sigma}{\partial u}(t,x)\right.\\
&\left.+\int_{\mathbb{R}}\frac{\partial\gamma}{\partial u}(t,x,\zeta)r(t,x,\zeta)\nu(\mathrm{d}\zeta)\right\}\mathrm{d}t\Bigg)\mathrm{d}x\Bigg]\\
&=E\left[\int_D\int_0^T-\eta(t,x)\frac{\partial f}{\partial y}\right\}\mathrm{d}t+\int_0^T\left\{\frac{\partial H}{\partial u}(t,x)-\frac{\partial f}{\partial u}(t,x)\right\}\beta(t,x)\mathrm{d}t\mathrm{d}x\Bigg]\\
&=-I_1+E\left[\int_D\int_0^T\frac{\partial H}{\partial u}(t,x)\beta(t,x)\mathrm{d}t\mathrm{d}x\right].
\end{aligned}
\tag{13.3.7}
$$

Adding (13.3.5) and (13.3.7) we get

$$
\frac{\mathrm{d}}{\mathrm{d}a}J(u+a\beta)|_{a=0}=I_1+I_2=E\left[\int_D\int_0^T\frac{\partial H}{\partial u}(t,x)\beta(t,x)\mathrm{d}t\mathrm{d}x\right].
$$

We conclude that

$$
\frac{\mathrm{d}}{\mathrm{d}a}J(u+a\beta)|_{a=0}=0
$$

if and only if

$$
E\left[\int_D\int_0^T\frac{\partial H}{\partial u}(t,x)\beta(t,x)\mathrm{d}t\mathrm{d}x\right]=0,
\tag{13.3.8}
$$

for all bounded $\beta\in\mathcal{A}$ of the form (13.3.2).

In particular, applying this to $\beta(t,x) = \theta(t,x)$ as in $A1$, we get that this is again equivalent to

$$\frac{\partial H}{\partial u}(t,x) = 0 \text{ for all } (t,x) \in [0,T] \times D. \tag{13.3.9}$$

□

13.4 Controls Which do not Depend on x

In some applications (see e.g. Sect. 13.5) it is important to consider only controls $u(t,x) = u(t)$ which do not depend on the space variable x. Thus we let the set $\mathcal{A}_0$ of admissible controls be defined by

$$\mathcal{A}_0 = \{u \in \mathcal{A};\, u(t,x) = u(t) \text{ does not depend on } x\}. \tag{13.4.1}$$

With the performance functional $J(u)$ as in Problem 13.1, the problem is now the following:

Problem 13.2 For each find $u_0^* \in \mathcal{A}_0$ such that

$$\sup_{u \in \mathcal{A}_0} J(u) = J(u_0^*). \tag{13.4.2}$$

It turns out that one can formulate an analog of Theorem 13.2 for this case, the main difference between this case and Theorem 13.2 is the integration with respect to $\mathrm{d}x$ in the maximum condition 3 below.

Theorem 13.7 (***Sufficient maximum principle for controls which do not depend on x***) *[DØ4] Suppose $\hat{u} \in \mathcal{A}_0$ with corresponding solutions $\hat{Y}(t,x)$ of (13.2.1) and $\hat{p}(t,x), \hat{q}(t,x), \hat{r}(t,x,\zeta)$ of (13.2.12) respectively. Assume that the following hold:*

1. *$y \to g(x,y)$ is concave for all x*
2. *$(\varphi, u) \to H(t,x,\varphi(x),\varphi,u,\widehat{p}(t,x),\widehat{q}(t,x),\hat{r}(t,x,\cdot))$ is concave for all t,x*
3. $$\sup_{w \in \mathbb{U}} \int_D H\big(t,x,\widehat{Y}(t,x),\widehat{Y}(t,\cdot),w,\widehat{p}(t,x),\widehat{q}(t,x),\hat{r}(t,x,\cdot)\big)\mathrm{d}x$$
   $$= \int_D H\big(t,x,\widehat{Y}(t,x),\widehat{Y}(t,\cdot),\widehat{u}(t),\widehat{p}(t,x),\widehat{q}(t,x),\hat{r}(t,x,\cdot)\big)\mathrm{d}x \text{ for all } t.$$

Then $\hat{u}(t)$ is an optimal control for the Problem 6.1.

Proof We proceed as in the proof of Theorem 13.2. Let $u \in \mathcal{A}_0$ with corresponding solution $Y(t,x)$ of (13.2.1). With $\hat{u} \in \mathcal{A}_0$, consider

$$J(u) - J(\hat{u}) = E\left[\int_0^T \int_D \{f - \hat{f}\}\mathrm{d}x\mathrm{d}t + \int_D \{g - \hat{g}\}\mathrm{d}x\right], \tag{13.4.3}$$

where
$\hat{f} = f(t, x, \hat{Y}(t, x), \hat{u}(t))$, $f = f(t, x, Y(t, x), u(t))$,
$\hat{g} = g(x, \hat{Y}(T, x))$ and $g = g(x, Y(T, x))$.
Using a similar shorthand notation for $\hat{b}$, b, $\hat{\sigma}$, σ and $\hat{\gamma}$, γ, and setting

$$\hat{H} = H(t, x, \hat{Y}(t, x), \hat{u}(t), \hat{p}(t, x), \hat{q}(t, x), \hat{r}(t, x, \cdot)) \tag{13.4.4}$$

and

$$H = H(t, x, Y(t, x), u(t), \hat{p}(t, x), \hat{q}(t, x), \hat{r}(t, x, \cdot)), \tag{13.4.5}$$

we see that (13.4.3) can be written

$$J(u) - J(\hat{u}) = I_1 + I_2, \tag{13.4.6}$$

where

$$I_1 = E\left[\int_0^T \left(\int_D \{f(t, x) - \widehat{f}(t, x)\}\mathrm{d}x\right)\mathrm{d}t\right], \quad I_2 = E\left[\int_D \{g(x) - \hat{g}(x)\}\mathrm{d}x\right]. \tag{13.4.7}$$

By the definition of H we have

$$I_1 = E\left[\int_0^T \int_D \left\{H(t, x) - \widehat{H}(t, x) - \widehat{p}(t, x)(L_u Y(t, x) - L_{\hat{u}}\hat{Y}(t, x) + \widetilde{b}(t, x)) \right.\right.$$
$$\left.\left. -\widehat{q}(t, x)\widetilde{\sigma}(t, x) - \int_{\mathbb{R}} \hat{r}(t, x, \zeta)\tilde{\gamma}(t, x, \zeta)\nu(\mathrm{d}\zeta)\right\}\mathrm{d}x\mathrm{d}t\right]. \tag{13.4.8}$$

Since g is concave with respect to y we have

$$g(x, Y(T, x)) - g(x, \hat{Y}(T, x))) \leq \frac{\partial g}{\partial y}(x, \hat{Y}(T, x))(Y(T, x) - \hat{Y}(T, x)). \tag{13.4.9}$$

Therefore, as in the proof of Theorem 13.2,

$$I_2 \leq E\left[\int_D \int_0^T \left\{\widehat{p}(t, x)[L_u Y(t, x) - L_{\tilde{u}}\tilde{Y}(t, x) + \widetilde{b}(t, x)] - \widetilde{Y}(t, x)\left[L_{\hat{u}}^* \hat{p}(t, x)\right.\right.\right.$$
$$\left.\left.\left. + \frac{\partial \widehat{H}(t, x)}{\partial y}(t, x)\right] + \widetilde{\sigma}(t, x)\widehat{q}(t, x) + \int_{\mathbb{R}} \tilde{\gamma}(t, x, \zeta)\hat{r}(t, x, \zeta)\nu(\mathrm{d}\zeta)\right\}\mathrm{d}t\mathrm{d}x\right], \tag{13.4.10}$$

where, as before,

$$\frac{\partial \widehat{H}(t, x)}{\partial y} = \frac{\partial H}{\partial y}(t, x, y, \hat{Y}(t, .)(x), \hat{u}(t), \hat{p}(t, x), \hat{q}(t, x), \hat{r}(t, x, .))\,|_{y=\hat{Y}(t,x)} \tag{13.4.11}$$

Adding (13.4.8)–(13.4.10) we get as in Eq. (13.2.29),

$$J(u) - J(\hat{u}) \leq \mathbb{E}\left[\int_0^T \left(\int_D \{H(t,x) - \hat{H}(t,x) - \nabla_{\hat{Y}}\hat{H}(\tilde{Y})(t,x)\}\mathrm{d}x\right)\mathrm{d}t\right]. \tag{13.4.12}$$

By the concavity assumption of H in (φ, u) we have:

$$H(t,x) - \hat{H}(t,x) \leq \nabla_{\hat{Y}}\hat{H}(Y - \hat{Y})(t,x) + \frac{\partial \widehat{H}}{\partial u}(t,x)(u(t) - \hat{u}(t)), \tag{13.4.13}$$

and the maximum condition implies that

$$\int_D \frac{\partial \widehat{H}}{\partial u}(t,x)(u(t) - \hat{u}(t))\mathrm{d}x \leq 0. \tag{13.4.14}$$

Hence

$$\int_D \{H(t,x) - \hat{H}(t,x) - \nabla_{\hat{Y}}\hat{H}(Y - \hat{Y})(t,x)\}\mathrm{d}x \leq 0, \tag{13.4.15}$$

and therefore we conclude by (10.1.17) that $J(u) \leq J(\hat{u})$. Since $u \in \mathcal{A}_0$ was arbitrary, this shows that $\hat{u}$ is optimal. □

We proceed as in Theorem 13.6 to establish a corresponding necessary maximum principle for controls which do not depend on x. As in Sect. 13.3 we assume the following:

- For all $t_0 \in [0,T]$ and all bounded $\mathcal{H}_{t_0}$-measurable random variables $\alpha(\omega)$, the control $\theta(t,\omega) := \mathbf{1}_{[t_0,T]}(t)\alpha(\omega)$ belongs to $\mathcal{A}_0$.
- For all $u, \beta_0 \in \mathcal{A}_0$ with $\beta_0(t) \leq K < \infty$ for all t define

$$\delta(t) = \frac{1}{2K}\mathrm{dist}((u(t), \partial V) \wedge 1 > 0 \tag{13.4.16}$$

and put

$$\beta(t) = \delta(t)\beta_0(t). \tag{13.4.17}$$

Then the control

$$\widetilde{u}(t) = u(t) + a\beta(t); \quad t \in [0,T]$$

belongs to $\mathcal{A}_0$ for all $a \in (-1,1)$.

- For all β as in (13.4.17) the derivative process

$$\eta(t,x) := \frac{\mathrm{d}}{\mathrm{d}a}Y^{u+a\beta}(t,x)|_{a=0}$$

exists, and belongs to $\mathbf{L}^2(\lambda \times \mathbf{P})$ and

$$\begin{cases} \mathrm{d}\eta(t,x) = \left[\dfrac{\mathrm{d}L}{\mathrm{d}u}(Y)(t,x)\beta(t) + L_u\eta(t,x) + \dfrac{\partial b}{\partial y}(t,x)\eta(t,x)\right. \\ \quad \left. + \dfrac{\partial b}{\partial u}(t,x)\beta(t)\right]\mathrm{d}t \\ \quad + \left[\dfrac{\partial \sigma}{\partial y}(t,x)\eta(t,x) + \dfrac{\partial \sigma}{\partial u}(t,x)\beta(t)\right] dB(t) \\ \quad + \displaystyle\int_{\mathbb{R}} \left[\frac{\partial \gamma}{\partial y}(t,x,\zeta)\eta(t,x) + \frac{\partial \gamma}{\partial u}(t,x,\zeta)\beta(t)\right]\tilde{N}(\mathrm{d}t,\mathrm{d}\zeta); \\ \qquad (t,x) \in [0,T]\times D, \\ \eta(0,x) = \dfrac{\mathrm{d}}{\mathrm{d}a}Y^{u+a\beta}(0,x)|_{a=0} = 0; \; x \in D \\ \eta(t,x) = 0; \; (t,x) \in [0,T]\times \partial D. \end{cases} \tag{13.4.18}$$

Then we have the following result:

Theorem 13.8 ([DØ4]) (***Necessary maximum principle for controls not depending on x***) *Let $\hat{u} \in \mathcal{A}_0$. Then the following are equivalent:*

1. $\dfrac{\mathrm{d}}{\mathrm{d}a}J(\hat{u}+a\beta)|_{a=0} = 0$ *for all bounded* $\beta \in \mathcal{A}_0$ *of the form* (13.4.17).
2. $\left[\displaystyle\int_D \frac{\partial H}{\partial u}(t,x)\mathrm{d}x\right]_{u=\hat{u}(t)} = 0$ *for all* $t \in [0,T]$.

Proof The proof is analogous to the proof of Theorem 13.6 and is omitted. □

13.5 Application to Partial (Noisy) Observation Optimal Control

For simplicity we consider only the one-dimensional case in the following. Suppose the signal process $X(t) = X^{(u)}(t)$ and its corresponding observation process $G(t)$ are given respectively by the following system of stochastic differential equations

- (Signal process)

$$\begin{aligned} &\mathrm{d}X(t) = \alpha(X(t), G(t), u(t))\mathrm{d}t + \beta(X(t), G(t), u(t))\mathrm{d}v(t) \\ &+ \int_{\mathbb{R}} \gamma(X(t), G(t), u(t), \zeta)\tilde{N}(\mathrm{d}t, \mathrm{d}\zeta); \; t \in [0,T], \\ &X(0) \text{ has density } F(\cdot), \text{ i.e. } E[\phi(X(0))] = \int_{\mathbb{R}} \phi(x)F(x)\mathrm{d}x; \quad \phi \in C_0(\mathbb{R}). \end{aligned} \tag{13.5.1}$$

  As before $T > 0$ is a fixed constant.
- (Observation process)

$$\begin{cases} \mathrm{d}G(t) = h(X(t))\mathrm{d}t + \mathrm{d}w(t); \quad t \in [0,T], \\ G(0) = 0. \end{cases} \tag{13.5.2}$$

Here $\alpha : \mathbb{R} \times \mathbb{R} \times \mathbb{U} \to \mathbb{R}$, $\beta : \mathbb{R} \times \mathbb{R} \times \mathbb{U} \to \mathbb{R}$, $\gamma : \mathbb{R} \times \mathbb{R} \times \mathbb{U} \times \mathbb{R} \to \mathbb{R}$ and $h : \mathbb{R} \to \mathbb{R}$ are given deterministic functions, with h bounded. The processes $v(t) = v(t,\omega)$ and $w(t) = w(t,\omega)$ are independent Brownian motions, and $\tilde{N}(\mathrm{d}t, \mathrm{d}\zeta)$ is a compensated Poisson random measure, independent of both v and w. We let $\mathbb{F}^v := \{\mathcal{F}_t^v\}_{0\le t\le T}$ and $\mathbb{F}^w := \{\mathcal{F}_t^w\}_{0\le t\le T}$ denote the filtrations generated by $(v, \tilde{N})$ and w, respectively.

The process $u(t) = u(t,\omega)$ is our control process, assumed to have values in a given closed set $\mathbb{U} \subset \mathbb{R}$. We require that $u(t)$ be adapted to the filtration

$$\mathbb{G} := \{\mathcal{G}_t\}_{0\le t\le T}, \tag{13.5.3}$$

where $\mathcal{G}_t$ is the sigma-algebra generated by the observations $G(s)$, $s \le t$.

We call $u(t)$ admissible if, in addition, (13.5.1) and (13.5.2) has a unique strong solution $(X(t), G(t))$ such that

$$E\left[\int_0^T |\ell(X(t), u(t))|\mathrm{d}t + |k(X(T))|\right] < \infty, \tag{13.5.4}$$

where $\ell : \mathbb{R} \times \mathbb{U} \to \mathbb{R}$ and $k : \mathbb{R} \to \mathbb{R}$ are given functions, called the profit rate and the bequest function, respectively. The set of all admissible controls is denoted by $\mathcal{A}_{\mathbb{G}}$. For $u \in \mathcal{A}_{\mathbb{G}}$ we define the performance functional

$$J(u) = E\left[\int_0^T \ell(X(t), u(t))\mathrm{d}t + k(X(T))\right]. \tag{13.5.5}$$

We consider the following problem:

Problem 13.3 (***The noisy observation stochastic control problem***) Find $u^* \in \mathcal{A}_{\mathbb{G}}$ such that

$$\sup_{u\in\mathcal{A}_{\mathbb{G}}} J(u) = J(u^*). \tag{13.5.6}$$

We now proceed to show that this partial observation SDE insider control problem can be transformed into a *full observation SPDE control problem* of the type discussed in the previous sections:
To this end, define the probability measure Q by

$$dQ(\omega) = M_t(\omega)\mathrm{d}P(\omega) \text{ on } \mathcal{F}_t^v \vee \mathcal{F}_t^w, \tag{13.5.7}$$

where

$$M_t(\omega) = M_t(\omega) = \exp\left(-\int_0^t h(X(s))\mathrm{d}w(s) - \frac{1}{2}\int_0^t h^2(X(s))\mathrm{d}s\right). \tag{13.5.8}$$

It follows by the Girsanov theorem that the observation process $G(t)$ defined by (13.5.2) is a Brownian motion with respect to Q. Moreover, we have

$$\mathrm{d}P(\omega) = K_t(\omega)dQ(\omega), \tag{13.5.9}$$

where

$$\begin{aligned} K_t = M_t^{-1} &= \exp\left(\int_0^t h(X(s))\mathrm{d}w(s) + \frac{1}{2}\int_0^t h^2(X(s))\mathrm{d}s\right) \\ &= \exp\left(\int_0^t h(X(s))\mathrm{d}G(s) - \frac{1}{2}\int_0^t h^2(X(s))\mathrm{d}s\right). \end{aligned} \tag{13.5.10}$$

For $\varphi \in C_0^2(\mathbb{R})$ and fixed $g \in \mathbb{R}, c \in \mathbb{U}$ define the integro-differential operator $A = A_{g,c}$ by

$$\begin{aligned} A_{g,c}\varphi(x) &= \alpha(x,g,c)\frac{\partial\varphi}{\partial x}(x) + \frac{1}{2}\beta^2(x,g,c)\frac{\partial^2\varphi}{\partial x^2} \\ &\quad + \int_{\mathbb{R}} \{\varphi(x+\gamma(x,g,c,\zeta)) - \varphi(x) - \nabla\varphi(x)\gamma(x,g,c,\zeta)\}\nu(\mathrm{d}\zeta), \end{aligned} \tag{13.5.11}$$

and let A^* be the adjoint of A, in the sense that

$$(A\varphi,\psi)_{L^2(\mathbb{R})} = (\varphi, A^*\psi)_{L^2(\mathbb{R})} \tag{13.5.12}$$

for all $\varphi, \psi \in C_0^2(\mathbb{R})$.

Suppose that there exists a stochastic process $y(t,x)$ such that

$$E_Q[\varphi(X(t))K_t|\mathcal{G}_t] = \int_{\mathbb{R}} \varphi(x)y(t,x)\mathrm{d}x \tag{13.5.13}$$

for all bounded measurable functions φ. Then $y(t,x)$ is called the *unnormalized* conditional density of $X(t)$ given the observation filtration $\mathcal{G}_t$. Note that by the Bayes rule we have

$$E[\varphi(X(t))|\mathcal{G}_t] = \frac{E_Q[\varphi(X(t))K_t|\mathcal{G}_t]}{E_Q[K_t|\mathcal{G}_t]}. \tag{13.5.14}$$

It is known that under certain conditions the process $y(t,x) = y(t,x,z)$ exists and satisfies the following integro-SPDE, called the Duncan–Mortensen–Zakai equation:

$$\begin{aligned} \mathrm{d}y(t,x) &= A^*_{G(t),u(t)}y(t,x)\mathrm{d}t + h(x)y(t,x)\mathrm{d}G(t); \quad t \geq 0 \\ y(0,x) &= F(x). \end{aligned} \tag{13.5.15}$$

See for example Theorem 7.17 in [BC]. If (13.5.13) holds, we get

$$\begin{aligned} J(u) &= E\left[\int_0^T \ell(X(t), u(t))\mathrm{d}t + k(X(T))\right] \\ &= E_Q\left[\int_0^T \ell(X(t), u(t))K_t\mathrm{d}t + k(X(T))K_T\right] \\ &= E_Q\left[\int_0^T E_Q[\ell(X(t), u(t))K_t|\mathcal{G}_t]\mathrm{d}t + E_Q[k(X(T))K_T|\mathcal{G}_T]\right] \\ &= E_Q\left[\int_0^T E_Q[\ell(X(t), u(t))K_t|\mathcal{G}_t]\mathrm{d}t + E_Q[k(X(T))K_T|\mathcal{G}_T]\right] \\ &= E_Q\left[\int_0^T E_Q[\ell(X(t), v)K_t|\mathcal{G}_t]_{v=u(t)}\mathrm{d}t + E_Q[k(X(T))K_T|\mathcal{G}_T]\right] \\ &= E_Q\left[\int_0^T \int_{\mathbb{R}} \ell(x, u(t))y(t, x)\mathrm{d}x\mathrm{d}t + \int_{\mathbb{R}} k(x)y(T, x)\mathrm{d}x\right] =: J_Q(u). \end{aligned} \tag{13.5.16}$$

This transforms the insider partial observation SDE control problem (13.5.6) into a full observation SPDE control problem of the type we have discussed in the previous sections.

We summarize what we have proved as follows:

Theorem 13.9 (***From partial observation SDE control to full observation SPDE control***) *Assume that* (13.5.13) *and* (13.5.15) *hold. Then the solution* $u^*(t)$ *of the partial observation SDE control problem* (13.5.6) *coincides with the solution* u^* *of the following (full observation) SPDE control problem:*

Problem 13.4 Find $u^* \in \mathcal{A}_{\mathbb{G}}$ such that

$$\sup_{u \in \mathcal{A}_{\mathbb{G}}} J_Q(u) = J_Q(u^*), \tag{13.5.17}$$

where

$$J_Q(u) = E_Q\left[\int_0^T \int_{\mathbb{R}} \ell(x, u(t))y(t, x)\mathrm{d}x\mathrm{d}t + \int_{\mathbb{R}} k(x)y(T, x)\mathrm{d}x\right], \tag{13.5.18}$$

and $y(t, x)$ solves the SPDE

$$\begin{aligned} \mathrm{d}y(t, x) &= A^*_{G(t),u(t)}y(t, x)\mathrm{d}t + h(x)y(t, x)\mathrm{d}G(t); \quad t \geq 0 \\ y(0, x) &= F(x). \end{aligned} \tag{13.5.19}$$

13.5.1 Optimal Portfolio with Noisy Observations

We now study an example illustrating an application of Theorem 13.9:
Suppose the signal process $X(t) = X^{\pi}(t)$ is given by

$$\begin{cases} \mathrm{d}X(t) = \pi(t)[\alpha_0(t)\mathrm{d}t + \beta_0(t)\mathrm{d}v(t)]; \quad 0 \le t \le T, \\ X(0) \text{ is a random variable independent of } \{v(t)\}_{t\ge 0} \text{ and with density } F. \end{cases} \tag{13.5.20}$$

Here $\pi(t)$ is the control, representing the portfolio in terms of the amount invested in the risky asset at time t, when the risky asset unit price $S(t)$ is given by

$$\begin{cases} \mathrm{d}S(t) = S(t)[\alpha_0(t)\mathrm{d}t + \beta_0(t)\mathrm{d}v(t)]; \quad 0 \le t \le T, \\ S(0) > 0, \end{cases} \tag{13.5.21}$$

and the safe investment unit price is $S_0(t) = 1$ for all t. The process $X(t)$ then represents the corresponding value of the investment at time t. For π to be admissible we require that $X(t) > 0$ for all t. Suppose the observations $G(t)$ of $X(t)$ at time t are not exact, but subject to uncertainty or noise, so that the observation process is given by

$$\begin{cases} \mathrm{d}G(t) = X(t)\mathrm{d}t + \mathrm{d}w(t); \quad 0 \le t \le T, \\ G(0) = 0. \end{cases} \tag{13.5.22}$$

Here, as above, the processes v and w are independent Brownian motions. Let $U : [0, \infty) \mapsto [-\infty, \infty)$ be a given C^1 (concave) utility function. The performance functional is assumed to be

$$J(\pi) = E[U(X^{\pi}(T))]. \tag{13.5.23}$$

By Theorem 13.9, the problem to maximize $J(\pi)$ over all $\pi \in \mathcal{A}_{\mathbb{G}}$ is equivalent to the following problem:

Problem 13.5 Find $\hat{\pi} \in \mathcal{A}_{\mathbb{G}}$ such that

$$\sup_{\pi\in\mathcal{A}_{\mathbb{G}}} J_Q(\pi) = J_Q(\hat{\pi}), \tag{13.5.24}$$

where

$$J_Q(\pi) = E_Q\left[\int_{\mathbb{R}^+} U(x)y(T,x)\mathrm{d}x\right], \tag{13.5.25}$$

and $y(t,x) = y^{\pi}(t,x)$ is the solution of the SPDE

$$\begin{cases} \mathrm{d}y(t,x) = (A^*_{\pi(t)}y)(t,x)\mathrm{d}t + xy(t,x)\mathrm{d}G(t); \quad 0 \le t \le T, \\ y(0,x) = F(x), \end{cases} \tag{13.5.26}$$

where

$$(A^*_{\pi(t)}y)(t,x) = -\pi(t)\alpha_0(t)y'(t,x) + \frac{1}{2}\pi^2(t)\beta_0^2(t)y''(t,x),$$

with $y'(t,x) = \dfrac{\partial y(t,x)}{\partial x}$, $y''(t,x) = \dfrac{\partial^2 y(t,x)}{\partial x^2}$.

Define the space

$$\mathbf{H}^1(\mathbb{R}^+) = \{y \in \mathbf{L}^2(\mathbb{R}^+), \frac{\partial y}{\partial x} \in \mathbf{L}^2(\mathbb{R}^+)\} \tag{13.5.27}$$

The $\mathbf{H}^1$-norm is given by:

$$\|y(t)\|^2_{\mathbf{H}^1(\mathbb{R}^+)} = \|y(t)\|^2_{\mathbf{L}^2(\mathbb{R}^+)} + \|y'(t)\|^2_{\mathbf{L}^2(\mathbb{R}^+)} \tag{13.5.28}$$

We have

$$\mathbf{H}^1(\mathbb{R}^+) \subset \mathbf{L}^2(\mathbb{R}^+) \subset \mathbb{H}^{-1}(\mathbb{R}^+) \tag{13.5.29}$$

We verify the coercivity condition of the operator $-A^*_{\pi(t)}$:

$$\begin{aligned}
&2\langle -A^*_{\pi(t)}y, y\rangle = 2\pi(t,z)\alpha_0(t)\langle y'(t,x), y(t,x)\rangle - \pi^2(t,z)\beta_0^2(t)\langle y''(t,x), y(t,x)\rangle \\
&= 2\pi(t)\alpha_0(t)\int_{\mathbb{R}^+} y'(t,x)y(t,x)\mathrm{d}x - \pi^2(t)\beta_0^2(t)\int_{\mathbb{R}^+} y''(t,x)y(t,x)\mathrm{d}x \\
&= \pi(t,z)\alpha_0(t)[y^2(t,x)]_{\partial\mathbb{R}^+} - \pi^2(t)\beta_0^2(t)[y(t,x)y'(t,x)]_{\partial\mathbb{R}^+} \\
&\quad + \pi^2(t)\beta_0^2(t)\int_{\mathbb{R}^+}(y'(t,x))^2\mathrm{d}x.
\end{aligned} \tag{13.5.30}$$

Suppose that $y(t,x) = 0$ for $x = 0$. Then we get

$$2\langle -A^*_{\pi(t)}y, y\rangle = \pi^2(t)\beta_0^2(t)\|y'(t)\|^2_{\mathbf{L}^2(\mathbb{R}^+)}. \tag{13.5.31}$$

Let

$$\mathbf{H}^1_0(\mathbb{R}^+) = \{y \in \mathbf{H}^1, y = 0 \text{ on } \partial\mathbb{R}^+\}. \tag{13.5.32}$$

We have $|y(t)|_{1,\mathbb{R}^+} = \|y'(t)\|_{\mathbf{L}^2(\mathbb{R}^+)}$ is a norm in $\mathbf{H}^1_0(\mathbb{R}^+)$, which is equivalent to the $\mathbf{H}^1(\mathbb{R}^+)$ norm; i.e. there exist $a, b > 0$ such that

$$a\|y(t)\|_{1,\mathbb{R}^+} \le |y(t)|_{1,\mathbb{R}^+} = \|y'(t)\|_{\mathbf{L}^2(\mathbb{R}^+)} \le b\|y(t)\|_{1,\mathbb{R}^+} \tag{13.5.33}$$

We conclude that the following coercivity condition is satisfied:

$$2\langle -A^*_{\pi(t)}y, y\rangle \ge a^2\pi^2(t,z)\beta_0^2(t)\|y(t,z)\|^2_{1,\mathbb{R}^+}. \tag{13.5.34}$$

Using Theorems 1.1 and 2.1 in Pardoux [Pa1], we obtain that (13.5.26) has a unique solution $y(t,x,z) \in \mathbf{L}^2(\Omega, \mathbf{C}(0,T,\mathbf{L}^2(\mathbb{R}^+)))$. Moreover, the first and second partial derivatives with respect to x, denoted by $y'(t,x,z)$ and $y''(t,x,z)$ respectively, exist and belong to $L^2(\mathbb{R})$.

The problem (13.5.24) is of the type discussed in Sect. 13.2 and we now apply the methods developed there to study it:
The Hamiltonian given in (13.2.11) now gets the form

$$H(t,x,y,\varphi,\pi,p,q) = (A_\pi^*\varphi)p + xyq, \tag{13.5.35}$$

and the adjoint BSDE (13.2.12) becomes

$$\begin{cases} \mathrm{d}p(t,x) = -[A_{\pi(t)}p(t,x) + xq(t,x)]\mathrm{d}t + q(t,x)\mathrm{d}G(t); \quad 0 \le t \le T, \\ p(T,x) = U(x). \end{cases} \tag{13.5.36}$$

where $\mathcal{G}_t$ is the sigma-algebra generated by $\{G(s)\}_{s\le t}$, for $0 \le t \le T$, and

$$A_{\pi(t)}p(t,x) = \pi(t)\alpha_0(t)p'(t,x) + \frac{1}{2}\pi^2(t)\beta_0^2(t)p''(t,x). \tag{13.5.37}$$

By [ØPZ, ZRW], this backward SPDE (BSPDE for short) admits a unique solution which belongs to $\mathbf{L}^2(\mathbb{R}^+)$.

The map

$$\pi \mapsto \int_{\mathbb{R}^+} H(t,x,y(t,x),y(t,.),\pi,p(t,x),q(t,x))\mathrm{d}x$$

is maximal when

$$\int_{\mathbb{R}^+} \{-\alpha_0(t)y'(t,x) + \pi\beta_0^2(t)y''(t,x)\}p(t,x)\mathrm{d}x = 0, \tag{13.5.38}$$

i.e. when $\pi = \hat{\pi}(t)$, given by

$$\hat{\pi}(t) = \frac{\alpha_0(t)\int_{\mathbb{R}_+} y'(t,x)p(t,x)\mathrm{d}x}{\beta_0^2(t)\int_{\mathbb{R}_+} y''(t,x)p(t,x)\mathrm{d}x}. \tag{13.5.39}$$

Using integration by parts and (13.5.13)–(13.5.14) we can rewrite this as follows:

$$\begin{aligned} \hat{\pi}(t) &= -\frac{\alpha_0(t)\int_{\mathbb{R}_+} y(t,x)p'(t,x)\mathrm{d}x}{\beta_0^2(t)\int_{\mathbb{R}_+} y(t,x)p''(t,x)\mathrm{d}x} \\ &= -\frac{\alpha_0(t)\mathbb{E}[p'(t,X(t))|\mathcal{G}_t]}{\beta_0^2(t)\mathbb{E}[p''(t,X(t))|\mathcal{G}_t]}. \end{aligned} \tag{13.5.40}$$

We summarise what we have proved as follows:

Theorem 13.10 ([DØ4]) *Assume that the conditions of Theorem 13.9 hold. The optimal portfolio $\hat{\pi}(t) \in \mathcal{A}$ for the noisy observation portfolio problem (13.5.24), is given in feedback form by*

$$\hat{\pi}(t) = -\frac{\alpha_0(t)E[p'(t, X(t))|\mathcal{G}_t]}{\beta_0^2(t)E[p''(t, X(t))|\mathcal{G}_t]}, \tag{13.5.41}$$

in the sense that it is expressed in terms of the corresponding solution

$$p(t, x) = p^{\pi}(t, x)$$

of the BSPDE

$$\begin{cases} \mathrm{d}p(t, x) = -[A_{\hat{\pi}(t)}p(t, x) + xq(t, x)]\mathrm{d}t + q(t, x)\mathrm{d}G(t); \, 0 \le t \le T, \\ p(T, x) = U(x), \end{cases} \tag{13.5.42}$$

with $\mathbb{E}[p''(t, X(t)) \mid \mathcal{G}_t] \neq 0$ for all t.

Remark 13.11 It is interesting to compare the above result with the well-known case when there is no noise in the observations. Then the optimal portfolio π^* is given in feedback form by

$$\pi^*(t) = \frac{\alpha_0(t)}{\beta_0^2(t)}X(t).$$

Thus, comparing with 13.5.41 we see that the expression

$$-\frac{\mathbb{E}[p'(t, X(t)) \mid \mathcal{G}_t]}{\mathbb{E}[p''(t, X(t)) \mid \mathcal{G}_t]}$$

reduces to $X(t)$ in this case.

13.6 Exercises

Exercise 13.1 Transform the following *partial observation SDE* control problems into *complete observation SPDE* control problems:

(a) (The partially observed linear-quadratic control problem)

$$\text{(signal):} \quad \begin{cases} \mathrm{d}X(t) = (\alpha X(t) + u(t))\mathrm{d}t + \sigma \mathrm{d}v(t);\ t > 0, \\ X(0) \text{ has density } F(x), \end{cases}$$

$$\text{(observations):} \quad \begin{cases} \mathrm{d}G(t) = h(x(t))\mathrm{d}t + \mathrm{d}w(t);\ t > 0, \\ G(0) = 0, \end{cases}$$

$$\text{performance:} \quad J(u) = E\left[\int_0^T \left\{X^2(t) + \theta u^2(t)\right\} \mathrm{d}t\right], \quad J^* = \inf_{u \in \mathcal{A}_{\mathbb{G}}} J(u).$$

Here $\alpha, \sigma \neq 0, \theta > 0$ are constants and F and h are given functions.

(b) (The partially observed optimal portfolio problem)

$$\text{(signal)}: \quad \begin{cases} \mathrm{d}X(t) = X(t)[\alpha u(t)\mathrm{d}t + \beta u(t)\mathrm{d}v(t)];\ t > 0 \\ X(0) \text{ has density } F(x), \end{cases}$$

$$\text{(observations)}: \quad \begin{cases} \mathrm{d}G(t) = X(t)\mathrm{d}t + \mathrm{d}w(t);\ t > 0, \\ G(0) = 0. \end{cases}$$

$$\text{performance:} \quad J(u) = E[X^{\gamma}(T)],\ J^* = \sup_{u \in \mathcal{A}_{\mathbb{G}}} J(u).$$

Here $\alpha > 0$, $\beta > 0$ and $\gamma \in (0, 1)$ are constants.

We may interpret $u(t)$ as the *portfolio* representing the fraction of the total wealth $x(t)$ invested in the risky asset with price dynamics

$$\begin{aligned} \mathrm{d}S_1(t) &= S_1(t)[\alpha \mathrm{d}t + \beta \mathrm{d}v(t)];\ t > 0, \\ S_1(0) &> 0. \end{aligned}$$

The remaining fraction $1 - \pi(t)$ is then invested in the other investment alternative, being a risk free asset with price $S_0(t) = 1$ for all t.

Exercise 13.2 (***Terminal Conditions***) Let $Y(t, x) = Y^{(u)}(t, x)$ be as in (13.1.1)–(13.1.3) and define

$$J(u) = E\left[\int_0^T \left(\int_D \ln u(t, x)\mathrm{d}x\right) \mathrm{d}t\right];\ u \in \tilde{\mathcal{A}}, \tag{13.6.1}$$

where $\tilde{\mathcal{A}}$ is the set of controls in $\mathcal{A}$ (see (13.1.3)) such that the *terminal constraint*

$$E\left[\int_D Y^{(u)}(T, x)\mathrm{d}x\right] \geq 0 \tag{13.6.2}$$

holds. We consider the constrained stochastic control problem to find $\tilde{J} \in \mathbb{R}$ and $\tilde{u} \in \tilde{\mathcal{A}}$ such that

$$\tilde{J} = \sup_{u \in \tilde{\mathcal{A}}} J(u) = J(\tilde{u}). \tag{13.6.3}$$

To solve this problem we use the *Lagrange multiplier method* as follows:
(a) Fix $\lambda > 0$ and solve the *unconstrained* control problem to find $J_\lambda^* \in \mathbb{R}$ and $u_\lambda^* \in \mathcal{A}$ such that

$$J_\lambda^* = \sup_{u \in \mathcal{A}} J_\lambda(u) = J_\lambda(u_\lambda^*), \tag{13.6.4}$$

where

$$J_\lambda(u) = E\left[\int_0^T \left(\int_D \ln u(t,x)\mathrm{d}x\right)\mathrm{d}t + \int_D \lambda Y(T,x)\mathrm{d}x\right]. \tag{13.6.5}$$

[Hint: Use the method in Example 13.1.]
(b) Suppose there exists $\hat{\lambda} > 0$ such that

$$E\left[\int_D Y^{(u_{\hat{\lambda}}^*)}(T,x)\mathrm{d}x\right] = 0, \tag{13.6.6}$$

where $u_{\hat{\lambda}}^*$ is the corresponding solution of the *unconstrained* problem (13.6.4)–(13.6.5) with $\lambda = \hat{\lambda}$. Show that then in fact $\tilde{u} := u_{\hat{\lambda}}^* \in \tilde{\mathcal{A}}$ solves the *constrained* problem (13.6.1)–(13.6.3) and hence $\tilde{J} = J_{\hat{\lambda}}^*$.
(c) Use the above to solve the constrained stochastic control problem (13.6.3).

Exercise 13.3 (***Controls Which do not Depend on x***) Consider Example 13.1 again, but this time we only allow controls $u(t,x) = u(t)$ which do not depend on x. Use Theorem 13.7 to find the optimal control $u^*(t)$ in this case.

Chapter 14
Solutions of Selected Exercises



14.1 Exercises of Chap. 1

Exercise 1.1

Choose $f \in C^2(\mathbb{R})$ and put $Y(t) = f(X(t))$. Then by the Itô formula

$$\begin{aligned} \mathrm{d}Y(t) &= f'(X(t))[\alpha\,\mathrm{d}t + \sigma\,\mathrm{d}B(t)] + \tfrac{1}{2}\sigma^2 f''(X(t))\mathrm{d}t \\ &\quad + \int_{|z|<R} \left\{ f(X(t^-) + \gamma(z)) - f(X(t^-)) - \gamma(z) f'(X(t^-)) \right\} \nu(\mathrm{d}z)\mathrm{d}t \\ &\quad + \int_{\mathbb{R}} \left\{ f(X(t^-) + \gamma(z)) - f(X(t^-)) \right\} \bar{N}(\mathrm{d}t, \mathrm{d}z). \end{aligned} \tag{14.1.1}$$

(i) In particular, if $f(x) = \exp(x)$ this gives

$$\begin{aligned} \mathrm{d}Y(t) &= Y(t)[\alpha\,\mathrm{d}t + \sigma\,\mathrm{d}B(t)] + \tfrac{1}{2}\sigma^2 Y(t)\mathrm{d}t \\ &\quad + \int_{|z|<R} \left\{ \exp(X(t^-) + \gamma(z)) - \exp(X(t^-)) - \gamma(z)\exp(X(t^-)) \right\} \nu(\mathrm{d}z)\mathrm{d}t \\ &\quad + \int_{\mathbb{R}} \left\{ \exp(X(t^-) + \gamma(z)) - \exp(X(t^-)) \right\} \bar{N}(\mathrm{d}t, \mathrm{d}z) \\ &= Y(t^-)\Bigg[\left(\alpha + \tfrac{1}{2}\sigma^2 + \int_{|z|<R} \left\{ \mathrm{e}^{\gamma(z)} - 1 - \gamma(z) \right\} \nu(\mathrm{d}z) \right) \mathrm{d}t \\ &\quad + \sigma\,\mathrm{d}B(t) + \int_{\mathbb{R}} \left\{ \mathrm{e}^{\gamma(z)} - 1 \right\} \bar{N}(\mathrm{d}t, \mathrm{d}z) \Bigg]. \end{aligned} \tag{14.1.2}$$

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(ii) By (i) we see that $Y(t)$ solves the equation

$$\mathrm{d}Y(t) = Y(t^-)\Big[\beta\,\mathrm{d}t + \theta\,\mathrm{d}B(t) + \lambda\int_{\mathbb{R}} z\bar{N}(\mathrm{d}t,\mathrm{d}z)\Big]$$

if and only if

$$\alpha + \tfrac{1}{2}\sigma^2 + \int_{|z|<R}\left\{\mathrm{e}^{\gamma(z)} - 1 - \gamma(z)\right\}\nu(\mathrm{d}z) = \beta, \quad \sigma = \theta$$

and $\mathrm{e}^{\gamma(z)} - 1 = \lambda z$ (i.e., $\gamma(z) = \ln(1+\lambda z)$) a.e. ν.

Exercise 1.2

We first make some general remarks:
Suppose $\mathrm{d}X_i(t) = \alpha_i(t,\omega)\mathrm{d}t + \sigma_i(t,\omega)\mathrm{d}B(t) + \int_{\mathbb{R}}\gamma_i(t,z,\omega)\bar{N}(\mathrm{d}t,\mathrm{d}z)$ for $i = 1, 2$. Define $Y(t) = X_1(t)\cdot X_2(t)$. Then, by the Itô formula with $f(x_1,x_2) = x_1\cdot x_2$,

$$\begin{aligned}
\mathrm{d}Y(t) &= X_2(t)[\alpha_1\mathrm{d}t + \sigma_1\mathrm{d}B(t)] + X_1(t)[\alpha_2\mathrm{d}t + \sigma_2\mathrm{d}B(t)] + \tfrac{1}{2}\cdot 2\sigma_1\sigma_2\mathrm{d}t\\
&\quad + \int_{|z|<R}\big\{(X_1(t^-)+\gamma_1(t,z))(X_2(t^-)+\gamma_2(t,z)) - X_1(t^-)X_2(t^-)\\
&\quad -X_2(t^-)\gamma_1(t,z) - X_1(t^-)\gamma_2(t,z)\big\}\nu(\mathrm{d}z)\mathrm{d}t\\
&\quad + \int_{\mathbb{R}}\big\{(X_1(t^-)+\gamma_1(t,z))(X_2(t^-)+\gamma_2(t,z)) - X_1(t^-)X_2(t^-)\big\}\bar{N}(\mathrm{d}t,\mathrm{d}z)\\
&= X_2(t)[\alpha_1\mathrm{d}t + \sigma_1\mathrm{d}B(t)] + X_1(t)[\alpha_2\mathrm{d}t + \sigma_2\mathrm{d}B(t)] + \sigma_1\sigma_2\mathrm{d}t\\
&\quad + \int_{|z|<R}\gamma_1(t,z)\gamma_2(t,z)\nu(\mathrm{d}z)\mathrm{d}t\\
&\quad + \int_{\mathbb{R}}\big\{\gamma_1(t,z)\gamma_2(t,z) + X_1(t^-)\gamma_2(t,z) + X_2(t^-)\gamma_1(t,z)\big\}\bar{N}(\mathrm{d}t,\mathrm{d}z). \quad (14.1.3)
\end{aligned}$$

In particular, if $\mathrm{d}X(t) = \alpha\,\mathrm{d}t + \sigma\,\mathrm{d}B(t) + \int_{\mathbb{R}}\gamma(t,z)\bar{N}(\mathrm{d}t,\mathrm{d}z)$, we get

$$\begin{aligned}\mathrm{d}(\mathrm{e}^{\lambda t}X(t)) &= X(t)\lambda\mathrm{e}^{\lambda t}\mathrm{d}t + \mathrm{e}^{\lambda t}[\alpha\,\mathrm{d}t + \sigma\,\mathrm{d}B(t)] + \int_{\mathbb{R}} \mathrm{e}^{\lambda t}\gamma(t,z)\bar{N}(\mathrm{d}t,\mathrm{d}z)\\ &= \mathrm{e}^{\lambda t}\mathrm{d}X(t) + \lambda\,X(t)\mathrm{e}^{\lambda t}\mathrm{d}t.\end{aligned}$$

(i) Now consider the equation

$$\mathrm{d}X(t) = (m - X(t))\mathrm{d}t + \sigma\,\mathrm{d}B(t) + \gamma\int_{\mathbb{R}} z\widetilde{N}(\mathrm{d}t,\mathrm{d}z),$$

where we assume that $\gamma\,z > -1$ for a.a. z (ν). It can be written

$$\mathrm{d}(\mathrm{e}^{t}X(t)) = m\,\mathrm{e}^{t}\mathrm{d}t + \sigma\,\mathrm{e}^{t}\mathrm{d}B(t) + \gamma\,\mathrm{e}^{t}\int_{\mathbb{R}} z\widetilde{N}(\mathrm{d}t,\mathrm{d}z).$$

This gives the solution

$$X(t) = X(0)\mathrm{e}^{-t} + m\int_0^t \mathrm{e}^{(s-t)}ds + \sigma\int_0^t \mathrm{e}^{(s-t)}\mathrm{d}B(s) + \gamma\int_0^t\int_{\mathbb{R}} z\mathrm{e}^{(s-t)}\widetilde{N}(\mathrm{d}t,\mathrm{d}z)$$

or

$$X(t) = m + (X_0 - m)\mathrm{e}^{-t} + \sigma\int_0^t \mathrm{e}^{s-t}\mathrm{d}B(s) + \gamma\int_0^t\int_{\mathbb{R}} z\mathrm{e}^{s-t}\widetilde{N}(\mathrm{d}t,\mathrm{d}z). \quad (14.1.4)$$

(ii) Next consider the equation

$$\mathrm{d}X(t) = \alpha\,\mathrm{d}t + \gamma\,X(t^-)\int_{\mathbb{R}} z\bar{N}(\mathrm{d}t,\mathrm{d}z); \qquad X(0) = x \in \mathbb{R}. \quad (14.1.5)$$

Define, for a given function $\theta(z)$,

$$G(t) = \exp\left(\int_0^t\int_{\mathbb{R}} \theta(z)\bar{N}(\mathrm{d}t,\mathrm{d}z) - \int_{|z|<R}\left\{\mathrm{e}^{\theta(z)} - 1 - \theta(z)\right\}\nu(\mathrm{d}z)\cdot t\right).$$

Then by Itô's formula (see Exercise 1.1)

$$\mathrm{d}G(t) = G(t^-)\int_{\mathbb{R}}\left\{\mathrm{e}^{\theta(z)} - 1\right\}\bar{N}(\mathrm{d}t,\mathrm{d}z).$$

Hence, if we put

$$\widetilde{X}(t) = X(0)G(t) + \alpha\, G(t) \int_0^t G^{-1}(s)\mathrm{d}s, \tag{14.1.6}$$

we have

$$\begin{aligned}
\mathrm{d}\widetilde{X}(t) &= X(0)\mathrm{d}G(t) + \alpha\, G(t)G^{-1}(t)\mathrm{d}t + \alpha \int_0^t G^{-1}(s)\mathrm{d}s \cdot \mathrm{d}G(t) \\
&= \alpha\, \mathrm{d}t + X(0)G(t^-) \int_{\mathbb{R}} \left\{\mathrm{e}^{\theta(z)} - 1\right\} \bar{N}(\mathrm{d}t, \mathrm{d}z) \\
&\quad + \alpha \cdot \int_0^t G^{-1}(s)\mathrm{d}s \cdot \left[G(t^-) \int_{\mathbb{R}} \left\{\mathrm{e}^{\theta(z)} - 1\right\} \bar{N}(\mathrm{d}t, \mathrm{d}z) \right] \\
&= \alpha\, \mathrm{d}t + \left[X(0)G(t^-) + \alpha\, G(t^-) \int_0^t G^{-1}(s)\mathrm{d}s \right] \int_{\mathbb{R}} \left\{\mathrm{e}^{\theta(z)} - 1\right\} \bar{N}(\mathrm{d}t, \mathrm{d}z) \\
&= \alpha\, \mathrm{d}t + \widetilde{X}(t^-) \int_{\mathbb{R}} \left\{\mathrm{e}^{\theta(z)} - 1\right\} \bar{N}(\mathrm{d}t, \mathrm{d}z).
\end{aligned}$$

So $X(t) := \widetilde{X}(t)$ solves (14.1.5) if we choose $\theta(z)$ such that

$$\mathrm{e}^{\theta(z)} - 1 = \gamma\, z \quad \text{a.s. } \nu$$

i.e.,

$$\theta(z) = \ln(1 + \gamma\, z) \quad \text{a.s. } \nu. \tag{14.1.7}$$

Exercise 1.6

By (1.2.5) (Example 1.15) we know that the equation

$$\mathrm{d}X(t) = X(t^-) \int_{\mathbb{R}} (\mathrm{e}^{\gamma(t,z)} - 1)\widetilde{N}(\mathrm{d}t, \mathrm{d}z); \quad X(0) = 1 \tag{14.1.8}$$

has the solution

$$X(t) = \exp\left\{\int_0^t \int_{\mathbb{R}} \gamma(s,z)N(\mathrm{d}s,\mathrm{d}z) - \int_0^t \int_{\mathbb{R}} (\mathrm{e}^{\gamma(s,z)} - 1)\nu(\mathrm{d}z)ds\right\}$$

$$= \exp\left\{\int_0^t \int_{\mathbb{R}} \gamma(s,z)\widetilde{N}(\mathrm{d}s,\mathrm{d}z) - \int_0^t \int_{\mathbb{R}} (\mathrm{e}^{\gamma(s,z)} - 1 - \gamma(s,t))\nu(\mathrm{d}z)\mathrm{d}s\right\}. \tag{14.1.9}$$

If we assume that

$$\int_0^t \int_{\mathbb{R}} (\mathrm{e}^{\gamma(s,z)} - 1)^2 \nu(\mathrm{d}z)\mathrm{d}s < \infty \tag{14.1.10}$$

then by (14.1.8) we see that $E[X(t)] = 1$ and hence by (14.1.9) we get

$$E\left[\exp\left\{\int_0^t \int_{\mathbb{R}} \gamma(s,z)\widetilde{N}(\mathrm{d}s,\mathrm{d}z)\right\}\right] = \exp\left\{\int_0^t \int_{\mathbb{R}} (\mathrm{e}^{\gamma(s,z)} - 1 - \gamma(s,z))\nu(\mathrm{d}z)\mathrm{d}s\right\}. \tag{14.1.11}$$

Exercise 1.7

By (14.1.3) in the solution of Exercise 1.2 we have (with $R = \infty$)

$$\begin{aligned}
\mathrm{d}(X_1(t)X_2(t)) &= \int_{\mathbb{R}} \gamma_1(t,z)\gamma_2(t,z)\nu(\mathrm{d}z)\mathrm{d}t \\
&\quad + \int_{\mathbb{R}} \left\{\gamma_1(t,z) + X_1(t^-)\gamma_2(t,z) + X_2(t^-)\gamma_1(t,z)\right\}\widetilde{N}(\mathrm{d}t,\mathrm{d}z) \\
&= X_1(t^-)\int_{\mathbb{R}} \gamma_2(t,z)\widetilde{N}(\mathrm{d}t,\mathrm{d}z) \\
&\quad + X_2(t^-)\int_{\mathbb{R}} \gamma_1(t,z)\widetilde{N}(\mathrm{d}t,\mathrm{d}z) + \int_{\mathbb{R}} \gamma_1(t,z)\gamma_2(t,z)\nu(\mathrm{d}z) \\
&= X_1(t^-)\mathrm{d}X_2(t) + X_2(t^-)\mathrm{d}X_1(t) + \int_{\mathbb{R}} \gamma_1(t,z)\gamma_2(t,z)N(\mathrm{d}t,\mathrm{d}z),
\end{aligned} \tag{14.1.12}$$

which is (1.5.1).

Exercise 1.9

$$Z(t) = \exp\left(\int_0^t \int_{\mathbb{R}} \ln(1-\theta(s,z))\widetilde{N}(\mathrm{d}s,\mathrm{d}z) + \int_0^t \int_{\mathbb{R}} \{\ln(1-\theta(s,z)) + \theta(s,z)\}\,\nu(\mathrm{d}z)\mathrm{d}s \right).$$

Define

$$\mathrm{d}X(t) = \alpha(t)\mathrm{d}t + \int_{\mathbb{R}} \gamma(t,z)\widetilde{N}(\mathrm{d}t,\mathrm{d}z)$$

where

$$\alpha(t) = \int_{\mathbb{R}} \{\ln(1-\theta(t,z)) + \theta(t,z)\}\,\nu(\mathrm{d}z)$$

and

$$\gamma(t,z) = \ln(1-\theta(t,z)).$$

Then by the Itô formula, with $Z(t) = f(X(t))$, $f(x) = \exp(x)$

$$\begin{aligned}
\mathrm{d}Z(t) &= f'(X(t))\alpha(t)\mathrm{d}t \\
&\quad + \int_{\mathbb{R}} \left\{ f(X(t^-) + \gamma(t,z)) - f(X(t^-)) - f'(X(t^-))\gamma(t,z) \right\} \nu(\mathrm{d}z)\mathrm{d}t \\
&\quad + \int_{\mathbb{R}} \left\{ f(X(t^-) + \gamma(t,z)) - f(X(t^-)) \right\} \widetilde{N}(\mathrm{d}t,\mathrm{d}z) \\
&= Z(t)\alpha(t)\mathrm{d}t \\
&\quad + \int_{\mathbb{R}} \left\{ \exp(X(t^-) + \gamma(t,z)) - \exp(X(t^-)) - \exp(X(t^-))\gamma(t,z) \right\} \nu(\mathrm{d}z)\mathrm{d}t \\
&\quad + \int_{\mathbb{R}} \left\{ \exp(X(t^-) + \gamma(t,z)) - \exp(X(t^-)) \right\} \widetilde{N}(\mathrm{d}t,\mathrm{d}z) \\
&= Z(t)\alpha(t)\mathrm{d}t + \int_{\mathbb{R}} Z(t^-) \{\exp\gamma(t,z) - 1 - \gamma(t,z)\}\,\nu(\mathrm{d}z)\mathrm{d}t \\
&\quad + \int_{\mathbb{R}} Z(t^-) \{\exp\gamma(t,z) - 1\}\,\widetilde{N}(\mathrm{d}t,\mathrm{d}z) \\
&= Z(t) \int_{\mathbb{R}} \{\ln(1-\theta(t,z)) + \theta(t,z)\}\,\nu(\mathrm{d}z)\mathrm{d}t \\
&\quad + Z(t^-) \int_{\mathbb{R}} \{1 - \theta(t,z) - 1 - \ln(1-\theta(t,z))\}\,\nu(\mathrm{d}z)\mathrm{d}t
\end{aligned}$$

$$+ Z(t^-) \int_{\mathbb{R}} \{1 - \theta(t,z) - 1\} \widetilde{N}(\mathrm{d}t, \mathrm{d}z)$$
$$= -Z(t) \int_{\mathbb{R}} \theta(t,z) \widetilde{N}(\mathrm{d}t, \mathrm{d}z).$$

14.2 Exercises of Chap. 2

Exercise 2.1

(a) In this case Eq. (2.1.17) gets the form

$$S(t^-) \int_{\mathbb{R}} z\theta(t,z)\nu(dz) = S(t^-)\alpha.$$

If $\int_{\mathbb{R}} |z|\nu(dz) > |\alpha|$ we can choose $A \subset \mathbb{R}_0$ such that

$$|\alpha| < \int_A |z|\nu(dz) := K < \infty$$

and then choose

$$\theta(t,z) = \frac{\alpha}{K} \chi_A(z) \mathrm{sign}(z).$$

Hence the market has no arbitrage, by Theorem 2.3.
(b) In this case Eq. (2.1.17) gets the form

$$\begin{bmatrix} -1 \\ 2 \end{bmatrix} u + \int_{\mathbb{R}} \begin{bmatrix} 3z \\ -6z \end{bmatrix} \theta(z)\nu(dz) = \begin{bmatrix} -1 \\ 4 \end{bmatrix},$$

which has no solution $(u, \theta(z))$. By Theorem 2.5(b) the market has an arbitrage. Indeed, if we choose $\varphi_1(t) = \frac{2}{S_1(t)}$, $\varphi_2(t) = \frac{1}{S_2(t)}$ and $\varphi_0(t)$ such that $\varphi = (\varphi_0, \varphi_1, \varphi_2)$ is self-financing and $X^\varphi(0) = 0$, then

$$X^\varphi(t) = \int_0^t \varphi_1(s) dS_1(s) + \int_0^t \varphi_2(s) dS_2(s) = \int_0^t 2ds = 2t > 0$$

for all $t > 0$. Hence φ is an arbitrage.

Exercise 2.2

(a) We apply Theorem 1.31 and try to find a solution $\theta(z) \leq 1$ of (1.4.4), which in this case gets the form

$$\int_{\mathbb{R}} z\theta(z)\nu(\mathrm{d}z) = \frac{\mu - r}{\gamma}. \tag{14.2.1}$$

By (2.41) we see that if $A \in \mathbb{B}_0$ with $\left|\frac{\mu-r}{\gamma}\right| < \int_A |z|\nu(\mathrm{d}z) := K < \infty$ then

$$\theta(z) = \frac{\mu - r}{\gamma K}\mathcal{X}_A(z)[z]$$

is a possible solution. If ν is concentrated on one point z_0, i.e., if

$$\nu(\mathbb{R}) = \nu(\{z_0\})$$

(which means that $\eta(t)$ is a Poisson process multiplied by z_0) then this is the only solution. On the other hand, if there exist two sets $A, B \in \mathbb{B}_0$ such that $A \cap B = \emptyset$ and

$$\nu(A) > 0, \quad \nu(B) > 0 \tag{14.2.2}$$

then we see that there are infinitely many solutions $\theta(z)$ of (14.2.1) such that $\theta(z) < 1$.

Fix a solution $\theta(z) < 1$ of (14.2.1) and define

$$Z(t) = Z^\theta(t) = \exp\left\{\int_0^t \int_{\mathbb{R}} \ln(1-\theta(z))\tilde{N}(\mathrm{d}s, \mathrm{d}z) + t\int_{\mathbb{R}} \{\ln(1-\theta(z)) + \theta(z)\}\nu(\mathrm{d}z)\right\}, \quad 0 \le t \le T$$

and

$$\mathrm{d}Q = \mathrm{d}Q^\theta = Z^\theta(T)\mathrm{d}P \quad \text{on } \mathbb{F}_T.$$

Then by Theorem 1.31 $\bar{S}_1(t)$ is a local martingale with respect to Q.

(b) We combine (2.4.3) and (2.4.4) to get

$$\phi_0(t) = \mathrm{e}^{-rt}(V^\phi(t) - \phi_1(t)S_1(t))$$

and

$$\mathrm{d}V^\phi(t) = rV^\phi(t)\mathrm{d}t + \phi_1(t)S_1(t^-)\left[(\mu - r)\mathrm{d}t + \gamma\int_{\mathbb{R}} z\widetilde{N}(\mathrm{d}t, \mathrm{d}z)\right].$$

From this we obtain

$$\mathrm{d}(\mathrm{e}^{-rt}V^\phi(t)) = \mathrm{e}^{-rt}\phi_1(t)S_1(t^-)\left[(\mu - r)\mathrm{d}t + \gamma\int_{\mathbb{R}} z\widetilde{N}(\mathrm{d}t, \mathrm{d}z)\right]$$

or

$$e^{-rt}V^{\phi}(t) = V^{\phi}(0) + \int_0^t e^{-rs}\phi_1(s)S_1(s^-)\left[(\mu - r)ds + \gamma\int_{\mathbb{R}} z\widetilde{N}(ds, dz)\right].$$

Therefore $e^{-rt}V^{\phi}(t)$ is a lower bounded local martingale, and hence a supermartingale, with respect to Q. But then

$$0 = E_Q[V^{\phi}(0)] \geq E_Q[e^{-rT}V^{\phi}(T)],$$

which shows that (2.4.6) cannot hold.

We conclude that *there is no arbitrage in this market* (if (2.4.1) holds).

This example illustrates the First Fundamental Theorem of Asset Pricing, which states the connection between (1) the existence of an ELMM and (2) the nonexistence of arbitrage, or No Free Lunch with Vanishing Risk (NFLVR) to be more precise. See, e.g., [DS, LS].

Exercise 2.3

To find Q we apply Theorem 1.31. So we must find a solution $(\theta_1(z), \theta_2(z))$ of the two equations:

(i) $\gamma_{11}\int_{\mathbb{R}} z\theta_1(z)\nu_1(dz) + \gamma_{12}\int_{\mathbb{R}} z\theta_2(z)\nu_2(dz) = \mu_1$

(ii) $\gamma_{21}\int_{\mathbb{R}} z\theta_1(z)\nu_1(dz) + \gamma_{22}\int_{\mathbb{R}} z\theta_2(z)\nu_2(dz) = \mu_2$

and such that $\theta_j(z) < 1$ for $j = 1, 2$.

This system is equivalent to

(iii) $\int_{\mathbb{R}} z\theta_1(z)\nu_1(dz) = \lambda_{11}\mu_1 + \lambda_{12}\mu_2$

(iv) $\int_{\mathbb{R}} z\theta_2(z)\nu_2(dz) = \lambda_{21}\mu_1 + \lambda_{22}\mu_2.$

By our assumption (2.4.7) we see that we can choose $A_i \in \mathbb{B}_0$ with

$$|\lambda_{i1}\mu_1 + \lambda_{i2}\mu_2| < \int_{A_i} |z|\nu_i(dz) = K_i < \infty$$

and then the functions

$$\theta_i(z) = \text{sign}z\frac{\lambda_{i1}\mu_1 + \lambda_{i2}\mu_2}{K_i}\mathcal{X}_{A_i}(z)\ ; \quad i = 1, 2$$

solve (i), (ii). With this choice of $\theta_i(z)$; $i = 1, 2$ we define

$$Z(t) = \exp\left\{\sum_{i=1}^{2}\left[\int_0^t\int_{\mathbb{R}} \ln(1-\theta_i(z_i))\tilde{N}_i(ds, \mathrm{d}z_i) + \int_0^t\int_{\mathbb{R}} \{\ln(1-\theta_i(z)) + \theta_i(z)\}\,\nu_i(\mathrm{d}z)\right]\right\}; \quad 0 \le t \le T$$

and we put

$$\mathrm{d}Q = Z(T)\mathrm{d}P \quad \text{on } \mathbb{F}_T.$$

Then Q is an equivalent local martingale measure for $(S_1(t), S_2(t))$. By Theorem 2.3 the market has no arbitrage.

14.3 Exercises of Chap. 3

Exercise 3.1

We seek

$$\Phi(s,x) = \sup_{\tau \ge 0} E^{(s,x)}\left[\mathrm{e}^{-\rho(s+\tau)}(X(\tau) - a)\right],$$

where

$$\mathrm{d}X(t) = \mathrm{d}B(t) + \gamma\int_{\mathbb{R}} z\tilde{N}(\mathrm{d}t, \mathrm{d}z); \quad X(0) = x \in \mathbb{R}.$$

We intend to apply Theorem 3.2 and start by putting

$$Y(t) = \begin{bmatrix} s+t \\ X(t) \end{bmatrix}; \quad Y(0) = \begin{bmatrix} s \\ x \end{bmatrix} = y \in \mathbb{R}^2 = \mathcal{S}.$$

The generator of Y is

$$A\phi(s,z) = \frac{\partial\phi}{\partial s} + \tfrac{1}{2}\frac{\partial^2\phi}{\partial x^2} + \int_{\mathbb{R}}\left\{\phi(s, x+\gamma z) - \phi(s,x) - \frac{\partial\phi}{\partial x}(s,x)\gamma z\right\}\nu(\mathrm{d}z).$$

According to Theorem 3.2 (ix) we should look for a function ϕ such that $A\phi(s,x) = 0$ in D. We try

$$\phi(s,x) = \mathrm{e}^{-\rho s}\psi(x) \quad \text{for some function } \psi.$$

Then

$$A\phi(s,x) = \mathrm{e}^{-\rho s} A_0 \psi(x),$$

where

$$A_0\psi(x) = -\rho\psi(x) + \tfrac{1}{2}\psi''(x) + \int_{\mathbb{R}} \left\{\psi(x+\gamma z) - \psi(x) - \psi'(x)\gamma z\right\} \nu(\mathrm{d}z).$$

Choose

$$\psi(x) = \mathrm{e}^{\lambda x} \quad \text{for some constant } \lambda > 0.$$

Then

$$\begin{aligned} A_0\psi(x) &= -\rho\mathrm{e}^{\lambda x} + \tfrac{1}{2}\lambda^2\mathrm{e}^{\lambda x} + \int_{\mathbb{R}} \left\{\mathrm{e}^{\lambda(x+\gamma z)} - \mathrm{e}^{\lambda x} - \lambda\mathrm{e}^{\lambda x}\cdot\gamma z\right\} \nu(\mathrm{d}z) \\ &= \mathrm{e}^{\lambda x}\Big[-\rho + \tfrac{1}{2}\lambda^2 + \int_{\mathbb{R}} \left\{\mathrm{e}^{\lambda\gamma z} - 1 - \lambda\gamma z\right\} \nu(\mathrm{d}z)\Big]. \end{aligned}$$

Put

$$h(\lambda) := -\rho + \tfrac{1}{2}\lambda^2 + \int_{\mathbb{R}} \left\{\mathrm{e}^{\lambda\gamma z} - 1 - \lambda\gamma z\right\} \nu(\mathrm{d}z).$$

Note that $h(0) = -\rho < 0$. Therefore, since

$$\mathrm{e}^{\lambda\gamma z} - 1 - \lambda\gamma z \geq 0 \quad \text{for all } x \in \mathbb{R},$$

we see that $\lim_{\lambda\to\infty} h(\lambda) = \infty$.

So the equation $h(\lambda) = 0$ has at least one solution $\lambda_1 > 0$. Define

$$\psi(x) = \begin{cases} x - a & \text{for } x \geq x^*, \\ C\,\mathrm{e}^{\lambda_1 x} & \text{for } x < x^*, \end{cases} \tag{14.3.1}$$

where $C > 0$, $x^* > 0$ are two constants to be determined.

If we require ψ to be continuous at $x = x^*$ we get the equation

$$C\,\mathrm{e}^{\lambda_1 x^*} = x^* - a. \tag{14.3.2}$$

If we require ψ to be differentiable at $x = x^*$ we get the additional equation

$$\lambda_1 C\,\mathrm{e}^{\lambda_1 x^*} = 1. \tag{14.3.3}$$

Dividing (14.3.1) by (14.3.2) we get

$$x^* = a + \frac{1}{\lambda_1}\,, \quad C = \frac{1}{\lambda_1}\,\mathrm{e}^{-(\lambda_1 a+1)}. \tag{14.3.4}$$

We now propose that the function

$$\phi(s,x) := \mathrm{e}^{-\rho s}\psi(x)$$

with $\psi(x)$ given by (14.3.1)–(14.3.3) satisfies all the requirements of Theorem 3.2 (possibly under some assumptions) and hence that

$$\phi(s,x) = \Phi(s,x)$$

and that

$$\tau^* := \inf\left\{t > 0;\, X(t) \geq x^*\right\}$$

is an optimal stopping time.

We proceed to check if the conditions (i)–(xi) of Theorem 3.2 hold. Many of these conditions are satisfied trivially or by construction of ϕ. We only discuss the remaining ones:
(ii) We know that $\phi = g$ for $x > x^*$, by construction. For $x < x^*$ we must check that

$$C\mathrm{e}^{\lambda_1 x} \geq x - a\,.$$

To this end, put

$$k(x) = C\mathrm{e}^{\lambda_1 x} - x + a\,; \quad x \leq x^*.$$

Then

$$\begin{aligned} &k(x^*) = k'(x^*) = 0 \quad \text{and} \\ &k''(x^*) = \lambda_1^2 C\mathrm{e}^{\lambda_1 x} > 0 \quad \text{for } x \leq x^*. \end{aligned}$$

Therefore $k'(x) < 0$ for $x < x^*$ and hence $k(x) > 0$ for $x < x^*$. Hence (ii) holds.
(vi) We know that $A\phi + f = A\phi = 0$ for $x < x^*$, by construction. For $x > x^*$ we have

$$\begin{aligned} A\phi(s,x) &= \mathrm{e}^{-\rho s}A_0(x-a) \\ &= \mathrm{e}^{-\rho s}(-\rho(x-a) + \int_{x+\gamma z < x^*} \left\{C\mathrm{e}^{\lambda_1(x+\gamma z)} - (x+\gamma z - a)\right\}\nu(\mathrm{d}z)) \\ &< \mathrm{e}^{-\rho s}(-\rho(x^*-a) + \int_{\mathbb{R}} \left\{C\mathrm{e}^{\lambda_1 x^*} - (x^*+\gamma z - a)\right\}\nu(\mathrm{d}z)). \end{aligned}$$

Therefore, using (14.3.2) and (14.3.4) we see that (vi) holds if

$$-\int_{\mathbb{R}} \{\lambda_1 \gamma z\}\, \nu(\mathrm{d}z) \le \rho. \tag{14.3.5}$$

(viii) In our case this condition gets the form

$$E\Big[\int_0^\infty \Big\{\sigma^2 \mathrm{e}^{-2\rho t} X^2(t) + \int_{\mathbb{R}} \mathrm{e}^{-2\rho t} \big|(X(t)+\gamma z)^2 - X^2(t)\big|^2 \nu(\mathrm{d}z)\Big\}\, \mathrm{d}t\Big] < \infty$$

i.e.,

$$E\Big[\int_0^\infty \mathrm{e}^{-2\rho t} \Big\{\sigma^2 X^2(t) + \int_{\mathbb{R}} \big|2X(t)\gamma z + \gamma^2 z^2\big| \nu(\mathrm{d}z)\Big\}\, \mathrm{d}t\Big] < \infty\,.$$

This will hold if

$$\gamma z \le 0 \quad \text{a.s. } \nu \tag{14.3.6}$$

or if

$$\sup_{\tau \in \mathcal{T}} E^x\Big[\mathrm{e}^{-2\rho\tau}\Big(\int_0^\tau \int_{\mathbb{R}} z N(\mathrm{d}s, \mathrm{d}z)\Big)^2\Big] < \infty\,. \tag{14.3.7}$$

We will not discuss this condition further here.
(x) With our proposed solution ϕ we have

$$D = \big\{(s,x) \in \mathbb{R}^2; x < x^*\big\}\,.$$

So condition (x) states that

$$\tau_D := \inf\big\{t > 0; X(t) > x^*\big\} < \infty \text{ a.s.} \tag{14.3.8}$$

Some conditions are needed on σ, γ, and ν for (14.3.8) to hold. For example, it suffices that

$$\overline{\lim_{t\to\infty}}\, X(t) = \overline{\lim_{t\to\infty}} \Big\{\sigma\, B(t) + \int_0^t \int_{\mathbb{R}} \gamma\, z N(\mathrm{d}s, \mathrm{d}z)\Big\} = \infty \quad \text{a.s.} \tag{14.3.9}$$

(xi) For (xi) to hold it suffices that

$$\sup_{\tau\in\mathcal{T}} E^x[\mathrm{e}^{-2\rho\tau}X^2(\tau)] < \infty\ . \tag{14.3.10}$$

Again it suffices to assume that (14.3.7) holds.

Conclusion

Assume that (14.3.5), (14.3.6), and (14.3.8) hold. Then the value function is

$$\Phi(s,x) = \mathrm{e}^{-\rho s}\psi(x),$$

where $\psi(x)$ is given by (14.3.1) and (14.3.4). An optimal stopping time is

$$\tau^* = \inf\left\{t > 0;\, X(t) \geq x^*\right\}.$$

Exercise 3.2

Define

$$\mathrm{d}Y(t) = \begin{bmatrix} \mathrm{d}t \\ \mathrm{d}P(t) \\ \mathrm{d}Q(t) \end{bmatrix} = \begin{bmatrix} 1 \\ \alpha\, P(t) \\ -\lambda\, Q(t) \end{bmatrix}\mathrm{d}t + \begin{bmatrix} 0 \\ \beta\, P(t) \\ 0 \end{bmatrix}\mathrm{d}B(t) + \begin{bmatrix} 0 \\ \gamma \int_{\mathbb{R}} P(t^-)z\widetilde{N}(\mathrm{d}t,\mathrm{d}z) \\ 0 \end{bmatrix}.$$

Then the generator A of $Y(t)$ is

$$\begin{aligned} A\phi(y) = A\phi(s,p,q) &= \frac{\partial\phi}{\partial s} + \alpha\, p\frac{\partial\phi}{\partial p} - \lambda\, q\frac{\partial\phi}{\partial q} + \tfrac{1}{2}\beta^2 p^2\frac{\partial^2\phi}{\partial p^2} \\ &+ \int_{\mathbb{R}} \left\{\phi(s,\, p+\gamma\, pz,\, q) - \phi(s,p,q) - \frac{\partial\phi}{\partial p}(s,p,q)\gamma\, zp\right\}\nu(\mathrm{d}z)\ . \end{aligned}$$

If we try

$$\phi(s,\,p,\,q) = \mathrm{e}^{-\rho s}\psi(w) \qquad \text{with } w = p\cdot q\ ,$$

then

$$A\phi(s,\,p,\,q) = \mathrm{e}^{-\rho s}A_0\psi(w),$$

where

$$\begin{aligned} A_0\psi(w) &= -\rho\psi(w) + (\alpha-\lambda)w\psi'(w) + \tfrac{1}{2}\beta^2 w^2\psi''(w) \\ &+ \int_{\mathbb{R}} \left\{\psi((1+\gamma\, z)w) - \psi(w) - \gamma\, wz\psi'(w)\right\}\nu(\mathrm{d}z). \end{aligned}$$

Consider the set U defined in Proposition 3.3:

$$\begin{aligned} U &= \{y;\, Ag(y) + f(y) > 0\} = \{(s,\, p,\, q);\, A_0(\theta\, w) + \lambda\, w - K > 0\} \\ &= \{(s,\, p,\, q);\, [\theta(\alpha - \rho - \lambda) + \lambda]w - K > 0\} \\ &= \begin{cases} \left\{(s,\, p,\, q) : w > \dfrac{K}{\theta(\alpha - \rho - \lambda) + \lambda}\right\} & \text{if } \theta(\alpha - \rho - \lambda) + \lambda > 0 \\ \emptyset & \text{if } \theta(\alpha - \rho - \lambda) + \lambda \leq 0. \end{cases} \end{aligned}$$

By Proposition 3.4 we therefore get

Case 1:

Assume $\lambda \leq \theta(\lambda + \rho - \alpha)$.
Then $\tau^* = 0$ is optimal and $\Phi(y) = g(y) = \mathrm{e}^{-\rho s} p \cdot q$ for all y.

Case 2:

Assume $\theta(\lambda + \rho - \alpha) < \lambda$.
Then $U = \left\{(s, w);\, w > \dfrac{K}{\lambda - \theta(\lambda + \rho - \alpha)}\right\} \subset D$.

In view of this it is natural to guess that the continuation region D has the form

$$D = \left\{(s, w);\, 0 < w^* < w\right\}$$

for some constant w^*; $0 < w^* < \dfrac{K}{\lambda - \theta(\lambda + \rho - \alpha)}$. In D we try to solve the equation

$$A_0\psi(w) + f(w) = 0.$$

The homogeneous equation $A_0\psi_0(w) = 0$ has a solution $\psi_0(w) = w^r$ if and only if

$$h(r) := -\rho + (\alpha - \lambda)r + \tfrac{1}{2}\beta^2 r(r-1) + \int_{\mathbb{R}} \left\{(1 + \gamma\, z)^r - 1 - r\, \gamma\, z\right\} \nu(\mathrm{d}z) = 0.$$

Since $h(0) = -\rho < 0$ and $\lim\limits_{|r| \to \infty} h(r) = \infty$, we see that the equation $h(r) = 0$ has two solutions r_1, r_2 such that

$$r_2 < 0 < r_1.$$

Let r be a solution of this equation. To find a particular solution $\psi_1(w)$ of the non-homogeneous equation

$$A_0\psi_1(w) + \lambda w - K = 0$$

we try

$$\psi_1(w) = aw + b$$

and find

$$a = \frac{\lambda}{\lambda + \rho - \alpha}, \qquad b = -\frac{K}{\rho}.$$

This gives that for all constants C the function

$$\psi(w) = C\, w^r + \frac{\lambda}{\lambda + \rho - \alpha} w - \frac{K}{\rho}$$

is a solution of

$$A_0\psi(w) + \lambda\, w - K = 0.$$

Therefore we try to put

$$\psi(w) = \begin{cases} \theta\, w; & 0 < w \le w^* \\ C\, w^r + \dfrac{\lambda}{\lambda + \rho - \alpha} w - \dfrac{K}{\rho}; & w \ge w^*, \end{cases} \tag{14.3.11}$$

where $w^* > 0$ and C remain to be determined.

Continuity and differentiability at $w = w^*$ give

$$\theta\, w^* = C(w^*)^r + \frac{\lambda}{\lambda + \rho - \alpha} w^* - \frac{K}{\rho} \tag{14.3.12}$$

$$\theta = C\, r(w^*)^{r-1} + \frac{\lambda}{\lambda + \rho - \alpha}. \tag{14.3.13}$$

Combining (14.3.12) and (14.3.13) we get

$$w^* = \frac{(-r)K(\lambda + \rho - \alpha)}{(1 - r)\rho(\lambda - \theta(\lambda + \rho - \alpha))} \tag{14.3.14}$$

and

$$C = \frac{\lambda - \theta(\lambda + \rho - \alpha)}{-r} \cdot (w^*)^{1-r}. \tag{14.3.15}$$

Since we need to have $w^* > 0$ we are led to the following condition:

Case 2a:

$\theta(\lambda + \rho - \alpha) < \lambda$ and $\lambda + \rho - \alpha > 0$.

Then we choose $r = r_2 < 0$, and with the corresponding values (14.3.14), (14.3.15) of w^* and C the function $\phi(s, p, q) = \mathrm{e}^{-\rho s}\psi(p \cdot q)$, with ψ given by (14.3.11), is the value function of the problem. The optimal stopping time τ^* is

$$\tau^* = \inf\left\{t > 0;\, P(t) \cdot Q(t) \le w^*\right\}, \tag{14.3.16}$$

provided that all the other conditions of Theorem 3.2 are satisfied. For condition (vi) to hold it suffices that

$$\begin{aligned} &w^*(\lambda - \theta(\lambda + \rho - \alpha)) - K \\ &+ \int_{\gamma z w > w^*} \left\{ C(w + \gamma z w)^r + \frac{\lambda}{\lambda + \rho - \alpha}(w + \gamma z w) - \frac{K}{\rho} - \theta(w + \gamma z w) \right\} \nu(\mathrm{d}z) \le 0. \end{aligned} \tag{14.3.17}$$

See also Remark 14.1.

Case 2b:

$\theta(\lambda + \rho - \alpha) < \lambda$ and $\lambda + \rho - \alpha \le 0$, i.e.,

$$\alpha \ge \lambda + \rho\,.$$

In this case we have $\Phi^*(y) = \infty$.

To see this note that since

$$P(t) = p + \int_0^t \alpha\, P(s)\mathrm{d}s + \int_0^t \beta\, P(s)\mathrm{d}B(s) + \int_0^t \int_{\mathbb{R}} \gamma\, P(s^-) z \widetilde{N}(\mathrm{d}s, \mathrm{d}z),$$

we have

$$E[P(t)] = p + \int_0^t \alpha\, E[P(s)]\mathrm{d}s,$$

which gives

$$E[P(t)] = p\, \mathrm{e}^{\alpha t}.$$

Therefore

$$E[\mathrm{e}^{-\rho t} P(t) Q(t)] = E[pq\, \mathrm{e}^{-\rho t}\mathrm{e}^{-\lambda t} P(t)] = pq \exp\{(\alpha - \lambda - \rho)t\}\,.$$

Hence

$$\lim_{T\to\infty} E\left[\int_0^T \mathrm{e}^{-\rho t} P(t)Q(t)\mathrm{d}t\right] = \lim_{T\to\infty} pq \int_0^T \exp\{(\alpha-\lambda-\rho)t\}\,\mathrm{d}t = \infty$$

if and only if $\alpha \geq \lambda + \rho$.

Remark 14.1 (***On condition (viii) of Theorem*** 3.2) Consider

$$\phi(Y(t)) = \mathrm{e}^{-\rho t}\psi(P(t)Q(t)),$$

where

$$P(t) = p\exp\left\{\left(\alpha - \tfrac{1}{2}\beta^2 - \gamma\int_{\mathbb{R}} z\,\nu(\mathrm{d}z)\right)t + \int_0^t\int_{\mathbb{R}} \ln(1+\gamma z)N(\mathrm{d}t,\mathrm{d}z) + \beta B(t)\right\}$$

and

$$Q(t) = q\exp(-\lambda t).$$

We have

$$P(t)Q(t) = pq\exp\left\{\left(\alpha-\lambda-\tfrac{1}{2}\beta^2-\gamma\int_{\mathbb{R}} z\,\nu(\mathrm{d}z)\right)t + \int_0^t\int_{\mathbb{R}} \ln(1+\gamma z)N(\mathrm{d}t,\mathrm{d}z) + \beta B(t)\right\}$$

and

$$\mathrm{e}^{-\rho t}P(t)Q(t) = pq\exp\left\{\left(\alpha-\lambda-\rho-\tfrac{1}{2}\beta^2-\gamma\int_{\mathbb{R}} z\,\nu(\mathrm{d}z)\right)t + \int_0^t\int_{\mathbb{R}} \ln(1+\gamma z)N(\mathrm{d}s,\mathrm{d}z) + \beta B(t)\right\}.$$

Hence

$$E[(\mathrm{e}^{-\rho t}P(t)Q(t))^2] = (pq)^2 E\left[\exp\left\{\left(2\alpha-2\lambda-2\rho-\beta^2-2\gamma\int_{\mathbb{R}} z\,\nu(\mathrm{d}z)\right)t\right.\right.$$

$$+2\int_0^t\int_{\mathbb{R}}\ln(1+\gamma z)N(\mathrm{d}s,\mathrm{d}z)+2\beta B(t)\Big\}\Big]$$

$$=(pq)^2\exp\left\{\left(2\alpha-2\lambda-2\rho-\beta^2-2\gamma\int_{\mathbb{R}}z\,\nu(\mathrm{d}z)\right)t+2\beta^2t\right\}$$

$$\cdot E\left[\exp\left(2\int_0^t\int_{\mathbb{R}}\ln(1+\gamma z)N(\mathrm{d}t,\mathrm{d}z)\right)\right].$$

Using Exercise 1.6 we get

$$E[(\mathrm{e}^{-\rho t}P(t)Q(t))^2]=p^2q^2\exp\left\{\left(2\alpha-2\lambda-2\rho+\beta^2-2\gamma\int_{\mathbb{R}}z\,\nu(\mathrm{d}z)\right.\right.$$

$$\left.\left.+\int_{\mathbb{R}}\left\{(1+\gamma z)^2-1-2\ln(1+\gamma z)\right\}\nu(\mathrm{d}z)\right)t\right\}.$$

So condition (viii) of Theorem 6.10 holds if

$$2\alpha-2\lambda-2\rho+\beta^2+\int_{\mathbb{R}}\left\{\gamma^2z^2-2\ln(1+\gamma z)\right\}\nu(\mathrm{d}z)<0.$$

Exercise 4.3

In this case we have

$$g(s,x)=\mathrm{e}^{-\rho s}|x|$$

and

$$\mathrm{d}X(t)=\mathrm{d}B(t)+\int_{\mathbb{R}}z\tilde{N}(\mathrm{d}t,\mathrm{d}z).$$

We look for a solution of the form

$$\phi(s,x)=\mathrm{e}^{-\rho s}\psi(x).$$

The continuation region is given by

$$D=\{(s,x)\in\mathbb{R}\times\mathbb{R}:\phi(s,x)>g(s,x)\}=\{(s,x)\in\mathbb{R}\times\mathbb{R}:\psi(x)\geq|x|\}.$$

Because of the symmetry we assume that D is of the form

$$D = \left\{(s, x) \in \mathbb{R} \times \mathbb{R};\ -x^* < x < x^*\right\},$$

where $x^* > 0$. It is trivial that D is a Lipschitz surface and $X(t)$ spends 0 time on ∂D. We must have

$$A\phi \equiv 0 \quad \text{on } D, \tag{14.3.18}$$

where the generator A is given by

$$A\phi = \frac{\partial \phi}{\partial s} + \frac{1}{2}\frac{\partial^2 \phi}{\partial x^2} + \int_{\mathbb{R}} \left\{\phi(s, x + z) - \phi(s, x) - \frac{\partial \phi}{\partial x}(s, x)z\right\} \nu(\mathrm{d}z).$$

Hence (14.3.18) becomes

$$-\rho\psi(x) + \tfrac{1}{2}\psi''(x) + \int_{\mathbb{R}} \left\{\psi(x + z) - \psi(x) - z\psi'(x)\right\} \nu(\mathrm{d}z) = 0. \tag{14.3.19}$$

For $|x| < \xi$ this equation becomes, by (3.4.1)

$$-\rho\psi(x) + \frac{1}{2}\psi''(x) = 0,$$

which has the general solution

$$\psi(x) = C_1 \mathrm{e}^{\sqrt{2\rho}x} + C_2 \mathrm{e}^{-\sqrt{2\rho}x},$$

where C_1, C_2 are arbitrary constants. Let $\lambda = \sqrt{2\rho}$ and $-\lambda$ be the two roots of the equation

$$F(\lambda) := -\rho + \frac{1}{2}\lambda^2 = 0.$$

Because of the symmetry we guess that

$$\psi(x) = \frac{C}{2}\left(\mathrm{e}^{\lambda x} + \mathrm{e}^{-\lambda x}\right) = C\cosh(\lambda x);\ x \in D = \left\{(s, x);\ |x| < x^*\right\}$$

for some constants $C > 0$ and $x^* \in (0, \xi)$. Therefore

$$\psi(x) = \begin{cases} C\cosh\left(\lambda x\right) & \text{for } |x| < x^* \\ |x| & \text{for } |x| \geq x^*. \end{cases}$$

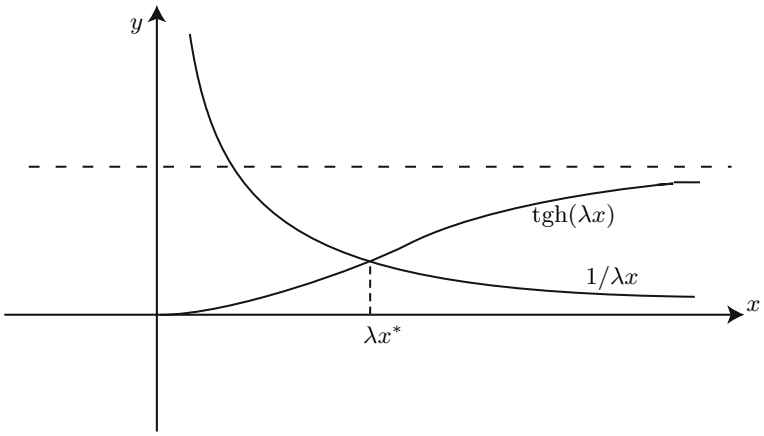


Fig. 14.1 The value of x^*

In order to find x^* and C, we impose the continuity and C^1-conditions on $\psi(x)$ at $x = x^*$:

- Continuity: $1 = |x^*| = C\cosh\left(\lambda x^*\right)$
- C^1 : $1 = C\lambda\sinh(\lambda x^*)$

It follows that:

$$C = \frac{x^*}{\cosh(\lambda x^*)} \tag{14.3.20}$$

and x^* is the solution of

$$\mathrm{tgh}\left(\lambda x^*\right) = \frac{1}{\lambda x^*}. \tag{14.3.21}$$

Figure 14.1 illustrates that there exists a unique solution for (14.3.21).

Finally we have to verify that the conditions of Theorem 3.2 hold. We check some:
(ii) $\psi(x) \geq |x|$ for $(s, x) \in D$.

Define

$$h(x) = C\cosh(\lambda x) - x \ ; \ x > 0.$$

Then $h(x^*) = h'(x^*) = 0$ and $h''(x) = C\lambda^2\cosh(\lambda x) > 0$ for all x. Hence $h(x) > 0$ for $0 < x < x^*$, so (ii) holds. See Fig. 14.2.
(vi) $A\psi \leq 0$ outside $\bar{D}$.

This holds since, by (3.4.2) and (3.4.3),

$$A\psi(x) = -\rho|x| + \int_{\mathbb{R}} \{|x + z| - x - z\}\,\nu(\mathrm{d}z) \leq 0 \text{ for all } x > x^*.$$

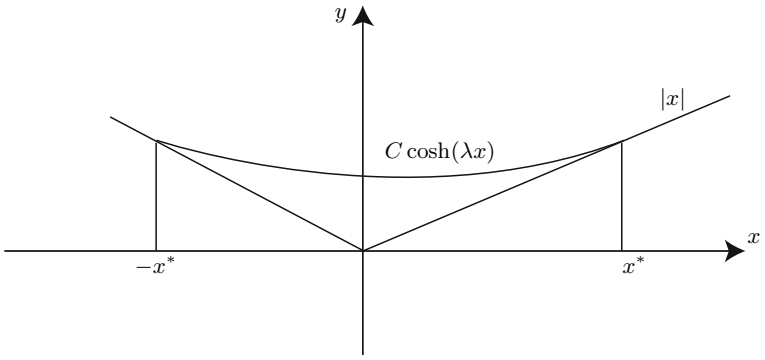


Fig. 14.2 The function ψ

Since all the conditions of Theorem 3.2 are satisfied, we conclude that

$$\phi(s, y) = \mathrm{e}^{-\rho s}\psi(y)$$

is the optimal value function and $\tau^* = \inf\{t > 0;\ |X(t)| = x^*\}$

Exercise 3.4

From Example 3.5 we know that in the *no delay case* ($\delta = 0$) the solution of the problem (3.4.8) is the following (under some additional assumptions on the Lévy measure ν):

$$\Phi_0(s, x) = \mathrm{e}^{-\rho s}\Psi_0(x), \tag{14.3.22}$$

where

$$\Psi_0(x) = \begin{cases} x - q & ; \quad x \geq x_0^* \\ C_0 x^{\lambda} & ; \quad 0 < x < x_0^*. \end{cases} \tag{14.3.23}$$

Here $\lambda > 1$ is uniquely determined by the equation

$$-\rho + \mu\lambda + \tfrac{1}{2}\sigma^2\lambda(\lambda - 1) + \int_{\mathbb{R}} \left\{(1 + z)^{\lambda} - 1 - \lambda z\right\} \nu(\mathrm{d}z) = 0, \tag{14.3.24}$$

and x_0^* and C_0 are given by

$$x_0^* = \frac{\lambda q}{\lambda - 1}, \tag{14.3.25}$$

$$C_0 = \frac{1}{\lambda}(x_0^*)^{1-\lambda}. \tag{14.3.26}$$

The corresponding optimal stopping time $\tau^* \in \mathcal{T}_0$ is

$$\tau^* = \inf\left\{t > 0;\, X(t) \geq x_0^*\right\}. \tag{14.3.27}$$

Thus it is optimal to sell at the first time the price $X(t)$ equals or exceeds the value x_0^*.

To find the solution in the *delay* case ($\delta > 0$) we note that we have $f = 0$ and

$$g(y) = g(s, x) = \mathrm{e}^{-\rho s}(x - q).$$

Hence, by (3.4.5),

$$\begin{aligned}\tilde{g}_\delta(y) &= E^y[g(Y(\delta))] = E^{s,x}[\mathrm{e}^{-\rho(s+\delta)}(X(\delta) - q)] \\ &= \mathrm{e}^{-\rho(s+\delta)}(E^x[X(\delta)] - q) = \mathrm{e}^{-\rho(s+\delta)}(x\,\mathrm{e}^{\mu\delta} - q) \\ &= \mathrm{e}^{-\rho s + \delta(\mu-\rho)}(x - q\,\mathrm{e}^{-\mu\delta}) = K\,\mathrm{e}^{-\rho s}(x - \tilde{q}),\end{aligned} \tag{14.3.28}$$

where

$$K = \mathrm{e}^{\delta(\mu-\rho)} \quad \text{and} \quad \tilde{q} = q\,\mathrm{e}^{-\mu\delta}. \tag{14.3.29}$$

Thus $\tilde{g}_\delta$ has the same form as g, so we can apply the results (14.3.22)–(14.3.27) to find $\tilde{\Phi}(y)$ and the corresponding optimal τ^*:

$$\tilde{\Phi}(y) = \tilde{\Phi}(s, x) = \mathrm{e}^{-\rho s}\tilde{\Psi}(x), \tag{14.3.30}$$

where

$$\tilde{\Psi}(x) = \begin{cases} K(x - \tilde{q}) & ; \quad x \geq \tilde{x}^* \\ \tilde{C}\,x^\lambda & ; \quad 0 < x < \tilde{x}^* \end{cases} \tag{14.3.31}$$

with λ as in (14.3.24). Here $\tilde{x}^*$ and $\tilde{C}$ are given by

$$\tilde{x}^* = \frac{\lambda\tilde{q}}{\lambda - 1}, \tag{14.3.32}$$

$$\tilde{C} = \frac{1}{\lambda}(\tilde{x}^*)^{1-\lambda}. \tag{14.3.33}$$

The corresponding optimal stopping time for problem (3.3.7) and (3.3.6), respectively, is

$$\tilde{\tau}^* = \inf\left\{t > 0;\, X(t) \geq \tilde{x}^*\right\} \tag{14.3.34}$$

$$\alpha^* = \tilde{\tau}^* + \delta. \tag{14.3.35}$$

Using Theorem 3.11 we conclude the following:

Conclusion

The value function $\Phi_\delta(y)$ for the delayed optimal stopping problem (3.4.7) is given by

$$\Phi_\delta(y) = \tilde{\Phi}(y),$$

where $\tilde{\Phi}$ is as in (14.3.30)–(14.3.33). The corresponding optimal stopping time $\alpha^* \in \mathcal{T}_\delta$ is

$$\alpha^* = \inf\left\{t > 0; X(t) \geq \tilde{x}^*\right\} + \delta.$$

Remark 14.2 Assume for example that

$$\mu > 0.$$

Then comparing (14.3.32) with the non-delayed case (14.3.25) we see that $\tilde{q} > q$ and hence

$$\tilde{x}^* < x_0^*$$

Thus, in terms of the *delayed effect of the stopping time* formulation (see 3.3.4), it is optimal to stop at the first time $t = \tilde{\tau}^*$ when $X(t) \geq \tilde{x}^*$. This is sooner than in

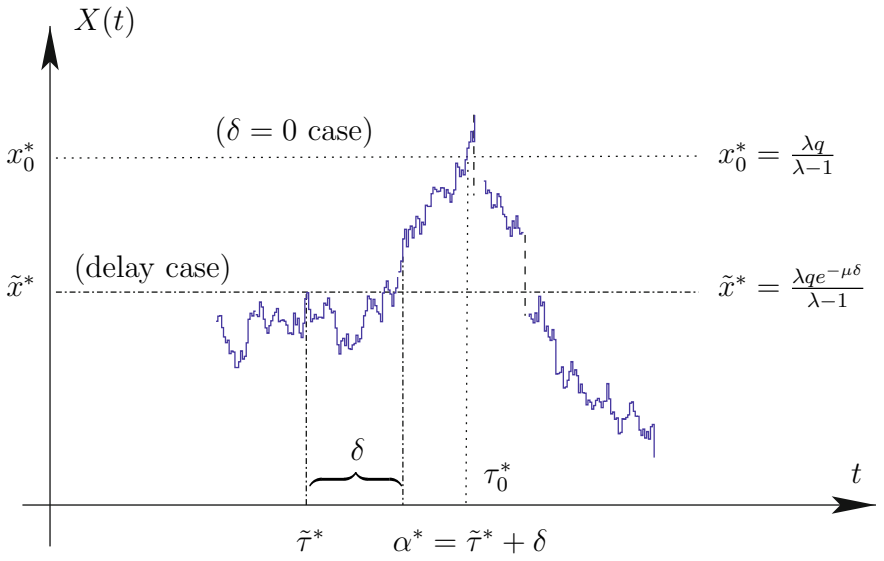


Fig. 14.3 The optimal stopping times for Exercise 3.4 ($\mu > 0$)

the non-delayed case, because of the anticipation that during the delay time interval $[\tau^*, \tau^* + \delta]$ $X(t)$ is likely to increase (since $\mu > 0$). See Fig. 14.3.

14.4 Exercises of Chap. 4

Exercise 4.3

(a) The measure Q_θ is an equivalent local martingale measure by Theorem 1.31. Since θ_0 and θ_1 are bounded, Q_θ is an EMM.
(b) By the Bayes rule for conditional expectation we have

$$\begin{aligned} Y(t) &= \frac{E\left[\exp\left(-\int_t^T r(s)ds\right) F G_\theta(T) \mid \mathcal{F}_t\right]}{G_\theta(t)} \\ &= -\frac{E\left[\exp\left(-\int_0^T r(s)ds\right) \exp\left(\int_0^t r(s)ds\right) G_\theta(T) F \mid \mathcal{F}_t\right]}{G_\theta(t)} \\ &= \frac{E\left[\exp\left(-\int_0^T r(s)ds\right) G_\theta(T) F \mid \mathcal{F}_t\right]}{\exp\left(-\int_0^t r(s)ds\right) G_\theta(t)} = \frac{E[\Gamma(T) F \mid \mathcal{F}_t]}{\Gamma(t)}, \end{aligned}$$

where

$$\Gamma(t) = \exp\left(-\int_0^t r(s)ds\right) G_\theta(t)\ ;\ 0 \le t \le T,$$

i.e.

$$\begin{cases} d\Gamma(t) = & \Gamma(t^-)\left[-r(t)dt + \theta_0(t)dB(t) + \int_{\mathbb{R}} \theta_1(t,\zeta)\tilde{N}(dt,d\zeta)\right] \\ \Gamma(0) = 1. \end{cases}$$

Comparing with (4.3.2)–(4.3.3) in Theorem 4.8 we see that $Y(t)$ is the solution of the linear BSDE

$$\begin{cases} dY(t) &= -\left[-r(t)Y(t) + \theta_0(t)Z(t) + \int_{\mathbb{R}} \theta_1(t,\zeta)K(t,\zeta)\nu(d\zeta)\right]dt \\ &\quad + Z(t)dB(t) + \int_{\mathbb{R}} K(t,\zeta)\tilde{N}(dt,d\zeta)\ ;\ 0 \le t \le T \\ Y(T) &= F, \end{cases}$$

as claimed.

14.5 Exercises of Chap. 5

Exercise 5.1

Put

$$Y(t) = \begin{bmatrix} s+t \\ X(t) \end{bmatrix}.$$

Note: If we put

$$\mathcal{S} = \{(s,x);\, s < T\}$$

then

$$\tau_{\mathcal{S}} = \inf\left\{t > 0;\ Y^{s,x}(t,x) \notin \mathcal{S}\right\} = T - s.$$

The generator of $Y(t)$ is

$$A^u\phi(y) = A^u\phi(s,x) = \frac{\partial\phi}{\partial s} + (\mu - \rho x - u)\frac{\partial\phi}{\partial x} + \tfrac{1}{2}\sigma^2\frac{\partial^2\phi}{\partial x^2} + \int_{\mathbb{R}} \left\{\phi(s, x+\theta z) - \phi(s,x) - \frac{\partial\phi}{\partial x}\cdot\theta z\right\}\nu(\mathrm{d}z).$$

So the conditions of Theorem 5.1 get the form

(i) $A^u\phi(s,x) + \mathrm{e}^{-\delta s}\dfrac{u^\gamma}{\gamma} \le 0$ for all $u \ge 0$, $s < T$,

(ii) $\lim\limits_{s\to T^-} \phi(s,x) = \lambda x$,

(iv) $\left\{\phi^-(Y(\tau))\right\}_{\tau\le\tau_{\mathcal{S}}}$ is uniformly integrable,

(v) $A^{\hat{u}}\phi(s,x) + \mathrm{e}^{-\delta s}\dfrac{\hat{u}^\gamma}{\gamma} = 0$ for $s < T$, in addition to requirements (iii) and (vi).

We try a function ϕ of the form

$$\phi(s,x) = h(s) + k(s)x$$

for suitable functions $h(s), k(s)$. Then the conditions above get the form

(i)' $h'(s) + k'(s)x + (\mu - \rho x - u)k(s) + \mathrm{e}^{-\delta s}\dfrac{u^\gamma}{\gamma}$
$+ \int_{\mathbb{R}} \{h(s) + k(s)(x+\gamma z) - h(s) - k(s)x - k(s)\gamma z\}\,\nu(\mathrm{d}z) \le 0,$
i.e.

$$\mathrm{e}^{-\delta s}\tfrac{u^\gamma}{\gamma} + h'(s) + k'(s)x + (\mu - \rho x - u)k(s) \le 0 \quad \text{for all } s < T,\ u \ge 0,$$

(ii)' $h(T) = 0,\ k(T) = \lambda,$
(iv)' $\{h(\tau) + k(\tau)X(\tau)\}_{\tau \le \tau_S}$ is uniformly integrable,
(v)' $h'(s) + k'(s)x + (\mu - \rho x - \hat{u})k(s) + \mathrm{e}^{-\delta s}\,\frac{\hat{u}^\gamma}{\gamma} = 0.$

From (i)' and (v)' we get

$$-k(s) + \mathrm{e}^{-\delta s}\,\hat{u}^{\gamma-1} = 0$$

or

$$\hat{u} = \hat{u}(s) = \left(\mathrm{e}^{\delta s}k(s)\right)^{\frac{1}{\gamma-1}}.$$

Combined with (v)' this gives

(1) $k'(s) - \rho\,k(s) = 0 \quad \text{so} \quad k(s) = \lambda\,\mathrm{e}^{\rho(s-T)},$
(2) $h'(s) = (\hat{u}(s) - \mu)k(s) - \mathrm{e}^{-\delta s}\,\frac{\hat{u}^\gamma(s)}{\gamma}, \quad h(T) = 0.$

Note that

$$\begin{aligned} h'(s) &= \left(\mathrm{e}^{\delta s}k(s)\right)^{\frac{1}{\gamma-1}}k(s) - \mu\,k(s) - \mathrm{e}^{-\delta s}\,\frac{\left(\mathrm{e}^{\delta s}k(x)\right)^{\frac{\gamma}{\gamma-1}}}{\gamma} \\ &= \mathrm{e}^{\frac{\delta s}{\gamma-1}}\,k(s)^{\frac{\gamma}{\gamma-1}} - \mu\,k(s) - \mathrm{e}^{-\delta s\left(1-\frac{\gamma}{\gamma-1}\right)} \cdot \tfrac{1}{\gamma} \cdot k(s)^{\frac{\gamma}{\gamma-1}} \\ &= \mathrm{e}^{\frac{\delta s}{\gamma-1}}\,k(s)^{\frac{\gamma}{\gamma-1}}\left[1 - \tfrac{1}{\gamma}\right] - \mu\,k(s) < 0\,. \end{aligned}$$

Hence, since $h(T) = 0$, we have $h(s) > 0$ for $s < T$. Therefore

$$\phi(s, x) = h(s) + k(s)x \ge 0\,.$$

Clearly ϕ satisfies (i), (ii), (iv) and (v). It remains to check (vi), i.e., that

$$\{h(\tau) + k(\tau)X(\tau)\}_{\tau \le T}$$

is uniformly integrable, and to check (iii).

For these properties to hold some conditions on ν must be imposed. We omit the details.

We conclude that if these conditions hold then

$$\hat{u}(s) = \lambda^{\frac{1}{\gamma-1}} \exp\left\{\frac{(\delta+\rho)s - \rho\,T}{\gamma - 1}\right\}; \quad s \le T \tag{14.5.1}$$

is the optimal control.

Exercise 5.2

Define

$$J(u) = E\Big[\int_0^{T_0} \mathrm{e}^{-\delta t}\,\frac{u^\gamma(t)}{\gamma}\mathrm{d}t + \lambda\, X(T_0)\Big],$$

where

$$\mathrm{d}X(t) = (\mu - \rho\, X(t) - u(t))\mathrm{d}t + \sigma\, B(t) + \gamma \int_{\mathbb{R}} z\,\widetilde{N}(\mathrm{d}t, \mathrm{d}z); \quad 0 \le t \le T_0.$$

The Hamiltonian is

$$H(t, x, u, p, q, r) = \mathrm{e}^{-\delta t}\,\tfrac{u^\gamma}{\gamma} + (\mu - \rho\, x - u)p + \sigma\, q + \int_{\mathbb{R}} \gamma\, z r(t, z)\nu(\mathrm{d}z).$$

The adjoint equation is

$$\begin{cases} \mathrm{d}\hat{p}(t) = \rho\,\hat{p}(t)\mathrm{d}t + \sigma\,\hat{q}(t)\mathrm{d}B(t) + \displaystyle\int_{\mathbb{R}} \hat{r}(t, z)\widetilde{N}(\mathrm{d}t, \mathrm{d}z); \quad t < T_0 \\ \hat{p}(T_0) = \lambda. \end{cases}$$

Since λ and ρ are deterministic, we guess that $\hat{q} = \hat{r} = 0$ and this gives

$$\hat{p}(t) = \lambda\, \mathrm{e}^{\rho(t - T_0)}.$$

Hence

$$H(t, \hat{X}(t), u, \hat{p}(t), \hat{q}(t), \hat{r}(t)) = \mathrm{e}^{-\delta t}\,\tfrac{u^\gamma}{\gamma} + (\mu - \rho\,\hat{X}(t) - u)\hat{p}(t),$$

which is maximal when

$$u = \hat{u}(t) = \big(\mathrm{e}^{\delta t}\,\hat{p}(t)\big)^{\frac{1}{\gamma - 1}} = \lambda^{\frac{1}{\gamma-1}} \exp\left\{\frac{(\delta + \rho)t - \rho\, T_0}{\gamma - 1}\right\}. \tag{14.5.2}$$

Exercise 5.3

In this case we have

$$\mathrm{d}X(t) = \begin{bmatrix} \displaystyle\int_{\mathbb{R}} u(t^-, \omega)z \\ \displaystyle\int_{\mathbb{R}} z^2 \end{bmatrix} \widetilde{N}(\mathrm{d}t, \mathrm{d}z) = \begin{bmatrix} \displaystyle\int_{\mathbb{R}} \gamma_1(t, X(t^-), u(t^-), z)\widetilde{N}(\mathrm{d}t, \mathrm{d}z) \\ \displaystyle\int_{\mathbb{R}} \gamma_2(t, X(t^-), u(t^-), z)\widetilde{N}(\mathrm{d}t, \mathrm{d}z) \end{bmatrix}$$

so the Hamiltonian is

$$H(t,x,u,p,q,r)=\int_{\mathbb{R}}\left\{u\,z\,r_1(t,z)+z^2r_2(t,z)\right\}\nu(\mathrm{d}z)$$

and the adjoint equations are ($g(x_1,x_2)=-(x_1-x_2)^2$)

$$\begin{cases}\mathrm{d}p_1(t)=\displaystyle\int_{\mathbb{R}} r_1(t^-,z)\widetilde{N}(\mathrm{d}t,\mathrm{d}z)\;;\quad t<T\\ p_1(T)=-2(X_1(T)-X_2(T))\end{cases}$$

$$\begin{cases}\mathrm{d}p_2(t)=\displaystyle\int_{\mathbb{R}} r_2(t^-,z)\widetilde{N}(\mathrm{d}t,\mathrm{d}z)\\ p_2(T)=2(X_1(T)-X_2(T)).\end{cases}$$

Now $X_1(T)-X_2(T)=\displaystyle\int_0^T\int_{\mathbb{R}}\left\{u(t^-)-z\right\}\widetilde{N}(\mathrm{d}t,\mathrm{d}z)$. So if $\hat{u}$ is a given candidate for an optimal control we get

$$\begin{aligned}\hat{r}_1(t,z)&=-2(\hat{u}(t)-z)z,\\ \hat{u}_2(t,z)&=2(\hat{u}(t)-z)z.\end{aligned}$$

This gives

$$\begin{aligned}H(t,x,u,\hat{p},\hat{q},\hat{r})&=\int_{\mathbb{R}}\left\{u\,z(-2(\hat{u}(t)-z)z)+z^2 2(\hat{u}(t)-z)z\right\}\nu(\mathrm{d}z)\\ &=-2u\int_{\mathbb{R}}\left\{\hat{u}(t)z^2-z^3\right\}\nu(\mathrm{d}z)+2\int_{\mathbb{R}}\left\{\hat{u}(t)z^3-z^4\right\}\nu(\mathrm{d}z).\end{aligned}$$

This is a linear expression in u, so we guess that the coefficient of u is 0, i.e., that

$$\hat{u}(t)=\frac{\displaystyle\int_{\mathbb{R}} z^3\nu(\mathrm{d}z)}{\displaystyle\int_{\mathbb{R}} z^2\nu(\mathrm{d}z)}\quad\text{for all } (t,\omega)\in[0,T]\times\Omega. \tag{14.5.3}$$

With this choice of $\hat{u}(t)$ all the conditions of the stochastic maximum principle are satisfied and we conclude that $\hat{u}$ is optimal.

Note that this implies that

$$\begin{aligned}\inf_u E\left[\left(F-\int_0^T u(t)\mathrm{d}S_1(t)\right)^2\right] &= E\left[\left(\int_0^T\int_{\mathbb{R}}\left\{z^2-\hat{u}(t)z\right\}\widetilde{N}(\mathrm{d}t,\mathrm{d}z)\right)^2\right]\\ &= \int_0^T\int_{\mathbb{R}} E[(z^2-\hat{u}(t)z)^2]\nu(\mathrm{d}z)\mathrm{d}t\\ &= T\int_{\mathbb{R}}\left[z^2-\frac{\int_{\mathbb{R}} z^3\nu(\mathrm{d}z)}{\int_{\mathbb{R}} z^2\nu(\mathrm{d}z)}z\right]^2\nu(\mathrm{d}z).\end{aligned}$$

We see that this is 0 if and only if

$$\int_{\mathbb{R}} z^3\nu(\mathrm{d}z) = z\int_{\mathbb{R}} z^2\nu(\mathrm{d}z) \quad \text{for a.a. } z(\nu), \tag{14.5.4}$$

i.e. iff ν *is supported on one point* $\{z_0\}$. *Only then is the market complete*! See [BDLØP] for more information.

Exercise 5.4

We try to find a, b such that the function

$$\varphi(s,x) = \mathrm{e}^{-\rho s}\psi(x) := \mathrm{e}^{-\rho s}(ax^2+b)$$

satisfies the conditions of (the minimum version of) Theorem 5.1. In this case the generator is

$$A^v_\varphi(s,x) = \mathrm{e}^{-\rho s}A^v_0\psi(x),$$

where

$$\begin{aligned}A^v_0\psi(x) =& -\rho\psi(x)+v\psi'(x)+\frac{1}{2}\sigma^2\psi''(x)\\ &+\int_{\mathbb{R}}\left\{\psi(x+z)-\psi(x)-z\psi'(x)\right\}\nu(\mathrm{d}z).\end{aligned}$$

Hence condition (i) of Theorem 5.1 becomes

$$\begin{aligned}&A^v_0\psi(x)+x^2+\theta v^2 = -\rho(ax^2+b)+v2ax+\frac{1}{2}\sigma^2 2a+a\int_{\mathbb{R}} z^2\nu(\mathrm{d}z)+x^2+\theta v^2\\ &=\theta v^2+2axv+x^2(1-\rho a)+a\left(\sigma^2+\int_{\mathbb{R}} z^2\nu(\mathrm{d}z)\right)-\rho b =: h(v).\end{aligned}$$

The function h is minimal when

$$v = u^*(x) = -\frac{ax}{\theta}. \tag{14.5.5}$$

With this value of v condition (v) becomes

$$x^2\left[1 - \rho a - \frac{a^2}{\theta}\right] + a\left(\sigma^2 + \int_{\mathbb{R}} z^2\nu(\mathrm{d}z)\right) - \rho b = 0.$$

Hence we choose $a > 0$ and b such that

$$a^2 + \rho\theta a - \theta = 0 \tag{14.5.6}$$

and

$$b = \frac{a}{\rho}\left(\sigma^2 + \int_{\mathbb{R}} z^2\nu(\mathrm{d}z)\right). \tag{14.5.7}$$

With these values of a and b we can easily check that

$$\varphi(s, x) := \mathrm{e}^{-\rho s}(ax^2 + b)$$

satisfies all the conditions of Theorem 5.1. The corresponding optimal control is given by (14.5.5).

Exercise 5.5

(b) The Hamiltonian for this problem is

$$H(t, x, u, p, q, r) = x^2 + \theta u^2 + up + \sigma q + \int_{\mathbb{R}} r(t^-, z)\tilde{N}(\mathrm{d}t, \mathrm{d}z).$$

The adjoint equation is

$$\begin{cases} \mathrm{d}p(t) = -2X(t)\mathrm{d}t + q(t)\mathrm{d}B(t) + \displaystyle\int_{\mathbb{R}} r(t^-, z)\tilde{N}(\mathrm{d}t, \mathrm{d}z)\ ;\ t < T \\ p(T) = 2\lambda X(T). \end{cases} \tag{14.5.8}$$

By imposing the first and second-order conditions, we see that $H(t, x, u, p, q, r)$ is minimal for

$$u = u(t) = \hat{u}(t) = -\frac{1}{2\theta}p(t). \tag{14.5.9}$$

In order to find a solution of (14.5.8), we consider $p(t) = h(t)X(t)$, where $h : \mathbb{R} \to \mathbb{R}$ is a deterministic function such that

$$h(T) = 2\lambda.$$

Note that $u(t) = -\dfrac{h(t)X(t)}{2\theta}$ and

$$\mathrm{d}X(t) = -\frac{h(t)X(t)}{2\theta}\mathrm{d}t + \sigma \mathrm{d}B(t) + \int_{\mathbb{R}} z\tilde{N}(\mathrm{d}t, \mathrm{d}z);\ X(0) = x.$$

Moreover, (14.5.8) turns into

$$\begin{aligned}\mathrm{d}p(t) &= h(t)\mathrm{d}X(t) + X(t)h'(t)\mathrm{d}t \\ &= X(t)\Big[-\frac{h(t)^2}{2\theta} + h'(t)\Big]\mathrm{d}t + h(t)\sigma \mathrm{d}B(t) + h(t)\int_{\mathbb{R}} z\tilde{N}(\mathrm{d}t, \mathrm{d}z).\end{aligned}$$

Hence $h(t)$ is the solution of

$$\begin{cases} h'(t) = \dfrac{h(t)^2}{2\theta} - 2;\ t < T \\ h(T) = 2\lambda. \end{cases} \tag{14.5.10}$$

The general solution of (14.5.10) is

$$h(t) = 2\sqrt{\theta}\,\frac{1 + \beta \mathrm{e}^{\frac{2t}{\sqrt{\theta}}}}{1 - \beta \mathrm{e}^{\frac{2T}{\sqrt{\theta}}}} \tag{14.5.11}$$

with $\beta = \frac{\lambda - \sqrt{\theta}}{\lambda + \sqrt{\theta}} \mathrm{e}^{-\frac{2T}{\sqrt{\theta}}}$. By using the stochastic maximum principle, we can conclude that

$$u^*(t) = -\frac{h(t)}{2\theta}X(t)$$

is the optimal control, $p(t) = h(t)X(t)$ and $q(t) = \sigma h(t)$, $r(t^-, z) = h(t)z$, where $h(t)$ is given by (14.5.11).

Exercise 5.6

If we try a function of the form

$$\varphi(s, x) = \mathrm{e}^{-\delta s}\psi(x)$$

then equations (i) and (v) for Theorem 5.1 combine to give the equation

$$\sup_{c \geq 0} \Big\{ \ln c - \delta\psi(x) + (\mu x - c)\psi'(x) + \frac{1}{2}\sigma^2 x^2 \psi''(x)$$

$$+\int_{\mathbb{R}}\left\{\psi(x+x\theta z)-\psi(x)-x\theta z\psi'(x)\right\}\nu(\mathrm{d}z)\Big\}=0.$$

The function

$$h(c):=\ln c-c\psi'(x);\ c>0$$

is maximal when

$$c=\hat{c}(x)=\frac{1}{\psi'(x)}.$$

If we set

$$\psi(x)=a\ \ln x+b$$

where a, b are constants, $a>0$, then this gives

$$\hat{c}(x)=\frac{x}{a},$$

and hence the above equation becomes

$$\ln x-\ln a-\delta(a\ln x+b)+\mu x\cdot\frac{a}{x}-1+\frac{1}{2}\sigma^2x^2\left(-\frac{a}{x^2}\right)$$
$$+a\int_{\mathbb{R}}\left\{\ln(x+x\theta z)-\ln x-x\theta z\cdot\frac{1}{x}\right\}\nu(\mathrm{d}z)=0$$

or

$$(1-\delta a)\ln x-\ln a-\delta b+\mu a-1-\frac{1}{2}\sigma^2a$$
$$+a\int_{\mathbb{R}}\{\ln(1+\theta z)-\theta z\}\,\nu(\mathrm{d}z)=0,\ \text{for all } x>0.$$

This is possible if and only if

$$a=\frac{1}{\delta}$$

and

$$b=\delta^{-2}\left[\delta\ln\delta-\delta+\mu-\frac{1}{2}\sigma^2+\int_{\mathbb{R}}\{\ln(1+\theta z)-\theta z\}\,\nu(\mathrm{d}z)\right].$$

One can now verify that if $\delta > \mu$ then with these values of a and b the function

$$\varphi(s,x) = \mathrm{e}^{-\delta t}(a \ln x + b)$$

satisfies all the conditions of Theorem 5.1. We conclude that

$$\Phi(s,x) = \mathrm{e}^{-\delta t}(a \ln x + b)$$

and that

$$c^*(x) = \hat{c} = \frac{x}{a}$$

(in feedback form) is an optimal consumption rate.

14.6 Exercises of Chap. 6

Exercise 6.1

(i)

$$A^{\alpha,\pi}\varphi(s,x) = \frac{\partial \varphi}{\partial s} + x[(1-\pi)r(s) + \pi\alpha]\frac{\partial \varphi}{\partial x} + \frac{1}{2}\beta^2\pi^2x^2\frac{\partial^2\varphi}{\partial x^2 3} + \int_{\mathbb{R}}\left\{\varphi(s, x + \pi \times \gamma(s,\zeta)) - \varphi(s,x) - \frac{\partial \varphi}{\partial x}(s,x)\right\}\nu(d\zeta).$$

(ii) The HJBI equation is

$$\begin{cases} \inf\limits_{\alpha\in A_1}\left(\sup\limits_{\pi\in A_2} A^{\alpha,\pi}\Phi(s,x)\right) = 0\ ;\ s > T \\ \Phi(T,x) = U(x). \end{cases}$$

(iii) With

$$\hat{\varphi}(s,x) = U\left(x\exp\left(\int_s^T r(t)dt\right)\right)$$

we get

$$A^{\alpha,\pi}\hat{\varphi}(s,x) = -U'(x\psi(s)) \times \psi(s)r(s) + x[(1-\pi)r(s) + \pi\alpha]U'(x\psi(s))\psi(s) + \frac{1}{2}\beta^2\pi^2x^2U''(x\psi(s))\psi^2 + \int_{\mathbb{R}}\{U((x + \pi \times \gamma(s,\zeta))\psi(s)) - U(x\psi(s)) - U'(x\psi(s))\psi(s)\pi \times \gamma(s,\zeta)\}\nu(d\zeta),$$

where

$$\psi(s) = \exp\left(\int_s^T r(t)dt\right).$$

(iv) The unique saddle point is found to be

$$\hat{\pi}(s,x) = 0, \hat{\alpha}(s,x) = r(s)\ ;\ (s,x) \in [-\infty, T] \times (0,\infty).$$

The interpretation of this result is the following: the worst scenario for the trader is to have $\alpha(s) = r(s)$, because then the underlying probability measure P is *martingale measure* for the market, and in that case, it is impossible for the trader to make any profit. She might as well choose not a trade, i.e. put $\pi = \hat{\pi} = 0$.

Exercise 6.2

From (5.4.10) we get

$$\begin{aligned} A_\pi(t,x) &= y'(t,x)\pi x b(t) + \frac{1}{2}y''(t,x)x^2\pi^2\sigma^2(t) + z'(t,x)\pi x\sigma(t) \\ &+ \int_{\mathbb{R}} \{y(t,x + x\pi\gamma(t,\zeta)) - y(t,x) - y'(t,x)x\pi\gamma(t,\zeta)\}\nu(d\zeta) \\ &+ \int_{\mathbb{R}} \{k(t,x + x\pi\gamma(t,\zeta),\zeta) - k(t,x,\zeta)\}\nu(d\zeta). \end{aligned} \tag{14.6.1}$$

This is maximal when $\pi = \hat{\pi}(t)$ satisfies the equation

$$\begin{aligned} &xy'(t,x)b(t) + y''(t,x)\sigma^2(t)\hat{\pi}(t)x^2 + z'(t,x)\sigma(t)x \qquad (14.6.2)\\ &\quad + \int_{\mathbb{R}} \{y'(t,x,+x\hat{\pi}(t)\gamma(t,\zeta)) - y'(t,x) + k'(t,x + x\hat{\pi}(t)\gamma(t,\zeta))\} \\ &\quad x\gamma(t,\zeta)\nu(d\zeta), \qquad (14.6.3) \end{aligned}$$

where (y,z,k) solves the partial BSDE

$$\begin{cases} dy(t,x) &= -A_{\hat{\pi}}(t,x)dt + z(t,x)dB(t) + \int_{\mathbb{R}} k(t,x,\zeta)\tilde{N}(dt,d\zeta) \\ y(T,x) &= \ln X^{\hat{\pi}}(T). \end{cases} \tag{14.6.4}$$

Since the coefficients $b(t)$, $\sigma(t)$ and $\gamma(t,\zeta)$ are deterministic, we guess that $\hat{\pi}(t)$ is deterministic also, and hence that we can in (14.6.4) choose

$$z(t,x) = k(t,x,\zeta) = 0.$$

Then let us try a solution of (14.6.4) of the form

$$y(t,x) = f(t) + \ln x \tag{14.6.5}$$

for some C^1-function f (to be determined).

Substituted into (14.6.1) this gives

$$A_\pi(t,x) = \pi(t)b(t) - \frac{1}{2}\pi^2(t)\sigma^2(t) \tag{14.6.6}$$

$$+ \int_{\mathbb{R}} \{\ln(1+\pi(t)\gamma(t,\zeta)) - \pi(t)\gamma(t,\zeta)\}\nu(d\zeta) = A_\pi(t), \tag{14.6.7}$$

and (14.6.2) becomes

$$b(t) - \sigma^2(t)\hat{\pi}(t) + \int_{\mathbb{R}} \frac{\hat{\pi}(t)\gamma^2(t,\zeta)}{1+\hat{\pi}(t)\gamma(t,\zeta)}\nu(d\zeta) = 0. \tag{14.6.8}$$

With $\hat{\pi}$ as in (14.6.8) the BSDE (14.6.4) reduces to

$$\begin{cases} f'(t) &= -A_{\hat{\pi}}(t)\ ;\ t \in [0,T] \\ f(T) &= 0, \end{cases}$$

which gives

$$f(t) = \int_t^T A_{\hat{\pi}}(s)ds\ ;\ t \in [0,T].$$

Therefore, the maximal expected logarithmic utility is

$$Y^{\hat{\pi}}(0) = y(0,x) = \int_0^T A_{\hat{\pi}}(s)ds + \ln x.$$

Exercise 6.3

The Hamiltonian for this problem is

$$H(t,x,\pi,\mu,p,q) = \frac{1}{2}\mu^2 + \pi x(\alpha(t)+\mu)p + \pi x\beta(t)q \tag{14.6.9}$$

and the BSDE for the adjoint processes p, q is

$$\begin{cases} dp(t) = -[\pi(t)\{(\alpha(t)+\mu(t))p(t) + \beta(t)q(t)\}]dt + q(t)dB(t); \quad 0 \le t \le T \\ p(T) = \dfrac{\theta}{X(T)}. \end{cases} \tag{14.6.10}$$

Maximizing H with respect to π gives the first-order equation

$$X(t)[(\alpha(t)+\mu(t))p(t)+\beta(t)q(t)]=0. \tag{14.6.11}$$

Since $X(t)>0$ and $\beta(t)\neq 0$, we deduce that

$$(\alpha(t)+\mu(t))p(t)+\beta(t)q(t)=0 \tag{14.6.12}$$

and

$$q(t)=-\frac{\alpha(t)+\mu(t)}{\beta(t)}p(t). \tag{14.6.13}$$

Hence (14.6.10) reduces to

$$\begin{cases} dp(t)=-\dfrac{\alpha(t)+\mu(t)}{\beta(t)}p(t)dB(t) \\ p(T)=\dfrac{\theta}{X(T)}. \end{cases} \tag{14.6.14}$$

Define

$$h(t)=p(t)X(t). \tag{14.6.15}$$

Then by the Itô formula we get

$$\begin{cases} dh(t) &= \left(\pi(t)\beta(t)-\dfrac{\alpha(t)+\mu(t)}{\beta(t)}\right)h(t)dB(t) \\ h(T) &= p(T)X(T)=\theta. \end{cases} \tag{14.6.16}$$

This BSDE has the solution

$$h(t)=\theta;\quad 0\leq t\leq T. \tag{14.6.17}$$

We deduce that

$$(\pi(t)\beta(t)-\frac{\alpha(t)+\mu(t)}{\beta(t)})h(t)=0, \tag{14.6.18}$$

from which we get the following expression for our candidate $\hat{\pi}(t)$ for the optimal portfolio

$$\hat{\pi}(t)=\frac{\alpha(t)+\hat{\mu}(t)}{\beta^2(t)}, \tag{14.6.19}$$

where $\hat{\mu}(t)$ is the corresponding candidate for the optimal perturbation.

Minimizing H with respect to μ gives the following first-order equation for the optimal $\hat{\mu}(t)$:

$$\hat{\mu}(t) + \hat{\pi}(t)\hat{X}(t)\hat{p}(t) = 0, \tag{14.6.20}$$

i.e.,

$$\hat{\mu}(t) = -\hat{\pi}(t)\hat{X}(t)\hat{p}(t) = -\theta\hat{\pi}(t). \tag{14.6.21}$$

We can now verify that $(\hat{\pi}, \hat{\mu})$ satisfies all the conditions of the sufficient maximum principle, and hence we conclude the following:

Proposition 14.3 (***Optimal portfolio under model uncertainty***) *The saddle point* $(\pi^*(t), \mu^*(t))$ *of the stochastic differential game* (6.4.2) *is given by*

$$\pi^*(t) = \frac{\alpha(t)}{\beta^2(t) + \theta}, \tag{14.6.22}$$

and

$$\mu^*(t) = -\frac{\alpha(t)\theta}{\beta^2(t) + \theta}. \tag{14.6.23}$$

14.7 Exercises of Chap. 7

Exercise 7.1

(a) The HJB equation, i.e., (vi) and (ix) of Theorem 7.2, for this problem gets the form

$$\begin{aligned} 0 = \sup_{u \geq 0} \Bigg\{ & e^{-\delta s} \frac{u^\gamma}{\gamma} + \frac{\partial \phi}{\partial s} + (\mu x - u)\frac{\partial \phi}{\partial x} + \frac{\sigma^2 x^2}{2}\frac{\partial^2 \phi}{\partial x^2} \\ & + \int_{\mathbb{R}} \left\{ \phi(s, x + \theta x z) - \phi(s, x) - \theta x z \frac{\partial \phi}{\partial x}(s, x) \right\} d\nu(z) \Bigg\} \end{aligned} \tag{14.7.1}$$

for $x > 0$.

We impose the first-order conditions to find the supremum, which is obtained for

$$u = u^*(s, x) = \left(e^{\delta s} \frac{\partial \phi}{\partial x} \right)^{1/(\gamma - 1)}. \tag{14.7.2}$$

We guess that $\phi(s,x) = K\mathrm{e}^{-\delta s}x^\gamma$ with $K > 0$ to be determined. Then

$$u^*(s,x) = \left(K\gamma\right)^{1/(\gamma-1)}x \tag{14.7.3}$$

and (14.7.1) turns into

$$\frac{1}{\gamma}(K\gamma)^{\gamma/(\gamma-1)} - K\delta + \left(\mu - (K\gamma)^{1/(\gamma-1)}\right)K\gamma + \frac{1}{2}\sigma^2 K\gamma(\gamma-1) + K\int_{\mathbb{R}}\{(1+\theta z)^\gamma - 1 - \gamma\theta z\}\,\nu(\mathrm{d}z) = 0$$

or

$$\gamma^{\gamma/(\gamma-1)}K^{1/(\gamma-1)} - \delta + \mu\gamma - \gamma^{\gamma/(\gamma-1)}K^{1/(\gamma-1)} + \frac{1}{2}\sigma^2\gamma(\gamma-1) + \int_{\mathbb{R}}\{(1+\theta z)^\gamma - 1 - \gamma\theta z\}\,\nu(\mathrm{d}z) = 0.$$

Hence

$$K = \frac{1}{\gamma}\left[\frac{1}{1-\gamma}\left(\delta - \mu\gamma + \frac{\sigma^2}{2}\gamma(1-\gamma) - \int_{\mathbb{R}}\{(1+\theta z)^\gamma - 1 - \gamma\theta z\}\,\nu(\mathrm{d}z)\right)\right]^{\gamma-1} \tag{14.7.4}$$

provided that

$$\delta - \mu\gamma + \frac{\sigma^2}{2}\gamma(1-\gamma) - \int_{\mathbb{R}}\{(1+\theta z)^\gamma - 1 - \gamma\theta z\}\,\nu(\mathrm{d}z) > 0.$$

With this choice of K the conditions of Theorem 7.2 are satisfied and we can conclude that $\phi = \Phi$ is the value function.

(b)

(i) First assume $\lambda \geq K$. Choose $\phi(s,x) = \lambda\mathrm{e}^{-\delta s}x^\gamma$. By the same computations as in a), condition (vi) of Theorem 7.2 gets the form

$$\lambda \geq \frac{1}{\gamma}\left[\frac{1}{\gamma-1}\left(\delta - \mu\gamma + \frac{1}{2}\sigma^2\gamma(1-\gamma) - \int_{\mathbb{R}}\{(1+\theta z)^\gamma - 1 - \gamma\theta z\}\,\nu(\mathrm{d}z)\right)\right]^{\gamma-1}. \tag{14.7.5}$$

Since $\lambda \geq K$, the inequality (14.7.5) holds by (14.7.4).

By Theorem 7.2(a), it follows that:

$$\phi(s,x) = \lambda e^{-\delta s} x^{\gamma} \geq \Phi(s,x)$$

where Φ is the value function for our problem. On the other hand, $\phi(s,x)$ is obtained by the (admissible) control of stopping immediately ($\tau = 0$). Hence we also have

$$\phi(s,x) \leq \Phi(s,x).$$

We conclude that

$$\Phi(s,x) = \lambda e^{-\delta s} x^{\gamma}$$

in this case and $\tau^* = 0$ is optimal. Note that $D = \emptyset$.

(ii) Assume now $\lambda < K$. Choose $\phi(s,x) = K e^{-\delta s} x^{\gamma}$. Then for all $(s,x) \in \mathbb{R} \times (0,\infty)$ we have

$$\phi(s,x) > \lambda e^{-\delta s} x^{\gamma}.$$

Hence we have $D = \mathbb{R} \times (0,\infty)$ and by Theorem 7.2(a) we conclude that

$$\Phi(s,x) \leq K e^{-\delta s} x^{\gamma}.$$

On the other hand, we have seen in (a) above that if we apply the control

$$u^*(s,x) = \big(K\gamma\big)^{1/(\gamma-1)} x$$

and never stop, then we achieve the performance $J^{(u^*)}(s,x) = K e^{-\delta s} x^{\gamma}$. Hence

$$\Phi(s,x) = K e^{-\delta s} x^{\gamma}$$

and it is optimal never to stop ($\tau^* = \infty$).

14.8 Exercises of Chap. 8

Exercise 8.1

In this case we put

$$\mathrm{d}Y(t) = \begin{bmatrix} \mathrm{d}t \\ \mathrm{d}X(t) \end{bmatrix} = \begin{bmatrix} 1 \\ \alpha \end{bmatrix} \mathrm{d}t + \begin{bmatrix} 0 \\ \sigma \end{bmatrix} \mathrm{d}B(t) + \begin{bmatrix} 0 \\ \beta \int\limits_{\mathbb{R}} z \, \widetilde{N}(\mathrm{d}t, \mathrm{d}z) \end{bmatrix} + \begin{bmatrix} 0 \\ -(1+\lambda) \end{bmatrix} d\xi(t).$$

The generator if $\xi = 0$ is

$$A\phi = \frac{\partial \phi}{\partial s} + \alpha \frac{\partial \phi}{\partial x} + \tfrac{1}{2}\sigma^2 \frac{\partial^2 \phi}{\partial x^2} + \int_{\mathbb{R}} \left\{ \phi(s, x + \beta z) - \phi(s, x) - \beta z \frac{\partial \phi}{\partial x}(s, x) \right\} \nu(\mathrm{d}z).$$

The non-intervention region D is described by (see (8.2.5))

$$\begin{aligned} D &= \left\{ (s, x); \sum_{i=1}^{k} \kappa_{ij} \frac{\partial \phi}{\partial y_i}(y) + \theta_j < 0 \quad \text{for all } j = 1, \ldots, p \right\} \\ &= \left\{ (s, x); -(1 + \lambda) \frac{\partial \phi}{\partial x}(s, x) + \mathrm{e}^{-\rho s} < 0 \right\}. \end{aligned}$$

If we guess that D has the form

$$D = \left\{ (s, x); 0 < x < x^* \right\} \quad \text{for some } x^* > 0$$

then by Theorem 8.2 we should have

$$A\phi(s, x) = 0 \quad \text{for } 0 < x < x^*.$$

We try a solution ϕ of the form

$$\phi(s, x) = \mathrm{e}^{-\rho s} \psi(x)$$

and get

$$\begin{aligned} A_0 \psi(x) := &-\rho \psi(x) + \alpha \psi'(x) + \tfrac{1}{2}\sigma^2 \psi''(x) + \int_{\mathbb{R}} \{ \psi(x + \beta z) - \psi(x) \\ &- \beta z \psi'(x) \} \nu(\mathrm{d}z) = 0. \end{aligned}$$

We now choose

$$\psi(x) = \mathrm{e}^{r x} \quad \text{for some constant } r \in \mathbb{R}$$

and get the equation

$$h(r) := -\rho + \alpha r + \tfrac{1}{2}\sigma^2 r^2 + \int_{\mathbb{R}} \left\{ \mathrm{e}^{r \beta z} - 1 - r \beta z \right\} \nu(\mathrm{d}z) = 0.$$

Since $h(0) < 0$ and $\lim_{r \to \infty} h(r) = \lim_{r \to -\infty} h(r) = \infty$, we see that the equation $h(r) = 0$ has two solutions r_1, r_2 such that

$$r_2 < 0 < r_1.$$

Outside D we require that

$$-(1+\lambda)\psi'(x) + 1 = 0$$

or

$$\psi(x) = \frac{x}{1+\lambda} + C_3, \quad C_3 \text{ constant.}$$

Hence we put

$$\psi(x) = \begin{cases} C_1 e^{r_1 x} + C_2 e^{r_2 x}; & 0 < x < x^* \\ \dfrac{x}{1+\lambda} + C_3; & x^* \le x \end{cases} \tag{14.8.1}$$

where C_1, C_2 are constants.

To determine C_1, C_2, C_3 and x^* we have the four equations:

$$\psi(0) = 0 \;\Rightarrow\; C_1 + C_2 = 0. \tag{14.8.2}$$

Put $C_2 = -C_1$

$$\psi \text{ continuous at } x = x^* \;\Rightarrow\; C_1(e^{r_1 x^*} - e^{r_2 x^*}) = \frac{x^*}{1+\lambda} + C_3 \tag{14.8.3}$$

$$\psi \in C^1 \text{ at } x = x^* \;\Rightarrow\; C_1(r_1 e^{r_1 x^*} - r_2 e^{r_2 x^*}) = \frac{1}{1+\lambda} \tag{14.8.4}$$

$$\psi \in C^2 \text{ at } x = x^* \;\Rightarrow\; C_1(r_1^2 e^{r_1 x^*} - r_2^2 e^{r_2 x^*}) = 0. \tag{14.8.5}$$

From (14.8.4) and (14.8.5) we deduce that

$$x^* = \frac{2(\ln|r_2| - \ln r_1)}{r_1 - r_2}. \tag{14.8.6}$$

Then by (14.8.4) we get the value for C_1, and hence the value of C_3 by (14.8.3).

With these values of C_1, C_2, C_3 and x^* we must verify that $\phi(s,x) = e^{-\rho s}\psi(x)$ satisfies all the requirements of Theorem 8.2:

(i) We have constructed ϕ such that $A\phi + f = 0$ in D. Outside D, i.e., for $x \ge x^*$, we have

$$\begin{aligned} e^{\rho s}(A\phi(s,x) + f(s,x)) &= A_0\psi(x) \\ &= -\rho\left(\frac{x}{1+\lambda} + C_3\right) \\ &\quad + \alpha \cdot \frac{1}{1+\lambda} + \int_{x+\beta z < x^*} \Big\{ C_1(e^{r_1(x+\beta z)} - e^{r_2(x+\beta z)}) \end{aligned}$$

$$
\begin{aligned}
&- \left(\frac{x+\beta z}{1+\lambda} + C_3\right)\Big\} \nu(\mathrm{d}z) \\
&\leq -\frac{\rho}{1+\lambda}x + \frac{\alpha}{1+\lambda} - \rho C_3 \ , \quad \text{which is decreasing in } x.
\end{aligned}
$$

So we need only to check that this holds for $x = x^*$, i.e., that

$$A_0\psi(x^*) \leq 0.$$

But this follows from the fact that $A_0\psi(x) = 0$ for all $x < x^*$ and $\psi \in C^2$.
(ii) By construction we have

$$-(1+\lambda)\psi'(x) + 1 = 0 \quad \text{for } x \geq x^*.$$

For $x < x^*$ the condition (ii) gets the form

$$F(x) := -(1+\lambda)C_1(r_1 \mathrm{e}^{r_1 x} - r_2 \mathrm{e}^{r_2 x}) + 1 \leq 0.$$

We know that $F(x^*) = 0$ by (14.8.4) and

$$F'(x) = -(1+\lambda)C_1(r_1^2 \mathrm{e}^{r_1 x} - r_2^2 \mathrm{e}^{r_2 x}).$$

So $F'(x^*) = 0$ by (14.8.5) and hence (since $C_1 > 0$)

$$F'(x) > F'(x^*) = 0 \qquad \text{for } x < x^*.$$

Hence

$$F(x) < 0 \quad \text{for } 0 < x < x^*.$$

The conditions (iii), (iv), and (v) are left to the reader to verify.
(vi) This holds by construction of ϕ.
(vii)–(x) These conditions claim the existence of an increasing process $\hat{\xi}$ such that $Y^{\hat{\xi}}(t)$ stays in $\bar{D}$ for all times t, $\hat{\xi}(t)$ is strictly increasing only when $Y(t) \notin D$, and if $Y(t) \notin \bar{D}$ then $\hat{\xi}(t)$ brings $Y(t)$ down to a point on ∂D. Such a singular control is called a *local time* at ∂D of the process $Y(t)$ reflected downwards at ∂D. The existence and uniqueness of such a local time is proved in [CEM].
(xi) This is left to the reader.

We conclude that the optimal dividend policy $\xi^*(t)$ is to take out exactly the amount of money needed to keep $X(t)$ on or below the value x^*. If $X(t) < x^*$ we take out nothing. If $X(t) > x^*$ we take out $X(t) - x^*$.

Exercise 8.2

It suffices to prove that the function

$$\Phi_0(s, x_1, x_2) := K\mathrm{e}^{-\delta s}(x_1 + x_2)^\gamma$$

satisfies conditions (i)–(iv) of Theorem 8.2. In this case we have (see Sect. 8.3)

$$A^{(v)}\Phi_0(y) = A^{(c)}\Phi_0(y) = \frac{\partial \Phi_0}{\partial s} + (rx_1 - c)\frac{\partial \Phi_0}{\partial x_1} + \alpha x_2 \frac{\partial \Phi_0}{\partial x_2} + \frac{1}{2}\beta^2 x_2^2 \frac{\partial^2 \Phi_0}{\partial x_2^2}$$
$$+ \int_{\mathbb{R}} \Big\{ \Phi_0(s, x_1, x_2 + x_2 z) - \Phi_0(s, x_1, x_2)$$
$$- x_2 z \frac{\partial \Phi_0}{\partial x_2}(s, x_1, x_2) \Big\} \nu(\mathrm{d}z)$$

and $f(s, x_1, x_2, c) = \mathrm{e}^{-\delta s}\dfrac{c^\gamma}{\gamma}$, so condition (i) becomes

(i)' $A^c\Phi_0(s, x_1, x_2) + \mathrm{e}^{-\delta s}\frac{c^\gamma}{\gamma} \le 0$ for all $c \ge 0$.

This holds because we know by Example 5.2 that (see (5.1.23))

$$\sup_{c \ge 0} \left\{ A^{(c)}\Phi(s, x_1, x_2) + \mathrm{e}^{-\delta s}\frac{c^\gamma}{\gamma} \right\} = 0.$$

Since in this case $\theta = 0$ and

$$\kappa = \begin{bmatrix} -(1+\lambda) & 1-\mu \\ 1 & -1 \end{bmatrix}$$

we see that condition (ii) of Theorem 8.2 becomes

(ii)' $-(1+\lambda)\frac{\partial \Phi_0}{\partial x_1} + \frac{\partial \Phi_0}{\partial x_2} \le 0$

(ii)" $(1-\mu)\frac{\partial \Phi_0}{\partial x_1} - \frac{\partial \Phi_0}{\partial x_2} \le 0.$

Since

$$\frac{\partial \Phi_0}{\partial x_1} = \frac{\partial \Phi_0}{\partial x_2} = K\mathrm{e}^{-\delta s}\gamma(x_1 + x_2)^{\gamma - 1}$$

we see that (ii)' and (ii)" hold trivially.

We leave the verification of conditions (iii)–(v) to the reader.

14.9 Exercises of Chap. 9

Exercise 9.1

By using the same notation as in Chap. 6, we have here

$$Y^{(v)}(t) = \begin{bmatrix} s+t \\ X^{(v)}(t) \end{bmatrix};\ t \geq 0;\ Y^{(v)}(0^-) = \begin{bmatrix} s \\ x \end{bmatrix} = y \in \mathbb{R}^2$$

$$\Gamma(y,\zeta) = \Gamma(s,x,\zeta) = \begin{bmatrix} s \\ x+\zeta \end{bmatrix};\ (s,x,\zeta) \in \mathbb{R}^3$$

$$K(y,\zeta) = K(s,x,\zeta) = \mathrm{e}^{-\rho s}(x + \lambda|\zeta|)$$

$$f(y) = f(s,x) = \mathrm{e}^{-\rho s}x^2,\ g(y) = 0.$$

By symmetry we expect the continuation region to be of the form

$$D = \{(s,x) : -\bar{x} < x < \bar{x}\}$$

for some $\bar{x} > 0$, to be determined.

As soon as $X(t)$ reaches the unknown value $\bar{x}$ or $-\bar{x}$, there is an intervention and $X(t)$ is brought down (or up) to a certain value $\hat{x}$ (or $-\hat{x}$) where $-\bar{x} < -\hat{x} < 0 < \hat{x} < \bar{x}$. We determine $\bar{x}$ and $\hat{x}$ in the following computations (Fig. 14.4).

We guess that the value function is of the form

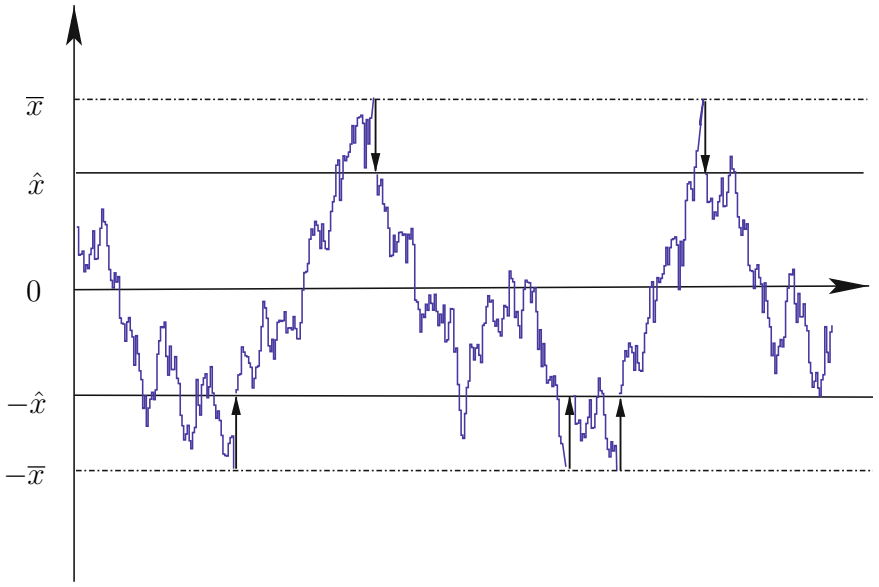


Fig. 14.4 The optimal strategy of Exercise 9.1

$$\phi(s,x) = \mathrm{e}^{-\rho s}\psi(x).$$

In the continuation region D, we have by Theorem 9.2 (x)

$$A\phi + f = 0, \tag{14.9.1}$$

where A is the generator of Y, i.e.,

$$A\phi(s,x) = \frac{\partial \phi}{\partial s} + \tfrac{1}{2}\frac{\partial^2 \phi}{\partial x^2} + \int_{\mathbb{R}} \left\{ \phi(s, x+\theta(x,z)) - \phi(s,x) - \theta(x,z)\frac{\partial \phi}{\partial x}(s,x) \right\} \nu(\mathrm{d}z).$$

In this case, if $|x| < \xi$, (14.9.1) becomes, by (9.3.1),

$$A_0\psi(x) + f(x) := -\rho\psi(x) + \frac{1}{2}\psi''(x) + x^2 = 0.$$

We try a solution of the form

$$\psi(x) = C\cosh(\gamma x) + \frac{1}{\rho}x^2 + \frac{1}{\rho^2}$$

where C is a constant (to be determined) and $\gamma = \sqrt{2\rho}$ is the positive solution of the equation

$$F(\gamma) := -\rho + \frac{1}{2}\gamma^2 = 0.$$

We try to set $C = -a$ where $a > 0$. Define

$$\psi_0(x) := \frac{1}{\rho}x^2 + \frac{1}{\rho^2} - a\cosh(\gamma x)$$

and put

$$\psi(x) = \psi_0(x);\ x \in D.$$

We recall that

$$\begin{aligned} D &= \{(s,x) : \phi(s,x) < \mathcal{M}\phi(s,x)\} \\ &= \{x : \psi(x) < \mathcal{M}\psi(x)\} \end{aligned}$$

and the intervention operator is in this case

$$\mathcal{M}\psi(x) = \inf\left\{\psi(x+\zeta) + c + \lambda|\zeta|;\ \zeta \in \mathbb{R}\right\}.$$

The first-order condition for a minimum $\hat{\zeta} = \hat{\zeta}(x)$ of the function

$$G(\zeta) = \begin{cases} \psi(x+\zeta) + c + \lambda\zeta & \zeta > 0 \\ \psi(x+\zeta) + c - \lambda\zeta & \zeta < 0 \end{cases}$$

is the following:

(i) $\zeta > 0$: $\psi'(x+\zeta) + \lambda = 0 \Rightarrow \psi'(x+\zeta) = -\lambda$
(ii) $\zeta < 0$: $\psi'(x+\zeta) - \lambda = 0 \Rightarrow \psi'(x+\zeta) = \lambda$.

Hence we look for points $\hat{x}, \bar{x}$ such that

$$-\bar{x} < -\hat{x} < 0 < \hat{x} < \bar{x}$$

and

$$\begin{aligned} \psi'(\hat{x}) &= -\lambda \\ \psi'(-\hat{x}) &= \lambda. \end{aligned} \tag{14.9.2}$$

Note that since $\hat{x} < \bar{x}$, $\psi'(\hat{x}) = \psi_0'(\hat{x})$.

Arguing as in Example 9.5, we put

$$\psi(x) = \begin{cases} \psi_0(x) & ;\ -\bar{x} \le x \le \bar{x} \\ \psi_0(\hat{x}) + c + \lambda(x - \hat{x}) & ;\ x > \bar{x} \\ \psi_0(-\hat{x}) + c - \lambda(x + \hat{x}) & ;\ x < -\bar{x}. \end{cases} \tag{14.9.3}$$

We have to show that there exist $0 < \hat{x} < \bar{x}$ and a value of a such that $\phi(s,x) := \mathrm{e}^{-\rho s}\psi(x)$ satisfies all the requirements of (the minimum version of) Theorem 9.2. By symmetry we may assume $x > 0$ and $\zeta > 0$ in the following.

Continuity at $x = \bar{x}$ gives the equation

$$\psi_0(\hat{x}) + c + \lambda(\bar{x} - \hat{x}) = \psi_0(\bar{x}).$$

Differentiability at $x = \bar{x}$ gives the equation

$$\lambda = \psi_0'(\bar{x}).$$

Substituting for ψ_0 these equations give

$$\frac{\hat{x}^2}{\rho} - a\cosh(\gamma\hat{x}) - \lambda\hat{x} + c = \frac{\bar{x}^2}{\rho} - a\cosh(\gamma\bar{x}) - \gamma\bar{x} \tag{14.9.4}$$

and

$$\lambda = \frac{2\bar{x}}{\rho} - a\gamma\sinh(\gamma\bar{x}). \tag{14.9.5}$$

In addition we have required

$$\lambda = \psi_0'(\hat{x}) = \frac{2\hat{x}}{\rho} - a\gamma \sinh(\gamma \hat{x}). \tag{14.9.6}$$

As in Example 9.5 one can prove that for each $c > 0$ there exist $a = a^*(c) > 0$, $\hat{x} = \hat{x}(x) > 0$ and $\bar{x} = \bar{x}(c) > \hat{x}$ such that (14.9.3)–(14.9.5) hold. With these values of $a, \hat{x}$ and $\bar{x}$ it remains to verify that the conditions of Theorem 9.2 hold. We check some of them:
(ii) $\psi \le \mathcal{M}\psi = \inf\{\psi(x-\zeta) + c + \lambda\zeta;\ \zeta > 0\}$.
First suppose $x \ge \bar{x}$.
If $x - \zeta \ge \bar{x}$ then

$$\psi(x-\zeta) + c + \lambda\zeta = \psi_0(\hat{x}) + c + \lambda(x - \zeta - \hat{x}) + c + \lambda\zeta = c + \psi(x) > \psi(x).$$

If $0 < x - \zeta < \bar{x}$ then

$$\psi(x-\zeta) + c + \lambda\zeta = \psi_0(x-\zeta) + c + \lambda\zeta,$$

which is minimal when

$$-\psi_0'(x-\zeta) + \lambda = 0$$

i.e., when

$$\zeta = \hat{\zeta} = x - \hat{x}.$$

This is the minimum point because

$$\psi_0''(\hat{x}) > 0.$$

See Fig. 14.5.
This shows that

$$\mathcal{M}\psi(x) = \psi(x - \hat{\zeta}) + c + \lambda\hat{\zeta} = \psi(\hat{x}) + c + \lambda(x - \hat{x}) = \psi(x)$$

for $x > \hat{x}$.
Next suppose $0 < x < \bar{x}$.
Then

$$\mathcal{M}\psi(x) = \psi_0(\hat{x}) + c + \lambda(x - \hat{x}) > \psi(x)$$

if and only if

$$\psi(x) - \lambda x < \psi(\hat{x}) - \lambda\hat{x} + c.$$

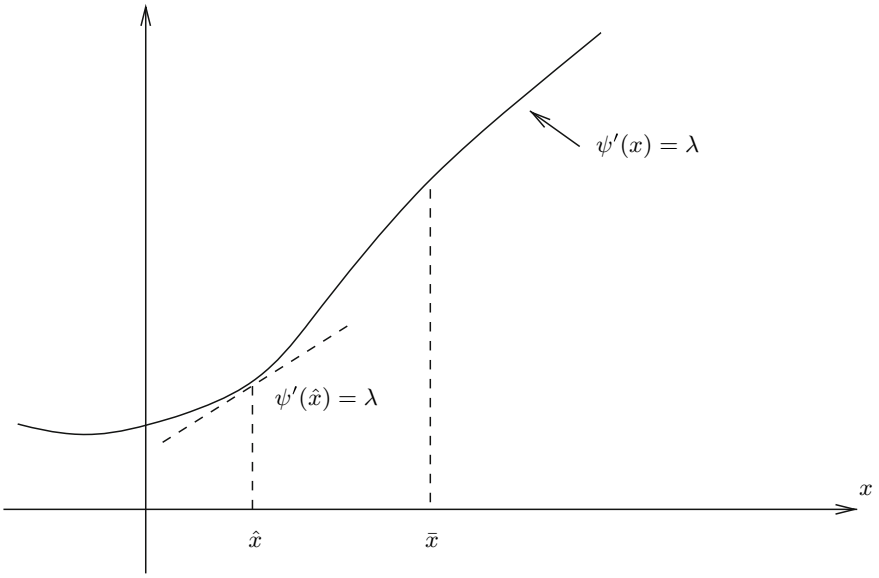


Fig. 14.5 The function $\psi(x)$ for $x > 0$

Now the minimum of

$$H(x) := \psi(x) - \lambda x \quad \text{for } 0 < x < \bar{x}$$

is attained when

$$\psi'(x) = \lambda$$

i.e. when $x = \hat{x}$.
Therefore

$$\psi(x) - \lambda x \leq \psi(\hat{x}) - \lambda\hat{x} < \psi(\hat{x}) - \lambda\hat{x} + c.$$

This shows that $\mathcal{M}\psi(x) > \psi(x) \quad \text{for all } 0 < x < \bar{x}$.
Combined with the above we can conclude that

$$\mathcal{M}\psi(x) \geq \psi(x) \quad \text{for all } x > 0,$$

which proves (ii). Moreover,

$$\mathcal{M}\psi(x) > \psi(x) \quad \text{if and only if } 0 < x < \bar{x}.$$

Hence

$$D \cap (0, \infty) = (0, \bar{x}).$$

Finally we verify
(vi) $A\phi + f \geq 0$ for $x > \bar{x}$.
For $x > \bar{x}$, we have, if $\bar{x} \leq \xi$ (using (9.3.2)),

$$A_0\psi(x) + f(x) = -\rho(\psi_0(\hat{x}) + c + \lambda(x - \hat{x})) + x^2.$$

This is nonnegative for all $x > \bar{x}$ iff it is nonnegative for $x = \bar{x}$, i.e., iff

$$-\rho\psi_0(\bar{x}) + \bar{x}^2 \geq 0. \tag{14.9.7}$$

By construction of ψ_0 we know that, for $x < \bar{x}$

$$-\rho\psi_0(x) + \frac{1}{2}\psi_0''(x) + \int_{\mathbb{R}} \left\{\psi_0(x + z) - \psi_0(x) - z\psi_0'(x)\right\} \nu(\mathrm{d}z) + x^2 = 0.$$

Therefore (14.9.7) holds iff

$$\frac{1}{2}\psi_0''(\bar{x}) + \int_{\mathbb{R}} \left\{\psi_0(\bar{x} + z) - \psi_0(\bar{x}) - z\psi_0'(\bar{x})\right\} \nu(\mathrm{d}z) \leq 0.$$

For this it suffices that

$$\int_{\mathbb{R}} z^2 \nu(\mathrm{d}z) \leq -\frac{\rho}{2}\psi_0''(\bar{x}). \tag{14.9.8}$$

Conclusion

Suppose $\bar{x} \leq \xi$ and that (14.9.8) holds. Then $\Phi(s, x) = \mathrm{e}^{-\rho s}\psi(x)$, with $\psi(x)$ given by (14.9.3) and $a, \hat{x}, \bar{x}$ given by (14.9.4)–(14.9.6). The optimal impulse control is to do nothing while $|X(t)| < \bar{x}$, then move $X(t)$ down to $\hat{x}$ (respectively, up to $-\hat{x}$) as soon as $X(t)$ reaches a value $\geq \bar{x}$ (respectively, a value $\leq -\bar{x}$).

Exercise 9.2

Here we put

$$\begin{aligned} Y^{(v)}(t) &= \begin{bmatrix} s + t \\ X^{(v)}(t) \end{bmatrix} \\ Y^{(v)}(0^-) &= \begin{bmatrix} s \\ x \end{bmatrix} = y \\ \Gamma(y, \zeta) &= x - c - (1 + \lambda)\zeta \\ K(y, \zeta) &= \mathrm{e}^{-\rho s}\zeta \\ f &\equiv g \equiv 0 \\ \mathcal{S} &= \{(s, x) : x > 0\}. \end{aligned}$$

We guess that the value function ϕ is of the form

$$\phi(s, x) = \mathrm{e}^{-\rho s}\psi(x)$$

and consider the intervention operator

$$\mathcal{M}\psi(x) = \sup\left\{\psi(x - c - (1 + \lambda)\zeta) + \zeta; 0 \le \zeta \le \frac{x - c}{1 + \lambda}\right\}. \tag{14.9.9}$$

Note that the condition on ζ is due to the fact that the impulse must be positive and $x - c - (1 + \lambda)\zeta$ must belong to $\mathcal{S}$. We distinguish between two cases:

(1) $\mu > \rho$.

In this case, suppose we wait until time t_1 and then take out

$$\zeta_1 = \frac{X(t_1) - c}{1 + \lambda}.$$

The corresponding value is

$$\begin{aligned} J^{(v_1)}(s, x) &= E^x\left[\frac{\mathrm{e}^{-\rho(t_1+s)}}{1 + \lambda}(X(t_1) - c)\right] \\ &= E^x\left[\frac{1}{1 + \lambda}\left(x\mathrm{e}^{-\rho s}\mathrm{e}^{(\mu-\rho)t_1} - c\mathrm{e}^{-\rho(s+t_1)}\right)\right] \\ &\to \infty \text{ as } t_1 \to \infty. \end{aligned}$$

Therefore we obtain $\Phi(s, x) = +\infty$ in this case.

(2) $\mu < \rho$.

We look for a solution by using the results of Theorem 9.2. In this case condition (x) becomes

$$\begin{aligned} A_0\psi(x) :=& -\rho\psi(x) + \mu x\psi'(x) + \tfrac{1}{2}\sigma^2\psi''(x) \\ &+ \int_{\mathbb{R}}\left\{\psi(x + \gamma xz) - \psi(x) - \gamma z\psi'(x)\right\}\nu(\mathrm{d}z) = 0 \text{ in } D. \end{aligned} \tag{14.9.10}$$

We try a solution of the form

$$\psi(x) = C_1 x^{\gamma_1} + C_2 x^{\gamma_2},$$

where $\gamma_1 > 1$, $\gamma_2 < 0$ are the solutions of the equation

$$F(\gamma) := -\rho + \mu\gamma + \frac{1}{2}\sigma^2\gamma(\gamma - 1) + \int_{\mathbb{R}}\{(1 + \theta z)^\gamma - 1 - \theta z\gamma\}\,\nu'(\mathrm{d}z) = 0.$$

We guess that the continuation region is of the form

$$D = \{(s, x) : 0 < x < \bar{x}\}$$

for some $\bar{x} > 0$ (to be determined).

We see that $C_2 = 0$, because otherwise $\lim_{x \to 0} |\psi(x)| = \infty$.

We guess that in this case it is optimal to wait till $X(t)$ reaches or exceeds a value $\bar{x} > c$ and then take out as much as possible, i.e., reduce $X(t)$ to 0. Taking the transaction costs into account this means that we should take out

$$\hat{\zeta}(x) = \frac{x - c}{1 + \lambda} \text{ for } x \geq \bar{x}.$$

We therefore propose that $\psi(x)$ has the form

$$\psi(x) = \begin{cases} C_1 x^{\gamma_1} \text{ for } 0 < x < \bar{x} \\ \dfrac{x - c}{1 + \lambda} \text{ for } x \geq \bar{x}. \end{cases}$$

Continuity and differentiability of $\psi(x)$ at $x = \bar{x}$ give the equations

$$C_1 \bar{x}^{\gamma_1} = \frac{\bar{x} - c}{1 + \lambda}$$

and

$$C_1 \gamma_1 \bar{x}^{\gamma_1 - 1} = \frac{1}{1 + \lambda}.$$

Combining these we get

$$\bar{x} = \frac{\gamma_1 c}{\gamma_1 - 1} \quad \text{and} \quad C_1 = \frac{\bar{x} - c}{1 + \lambda} \bar{x}^{-\gamma_1}.$$

With these values of $\bar{x}$ and C_1, we have to verify that ψ satisfies all the requirements of Theorem 3.2. We check some of them:

(ii) $\psi \geq \mathcal{M}\psi$ on $\mathcal{S}$.

Here $\mathcal{M}\psi = \sup \{\psi(x - c - (1 + \lambda)\zeta) + \zeta\}; \ 0 \leq \zeta \leq \left\{ \dfrac{x - c}{1 + \lambda} \right\}$.

If $x - c - (1 + \lambda)\zeta \geq \bar{x}$, then

$$\psi(x - c - (1 + \lambda)\zeta) + \zeta = \frac{x - 2c}{1 + \lambda} < \frac{x - c}{1 + \lambda} = \psi(x)$$

and if $x - c - (1 + \lambda)\zeta < \bar{x}$ then

$$h(\zeta) := \psi(x - c - (1 + \lambda)\zeta) + \zeta = C_1 (x - c - (1 + \lambda)\zeta)^{\gamma_1} + \zeta.$$

Since

$$h'\left(\frac{x-c}{1+\lambda}\right) = 1 \text{ and } h''(\zeta) > 0$$

we see that the maximum value of $h(\zeta)$; $0 \le \zeta \le \dfrac{x-c}{1+\lambda}$, is attained at $\zeta = \hat{\zeta}(x) = \dfrac{x-c}{1+\lambda}$.

Therefore

$$\mathcal{M}\psi(x) = \max\left(\frac{x-2c}{1+\lambda}, \frac{x-c}{1+\lambda}\right) = \frac{x-c}{1+\lambda} \text{ for all } x > c.$$

Hence $\mathcal{M}\psi(x) = \psi(x)$ for $x \ge \bar{x}$.

For $0 < x < \bar{x}$ consider

$$k(x) := C_1 x^{\gamma_1} - \frac{x-c}{1+\lambda}.$$

Since

$$k(\bar{x}) = k'(\bar{x}) = 0 \quad \text{and} \quad k''(x) > 0 \text{ for all } x,$$

we conclude that

$$k(x) > 0 \quad \text{for } 0 < x < \bar{x}.$$

Hence

$$\psi(x) > \mathcal{M}\psi(x) \quad \text{for } 0 < x < \bar{x}.$$

(vi) $A_0\psi(x) \le 0$ for $x \in \mathcal{S}\backslash\bar{D}$ i.e., for $x > \bar{x}$.

For $x > \bar{x}$, we have

$$\begin{aligned} A_0\psi(x) &= -\rho\frac{x-c}{1+\lambda} + \mu x \cdot \frac{1}{1+\lambda} \\ &\quad + \int_{x+\gamma xz<\bar{x}} \left\{C_1(x+\gamma xz)^{\gamma_1} - \frac{x+\gamma xz-c}{1+\lambda}\right\}\nu(\mathrm{d}z) \\ &\le (1+\lambda)^{-1}[(\mu-\rho)x + (\rho+\|\nu\|)c]. \end{aligned}$$

Therefore we see that

$$\begin{aligned} A_0\psi(x) &\le 0 \text{ for all } x > \bar{x} \\ &\Leftrightarrow (\mu-\rho)x + (\rho+\|\nu\|)c \le 0 \text{ for all } x > \bar{x} \\ &\Leftrightarrow (\mu-\rho)\bar{x} + (\rho+\|\nu\|)c \le 0 \end{aligned}$$

$$\Leftrightarrow \bar{x} \geq \frac{(\rho + \|\nu\|)c}{\rho - \mu}$$
$$\Leftrightarrow \frac{\gamma_1 c}{\gamma_1 - 1} \geq \frac{(\rho + \|\nu\|)c}{\rho - \mu}$$
$$\Leftrightarrow \gamma_1 \leq \frac{\rho + \|\nu\|}{\mu + \|\nu\|}.$$

Since

$$F\left(\frac{\rho}{\mu}\right) \geq -\rho + \mu\frac{\rho}{\mu} + \frac{1}{2}\sigma^2\frac{\rho}{\mu}\left(\frac{\rho}{\mu} - 1\right) > 0$$

and $F(\gamma_1) = 0, \gamma_1 > 1$ we conclude that $\gamma_1 < \frac{\rho}{\mu}$ and hence (vi) holds if $\|\nu\|$ is small enough.

Exercise 9.3

Here $f = g = 0, \Gamma(y, \zeta) = (s, 0), K(y, \zeta) = -c + (1 - \lambda)x$ and $\mathcal{S} = \mathbb{R}^2$; $y = (s, x)$. If there are no interventions, the process $Y(t)$ defined by

$$\mathrm{d}Y(t) = \begin{bmatrix} \mathrm{d}t \\ \mathrm{d}X(t) \end{bmatrix} = \begin{bmatrix} 1 \\ \mu \end{bmatrix}\mathrm{d}t + \begin{bmatrix} 0 \\ \sigma \end{bmatrix}\mathrm{d}B(t) + \begin{bmatrix} 0 \\ \int\limits_{\mathbb{R}} \theta\, z\, \widetilde{N}(\mathrm{d}t, \mathrm{d}z) \end{bmatrix}$$

has the generator

$$A\phi(y) = \frac{\partial\phi}{\partial s} + \mu\frac{\partial\phi}{\partial x} + \tfrac{1}{2}\sigma^2\frac{\partial^2\phi}{\partial x^2} + \int\limits_{\mathbb{R}} \left\{\phi(s, x + \theta\, z) - \phi(s, x) - \theta\, z\frac{\partial\phi}{\partial x}(s, z)\right\}\nu(\mathrm{d}z)\ ;$$
$$y = (s, x).$$

The intervention operator $\mathcal{M}$ is given by

$$\begin{aligned}\mathcal{M}\,\phi(y) &= \sup\{\phi(\Gamma(y, \zeta)) + K(y, \zeta); \zeta \in Z \text{ and } \Gamma(y, \zeta) \in \mathcal{S}\} \\ &= \phi(s, 0) + (1 - \lambda)x - c.\end{aligned}$$

If we try

$$\phi(s, x) = \mathrm{e}^{-\rho s}\psi(x),$$

we get that

$$A\phi(s, x) = \mathrm{e}^{-\rho s}A_0\psi(x),$$

where

$$A_0\psi(x) = -\rho\psi + \mu\psi'(x) + \tfrac{1}{2}\sigma^2\psi''(x) + \int_{\mathbb{R}} \left\{\psi(x+\theta\, z) - \psi(x) - \theta\, z\psi'(x)\right\} \nu(\mathrm{d}z)$$

and

$$\mathcal{M}\,\phi(s,x) = \mathrm{e}^{-\rho s}\mathcal{M}_0\psi(x),$$

where

$$\mathcal{M}_0\psi(x) = \psi(0) + (1-\lambda)x - c.$$

We guess that the continuation region D has the form

$$D = \left\{(s,x);\, x < x^*\right\}$$

for some $x^* > 0$ to be determined. To find a solution ψ_0 of $A_0\psi_0 + f = A_0\psi_0 = 0$, we try

$$\psi_0(x) = \mathrm{e}^{r\,x} \quad (r \text{ constant})$$

and get

$$\begin{aligned} A_0\psi_0(x) &= -\rho\,\mathrm{e}^{r\,x} + \mu\, r\,\mathrm{e}^{r\,x} + \tfrac{1}{2}\sigma^2 r^2 \mathrm{e}^{r\,x} \\ &\quad + \int_{\mathbb{R}} \left\{\mathrm{e}^{r(x+\theta\, z)} - \mathrm{e}^{r\,x} - r\,\theta\, z\,\mathrm{e}^{r\,x}\right\} \nu(\mathrm{d}z) \\ &= \mathrm{e}^{r\,x} h(r) = 0, \end{aligned}$$

where

$$h(r) = -\rho + \mu\, r + \tfrac{1}{2}\sigma^2 r^2 + \int_{\mathbb{R}} \left\{\mathrm{e}^{r\,\theta\, z} - 1 - r\,\theta\, z\right\} \nu(\mathrm{d}z).$$

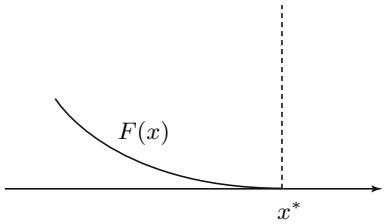
Choose $r_1 > 0$ such that

$$h(r_1) = 0 \quad \text{(see the solution of Exercise 3.1).}$$

Then we define

$$\psi(x) = \begin{cases} M\,\mathrm{e}^{r_1 x}; & x < x^* \\ \psi(0) + (1-\lambda)x - c = M + (1-\lambda)x - c; & x \geq x^* \end{cases} \tag{14.9.11}$$

Fig. 14.6 The function F



for some constant $M = \psi(0) > 0$. If we require continuity and differentiability at $x = x^*$ we get the equations

$$M\,\mathrm{e}^{r_1 x^*} = M + (1-\lambda)x^* - c \tag{14.9.12}$$

and

$$M\,r_1 \mathrm{e}^{r_1 x^*} = 1 - \lambda. \tag{14.9.13}$$

This gives the following equations for x^* and M:

$$k(x^*) := \mathrm{e}^{-r_1 x^*} + r_1 x^* - 1 - \frac{r_1 c}{1-\lambda} = 0, \quad M = \frac{1-\lambda}{r_1}\mathrm{e}^{-r_1 x^*} > 0. \tag{14.9.14}$$

Since $k(0) = -\dfrac{r_1 c}{1-\lambda} < 0$ and $\lim\limits_{x\to\infty} k(x) = \infty$, we see that there exists $x^* > 0$ s.t. $k(x^*) = 0$.

We must verify that with these values of x^* and M the conditions of Theorem 9.2 are satisfied. We consider some of them:
(ii) $\psi(x) \geq \mathcal{M}_0\psi(x)$.
For $x \geq x^*$ we have $\psi(x) = \mathcal{M}_0\psi(x) = M + (1-\lambda)x - c$. For $x < x^*$ we have $\psi(x) = M\,\mathrm{e}^{r_1 x}$ and $\mathcal{M}_0\psi(x) = M + (1-\lambda)x - c$. Define

$$F(x) = M\,\mathrm{e}^{r_1 x} - (M + (1-\lambda)x - c); \quad x \leq x^*.$$

See Fig. 14.6. We have

$$F(x^*) = F'(x^*) = 0 \text{ and } F''(x) = M\,r_1^2 \mathrm{e}^{r_1 x} > 0.$$

Hence $F'(x) < 0$ and so $F(x) > 0$ for $x < x^*$. Therefore

$$\psi(x) \geq \mathcal{M}_0\psi(x) \text{ for all } x.$$

(vi) $A_0\psi \le 0$ for $x > x^*$:
For $x > x^*$ we have

$$\begin{aligned} A_0\psi(x) &= -\rho[M + (1-\lambda)x - c] + \mu(1-\lambda) \\ &\quad + \int_{x+\theta z < x^*} \left\{ M\mathrm{e}^{r_1(x+\theta z)} - (M + (1-\lambda)(x+\theta z) - c) \right\} \nu(\mathrm{d}z) \\ &\le -\rho(1-\lambda)x + \rho(c-M) + \mu(1-\lambda) + c\|\nu\|. \end{aligned}$$

So

$$\begin{aligned} A_0\psi(x) \le 0 \text{ for all } x > x^* &\Leftrightarrow x \ge \frac{\mu}{\rho} + \frac{c-M}{1-\lambda} + \frac{c\|\nu\|}{\rho(1-\lambda)} \quad \text{for all } x > x^* \\ &\Leftrightarrow x^* \ge \frac{\mu}{\rho} + \frac{c-M}{1-\lambda} + \frac{c\|\nu\|}{\rho(1-\lambda)} \\ &\Leftrightarrow x^* \ge \frac{\mu}{\rho} + \frac{c}{1-\lambda} - \frac{1}{r_1}\mathrm{e}^{-r_1 x^*} + \frac{c\|\nu\|}{\rho(1-\lambda)} \\ &\Leftrightarrow \mathrm{e}^{-r_1 x^*} + r_1 x^* - \frac{c}{1-\lambda} \ge \frac{\mu}{\rho} + \frac{c\|\nu\|}{\rho(1-\lambda)} \\ &\Leftrightarrow 1 \ge \frac{\mu}{\rho} + \frac{c\|\nu\|}{\rho(1-\lambda)} \\ &\Leftrightarrow \mu + \frac{c\|\nu\|}{1-\lambda} \le \rho. \end{aligned}$$

So we need to assume that $\mu + \dfrac{c\|\nu\|}{1-\lambda} \le \rho$ for (vi) to hold.

Conclusion

Let

$$\psi(s,x) = \mathrm{e}^{-\rho s}\psi(x)$$

where ψ is given by (14.9.11). Assume that

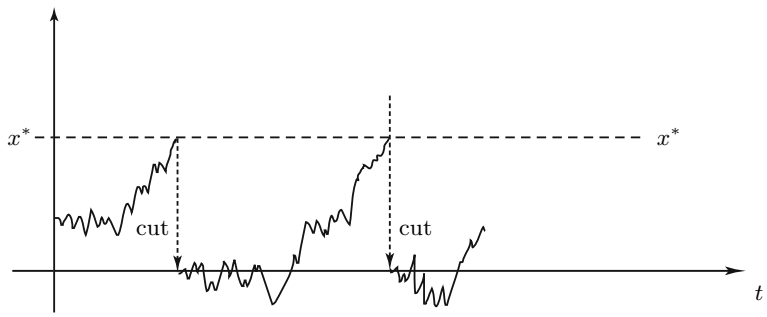


Fig. 14.7 The optimal forest management of Exercise 9.3

$$\mu + \frac{c\|\nu\|}{1-\lambda} \leq \rho.$$

Then

$$\phi(s,x) = \sup_v J^{(v)}(s,x)$$

and the optimal strategy is to cut the forest every time the biomass reaches the value x^* (see Fig. 14.7).

14.10 Exercises of Chap. 10

Exercise 10.1

As in Exercise 9.3, we have

$$\begin{aligned} f &= g = 0, \\ \Gamma(y,\zeta) &= (s,0);\ \ y = (s,x), \\ K(y,\zeta) &= (1-\lambda)x - c, \\ \mathcal{S} &= [0,\infty) \times \mathbb{R}. \end{aligned}$$

If there is no intervention, then $\phi_0 \equiv 0$ and

$$\mathcal{M}\phi_0 = \sup\{(1-\lambda)\zeta - c;\ \zeta = x\} = (1-\lambda)x - c.$$

Hence

$$\phi_1(y) = \sup_{\tau \leq \tau_{\mathcal{S}}} E^y[\mathcal{M}\phi_0(Y(\tau))] = \sup_{\tau \leq \tau_{\mathcal{S}}} E^y\left[\mathrm{e}^{-\rho(s+\tau)}((1-\lambda)X(\tau) - c)\right]. \quad (14.10.1)$$

This is an optimal stopping problem that can be solved by exploiting the three basic variational inequalities. We assume that the continuation region

$$D_1 = \{\phi_1 > \mathcal{M}\phi_0\}$$

is of the form

$$D_1 = \{(s,x);\ x < x_1\} \text{ for some } x_1 > 0$$

and that the value function has the form $\phi_1(s,x) = \mathrm{e}^{-\rho s}\psi_1(x)$ for some function ψ_1. On D_1, ψ_1 is the solution of

$$-\rho\psi_1(x)+\mu\psi_1'(x)+\tfrac{1}{2}\sigma^2\psi_1''(x)+\int_{\mathbb{R}}\left\{\psi_1(x+\theta z)-\psi_1(x)-\theta z\psi_1'(x)\right\}\nu(\mathrm{d}z)=0. \tag{14.10.2}$$

A solution of (14.10.2) is

$$\psi_1(x)=A\mathrm{e}^{\gamma_1 x}+B\mathrm{e}^{\gamma_2 x}$$

where $\gamma_2<0$ and $\gamma_1>1$, A and B arbitrary constants to be determined.

We choose $B=0$ and put $A_1=A>0$. We get

$$\psi_1(x)=\begin{cases}A_1\mathrm{e}^{\gamma_1 x} & x<x_1\\ (1-\lambda)x-c & x\geq x_1.\end{cases}$$

We impose the continuity and differentiability conditions of ψ_1 at $x=x_1$.

(i) Continuity: $A_1\mathrm{e}^{\gamma_1 x_1}=(1-\lambda)x_1-c$.
(ii) Differentiability: $A_1\gamma_1\mathrm{e}^{\gamma_1 x_1}=1-\lambda$.

We get $A_1=\dfrac{(1-\lambda)}{\gamma_1}\mathrm{e}^{-\gamma_1 x_1}$ and $x_1=\dfrac{1}{\gamma_1}+\dfrac{c}{1-\lambda}$.

As a second step, we evaluate

$$\phi_2(y)=\sup_{\tau}E^y[\mathcal{M}\phi_1(Y(\tau))].$$

We suppose $\phi_2(s,x)=\mathrm{e}^{-\rho s}\psi_2(x)$ and consider

$$\begin{aligned}\mathcal{M}\psi_1(x)&=\sup\{\psi_1(0)+(1-\lambda)\zeta-c;\,\zeta\leq x\}=\psi_1(0)+(1-\lambda)x-c\\&=(1-\lambda)x+A_1-c.\end{aligned}$$

Hence

$$\phi_2(y)=\sup_{\tau\leq\tau_S}E^y\left[\mathrm{e}^{-\rho(s+\tau)}\left((1-\lambda)X(\tau)-(c-A_1)\right)\right]. \tag{14.10.3}$$

By the same argument as before, we get $\Phi_2(s,x)=\mathrm{e}^{-\rho s}\psi_2(x)$, where

$$\psi_2(x)=\begin{cases}A_2\mathrm{e}^{\gamma_1 x} & x<x_2\\ (1-\lambda)x+A_1-c & x\geq x_2\end{cases}$$

where $x_2=\dfrac{1}{\gamma_1}+\dfrac{c-A_1}{1-\lambda}$ and $A_2=\dfrac{1-\lambda}{\gamma_1}\mathrm{e}^{-\gamma_1 x_2}$. Note that $x_2<x_1$ and $A_2>A_1$.

Since $\mathcal{M}\phi_0$ and $\mathcal{M}\phi_1$ have linear growth, the conditions of Theorem 10.2 are satisfied. Hence ϕ_1 and ϕ_2 are the solutions for our impulse control problems when respec-

tively one intervention and two interventions are allowed. The impulses are given by $\zeta_1 = \zeta_2 = x$ and $\tau_1 = \inf\{t : X(t) \geq x_2\}$ and $\tau_2 = \inf\{t > \tau_1 : X(t) \geq x_1\}$.

Exercise 10.2

Here we have (see the notation of Chap. 9)

$$\begin{aligned} f = g &\equiv 0 \\ K(x,\zeta) &= \zeta \\ \Gamma(x,\zeta) &= x - (1+\lambda)\zeta - c \\ \mathcal{S} &= \{(s,x);\ x > 0\}. \end{aligned}$$

We put $y = (s,x)$ and suppose $\phi_0(s,x) = \mathrm{e}^{-\rho s}\psi_0(x)$. Since $f = g = 0$ we have

$$\phi_0(y) = 0$$

and $\mathcal{M}\psi_0(y) = \sup\left\{\zeta : 0 \leq \zeta \leq \dfrac{x-c}{1+\lambda}\right\} = \left(\dfrac{x-c}{1+\lambda}\right)^+$. As a second step, we consider

$$\phi_1(s,x) = \sup_{\tau \leq \tau_{\mathcal{S}}} E^x[\mathcal{M}\phi_0(X(\tau))] = \sup_{\tau \leq \tau_{\mathcal{S}}} E^x\left[\mathrm{e}^{-\rho(\tau+s)}\,\frac{(X(\tau+s)-c)^+}{1+\lambda}\right]. \tag{14.10.4}$$

We distinguish between three cases

(a) $\mu > \rho$

Then

$$\phi_1(s,x) \geq \frac{x\mathrm{e}^{(\mu-\rho)(t+s)} - c\mathrm{e}^{-\rho(t+s)}}{1+\lambda}.$$

Hence if $t \to +\infty$

$$\phi_1(s,x) \to +\infty.$$

We obtain $\mathcal{M}\phi_1(s,x) = +\infty$ and clearly $\phi_n = +\infty$ for all n. In this case, the optimal stopping time does not exist.

(b) $\mu < \rho$

In this case we try to put $\phi_1(s,x) = \mathrm{e}^{-\rho s}\psi_1(x)$ and solve the optimal stopping problem (14.10.4) by using Theorem 3.2.

We guess that the continuation region is of the form $D = \left\{0 < x < x_1^*\right\}$ and solve

$$\begin{aligned} L\psi_1(x) := &-\rho\psi_1(x) + \mu x\psi_1'(x) + \frac{\sigma^2x^2}{2}\psi_1''(x) \\ &+ \int_{\mathbb{R}} \left\{\psi_1(x+\theta xz) - \psi_1(x) - \theta xz\psi_1'(x)\right\}\nu(dx) = 0. \end{aligned} \tag{14.10.5}$$

A solution of (14.10.5) is

$$\psi_1(x) = c_1 x^{\gamma_1} + c_2 x^{\gamma_2}$$

where $\gamma_2 < 0$ and $\gamma_1 > 1$ are solutions of the equation

$$k(\gamma) := -\rho + \mu\gamma + \frac{1}{2}\sigma^2\gamma(\gamma-1) + \int_{\mathbb{R}} \{(1+\theta z)^\gamma - 1 - \gamma\theta z\}\, \nu(\mathrm{d}z) = 0$$

and c_1, c_2 are arbitrary constants. Since $\gamma_2 < 0$, we put $c_2 = 0$. We obtain

$$\psi_1(x) = \begin{cases} c_1 x^{\gamma_1} & 0 < x < x_1^* \\ \dfrac{x-c}{1+\lambda} & x \geq x_1^*. \end{cases}$$

By imposing the condition of continuity and differentiability, we can compute c_1 and x_1^*. The result is:

1. $x_1^* = \dfrac{\gamma_1 c}{\gamma_1 - 1}$.
2. $c_1 = \dfrac{1}{\gamma_1(1+\lambda)} \Big(\dfrac{\gamma_1 c}{\gamma_1 - 1}\Big)^{1-\gamma_1}$.

Note that $\gamma_1 > 1$ and $x_1^* > c$. We check some of the conditions of Theorem 3.2:
(ii) $\psi_1(x) \geq \mathcal{M}\psi_0(x)$ for all x:
We know that $\phi_1(x) = \mathcal{M}\phi_0(x)$ for $x > x_1^*$. Consider $h_1(x) := \psi_1(x) - \mathcal{M}\psi_0(x)$. We have

$$\begin{aligned} h_1(x_1^*) &= 0, \\ h_1'(x_1^*) &= 0, \\ h_1''(x_1^*) &= c_1\gamma_1(\gamma_1 - 1)(x_1^*)^{\gamma_1 - 2} > 0. \end{aligned}$$

Hence x_1^* is a minimum for h_1 and $\psi_1(x) \geq \mathcal{M}\psi_0(x)$ for every $0 < x < x_1^*$.
(vi) $L\psi_1 \leq 0$ for all $x > 0$:
Clearly $L\psi_1 = 0$ for $0 < x < x_1^*$.
If $x > x_1^*$ then

$$\begin{aligned} L\psi_1(x) = ((\mu - \rho)x + c\rho)\frac{1}{1+\lambda} \\ + \int_{x+\theta x z < x_1^*} \left\{ c_1(x + \theta x z)^{\gamma_1} - \frac{x + \theta x z - c}{1+\lambda} \right\} \nu(\mathrm{d}z) \leq 0 \end{aligned}$$

if

$$x \geq \frac{c(\rho + \|\nu\|)}{\rho - \mu}.$$

Define

$$F(\gamma) := x^{-\gamma} L(x^{\gamma}) = -\rho + \mu\gamma + \frac{1}{2}\gamma(\gamma - 1) + \int_{\mathbb{R}} \{(1 + \theta z)^{\gamma} - 1 - \gamma\theta z\}\, \nu(\mathrm{d}z).$$

Then we know that $F(\gamma_2) = F(\gamma_1) = 0$ and $F'(\gamma) > 0$ for $\gamma \geq \gamma_1$. Since $F\left(\frac{\rho}{\mu}\right) > 0$ we have that $\frac{\rho}{\mu} > \gamma_1$, which implies that $x_1^* = \dfrac{\gamma_1 c}{\gamma_1 - 1} > \dfrac{\rho c}{\rho - \mu}$. Hence $L\psi_1(x) \leq 0$ for all $x \geq x_1^*$ if $\|\nu\|$ is small enough.

We conclude that under these conditions $\phi_1(s, x) = \mathrm{e}^{-\rho s}\psi_1(x)$ actually solves the optimal stopping problem (14.10.4).

Next we consider

$$\mathcal{M}\psi_1(x) = \sup\left\{\psi_1(x - (1 + \lambda)\zeta - c) + \zeta; 0 \leq \zeta \leq \frac{x - c}{1 + \lambda}\right\} = \frac{(x - c)^+}{1 + \lambda}$$

and repeat the same procedure to find ψ_2.

By induction, we obtain $\mathcal{M}\psi_n = \mathcal{M}\psi_{n-1} = \mathcal{M}\psi_1 = \mathcal{M}\psi_0$. Consequently, we also have

$$\Phi = \Phi_1$$

and $\Phi(s, x) = \Phi_n(s, x)$ for every n. Moreover, we achieve the optimal result with just one intervention.

(c) $\mu = \rho$

This case is left to the reader.

14.11 Exercises of Chap. 11

Exercise 11.1

In this case we have

$$\begin{aligned} \Gamma_1(\zeta, x_1, x_2) &= x_1 - \zeta - c - \lambda|\zeta| \\ \Gamma_2(\zeta, x_1, x_2) &= x_2 + \zeta \\ K(\zeta, x_1, x_2) &= c + \lambda|\zeta| \\ g &= 0 \end{aligned}$$

$$f(s, x_1, x_2, u) = \frac{\mathrm{e}^{-\rho s}}{\gamma} u^{\gamma}.$$

The generator is given by

$$L^u = \frac{\partial}{\partial t} + (rx_1 - u)\frac{\partial}{\partial x_1} + \alpha x_2 \frac{\partial}{\partial x_2} + \frac{\sigma^2}{2} x_2^2 \frac{\partial^2}{\partial x_2^2}.$$

Let $\phi(s, x_1, x_2)$ be the value function of the optimal consumption problem

$$\sup_{w \in \mathcal{W}} E^y \left[\int_0^{\infty} \mathrm{e}^{-\rho(s+t)} \frac{u(t)^{\gamma}}{\gamma} \mathrm{d}t \right]; \ y = (s, x_1, x_2)$$

with $c, \lambda > 0$ and $\phi_0(s, x_1, x_2)$ the corresponding value function in the case when there are no transaction costs, i.e.

$$c = \lambda = 0.$$

In order to prove that

$$\Phi(s, x_1, x_2) \leq K \mathrm{e}^{-\rho s}(x_1 + x_2)^{\gamma} = \Phi_0(s, x_1, x_2)$$

we check the hypotheses of Theorem 11.1(a):
(vi) $L^u \phi_0 + f \leq 0$
Since ϕ_0 is the value function in the absence of transaction costs, we have

$$\sup_{u \geq 0} \left\{ L^u \phi_0 + f \right\} = 0 \qquad \text{in } \mathbb{R}^3.$$

Note that

$$\begin{aligned} \mathcal{M}\phi_0(s, x_1, x_2) &= \sup_{\zeta \in \mathbb{R} \setminus \{0\}} \phi_0(s, x_1 - \zeta - \lambda|\zeta| - c, x_2 + \zeta) \\ &= \sup_{\zeta \in \mathbb{R} \setminus \{0\}} K \mathrm{e}^{-\rho s}(x_1 + x_2 - c - \lambda|\zeta|)^{\gamma} \\ &= K \mathrm{e}^{-\rho s}(x_1 + x_2 - c)^{\gamma}. \end{aligned}$$

Therefore

$$D = \{\phi_0 > \mathcal{M}\phi_0\} = \mathbb{R}^3.$$

Hence we can conclude that

$$\mathrm{e}^{-\rho s}(x_1 + x_2)^{\gamma} \geq \Phi(s, x_1, x_2).$$

Exercise 11.2

The HJBQVI's for this problem can be formulated in one equation as follows:

$$\min\left(\inf_{u\in\mathbb{R}}\left\{\frac{\partial\varphi}{\partial s}+u\frac{\partial\varphi}{\partial x}+\frac{1}{2}\sigma^2\frac{\partial^2\varphi}{\partial x^2}+x^2+u^2\right\},\varphi-\mathcal{M}\varphi\right)=0,$$

where

$$\mathcal{M}\varphi(s,x)=\inf_{\zeta\in\mathbb{R}}\{\varphi(s,x+\zeta)+c\}.$$

Since φ is a candidate for the value function Φ it is reasonable to guess that, for each s, $\varphi(s,z)$ is minimal for $z=0$. Hence

$$\mathcal{M}\varphi(s,x)=\varphi(s,0)+c,\ \text{attained for}\ \zeta=\zeta^*(x)=-x.$$

The minimum of the function

$$k(u):=\frac{\partial\varphi}{\partial s}+u\frac{\partial\varphi}{\partial x}+\frac{1}{2}\sigma^2\frac{\partial^2\varphi}{\partial x^2}+x^2+u^2\ \ ;\quad u\in\mathbb{R}$$

is attained when

$$u=\hat{u}(s,x):=-\frac{1}{2}\frac{\partial\varphi}{\partial x}(s,x)$$

and this gives the minimum value

$$k_{\min}=k(\hat{u}(s,x))=\frac{\partial\varphi}{\partial s}-\frac{1}{4}\left(\frac{\partial\varphi}{\partial x}\right)^2+\frac{1}{2}\sigma^2\frac{\partial^2\varphi}{\partial x^2}+x^2.$$

Hence the HJBQVI gets the form

$$\min\left(\frac{\partial\varphi}{\partial s}-\frac{1}{4}\left(\frac{\partial\varphi}{\partial x}\right)^2+\frac{1}{2}\sigma^2\frac{\partial^2\varphi}{\partial x^2}+x^2,\varphi(s,0)+c\right)=0.$$

If we guess that the continuation region

$$D:=\{(s,x);\ \varphi(s,x)<\varphi(s,0)+c\}$$

has the form

$$D=\left\{(s,x);\ |x|<x^*(s),\ 0\le s\le T\right\}$$

for some function $x^*(s)$, then the HJBQVI can be split into the two equations

$$\frac{\partial\varphi}{\partial s} = -\frac{1}{2}\sigma^2\frac{\partial^2\varphi}{\partial x^2} + \frac{1}{4}\left(\frac{\partial\varphi}{\partial x}\right)^2 + \frac{1}{2}\sigma^2\frac{\partial^2\varphi}{\partial x^2} - x^2 \text{ for } |x| < x^*(s) \tag{14.11.1}$$

$$\varphi(s,x) = \varphi(s,0) + c \text{ for } |x| \geq x^*(s). \tag{14.11.2}$$

Equation (14.11.1) is known as a *Burgers* equation, which can be linearized by the transformation

$$w(s,x) := \exp\left(-\frac{1}{2\sigma^2}\varphi(s,x)\right). \tag{14.11.3}$$

This transforms (14.11.1) into the equation

$$\frac{\partial w}{\partial s} = -\frac{1}{2}\sigma^2\frac{\partial^2 w}{\partial x^2} + \frac{x^2}{2\sigma^2}w\,; \quad |x| < x^*(s) \tag{14.11.4}$$

with boundary values

$$w(s,x) = b(s) := w(s,0)\exp\left(-\frac{c}{2\sigma^2}\right);\ x = \pm x^*(s),\ s < T \tag{14.11.5}$$

$$w(T,x) = 1. \tag{14.11.6}$$

By the Feynman–Kac formula, the solution of (14.11.4)–(14.11.6) can be written

$$w(s,x) = E_{\hat{P}}\left(\exp(-\frac{1}{2}\int_0^{\hat{\tau}_D - s}\hat{B}^2(t)\mathrm{d}t)\ b(\hat{B}(\hat{\tau}_D - s))\right);\ (s,x) \in D \tag{14.11.7}$$

where $\hat{B}(t) = \hat{B}(t,\hat{\omega})$ is an auxiliary Brownian motion with law $\hat{P}$ starting at x and

$$\hat{\tau}_D = \hat{\tau}_D(\hat{\omega}) = \inf\left\{t > 0;\ |\hat{B}(t)| \geq \frac{1}{\sigma}x^*(t)\right\} \wedge T. \tag{14.11.8}$$

The high contact condition for determination of the curve $x^*(t)$ is then

$$\left(\frac{\partial w}{\partial s}, \frac{\partial w}{\partial x}\right)(s, x^*(s)) = \exp\left(-\frac{c}{2c^2}\right)\left(\frac{\partial w}{\partial s}(s,0), 0\right). \tag{14.11.9}$$

The suggested shape of $x^*(t)$ is shown in Fig. 14.8.

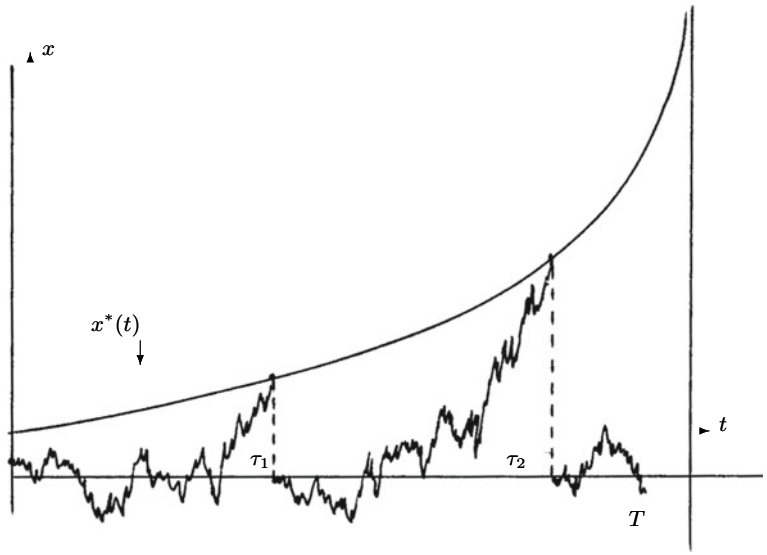


Fig. 14.8 The suggested optimal combined control of Exercise 11.2

14.12 Exercises of Chap. 12

Exercise 12.1

Because of the symmetry of h, we assume that the continuation region is of the form

$$D = \left\{(s, x) : -x^* < x < x^*\right\}$$

with $x^* > 0$.

We assume that the value function $\phi(s, x) = \mathrm{e}^{-\rho s}\psi(x)$. On D, ϕ is the solution of

$$L\phi(s, x) = 0, \tag{14.12.1}$$

where $L = \frac{\partial}{\partial s} + \frac{1}{2}\frac{\partial^2}{\partial x^2}$. We obtain

$$L_0\psi(x) := -\rho\psi(x) + \tfrac{1}{2}\psi''(x) = 0. \tag{14.12.2}$$

The general solution of (14.12.2) is

$$\psi(x) = c_1\mathrm{e}^{\sqrt{2\rho}x} + c_2\mathrm{e}^{-\sqrt{2\rho}x}.$$

We must have $\psi(x) = \psi(-x)$, hence $c_1 = c_2$. We put $c_1 = \frac{1}{2}c$ and assume $c > 0$. We impose continuity and differentiability conditions at $x = x^*$ (Fig. 14.9):

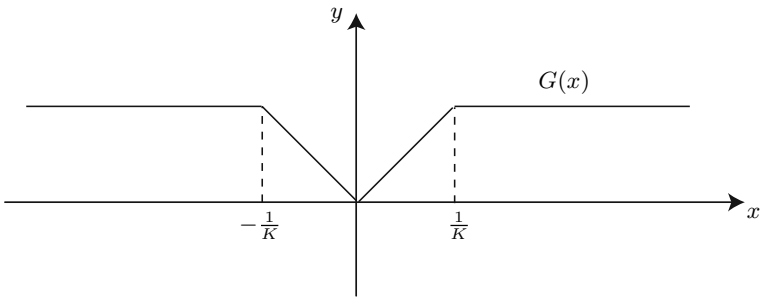


Fig. 14.9 The function $G(x)$

(i) Continuity at $x = x^*$

$$\frac{1}{2}c\left(\mathrm{e}^{\sqrt{2\rho}\,x^*} + \mathrm{e}^{-\sqrt{2\rho}\,x^*}\right) = Kx^*.$$

(ii) Differentiability at $x = x^*$

$$\frac{1}{2}c\sqrt{2\rho}\left(\mathrm{e}^{x^*\sqrt{2\rho}}\frac{1}{2} - \mathrm{e}^{-\sqrt{2\rho}\,x^*}\right) = K.$$

Then x^* is the solution of

$$\frac{1}{x^*\sqrt{2\rho}} = \frac{\mathrm{e}^{x^*\sqrt{2\rho}} - \mathrm{e}^{-\sqrt{2\rho}\,x^*}}{\mathrm{e}^{x^*\sqrt{2\rho}} + \mathrm{e}^{-x^*\sqrt{2\rho}}} = \mathrm{tgh}(x^*\sqrt{2\rho})$$

and

$$c = \frac{K}{\sqrt{2\rho}}\frac{1}{+\sinh(x^*\sqrt{2\rho})}.$$

We must check if $x^* < \frac{1}{K}$. If we put $z^* = x^*\sqrt{2\rho}$, then z^* is the solution of

$$\frac{1}{z^*} = \mathrm{tgh}(z^*).$$

We distinguish between two cases:

Case 1: For $K < \dfrac{1}{x^*} = \dfrac{\sqrt{2\rho}}{z^*}$ we have (Fig. 14.10)

$$\psi(x) = \begin{cases} 1 & |x| > \frac{1}{K} \\ K|x| & x^* < |x| < \frac{1}{K} \\ c\cosh(x\sqrt{2\rho}\,) & |x| < x^* \end{cases}$$

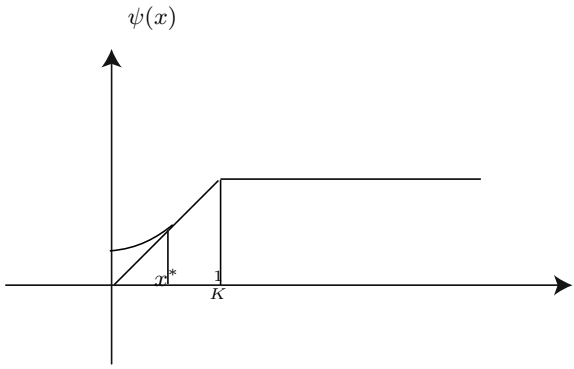


Fig. 14.10 The function $\psi(x)$ in Case 1 for $x \geq 0$

Since ψ is not C^2 at $x = x^*$ we prove that ψ is a viscosity solution for our optimal stopping problem.

(i) We first prove that ψ is a viscosity subsolution.
Let u belong to $C^2(\mathbb{R})$ and $u(x) \geq \psi(x)$ for all $x \in \mathbb{R}$ and let $y_0 \in \mathbb{R}$ be such that

$$u(y_0) = \psi(y_0).$$

Then ψ is a viscosity subsolution if and only if

$$\max(L_0 u(y_0), G(y_0) - u(y_0)) \geq 0 \text{ for all such } u, y_0. \tag{14.12.3}$$

We need to check (14.12.3) only for $y_0 = x^*$. We have

$$u(x^*) = \psi(x^*) = G(x^*)$$

i.e., $(G - u)(x^*) = 0$. Hence $\max(L_0 u(x^*), G(x^*) - u(x^*)) \geq (G - u)(x^*) = 0$.

(ii) We prove that ψ is a viscosity supersolution.
Let v belong to $C^2(\mathbb{R})$ and $v(x) \leq \psi(x)$ for every $x \in \mathbb{R}$ and let $y_0 \in \mathbb{R}$ be such that $v(y_0) = \psi(y_0)$. Then ψ is a viscosity supersolution if and only if

$$\max(L_0 v(y_0), G(y_0) - v(y_0)) \leq 0 \text{ for all such } v, y_0.$$

We check it only for $x = x^*$. Then

$$G(x^*) = \psi(x^*) = v(x^*).$$

Since $v \leq \psi$, $x = x^*$ is a maximum point for $H := v - \psi$. We have

$$H(x^*) = 0,$$

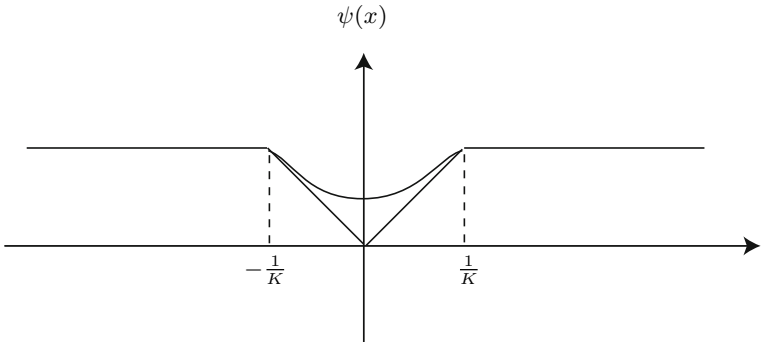


Fig. 14.11 The function $\psi(x)$ in Case 2

$$H'(x^*) = 0,$$
$$H''(x^*) = v''(x^*) - \psi''(x_-^*) \leq 0.$$

Hence $L_0 v(x^*) \leq L_0 \psi(x_-^*) \leq 0$. Therefore ψ is a viscosity supersolution. Since ψ is both a viscosity supersolution and subsolution, ψ is a viscosity solution.

Case 2: We consider now the case when $K \geq \dfrac{\sqrt{2\rho}}{z^*}$.

In this case, the continuation region is given by

$$D = \left\{ -\frac{1}{K} < x < \frac{1}{K} \right\}$$

i.e., $x^* = \frac{1}{K}$.

We have

$$\psi(x) = \begin{cases} 1 & |x| \geq \frac{1}{K} \\ c\cosh(z\sqrt{2\rho}) & |x| < \frac{1}{K} \end{cases}$$

ψ is not C^1 at $|x| = \frac{1}{K}$. See Fig. 14.11. We prove that it is a viscosity solution.

(i) ψ is a viscosity subsolution.

Let u belong to $C^2(\mathbb{R})$ and $u(x) \geq \psi(x)$ for every $x \in \mathbb{R}$. Let y_0 be such that $u(y_0) = \psi(y_0)$. Then ψ is a viscosity subsolution if and only if

$$\max(L_0 u(y_0), G(y_0) - u(y_0)) \geq 0. \tag{14.12.4}$$

For $y_0 = x^*$, $G(x^*) = \psi(x^*) = u(x^*)$. Hence (14.12.4) is trivially satisfied.

(ii) ψ is a viscosity supersolution.

Actually there does not exist any $v \in C^2(\mathbb{R})$ such that $v(\frac{1}{K}) = \psi(\frac{1}{K})$ and $v \le \psi$. Heuristically, this happens because ψ has an angle at $x = \frac{1}{K}$:

Suppose that there exists $v \in C^2(\mathbb{R})$ such that $v \le \psi$, $v(\frac{1}{K}) = \psi(\frac{1}{K})$. We consider $H := \psi - v$. We have $H(\frac{1}{K}) = 0$ and $H \ge 0$. Hence we must have

$$H'\left(\left(\frac{1}{K}\right)^-\right) := \lim_{x \to (\frac{1}{K})^-} H'(x) \le 0 \text{ and } H'\left(\left(\frac{1}{K}\right)^+\right) := \lim_{x \to (\frac{1}{K})^+} H'(x) \ge 0.$$

Therefore

$$\psi'\left(\left(\frac{1}{K}\right)^-\right) - v'\left(\left(\frac{1}{K}\right)^-\right) \le \psi'\left(\left(\frac{1}{K}\right)^+\right) - v'\left(\left(\frac{1}{K}\right)^+\right),$$

which implies that $\psi'\left(\left(\frac{1}{K}\right)^-\right) \le \psi'\left(\left(\frac{1}{K}\right)^+\right)$ since $v \in C^2(\mathbb{R})$.

Since ψ' does not satisfy this inequality, we conclude that there does not exist any $v \in C^2(\mathbb{R})$ such that $\psi \ge v$ and $\psi(\frac{1}{K}) = v(\frac{1}{K})$. ($v$ cannot be even C^1!).

We can conclude that ψ is a viscosity solution for our problem.

Exercise 12.2

We intend to apply (the minimum version of) Theorem 12.8 and note that in this case we have, with $y = (s, x) \in \mathbb{R}^2$,

$$\begin{aligned}
L^u\varphi(y) &= \frac{\partial\varphi}{\partial s} + u\frac{\partial\varphi}{\partial x} + \frac{1}{2}\frac{\partial^2\varphi}{\partial x^2} \\
&\quad + \int_{\mathbb{R}} \left\{\varphi(s, x+z) - \varphi(s,x) - z\frac{\partial\varphi}{\partial x}(s,x)\right\}\nu(\mathrm{d}z) \\
\Gamma(y,\zeta) &= y + (0,\zeta),\ \ K(y,\zeta) = c\mathrm{e}^{-\rho s},\ \ f(y,u) = \mathrm{e}^{-\rho s}(x^2 + \theta u^2), \\
g &= 0 \text{ and} \\
\mathcal{M}\varphi(y) &= \inf\left\{\varphi(s, x+\zeta) + c\mathrm{e}^{-\rho s};\ \zeta \in \mathbb{R}\backslash\{0\}\right\} = \varphi(s,0) + c\mathrm{e}^{-\rho s}.
\end{aligned}$$

and the corresponding HJBQVI becomes

$$\min\left(\inf_{\alpha\in U}\left\{L^\alpha\varphi(y) + f(y,\alpha)\right\}, \mathcal{M}\varphi(y) - \varphi(y)\right) = 0. \tag{14.12.5}$$

We first prove that

$$\Phi(x) < \Phi_1(x) = \mathrm{e}^{-\rho s}(ax^2 + b)$$

(see the solution of Exercise 5.4). Since clearly

$$\Phi(x) \le \Phi_1(x)$$

it suffices, by Theorem 12.8, to prove that $\Phi_1(x)$ does not satisfy Eq. (14.12.5) in the viscosity sense. In particular, (14.12.5) implies that

$$e^{-\rho s}(ax^2 + b) = \Phi_1(s, x) \leq \mathcal{M}\Phi_1(s, x) = e^{-\rho s}(b + c)\ ;\ x \in \mathbb{R}.$$

Since $a > 0$ this is impossible.

Next we prove that

$$\Phi(x) < \Phi_2(x)$$

where $\Phi_2(x)$ is the solution of Exercise 9.1. It has the form $\Phi_2(s, x) = e^{-\rho s}\psi_2(x)$, with

$$\psi_2(x) = \begin{cases} \psi_0(x);\ |x| \leq \bar{x} \\ \psi_0(\hat{x}) + c;\ |x| > \bar{x} \end{cases}$$

where

$$\psi_0(x) = \frac{1}{\rho}x^2 + \frac{b}{\rho^2} - a\cosh(\gamma x)$$

is a solution of

$$-\rho\psi_0(x) + \frac{1}{2}\psi_0''(x) + \int_{\mathbb{R}} \left\{\psi_0(x + z) - \psi_0(x) - z\psi_0'(x)\right\} \nu(\mathrm{d}z) + x^2 = 0.$$

Since clearly $\Phi(s, x) \leq \Phi_2(s, x)$, it suffices to prove that $\Phi_2(s, x)$ does not satisfy (14.12.5) in viscosity sense. In particular, (14.12.5) implies that

$$L^u\Phi_2(s, x) + e^{-\rho s}(x^2 + \theta u^2) \geq 0 \text{ for all } u \in \mathbb{R}.$$

For $|x| < \bar{x}$ this reads

$$u\psi_0'(x) + \theta u^2 \geq 0 \text{ for all } u \in \mathbb{R},\ |x| < \bar{x}. \tag{14.12.6}$$

The function

$$h(u) := \left(\frac{2x}{\rho} - a\gamma\sinh(\gamma x)\right)u + \theta u^2\ ;\ u \in \mathbb{R}$$

is minimal when

$$u = \hat{u} = \frac{1}{2\theta}\left(a\gamma\sinh(\gamma x) - \frac{2x}{\rho}\right)$$

with corresponding minimum value

$$h(\hat{u}) = -\frac{1}{4\theta}\left(a\gamma\sinh(\gamma x) - \frac{2x}{\rho}\right)^2.$$

Hence (14.12.6) cannot possibly hold and we conclude that Φ_2 cannot be a viscosity solution of (14.12.5). Hence $\Phi \neq \Phi_2$ and hence $\Phi(x) < \Phi_2(x)$ for some x.

14.13 Exercises of Chap. 13

Exercise 13.6

(a) Here $A\phi(x) = (\alpha x + u)\phi'(x) + \frac{1}{2}\sigma^2\phi''(x)$ and hence

$$\begin{aligned} A^*\psi(x) &= -\frac{\mathrm{d}}{\mathrm{d}x}((\alpha x + u)\psi(x)) + \frac{1}{2}\sigma^2\psi''(x) \\ &= -\alpha\psi(x) - (\alpha x + u)\psi'(x) + \frac{1}{2}\sigma^2\psi''(x). \end{aligned} \tag{14.13.1}$$

Therefore the corresponding complete observation controlled SPDE is

$$\begin{cases} \mathrm{d}y(t,x) &= \left[-\alpha y(t,x) - (\alpha x + u(t))\dfrac{\partial y}{\partial x}(t,x) + \dfrac{1}{2}\sigma^2\dfrac{\partial^2}{\partial x^2}y(t;x)\right]\mathrm{d}t \\ &\quad + h(x)y(t,x)\mathrm{d}\zeta(t)\ ;\ t > 0 \\ y(0,x) &= F(x) \end{cases} \tag{14.13.2}$$

with performance criterion

$$J(u) = E_Q\left[\int_0^T \left(\int_{\mathbb{R}}(x^2 + u^2(t))y(t,x)dx\right)\mathrm{d}t\right]. \tag{14.13.3}$$

(b) Here $A\phi(x) = \alpha x u\phi'(x) + \frac{1}{2}\beta^2x^2u^2\phi''(x)$ and hence

$$\begin{aligned} A^*\psi(x) &= -\frac{\mathrm{d}}{\mathrm{d}x}(\alpha u x\psi(x)) + \frac{1}{2}\frac{\mathrm{d}^2}{\mathrm{d}x^2}(\beta^2u^2x^2\psi(x)) \\ &= (\beta^2u^2 - \alpha u)\psi(x) + (2\beta^2u^2x - \alpha ux)\psi'(x) + \frac{1}{2}\beta^2u^2x^2\psi''(x). \end{aligned} \tag{14.13.4}$$

Hence the corresponding controlled complete observation SPDE is

$$\begin{cases} \mathrm{d}y(t,x) &= \Big[(\beta^2u^2(t)-\alpha u(t))y(t,x)+(2\beta^2u^2(t)x-\alpha u(t)x)\dfrac{\partial}{\partial x}y(t,x) \\ &\qquad +\dfrac{1}{2}\beta^2u^2(t)x^2\dfrac{\partial^2}{\partial x^2}y(t,x)\Big]\mathrm{d}t+h(x)y(t,x)d\zeta(t) \\ y(0,x) &= F(x), \end{cases} \tag{14.13.5}$$

with performance criterion

$$J(u)=E_Q\left[\int_{\mathbb{R}}x^\gamma y(T,x)\mathrm{d}x\right]. \tag{14.13.6}$$

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Notation and Symbols

$\mathbb{R}^n$	n-dimensional Euclidean space
$\mathbb{R}^+$	the nonnegative real numbers
$\mathbb{R}^{n\times m}$	the $n \times m$ matrices (real entries)
$\mathbb{Z}$	the integers
$\mathbb{N}$	the natural numbers
$\mathbf{B}_0$	the family of Borel sets $U \subset \mathbb{R}$ whose closure $\bar{U}$ does not contain 0
$\mathbb{R}^n \simeq \mathbb{R}^{n\times 1}$	i.e., vectors in $\mathbb{R}^n$ are regarded as $n \times 1$ matrices
I_n	the $n \times n$ identity matrix
A^T	the transposed of the matrix A
$\mathcal{P}(\mathbb{R}^k)$	set of functions $f : \mathbb{R}^k \to \mathbb{R}$ of at must polynomial growth, i.e., there exists constants C, m such that: $\|f(y)\| \leq C(1 + \|y\|^m)$ for all $y \in \mathbb{R}^k$
$C(U, V)$	the continuous functions from U into V
$C(U)$	the same as $C(U, \mathbb{R})$
$C_0(U)$	the functions in $C(U)$ with compact support
$C^k = C^k(U)$	the functions in $C(U, \mathbb{R})$ with continuous derivatives up to order k
$C_0^k = C_0^k(U)$	the functions in $C^k(U)$ with compact support in U
$C^{k+\alpha}$	the functions in C^k whose kth derivatives are Lipschitz continuous with exponent α
$C^{1,2}(\mathbb{R} \times \mathbb{R}^n)$	the functions $f(t, x)$; $\mathbb{R} \times \mathbb{R}^n \to \mathbb{R}$ which are C^1 w.r.t. $t \in \mathbb{R}$ and C^2 w.r.t. $x \in \mathbb{R}^n$
$C_b(U)$	the bounded continuous functions on U
$\|x\|^2 = x^2$	$\sum_{i=1}^n x_i^2$ if $x = (x_1, \ldots, x_n)$
$x \cdot y$	the dot product $\sum_{i=1}^n x_i y_i$ if $x = (x_1, \ldots, x_n)$, $y = (y_1, \ldots, y_n)$
x^+	$\max(x, 0)$ if $x \in \mathbb{R}$
x^-	$\max(-x, 0)$ if $x \in \mathbb{R}$
$\operatorname{sign} x$	$\begin{cases} 1 & \text{if } x \geq 0 \\ -1 & \text{if } x > 0 \end{cases}$

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$\sinh(x)$	hyperbolic sine of x $\left(= \frac{e^x - e^{-x}}{2}\right)$
$\cosh(x)$	hyperbolic cosine of x $\left(= \frac{e^x + e^{-x}}{2}\right)$
$\tgh(x)$	$\frac{\sinh(x)}{\cosh(x)}$
$s \wedge t$	the minimum of s and t $(= \min(s, t))$
$s \vee t$	the maximum of s and t $(= \max(s, t))$
δ_x	the unit point mass at x
$\operatorname{argmax}_{u \in U} f(u)$	$\{u^* \in U;\ f(u^*) \geq f(u), \forall u \in U\}$
$:=$	equal to by definition
$\overline{\lim}$, $\underline{\lim}$	the same as lim inf, lim sup
$\operatorname{supp} f$	the support of the function f
∇f	the same as $Df = \left[\frac{\partial f}{\partial x_i}\right]_{i=1}^n$
$\nabla_x F$	the Fréchet derivative of F at x (see the beginning of Section 6.2)
∂G	the boundary of the set G
$\bar{G}$	the closure of the set G
G^0	the interior of the set G
χ_G	the indicator function of the set G; $\chi_G(x) = 1$ if $x \in G$, $\chi_G(x) = 0$ if $x \notin G$
$(\Omega, \mathbb{F}, (\mathbb{F}_t)_{t \geq 0}, P)$	filtered probability space
$\Delta\eta_t$	the jump of η_t defined by $\Delta\eta_t = \eta_t - \eta_t^-$
P	the probability law of η_t
$N(t, U)$	see (1.1.2)
$\nu(U)$	$E[N(1, U)]$ see (1.1.3)
$\|\nu\|$	the norm (total mass) of the measure ν, i.e., $\nu(\mathbb{R})$
$\tilde{N}(dt, dz)$	see (1.1.7)
$B(t)$	Brownian motion
$P \ll Q$	the measure P is absolutely continuous w.r.t. the measure Q
$P \sim Q$	P is equivalent to Q, i.e., $P \ll Q$ and $Q \ll P$
E_Q	the expectation w.r.t. the measure Q
E	the expectation w.r.t. a measure which is clear from the context (usually P)
$E[Y] = E^\mu[Y] = \int Y d\mu$	the expectation of the random variable Y w.r.t. the measure μ
$[X, Y]$	quadratic covariation of X and Y, see Definition 1.28
$\mathcal{T}$	set of all stopping times $\leq \tau_S$ see (3.1.1)
τ_G	the first exit time from the set G of a process X_t : $\tau_G = \inf\{t > 0;\ X_t \notin G\}$
$\Delta_N Y(t)$	the jump of Y caused by the jump of N, see (8.2.2)
$\check{Y}(t^-)$	$Y(t^-) + \Delta_N Y(t)$ (see (9.1.5))
$\Delta_\xi Y(t)$	the jump of Y caused by the singular control ξ
$\Delta_\xi \phi$	see (8.2.3)

$\xi^{\mathrm{c}}(t)$	continuous part of $\xi(t)$, i.e., the process obtained by removing the jumps of $\xi(t)$
π/K	the restriction of the measure π to the set K
$A = A_Y$	the generator of jump diffusion Y
$\mathcal{M}$	intervention operator, see Definition 9.1
VI	variational inequality
QVI	quasivariational inequality
HJB	Hamilton–Jacobi–Bellman equation
HJBVI	Hamilton–Jacobi–Bellman variational inequality
HJBQVI	Hamilton–Jacobi–Bellman quasivariational inequality
SDE	Stochastic differential equation
càdlàg	right continuous with left limits
càglàd	left continuous with right limits
i.i.d.	independent identically distributed
iff	if and only if
a.a., a.e., a.s.	almost all, almost everywhere, almost surely
w.r.t.	with respect to
s.t.	such that

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